

Litmus Rule Reference

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MESSAGE	1015

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CONST_HIER_CG2	1018
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DEFAULT SEVERITY	1018
DESCRIPTION	1018
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CONST_HIER_CLK1	1021
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DEFAULT SEVERITY	1021
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CONST_HIER_CLK2a	1023
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DESCRIPTION	1023
CONST_HIER_CLK2b	1026
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DEFAULT SEVERITY	1026
DESCRIPTION	1026
CONST_HIER_CLK2c	1029
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CONST_HIER_CLK2d	1032
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CONST_HIER_CLK2e	1035
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MESSAGE	1035

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DESCRIPTION	1038
CONST_HIER_CLK3	1041
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MESSAGE	1046
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DESCRIPTION	1046
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NAME	1055

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DESCRIPTION	1059
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DESCRIPTION	1079
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DESCRIPTION	1095
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CONST_HIER_EXC12	1099
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DEFAULT SEVERITY	1099
DESCRIPTION	1099

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DESCRIPTION	1103
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DEFAULT SEVERITY	1111
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MESSAGE	1119
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DESCRIPTION	1119
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CONST_HIER_GRP1	1124
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DESCRIPTION	1134
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MESSAGE	1138
DEFAULT SEVERITY	1138
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CONST_LINT_CG

- [CONST_LINT_CG1](#)
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CONST_LINT_CG1

NAME

CONST_LINT_CG1 (CONSTRAINT_LINT)

MESSAGE

`set_clock_gating_check` applied to an object that is not a cell or a pin

DEFAULT SEVERITY

Warning

DESCRIPTION

The `set_clock_gating_check` command can be applied to clocks, ports, pins, or cells. This rule is checked during design initialization by `init_design` and reported by `report_sdc` command when the object is a clock or a port. It is useful for certain methodologies and can be disabled safely.

For other objects that are not pins or cells, rules [TCLCMD-917](#) and [TCLCMD-513](#) are reported.

Specific rule occurrence attributes for the details about the violation are listed below.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
file_line	unsigned	File number
file_name	string	File name
clock	object	Clock
pin	object	Pin

EXAMPLE

- This rule will be reported for the following commands:

```
set_clock_gating_check [get_clocks CLK]  
set_clock_gating_check [get_ports clk1]
```

CONST_LINT_CG2

NAME

CONST_LINT_CG2 (CONSTRAINT_LINT)

MESSAGE

`set_clock_gating_check` is applied to a clock gating cell

DEFAULT SEVERITY

Warning

DESCRIPTION

A redundant `set_clock_gating_check` command is applied to a clock gating cell or the input of a clock gating cell. This rule is checked during design initialization by `init_design` command and the results can be reported by `report_sdc` command.

A clock gating cell refers to an instance of a technology library cell which has Liberty attribute `clock_gating_integrated_cell`.

A technology library contains clock gating check requirements for a clock gating cell. User-defined clock gating check constraints are not required and is considered redundant.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
file_line	unsigned	File number
file_name	string	File name
scope	object	Inst
pin	object	Pin

EXAMPLE

- The following example contains clock gating cells and clock gating check constraints. This rule is reported for the `set_clock_gating_check` commands, because they are applied to the clock gating cell `i_cg0` and the inputs of the clock gating cells `i_cg1/E` and `i_cg2/CK`.

```
set_clock_gating_check -setup 0.5 -hold 0.25 [get_cells i_cg0]
set_clock_gating_check -setup 0.5 -hold 0.25 [get_pins i_cg1/E]
set_clock_gating_check -setup 0.5 -hold 0.25 [get_pins i_cg2/CK]
```

CONST_LINT_CLK

- [CONST_LINT_CLK1](#)
- [CONST_LINT_CLK2](#)
- [CONST_LINT_CLK3](#)
- [CONST_LINT_CLK4](#)
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- [CONST_LINT_CLK8](#)
- [CONST_LINT_CLK9](#)

CONST_LINT_CLK1

NAME

CONST_LINT_CLK1 (CONSTRAINT_LINT)

MESSAGE

The `-source source_pin` of a generated clock is not defined as source object of another clock prior to the specified generated clock constraint

DEFAULT SEVERITY

Warning

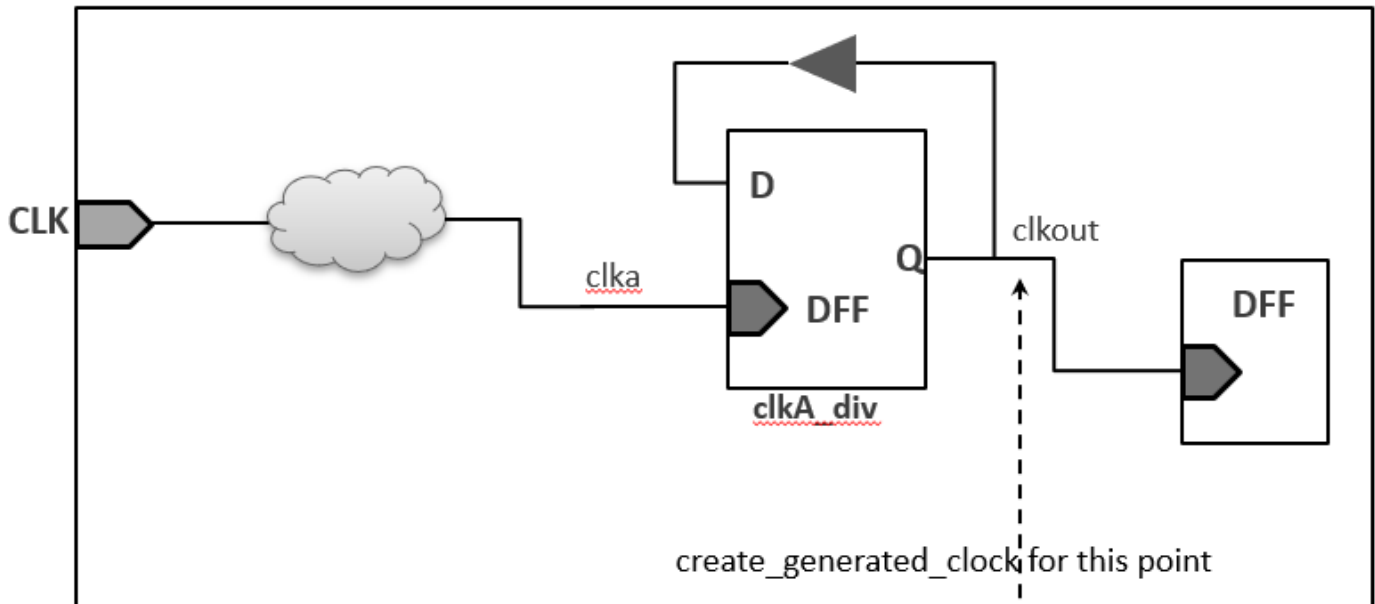
DESCRIPTION

Indicates that there is a master pin (specified with `-source`) of a generated clock that is not the source of another clock. This rule does not point to an incorrect command, but this can be used to enforce a methodology that requires specifying master clock source objects as the generated clock's source pins. This rule is reported even if a clock propagates to the pin specified in `-source`. This rule is checked during design initialization by `init_design` command and the results can be reported by `report_sdc` command.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
file_line	unsigned	File number
file_name	string	File name
pin	object	Pin

EXAMPLE



- The following generates a warning as the `-source <pin>` as indicated by the `create_generated_clock` command points to `clkA` which is not the source of another clock. The `-source` should instead point to `CLK`.

```
create_clock -name MCLK [get_ports CLK]
create_generated_clock -name GCLK -source clkA -divide_by 2 [get_pins
clkA_div/Q]
```

This rule will not be reported if the command is changed to:

```
create_generated_clock -name GCLK -source CLK -divide_by 2 [get_pins clkA_div/Q]
```

CONST_LINT_CLK2

NAME

CONST_LINT_CLK2 (CONSTRAINT_LINT)

MESSAGE

Source object of a generated clock is a port

DEFAULT SEVERITY

Warning

DESCRIPTION

Indicates that a generated clock has been defined on a port. This rule is checked during design initialization by [init_design](#) command and the results can be reported by [report_sdc](#) command.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
file_line	unsigned	File number
file_name	string	File name
pin	object	Pin

EXAMPLE

- This rule will be reported for the following `create_generated_clock` command for generating a clock defined on a port:

```
create_clock -name CK1 -period 20 -waveform {0 10} [get_ports {clk1}]
create_generated_clock -name GCK1 -source [get_ports {clk1}] -divide_by 2
[get_ports {clk2}]
```

This rule will not be reported if the command is changed to the following:

```
create_generated_clock -name GCK1 -source [get_ports {clk1}] -divide_by 2
[get_pins {u10/Q}]
```

CONST_LINT_CLK3

NAME

CONST_LINT_CLK3 (CONSTRAINT_LINT)

MESSAGE

Incomplete clock definition: missing `-waveform`

DEFAULT SEVERITY

Warning

DESCRIPTION

Indicates that there is a clock (real or virtual) definition that does not have the `-waveform` argument. This rule is checked during design initialization by `init_design` command and the results can be reported by `report_sdc` command.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
file_line	unsigned	File number
file_name	string	File name

EXAMPLE

- This rule will be reported for the following command:

```
create_clock -name VCK1 -period 10
```

CONST_LINT_CLK4

NAME

CONST_LINT_CLK4 (CONSTRAINT_LINT)

MESSAGE

Divided/multiplied clock

DEFAULT SEVERITY

Warning

DESCRIPTION

Indicates that you have a divided clock. Specifically, the rule checks for generated clocks that were created using the `-divide_by` or `-multiply_by` options.

This is checked when you read in your SDC files using the READ SDC command.

This rule has specific rule occurrence attributes for the details about the violation:

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
file_line	unsigned	File number
file_name	string	File name
command	string	Constraint command

EXAMPLE

For example, the Litmus flags the following code because gCLK1 is a divided clock:

```
create_generated_clock -name gCLK1 -source CLK1 -divide_by 2 [get_pins ff1/Q]
```

CONST_LINT_CLK5

NAME

CONST_LINT_CLK5 (CONSTRAINT_LINT)

MESSAGE

Edge-derived clock

DEFAULT SEVERITY

Warning

DESCRIPTION

Indicates that you have a generated clock that was created with the -edges or -edge_shift options. This is checked when you read in your SDC files using the READ SDC command.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
file_line	unsigned	File number
file_name	string	File name
command	string	Constraint command

EXAMPLE

The following command causes a rule check violation:

```
create_generated_clock -edges {1 3 7}-edge_shift {2 2 2} -source SYCLK {get_pins  
DIVIDE}
```

CONST_LINT_CLK6

NAME

CONST_LINT_CLK6 (CONSTRAINT_LINT)

MESSAGE

Unusual generated clock option (-add, -master_clock)

DEFAULT SEVERITY

Warning

DESCRIPTION

Indicates that you have a `create_generated_clock` command with either the `-add` or `-master_clock` option.

This is checked when you read in your SDC files using the READ SDC command.

This rule has specific rule occurrence attributes for the details about the violation:

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
file_line	unsigned	File number
file_name	string	File name
command	string	Constraint command

EXAMPLE

For example, the Litmus flags the following command because of the `-add` option.

```
create_generated_clock -name gCLK1 -source CLK1 -divide_by 2 -add [get_pins FF1/Q]
```

CONST_LINT_CLK7

NAME

CONST_LINT_CLK7 (CONSTRAINT_LINT)

MESSAGE

Clock definition point is not an input port, an output pin of a sequential cell or an output pin of a blackbox

DEFAULT SEVERITY

Warning

DESCRIPTION

Indicates that there is a clock source that is neither an input port nor an output pin of a sequential element. This rule is checked during design initialization by `init_design` command and the results can be reported by `report_sdc` command.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
file_line	unsigned	File number
file_name	string	File name
pin	object	Pin
command	string	Constraint command

EXAMPLE

- This rule will be reported for the following command because of U2 being a combinational element:

```
create_clock -name CK1 -period 10 -waveform {0 5} [get_pins {U2/Z}]
```

This will not be reported if the command is changed to the following:

```
create_clock -name CK1 -period 10 -waveform {0 5} [get_ports {clk1}]
```

CONST_LINT_CLK8

NAME

CONST_LINT_CLK8 (CONSTRAINT_LINT)

MESSAGE

Clock has same name as port or pin

DEFAULT SEVERITY

Warning

DESCRIPTION

Indicates that you have a clock with the same name as a port or a pin.

This is checked when you read in your SDC files using the READ SDC command. This can happen when the `create_clock` or `create_generated_clock` command is missing the `-name` option.

 It is not recommended to define clocks whose names coincide with ports or pins, because implicit references to such objects (that is, commands that refer to them without an appropriate `get_clocks/get_ports/get_pins` command) would be ambiguous.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
file_line	unsigned	File number
file_name	string	File name
command	string	Constraint command

clock_name	string	Clock name

EXAMPLE

The first command produces a warning because the `-name` argument is missing, which causes the new clock to have the same name as the port on which it is defined:

```
create_clock -period 10 -waveform {0 5} clk1
```

Where `clk1` is a port.

CONST_LINT_CLK9

NAME

CONST_LINT_CLK9 (CONSTRAINT_LINT)

MESSAGE

Virtual clock name does not match the defined standard

DEFAULT SEVERITY

Warning

DESCRIPTION

Indicates that you have a virtual clock name that does not match any of the wildcard patterns defined by `set_db ltms_const_virtual_clock_naming_style`. Specifically, Litmus checks the `-name` value of the `create_clock` commands.

 A clock is virtual when you use `create_clock` without a defined source list.

This is checked when you read in your SDC files using the READ SDC command.

This rule has specific rule occurrence attributes for the details about the violation:

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
file_line	unsigned	File number
file_name	string	File name
command	string	Constraint command

EXAMPLE

If you do set_db parameter ltms_const_virtual_clock_naming_style *_v, then the following command causes a warning:

```
create_clock -name vir_clk1 -period 10.0
```

CONST_LINT_CMD

- [CONST_LINT_CMD1](#)
- [CONST_LINT_CMD3](#)
- [CONST_LINT_CMD4](#)

CONST_LINT_CMD1

NAME

CONST_LINT_CMD1 (CONSTRAINT_LINT)

MESSAGE

Command is overwritten

DEFAULT SEVERITY

Warning

DESCRIPTION

Indicates that a command has been invalidated by a different command that replaces it. This rule is checked during design initialization by `init_design` command and the results can be reported by `report_sdc` command.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
file_line	unsigned	File number
file_name	string	File name
command	string	command
previous_command	string	Previous constraint command
clocks	object_list	Clocks
pin	object_list	Pins

previous_file_line	object_list	File number
previous_file_name	object_list	File name

EXAMPLE

- In the following pair of commands, the `set_clock_groups` command replaces the `set_false_path` exception already defined on timing paths from CLK1 to CLK2:

```
set_false_path -from [get_clocks CLK1] -to [get_clocks CLK2]
set_clock_groups -asynchronous -name CLK1_CLK2 -group [get_clocks CLK1] -group
[get_clocks CLK2]
```

 The order of these two commands does not affect this check.

CONST_LINT_CMD3

NAME

CONST_LINT_CMD3 (CONSTRAINT_LINT)

MESSAGE

Command is reset

DEFAULT SEVERITY

Warning

DESCRIPTION

Indicates that a command is removed by a `reset_*` command. This rule is checked during design initialization by `init_design` command and the result can be reported by `report_sdc` command.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
file_line	unsigned	File number
file_name	string	File name
command	string	command

EXAMPLE

- The following example contains false path constraint and a `reset_path_exception` command. This rule is reported for the `set_false_path` command, because it is removed by the `reset_path_exception` command.

```
set_false_path -from [get_ports din] -to [get_ports dout]
reset_path_exception -from [get_ports din] -to [get_ports dout]
```


CONST_LINT_CMD4

NAME

CONST_LINT_CMD4 (CONSTRAINT_LINT)

MESSAGE

Conflicting constraints

DEFAULT SEVERITY

Warning

DESCRIPTION

Indicates that conflicting constraints have been specified. The following scenarios can trigger this rule check:

- When more than one clock have been defined on a single pin, unless the second invocation of `create_clock` has the `-add` option
- When more than one of the following commands used for any port: `set_input_transition`, `set_driving_cell`, `set_clock_transition`

This rule is checked during design initialization by `init_design` command and the results can be reported by `report_sdc` command.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
file_line	unsigned	Conflicting constraint line number
file_name	string	Conflicting constraint file name

conflicting_command	string	conflicting_command
existing_command	string	Previous constraint command
object	string	Object
is_clock_overlap	bool	Indicates if the clock pin is redefined

EXAMPLE

- The following causes a warning, because CLK1 and CLK2 share the clk port. To avoid this warning, the CLK2 definition should have an `-add` option.

```
create_clock -name CLK1 [get_ports {clk}] -period 10 -waveform {0 4}  
create_clock -name CLK2 [get_ports {clk}] -period 17 -waveform {0 4}
```

- The following causes a warning, because `set_input_transition` and `set_driving_cell` are both used on port pi[1]:

```
set_input_transition 0.00 [get_ports {pi[1]}]  
set_driving_cell [get_ports {pi[1]}]
```

CONST_LINT_CTR

- [CONST_LINT_CTR1](#)
- [CONST_LINT_CTR2](#)
- [CONST_LINT_CTR3](#)
- [CONST_LINT_CTR4](#)
- [CONST_LINT_CTR5](#)
- [CONST_LINT_CTR7](#)

CONST_LINT_CTR1

NAME

CONST_LINT_CTR1 (CONSTRAINT_LINT)

MESSAGE

Clock transition is set on a virtual clock

DEFAULT SEVERITY

Warning

DESCRIPTION

Indicates that `set_clock_transition` is set on a virtual clock. For other object types, rule [TCLCMD-917](#) and [TCLCMD-513](#) will be reported. This rule is checked during design initialization by `init_design` command and the results can be reported by `report_sdc` command.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
file_line	unsigned	Constraint line number
file_name	string	Constraint file name
clock	object	Clock

EXAMPLE

- The Litmus issues a violation for the following because the clock transition is set on a virtual clock:

```
create_clock -name CLK1_w -period 10  
set_clock_transition -rise 0.75 CLK1_w
```

CONST_LINT_CTR2

NAME

CONST_LINT_CTR2 (CONSTRAINT_LINT)

MESSAGE

Clock transition value outside the characterization range (pre-layout)

DEFAULT SEVERITY

Warning

DESCRIPTION

Indicates that the value for `set_clock_transition` is outside the characterization range. If the ports of the clock specified as an argument to `set_clock_transition` drive a pin of a liberty cell with `min_transition` or `max_transition` defined, the characterization range is defined by these values. If not, the characterization range is defined by the attributes `ltms_const_min_clock_transition` and `ltms_const_max_clock_transition`. This rule is checked during design initialization by `init_design` command and the results can be reported by `report_sdc` command.

This rule is enabled when root attribute `ltms_const_constraint_analysis_type` is set to pre-layout (default: pre-layout).

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
file_line	unsigned	Constraint line number
file_name	string	Constraint file name

clock	object	Clock object
value	double	Transition value
limit	double	Limit
pin	object	Liberty pin with constraint
clock_port	object	Clock port
source	string	Source

EXAMPLE

- If the dofile contains these commands:

```
set_db ltms_const_min_clock_transition 0.1
set_db ltms_const_max_clock_transition 0.5
```

then the following SDC command generates a warning because the clock transition time is not within the specified range:

```
set_clock_transition -min 0.75 CLK1
```

CONST_LINT_CTR3

NAME

CONST_LINT_CTR3 (CONSTRAINT_LINT)

MESSAGE

`set_annotated_transition` outside the characterization range

DEFAULT SEVERITY

Warning

DESCRIPTION

Indicates that the value for `set_annotated_transition` is outside the characterization range. The characterization range is defined by the attributes `ltms_const_min_clock_transition` and `ltms_const_max_clock_transition`. This rule is checked during design initialization by `init_design` command and the results can be reported by `report_sdc` command.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
file_line	unsigned	Command line number
file_name	string	Command file name

EXAMPLE

- If the dofile contains these commands:

```
set_db ltms_const_min_clock_transition 0.1  
set_db ltms_const_max_clock_transition 0.5
```

then the following SDC command will generate a warning:

```
set_annotated_transition -min 0.75 CLK1
```

CONST_LINT_CTR4

NAME

CONST_LINT_CTR4 (CONSTRAINT_LINT)

MESSAGE

Clock transition value outside the characterization range (post-layout)

DEFAULT SEVERITY

Warning

DESCRIPTION

Indicates that the value for `set_input_transition` is outside the characterization range. If the port specified as argument to `set_input_transition` drives a pin of a liberty cell with `min_transition` or `max_transition` defined, the characterization range is defined by these values. If not, the characterization range is defined by the attributes `ltms_const_min_clock_transition` and `ltms_const_max_clock_transition`. This rule is checked during design initialization by `init_design` command and the results can be reported by `report_sdc` command.

This rule is enabled when root attribute `ltms_const_constraint_analysis_type` sets to post-layout (default: pre-layout).

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
file_line	unsigned	File number
file_name	string	File name

EXAMPLE

- If the dofile contains this command:

```
set_db ltms_const_min_clock_transition 0.1  
set_db ltms_const_max_clock_transition 0.5
```

then, the following SDC command generates a warning because the clock transition time is not within the specified range:

```
set_input_transition -min 0.75 CLK1
```

CONST_LINT_CTR5

NAME

CONST_LINT_CTR5 (CONSTRAINT_LINT)

MESSAGE

`set_clock_transition` is set on a clock in post-layout

DEFAULT SEVERITY

Warning

DESCRIPTION

Indicates that `set_clock_transition` is set on a clock in post-layout. This rule is checked during design initialization by `init_design` command and the results can be reported by `report_sdc` command.

This rule is enabled when root attribute `ltms_const_constraint_analysis_type` sets to post-layout (default: pre-layout).

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
file_line	unsigned	File number
file_name	string	File name
clock	object	Clock

EXAMPLE

- The following generates an error only when in post-layout:

```
set_clock_transition -rise 0.75 CLK1
```

CONST_LINT_CTR7

NAME

CONST_LINT_CTR7 (CONSTRAINT_LINT)

MESSAGE

Negative clock transition value

DEFAULT SEVERITY

Warning

DESCRIPTION

Indicates that a negative clock transition value has been specified. This rule is checked during design initialization by `init_design` command and the results can be reported by `report_sdc` command.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
file_line	unsigned	Constraint line number
file_name	string	Constraint file name

EXAMPLE

- The following produces a warning because the clock transition is a negative number:

```
set_clock_transition -fall -0.75 CLK1
```

CONST_LINT_DAT

- [CONST_LINT_DAT1](#)

CONST_LINT_DAT1

NAME

CONST_LINT_DAT1 (CONSTRAINT_LINT)

MESSAGE

`set_max_time_borrow` set on a cell that is not a latch or on a pin other than a data/enable pin of a latch

DEFAULT SEVERITY

Warning

DESCRIPTION

The `set_max_time_borrow` command can be applied to clocks, cells or pins. For other object types, rule [TCLCMD-917](#) and [TCLCMD-513](#) will be reported. This rule is reported when the command is applied to cells that are not level-sensitive or pins that are not data/enable pins of level-sensitive sequential cells. This rule is checked during design initialization by `init_design` command and the results can be reported by `report_sdc` command.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
file_line	unsigned	File number
file_name	string	File name
scope	object	Inst

EXAMPLE

- If cell L1 is a latch, then this rule will be reported for the following command:

```
set_max_time_borrow 5 [get_cells MUX1]  
set_max_time_borrow 5 [get_pins L1/Q]
```

CONST_LINT_EXC

- [CONST_LINT_EXC1](#)
- [CONST_LINT_EXC2](#)

CONST_LINT_EXC1

NAME

CONST_LINT_EXC1 (CONSTRAINT_LINT)

MESSAGE

Duplicate object is specified

DEFAULT SEVERITY

Warning

DESCRIPTION

Indicates that the list of through points specified in `set_false_path`, `set_multicycle_path`, `set_min_delay` or `set_max_delay` command contains a duplicate object. A through point is specified with `-through`, `-rise_through` or `-fall_through` option of the command. This rule is checked during design initialization by `init_design` command and the results can be reported by `report_sdc` command.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
file_line	unsigned	File number
file_name	string	File name
pin	object	Pin

EXAMPLE

- The following example contains false path constraint. This rule is reported for the `set_false_path` command, because the pin `i_buf/I` is specified in both `-through` options.

```
set_false_path -through [get_pins i_buf/*] -through [get_pins i_buf/I]
```

CONST_LINT_EXC2

NAME

CONST_LINT_EXC2 (CONSTRAINT_LINT)

MESSAGE

Exception is specified with clock only

DEFAULT SEVERITY

Warning

DESCRIPTION

Indicates that an exception command (`set_false_path`, `set_multicycle_path`, `set_min_delay` or `set_max_delay`) is specified with clock object only. This rule is checked during design initialization by `init_design` command and the results can be reported by `report_sdc` command.

EXAMPLE

- The following example contains false path constraint. This rule is reported for the `set_false_path` command, because it is specified with clock object only.

```
set_false_path -from [get_clocks clk1] -to [get_clocks clk2]
```

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence Id
file_line	unsigned	File number
file_name	string	File name

command	string	command
---------	--------	---------

CONST_LINT_FLP

- [CONST_LINT_FLP1](#)

CONST_LINT_FLP1

NAME

CONST_LINT_FLP1 (CONSTRAINT_LINT)

MESSAGE

False path does not have both `-from` and `-to`

DEFAULT SEVERITY

Warning

DESCRIPTION

Indicates a `set_false_path` command that does not have both `-from` and `-to` arguments. This rule is checked during design initialization by `init_design` command and the results can be reported by `report_sdc` command.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
file_line	unsigned	File number
file_name	string	File name

EXAMPLE

- This rule will be reported for the following command:

```
set_false_path -from [get_cells conf_reg_1]
```

CONST_LINT_HFN

- [CONST_LINT_HFN1](#)
- [CONST_LINT_HFN2](#)

CONST_LINT_HFN1

NAME

CONST_LINT_HFN1 (CONSTRAINT_LINT)

MESSAGE

Invalid constraint in current context (post-layout)

DEFAULT SEVERITY

Warning

DESCRIPTION

Indicates that an invalid constraint in the current context is used. Specifically, the Litmus flags `set_dont_touch`, `set_dont_touch_network`, `set_ideal_net`, `set_ideal_network`, `set_ideal_latency`, and `set_ideal_transition`. This rule is checked during design initialization by `init_design` command and the results can be reported by `report_sdc` command.

This rule is enabled when root attribute `ltms_const_constraint_analysis_type` sets to post-layout (default: pre-layout).

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
file_line	unsigned	File number
file_name	string	File name
command	string	Constraint command

EXAMPLE

- This rule will be reported for any of the following commands:

```
set_dont_touch_network [all_clocks]  
set_dont_touch [get_cells gc1]
```

CONST_LINT_HFN2

NAME

CONST_LINT_HFN2 (CONSTRAINT_LINT)

MESSAGE

`set_dont_touch` should not be used on library cells

DEFAULT SEVERITY

Warning

DESCRIPTION

Indicates that the `set_dont_touch` command was specified for a libcell object. This rule is checked during design initialization by `init_design` command and the results can be reported by `report_sdc` command.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
file_line	unsigned	File number
file_name	string	File name

EXAMPLE

- This rule will be reported in a case like the following:

```
set_dont_touch [get_lib_cells {slow/OR2LLX05}]
```

CONST_LINT_IDL

- [CONST_LINT_IDL1](#)
- [CONST_LINT_IDL2](#)
- [CONST_LINT_IDL3](#)
- [CONST_LINT_IDL4](#)
- [CONST_LINT_IDL5](#)
- [CONST_LINT_IDL6](#)
- [CONST_LINT_IDL7](#)

CONST_LINT_IDL1

NAME

CONST_LINT_IDL1 (CONSTRAINT_LINT)

MESSAGE

Incorrect input delay constraint vs. several clocks (missing `-add_delay`)

DEFAULT SEVERITY

Warning

DESCRIPTION

Indicates that, for an input port that is driven by more than one clock, `set_input_delay` command is missing the `-add_delay` option, which is required for each additional clock that drives the input port. This rule is checked during design initialization by `init_design` command and the results can be reported by `report_sdc` command.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
file_line	unsigned	File number
file_name	string	File name
pin	object	Pin

EXAMPLE

- The following causes a rule check violation because there is a missing `-add_delay` option for CLK1:

```
set_input_delay 2.0 -clock CLK0 {IN1}  
set_input_delay 3.0 -clock CLK1 {IN1}
```


CONST_LINT_IDL2

NAME

CONST_LINT_IDL2 (CONSTRAINT_LINT)

MESSAGE

Detected input delay vs. clock that is not virtual

DEFAULT SEVERITY

Warning

DESCRIPTION

Indicates that there is an input delay specified for a clock that is not virtual. This rule is checked during design initialization by `init_design` command and the results can be reported by `report_sdc` command.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
file_line	unsigned	File number
file_name	string	File name
clock	object	Real/Generated clock's definition

EXAMPLE

- The following produces a warning message if CLK1 is a real clock (rather than a virtual clock):

```
set_input_delay 2.75 -min -clock CLK1 {IN1}
```

CONST_LINT_IDL3

NAME

CONST_LINT_IDL3 (CONSTRAINT_LINT)

MESSAGE

Detected level sensitive input

DEFAULT SEVERITY

Warning

DESCRIPTION

Indicates that is a level-sensitive input port. Specifically, you have a `set_input_delay` command that uses `-level_sensitive`.

This is checked when you read in your SDC files using the `READ SDC` command.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
file_line	unsigned	File number
file_name	string	File name
command	string	Constraint command

EXAMPLE

The following command causes a warning because it uses a level-sensitive delay.

```
set_input_delay 2.0 -clock CLK1 -level_sensitive {IN1}
```

CONST_LINT_IDL4

NAME

CONST_LINT_IDL4 (CONSTRAINT_LINT)

MESSAGE

Detected edge sensitive input relative to clock falling edge

DEFAULT SEVERITY

Warning

DESCRIPTION

Indicates that there is an edge-sensitive input in relation to the clock falling edge. Specifically, you have a `set_input_delay` command that uses `-clk_fall`.

This is checked when you read in your SDC files using the `READ SDC` command.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
file_line	unsigned	File number
file_name	string	File name
command	string	Constraint command

EXAMPLE

The following command causes a warning because its edge-sensitive delay input uses a clock falling edge.

```
set_input_delay 2.0 -clock CLK1 -clock_fall {IN1}
```

CONST_LINT_IDL5

NAME

CONST_LINT_IDL5 (CONSTRAINT_LINT)

MESSAGE

Unusual input delay option

DEFAULT SEVERITY

Warning

DESCRIPTION

Indicates that you have a set_input_delay command that uses either -source_latency_included or -network_latency_included.

This is checked when you read in your SDC files using the READ SDC command.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
file_line	unsigned	File number
file_name	string	File name
command	string	Constraint command
is_source	bool	Is source?

EXAMPLE

The following command causes a violation because it uses -source_latency_included.

```
set_input_delay 2.0 -clock CLK1 -source_latency_included {IN1}
```

CONST_LINT_IDL6

NAME

CONST_LINT_IDL6 (CONSTRAINT_LINT)

MESSAGE

Negative input delay value

DEFAULT SEVERITY

Warning

DESCRIPTION

Indicates that a `set_input_delay` command has a negative value. This rule is checked during design initialization by `init_design` command and the results can be reported by `report_sdc` command.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
file_line	unsigned	File number
file_name	string	File name

EXAMPLE

- The following code causes a violation because it has a negative delay value:

```
set_input_delay -2.0 -clock CLK1 {IN1}
```


CONST_LINT_IDL7

NAME

CONST_LINT_IDL7 (CONSTRAINT_LINT)

MESSAGE

`set_input_delay` defined on an internal pin

DEFAULT SEVERITY

Warning

DESCRIPTION

Indicates that the SDC file specifies `set_input_delay` on a pin internal to the current design (not a port). In some cases, this can cause timing path splitting and rule `CONST_LINT_REF4` to be reported. This rule does not necessarily indicate a problem depending on the methodology used. This rule is checked during design initialization by `init_design` command and the results can be reported by `report_sdc` command.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
file_line	unsigned	File number
file_name	string	File name
pin	object	Pin

EXAMPLE

- This rule will be reported for the following command:

```
set_input_delay 1 [get_pins blk/d1]
```

CONST_LINT_INFO

- [CONST_LINT_INFO1](#)
- [CONST_LINT_INFO2](#)
- [CONST_LINT_INFO3](#)

CONST_LINT_INFO1

NAME

CONST_LINT_INFO1 (CONSTRAINT_LINT)

MESSAGE

Constraint is not found

DEFAULT SEVERITY

Warning

DESCRIPTION

Indicates that no constraint is present at top or block. Constraint files are setup by `read_sdc`, `set_target_check_constraints`, or `read_mmmc` command and loaded during design initialization by `init_design` command. This rule is checked during design initialization by `init_design` command and the results can be reported by `report_sdc` command.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
file_line	unsigned	File number
file_name	string	File name
scope	string	scope

EXAMPLE

- The following example loads the constraints specified in null.sdc into the default constraint mode. This rule is reported for the default constraint mode, because null.sdc does not contain any constraint.

```
read_netlist -top top top1.v
read_sdc null.sdc
init_design
```

CONST_LINT_INFO2

NAME

CONST_LINT_INFO2 (CONSTRAINT_LINT)

MESSAGE

SDC version (sdc_version) is not specified in a constraint file

DEFAULT SEVERITY

Ignore (default Warning)

DESCRIPTION

Indicates that no SDC version is specified in the constraint file. Constraint files are setup by `read_sdc`, `set_target_check_constraints`, or `read_mmmc` command and loaded during design initialization by `init_design` command. This rule is checked during design initialization by `init_design` command and the results can be reported by `report_sdc` command.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
file_line	unsigned	File number
file_name	string	File name
sdc_file_name	string	SDC file name

CONST_LINT_INFO3

NAME

CONST_LINT_INFO3 (CONSTRAINT_LINT)

MESSAGE

SDC version (sdc_version) specified in a constraint file is under minimum limit

DEFAULT SEVERITY

Ignore (default Warning)

DESCRIPTION

Indicates that SDC version specified in the constraint file is under minimum limit. Constraint files are setup by [read_sdc](#), [set_target_check_constraints](#), or [read_mmmc](#) command and loaded during design initialization by [init_design](#) command. This rule is checked during design initialization by [init_design](#) command and the results can be reported by [report_sdc](#) command.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
file_line	unsigned	File number
file_name	string	File name
sdc_file_name	string	SDC file name
sdc_version	string	SDC Version

CONST_LINT_ITR

- [CONST_LINT_ITR1](#)
- [CONST_LINT_ITR2](#)
- [CONST_LINT_ITR3](#)
- [CONST_LINT_ITR4](#)

CONST_LINT_ITR1

NAME

CONST_LINT_ITR1 (CONSTRAINT_LINT)

MESSAGE

Input transition value outside characterization range

DEFAULT SEVERITY

Warning

DESCRIPTION

Indicates that the input transition value is outside the characterization range. If the port specified as an argument to `set_input_transition` drives a pin of a liberty cell with `min_transition` or `max_transition` defined, the characterization range is defined by these values. If not, the characterization range is defined by the attributes `ltms_const_min_input_transition` and `ltms_const_max_input_transition`. This rule is checked during design initialization by `init_design` command and the results can be reported by `report_sdc` command.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
file_line	unsigned	File number
file_name	string	File name
port_pin	object	Port
value	double	Transition value

limit	double	Limit

EXAMPLE

- If the following commands are in the dofile:

```
set_db ltms_const_min_input_transition 0.1
set_db ltms_const_max_input_transition 0.5
```

Then the following SDC command will generate a warning because the value is not within the specified range:

```
set_input_transition 0.8 [get_ports {in1}]
```

CONST_LINT_ITR2

NAME

CONST_LINT_ITR2 (CONSTRAINT_LINT)

MESSAGE

Unusual set driving cell options

DEFAULT SEVERITY

Warning

DESCRIPTION

Indicates that you have instances of `set_input_transition -clock/-clock_fall` and `set_driving_cell -min/-max/-clock/-clock_fall`. Litmus issues a warning when it detects these options, because they are not present in all versions of SDC.

This is checked when you read in your SDC files using the READ SDC command.

This rule has specific rule occurrence attributes for the details about the violation:

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
file_line	unsigned	File number
file_name	string	File name
command	string	Constraint command
unusual_option	string	Unusual option

CONST_LINT_ITR3

NAME

CONST_LINT_ITR3 (CONSTRAINT_LINT)

MESSAGE

`set_drive` is not recommended

DEFAULT SEVERITY

Warning

DESCRIPTION

Indicates that the `set_drive` command has been used. This command works with a linear delay model. It is inaccurate when using non-linear delay models. This command has been superseded by `set_driving_cell`, which works with all delay models accurately. This rule is checked during design initialization by `init_design` command and the results can be reported by `report_sdc` command.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
file_line	unsigned	File number
file_name	string	File name

EXAMPLE

- This rule will be reported for the following command:

```
set_drive -rise 0.75 [get_ports {IN1}]
```

CONST_LINT_ITR4

NAME

CONST_LINT_ITR4 (CONSTRAINT_LINT)

MESSAGE

Detected driving cell that is not recommended

DEFAULT SEVERITY

Warning

DESCRIPTION

Indicates that a `set_driving_cell` command has specified a library cell that is not one of the cells listed in the attribute `ltms_const_allowed_driving_cell`. If the value of this attribute is empty (the default), then all library cells are allowed. This rule is checked during design initialization by `init_design` command and the results can be reported by `report_sdc` command.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
file_line	unsigned	File number
file_name	string	File name
command	string	Constraint command

EXAMPLE

- If the dofile contains this command:

```
set_db ltms_const_allowed_driving_cell "BFLX16 BFLX12"
```

then this rule would report a warning for the following SDC command:

```
set_driving_cell -lib_cell [get_lib_cells OR2LLX05] [get_ports {in1}]
```

CONST_LINT_LAT

- [CONST_LINT_LAT1](#)
- [CONST_LINT_LAT2](#)
- [CONST_LINT_LAT3](#)
- [CONST_LINT_LAT4](#)

CONST_LINT_LAT1

NAME

CONST_LINT_LAT1 (CONSTRAINT_LINT)

MESSAGE

Clock latency is set on an object that is not a clock (pre-layout)

DEFAULT SEVERITY

Warning

DESCRIPTION

Indicates that clock latency is set on an object that is not a clock (such as a port or a pin). This rule is checked during design initialization by `init_design` command and the results can be reported by `report_sdc` command.

This rule is enabled when root attribute `ltms_const_constraint_analysis_type` sets to pre-layout (default: pre-layout).

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
file_line	unsigned	File number
file_name	string	File name

EXAMPLE

- The following code causes a violation because clock latency should be set on CLK2_w, not on CLK2:

```
create_clock -name CLK2_w -period 10 CLK2
set_clock_latency 2.0 -source CLK2
```

The following is the correct version:

```
set_clock_latency 2.0 -source [get_clocks CLK2_w]
```

CONST_LINT_LAT2

NAME

CONST_LINT_LAT2 (CONSTRAINT_LINT)

MESSAGE

Detected `set_clock_latency` (in post-layout stage)

DEFAULT SEVERITY

Warning

DESCRIPTION

Indicates that there is clock network latency (`set_clock_latency` without -source) defined on a non-virtual clock during post-layout. This rule is checked during design initialization by `init_design` command and the results can be reported by `report_sdc` command.

This rule is enabled when root attribute `ltms_const_constraint_analysis_type` sets to post-layout (default: pre-layout).

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
file_line	unsigned	File number
file_name	string	File name
command	string	Constraint command
object_command	string	Object command

EXAMPLE

- In the following example, the Litmus issues a warning because the clock should not have latency in the post-layout stage:

```
set_clock_latency 1.5 -rise CLK1
```

- The following commands will not be reported since they specify source latency or latency on virtual clock:

```
set_clock_latency 2.0 -source CLK2  
set_clock_latency 1.0 -min CLK2_vir
```

CONST_LINT_LAT3

NAME

CONST_LINT_LAT3 (CONSTRAINT_LINT)

MESSAGE

Negative clock latency value

DEFAULT SEVERITY

Warning

DESCRIPTION

Indicates that there is a negative value for `set_clock_latency`. This rule is checked during design initialization by `init_design` command and the results can be reported by `report_sdc` command.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
file_line	unsigned	File number
file_name	string	File name

EXAMPLE

- The following command causes a violation because `-min` is set to -2.0:

```
set_clock_latency -2.0 -min CLK1
```

CONST_LINT_LAT4

NAME

CONST_LINT_LAT4 (CONSTRAINT_LINT)

MESSAGE

In `set_clock_latency`, `-clock` does not affect clocks in the `object_list`

DEFAULT SEVERITY

Warning

DESCRIPTION

The `-clock` option is used to specify latencies on objects that are not clocks, when applying different latency values to paths that are constrained by different clocks. When the latency is specified on a clock object, the `-clock` option has no effect. This rule is checked during design initialization by `init_design` command and the results can be reported by `report_sdc` command.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
file_line	unsigned	File number
file_name	string	File name

EXAMPLE

- The Litmus issues a warning for lines such as:

```
set_clock_latency 2.0 -clock CK1 -rise [get_clocks CK2]  
set_clock_latency 1.0 -clock CK2 [list [get_clocks CK3] [get_ports clk4]]
```

CONST_LINT_MCP

- [CONST_LINT_MCP1](#)

CONST_LINT_MCP1

NAME

CONST_LINT_MCP1 (CONSTRAINT_LINT)

MESSAGE

Multi-cycle path does not have both `-from` and `-to`

DEFAULT SEVERITY

Warning

DESCRIPTION

Indicates a `set_multicycle_path` command that does not have both `-from` and `-to` arguments. This rule is checked during design initialization by `init_design` command and the results can be reported by `report_sdc` command.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
file_line	unsigned	File number
file_name	string	File name

EXAMPLE

- This rule will be reported for the following command:

```
set_multicycle_path 2 -through [get_pins mul/and2/Z]
```

CONST_LINT_ODL

- [CONST_LINT_ODL1](#)
- [CONST_LINT_ODL2](#)
- [CONST_LINT_ODL3](#)
- [CONST_LINT_ODL4](#)
- [CONST_LINT_ODL5](#)
- [CONST_LINT_ODL6](#)
- [CONST_LINT_ODL7](#)
- [CONST_LINT_ODL8](#)

CONST_LINT_ODL1

NAME

CONST_LINT_ODL1 (CONSTRAINT_LINT)

MESSAGE

Incorrect output delay constraint vs. several clocks (missing `-add_delay`)

DEFAULT SEVERITY

Warning

DESCRIPTION

Indicates that there is an incorrect output delay constraint versus multiple clocks. Specifically, the Litmus checks output ports that are driven by more than one clock; `set_output_delay` commands must have an `-add_delay` option defined for each additional clock that drives the output port. This rule is checked during design initialization by `init_design` command and the results can be reported by `report_sdc` command.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
file_line	unsigned	File number
file_name	string	File name
pin	object	Pin

EXAMPLE

- The following command causes a rule check violation because there is a missing `-add_delay` option for CLK2:

```
set_output_delay 3.0 -clock CLK1 {OUT2}  
set_output_delay 2.75 -clock CLK2 {OUT2}
```

CONST_LINT_ODL2

NAME

CONST_LINT_ODL2 (CONSTRAINT_LINT)

MESSAGE

Detected output delay vs. clock that is not virtual

DEFAULT SEVERITY

Warning

DESCRIPTION

Indicates that there is an output delay specified versus a clock that is not virtual. This rule is checked during design initialization by `init_design` command and the results can be reported by `report_sdc` command.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
file_line	unsigned	File number
file_name	string	File name
clock	object	Real/Generated clock's definition

EXAMPLE

- The following produces a warning because `set_output_delay` should specify the virtual clock, vCLK3, rather than the real clock, CLK3:

```
create_clock -period 10 CLK3
create_clock -period 10 -name vCLK3
set_output_delay 0.75 -min -clock CLK3 {OUT1}
```

CONST_LINT_ODL3

NAME

CONST_LINT_ODL3 (CONSTRAINT_LINT)

MESSAGE

Detected level sensitive output

DEFAULT SEVERITY

Warning

DESCRIPTION

Indicates level-sensitive outputs. Specifically, there is a `set_output_delay` command that uses `-level_sensitive`.

This is checked when you read in your SDC files using the READ SDC command.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
file_line	unsigned	File number
file_name	string	File name
command	string	Constraint command

EXAMPLE

The following command causes a warning because it uses a level-sensitive output delay.

```
set_output_delay 2.0 -clock CLK1 -level_sensitive {OUT1A}
```

CONST_LINT_ODL4

NAME

CONST_LINT_ODL4 (CONSTRAINT_LINT)

MESSAGE

Detected edge sensitive output relative to clock falling edge

DEFAULT SEVERITY

Warning

DESCRIPTION

Indicates that there is an edge-sensitive output in relation to a clock-falling edge. Specifically, there is a `set_output_delay` command that uses `-clock_fall`.

This is checked when you read in your SDC files using the `READ SDC` command.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
file_line	unsigned	File number
file_name	string	File name
command	string	Constraint command

EXAMPLE

The following command causes a warning because its edge-sensitive output delay uses a clock falling edge.

```
set_output_delay 2.0 -clock CLK1 -clock_fall {OUT1A}
```


CONST_LINT_ODL5

NAME

CONST_LINT_ODL5 (CONSTRAINT_LINT)

MESSAGE

Unusual output delay option

DEFAULT SEVERITY

Warning

DESCRIPTION

Indicates that you have a set_output_delay command that uses either -source_latency_included or -network_latency_included.

This is checked when you read in your SDC files using the READ SDC command.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
file_line	unsigned	File number
file_name	string	File name
command	string	Constraint command
is_source	bool	Is source ?

EXAMPLE

The following command causes a violation because it uses -source_latency_included.

```
set_output_delay 2.0 -clock CLK1 -source_latency_included {OUT1A}
```

CONST_LINT_ODL6

NAME

CONST_LINT_ODL6 (CONSTRAINT_LINT)

MESSAGE

Negative `-max` output delay value

DEFAULT SEVERITY

Warning

DESCRIPTION

Indicates that there is a negative maximum output delay value. Specifically, this rule check looks for `set_output_delay` commands that have a negative value, except for those where `-min` is specified and `-max` is not. This rule is checked during design initialization by `init_design` command and the results can be reported by `report_sdc` command.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
file_line	unsigned	File number
file_name	string	File name

EXAMPLE

- The following command causes a violation because it has a negative delay value:

```
set_output_delay -2.0 -clock CLK1 {OUT1A}
```

CONST_LINT_ODL7

NAME

CONST_LINT_ODL7 (CONSTRAINT_LINT)

MESSAGE

Positive/zero `-min` output delay value

DEFAULT SEVERITY

Warning

DESCRIPTION

Indicates that there is a positive/zero minimum delay value. In a design methodology where there is no glue logic between blocks and all block inputs are registered, the delay from a block's output to a register in another block is zero.

Because the register is outside the block, the hold check is done versus the ideal clock edge. To model the real hold check, the output delay must be negative, and its absolute value must be equal to the expected hold time requirement. This is the reason this rule reports non-negative `-min` output delays.

In a design methodology using glue logic, or where output delays are set on the top-level output ports, the external delay from the output to the capturing register could be positive and bigger than the hold requirement. In which case, the correct output delay value would be positive. For this methodology, this rule is not as useful and might be disabled.

This rule is checked during design initialization by `init_design` command and the results can be reported by `report_sdc` command.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
-----------	------	-------------

occurrence_id	unsigned	Occurrence id
file_line	unsigned	File number
file_name	string	File name

EXAMPLE

- The following command causes a violation because it does not have a negative `-min` delay value:

```
set_output_delay -min 0 -clock CLK1 {OUT1A}
```

CONST_LINT_ODL8

NAME

CONST_LINT_ODL8 (CONSTRAINT_LINT)

MESSAGE

`set_output_delay` defined on an internal pin

DEFAULT SEVERITY

Warning

DESCRIPTION

Indicates that the SDC file specifies `set_output_delay` on a pin internal to the current design (not a port). In some cases, this can cause timing path splitting and rule `CONST_LINT_REF4` to be reported. This rule does not necessarily indicate a problem depending on the methodology used. This rule is checked during design initialization by `init_design` command and the results can be reported by `report_sdc` command.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
file_line	unsigned	File number
file_name	string	File name
pin	object	Pin

EXAMPLE

- This rule will be reported for the following command:

```
set_output_delay 1 [get_pins blk/d1]
```

CONST_LINT_OLD

- [CONST_LINT_OLD1](#)
- [CONST_LINT_OLD2](#)

CONST_LINT_OLD1

NAME

CONST_LINT_OLD1 (CONSTRAINT_LINT)

MESSAGE

Load value outside characterization range

DEFAULT SEVERITY

Warning

DESCRIPTION

Indicates that the load value is outside the characterization range. If the port specified as an argument to `set_load` is driven by a pin of a liberty cell with `min_capacitance` or `max_capacitance` defined, the characterization range is defined by these values. If not, the characterization range is defined by the attributes `ltms_const_min_capacitance` and `ltms_const_max_capacitance`. This rule is checked during design initialization by `init_design` command and the results can be reported by `report_sdc` command.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
file_line	unsigned	File number
file_name	string	File name
port_pin	object	Port
value	double	Capacitance value
limit	double	Limit
pin	object	Liberty pin with constraint

EXAMPLE

- If the attribute `ltms_const_min_capacitance` is set to 0.1 and the attribute `ltms_const_max_capacitance` is set to 0.5, then the following command causes a warning because 0.75 is outside the characterization range:

```
set_load 0.75 {OUT1}
```

CONST_LINT_OLD2

NAME

CONST_LINT_OLD2 (CONSTRAINT_LINT)

MESSAGE

Used `set_load -min/-max` when not in min-max mode

DEFAULT SEVERITY

Ignore (default Warning)

DESCRIPTION

Indicates that `set_load` is used with `-min/-max` option but the design has not been previously set to be timed in min-max mode. To set min-max mode, `timing_analysis_type` attribute needs to be either `best_case_worst_case` or `ocv` or set to default. This rule is checked during design initialization by `init_design` command and the results can be reported by `report_sdc` command.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id 123
file_line	unsigned	File number
file_name	string	File name

EXAMPLE

- This rule will be reported for the following commands:

```
set_db timing_analysis_type single
set_load -min -pin_load 0.25 [get_ports Z]
```

CONST_LINT_REF

- [CONST_LINT_REF1](#)
- [CONST_LINT_REF2](#)
- [CONST_LINT_REF4](#)

CONST_LINT_REF1

NAME

CONST_LINT_REF1 (CONSTRAINT_LINT)

MESSAGE

Constraints is applied on hierarchical pin

DEFAULT SEVERITY

Warning

DESCRIPTION

Indicates that `constraints` is applied on hierarchical pin. This rule is checked during design initialization by `init_design` command and the results can be reported by `report_sdc` command.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
file_line	unsigned	File number
file_name	string	File name
pin	object_list	Pins
command	string	Constraint command

EXAMPLE

- The following command causes a rule check violation if testkey/test_clk is a logical pin:

```
create_generated_clock -edges {1 2 3} -source [get_ports clka] [get_pins  
testkey/test_clk]
```

CONST_LINT_REF2

NAME

CONST_LINT_REF2 (CONSTRAINT_LINT)

MESSAGE

Invalid object specified

DEFAULT SEVERITY

Warning

DESCRIPTION

Indicates that an object specified in a command argument has the wrong object type, is not a valid reference for the specific argument, or is an invalid object by itself (for example, a generated clock that has no master clock). The type of an object referenced by an SDC command is one of the following:

- clock
- cell
- design
- lib
- lib_cell
- lib_pin
- net
- pin
- port

This rule is checked during design initialization by `init_design` command and the results can be reported by `report_sdc` command.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
file_line	unsigned	File number
file_name	string	File name
command	string	Constraint command
pin	object	Pin

EXAMPLE

- The following command causes a rule check violation because output port is not a valid object for -from option:

```
set_false_path -from [get_ports OUT]
```


CONST_LINT_REF4

NAME

CONST_LINT_REF4 (CONSTRAINT_LINT)

MESSAGE

Object is forced to be timing start/end point

DEFAULT SEVERITY

Warning

DESCRIPTION

Indicates that an object specified in `set_input_delay` command or with `-from` option of `set_min_delay` or `set_max_delay` command is forced to be a timing start point. Moreover, an object specified in `set_output_delay` command or with `-to` option of `set_min_delay` or `set_max_delay` command is forced to be a timing end point. Thereafter, it causes timing path segmentation. This rule is checked during design initialization by `init_design` command and the results can be reported by `report_sdc` command.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
file_line	unsigned	File number
file_name	string	File name
start_points	object_list	Start points
end_points	object_list	End points
command	string	delay_constraints

EXAMPLE

- The following example contains commands causing timing path segmentation. This rule is reported for the `set_input_delay`, `set_output_delay`, `set_min_delay`, and `set_max_delay` commands, because they force the pins `i_buf0/I` and `i_buf1/I` to be timing start points, moreover, force the pins `i_buf0/O` and `i_buf1/O` to be timing end points.

```
set_input_delay -clock [get_clocks vclk] 1 [get_pins i_buf0/I]
set_output_delay -clock [get_clocks vclk] 1 [get_pins i_buf0/O]
set_max_delay 4 -from [get_pins i_buf1/I] -to [get_pins i_buf1/O]
set_min_delay 2 -from [get_pins i_buf1/I] -to [get_pins i_buf1/O]
```

CONST_LINT_SCA

- [CONST_LINT_SCA2](#)
- [CONST_LINT_SCA3](#)

CONST_LINT_SCA2

NAME

CONST_LINT_SCA2 (CONSTRAINT_LINT)

MESSAGE

[set_case_analysis](#) defined on a port

DEFAULT SEVERITY

Ignore (default Warning)

DESCRIPTION

Indicates that [set_case_analysis](#) is applied on a port. This rule is checked during design initialization by [init_design](#) command and the results can be reported by [report_sdc](#) command.

 This rule is disabled by default. Use [set_rule_handling -unsuppress](#) command to enable the rule.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
file_line	unsigned	File line number
file_name	string	File name
pin	object	Port

EXAMPLE

- The following command causes a violation because `set_case_analysis` is applied on a port:

```
set_case_analysis 1 [get_ports my_port]
```

CONST_LINT_SCA3

NAME

CONST_LINT_SCA3 (CONSTRAINT_LINT)

MESSAGE

[set_case_analysis](#) defined on a pin

DEFAULT SEVERITY

Ignore (default Warning)

DESCRIPTION

Indicates that [set_case_analysis](#) is applied on a pin. This rule is checked during design initialization by [init_design](#) command and the results can be reported by [report_sdc](#) command.

 This rule is disabled by default. Use [set_rule_handling -unsuppress](#) command to enable the rule.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
file_line	unsigned	File line number
file_name	string	File name
pin	object	Pin

EXAMPLE

- The following command causes a violation because `set_case_analysis` is applied on a pin:

```
set_case_analysis 1 [get_pins my_pin]
```

CONST_LINT_SCOPE

- [CONST_LINT_SCOPE2](#)

CONST_LINT_SCOPE2

NAME

CONST_LINT_SCOPE2 (CONSTRAINT_LINT)

MESSAGE

Return value of full_name attribute is out of scope

DEFAULT SEVERITY

Warning

DESCRIPTION

Indicates that the return value of full_name attribute of block level constraint is out of block-level scope. The rule check is only effective for block level constraint in hierarchical flow. Not effective in non-hierarchical flow.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence Id
file_line	unsigned	File number
file_name	string	File name
objType	string	objType

CONST_LINT_SKIPPED

NAME

CONST_LINT_SKIPPED (CONSTRAINT_LINT)

MESSAGE

SDC command is skipped

DEFAULT SEVERITY

Warning

DESCRIPTION

Indicates that a command in block level constraints is skipped. In Constraint Hierarchical Signoff flow, block level constraints are loaded into block scope, and all the objects referred in the block level constraints need to be accessible by the commands in the top level.

A command in block level constraints is skipped if it accepts port object and cannot accept pin object. The following are the possible causes for this violation:

- `set_load` command applied to port at block level
- `set_max_fanout` command applied to input port at block level
- `set_min_fanout` command applied to input port at block level
- `set_wire_load_model` command applied to input port at block level
- `set_drive` command applied to port at block level
- `set_driving_cell` command applied to port at block level
- `set_fanout_load` command applied to port at block level
- `set_input_transition` command applied to port at block level

This rule is checked during design initialization by `init_design` command and the results can be

reported by `report_sdc` command.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
file_line	unsigned	File number
file_name	string	File name
command	string	Constraint command

EXAMPLE

- The following example contains `set_input_transition` command in block level constraints. This rule is reported for the `set_input_transition` command, because it is applied to the port `blk_din` in the block level which cannot be accessible by the `set_input_transition` command in the top level. Note that `set_input_transition` command accepts port object only.

Block level constraints

```
set_input_transition 2 [get_ports blk_din]
```

CONST_LINT_UNC

- [CONST_LINT_UNC1](#)
- [CONST_LINT_UNC2](#)

CONST_LINT_UNC1

NAME

CONST_LINT_UNC1 (CONSTRAINT_LINT)

MESSAGE

Clock uncertainty is set for an object that is not a clock

DEFAULT SEVERITY

Warning

DESCRIPTION

Indicates that clock uncertainty is set for an object that is not a clock. This rule is checked during design initialization by `init_design` command and the results can be reported by `report_sdc` command.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
file_line	unsigned	File number
file_name	string	File name
pin	object	Pin

EXAMPLE

- In the following example, the Litmus issues a warning because IN1 is not a clock:

```
set_clock_uncertainty 2.5 -setup IN1
```


CONST_LINT_UNC2

NAME

CONST_LINT_UNC2 (CONSTRAINT_LINT)

MESSAGE

Negative clock uncertainty value

DEFAULT SEVERITY

Warning

DESCRIPTION

Indicates that there is a negative value for clock uncertainty. This rule is checked during design initialization by `init_design` command and the results can be reported by `report_sdc` command.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
file_line	unsigned	File number
file_name	string	File name

EXAMPLE

- The following example generates a warning because it specifies a negative clock uncertainty:

```
set_clock_uncertainty -1.5 -setup CLK1
```

CONST_LINTCFG_CMD

- [CONST_LINTCFG_CMD2](#)
- [CONST_LINTCFG_CMD3](#)
- [CONST_LINTCFG_CMD4](#)

CONST_LINTCFG_CMD2

NAME

CONST_LINTCFG_CMD2 (CONSTRAINT_LINT)

MESSAGE

Unsupported command used

DEFAULT SEVERITY

Warning

DESCRIPTION

Indicates that an unsupported command is used in an SDC file. The software recognizes this command, but ignores it. Commands can also be configured as unsupported by setting the `usage` command parameter to a value of `unsupported`. This rule is checked during design initialization by `init_design` command and the results can be reported by `report_sdc` command.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
file_line	unsigned	File number
file_name	string	File name
command	string	Constraint command
msg	string	Message

EXAMPLE

- The following command causes a rule check violation because the software does not recognize the `read_verilog` command:

```
read_verilog
```

CONST_LINTCFG_CMD3

NAME

CONST_LINTCFG_CMD3 (CONSTRAINT_LINT)

MESSAGE

User defined forbidden command used

DEFAULT SEVERITY

Warning

DESCRIPTION

Indicates that a forbidden command is used in an SDC file. To configure a command as forbidden, the `configure_lint_check` command for that command is used. This command has a parameter called `usage`, where the command can be specified as `forbidden`. This rule is checked during design initialization by `init_design` command and the results can be reported by `report_sdc` command.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
file_line	unsigned	File number
file_name	string	File name

EXAMPLE

- To configure `set_dont_touch` as forbidden, the following command in a lint configuration file can be used:

```
configure_lint_check set_dont_touch { command { usage forbidden } }
```

After reading that configuration file, the following command causes a rule check violation because the `set_dont_touch` command is used:

```
set_dont_touch [get_nets n1]
```

CONST_LINTCFG_CMD4

NAME

CONST_LINTCFG_CMD4 (CONSTRAINT_LINT)

MESSAGE

Non-SDC command used

DEFAULT SEVERITY

Warning

DESCRIPTION

Indicates that a command that does not comply with currently supported SDC version is used.

This rule is checked during design initialization by *init_design* command and the results can be reported by *report_rule_check* command.

To configure a command as non-SDC compliant, use the *configure_lint_check* command for that command. This command has *is_sdc_compliant* parameter that you can set to no.

Note: If a command is not SDC compliant, it can still be among the commands/options supported by the tool.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
file_line	unsigned	File number
file_name	string	File name

command	string	Constraint command
msg	string	Message

EXAMPLE

The following command causes a rule check violation if `is_sdc_compliant` parameter of `set_sense` command is set to no by lint configuration.

```
set_sense -stop_propagation [get_pins my_pin]
```

CONST_LINTCFG_OPT

- [CONST_LINTCFG_OPT2](#)
- [CONST_LINTCFG_OPT3](#)
- [CONST_LINTCFG_OPT4](#)
- [CONST_LINTCFG_OPT5](#)
- [CONST_LINTCFG_OPT6](#)
- [CONST_LINTCFG_OPT7](#)
- [CONST_LINTCFG_OPT8](#)
- [CONST_LINTCFG_OPT10](#)

CONST_LINTCFG_OPT2

NAME

CONST_LINTCFG_OPT2 (CONSTRAINT_LINT)

MESSAGE

Unsupported option used

DEFAULT SEVERITY

Warning

DESCRIPTION

Indicates that an unsupported command option is used in an SDC file. The software recognizes this option, but ignores it. To configure an option as unsupported, the `configure_lint_check` command is used. This command has a *usage* parameter that can be set to a value of `unsupported`. This rule is checked during design initialization by `init_design` command and the results can be reported by `report_sdc` command.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
file_line	unsigned	File number
file_name	string	File name
command	string	Constraint command
msg	string	Message

EXAMPLE

- The following command causes a rule check violation because the software does not support the `-exact` option in the `get_nets` command:

```
get_nets -exact clka
```

CONST_LINTCFG_OPT3

NAME

CONST_LINTCFG_OPT3 (CONSTRAINT_LINT)

MESSAGE

User defined forbidden option used

DEFAULT SEVERITY

Warning

DESCRIPTION

Indicates that a forbidden command option is used in an SDC file. To configure an option as forbidden, the `configure_lint_check` command for that option is used. This command has a parameter called `option`, which can be set to a value of `forbidden`. This rule is checked during design initialization by `init_design` command and the results can be reported by `report_sdc` command.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
file_line	unsigned	File number
file_name	string	File name

EXAMPLE

- To forbid the using of the `-hsc` option in the `get_cells` command, the following command is used in a lint configuration file:

```
configure_lint_check get_cells { -hsc { usage forbidden } }
```

After reading that configuration file, the following command causes a rule check violation because the `-hsc` option is used:

```
get_cells -hsc @ mod1/a_reg[1]
```

CONST_LINTCFG_OPT4

NAME

CONST_LINTCFG_OPT4 (CONSTRAINT_LINT)

MESSAGE

Non-SDC option used

DEFAULT SEVERITY

Warning

DESCRIPTION

Indicates that an option that does not comply with currently supported SDC version is used.

This rule is checked during design initialization by *init_design* command and the results can be reported by *report_rule_check* command.

To configure an option as non-SDC compliant, use the *configure_lint_check* command for that option. This option has *is_sdc_compliant* parameter that you can set to no.

Note: If an option is not SDC compliant, it can still be among the commands/options supported by the tool.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
file_line	unsigned	File number
file_name	string	File name

EXAMPLE

The following command causes a rule check violation if `is_sdc_compliant` parameter of `-reference_pin` option is set to no by lint configuration.

```
set_input_delay 0 -reference_pin [get_pins my_pin] [get_ports my_port]
```

CONST_LINTCFG_OPT5

NAME

CONST_LINTCFG_OPT5 (CONSTRAINT_LINT)

MESSAGE

Missing required option

DEFAULT SEVERITY

Warning

DESCRIPTION

Indicates that a command option has been used without one of its required options. To configure an option as required, the `configure_lint_check` command for that option is used. This command has a parameter called `required_options`, where a list of required options can be specified. This rule is checked during design initialization by `init_design` command and the results can be reported by `report_sdc` command.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
file_line	unsigned	File number
file_name	string	File name
command	string	Constraint command
msg	string	options

EXAMPLE

- The following command causes a rule check violation because using the `-clock_fall` option requires the presence of the `-clock` option:

```
set_input_delay 2 -clock_fall [get_ports IN1]
```

CONST_LINTCFG_OPT6

NAME

CONST_LINTCFG_OPT6 (CONSTRAINT_LINT)

MESSAGE

Missing mandatory option

DEFAULT SEVERITY

Warning

DESCRIPTION

Indicates that a mandatory option of a command is not specified. This also applies to instances where one of several options is mandatory. To configure an option as mandatory, the `configure_lint_check` command for that option is used. This command has a parameter called `usage`, which it can be set to `mandatory`. This rule is checked during design initialization by `init_design` command and the results can be reported by `report_sdc` command.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
file_line	unsigned	File number
file_name	string	File name
command	string	Constraint command
msg	string	Message

EXAMPLE

- The following command causes a rule check violation because using the `-clock_fall` option requires the presence of the `-clock` option:

```
set_input_delay 2 -clock_fall [get_ports IN1]
```

CONST_LINTCFG_OPT7

NAME

CONST_LINTCFG_OPT7 (CONSTRAINT_LINT)

MESSAGE

Missing recommended option

DEFAULT SEVERITY

Warning

DESCRIPTION

Indicates that a recommended option of a command is not specified. To configure an option as recommended, the `configure_lint_check` command for that option is used. This command has a parameter called `usage`, which can be set to a value of `recommended`. This rule is checked during design initialization by `init_design` command and the results can be reported by `report_sdc` command.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
file_line	unsigned	File number
file_name	string	File name

EXAMPLE

- To recommend using the `-waveform` option in the `create_clock` command, the following command in a lint configuration file is used:

```
configure_lint_check create_clock { -waveform { usage recommended } }
```

After reading that configuration file, the following command causes a rule check violation because it is missing the recommended `-waveform` option:

```
create_clock -name CLK -period 10
```

CONST_LINTCFG_OPT8

NAME

CONST_LINTCFG_OPT8 (CONSTRAINT_LINT)

MESSAGE

Using mutually exclusive options

DEFAULT SEVERITY

Warning

DESCRIPTION

Indicates that options that cannot appear together are used in the same command. To configure options as mutually exclusive, the `configure_lint_check` command is used. This command has a parameter called `mutex_options`, where a list of options that are mutually exclusive can be specified. This rule is checked during design initialization by `init_design` command and the results can be reported by `report_sdc` command.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
file_line	unsigned	File number
file_name	string	File name
command	string	Constraint command

command_options	string	Related command options

EXAMPLE

- The following command causes a rule check violation because the `-edges` and `-duty_cycle` cannot appear together in the same `create_generated_clock` command:

```
create_generated_clock -edges {1 2 3} -duty_cycle 50 -source [get_ports clk]
[get_pins div_reg/Q]
```

CONST_LINTCFG_OPT10

NAME

CONST_LINTCFG_OPT10 (CONSTRAINT_LINT)

MESSAGE

User defined unhandled option used

DEFAULT SEVERITY

Warning

DESCRIPTION

Indicates that an unhandled command option is used in an SDC file. To configure an option as unhandled, the `configure_lint_check` command for that option is used. This command has a parameter called `usage`, which can be set to a value of `unhandled`. This rule is checked during design initialization by `init_design` command and the results can be reported by `report_sdc` command.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
file_line	unsigned	File number
file_name	string	File name

EXAMPLE

- To set `-hsc` option in the `get_cells` command as unhandled, the following command in a lint configuration file is used:

```
configure_lint_check get_cells { -hsc { usage unhandled } }
```

After reading that configuration file, the following command causes a rule check violation because the `-hsc` option is used:

```
get_cells -hsc @ mod1/a_reg[1]
```

CONST_LINTCFG_REF

- [CONST_LINTCFG_REF4](#)
- [CONST_LINTCFG_REF5](#)

CONST_LINTCFG_REF4

NAME

CONST_LINTCFG_REF4 (CONSTRAINT_LINT)

MESSAGE

Invalid object specified

DEFAULT SEVERITY

Warning

DESCRIPTION

Indicates that an object specified in a command argument has the wrong object type, is not a valid reference for the specific argument, or is an invalid object by itself (for example, a generated clock that has no master clock). The type of an object referenced by a SDC command is one of the following:

- clock
- cell
- design
- lib
- lib_cell
- lib_pin
- net
- pin
- port

This rule is checked during design initialization by `init_design` command and the results can be reported by `report_sdc` command. To configure a list of valid object types for an option, the `configure_lint_check` command for that option is used. This command has a parameter called `valid_object_types`, where a list of valid object types can be specified.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
file_line	unsigned	File number
file_name	string	File name
command	string	Constraint command
option	string	option
object	string	object
valid_object_types	string	valid_object_types

EXAMPLE

- The following command causes a rule check violation because the object list cannot be of type clock:

```
set_disable_timing [get_clocks CLK1]
```

CONST_LINTCFG_REF5

NAME

CONST_LINTCFG_REF5 (CONSTRAINT_LINT)

MESSAGE

Implicit object specified

DEFAULT SEVERITY

Warning

DESCRIPTION

Indicates that a potentially ambiguous implicit object reference is used, the appropriate `get_*` command has not been used, and there is more than one allowed object type for this argument. This rule is checked during design initialization by `init_design` command and the results can be reported by `report_sdc` command.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
file_line	unsigned	File number
file_name	string	File name
command	string	Constraint command
msg	string	Message

EXAMPLE

- The following command causes a rule check violation because the `-to` option allows a clock, cell, port, or pin argument:

```
set_false_path -to CLK1
```

CONST_LINTCFG_VAL

- [CONST_LINTCFG_VAL3](#)
- [CONST_LINTCFG_VAL4](#)

CONST_LINTCFG_VAL3

NAME

CONST_LINTCFG_VAL3 (CONSTRAINT_LINT)

MESSAGE

Invalid value specified

DEFAULT SEVERITY

Warning

DESCRIPTION

Indicates that the value specified for a command or option is invalid. This usually happens in two cases: the given value does not match one of the expected enumerated types or the value does not match one of the patterns specified as valid values. To configure the valid values for an option, the `configure_lint_check` command for that option can be used. This command has a parameter called `valid_values`, where a list of valid values for the option can be specified.

 This rule is checked during design initialization by `init_design` command and the results can be reported by `report_sdc` command.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
file_line	unsigned	File number
file_name	string	File name
command	string	Constraint command
msg	string	options

EXAMPLE

- The following command causes a rule check violation because block is not a valid mode name for the `set_wire_load_mode` command:

```
set_wire_load_mode block
```

CONST_LINTCFG_VAL4

NAME

CONST_LINTCFG_VAL4 (CONSTRAINT_LINT)

MESSAGE

Value is out of range

DEFAULT SEVERITY

Warning

DESCRIPTION

Indicates that a value specified to a command option is out of range. That is, it does not satisfy the range expression specified in the SDC language or in the lint configuration.

To configure the valid range for an option value, use the `configure_lint_check` command for that option. This command has a parameter called `valid_range`, where you can specify valid range for the option.

 This rule is checked during design initialization by `init_design` command and the results can be reported by `report_sdc` command.

This rule has the following specific rule occurrence attributes for the details about the violation:

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
file_line	unsigned	File number
file_name	string	File name
command	string	Constraint command

msg	string	Message

EXAMPLE

The following command causes a rule check violation because the value specified for minimum porosity is outside the valid range of 0-90:

```
set_min_porosity 90.001 [current_design]
```

TA

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TA-100

NAME

TA-100

SUMMARY

Data signal feeds clock pin . The pin expects a clock signal but is fed by a data signal.

DESCRIPTION

May be a problem with generated clocks or with the `set_clock_root` command or the `set_clock_arrival_time` command. Address any associated TA-101 or TA-102 messages first. Trace the signal back to its source, and make sure the clock is defined properly. Remove the `set_input_delay` command using the `reset_input_delay` command (if present) from the clock root. Make sure wild card constraints do not pick up unintended ports and make sure the clock signal is not converted to a data signal before feeding this pin. If necessary, use the `set_clock_info_change` command to convert the data signal to a clock signal.

TA-101

NAME

TA-101

SUMMARY

Pin expects a clock signal but is not fed by a clock signal. The pin will not be timed or optimized, and the fanout and fanin cones will not be optimized.

DESCRIPTION

Possible causes are that the clock is not defined or the clock signal is converted to a data signal before feeding this pin. Use the `get_clock_source`, `get_fanin`, or the `get_drive_pin` command to identify the signal source, and make sure the clock is defined. Define all clocks before setting any constraints on the design. Make sure wild card constraints do not pick up unintended ports, and make sure the clock signal is not converted to a data signal before feeding the pin. If necessary, use the `set_clock_info_change` command to convert the data signal to a clock signal.

TA-102

NAME

TA-102

SUMMARY

Multiple signals arrive at pin .

DESCRIPTION

Indicates that multiple signals arrive at this pin. This can happen when the fanin cone contains two different clock domains or when two clocks arrive at the clock pin of a sequential element.

1. Use the `report_cell_instance_timing` command and the `report_net` or the `get_fanin` command to trace back to the point where the clock phases merged.
2. Check for clocks propagating through clock sources and, if necessary, disable the path from the clock source using the `set_false_path` command.
3. Check if there is a mux on the clocks without the `set_constant_for_timing` command applied on its select line. 4. Use the `set_false_path` command to disable non-functional paths. Sometimes, a small number of `set_false_paths` commands will fix a large number of TA-102 warnings.

TA-103

NAME

TA-103

SUMMARY

Required time assertion for signal does not match any incoming signal at pin .An assertion applied to the pin is referenced to a clock phase that is not present in any arriving signal at the pin.

DESCRIPTION

1. Check for bad constraints. Use the `set_external_delay` command instead of the

set_data_required_time command.

2. Check if the constraint is using an incorrect ideal clock.
3. Use the report_cell_instance_timing command to see what phases are arriving at the pin.

TA-104

NAME

TA-104

SUMMARY

No arrival time or input delay assertion found at port . You have not specified the set_input_delay command or the set_data_arrival_time command at the port. Use the set_input_delay command to specify arrival time on an input port. If the pin is truly unconstrained, use the -from option with the set_false_path command.

DESCRIPTION

TA-105

NAME

TA-105

SUMMARY

No arrival time assertion found for signal at port . The pin is only partially constrained. Specify all four constraint pairs: early rise, early fall, late rise, and late fall.

DESCRIPTION

TA-106

NAME

TA-106

SUMMARY

Arrival time assertion with no clock information found at port . There is a set_data_arrival_time assertion at the port, but there is no ideal clock associated with it. The ideal clock was not specified when the assertion was given. The asynchronous clocks (indicated by @) are not considered defined ideal clocks.

DESCRIPTION

TA-107

NAME

TA-107

SUMMARY

No drive assertion found at port . To specify drive strength, use the set_drive_cell command or the set_drive_resistance command.

DESCRIPTION

TA-108

NAME

TA-108

SUMMARY

No drive time assertion found for signal at port . The pin is only partially constrained. Specify all four constraint pairs: early rise, early fall, late rise, and late fall.

DESCRIPTION

TA-109

NAME

TA-109

SUMMARY

No required time or external delay assertion found at port . Use the `set_external_delay` command to set the external delay or the `set_data_required_time` command to set the required time for the pin. If the pin is truly unconstrained, use the `-to` option with the `set_false_path` command.

DESCRIPTION

TA-110

NAME

TA-110

SUMMARY

No required time or external delay assertion found for signal at port .The pin is only partially constrained. Specify all four constraint pairs: early rise, early fall, late rise, and late fall.

DESCRIPTION

TA-111

NAME

TA-111

SUMMARY

Data gating clock at pin .The clock is a gated clock so verify that clock gating is intended at this pin. If clock gating is intended, make sure that CTPKS/CTGEN is aware of the clock gating. If gating is not intended, the problem must be corrected.

DESCRIPTION

TA-112

NAME

TA-112

SUMMARY

A timing loop was found in the design. The arc from to has been cut . A description of the timing loop follows

DESCRIPTION

A timing loop was found in the design and was cut at an arbitrary point - possibly within a clock tree. Timing analysis along the loop path will stop at the loop break point. If the design changes, the cut may move to a different place - causing different results. You can examine the detailed loop path in the file CTE_loops.rpt. To break the loop at more optimal point or to ensure that the loop break is always at the same point, you can use the `set_disable_timing` command and then run `report_timing` again.

TA-115

NAME

TA-115

SUMMARY

Internal Error: .

DESCRIPTION

TA-116

NAME

TA-116

SUMMARY

Clock not propagated beyond pin .Possible causes are that the clock is connected to a black-box, the clock is connected to an output port, or the clock is connected to a data input of a tri-state buffer or a sequential element (in which case, the output of the cell is no longer a clock).If the clock is connected to a black box, use the `set_generated_clock` command at the black box output.

DESCRIPTION

TA-117

NAME

TA-117

SUMMARY

Relaxing constraint from %f to %f.

DESCRIPTION

TA-118

NAME

TA-118

SUMMARY

To change the coverage ratio, use the -coverage_ratio option.

DESCRIPTION

TA-119

NAME

TA-119

SUMMARY

There are constraint errors. Please see the `check_timing` command for more details.

DESCRIPTION

TA-120

NAME

TA-120

SUMMARY

SDF error in file on line : .

DESCRIPTION

TA-122

NAME

TA-122

SUMMARY

Redefining clock . Some of the timing assertions may have to be redefined because they all use absolute timing. This message occurs when an ideal clock is defined more than once.

DESCRIPTION

TA-125

NAME

TA-125

SUMMARY

No input delay assertion found for signal at port .The pin is only partially constrained. Specify all four constraint pairs: early rise, early fall, late rise, and late fall.

DESCRIPTION

TA-126

NAME

TA-126

SUMMARY

Input delay assertion with no clock information found at port .

DESCRIPTION

TA-127

NAME

TA-127

SUMMARY

Asynchronous signal arriving at pin , where synchronous signal is expected.

DESCRIPTION

TA-128

NAME

TA-128

SUMMARY

An array of size by by was passed into 3d lookup.

DESCRIPTION

TA-129

NAME

TA-129

SUMMARY

An array of size by was passed into 2d lookup.

DESCRIPTION

TA-130

NAME

TA-130

SUMMARY

An array of size was passed into 1d lookup.

DESCRIPTION

TA-131

NAME

TA-131

SUMMARY

Non-linear template has a variable that is neither input net transition nor output total net capacitance.

DESCRIPTION

TA-132

NAME

TA-132

SUMMARY

2D, non-linear template has variable combination other than input net transition and output total capacitance.

DESCRIPTION

TA-133

NAME

TA-133

SUMMARY

3D, non-linear template has variable combination other than input net transition, output total capacitance, and related out net capacitance.

DESCRIPTION

TA-134

NAME

TA-134

SUMMARY

Non-linear template uses variables. A maximum of 3 variables is allowed.

DESCRIPTION

TA-135

NAME

TA-135

SUMMARY

Non-linear template has a variable that is neither constrained nor related pin transition.

DESCRIPTION

TA-136

NAME

TA-136

SUMMARY

2D, non-linear template has variable combination other than constrained and related pin transition.

DESCRIPTION

TA-137

NAME

TA-137

SUMMARY

3D, non-linear template has variable combination other than constrained pin transition, related pin transition, and related out net capacitance.

DESCRIPTION

TA-138

NAME

TA-138

SUMMARY

Disabling test signal arc will disable timing delay/check arcs on the test pins. Enable test signal arcs to fix setuphold violation on test pins. Scan chain most likely will give setuphold violation if not buffered.

DESCRIPTION

TA-139

NAME

TA-139

SUMMARY

2D, non-linear template has variable combination other than and .

DESCRIPTION

TA-140

NAME

TA-140

SUMMARY

Both positive and negative phases of clock are found at one or more pins. One example of pin with non-unate clock phases is . The phase of the clock from such pins is ignored because the global `timing_clock_phase_propagation` is set to

DESCRIPTION

An example of when both positive and negative phases of a clock are present is a 2-input XOR gate whose output drives a clock pin of a flip-flop. One input to the XOR gate is a clock signal and the other is a data signal. If the value of the data signal is unknown, both the positive and the negative phases of the clock can be present at the output of XOR. Use the global `timing_clock_phase_propagation` to control which clock phase to select in such situations

TA-141

NAME

TA-141

SUMMARY

Ignoring one or more timing checks because the launching clock and the capturing clock are mutually exclusive. This condition was detected automatically because the `timing_clock_phase_propagation` global is set to `auto`. An example of an ignored timing check is a check with the signal pin launched by polarity of clock and the reference pin captured by polarity of clock

DESCRIPTION

Consider a 2-input XOR gate whose output drives a clock pin of a flip-flop. One input to the XOR gate is a clock signal and the other is a data signal. If the value of the data signal is unknown, both the positive and the negative phases of the clock can be present at the output of XOR. However, the positive polarity clock and the negative polarity clock cannot be present simultaneously because they are mutually exclusive. When the `timing_clock_phase_propagation` global is set to `auto`, mutually exclusive clocks are detected automatically and they are used to ignore certain timing checks.

TA-142

NAME

TA-142

SUMMARY

Ignoring one or more timing checks because the launching clock condition and the check arc condition are mutually exclusive. This condition was detected automatically because the `timing_clock_phase_propagation` global is set to `auto` and the `timing_remove_false_paths_for_check_arcs` global is set to `true`. An example of an ignored timing check is a check with the signal pin launched by polarity of clock and the reference pin captured by polarity of clock

DESCRIPTION

Consider a flip-flop with two input pins D and SD and one clock pin CP. D is used in non-scan mode and SD is used in scan mode. Consider a clock SCAN_CLK that is active in scan mode. If SCAN_CLK is the launching clock for a timing check between D and CP, the timing check is ignored because the non-scan mode and the scan mode are mutually exclusive.

TA-143

NAME

TA-143

SUMMARY

pulsed latch instance is not clocked by a pulse generator, ignoring timing checks for this instance

DESCRIPTION

If none of sequential elements are clocked by pulse generators, turn off pulsed latch analysis by `set_global timing_enable_pulsed_latch` off. If pulsed latch analysis is needed, make sure that all pulsed latches are clocked by pulse generators

TA-144

NAME

TA-144

SUMMARY

pulsed latch instance is an active latch but is clocked by a pulse, ignoring timing checks for this instance

DESCRIPTION

If none of sequential elements are clocked by pulse generators, turn off pulsed latch analysis by set_global timing_enable_pulsed_latch off. If pulsed latch analysis is needed, check for the polarity at the output of the relevant pulse generator and insert inverters if needed. Also, if the global timing_infer_pulsed_latch is set to on, you can turn off auto polarity inferencing by set_global timing_infer_pulsed_latch off and manually specify pulse generator cells, pulsed latches and polarities in your library

TA-145

NAME

TA-145

SUMMARY

pulse switching violation at pin

DESCRIPTION

TA-147

NAME

TA-147

SUMMARY

max pulse width violation. slack = %f, maxPW = %f, actualPW = %f, on pin .

DESCRIPTION

TA-148

NAME

TA-148

SUMMARY

A pulse generator of polarity detected on pin .

DESCRIPTION

TA-149

NAME

TA-149

SUMMARY

set_drive constraint on port is skipped. It cannot overwrite an existing set_driving_cell constraint.

DESCRIPTION

TA-150

NAME

TA-150

SUMMARY

constraint on port is overwriting an existing constraint.

DESCRIPTION

TA-151

NAME

TA-151

SUMMARY

The views for which the arc is soft-broken:

DESCRIPTION

TA-152

NAME

TA-152

SUMMARY

A latency path from the edge of the master clock at source pin to the edge of generated clock at pin cannot be found. You must modify your `create_generated_clock` constraint to be consistent with the network topology. The analysis will continue using 0ns source latency for generated clock. For backward compatibility with earlier releases or to remove the edge-to-edge sufficiency checking, you should set the global `timing_enable_genclock_edge_based_source_latency` to false in the Legacy UI or set the Stylus root attribute `timing_enable_generated_clock_edge_based_source_latency` to false in the Stylus UI.

DESCRIPTION

One aspect of generated clock handling is which path to select (among all paths from the source to the target) to determine the source latency of the generated clock. At run time, the Common Timing Engine (CTE) checks whether a path being analyzed between a generated clock and its source has the correct cause-effect relationship consistent with the assertion. Paths found to be inconsistent are ignored. Only the paths that are in agreement with the edge relationship are considered for source latency. If there is no such path, zero source latency is used. This selection is done independently for each edge of the generated clock, so the result might be different for each edge. For example, there might be no latency path satisfying the edge relationship for the rise transition (and hence zero source latency), while there might be several candidate paths for the fall transition.

The edge relationship between a generated clock and its source is specified by the `create_generated_clock` assertion options `-divide_by`, `-multiply_by`, `-edges` and `-invert`.

For the `-divide_by` option with a factor which is power of two, both edges of the generated clock are assumed to be generated from the leading edge of the master clock at the source pin. This normally corresponds to the rise transition unless the clock is inverted at the source pin. If the divide factor is not power of two, the assumed relationship is rise-to-rise, fall-to-fall, which means the rise (fall) edge of the master clock maps to the rise (fall) edge of the generated clock.

The `-multiply_by` option assumes the same rise->rise and fall->fall relationships as `-divide_by` that are not a power-of-two factor.

For the `-edges` option, the master clock edges are specified explicitly. Note that the edge numbering always starts with the rise transition being #1, and continues with the next fall being #2 and so on. Even if the clock is inverted at the source pin, #1 is the rise transition. For example, assuming positive polarities, the edge specification {1 4 7} indicates that the relationship is rise-to-rise, fall-to-fall. Similarly, {2 5 8} indicates fall-to-rise, rise-to-fall. On the other hand, {1 3 5} indicates rise-to-rise, rise-to-fall, which is the same as `-divide_by 2`. Finally, the `-invert` option reverses the generated clock waveform, so the target edges are swapped during mapping.

Given the edge relationship established from the assertion, if a generated clock edge cannot be produced from the corresponding master clock edge (there is no path satisfying the relationship),

CTE issues an error to indicate this. To disable this checking, you can set `timing_enable_genclk_edge_based_source_latency` to false.

This error indicates the generated clock constraint or logic has some issues and should be recoded so the parent-to-generated clock relationship can be satisfied by the logic. If it is not correct, zero latency will be used for the generated clocks that do not have a satisfiable latency path.

TA-153

NAME

TA-153

SUMMARY

Simultaneous Input Switching analysis should be done with both Early and Late delays. In the current analysis, Late arrival delays are not available for some views. As a result, input arrival times will be checked using early delays only. For the most accurate analysis, you should make sure the same set of views are defined for both setup and hold.

DESCRIPTION

TA-200

NAME

TA-200

SUMMARY

The number of columns has been truncated. Through pin lists may not be reported.

DESCRIPTION

TA-201

NAME

TA-201

SUMMARY

The software could not find a defined path group matching the name . You can use the `report_path_groups` or `get_path_groups` commands to see the currently defined path groups.

DESCRIPTION

This is a generic message issued when commands in the user script reference a path group that is currently not registered in the software. Typically, this situation occurs when either the `report_path_groups -name <argument>` or the `report_timing -path_group <argument>` are invoked with an undefined path group name. In these cases, the associated reporting command is aborted. To resolve this issue, you should first check that the requested path group has been defined properly. To see the currently defined set of path groups, you can use the `report_path_groups` command or the `get_path_groups` query. Use the `group_path` command to define any missing path groups.

TA-300

NAME

TA-300

SUMMARY

Cannot open file for reading.

DESCRIPTION

TA-301

NAME

TA-301

SUMMARY

Cannot open file for reading.

DESCRIPTION

TA-302

NAME

TA-302

SUMMARY

Cannot open file for writing.

DESCRIPTION

TA-303

NAME

TA-303

SUMMARY

Cannot open file for writing.

DESCRIPTION

TA-500

NAME

TA-500

SUMMARY

Error in loading DCM Rule.

DESCRIPTION

TA-501

NAME

TA-501

SUMMARY

The timing model for cell is absent in both DCM and .lib.

DESCRIPTION

TA-502

NAME

TA-502

SUMMARY

Error in delay calculation for cell .

DESCRIPTION

TA-503

NAME

TA-503

SUMMARY

Error in slew calculation for cell .

DESCRIPTION

TA-504

NAME

TA-504

SUMMARY

Error in check calculation for cell .

DESCRIPTION

TA-505

NAME

TA-505

SUMMARY

Error in function .

DESCRIPTION

TA-506

NAME

TA-506

SUMMARY

External function , called

DESCRIPTION

TA-507

NAME

TA-507

SUMMARY

DCM rule is already loaded.

DESCRIPTION

TA-508

NAME

TA-508

SUMMARY

Cross mode delay arc in DCL model is ignored.

DESCRIPTION

TA-509

NAME

TA-509

SUMMARY

expose function not found.

DESCRIPTION

TA-510

NAME

TA-510

SUMMARY

DCM rule has not been loaded.

DESCRIPTION

TA-511

NAME

TA-511

SUMMARY

DCM not compliant with IEEE 1481-1997.

DESCRIPTION

TA-512

NAME

TA-512

SUMMARY

The mode specified, , is invalid.

DESCRIPTION

TA-516

NAME

TA-516

SUMMARY

DCM Loading Error - not compliant with IEEE 1481.

DESCRIPTION

TA-517

NAME

TA-517

SUMMARY

Units mismatch - 1e ohms, 1e farads, 1e seconds.

DESCRIPTION

TA-518

NAME

TA-518

SUMMARY

Output transition occurs on both rise and fall at enable.

DESCRIPTION

TA-519

NAME

TA-519

SUMMARY

The top level current module/design for the session does not match the owning boundary/module for the object. Please ensure all object names are specified relative to the current instance and that the current instance maps all objects to the current top-level module.

DESCRIPTION

TA-520

NAME

TA-520

SUMMARY

Arrival Recalc node has no AT.

DESCRIPTION

TA-521

NAME

TA-521

SUMMARY

Arrival Inc node has valid AT.

DESCRIPTION

TA-522

NAME

TA-522

SUMMARY

TraceArrivalLoop finds a previously snipped delay arc.

DESCRIPTION

TA-523

NAME

TA-523

SUMMARY

The software has encountered unexpected internal error condition TA-523 - for and may not perform reliably in this state. You should contact Cadence support for assistance in resolving this problem and refer to the error code mentioned above. There is no current user-level workaround for resolving this problem.

DESCRIPTION

TA-524

NAME

TA-524

SUMMARY

In TraceClockFlagLoop cannot back up.

DESCRIPTION

TA-525

NAME

TA-525

SUMMARY

TraceEndTagLoop finds a previously snipped delay arc.

DESCRIPTION

TA-526

NAME

TA-526

SUMMARY

Internal error happened during Timing Analysis, please contact Cadence customer support for assistance.

DESCRIPTION

This error generally indicates a problem related to multi-threading of the analysis and will require more detailed analysis and debug by Cadence.

TA-527

NAME

TA-527

SUMMARY

TraceEndTagLoop finds an unvisited node.

DESCRIPTION

TA-528

NAME

TA-528

SUMMARY

Missing data on TAInstancePin.

DESCRIPTION

TA-529

NAME

TA-529

SUMMARY

Required Recalc node has no RT.

DESCRIPTION

TA-530

NAME

TA-530

SUMMARY

Required Inc node has valid RT.

DESCRIPTION

TA-531

NAME

TA-531

SUMMARY

The software has encountered unexpected internal error condition TA-531 - and may not perform reliably in this state. You should contact Cadence support for assistance in resolving this problem and refer to the error code mentioned above. There is no current user-level workaround for resolving this problem.

DESCRIPTION

TA-532

NAME

TA-532

SUMMARY

The software has encountered unexpected internal error condition TA-532 - and may not perform reliably in this state. You should contact Cadence support for assistance in resolving this problem and refer to the error code mentioned above. There is no current user-level workaround for resolving this problem.

DESCRIPTION

TA-533

NAME

TA-533

SUMMARY

The software has encountered unexpected internal error condition TA-533 - and may not perform reliably in this state. You should contact Cadence support for assistance in resolving this problem and refer to the error code mentioned above. There is no current user-level workaround for resolving this problem.

DESCRIPTION

TA-534

NAME

TA-534

SUMMARY

The software has encountered unexpected internal error condition TA-534 - and may not perform reliably in this state. You should contact Cadence support for assistance in resolving this problem and refer to the error code mentioned above. There is no current user-level workaround for resolving this problem.

DESCRIPTION

TA-535

NAME

TA-535

SUMMARY

This cursor is not implemented.

DESCRIPTION

TA-536

NAME

TA-536

SUMMARY

Ran out of memory

DESCRIPTION

TA-537

NAME

TA-537

SUMMARY

Pin is driven and simultaneously, preserving .

DESCRIPTION

TA-538

NAME

TA-538

SUMMARY

Constant assertion propagation to pin leads to and values simultaneously, preserving .

DESCRIPTION

TA-539

NAME

TA-539

SUMMARY

Mixed instance and net arcs detected at pin .

DESCRIPTION

TA-540

NAME

TA-540

SUMMARY

Pin was expected to be an instance output pin.

DESCRIPTION

TA-541

NAME

TA-541

SUMMARY

Timing Model for cell absent in DCM; will use .lib.

DESCRIPTION

TA-542

NAME

TA-542

SUMMARY

Three-state enable and disable combined on the arc from pin to pin .

DESCRIPTION

TA-543

NAME

TA-543

SUMMARY

Non-unate arc between pins and is not handled properly by CPPR analysis. Path analysis ignored.

DESCRIPTION

TA-544

NAME

TA-544

SUMMARY

Unsupported clock network found in design during CPPR analysis.

DESCRIPTION

TA-545

NAME

TA-545

SUMMARY

Unsupported clock network point found at pin .

DESCRIPTION

TA-546

NAME

TA-546

SUMMARY

Original library units will be used even for DCL data.

DESCRIPTION

TA-547

NAME

TA-547

SUMMARY

Defaults units of ns, Kohm and pF will be used.

DESCRIPTION

TA-548

NAME

TA-548

SUMMARY

Units for mismatched: %g .lib and %g DCL.

DESCRIPTION

TA-549

NAME

TA-549

SUMMARY

Pin is driven and simultaneously, preserving .

DESCRIPTION

TA-550

NAME

TA-550

SUMMARY

Searching for true path.

DESCRIPTION

TA-551

NAME

TA-551

SUMMARY

Clock clipping at pin is possible due to wrong gate type or wrong trigger for input .

DESCRIPTION

TA-552

NAME

TA-552

SUMMARY

Clock clipping at pin is possible due to incompatible frequencies between clock signal on and data signal on .

DESCRIPTION

TA-553

NAME

TA-553

SUMMARY

edge of generated clock is out of synch with corresponding edge of source clock : %f vs. %f.

DESCRIPTION

TA-554

NAME

TA-554

SUMMARY

Cannot convert generated clock to a regular clock.

DESCRIPTION

TA-555

NAME

TA-555

SUMMARY

Cannot convert to a generated clock.

DESCRIPTION

TA-556

NAME

TA-556

SUMMARY

At pin edges of the generated clock do not appear to be derived from the edges of the source clock .
Please check the generated clock assertion on the pin.

DESCRIPTION

TA-557

NAME

TA-557

SUMMARY

Ceffective -> : %g (late) %g (early)

DESCRIPTION

TA-558

NAME

TA-558

SUMMARY

Function string is not defined in cell

DESCRIPTION

TA-559

NAME

TA-559

SUMMARY

Error in obtaining footprint for cell .

DESCRIPTION

TA-560

NAME

TA-560

SUMMARY

slack allocation .

DESCRIPTION

TA-561

NAME

TA-561

SUMMARY

Pass : maximum slack minimum slack .

DESCRIPTION

TA-562

NAME

TA-562

SUMMARY

Generating SDF file ..

DESCRIPTION

TA-563

NAME

TA-563

SUMMARY

Deriving context .

DESCRIPTION

TA-564

NAME

TA-564

SUMMARY

Essential OLA expose function is missing. Please consult OLA library vendor.

DESCRIPTION

TA-565

NAME

TA-565

SUMMARY

Error in function on cell .

DESCRIPTION

TA-566

NAME

TA-566

SUMMARY

Error in function pin

DESCRIPTION

TA-567

NAME

TA-567

SUMMARY

Error in function on cell pin .

DESCRIPTION

TA-568

NAME

TA-568

SUMMARY

Error in function on pin-pair

DESCRIPTION

TA-569

NAME

TA-569

SUMMARY

function on cell returned negative value. The cell is not used for synthesis.

DESCRIPTION

TA-570

NAME

TA-570

SUMMARY

Essential OLA Expose function: is missing. The expose functions are essential to recognize the OLA function graph. When the expose functions are missing, BuildGates Synthesis cannot understand the function of the cell and cannot use it for synthesis. However, if you start from the gate-level netlist, missing this particular expose function should be okay.

DESCRIPTION

TA-571

NAME

TA-571

SUMMARY

Unable to perform calculation for arc: ->.

DESCRIPTION

TA-572

NAME

TA-572

SUMMARY

Incorrect use of the dcl_use_ssm_library global. You cannot use this global when the dcl performance level is one. You are trying to set the global to true when the dcl performance level is set to one. Setting the dcl performance level to one passes the detail parasitic data so that the the OLA library can reduce them. Setting the global to true computes the reduced model; thus, the two assertions are conflicting and are not allowed.

DESCRIPTION

TA-580

NAME

TA-580

SUMMARY

The timer has encountered an unexpected error while trying to compute a CPPR adjustment for clock in view for the endpoint . This issue cannot be debugged at the user-level. You should contact Cadence for assistance resolving this problem. The timing analysis will continue using a conservative adjustment value of 0ns.

DESCRIPTION

CPPR credit is computed as part of OCV analysis to provide some credit back to the paths slack for parts of the path that are shared between launch and capture. In cases where the software encounters an unforeseen problem in calculating the common credit - this error message will be issued. There is no user-level debug or adjustments that can be recommended to workaround this issue. A more detailed investigation as to the cause will need to be done on the Cadence side. Please contact Cadence support for help and advice resolving this issue.

CPPR is a credit that reduces pessimism. By falling back to a credit of 0ns, the analysis will be conservative. If this added pessimism does not cause paths to fail, then it is a reasonable workaround - and you can continue your analyses while the problem is being addressed.

TA-581

NAME

TA-581

SUMMARY

The timer has encountered an unexpected error: . You should contact Cadence for assistance resolving this problem.

DESCRIPTION

TA-800

NAME

TA-800

SUMMARY

Ignoring internal pin at creating spice.

DESCRIPTION

TA-801

NAME

TA-801

SUMMARY

Number of spice-subckt pins of cell is less than the logical ones. Spice writer will use the spice-subckt pin list.

DESCRIPTION

Using the -spice option with the read_library_update command allows BuildGates Synthesis to define the pin order in the spice subckt definition. If there is a difference between the spice-subckt pin order and the TLF pin order, the spice-subckt pin order will be used.

TA-802

NAME

TA-802

SUMMARY

Logical pin of cell is not found in spice-subckt definition. Spice writer will use the spice-subckt pin list. Using the -spice option with the read_library_update command allows BuildGates Synthesis to define the pin order in the spice subckt definition. If there is a difference between the spice-subckt pin order and the TLF pin order, the spice-subckt pin order will be used.

DESCRIPTION

TA-803

NAME

TA-803

SUMMARY

Spice file name is not specified, spicewriter will use the default filename: .

DESCRIPTION

TA-804

NAME

TA-804

SUMMARY

Unable to build spice properties for net . Net is Ideal Net Mode. The BuildGates spicewriter can only write out spice netlist for nets that are reduced using the BuildGates spef reducer only. The net arc may be modeled as an ideal net.

DESCRIPTION

TA-805

NAME

TA-805

SUMMARY

Unable to build spice properties for the net . The net is not Elmore type. The spicewriter can only write out a spice netlist for nets that are reduced using the native spef reducer only.

DESCRIPTION

TA-806

NAME

TA-806

SUMMARY

Unable to build spice properties for net . Library does not contain the wire load model. Please check the timing library and find out if it has a wire load model. The BuildGates spicewriter can only write out a spice netlist for nets that are reduced using the BuildGates spef reducer only. The net arc may be modeled as an ideal net

DESCRIPTION

TA-900

NAME

TA-900

SUMMARY

Cannot find inst/boundary to .

DESCRIPTION

TA-901

NAME

TA-901

SUMMARY

Internal Error Analysis Id does not correspond with boundary.

DESCRIPTION

TA-902

NAME

TA-902

SUMMARY

, cluster not an area cluster.

DESCRIPTION

TA-903

NAME

TA-903

SUMMARY

Cannot find instance to pull into.

DESCRIPTION

TA-904

NAME

TA-904

SUMMARY

Cannot find top level boundary to pull from.

DESCRIPTION

TA-905

NAME

TA-905

SUMMARY

Cannot find top level boundary to pull to. Check to see if the instance is a hierarchical module.

DESCRIPTION

TA-906

NAME

TA-906

SUMMARY

Pin is not a valid start point; constraint originating from this pin will be ignored. Valid start points for path exceptions are 1) input port, 2) bidirectional port, 3) clock pin of a sequential cell, 4) pin associated with the set_input_delay command, or 5) a pin associated with the set_path_delay command using the -from option.

DESCRIPTION

TA-907

NAME

TA-907

SUMMARY

Pin is not a valid end point; constraint terminating at this pin will be ignored. Valid end points for path exceptions are 1) output port, 2) bidirectional port, 3) data pin of a sequential cell, 4) pin associated with the `set_external_delay` command or 5) a pin associated with the `set_path_delay` command using the `-to` option.

DESCRIPTION

TA-908

NAME

TA-908

SUMMARY

The `set_input_delay` on pin references `generated_clock`, on pin before it is created

DESCRIPTION

The `set_input_delay` command references a `generated_clock` which is in the fanout cone of the `set_input_delay` pin. Arrival time computation on the target pin of the `generated_clock` depends on arrival time of all the pins in the fanin cone. Therefore, there is a cyclic dependency. CTE breaks this cycle by ignoring the effect of `set_input_delay` on the `generated_clock`.

TA-950

NAME

TA-950

SUMMARY

No timing arc between and .

DESCRIPTION

Use report_cell_instance -timing for cell arcs or report_net for net arcs to check that a valid timing arc exists between the specified pins.

TA-951

NAME

TA-951

SUMMARY

No timing arc from pin .

DESCRIPTION

Use report_cell_instance -timing for cell arcs or report_net for net arcs to check that a valid timing arc from the specified pin exists.

TA-952

NAME

TA-952

SUMMARY

No timing arc to pin .

DESCRIPTION

Use report_cell_instance -timing for cell arc or report_net for net arc to check that a valid timing arc to the specified pin exists.

TA-953

NAME

TA-953

SUMMARY

No timing arc between and with condition .

DESCRIPTION

Use report_cell_instance -timing to check that a valid timing arc exists between the specified pin with the specified condition.

TA-955

NAME

TA-955

SUMMARY

No check arc between clock pin and data pin . Use report_cell_instance -timing to make sure that a valid check arc exists between the specified pins.

DESCRIPTION

TA-956

NAME

TA-956

SUMMARY

No check arc with clock pin . Use report_cell_instance -timing to make sure that a valid check arc with the specified clock pin exists.

DESCRIPTION

TA-957

NAME

TA-957

SUMMARY

No check arc with data pin . Use report_cell_instance -timing make sure that a valid check arc with the specified data pin exists.

DESCRIPTION

TA-958

NAME

TA-958

SUMMARY

Worst slack reported in the design = %f (late) %f (early)

DESCRIPTION

TA-959

NAME

TA-959

SUMMARY

.

DESCRIPTION

TA-960

NAME

TA-960

SUMMARY

.

DESCRIPTION

TA-961

NAME

TA-961

SUMMARY

Can not find object of name

DESCRIPTION

TA-962

NAME

TA-962

SUMMARY

Trying to create a new with a name matching an existing .

DESCRIPTION

This error comes when a lib set op cond, constraint mode, rc corner dc corner or analysis view is being created using MMMC create_library_set, create_op_cond, create_constraint_mode, create_rc_corner, create_delay_corner or create_analysis_view commands, with a name that already exists. Use unique names to avoid this error.

TA-963

NAME

TA-963

SUMMARY

Worst slack reported in the design = %f

DESCRIPTION

TA-964

NAME

TA-964

SUMMARY

SSTA: STA specific function call in SSTA mode

DESCRIPTION

TA-965

NAME

TA-965

SUMMARY

An internal error has occurred due to the timing system not being properly initialized before being called internally. You should contact Cadence for assistance in resolving this issue.

DESCRIPTION

The software is composed of multiple module which together deliver full functional solution to the user. In the interaction of these modules, the timing system module may be initialized and reset as needed by other modules requiring timing services. If a timing service is requested and the timing system has not been properly initialized internally then this error message will be issued.

This error does not represent an incorrect action by the user. Assistance from Cadence will be required to resolve this issue.

TA-967

NAME

TA-967

SUMMARY

Constraint mode was previously specified in `set_interactive_constraint_modes` command but becomes inactive. It will be removed from the interactive constraint modes set.

DESCRIPTION

TA-968

NAME

TA-968

SUMMARY

Reading timing constraints file ...

DESCRIPTION

TA-970

NAME

TA-970

SUMMARY

Simultaneous setup hold analysis is currently not supported for . Result may not match with non simultaneous analysis. Please use global timing_enable_simultaneous_setup_hold_mode to disable simultaneous analysis.

DESCRIPTION

TA-971

NAME

TA-971

SUMMARY

SSTA: asserting fixed delay values...may overwrite original delay sensitivities in SSTA mode

DESCRIPTION

TA-975

NAME

TA-975

SUMMARY

pulse detected on enable pin of active level latch, timing check reported are not valid.

DESCRIPTION

TA-976

NAME

TA-976

SUMMARY

Path groups asserted by the group_path/set_path_adjust_group constraint are considered by the software to be global in nature - they cannot be restricted to specific constraint modes. Due to this restriction, the group_path/set_path_adjust_group constraint is not supported directly in the constraint files which are configured in mode-specific commands such as create_constraint_mode. The current group_path/set_path_adjust_group entry will be ignored. You can enter the group_path/set_path_adjust_group constraint interactively or as part of a general command script - where it will be interpreted as global and will be in effect for all modes.

DESCRIPTION

TA-977

NAME

TA-977

SUMMARY

The current set of group_path constraints contain references to clock objects which are no longer defined in the current set of active views. Path groups containing invalid clock references have been removed. You will need to redefine those path groups if you switch to a new set of views where they are again active.

DESCRIPTION

TA-978

NAME

TA-978

SUMMARY

A skew check is being performed between clocks and which have different periods. Skew checks between multi-frequency clocks are not supported. The reported slack for this check may be inaccurate.

DESCRIPTION

TA-979

NAME

TA-979

SUMMARY

INFO: New setup and hold views overwrite old settings during design initialization

DESCRIPTION

TA-1000

NAME

TA-1000

SUMMARY

In order to set the global `timing_path_based_recompute_use_incr_slew` to true, first the global `timing_path_based_recompute_noise` must be set to true using `set_global` command.

DESCRIPTION

TA-1005

NAME

TA-1005

SUMMARY

SDF map file parsing error on line :

DESCRIPTION

TA-1006

NAME

TA-1006

SUMMARY

The software has determined that multi-drive net is comprised of %ld timing arcs which is above the threshold of specified by the timing_reduce_multi_drive_net_arcs_threshold setting. The net will be reduced to a single-driver model. The reduction can affect SDF generation and annotation - you should consult the read_sdf documentation in the Command Reference for more information on controlling multi-drive net reduction and related SDF flow issues.

DESCRIPTION

TA-1007

NAME

TA-1007

SUMMARY

The software has determined that multi-drive net is comprised of %ld timing arcs which is above the arc threshold of . The timing global timing_reduce_multi_drive_net_arcs is currently disabled - preventing simplification of the multi-drive net structure. Maintaining the current multi-drive net structure will have a significant impact on the memory requirements and performance of the analysis.

DESCRIPTION

TA-1008

NAME

TA-1008

SUMMARY

The software has determined that multi-drive net is comprised of %ld timing arcs which is above the arc threshold of . The timing global timing_reduce_multi_drive_net_arcs is currently on but the net is not getting reduction. Maintaining the current multi-drive net structure will have a significant impact on the memory requirements and performance of the analysis.

DESCRIPTION

TA-1009

NAME

TA-1009

SUMMARY

Internal Multi-threading Error:

DESCRIPTION

TA-1010

NAME

TA-1010

SUMMARY

The requested multi-mode/multi-corner (MMMC) database could not be fully initialized with the current license. The design will be initialized only with the primary Setup and Hold analysis views. If you are restoring the design from a saved Tempus session - timing graph data will not be loaded. To fully initialize this design, you must use a license that supports concurrent MMMC analysis.

DESCRIPTION

TA-1011

NAME

TA-1011

SUMMARY

Unable to calculate PLL adjustment for generated clock in view defined on pin , zero PLL adjustment will be used for this clock

DESCRIPTION

TA-1012

NAME

TA-1012

SUMMARY

Detected a missing clock arrival for one transition in view at pin which is a pulse generator output. No clock will propagate beyond this pin. To analyze designs with pulsed latch cells, the global `timing_enable_pulsed_latch` should be set to true.

DESCRIPTION

TA-1013

NAME

TA-1013

SUMMARY

Detected a missing clock arrival for one transition in view at pin because a rise or fall case analysis is applied. No clock will propagate beyond this pin.

DESCRIPTION

TA-1015

NAME

TA-1015

SUMMARY

The software has detected a missing clock arrival in view at pin . An internal error has prevented delays from being calculated properly for all of the timing arcs relating to this pin. As a result, clock propagation has been stopped at this point. You will need to contact Cadence customer support for assistance in resolving this issue.

DESCRIPTION

The software requires that all timing arcs receive an appropriate delay annotation from the delay calculation system. If there is a problem in the software that prevents delay annotation from occurring - the related timing arc is treated as a disabled arc.

In the context of clock propagation, the software requires both valid rising and falling transitions of the clock to be available in order to propagate the clock forward. The TA-1015 error is issued if one of the required transitions is not available due to a timing arc being disabled due to a missing delay annotation.

A missing delay annotation is an internal error condition that is generally not related to the customers configuration, constraints, etc - and cannot be debugged in the field. Users should contact Cadence support for assistance.

TA-1016

NAME

TA-1016

SUMMARY

Disable latch loops asserted by the `set_latch_loop_disable` constraint are considered by the software to be global in nature - they cannot be restricted to specific constraint modes. Due to this restriction, the `set_latch_loop_disable` constraint is not supported directly in the constraint files which are configured in mode-specific commands such as `create_constraint_mode`. The current `set_latch_loop_disable` entry will be ignored. You can enter the `set_latch_loop_disable` constraint interactively or as part of a general command script - where it will be interpreted as global and will be in effect for all modes.

DESCRIPTION

TA-1017

NAME

TA-1017

SUMMARY

The `multi_drive_preferred_driver` property has been specified on more than one of the driver pins of net . The software will retain pin as the preserved driver and continue with multi-drive reduction of the net.

DESCRIPTION

TA-1019

NAME

TA-1019

SUMMARY

While performing multi-driver reduction on net , the software was not able to identify a driver instance that was not disabled across all analysis views. Instance pin was selected as the preserved driver. No timing paths will propagate to this net. Review your constraints if this does not match the timing intent.

DESCRIPTION

TA-1020

NAME

TA-1020

SUMMARY

While performing multi-driver reduction on net , the software has determined that the user-specified preserved driver pin is not enabled across all analysis views. This driver will continue to be maintained as the preserved driver. Review your constraints if this does not match the timing intent.

DESCRIPTION

TA-1022

NAME

TA-1022

SUMMARY

Top clocks successfully mapped to block clocks:

DESCRIPTION

TA-1023

NAME

TA-1023

SUMMARY

Top clocks NOT mapped to any block clocks:

DESCRIPTION

TA-1024

NAME

TA-1024

SUMMARY

During read_timing_context processing, there were either top-level clocks found which could not be mapped to block-level clocks, or unmapped block clocks found. Review timing_context_clock_mapping.rpt file for list of such clocks. User should either change the clock definition to allow auto-mapping of such clocks or manually map them by providing clock map input file to read_timing_context -clock_mapping_file.

DESCRIPTION

TA-1025

NAME

TA-1025

SUMMARY

Block input port clocks NOT mapped to any top clocks:

DESCRIPTION

TA-1027

NAME

TA-1027

SUMMARY

During read_timing_context processing, there were top-level clocks found that are entering the block input ports with both positive and negative polarities. Such clocks can not be mapped to any single block-level clock. Please check your block level constraints. Examine timing_context_clock_mapping.rpt file for list of such clocks.

DESCRIPTION

TA-1028

NAME

TA-1028

SUMMARY

Reading latency file ...

DESCRIPTION

TA-1029

NAME

TA-1029

SUMMARY

A reporting error(s) has occurred while printing some paths in the previous report_timing output. The problem paths can be identified in the report by searching for the *DIAG:(Missing Adj) keyword. The slacks shown for these paths and the detailed path sections are correct, but the report is missing information to allow the steps of the slack computation section to sum properly. You should contact Cadence for assistance in resolving this problem.

DESCRIPTION

TA-1030

NAME

TA-1030

SUMMARY

No matching block-level clocks found on input ports:

DESCRIPTION

TA-1031

NAME

TA-1031

SUMMARY

No matching block-level clocks found on output ports:

DESCRIPTION

TA-1032

NAME

TA-1032

SUMMARY

There is a mismatch in the active views provided for block-level analysis compared with data provided from the top-level context. Block view(s) have no corresponding views in the top-level analysis. You should either inactivate the mismatching block-level views, or adjust the view names to match the related top-level views.

DESCRIPTION

TA-1033

NAME

TA-1033

SUMMARY

Read failed for delay CPPR adjustment for instance for clocks and in view from context db. The lookup failed at ports and for endpoints and respectively. The timing analysis will continue using a conservative adjustment value of 0ns.

DESCRIPTION

TA-1034

NAME

TA-1034

SUMMARY

Found invalid CPPR adjustment for instance for clocks and in view in context db. Invalid adjustment found for endpoint on port for , and endpoint on port for , . The timing analysis will continue using a conservative adjustment value of 0ns.

DESCRIPTION

TA-1035

NAME

TA-1035

SUMMARY

Read failed for arrival time for , for view at port for instance for clock in context db.

DESCRIPTION

TA-1036

NAME

TA-1036

SUMMARY

Clockset present on pin is inconsistent with the phases. Clock is missing in the clockset in view .

DESCRIPTION

TA-1037

NAME

TA-1037

SUMMARY

Clockset on pin is empty whereas clock phase for clock is present in view .

DESCRIPTION

TA-1038

NAME

TA-1038

SUMMARY

Mapping failed while trying to map , transition at pin for clock for view to port terminal.

DESCRIPTION

TA-1039

NAME

TA-1039

SUMMARY

Max input delay has been replaced by min value because min value was larger than max for input for edge.

DESCRIPTION

TA-1040

NAME

TA-1040

SUMMARY

A -ignore_clock_latency constraint has been asserted which results in port being of a path. This port also has a asserted on it with respect to the reference clock . After adjusting the delay to remove the current combined (source + network) clock latency of %.3f - the delay has become negative (%.3f). Analysis will continue use the negative delay time. You should review the boundary timing to ensure this is correct. If the sourcetwork latency embedded in the is inconsistent with the currently asserted set_clock_latency on the clock waveform - then this problem may arise.

DESCRIPTION

TA-1041

NAME

TA-1041

SUMMARY

Virtual block clocks NOT mapped to any top clocks:

DESCRIPTION

TA-1042

NAME

TA-1042

SUMMARY

Enabling global timing_cppr_enable_mismatch_transition_mode has precedence over disabling global timing_disable_backward_compatible_cppr_mode. Disable setting for global timing_disable_backward_compatible_cppr_mode ignored.

DESCRIPTION

TA-1043

NAME

TA-1043

SUMMARY

Number of block flat sinks for port is not equal to number of top flat sinks .

DESCRIPTION

TA-1044

NAME

TA-1044

SUMMARY

Context data was generated for different instances in merge directories and .

DESCRIPTION

TA-1045

NAME

TA-1045

SUMMARY

Context data was generated with different settings in merge directories and .

DESCRIPTION

TA-1046

NAME

TA-1046

SUMMARY

Context data merge directories have duplicated view . The views data will be taken from the first directory in the list which has the view.

DESCRIPTION

TA-1047

NAME

TA-1047

SUMMARY

Merged context (/) for port are different for view (orig = %.4f new = %.4f). The from first directory will be retained.

DESCRIPTION

TA-1048

NAME

TA-1048

SUMMARY

To provide block-level analysis that is consistent with the top-level analysis, the software expects most system settings to be set in a similar manner. The `read_timing_context` command has found that the setting of global variable with value is inconsistent with that of the current context database with value . The analysis will continue with the current setting - but to provide the most consistent results between top and block-level analyses, you should consider aligning the settings as closely as possible.

DESCRIPTION

TA-1049

NAME

TA-1049

SUMMARY

To provide block-level analysis that is consistent with the top-level analysis, the software expects most system settings to be set in a similar manner. The `read_timing_context` command has found that the default setting of global variable in the current session with value is inconsistent with the default value in the current context database with value . The analysis will continue with session current setting - but to provide the most consistent results between top and block-level analyses, you should consider aligning the settings as closely as possible.

DESCRIPTION

TA-1050

NAME

TA-1050

SUMMARY

To provide block-level analysis that is consistent with the top-level analysis, the software expects most system settings to be set in a similar manner. The `read_timing_context` command has found that some internal setting of global variable in the current session is inconsistent with the default value in the current context database. The analysis will continue with session current setting - but to provide the most consistent results between top and block-level analyses, you can contact Cadence Design Systems to correct the discrepancy.

DESCRIPTION

TA-1051

NAME

TA-1051

SUMMARY

To provide block-level analysis that is consistent with the top-level analysis, the software requires critical system global settings to be set in a similar manner. The `read_timing_context` command has failed because the setting of is inconsistent with the current context database. You should look at settings of which might have changed. You should make these settings consistent and then rerun the software.

DESCRIPTION

TA-1052

NAME

TA-1052

SUMMARY

To provide block-level analysis that is consistent with the top-level analysis, the software requires critical system global settings to be set in a similar manner. The `read_timing_context` command has failed because the setting of global variable with value is inconsistent with the current context database with value . You should make these settings consistent and then rerun the software.

DESCRIPTION

TA-1053

NAME

TA-1053

SUMMARY

Multi-Threaded Inter Power Domain Network identification for clock network voltage crossings initiated with threads.

DESCRIPTION

TA-1054

NAME

TA-1054

SUMMARY

Glitch context model can only be read after reading the block timing context model. Ignoring read_timing_context -glitch_incremental as it is specified without reading the block timing context model. Rerun the command after reading timing context model for module .

DESCRIPTION

TA-1057

NAME

TA-1057

SUMMARY

Timing context model contains port which is not found in current block. The analysis will continue by ignoring the context data for this port. For accurate context analysis, ensure block netlist consistency while writing and reading timing context.

DESCRIPTION

TA-1060

NAME

TA-1060

SUMMARY

Failed to read the block context model generated from which is newer software version than the current . Use or newer software version to read this context model.

DESCRIPTION

TA-1061

NAME

TA-1061

SUMMARY

Reading the block context model generated from which is older or same software version than the current .

DESCRIPTION

TA-1062

NAME

TA-1062

SUMMARY

Direction of block port was while writing the context, but it is in current design. Context data for this port will be ignored. For accurate analysis with timing context, ensure block netlist consistency between context writing and reading sessions.

DESCRIPTION

TA-1064

NAME

TA-1064

SUMMARY

The timer has encountered an unexpected error while trying to lookup delay CPPR adjustment for instance for top level clocks and in view from context db. The lookup failed at ports and for endpoints and respectively. You should contact Cadence for assistance resolving this problem. The timing analysis will continue using a conservative adjustment value of 0ns.

DESCRIPTION

TA-1065

NAME

TA-1065

SUMMARY

The timer has found invalid CPPR adjustment for instance for top level clocks and in view in context db. Invalid adjustment found for endpoint on port for , and endpoint on port for , . You should contact Cadence for assistance resolving this problem. The timing analysis will continue using a conservative adjustment value of 0ns.

DESCRIPTION

TA-1066

NAME

TA-1066

SUMMARY

Context external delay assertion at port with reference failed. Reference clock clock phase with polarity not found for , transition at the reference pin. You should contact Cadence for assistance resolving this problem.

DESCRIPTION

TA-1070

NAME

TA-1070

SUMMARY

Failed to read the block context model generated from which is a different release number than the current software version . Use the software of the same release number to read this context model.

DESCRIPTION

TA-1071

NAME

TA-1071

SUMMARY

Failed to read module as context generation for module was not successful. Generate context again for module .

DESCRIPTION

TA-1072

NAME

TA-1072

SUMMARY

Inconsistent die specification (/) across multiple views

DESCRIPTION

TA-1073

NAME

TA-1073

SUMMARY

Setting the global to value is disallowed after automated inter-die analysis infrastructure is built.

DESCRIPTION

TA-1075

NAME

TA-1075

SUMMARY

Timing context could not be read successfully from dir. Review the context generation log to check whether context write was successful or not. Reread the successfully generated context to ensure accurate analysis with timing context.

DESCRIPTION

TA-1076

NAME

TA-1076

SUMMARY

Timing context has been read successfully from dir.

DESCRIPTION

TA-1077

NAME

TA-1077

SUMMARY

Timing context could not be generated successfully in dir. Rerun the command to ensure successful generation of context.

DESCRIPTION

TA-1078

NAME

TA-1078

SUMMARY

Timing context could not be generated successfully. Rerun the command to ensure successful generation of context.

DESCRIPTION

TA-1079

NAME

TA-1079

SUMMARY

Timing Context has been generated successfully in dir.

DESCRIPTION

TA-1080

NAME

TA-1080

SUMMARY

Analysis has detected a missing clock arrival for one transition in view at pin . The existing clock edge will be propagated, but may result in issues in downstream logic if a fully formed clock is required. Pin is also a pulse-clock generator output. To analyze designs with pulsed latch cells, the global timing_enable_pulsed_latch should be set to true.

DESCRIPTION

TA-1081

NAME

TA-1081

SUMMARY

Detected a missing clock arrival for one transition in view at pin because a rise or fall case analysis is applied. The existing clock edge will be propagated, but may result in issues in downstream logic if a fully formed clock is required.

DESCRIPTION

TA-1083

NAME

TA-1083

SUMMARY

The software has detected a missing clock arrival in view at pin . An internal error has prevented delays from being calculated properly for all of the timing arcs relating to this pin. A single clock edge will continue to propagate beyond this pin, but may result in additional errors if the missing edge is required for downstream analysis. You will need to contact Cadence customer support for assistance in resolving this issue.

DESCRIPTION

The software requires that all timing arcs receive an appropriate delay annotation from the delay calculation system. If there is a problem in the software that prevents delay annotation from occurring - the related timing arc is treated as a disabled arc.

In the context of clock propagation, having both clock edges is generally a critical requirement. The TA-1083 error is issued if one of the required transitions is not available due to a timing arc being disabled due to a missing delay annotation. The software will continue to propagate the valid clock edge, but this may result in additional downstream errors where the missing clock edge is required for proper analyses.

A missing delay annotation is an internal error condition that is generally not related to the customers configuration, constraints, etc - and cannot be debugged in the field. Users should contact Cadence support for assistance.

TA-1084

NAME

TA-1084

SUMMARY

The software has detected that changes in the design have resulted in unexpected changes to previously identified loops in the design at pin . The analysis will continue, but may be unreliable due to these changes. This is an internal error condition - you should contact Cadence for additional support and guidance on this problem.

DESCRIPTION

TA-1085

NAME

TA-1085

SUMMARY

The software has detected that the user has changed the `timing_context_retain_port_constraints` global setting after the `read_timing_context` got completed. The software will ignore the last changed global setting for future analysis.

DESCRIPTION

TA-1086

NAME

TA-1086

SUMMARY

The software has detected that the block context analysis has different skewness setting than the top context generation. The timing results may not correlate between the top context generation and the block context analysis.

DESCRIPTION

TA-1087

NAME

TA-1087

SUMMARY

The instances of the module are not a part of the same cluster for MIM context analysis. The reason causing the difference can be found in the multi_instance_cluster_info.rpt file

DESCRIPTION

TA-1088

NAME

TA-1088

SUMMARY

The software has detected an invalid value of cpr threshold for view . Please provide non-negative value of cpr threshold for this view.

DESCRIPTION

TA-1089

NAME

TA-1089

SUMMARY

Merged context external delay constraint reference pin information found to be different for port in view . Previous reference pin is , while new reference pin is . from first directory will be retained.

DESCRIPTION

TA-1090

NAME

TA-1090

SUMMARY

Merged context input delay constraint reference pin information found to be different for port for , in view . Previous reference pin is , while new reference pin is . from first directory will be retained.

DESCRIPTION

TA-1091

NAME

TA-1091

SUMMARY

Merged context input delay constraints reference pin phase information found to be different for port for , in view . Information from first directory will be retained.

DESCRIPTION

TA-1092

NAME

TA-1092

SUMMARY

Merged context input delay constraints reference pin (/) for port are different for view (orig = %.4f new = %.4f). The from first directory will be retained.

DESCRIPTION

TA-1093

NAME

TA-1093

SUMMARY

The timing context model was not loaded because of an unknown clock stored in the context data for instance . This is likely because all instances of the block could not be put into a single multi-instance cluster in the hierarchical context saved for the block. Please check the multi_instance_cluster_info.rpt for the block in the session being loaded to see which block instances can be loaded together and specify them with the -instances option when reading the session.

DESCRIPTION

TA-1094

NAME

TA-1094

SUMMARY

Context data was generated with different setting in merge directories and . Software will use the maximum value of for analysis.

DESCRIPTION

TA-1095

NAME

TA-1095

SUMMARY

The software has detected that context data was generated with different settings in the merge dbs and , therefore timing context could not be read successfully. Regenerate the context data with consistent global settings across all dbs to successfully perform read_timing_context merge.

DESCRIPTION

TA-1096

NAME

TA-1096

SUMMARY

Hierarchical Constraint Checking: Clock mapping between reference clock and target clock established at data port of target instance , but no such clock port based mapping established.

DESCRIPTION

TA-1097

NAME

TA-1097

SUMMARY

Timing context is not supported in analysis mode. Please use setAnalysisMode command to switch analysis mode to onChipVariation mode to generate timing context.

DESCRIPTION

TA-1098

NAME

TA-1098

SUMMARY

Timing context is not supported in analysis mode. Please use setAnalysisMode command to switch analysis mode to onChipVariation mode to read timing context.

DESCRIPTION

TA-1099

NAME

TA-1099

SUMMARY

The bounding library needed for derate interpolation of library cell in delay_corner does not have derate factor asserted. Interpolation will proceed using a derate value of 1.0 for this library. To avoid incorrect analysis and results, you should assert derate factors for all libraries involved in interpolation.

DESCRIPTION

TA-1100

NAME

TA-1100

SUMMARY

No bounding libraries could be found for library cell in delay_corner when trying to interpolate the derate factor based on the current analysis voltage. A derate value of 1.0 will be used.

DESCRIPTION

TA-1101

NAME

TA-1101

SUMMARY

The software has detected block clock , in view , is entering port via multiple top level paths. This can happen if the port has multiple drivers at the top level. Slacks for timing paths driven by these ports may not co-relate well with the top level slacks due to multi-driver situation.

DESCRIPTION

TA-1102

NAME

TA-1102

SUMMARY

The software has detected that the block context analysis has different socv nsigma multiplier setting than the top context generated db in view in analysis mode, therefore the timing results may not correlate between the top context generation and the block context analysis.

DESCRIPTION

TA-1103

NAME

TA-1103

SUMMARY

Multi-Threaded Inter Power Domain Clock Network building initiated with threads.

DESCRIPTION

TA-1104

NAME

TA-1104

SUMMARY

The software has determined that the hierarchical net structure is too complex to handle constraints set of hierarchical pins connected to it. Those constraints will be ignored. This will likely impact the accuracy of the analysis. Please check the net.

DESCRIPTION

TA-1105

NAME

TA-1105

SUMMARY

The software has determined that the hierarchical context nested depth of context db is different from the nested depth of context db . Only hierarchical context dbs from the same nested depth can be read together in a single read_timing_context command.

DESCRIPTION

IMPTCM

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IMPTCM-1

NAME

IMPTCM-1

SUMMARY

Option name cannot be empty. Specify a valid option name.

DESCRIPTION

IMPTCM-2

NAME

IMPTCM-2

SUMMARY

"" is not a valid option name. Option name must have a leading -.

DESCRIPTION

IMPTCM-3

NAME

IMPTCM-3

SUMMARY

"" is not a valid value for the th argument.

DESCRIPTION

IMPTCM-4

NAME

IMPTCM-4

SUMMARY

The value "" specified for the type of argument "" is not a valid . Review the command specification and remove the argument or specify a legal value.

DESCRIPTION

IMPTCM-5

NAME

IMPTCM-5

SUMMARY

Missing value for th argument.

DESCRIPTION

IMPTCM-6

NAME

IMPTCM-6

SUMMARY

Missing value for option "".

DESCRIPTION

IMPTCM-7

NAME

IMPTCM-7

SUMMARY

Attempt to retrieve value from option "".

DESCRIPTION

IMPTCM-8

NAME

IMPTCM-8

SUMMARY

Attempt to set value for option "".

DESCRIPTION

IMPTCM-9

NAME

IMPTCM-9

SUMMARY

Cannot add twoArgs alias "" to option "" because it already has an alias.

DESCRIPTION

IMPTCM-10

NAME

IMPTCM-10

SUMMARY

Failed to set value for option "".

DESCRIPTION

IMPTCM-11

NAME

IMPTCM-11

SUMMARY

Cannot add option "" for command <> because "" is a reserved option.

DESCRIPTION

IMPTCM-12

NAME

IMPTCM-12

SUMMARY

Cannot add alias "" to option "" because the option already has another type of alias.

DESCRIPTION

IMPTCM-13

NAME

IMPTCM-13

SUMMARY

Cannot add alias "" to option "" because "" is not a valid bool alias.

DESCRIPTION

IMPTCM-14

NAME

IMPTCM-14

SUMMARY

Cannot add alias "" to option "" because it is not a option.

DESCRIPTION

IMPTCM-15

NAME

IMPTCM-15

SUMMARY

"" is not in the legal range of integer values for "". The value must be $\geq$ and $\leq$.

DESCRIPTION

IMPTCM-16

NAME

IMPTCM-16

SUMMARY

Cannot add alias "" to option "" because "" contains more aliases than enums.

DESCRIPTION

IMPTCM-17

NAME

IMPTCM-17

SUMMARY

"" is not a valid Boolean value for option "".

DESCRIPTION

IMPTCM-18

NAME

IMPTCM-18

SUMMARY

"" is not in the legal range of float values for "". The value must be $\geq$ and $\leq$

DESCRIPTION

IMPTCM-19

NAME

IMPTCM-19

SUMMARY

Command is obsolete and will be removed in future releases. Use instead. You should update to in your script.

DESCRIPTION

IMPTCM-20

NAME

IMPTCM-20

SUMMARY

Command is no longer supported. It has been replaced by .

DESCRIPTION

IMPTCM-21

NAME

IMPTCM-21

SUMMARY

Ambiguous option name "", found "" and "".

DESCRIPTION

IMPTCM-22

NAME

IMPTCM-22

SUMMARY

Failed to register command because it does not have "set" as prefix.

DESCRIPTION

IMPTCM-23

NAME

IMPTCM-23

SUMMARY

"" is not a valid enum for "", the allowed values are {}.

DESCRIPTION

IMPTCM-24

NAME

IMPTCM-24

SUMMARY

The associate option "" of alias "" does not exist.

DESCRIPTION

IMPTCM-25

NAME

IMPTCM-25

SUMMARY

"" is not a valid alias. The valid aliases are "".

DESCRIPTION

IMPTCM-26

NAME

IMPTCM-26

SUMMARY

Only string type option is allowed to have two arguments. Option "" is not a string option.

DESCRIPTION

IMPTCM-27

NAME

IMPTCM-27

SUMMARY

"" is not a valid Boolean value for option "". The allowed values are {true false}.

DESCRIPTION

IMPTCM-28

NAME

IMPTCM-28

SUMMARY

"" is not a valid ternary value for option "". The valid values are {auto true false}.

DESCRIPTION

IMPTCM-29

NAME

IMPTCM-29

SUMMARY

Command "" has already been registered.

DESCRIPTION

IMPTCM-30

NAME

IMPTCM-30

SUMMARY

Command "" is not registered in command manager.

DESCRIPTION

IMPTCM-31

NAME

IMPTCM-31

SUMMARY

Command "" does not take any arguments.

DESCRIPTION

IMPTCM-32

NAME

IMPTCM-32

SUMMARY

Wrong number of arguments specified for command "".

DESCRIPTION

IMPTCM-33

NAME

IMPTCM-33

SUMMARY

Failed to register arguments for command "".

DESCRIPTION

IMPTCM-34

NAME

IMPTCM-34

SUMMARY

Option "" does not have an argument name which is required for all mode options.

DESCRIPTION

IMPTCM-35

NAME

IMPTCM-35

SUMMARY

Option requires two floating numbers.

DESCRIPTION

IMPTCM-36

NAME

IMPTCM-36

SUMMARY

Must specify two floating numbers for argument "".

DESCRIPTION

IMPTCM-37

NAME

IMPTCM-37

SUMMARY

Option "" requires four floating numbers.

DESCRIPTION

IMPTCM-38

NAME

IMPTCM-38

SUMMARY

Argument "" is expected to be type argument, but it is specified as type.

DESCRIPTION

IMPTCM-39

NAME

IMPTCM-39

SUMMARY

Command "" already has "" as an alias argument.

DESCRIPTION

IMPTCM-40

NAME

IMPTCM-40

SUMMARY

Must specify four floating numbers for argument "".

DESCRIPTION

IMPTCM-41

NAME

IMPTCM-41

SUMMARY

Command "" already has an argument named "".

DESCRIPTION

IMPTCM-42

NAME

IMPTCM-42

SUMMARY

"" is not a legal option for command "". Check whether the option is an existing one through <command> -help.

DESCRIPTION

This message is displayed when a command is executed with incorrect option(s). The message will also display the list of switches which are supported by the command. Review your command and make sure that all the options you specified are legal.

IMPTCM-43

NAME

IMPTCM-43

SUMMARY

The associate option "" of alias "" is already specified.

DESCRIPTION

IMPTCM-44

NAME

IMPTCM-44

SUMMARY

Argument "" is already specified.

DESCRIPTION

IMPTCM-45

NAME

IMPTCM-45

SUMMARY

The alias "" for option "" is already specified.

DESCRIPTION

IMPTCM-46

NAME

IMPTCM-46

SUMMARY

Argument "" is required for command "", either this option is not specified or an option prior to it is not specified correctly.

DESCRIPTION

IMPTCM-47

NAME

IMPTCM-47

SUMMARY

Failed to create command "" because public command cannot be in namespace.

DESCRIPTION

IMPTCM-48

NAME

IMPTCM-48

SUMMARY

"" is not a legal option for command "". Either the current option or an option prior to it is not specified correctly.

DESCRIPTION

IMPTCM-49

NAME

IMPTCM-49

SUMMARY

Failed to create beta command "" because beta feature is not specified.

DESCRIPTION

IMPTCM-50

NAME

IMPTCM-50

SUMMARY

Cannot verify beta feature for beta command "" because beta checker is not available.

DESCRIPTION

IMPTCM-51

NAME

IMPTCM-51

SUMMARY

No beta feature registered for beta command "".

DESCRIPTION

IMPTCM-52

NAME

IMPTCM-52

SUMMARY

Option "" in the exclusive list "" is not a valid option.

DESCRIPTION

IMPTCM-53

NAME

IMPTCM-53

SUMMARY

At least one option in the exclusive option list "" needs to be specified.

DESCRIPTION

IMPTCM-54

NAME

IMPTCM-54

SUMMARY

"-help" failed for command "" because argument definitions have not been defined for the command.

DESCRIPTION

IMPTCM-55

NAME

IMPTCM-55

SUMMARY

"" is an internal command. "-help" is only available for public commands.

DESCRIPTION

IMPTCM-56

NAME

IMPTCM-56

SUMMARY

Failed to register options for command "".

DESCRIPTION

IMPTCM-57

NAME

IMPTCM-57

SUMMARY

Failed to run command "" because no option is registered with this command.

DESCRIPTION

IMPTCM-58

NAME

IMPTCM-58

SUMMARY

Option "" must be specified as the first option.

DESCRIPTION

IMPTCM-59

NAME

IMPTCM-59

SUMMARY

Option "" for command is obsolete and has been replaced by "". All options will now require "-". The obsolete option still works in this release but to avoid this warning and to ensure compatibility with future releases, update your script to use "".

DESCRIPTION

IMPTCM-60

NAME

IMPTCM-60

SUMMARY

No option specified for command "". Must specify at least one valid option.

DESCRIPTION

IMPTCM-61

NAME

IMPTCM-61

SUMMARY

Command "" is not a valid mode command.

DESCRIPTION

IMPTCM-62

NAME

IMPTCM-62

SUMMARY

"" is not a valid argument name. Argument name must not have a leading -.

DESCRIPTION

IMPTCM-63

NAME

IMPTCM-63

SUMMARY

Command "" does not have an older version proc.

DESCRIPTION

IMPTCM-64

NAME

IMPTCM-64

SUMMARY

Option "" requires two integer numbers.

DESCRIPTION

IMPTCM-65

NAME

IMPTCM-65

SUMMARY

Must specify two integer numbers for argument "".

DESCRIPTION

IMPTCM-66

NAME

IMPTCM-66

SUMMARY

Alias "" and "" are mutually exclusive. Do not specify both at the same time.

DESCRIPTION

IMPTCM-67

NAME

IMPTCM-67

SUMMARY

Option and are mutually exclusive. Check if more than one options that are mutually exclusive were given or if an invalid option was given.

DESCRIPTION

IMPTCM-68

NAME

IMPTCM-68

SUMMARY

"NULL" is not a valid value for option "".

DESCRIPTION

IMPTCM-69

NAME

IMPTCM-69

SUMMARY

"NULL" is not a valid argument name.

DESCRIPTION

IMPTCM-70

NAME

IMPTCM-70

SUMMARY

Option "" for command is obsolete and has been replaced by "". The obsolete option still works in this release but to avoid this warning and to ensure compatibility with future releases, update your script to use "".

DESCRIPTION

IMPTCM-71

NAME

IMPTCM-71

SUMMARY

Option "" for command is no longer supported. It has been replaced by "".

DESCRIPTION

IMPTCM-72

NAME

IMPTCM-72

SUMMARY

"" is not a valid alias for option "".

DESCRIPTION

IMPTCM-73

NAME

IMPTCM-73

SUMMARY

Option "" must have an argument, "" without an argument is the same as "". "" without argument still works in this release, but to avoid this warning and to ensure compatibility with future releases, update your script to use "".

DESCRIPTION

IMPTCM-74

NAME

IMPTCM-74

SUMMARY

"NULL" is not a valid option name.

DESCRIPTION

IMPTCM-75

NAME

IMPTCM-75

SUMMARY

Option "" for command is no longer supported.

DESCRIPTION

IMPTCM-76

NAME

IMPTCM-76

SUMMARY

"" is not a valid default value for enum option "". Check the valid enum values defined for the option.

DESCRIPTION

IMPTCM-77

NAME

IMPTCM-77

SUMMARY

Option "" for command is obsolete and will be removed in a future release. The obsolete option still works in this release but to avoid this warning and to ensure compatibility with future releases, remove the obsolete option from your script.

DESCRIPTION

IMPTCM-78

NAME

IMPTCM-78

SUMMARY

Option "" cannot be used with other options.

DESCRIPTION

IMPTCM-79

NAME

IMPTCM-79

SUMMARY

Failed to register aliases "" for command "" because an option of the same name exists already.

DESCRIPTION

IMPTCM-80

NAME

IMPTCM-80

SUMMARY

Attempt to add convert callback for option "".

DESCRIPTION

IMPTCM-81

NAME

IMPTCM-81

SUMMARY

Attempt to call parseArg for mode command "". Must use addOpt to register options for a mode command.

DESCRIPTION

IMPTCM-82

NAME

IMPTCM-82

SUMMARY

Failed to run command "" due to the above error.

DESCRIPTION

IMPTCM-84

NAME

IMPTCM-84

SUMMARY

Must specify a file name with option "" for command .

DESCRIPTION

IMPTCM-85

NAME

IMPTCM-85

SUMMARY

Failed to write out command to mode file because it is not a mode command.

DESCRIPTION

IMPTCM-86

NAME

IMPTCM-86

SUMMARY

Must specify at least one command name with option "" for command .

DESCRIPTION

IMPTCM-87

NAME

IMPTCM-87

SUMMARY

Option "" requires four integers.

DESCRIPTION

IMPTCM-88

NAME

IMPTCM-88

SUMMARY

Must specify four integers for argument "".

DESCRIPTION

IMPTCM-89

NAME

IMPTCM-89

SUMMARY

Command requires at least the command name.

DESCRIPTION

IMPTCM-90

NAME

IMPTCM-90

SUMMARY

Argument definition "" of command "" does not contain a valid argument type. The valid argument types are "".

DESCRIPTION

IMPTCM-91

NAME

IMPTCM-91

SUMMARY

Command "" is given an invalid argument definition "". Must specify at least argument type, argument name, and argument description.

DESCRIPTION

IMPTCM-92

NAME

IMPTCM-92

SUMMARY

Must specify the allowed enums in argument definition "" for command ""

DESCRIPTION

IMPTCM-93

NAME

IMPTCM-93

SUMMARY

Command "" has been given an invalid argument definition "". The isRequired field must be 1 or 0, but "" is specified.

DESCRIPTION

IMPTCM-94

NAME

IMPTCM-94

SUMMARY

Command "" has been given an invalid argument definition "". The isPublic field must be 1 or 0, but "" is specified.

DESCRIPTION

IMPTCM-95

NAME

IMPTCM-95

SUMMARY

Argument definition "" for command "" contains invalid min and max values - "" and "".

DESCRIPTION

IMPTCM-96

NAME

IMPTCM-96

SUMMARY

Failed to parse arguments for command "" because arguments have not been defined for the command.

DESCRIPTION

IMPTCM-97

NAME

IMPTCM-97

SUMMARY

Argument "" in the exclusive argument list "" is not a valid option for command "".

DESCRIPTION

IMPTCM-98

NAME

IMPTCM-98

SUMMARY

Exclusive arguments must be either all required or all optional. Argument "" is registered as required, but "" is registered as optional.

DESCRIPTION

IMPTCM-99

NAME

IMPTCM-99

SUMMARY

Exclusive arguments must be options. Argument "" is not registered as an option.

DESCRIPTION

IMPTCM-100

NAME

IMPTCM-100

SUMMARY

Invalid range values (min = , max =) specified for option "".

DESCRIPTION

IMPTCM-101

NAME

IMPTCM-101

SUMMARY

Invalid range values (min = , max =) specified for option "".

DESCRIPTION

IMPTCM-102

NAME

IMPTCM-102

SUMMARY

Default value for option "" is out of the specified range (min = , max =).

DESCRIPTION

IMPTCM-103

NAME

IMPTCM-103

SUMMARY

Default value for option "" is out of the specified range (min = , max =).

DESCRIPTION

IMPTCM-104

NAME

IMPTCM-104

SUMMARY

Attempt to register invalid option name "" for command "".

DESCRIPTION

IMPTCM-105

NAME

IMPTCM-105

SUMMARY

"" is not recognized as a legal point list.

DESCRIPTION

IMPTCM-106

NAME

IMPTCM-106

SUMMARY

Option "" requires integer numbers.

DESCRIPTION

IMPTCM-107

NAME

IMPTCM-107

SUMMARY

Must specify integer numbers for argument "".

DESCRIPTION

IMPTCM-108

NAME

IMPTCM-108

SUMMARY

Option "" requires floating numbers.

DESCRIPTION

IMPTCM-109

NAME

IMPTCM-109

SUMMARY

Must specify floating numbers for argument "".

DESCRIPTION

IMPTCM-110

NAME

IMPTCM-110

SUMMARY

Failed to find matching "" for "".

DESCRIPTION

IMPTCM-111

NAME

IMPTCM-111

SUMMARY

Attempt to register both tcmArgExclusive and tcmArgGroup for command "".

DESCRIPTION

IMPTCM-112

NAME

IMPTCM-112

SUMMARY

Group "" contains invalid option "".

DESCRIPTION

IMPTCM-113

NAME

IMPTCM-113

SUMMARY

Option "" is required when option "" is specified. Type man for more details.

DESCRIPTION

IMPTCM-114

NAME

IMPTCM-114

SUMMARY

"" is an invalid value for option "". Make sure the value is in hex format.

DESCRIPTION

IMPTCM-115

NAME

IMPTCM-115

SUMMARY

Option "" was not specified correctly.

DESCRIPTION

IMPTCM-116

NAME

IMPTCM-116

SUMMARY

"-help" is currently not available. Use "help " to get the usage.

DESCRIPTION

IMPTCM-117

NAME

IMPTCM-117

SUMMARY

"-keepExtraOption" cannot be specified for command "" because it has none -option argument that is optional.

DESCRIPTION

IMPTCM-118

NAME

IMPTCM-118

SUMMARY

Command "" was not invoked with "-reset".

DESCRIPTION

IMPTCM-119

NAME

IMPTCM-119

SUMMARY

Failed to parse arguments because"" is not a valid command name.

DESCRIPTION

IMPTCM-120

NAME

IMPTCM-120

SUMMARY

Illegal argument definition: Both non-option arguments, "" and "", allow variable arguments.

DESCRIPTION

IMPTCM-121

NAME

IMPTCM-121

SUMMARY

Illegal argument definition: Option "" allows variable arguments and non-option argument "".

DESCRIPTION

IMPTCM-122

NAME

IMPTCM-122

SUMMARY

Illegal argument definition: Non-option argument that allows variable arguments and optional non-option argument.

DESCRIPTION

IMPTCM-123

NAME

IMPTCM-123

SUMMARY

Options in the repeat option list are not valid options for command .

DESCRIPTION

IMPTCM-124

NAME

IMPTCM-124

SUMMARY

Option in the repeat option list is a type option which cannot be repeated.

DESCRIPTION

IMPTCM-125

NAME

IMPTCM-125

SUMMARY

Option "" for command is obsolete and will be ignored. It no longer has any effect and should be removed from your script.

DESCRIPTION

IMPTCM-126

NAME

IMPTCM-126

SUMMARY

Option "" no longer requires a true/false value.

DESCRIPTION

IMPTCM-127

NAME

IMPTCM-127

SUMMARY

Invalid unit . The first letter represents scale and it must be k, m, u, n, p, f, or a. The default unit is .

DESCRIPTION

IMPTCM-128

NAME

IMPTCM-128

SUMMARY

"" and "" are mutually exclusive. Do not specify both at the same time.

DESCRIPTION

IMPTCM-129

NAME

IMPTCM-129

SUMMARY

Ambiguous option name "", found "".

DESCRIPTION

IMPTCM-130

NAME

IMPTCM-130

SUMMARY

Option "" requires arguments, only is given.

DESCRIPTION

IMPTCM-131

NAME

IMPTCM-131

SUMMARY

Argument "" requires arguments, only is given.

DESCRIPTION

IMPTCM-132

NAME

IMPTCM-132

SUMMARY

"" requires arguments, but the given argument value "" has .

DESCRIPTION

IMPTCM-133

NAME

IMPTCM-133

SUMMARY

Argument number for "alias" type is not correct, 4 is required while is provided.

DESCRIPTION

IMPTCM-134

NAME

IMPTCM-134

SUMMARY

Alias type is not recognized: "". The valid alias types are "".

DESCRIPTION

IMPTCM-135

NAME

IMPTCM-135

SUMMARY

Illegal argument definition: More than one non-option arguments and one of them allows variable arguments.

DESCRIPTION

IMPTCM-136

NAME

IMPTCM-136

SUMMARY

" -help" is not available. The command is obsolete.

DESCRIPTION

IMPTCM-137

NAME

IMPTCM-137

SUMMARY

Object in command "" is not a list, does not support object list usage.

DESCRIPTION

IMPTCM-139

NAME

IMPTCM-139

SUMMARY

Object in command "" is a list, does not support single object usage.

DESCRIPTION

IMPTCM-140

NAME

IMPTCM-140

SUMMARY

"" type is not supported in object parse.

DESCRIPTION

IMPTCM-141

NAME

IMPTCM-141

SUMMARY

Argument definition "" for command "" does not contain object type.

DESCRIPTION

IMPTCM-142

NAME

IMPTCM-142

SUMMARY

Object in command "" does not support "" type object.

DESCRIPTION

IMPTCM-143

NAME

IMPTCM-143

SUMMARY

Bad object add "".

DESCRIPTION

IMPTCM-144

NAME

IMPTCM-144

SUMMARY

Option "" for command "" contains invalid flag "".

DESCRIPTION

IMPTCM-145

NAME

IMPTCM-145

SUMMARY

Argument definition "" for command "" is invalid.

DESCRIPTION

IMPTCM-146

NAME

IMPTCM-146

SUMMARY

"" and "" are mutually exclusive. Do not specify both at the same time in the option "" definition for command "".

DESCRIPTION

IMPTCM-148

NAME

IMPTCM-148

SUMMARY

Unable to create new manager on current thread because there is already TCM manager created on current thread.

DESCRIPTION

IMPTCM-149

NAME

IMPTCM-149

SUMMARY

Mode commands are global settings. Must be defined on the global Tcl interpreter in this session. Ignore the given Tcl interpreter, create and return one on the global Tcl Interpreter.

DESCRIPTION

IMPTCM-150

NAME

IMPTCM-150

SUMMARY

Unable to create TCM manager because the specified interpreter has already created a TCM manager on it

DESCRIPTION

IMPTCM-151

NAME

IMPTCM-151

SUMMARY

Could not get TCM manager with NULL Tcl interpreter. Will return the first created TCM manager in this session.

DESCRIPTION

IMPTCM-152

NAME

IMPTCM-152

SUMMARY

Invalid interpreter specified when creating TCM command . The command will not be created.

DESCRIPTION

IMPTCM-153

NAME

IMPTCM-153

SUMMARY

Invalid interpreter specified when finding TCM command.

DESCRIPTION

IMPTCM-154

NAME

IMPTCM-154

SUMMARY

Unable to delete main TCM manager with the specified Tcl interpreter.

DESCRIPTION

IMPTCM-155

NAME

IMPTCM-155

SUMMARY

Unable to get TCM manager through the given Tcl interpreter. Will return the corresponding manager in current thread.

DESCRIPTION

IMPTCM-160

NAME

IMPTCM-160

SUMMARY

"" and "" are mutually exclusive. Do not specify both at the same time in object definition.

DESCRIPTION

IMPTCM-161

NAME

IMPTCM-161

SUMMARY

Object for command "" is not a list, "" matches more than one objects for specified type "" object in design.

DESCRIPTION

IMPTCM-162

NAME

IMPTCM-162

SUMMARY

"" does not match any object in design for specified type "" object in command "".

DESCRIPTION

IMPTCM-163

NAME

IMPTCM-163

SUMMARY

Option "" for command is obsolete and will be ignored. It no longer has any effect and should be removed from your script.

DESCRIPTION

IMPTCM-164

NAME

IMPTCM-164

SUMMARY

Option "" for command is obsolete and will be removed in a future release. The obsolete option still works in this release but to avoid this message and to ensure compatibility with future releases,

remove the obsolete option from your script.

DESCRIPTION

IMPTCM-165

NAME

IMPTCM-165

SUMMARY

"" does not match any object for specified type "" object in command "".

DESCRIPTION

IMPTCM-166

NAME

IMPTCM-166

SUMMARY

Alias define of argument "" for command "" is not correct, should be "alias_name alias_type".

DESCRIPTION

IMPTCM-167

NAME

IMPTCM-167

SUMMARY

Object parse failed in command "" due to previous warning.

DESCRIPTION

IMPTCM-168

NAME

IMPTCM-168

SUMMARY

Ambiguous option name "" matches more than one option: "". You must use the full option name, or a unique prefix of one of the legal options.

DESCRIPTION

IMPTCM-169

NAME

IMPTCM-169

SUMMARY

Failed to init options for command .

DESCRIPTION

IMPTCM-170

NAME

IMPTCM-170

SUMMARY

"" is a partial match option to "" for command "".

DESCRIPTION

IMPTCM-171

NAME

IMPTCM-171

SUMMARY

Cannot output to "".

DESCRIPTION

IMPTCM-172

NAME

IMPTCM-172

SUMMARY

Cannot specify non-option argument "" after non-option argument "" which allows variable values.

DESCRIPTION

IMPTCM-173

NAME

IMPTCM-173

SUMMARY

"" is not recognized as a legal box list.

DESCRIPTION

IMPTCM-174

NAME

IMPTCM-174

SUMMARY

Argument definition "" of command "" has "" which is not valid. The valid flag types are "".

DESCRIPTION

IMPTCM-175

NAME

IMPTCM-175

SUMMARY

Argument definition "" of command "" has "" which is already defined. Change the name to one that does not exist.

DESCRIPTION

IMPTCM-176

NAME

IMPTCM-176

SUMMARY

Argument definition "" of command "" has "" which is not valid. The valid usage is "".

DESCRIPTION

IMPTCM-177

NAME

IMPTCM-177

SUMMARY

Boolean argument "" of command "" is value-allowed, it cannot be repeated.

DESCRIPTION

IMPTCM-178

NAME

IMPTCM-178

SUMMARY

Enum "" of option "" is partial-matched to "".

DESCRIPTION

IMPTCM-179

NAME

IMPTCM-179

SUMMARY

Ambiguous enums found for given partial enum "": "", "".

DESCRIPTION

IMPTCM-180

NAME

IMPTCM-180

SUMMARY

"" is not recognized as a legal polygon list.

DESCRIPTION

IMPTCM-181

NAME

IMPTCM-181

SUMMARY

"" is not a recognized rename option.

DESCRIPTION

IMPTCM-182

NAME

IMPTCM-182

SUMMARY

"" is an existed option name.

DESCRIPTION

IMPTCM-183

NAME

IMPTCM-183

SUMMARY

"" is not allowed to use in current Innovus session.

DESCRIPTION

IMPTCM-184

NAME

IMPTCM-184

SUMMARY

Need string value after "" option.

DESCRIPTION

IMPTCM-185

NAME

IMPTCM-185

SUMMARY

"" is not a valid name for "" option.

DESCRIPTION

IMPTCM-186

NAME

IMPTCM-186

SUMMARY

"" is not a valid die name.

DESCRIPTION

IMPTCM-187

NAME

IMPTCM-187

SUMMARY

"" option need a non empty value.

DESCRIPTION

IMPTCM-188

NAME

IMPTCM-188

SUMMARY

Need to specify type character: e, s, f, d, n, or b.

DESCRIPTION

IMPTCM-189

NAME

IMPTCM-189

SUMMARY

Invalid type character "". Allowed types are : e, s, f, d, n, or b.

DESCRIPTION

IMPTCM-190

NAME

IMPTCM-190

SUMMARY

Boolean options must be flagged.

DESCRIPTION

IMPTCM-191

NAME

IMPTCM-191

SUMMARY

Need to specify required/optional/obsolete character.

DESCRIPTION

IMPTCM-192

NAME

IMPTCM-192

SUMMARY

Need to specify single/multiple character.

DESCRIPTION

IMPTCM-193

NAME

IMPTCM-193

SUMMARY

"" specified is not valid for unflagged options.

DESCRIPTION

IMPTCM-194

NAME

IMPTCM-194

SUMMARY

Only directory and string options can be multi-instanced.

DESCRIPTION

IMPTCM-195

NAME

IMPTCM-195

SUMMARY

Boolean options cannot be multi-valued.

DESCRIPTION

IMPTCM-196

NAME

IMPTCM-196

SUMMARY

Boolean options can not be required.

DESCRIPTION

IMPTCM-197

NAME

IMPTCM-197

SUMMARY

Configuration string too long (should be 3 characters).

DESCRIPTION

IMPTCM-198

NAME

IMPTCM-198

SUMMARY

Type-list string must be bounded by parentheses.

DESCRIPTION

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TCLCMD-100

NAME

TCLCMD-100

SUMMARY

Target technology library not set. Use set_global

DESCRIPTION

TCLCMD-101

NAME

TCLCMD-101

SUMMARY

No view selected

DESCRIPTION

You must set a current view by setting a current module

TCLCMD-102

NAME

TCLCMD-102

SUMMARY

Attempt to map the netlist failed

DESCRIPTION

TCLCMD-103

NAME

TCLCMD-103

SUMMARY

No critical region extracted

DESCRIPTION

TCLCMD-104

NAME

TCLCMD-104

SUMMARY

Attempt to unmap the netlist failed

DESCRIPTION

TCLCMD-105

NAME

TCLCMD-105

SUMMARY

No non-critical region extracted

DESCRIPTION

TCLCMD-106

NAME

TCLCMD-106

SUMMARY

Invalid epsilon value

DESCRIPTION

TCLCMD-107

NAME

TCLCMD-107

SUMMARY

Invalid fanin_depth value

DESCRIPTION

TCLCMD-108

NAME

TCLCMD-108

SUMMARY

Invalid fanout_depth value

DESCRIPTION

TCLCMD-109

NAME

TCLCMD-109

SUMMARY

Invalid add_delay value

DESCRIPTION

TCLCMD-110

NAME

TCLCMD-110

SUMMARY

Invalid type value

DESCRIPTION

TCLCMD-111

NAME

TCLCMD-111

SUMMARY

Invalid max_size value

DESCRIPTION

TCLCMD-112

NAME

TCLCMD-112

SUMMARY

Unknown option

DESCRIPTION

TCLCMD-113

NAME

TCLCMD-113

SUMMARY

Could not open file to write

DESCRIPTION

TCLCMD-114

NAME

TCLCMD-114

SUMMARY

Invalid level value

DESCRIPTION

TCLCMD-115

NAME

TCLCMD-115

SUMMARY

Invalid threshold value

DESCRIPTION

TCLCMD-116

NAME

TCLCMD-116

SUMMARY

Invalid type of object

DESCRIPTION

TCLCMD-117

NAME

TCLCMD-117

SUMMARY

Invalid object id

DESCRIPTION

TCLCMD-118

NAME

TCLCMD-118

SUMMARY

Module not found in database

DESCRIPTION

TCLCMD-119

NAME

TCLCMD-119

SUMMARY

No module selected

DESCRIPTION

You must set a current module

TCLCMD-120

NAME

TCLCMD-120

SUMMARY

View not found in module

DESCRIPTION

TCLCMD-121

NAME

TCLCMD-121

SUMMARY

Optimization script not set yet

DESCRIPTION

TCLCMD-122

NAME

TCLCMD-122

SUMMARY

Attempt to optimize the netlist failed

DESCRIPTION

TCLCMD-123

NAME

TCLCMD-123

SUMMARY

Could not find file to open

DESCRIPTION

TCLCMD-124

NAME

TCLCMD-124

SUMMARY

Primitive gate type not specified

DESCRIPTION

TCLCMD-125

NAME

TCLCMD-125

SUMMARY

Unknown primitive type (nor or inv only)

DESCRIPTION

TCLCMD-126

NAME

TCLCMD-126

SUMMARY

Invalid block_rise value

DESCRIPTION

TCLCMD-127

NAME

TCLCMD-127

SUMMARY

Invalid block_fall value

DESCRIPTION

TCLCMD-128

NAME

TCLCMD-128

SUMMARY

Invalid drive_rise value

DESCRIPTION

TCLCMD-129

NAME

TCLCMD-129

SUMMARY

Invalid drive_fall value

DESCRIPTION

TCLCMD-130

NAME

TCLCMD-130

SUMMARY

Invalid load value

DESCRIPTION

TCLCMD-131

NAME

TCLCMD-131

SUMMARY

Invalid max_load value

DESCRIPTION

TCLCMD-132

NAME

TCLCMD-132

SUMMARY

Invalid phase value (inv, noninv and unknown only)

DESCRIPTION

TCLCMD-133

NAME

TCLCMD-133

SUMMARY

Invalid user_rise value

DESCRIPTION

TCLCMD-134

NAME

TCLCMD-134

SUMMARY

Invalid user_fall value

DESCRIPTION

TCLCMD-135

NAME

TCLCMD-135

SUMMARY

Invalid distance value

DESCRIPTION

TCLCMD-136

NAME

TCLCMD-136

SUMMARY

Invalid weight value

DESCRIPTION

TCLCMD-137

NAME

TCLCMD-137

SUMMARY

Invalid epsilon value

DESCRIPTION

TCLCMD-138

NAME

TCLCMD-138

SUMMARY

Invalid iteration value

DESCRIPTION

TCLCMD-139

NAME

TCLCMD-139

SUMMARY

Attempt to restructure the netlist failed

DESCRIPTION

TCLCMD-140

NAME

TCLCMD-140

SUMMARY

Invalid delay coefficient value

DESCRIPTION

TCLCMD-141

NAME

TCLCMD-141

SUMMARY

Conflicting options -footprint and -buffer

DESCRIPTION

TCLCMD-142

NAME

TCLCMD-142

SUMMARY

Conflicting options -footprint and -clone

DESCRIPTION

TCLCMD-143

NAME

TCLCMD-143

SUMMARY

Conflicting options -design_rule_only and -no_design_rule

DESCRIPTION

TCLCMD-144

NAME

TCLCMD-144

SUMMARY

Conflicting options -min_hold and -fix_hold

DESCRIPTION

TCLCMD-145

NAME

TCLCMD-145

SUMMARY

Conflicting options -design_rule_only and -min_hold

DESCRIPTION

TCLCMD-146

NAME

TCLCMD-146

SUMMARY

Conflicting options -design_rule_only and -fix_hold

DESCRIPTION

TCLCMD-147

NAME

TCLCMD-147

SUMMARY

Invalid iter value

DESCRIPTION

TCLCMD-148

NAME

TCLCMD-148

SUMMARY

Invalid speed_depth value

DESCRIPTION

TCLCMD-149

NAME

TCLCMD-149

SUMMARY

Invalid eval_factor value

DESCRIPTION

TCLCMD-150

NAME

TCLCMD-150

SUMMARY

Invalid critical_ratio value

DESCRIPTION

TCLCMD-151

NAME

TCLCMD-151

SUMMARY

Unknown property name

DESCRIPTION

TCLCMD-152

NAME

TCLCMD-152

SUMMARY

Unknown property value

DESCRIPTION

TCLCMD-153

NAME

TCLCMD-153

SUMMARY

Invalid cell name or id

DESCRIPTION

TCLCMD-154

NAME

TCLCMD-154

SUMMARY

Object is not a library cell

DESCRIPTION

TCLCMD-155

NAME

TCLCMD-155

SUMMARY

Attempt to set property on cell failed

DESCRIPTION

TCLCMD-156

NAME

TCLCMD-156

SUMMARY

Conflicting options -no_partition and -max_size

DESCRIPTION

TCLCMD-157

NAME

TCLCMD-157

SUMMARY

Invalid max_size value

DESCRIPTION

TCLCMD-158

NAME

TCLCMD-158

SUMMARY

Unrecognized restructure option

DESCRIPTION

TCLCMD-159

NAME

TCLCMD-159

SUMMARY

Conflicting options -no_partition and -max_size

DESCRIPTION

TCLCMD-160

NAME

TCLCMD-160

SUMMARY

Conflicting options -no_hill_climb and -max_uphill

DESCRIPTION

TCLCMD-161

NAME

TCLCMD-161

SUMMARY

Invalid fanin_depth value

DESCRIPTION

TCLCMD-162

NAME

TCLCMD-162

SUMMARY

Invalid fanout_depth value

DESCRIPTION

TCLCMD-163

NAME

TCLCMD-163

SUMMARY

Invalid max_uphill_moves value

DESCRIPTION

TCLCMD-164

NAME

TCLCMD-164

SUMMARY

Number of end points not specified

DESCRIPTION

TCLCMD-165

NAME

TCLCMD-165

SUMMARY

Invalid number of end points

DESCRIPTION

TCLCMD-166

NAME

TCLCMD-166

SUMMARY

No value provided for option . The option requires a value to be specified for it. Run -help to see the basic syntax for this command. Run man to see a complete description of the command and its options. Review the syntax and specify a legal value for option .

DESCRIPTION

TCLCMD-167

NAME

TCLCMD-167

SUMMARY

Invalid min_slack value

DESCRIPTION

TCLCMD-168

NAME

TCLCMD-168

SUMMARY

Invalid max_slack value

DESCRIPTION

TCLCMD-169

NAME

TCLCMD-169

SUMMARY

The current module is not unique with respect to the top timing module.

DESCRIPTION

You must set a top timing module and set a unique current module or set a current instance (using set_current_instance) before performing any timing commands.

TCLCMD-170

NAME

TCLCMD-170

SUMMARY

The report column widths is not an integer

DESCRIPTION

TCLCMD-171

NAME

TCLCMD-171

SUMMARY

Verbosity level must be an integer from 0 to 9 or on or off

DESCRIPTION

TCLCMD-172

NAME

TCLCMD-172

SUMMARY

Verbosity level must be an integer from 3 to 9 or on or off

DESCRIPTION

TCLCMD-173

NAME

TCLCMD-173

SUMMARY

Unrecognized module

DESCRIPTION

TCLCMD-174

NAME

TCLCMD-174

SUMMARY

Must specify messages as on or off

DESCRIPTION

TCLCMD-175

NAME

TCLCMD-175

SUMMARY

Unrecognized table

DESCRIPTION

TCLCMD-176

NAME

TCLCMD-176

SUMMARY

Mandatory option not specified

DESCRIPTION

TCLCMD-177

NAME

TCLCMD-177

SUMMARY

Invalid indent value

DESCRIPTION

TCLCMD-178

NAME

TCLCMD-178

SUMMARY

Invalid truncate value

DESCRIPTION

TCLCMD-179

NAME

TCLCMD-179

SUMMARY

Invalid style value

DESCRIPTION

TCLCMD-180

NAME

TCLCMD-180

SUMMARY

Invalid major_sort value

DESCRIPTION

TCLCMD-181

NAME

TCLCMD-181

SUMMARY

Invalid minor_sort value

DESCRIPTION

TCLCMD-182

NAME

TCLCMD-182

SUMMARY

More than values specified for option max_widths

DESCRIPTION

TCLCMD-183

NAME

TCLCMD-183

SUMMARY

Invalid max_widths value

DESCRIPTION

TCLCMD-184

NAME

TCLCMD-184

SUMMARY

Option max_widths must contain at least one positive column

DESCRIPTION

TCLCMD-185

NAME

TCLCMD-185

SUMMARY

More than values specified for option min_widths

DESCRIPTION

TCLCMD-186

NAME

TCLCMD-186

SUMMARY

Invalid min_widths value

DESCRIPTION

TCLCMD-187

NAME

TCLCMD-187

SUMMARY

Attempt to read file(s) failed

DESCRIPTION

TCLCMD-188

NAME

TCLCMD-188

SUMMARY

No module found in parse tree

DESCRIPTION

TCLCMD-189

NAME

TCLCMD-189

SUMMARY

Unknown global attribute

DESCRIPTION

TCLCMD-190

NAME

TCLCMD-190

SUMMARY

Invalid begin value

DESCRIPTION

TCLCMD-191

NAME

TCLCMD-191

SUMMARY

Invalid end value

DESCRIPTION

TCLCMD-192

NAME

TCLCMD-192

SUMMARY

Invalid boolean value

DESCRIPTION

TCLCMD-193

NAME

TCLCMD-193

SUMMARY

Attempt to create type failed

DESCRIPTION

TCLCMD-194

NAME

TCLCMD-194

SUMMARY

Could not create module

DESCRIPTION

TCLCMD-195

NAME

TCLCMD-195

SUMMARY

Could not create view

DESCRIPTION

TCLCMD-196

NAME

TCLCMD-196

SUMMARY

Could not find module in database

DESCRIPTION

TCLCMD-197

NAME

TCLCMD-197

SUMMARY

Attempt to select module failed

DESCRIPTION

TCLCMD-198

NAME

TCLCMD-198

SUMMARY

Could not find view in current module

DESCRIPTION

TCLCMD-199

NAME

TCLCMD-199

SUMMARY

Attempt to select view failed

DESCRIPTION

TCLCMD-200

NAME

TCLCMD-200

SUMMARY

Attempt to create cell failed

DESCRIPTION

TCLCMD-201

NAME

TCLCMD-201

SUMMARY

Attempt to create port failed

DESCRIPTION

TCLCMD-202

NAME

TCLCMD-202

SUMMARY

Attempt to write Verilog file failed

DESCRIPTION

TCLCMD-203

NAME

TCLCMD-203

SUMMARY

Attempt to write bdf file failed

DESCRIPTION

TCLCMD-204

NAME

TCLCMD-204

SUMMARY

Attempt to write library file failed

DESCRIPTION

TCLCMD-205

NAME

TCLCMD-205

SUMMARY

Attempt to write blif file failed

DESCRIPTION

TCLCMD-206

NAME

TCLCMD-206

SUMMARY

Unknown command argument

DESCRIPTION

TCLCMD-207

NAME

TCLCMD-207

SUMMARY

Attempt to create enum type failed

DESCRIPTION

TCLCMD-208

NAME

TCLCMD-208

SUMMARY

Attempt to create port reference failed

DESCRIPTION

TCLCMD-209

NAME

TCLCMD-209

SUMMARY

Attempt to set net generator failed

DESCRIPTION

TCLCMD-210

NAME

TCLCMD-210

SUMMARY

Attempt to set bus component generator failed

DESCRIPTION

TCLCMD-211

NAME

TCLCMD-211

SUMMARY

Attempt to set instance generator failed

DESCRIPTION

TCLCMD-212

NAME

TCLCMD-212

SUMMARY

Unknown wired logic function (wand and wor only)

DESCRIPTION

TCLCMD-213

NAME

TCLCMD-213

SUMMARY

Attempt to add member to enum type failed

DESCRIPTION

TCLCMD-214

NAME

TCLCMD-214

SUMMARY

Attempt to create instance failed

DESCRIPTION

TCLCMD-215

NAME

TCLCMD-215

SUMMARY

Attempt to create net failed

DESCRIPTION

TCLCMD-216

NAME

TCLCMD-216

SUMMARY

Attempt to get pin failed

DESCRIPTION

TCLCMD-217

NAME

TCLCMD-217

SUMMARY

Attempt to add connection failed

DESCRIPTION

TCLCMD-218

NAME

TCLCMD-218

SUMMARY

Attempt to add net connection failed

DESCRIPTION

TCLCMD-219

NAME

TCLCMD-219

SUMMARY

Attempt to set attribute on object failed

DESCRIPTION

TCLCMD-220

NAME

TCLCMD-220

SUMMARY

Attempt to set attribute on context failed

DESCRIPTION

TCLCMD-221

NAME

TCLCMD-221

SUMMARY

: Could not find driving cell in any library

DESCRIPTION

TCLCMD-222

NAME

TCLCMD-222

SUMMARY

Could not find instance in current module

DESCRIPTION

TCLCMD-223

NAME

TCLCMD-223

SUMMARY

Could not find net in current module

DESCRIPTION

TCLCMD-224

NAME

TCLCMD-224

SUMMARY

Could not find port in current module

DESCRIPTION

TCLCMD-225

NAME

TCLCMD-225

SUMMARY

Could not find type

DESCRIPTION

TCLCMD-226

NAME

TCLCMD-226

SUMMARY

Could not find type of object

DESCRIPTION

TCLCMD-227

NAME

TCLCMD-227

SUMMARY

Object is neither bus port nor bus net

DESCRIPTION

TCLCMD-228

NAME

TCLCMD-228

SUMMARY

Attempt to read library failed

DESCRIPTION

TCLCMD-229

NAME

TCLCMD-229

SUMMARY

Could not find the target technology to update

DESCRIPTION

TCLCMD-230

NAME

TCLCMD-230

SUMMARY

Attempt to read bdf file failed

DESCRIPTION

TCLCMD-231

NAME

TCLCMD-231

SUMMARY

Attempt to free object failed

DESCRIPTION

TCLCMD-232

NAME

TCLCMD-232

SUMMARY

Attempt to load ATL library failed

DESCRIPTION

TCLCMD-233

NAME

TCLCMD-233

SUMMARY

Attempt to duplicate view failed

DESCRIPTION

TCLCMD-234

NAME

TCLCMD-234

SUMMARY

Module already exists

DESCRIPTION

TCLCMD-235

NAME

TCLCMD-235

SUMMARY

Module is empty

DESCRIPTION

TCLCMD-236

NAME

TCLCMD-236

SUMMARY

Attempt to duplicate module failed

DESCRIPTION

TCLCMD-237

NAME

TCLCMD-237

SUMMARY

Attempt to create module reference for view failed

DESCRIPTION

TCLCMD-238

NAME

TCLCMD-238

SUMMARY

No such object found

DESCRIPTION

TCLCMD-239

NAME

TCLCMD-239

SUMMARY

Invalid object to uniquely instantiate

DESCRIPTION

TCLCMD-240

NAME

TCLCMD-240

SUMMARY

Invalid object to dissolve

DESCRIPTION

TCLCMD-241

NAME

TCLCMD-241

SUMMARY

Attempt to link ACL module(s) failed

DESCRIPTION

TCLCMD-242

NAME

TCLCMD-242

SUMMARY

Attempt to move net connections failed

DESCRIPTION

TCLCMD-243

NAME

TCLCMD-243

SUMMARY

Attempt to flatten selected view(s) failed

DESCRIPTION

TCLCMD-244

NAME

TCLCMD-244

SUMMARY

Could not open file to read

DESCRIPTION

TCLCMD-245

NAME

TCLCMD-245

SUMMARY

Attempt to load library failed

DESCRIPTION

TCLCMD-246

NAME

TCLCMD-246

SUMMARY

Attempt to read database file failed

DESCRIPTION

TCLCMD-247

NAME

TCLCMD-247

SUMMARY

Invalid threshold value

DESCRIPTION

TCLCMD-248

NAME

TCLCMD-248

SUMMARY

Invalid floating point number

DESCRIPTION

TCLCMD-249

NAME

TCLCMD-249

SUMMARY

Invalid number of bits

DESCRIPTION

TCLCMD-250

NAME

TCLCMD-250

SUMMARY

Invalid deviation value

DESCRIPTION

TCLCMD-251

NAME

TCLCMD-251

SUMMARY

Invalid bit width

DESCRIPTION

TCLCMD-252

NAME

TCLCMD-252

SUMMARY

Attempt to set view name on view failed

DESCRIPTION

TCLCMD-253

NAME

TCLCMD-253

SUMMARY

Invalid max_line value

DESCRIPTION

TCLCMD-254

NAME

TCLCMD-254

SUMMARY

No unique parent instance

DESCRIPTION

TCLCMD-255

NAME

TCLCMD-255

SUMMARY

Attempt to set target technology to library failed

DESCRIPTION

TCLCMD-256

NAME

TCLCMD-256

SUMMARY

Invalid verbosity level (0 - 100 only)

DESCRIPTION

TCLCMD-257

NAME

TCLCMD-257

SUMMARY

Attempt to create hierarchy failed

DESCRIPTION

TCLCMD-258

NAME

TCLCMD-258

SUMMARY

Could not find view in module

DESCRIPTION

TCLCMD-259

NAME

TCLCMD-259

SUMMARY

Attempt to resolve wired logic failed

DESCRIPTION

TCLCMD-260

NAME

TCLCMD-260

SUMMARY

Attempt to check netlist failed

DESCRIPTION

TCLCMD-261

NAME

TCLCMD-261

SUMMARY

Cell has different port boundary from cell , cant rebind

DESCRIPTION

TCLCMD-262

NAME

TCLCMD-262

SUMMARY

Invalid line length

DESCRIPTION

TCLCMD-263

NAME

TCLCMD-263

SUMMARY

The view to be compared with not specified

DESCRIPTION

TCLCMD-264

NAME

TCLCMD-264

SUMMARY

Pin is not a port

DESCRIPTION

TCLCMD-265

NAME

TCLCMD-265

SUMMARY

No matching clock found for

DESCRIPTION

TCLCMD-266

NAME

TCLCMD-266

SUMMARY

Top timing module has no view

DESCRIPTION

TCLCMD-267

NAME

TCLCMD-267

SUMMARY

Waveform and lead/trail options are mutually incompatible

DESCRIPTION

TCLCMD-268

NAME

TCLCMD-268

SUMMARY

Waveform must have at least two edges

DESCRIPTION

TCLCMD-269

NAME

TCLCMD-269

SUMMARY

Invalid waveform value

DESCRIPTION

TCLCMD-270

NAME

TCLCMD-270

SUMMARY

Either the period or waveform must be specified

DESCRIPTION

TCLCMD-271

NAME

TCLCMD-271

SUMMARY

Invalid period value

DESCRIPTION

TCLCMD-272

NAME

TCLCMD-272

SUMMARY

Conflicting options -is_clock and -is_data

DESCRIPTION

TCLCMD-273

NAME

TCLCMD-273

SUMMARY

Conflicting options -lead and -trail

DESCRIPTION

TCLCMD-274

NAME

TCLCMD-274

SUMMARY

Invalid transition value .

DESCRIPTION

TCLCMD-275

NAME

TCLCMD-275

SUMMARY

Invalid rise value

DESCRIPTION

TCLCMD-276

NAME

TCLCMD-276

SUMMARY

Invalid fall value

DESCRIPTION

TCLCMD-277

NAME

TCLCMD-277

SUMMARY

Conflicting options -early and -late

DESCRIPTION

TCLCMD-278

NAME

TCLCMD-278

SUMMARY

Attempt to get slack failed

DESCRIPTION

TCLCMD-279

NAME

TCLCMD-279

SUMMARY

Attempt to get worst slack failed

DESCRIPTION

TCLCMD-280

NAME

TCLCMD-280

SUMMARY

Attempt to set top timing module failed

DESCRIPTION

TCLCMD-281

NAME

TCLCMD-281

SUMMARY

Top timing module not set yet

DESCRIPTION

TCLCMD-282

NAME

TCLCMD-282

SUMMARY

Attempt to get arrival time failed

DESCRIPTION

TCLCMD-283

NAME

TCLCMD-283

SUMMARY

Attempt to get required time failed

DESCRIPTION

TCLCMD-284

NAME

TCLCMD-284

SUMMARY

Attempt to mark critical output failed

DESCRIPTION

TCLCMD-285

NAME

TCLCMD-285

SUMMARY

Invalid capacitance value

DESCRIPTION

TCLCMD-286

NAME

TCLCMD-286

SUMMARY

Invalid number of sinks

DESCRIPTION

TCLCMD-287

NAME

TCLCMD-287

SUMMARY

Invalid number of sources

DESCRIPTION

TCLCMD-288

NAME

TCLCMD-288

SUMMARY

Invalid fanout load value

DESCRIPTION

TCLCMD-289

NAME

TCLCMD-289

SUMMARY

Invalid load limit value

DESCRIPTION

TCLCMD-290

NAME

TCLCMD-290

SUMMARY

Could not find technology library

DESCRIPTION

TCLCMD-291

NAME

TCLCMD-291

SUMMARY

Could not find operating condition in technology library

DESCRIPTION

TCLCMD-292

NAME

TCLCMD-292

SUMMARY

Could not find wire load model in technology library

DESCRIPTION

TCLCMD-293

NAME

TCLCMD-293

SUMMARY

Invalid wire load mode(top, enclosed or segmented)

DESCRIPTION

TCLCMD-294

NAME

TCLCMD-294

SUMMARY

Invalid resistance value

DESCRIPTION

TCLCMD-295

NAME

TCLCMD-295

SUMMARY

Invalid source capacitance value

DESCRIPTION

TCLCMD-296

NAME

TCLCMD-296

SUMMARY

Invalid sink capacitance value

DESCRIPTION

TCLCMD-297

NAME

TCLCMD-297

SUMMARY

Invalid source resistance value

DESCRIPTION

TCLCMD-298

NAME

TCLCMD-298

SUMMARY

Invalid sink resistance value

DESCRIPTION

TCLCMD-299

NAME

TCLCMD-299

SUMMARY

Cannot determine timing context

DESCRIPTION

TCLCMD-300

NAME

TCLCMD-300

SUMMARY

Invalid slew limit value

DESCRIPTION

TCLCMD-301

NAME

TCLCMD-301

SUMMARY

Module must be unique in top timing module that must be set first

DESCRIPTION

TCLCMD-302

NAME

TCLCMD-302

SUMMARY

Attempt to read SDF file failed

DESCRIPTION

TCLCMD-303

NAME

TCLCMD-303

SUMMARY

Invalid analysis mode

DESCRIPTION

TCLCMD-306

NAME

TCLCMD-306

SUMMARY

Wrong source edge selected

DESCRIPTION

TCLCMD-307

NAME

TCLCMD-307

SUMMARY

: No timing arc exists between specified source and sink pins.

DESCRIPTION

TCLCMD-310

NAME

TCLCMD-310

SUMMARY

: Could not find pin on cell

DESCRIPTION

TCLCMD-311

NAME

TCLCMD-311

SUMMARY

No cell specified for either rising or falling edge

DESCRIPTION

TCLCMD-312

NAME

TCLCMD-312

SUMMARY

Conflicting options -signal and -reference

DESCRIPTION

TCLCMD-313

NAME

TCLCMD-313

SUMMARY

Invalid value

DESCRIPTION

TCLCMD-314

NAME

TCLCMD-314

SUMMARY

Invalid arrival value

DESCRIPTION

TCLCMD-315

NAME

TCLCMD-315

SUMMARY

set_multicycle_path option number_of_cycles requires an integer value instead of . The command will be ignored.

DESCRIPTION

TCLCMD-316

NAME

TCLCMD-316

SUMMARY

is not defined

DESCRIPTION

TCLCMD-317

NAME

TCLCMD-317

SUMMARY

No arguments supplied to option

DESCRIPTION

TCLCMD-318

NAME

TCLCMD-318

SUMMARY

edge_from is not lead or trail

DESCRIPTION

TCLCMD-319

NAME

TCLCMD-319

SUMMARY

edge_to is not lead or trail

DESCRIPTION

TCLCMD-320

NAME

TCLCMD-320

SUMMARY

Invalid clock uncertainty value

DESCRIPTION

TCLCMD-321

NAME

TCLCMD-321

SUMMARY

Invalid object flag value

DESCRIPTION

TCLCMD-322

NAME

TCLCMD-322

SUMMARY

The type of object is neither integer nor string

DESCRIPTION

TCLCMD-323

NAME

TCLCMD-323

SUMMARY

Object does not match type

DESCRIPTION

TCLCMD-324

NAME

TCLCMD-324

SUMMARY

Module does not have unique timing context

DESCRIPTION

TCLCMD-325

NAME

TCLCMD-325

SUMMARY

HPT initialization failed

DESCRIPTION

TCLCMD-326

NAME

TCLCMD-326

SUMMARY

Attempt to load XATL library failed

DESCRIPTION

TCLCMD-327

NAME

TCLCMD-327

SUMMARY

Couldnt get unique timing analysis id from module

DESCRIPTION

TCLCMD-328

NAME

TCLCMD-328

SUMMARY

: wrong value passed

DESCRIPTION

TCLCMD-329

NAME

TCLCMD-329

SUMMARY

Unknown effort level

DESCRIPTION

TCLCMD-330

NAME

TCLCMD-330

SUMMARY

Unknown flatten option

DESCRIPTION

TCLCMD-331

NAME

TCLCMD-331

SUMMARY

Unknown priority option

DESCRIPTION

TCLCMD-332

NAME

TCLCMD-332

SUMMARY

Unknown minimize option

DESCRIPTION

TCLCMD-333

NAME

TCLCMD-333

SUMMARY

set_global report_column_width is obsolete, use set_table_style

DESCRIPTION

TCLCMD-334

NAME

TCLCMD-334

SUMMARY

set_global verilog_naming_style is obsolete

DESCRIPTION

TCLCMD-335

NAME

TCLCMD-335

SUMMARY

Unknown cell area option (combinational, sequential and both only)

DESCRIPTION

TCLCMD-336

NAME

TCLCMD-336

SUMMARY

is not defined as a clock

DESCRIPTION

TCLCMD-337

NAME

TCLCMD-337

SUMMARY

Both period and waveform options are given for clock

DESCRIPTION

TCLCMD-338

NAME

TCLCMD-338

SUMMARY

Clock name not specified

DESCRIPTION

TCLCMD-339

NAME

TCLCMD-339

SUMMARY

Both lead and trail options are given for clock

DESCRIPTION

TCLCMD-340

NAME

TCLCMD-340

SUMMARY

Both waveform and lead/trail options are given for clock

DESCRIPTION

TCLCMD-341

NAME

TCLCMD-341

SUMMARY

segmented wire load mode not supported yet(top and enclosed only)

DESCRIPTION

TCLCMD-342

NAME

TCLCMD-342

SUMMARY

Attempt to write equation file failed

DESCRIPTION

TCLCMD-343

NAME

TCLCMD-343

SUMMARY

Invalid mode (tristate, latch or both only)

DESCRIPTION

TCLCMD-344

NAME

TCLCMD-344

SUMMARY

Window id not specified

DESCRIPTION

TCLCMD-345

NAME

TCLCMD-345

SUMMARY

Call-back function name not specified

DESCRIPTION

TCLCMD-346

NAME

TCLCMD-346

SUMMARY

Either -fanin or -fanout must be specified

DESCRIPTION

TCLCMD-347

NAME

TCLCMD-347

SUMMARY

Both -fanin and -fanout specified

DESCRIPTION

TCLCMD-348

NAME

TCLCMD-348

SUMMARY

Unrecognized value of option -type

DESCRIPTION

TCLCMD-349

NAME

TCLCMD-349

SUMMARY

Value of option -line must be an integer

DESCRIPTION

TCLCMD-350

NAME

TCLCMD-350

SUMMARY

Value of option -message_id must be an integer from 0 to 999

DESCRIPTION

TCLCMD-351

NAME

TCLCMD-351

SUMMARY

Invalid number for max_slack

DESCRIPTION

TCLCMD-352

NAME

TCLCMD-352

SUMMARY

Invalid number for max_paths

DESCRIPTION

TCLCMD-353

NAME

TCLCMD-353

SUMMARY

Invalid net object id

DESCRIPTION

TCLCMD-354

NAME

TCLCMD-354

SUMMARY

Cannot specify option with option.

DESCRIPTION

TCLCMD-355

NAME

TCLCMD-355

SUMMARY

Invalid max_fanout value

DESCRIPTION

TCLCMD-356

NAME

TCLCMD-356

SUMMARY

Invalid min_fanout value

DESCRIPTION

TCLCMD-357

NAME

TCLCMD-357

SUMMARY

Invalid pin object id

DESCRIPTION

TCLCMD-358

NAME

TCLCMD-358

SUMMARY

Could not find pin on instance

DESCRIPTION

TCLCMD-359

NAME

TCLCMD-359

SUMMARY

Unrecognized message type

DESCRIPTION

TCLCMD-360

NAME

TCLCMD-360

SUMMARY

No such object with the id found

DESCRIPTION

TCLCMD-361

NAME

TCLCMD-361

SUMMARY

Mismatch in lengths of object and analysis id lists

DESCRIPTION

TCLCMD-362

NAME

TCLCMD-362

SUMMARY

is neither valid net nor pin id

DESCRIPTION

TCLCMD-363

NAME

TCLCMD-363

SUMMARY

Invalid module id

DESCRIPTION

TCLCMD-364

NAME

TCLCMD-364

SUMMARY

Invalid view id

DESCRIPTION

TCLCMD-365

NAME

TCLCMD-365

SUMMARY

Ambiguous command option

DESCRIPTION

TCLCMD-366

NAME

TCLCMD-366

SUMMARY

Both option -all and the list of module argument specified

DESCRIPTION

TCLCMD-367

NAME

TCLCMD-367

SUMMARY

Attempt to write VHDL file failed

DESCRIPTION

TCLCMD-368

NAME

TCLCMD-368

SUMMARY

Syntax error in the instance mapping file

DESCRIPTION

TCLCMD-369

NAME

TCLCMD-369

SUMMARY

Invalid format:

DESCRIPTION

TCLCMD-370

NAME

TCLCMD-370

SUMMARY

Invalid to pin list:

DESCRIPTION

TCLCMD-371

NAME

TCLCMD-371

SUMMARY

No to pins found in list

DESCRIPTION

TCLCMD-372

NAME

TCLCMD-372

SUMMARY

Internal Error:

DESCRIPTION

TCLCMD-373

NAME

TCLCMD-373

SUMMARY

Invalid instance

DESCRIPTION

TCLCMD-374

NAME

TCLCMD-374

SUMMARY

Only one instance can be specified

DESCRIPTION

TCLCMD-375

NAME

TCLCMD-375

SUMMARY

Invalid argument supplied to option.

DESCRIPTION

TCLCMD-376

NAME

TCLCMD-376

SUMMARY

Attempt to change the name of module to failed

DESCRIPTION

TCLCMD-377

NAME

TCLCMD-377

SUMMARY

The object to be renamed not specified

DESCRIPTION

TCLCMD-378

NAME

TCLCMD-378

SUMMARY

More than one objects to be renamed are specified

DESCRIPTION

Only one object to be renamed can be specified at a time

TCLCMD-379

NAME

TCLCMD-379

SUMMARY

Attempt to change the name of net to failed

DESCRIPTION

TCLCMD-380

NAME

TCLCMD-380

SUMMARY

Invalid bus object id

DESCRIPTION

TCLCMD-381

NAME

TCLCMD-381

SUMMARY

Attempt to change the name of bus to failed

DESCRIPTION

TCLCMD-382

NAME

TCLCMD-382

SUMMARY

Attempt to change the name of instance to failed

DESCRIPTION

TCLCMD-383

NAME

TCLCMD-383

SUMMARY

Invalid critical_offset value

DESCRIPTION

TCLCMD-384

NAME

TCLCMD-384

SUMMARY

Invalid max_area value

DESCRIPTION

TCLCMD-385

NAME

TCLCMD-385

SUMMARY

Conflicting options -dont_reclaim_area and -dont_reclaim_violators

DESCRIPTION

TCLCMD-386

NAME

TCLCMD-386

SUMMARY

Conflicting options -dont_reclaim_area and -reclaim_maximum_area

DESCRIPTION

TCLCMD-387

NAME

TCLCMD-387

SUMMARY

Conflicting options -reclaim_maximum_area and -dont_reclaim_violators

DESCRIPTION

TCLCMD-388

NAME

TCLCMD-388

SUMMARY

Conflicting options -dont_reclaim_area and -reclaim_area_only

DESCRIPTION

TCLCMD-389

NAME

TCLCMD-389

SUMMARY

Conflicting options -no_pinswap and -pinswap_only

DESCRIPTION

TCLCMD-390

NAME

TCLCMD-390

SUMMARY

Given argument must be a positive integer

DESCRIPTION

TCLCMD-391

NAME

TCLCMD-391

SUMMARY

Unable to set count of message

DESCRIPTION

TCLCMD-392

NAME

TCLCMD-392

SUMMARY

is not a valid message id or message type

DESCRIPTION

TCLCMD-393

NAME

TCLCMD-393

SUMMARY

Incompatible options: and

DESCRIPTION

TCLCMD-394

NAME

TCLCMD-394

SUMMARY

Cannot use the -master_clock parameter of the create_generated_clock command without the -add parameter. Ensure that these options are specified together.

DESCRIPTION

This error is displayed when you use the -master_clock parameter of the create_generated_clock command without the -add parameter. It is an SDC requirement that the -master_clock parameter should be used with the -add parameter.

.P

If there are multiple clocks reaching the source-point, it is assumed that you will be creating multiple generated clocks at the target pin using each source-clock. Therefore, whenever the -master_clock parameter is used, the -add parameter is needed. Note that these parameters should be used together even if you decide to create only one generated clock with one master clock.

TCLCMD-395

NAME

TCLCMD-395

SUMMARY

Options and require option

DESCRIPTION

TCLCMD-396

NAME

TCLCMD-396

SUMMARY

: Missing parameter

DESCRIPTION

TCLCMD-397

NAME

TCLCMD-397

SUMMARY

: Bad parameter

DESCRIPTION

TCLCMD-398

NAME

TCLCMD-398

SUMMARY

: Only one pin allowed

DESCRIPTION

TCLCMD-399

NAME

TCLCMD-399

SUMMARY

Attempting to set _opt_quick_sweep to . It must be 1 or 0

DESCRIPTION

TCLCMD-400

NAME

TCLCMD-400

SUMMARY

Library not specified

DESCRIPTION

TCLCMD-401

NAME

TCLCMD-401

SUMMARY

Could not find cell in library

DESCRIPTION

TCLCMD-402

NAME

TCLCMD-402

SUMMARY

Failed to open temporary file

DESCRIPTION

TCLCMD-403

NAME

TCLCMD-403

SUMMARY

Verification of FNL netlist failed with output pin in cell

DESCRIPTION

TCLCMD-404

NAME

TCLCMD-404

SUMMARY

Verification of FNL netlist failed with pin in sequential cell

DESCRIPTION

TCLCMD-405

NAME

TCLCMD-405

SUMMARY

Invalid module id: , cant create module reference

DESCRIPTION

TCLCMD-406

NAME

TCLCMD-406

SUMMARY

Attempt to load ctlf library failed

DESCRIPTION

TCLCMD-407

NAME

TCLCMD-407

SUMMARY

: unknown constraint format

DESCRIPTION

TCLCMD-408

NAME

TCLCMD-408

SUMMARY

Invalid operating parameter value

DESCRIPTION

TCLCMD-409

NAME

TCLCMD-409

SUMMARY

Can not update with library since that library is already loaded; an update library should not be loaded by itself first

DESCRIPTION

TCLCMD-410

NAME

TCLCMD-410

SUMMARY

Attempt to write EDIF file failed

DESCRIPTION

TCLCMD-411

NAME

TCLCMD-411

SUMMARY

Invalid instance object id

DESCRIPTION

TCLCMD-412

NAME

TCLCMD-412

SUMMARY

Attempt to change the name of port to failed

DESCRIPTION

TCLCMD-413

NAME

TCLCMD-413

SUMMARY

Attempt to disable arc from pin to on cell failed

DESCRIPTION

TCLCMD-414

NAME

TCLCMD-414

SUMMARY

Attempt to read Berkeley PLA file failed

DESCRIPTION

TCLCMD-415

NAME

TCLCMD-415

SUMMARY

Invalid value of attribute

DESCRIPTION

TCLCMD-416

NAME

TCLCMD-416

SUMMARY

Invalid MTM value (minimum, typical or maximum only)

DESCRIPTION

TCLCMD-417

NAME

TCLCMD-417

SUMMARY

Attempt to update library failed

DESCRIPTION

TCLCMD-418

NAME

TCLCMD-418

SUMMARY

Requested object (id) not unique with respect to top timing module or current instance

DESCRIPTION

Use set_current_instance from the top timing module so the requested object is unique, or use a full hierarchical name. to specify the object.

TCLCMD-419

NAME

TCLCMD-419

SUMMARY

cannot be applied on hierarchical pin . The constraint will be ignored.

DESCRIPTION

TCLCMD-420

NAME

TCLCMD-420

SUMMARY

Invalid floating point number for option

DESCRIPTION

TCLCMD-421

NAME

TCLCMD-421

SUMMARY

Unknown value for option in command (min, typ or max only)

DESCRIPTION

TCLCMD-422

NAME

TCLCMD-422

SUMMARY

Attempt to read EDIF file failed

DESCRIPTION

TCLCMD-423

NAME

TCLCMD-423

SUMMARY

Invalid case analysis value . Specify one of the following: 0, 1 , zero, one, rising, falling, rise, fall

DESCRIPTION

TCLCMD-424

NAME

TCLCMD-424

SUMMARY

No constant asserted on pin

DESCRIPTION

TCLCMD-425

NAME

TCLCMD-425

SUMMARY

Invalid max_change value , must be integer

DESCRIPTION

TCLCMD-426

NAME

TCLCMD-426

SUMMARY

Incomplete values specified in MTM triplet

DESCRIPTION

TCLCMD-427

NAME

TCLCMD-427

SUMMARY

IPO Change file, , cant be opened for writing. May be problem with write permission

DESCRIPTION

TCLCMD-428

NAME

TCLCMD-428

SUMMARY

There are multiple top modules. Use -module <module_name> or -all to build all the modules

DESCRIPTION

Ambiguity in command due to multiple top modules. Use get_hdl_top_level to get the names of all top modules. Use -module <module_name> to build the desired top module. Use -all to build all the

top modules

TCLCMD-429

NAME

TCLCMD-429

SUMMARY

Mixing of clocks and pins is not permitted

DESCRIPTION

TCLCMD-430

NAME

TCLCMD-430

SUMMARY

Can not find wire load selection table in technology library

DESCRIPTION

TCLCMD-431

NAME

TCLCMD-431

SUMMARY

Invalid min_area value

DESCRIPTION

TCLCMD-432

NAME

TCLCMD-432

SUMMARY

Can not wire load on instance (not a module)

DESCRIPTION

TCLCMD-433

NAME

TCLCMD-433

SUMMARY

Unknown option in set_global naming_style. Valid options are none, vhdl and verilog

DESCRIPTION

TCLCMD-434

NAME

TCLCMD-434

SUMMARY

Object is a view other than library. Only view of library can be renamed

DESCRIPTION

TCLCMD-435

NAME

TCLCMD-435

SUMMARY

Library already exists, failed to change the name of library

DESCRIPTION

TCLCMD-436

NAME

TCLCMD-436

SUMMARY

Invalid Number Of Groups

DESCRIPTION

TCLCMD-437

NAME

TCLCMD-437

SUMMARY

Invalid number of Cells Per Group

DESCRIPTION

TCLCMD-438

NAME

TCLCMD-438

SUMMARY

DEF Filename Not Specified

DESCRIPTION

TCLCMD-439

NAME

TCLCMD-439

SUMMARY

Not all cells in the design exist in the physical library

DESCRIPTION

TCLCMD-450

NAME

TCLCMD-450

SUMMARY

Invalid net found

DESCRIPTION

TCLCMD-451

NAME

TCLCMD-451

SUMMARY

Could not rename net to

DESCRIPTION

TCLCMD-452

NAME

TCLCMD-452

SUMMARY

Invalid module found

DESCRIPTION

TCLCMD-453

NAME

TCLCMD-453

SUMMARY

Could not rename module to

DESCRIPTION

TCLCMD-454

NAME

TCLCMD-454

SUMMARY

Invalid instance found

DESCRIPTION

TCLCMD-455

NAME

TCLCMD-455

SUMMARY

Could not rename instance to

DESCRIPTION

TCLCMD-456

NAME

TCLCMD-456

SUMMARY

Expecting Placement option - one of all or io or cell

DESCRIPTION

TCLCMD-457

NAME

TCLCMD-457

SUMMARY

Missing do_change_name conversion rule - cannot continue

DESCRIPTION

TCLCMD-458

NAME

TCLCMD-458

SUMMARY

Cannot open do_change_name log file for writing

DESCRIPTION

TCLCMD-459

NAME

TCLCMD-459

SUMMARY

Cannot open do_change_name rule file for reading

DESCRIPTION

TCLCMD-460

NAME

TCLCMD-460

SUMMARY

Invalid bus net found

DESCRIPTION

TCLCMD-461

NAME

TCLCMD-461

SUMMARY

Could not rename bus net to

DESCRIPTION

TCLCMD-462

NAME

TCLCMD-462

SUMMARY

Constants on hierarchical ports are not computed, try instance pins/I/O ports instead

DESCRIPTION

TCLCMD-463

NAME

TCLCMD-463

SUMMARY

Constants with tags are only permitted on I/O ports

DESCRIPTION

TCLCMD-464

NAME

TCLCMD-464

SUMMARY

Invalid RC tree type (best_case_tree, balanced_tree or worst_case_tree)

DESCRIPTION

TCLCMD-465

NAME

TCLCMD-465

SUMMARY

Invalid port object id

DESCRIPTION

TCLCMD-466

NAME

TCLCMD-466

SUMMARY

Invalid port found

DESCRIPTION

TCLCMD-467

NAME

TCLCMD-467

SUMMARY

Could not rename port to

DESCRIPTION

TCLCMD-468

NAME

TCLCMD-468

SUMMARY

Invalid bus port found

DESCRIPTION

TCLCMD-469

NAME

TCLCMD-469

SUMMARY

Could not rename bus port to

DESCRIPTION

TCLCMD-470

NAME

TCLCMD-470

SUMMARY

Number of cycle additions for rise and fall at pin are different, cant be converted to GCF

DESCRIPTION

TCLCMD-471

NAME

TCLCMD-471

SUMMARY

Unknown combination of edge identifier and timing check type on multi-cycle path assertion at pin , cant be converted to GCF

DESCRIPTION

TCLCMD-472

NAME

TCLCMD-472

SUMMARY

Ideal clock not found in clock polarity table

DESCRIPTION

TCLCMD-473

NAME

TCLCMD-473

SUMMARY

Option must be specified for command

DESCRIPTION

TCLCMD-474

NAME

TCLCMD-474

SUMMARY

Invalid slack limit value

DESCRIPTION

TCLCMD-475

NAME

TCLCMD-475

SUMMARY

Invalid number of arguments, Please Refer Command Usage

DESCRIPTION

TCLCMD-476

NAME

TCLCMD-476

SUMMARY

Invalid Pin Name specified

DESCRIPTION

TCLCMD-477

NAME

TCLCMD-477

SUMMARY

Invalid Pin Side specified

DESCRIPTION

Expecting one of left,right,top,bottom or unassigned

TCLCMD-478

NAME

TCLCMD-478

SUMMARY

Invalid Pin Index specified

DESCRIPTION

Expecting a positive integer index

TCLCMD-479

NAME

TCLCMD-479

SUMMARY

Invalid number of arguments, Please Refer Command Usage

DESCRIPTION

TCLCMD-480

NAME

TCLCMD-480

SUMMARY

Unable to set pin location

DESCRIPTION

TCLCMD-481

NAME

TCLCMD-481

SUMMARY

Unable to set Floorplan Parameters

DESCRIPTION

Check if current top module is set

TCLCMD-482

NAME

TCLCMD-482

SUMMARY

Unable to get Floorplan Parameters

DESCRIPTION

Check if current top module is set

TCLCMD-483

NAME

TCLCMD-483

SUMMARY

Current View not Valid

DESCRIPTION

Check if the top module has been set

TCLCMD-484

NAME

TCLCMD-484

SUMMARY

Two points(minimum and maximum) must be specified in voltage_threshold

DESCRIPTION

TCLCMD-485

NAME

TCLCMD-485

SUMMARY

Invalid voltage threshold value %.4f(must be between 0 and 1)

DESCRIPTION

TCLCMD-486

NAME

TCLCMD-486

SUMMARY

Only one of and options may be specified

DESCRIPTION

TCLCMD-487

NAME

TCLCMD-487

SUMMARY

Verification of truth table failed with output pin in cell

DESCRIPTION

TCLCMD-488

NAME

TCLCMD-488

SUMMARY

Verification of truth table failed with pin in sequential cell

DESCRIPTION

TCLCMD-489

NAME

TCLCMD-489

SUMMARY

Unknown PLA type (valid type: f,d,r,fd,fr,dr,fdr)

DESCRIPTION

TCLCMD-490

NAME

TCLCMD-490

SUMMARY

Pin Name Not Specified For Special Pin

DESCRIPTION

TCLCMD-491

NAME

TCLCMD-491

SUMMARY

Pin Type should be power or ground

DESCRIPTION

TCLCMD-492

NAME

TCLCMD-492

SUMMARY

This command requires a PKS license

DESCRIPTION

TCLCMD-493

NAME

TCLCMD-493

SUMMARY

This command requires a physical library to be loaded

DESCRIPTION

TCLCMD-494

NAME

TCLCMD-494

SUMMARY

Report placement area failed.

DESCRIPTION

TCLCMD-495

NAME

TCLCMD-495

SUMMARY

Option has invalid value

DESCRIPTION

TCLCMD-496

NAME

TCLCMD-496

SUMMARY

Attempt to delete net connection failed for

DESCRIPTION

TCLCMD-497

NAME

TCLCMD-497

SUMMARY

Physical and Logical library dont match

DESCRIPTION

TCLCMD-498

NAME

TCLCMD-498

SUMMARY

Invalid value for set_distribution_point command

DESCRIPTION

Expecting a value on, auto or off

TCLCMD-499

NAME

TCLCMD-499

SUMMARY

Both -gns_view and -gns_library options must be specified

DESCRIPTION

TCLCMD-500

NAME

TCLCMD-500

SUMMARY

Negative epsilon %.2f ignored. Default of .5 used

DESCRIPTION

TCLCMD-501

NAME

TCLCMD-501

SUMMARY

Empty argument list: no object to set dont_modify

DESCRIPTION

TCLCMD-502

NAME

TCLCMD-502

SUMMARY

This function is obsolete: use write_bdf -hierarchical

DESCRIPTION

TCLCMD-503

NAME

TCLCMD-503

SUMMARY

Both options -net and -cell are given, total area assumed

DESCRIPTION

TCLCMD-504

NAME

TCLCMD-504

SUMMARY

Unknown type of object filter

DESCRIPTION

TCLCMD-505

NAME

TCLCMD-505

SUMMARY

The net to which the pin(or port) connects is dangling

DESCRIPTION

TCLCMD-506

NAME

TCLCMD-506

SUMMARY

This command is obsolete; use do_copy_module instead

DESCRIPTION

TCLCMD-507

NAME

TCLCMD-507

SUMMARY

This command is obsolete; use do_remove_design instead

DESCRIPTION

TCLCMD-508

NAME

TCLCMD-508

SUMMARY

The mapping for instance already specified, will be overridden

DESCRIPTION

TCLCMD-509

NAME

TCLCMD-509

SUMMARY

Instance already bound to

DESCRIPTION

TCLCMD-510

NAME

TCLCMD-510

SUMMARY

The name of this command is changed; use do_create_hierarchy instead

DESCRIPTION

TCLCMD-511

NAME

TCLCMD-511

SUMMARY

Obsolete option -critical_range. Use -critical_ratio

DESCRIPTION

TCLCMD-512

NAME

TCLCMD-512

SUMMARY

Obsolete option -dont_reclaim_violators. This is now the default

DESCRIPTION

TCLCMD-513

NAME

TCLCMD-513

SUMMARY

The software could not find a matching object of the specified type for the pattern

DESCRIPTION

The get_* commands are used for retrieving object and collection references from the timing system. If no objects could be returned that match the specified pattern, possible causes include:

.P

- The object truly does not exist in the design.

.sp .5

- You have specified an incorrect wildcard matching pattern - refer to the documentation for more information on proper wildcarding syntax.

TCLCMD-514

NAME

TCLCMD-514

SUMMARY

Command is obsolete and will be removed in a future release. Please use instead

DESCRIPTION

TCLCMD-515

NAME

TCLCMD-515

SUMMARY

Option is obsolete and will be removed in a future release. Option ignored.

DESCRIPTION

TCLCMD-516

NAME

TCLCMD-516

SUMMARY

The file has been created

DESCRIPTION

TCLCMD-517

NAME

TCLCMD-517

SUMMARY

Constraint of clock required time on pin will not be converted in GCF

DESCRIPTION

The command `set_clock_required_time` should be replaced by the command `set_external_delay -ref` in order to specify required times for output ports on the clock network. `set_clock_required_time` should only generated internally by timing analysis during time budgeting and is an obsolete command.

TCLCMD-518

NAME

TCLCMD-518

SUMMARY

Unrecognized string type for object (integer or string only)

DESCRIPTION

TCLCMD-519

NAME

TCLCMD-519

SUMMARY

Clock arrival time of specified on edge only, assuming that the same insertion delay is applied to edge, too

DESCRIPTION

TCLCMD-520

NAME

TCLCMD-520

SUMMARY

The design contains blackbox(es)

DESCRIPTION

TCLCMD-521

NAME

TCLCMD-521

SUMMARY

Required time on pin with respect to clock cannot be converted to GCF

DESCRIPTION

Currently there is no corresponding construct defined in GCF which can represent set_required_time constraint

TCLCMD-522

NAME

TCLCMD-522

SUMMARY

Edge identifier and/or timing check type at target pin doesnt match the type at source pin in false path assertion

DESCRIPTION

Due to the mismatch, the path may be over-constrained when converted to GCF

TCLCMD-523

NAME

TCLCMD-523

SUMMARY

Edge identifier and/or timing check type at target pin doesn't match the type at source pin in multi-cycle path assertion

DESCRIPTION

Due to the mismatch, the path may be over-constrained when converted to GCF

TCLCMD-524

NAME

TCLCMD-524

SUMMARY

Clock uncertainty specified with clock pair and cannot be converted to GCF1.3, ignored

DESCRIPTION

TCLCMD-525

NAME

TCLCMD-525

SUMMARY

Empty argument list: no object to set cell property

DESCRIPTION

TCLCMD-526

NAME

TCLCMD-526

SUMMARY

Option -minimize_area has been changed to -reclaim_maximum_area. Use the new option instead

DESCRIPTION

TCLCMD-527

NAME

TCLCMD-527

SUMMARY

Option -force is obsolete and neednt be specified

DESCRIPTION

TCLCMD-528

NAME

TCLCMD-528

SUMMARY

Ignoring -no_acl option due to the presence of -hierarchy option

DESCRIPTION

Use of the -hierarchy option during netlisting suppresses the -no_acl option.

TCLCMD-529

NAME

TCLCMD-529

SUMMARY

More than three values are specified for , the first three are taken and rest of them are ignored

DESCRIPTION

TCLCMD-531

NAME

TCLCMD-531

SUMMARY

Disabling clocks and in different domains using set_clock_uncertainty with large negative number cannot be converted to GCF; Use set_false_path instead

DESCRIPTION

TCLCMD-532

NAME

TCLCMD-532

SUMMARY

Different number of extra cycles specified for data signal between clock pair and , cant be converted to GCF

DESCRIPTION

TCLCMD-533

NAME

TCLCMD-533

SUMMARY

both cells_per_group and number_of_groups cannot be present simultaneously, Assuming default of number of groups = 7

DESCRIPTION

TCLCMD-534

NAME

TCLCMD-534

SUMMARY

Command option -noncombinational is obsolete; Use -tristate and/or -sequential instead

DESCRIPTION

TCLCMD-535

NAME

TCLCMD-535

SUMMARY

set_global hdl_verilog_escaped_naming_style is obsolete

DESCRIPTION

TCLCMD-536

NAME

TCLCMD-536

SUMMARY

No timing arc exists between pins and

DESCRIPTION

TCLCMD-537

NAME

TCLCMD-537

SUMMARY

Option ignored

DESCRIPTION

TCLCMD-539

NAME

TCLCMD-539

SUMMARY

Begin tag found on no pin

DESCRIPTION

TCLCMD-540

NAME

TCLCMD-540

SUMMARY

Changing DCL calc mode; this may invalidate library specific assertions such as set_wire_load and they need to be re-issued

DESCRIPTION

TCLCMD-541

NAME

TCLCMD-541

SUMMARY

Target library is not set, cant get default operating condition information for GCF

DESCRIPTION

TCLCMD-542

NAME

TCLCMD-542

SUMMARY

Target library is not set, cant get scale of physical property information for GCF

DESCRIPTION

TCLCMD-543

NAME

TCLCMD-543

SUMMARY

Path exception with -through pin to target pin is not supported by back-end tool

DESCRIPTION

The generated GCF output may contain unsupported GCF v1.3 construct(s)

TCLCMD-544

NAME

TCLCMD-544

SUMMARY

Path exception with pin asserted on port is not supported by back-end tool

DESCRIPTION

The generated GCF output may contain unsupported GCF v1.3 construct(s)

TCLCMD-545

NAME

TCLCMD-545

SUMMARY

Path exception asserted on hierarchical pin is not supported by back-end tool

DESCRIPTION

The generated GCF output may contain unsupported GCF v1.3 construct(s)

TCLCMD-546

NAME

TCLCMD-546

SUMMARY

Path exception with TO pin asserted on instance output is not supported by back-end tool

DESCRIPTION

The generated GCF output may contain unsupported GCF v1.3 construct(s)

TCLCMD-547

NAME

TCLCMD-547

SUMMARY

Path exception with FROM pin asserted on instance output other than register data output is not supported by back-end tool

DESCRIPTION

The generated GCF output may contain unsupported GCF v1.3 construct(s)

TCLCMD-548

NAME

TCLCMD-548

SUMMARY

Edge identifier in false path assertion at pin is not supported by back-end tool. Path may be over-constrained when converted to GCF

DESCRIPTION

TCLCMD-549

NAME

TCLCMD-549

SUMMARY

Edge identifier in multi-cycle path assertion at pin is not supported by back-end tool. Both edges will be applied

DESCRIPTION

TCLCMD-550

NAME

TCLCMD-550

SUMMARY

Path exception asserted on bi-directional pin is not supported by back-end tool

DESCRIPTION

The generated GCF output may contain unsupported GCF v1.3 construct(s)

TCLCMD-551

NAME

TCLCMD-551

SUMMARY

Path exception with FROM pin asserted on instance input other than register clock input is not supported by back-end tool

DESCRIPTION

The generated GCF output may contain unsupported GCF v1.3 construct(s)

TCLCMD-552

NAME

TCLCMD-552

SUMMARY

Path exception with TO pin asserted on instance input other than register clock or data input is not supported by back-end tool

DESCRIPTION

The generated GCF output may contain unsupported GCF v1.3 construct(s)

TCLCMD-553

NAME

TCLCMD-553

SUMMARY

Clock uncertainty for early and late at clock are different. Only late value is converted to GCF

DESCRIPTION

TCLCMD-554

NAME

TCLCMD-554

SUMMARY

Edge identifier or timing check in false path assertion between pin and is not supported for ARC construct by back-end tool. Path may be over-constrained when converted to GCF

DESCRIPTION

TCLCMD-555

NAME

TCLCMD-555

SUMMARY

Edge identifier in false path assertion between clocks and is not supported by back-end tool. Path may be over-constrained when converted to GCF

DESCRIPTION

TCLCMD-556

NAME

TCLCMD-556

SUMMARY

Hold timing check for multi-cycle path from clock to between BuildGates and back-end tool are different, which may cause timing correlation issue if no corresponding cycles for setup timing check is specified

DESCRIPTION

TCLCMD-557

NAME

TCLCMD-557

SUMMARY

Hold timing check for multi-cycle path at pin between BuildGates and back-end tool are different, which may cause timing correlation issue if no corresponding cycles for setup timing check is specified

DESCRIPTION

TCLCMD-558

NAME

TCLCMD-558

SUMMARY

Hold timing check for multi-cycle path from pin to between BuildGates and back-end tool are different, which may cause timing correlation issue if no corresponding cycles for setup timing check is specified

DESCRIPTION

TCLCMD-559

NAME

TCLCMD-559

SUMMARY

Clock uncertainty specified with from clock is not supported by back-end tool, ignored

DESCRIPTION

TCLCMD-560

NAME

TCLCMD-560

SUMMARY

set_clock_info_change not supported in written GCF. Assertion ignored on pin

DESCRIPTION

TCLCMD-561

NAME

TCLCMD-561

SUMMARY

Obsolete global . Global ignored

DESCRIPTION

TCLCMD-562

NAME

TCLCMD-562

SUMMARY

Ignoring -distributed option with command because -hier option is NOT supplied

DESCRIPTION

TCLCMD-563

NAME

TCLCMD-563

SUMMARY

Non-integer cycle adjustments are not supported in GCF. Converting non-integer cycle adjustment on pin to an integer.

DESCRIPTION

TCLCMD-564

NAME

TCLCMD-564

SUMMARY

Hierarchical instance in module found, cant be unmapped

DESCRIPTION

TCLCMD-565

NAME

TCLCMD-565

SUMMARY

preserve_busses is now default behavior of do_create_hierarchy

DESCRIPTION

TCLCMD-566

NAME

TCLCMD-566

SUMMARY

Option for command is obsolete and has been replaced by . The obsolete option still works in this release, but to avoid this warning and to ensure compatibility with future releases, update your script to use .

DESCRIPTION

TCLCMD-567

NAME

TCLCMD-567

SUMMARY

Cannot use multi-bit register as the register type

DESCRIPTION

TCLCMD-568

NAME

TCLCMD-568

SUMMARY

Cell is dont use, pad cell, or has no function. It cannot be used as the register type

DESCRIPTION

TCLCMD-569

NAME

TCLCMD-569

SUMMARY

Cannot set register type on hierarchical or mapped instance

DESCRIPTION

TCLCMD-570

NAME

TCLCMD-570

SUMMARY

Latch register type not set for instance . Use option -latch

DESCRIPTION

TCLCMD-571

NAME

TCLCMD-571

SUMMARY

Flip-flop register type not set for instance . Use option -flip_flop

DESCRIPTION

TCLCMD-572

NAME

TCLCMD-572

SUMMARY

Cannot set register type on non-register instance

DESCRIPTION

TCLCMD-573

NAME

TCLCMD-573

SUMMARY

No register type was previously set

DESCRIPTION

TCLCMD-574

NAME

TCLCMD-574

SUMMARY

No register type was previously set on instance

DESCRIPTION

TCLCMD-575

NAME

TCLCMD-575

SUMMARY

Option -create_ports is not allowed for the technology cells. This option will be ignored

DESCRIPTION

TCLCMD-576

NAME

TCLCMD-576

SUMMARY

Non float value encountered in option , ignored

DESCRIPTION

TCLCMD-577

NAME

TCLCMD-577

SUMMARY

Missing from _pin option. Cell has multiple input pins. Selected the first input pin

DESCRIPTION

TCLCMD-578

NAME

TCLCMD-578

SUMMARY

option may significantly degrade model accuracy. Use this option only if model size or model extraction time is of utmost concern.

DESCRIPTION

TCLCMD-579

NAME

TCLCMD-579

SUMMARY

option is obsolete. Use option

DESCRIPTION

TCLCMD-580

NAME

TCLCMD-580

SUMMARY

Changing the global will have no effect on behavior since appropriate license is not available

DESCRIPTION

TCLCMD-581

NAME

TCLCMD-581

SUMMARY

Option is now the default and no longer necessary. It will be removed in a future release. Option ignored

DESCRIPTION

TCLCMD-582

NAME

TCLCMD-582

SUMMARY

Command `do_xform_timing_correction` is obsolete and will be removed in a future release. Please use `do_optimize -effort low` or other appropriate `do_xform_` commands instead

DESCRIPTION

TCLCMD-583

NAME

TCLCMD-583

SUMMARY

Design is mapped, no transformation performed unless `-force` or `-in_place` is specified

DESCRIPTION

TCLCMD-584

NAME

TCLCMD-584

SUMMARY

Design is not mapped, cant perform power-driven constant propagation

DESCRIPTION

TCLCMD-585

NAME

TCLCMD-585

SUMMARY

Stop point lies before start point . Nothing to optimize.

DESCRIPTION

TCLCMD-586

NAME

TCLCMD-586

SUMMARY

No corresponding module found for the primitve instance .

DESCRIPTION

TCLCMD-587

NAME

TCLCMD-587

SUMMARY

-unroll_loops option cannot be used with -blackbox option; ignored

DESCRIPTION

TCLCMD-588

NAME

TCLCMD-588

SUMMARY

-keep_trigger_arcs option cannot be used with -blackbox option; ignored

DESCRIPTION

TCLCMD-589

NAME

TCLCMD-589

SUMMARY

Option -hierarchical is default for net and pin ids. Option Ignored

DESCRIPTION

TCLCMD-590

NAME

TCLCMD-590

SUMMARY

Option -no_hierarchical is default for module ids. Option Ignored

DESCRIPTION

TCLCMD-591

NAME

TCLCMD-591

SUMMARY

The command is not supported on the placed design

DESCRIPTION

TCLCMD-592

NAME

TCLCMD-592

SUMMARY

Option is obsolescent and will be removed in a future release. Option still functional in current release.

DESCRIPTION

TCLCMD-593

NAME

TCLCMD-593

SUMMARY

Option is obsolescent and will be removed in a future release.

DESCRIPTION

TCLCMD-594

NAME

TCLCMD-594

SUMMARY

Invalid check_type specified for -check_type

DESCRIPTION

TCLCMD-595

NAME

TCLCMD-595

SUMMARY

Removing all path group assertions. If path group assertions are present and if further optimization is needed for this block, please re-apply the path group assertions after model extraction. The need to re-apply path group assertions will be removed in the next hot fix release.

DESCRIPTION

TCLCMD-596

NAME

TCLCMD-596

SUMMARY

With -flatten on it may take infinite amount of run time in structuring if a module is not collapsible into sum-of-product form. -flatten auto is advised in this case

DESCRIPTION

TCLCMD-597

NAME

TCLCMD-597

SUMMARY

Invalid compression level for -gzip:

DESCRIPTION

gzip level has to be between 0 and 9

TCLCMD-598

NAME

TCLCMD-598

SUMMARY

DEBUG ALLOCATING blocks of type . byte per block

DESCRIPTION

Only for debugging purpose. Tells me how many blocks is allocated within a function that might cause MSE

TCLCMD-599

NAME

TCLCMD-599

SUMMARY

-no_partition is by default on, to turn on partition, reset _struct_no_partition_on to 0.

DESCRIPTION

TCLCMD-600

NAME

TCLCMD-600

SUMMARY

Option is not valid for unplaced designs. Option ignored

DESCRIPTION

TCLCMD-601

NAME

TCLCMD-601

SUMMARY

set_flow_compatible_path_exceptions_mode cannot be off because set_flow_compatible_mode is on, command ignored

DESCRIPTION

To turn off set_flow_compatible_path_exceptions_mode, turn off set_flow_compatible_mode first

TCLCMD-602

NAME

TCLCMD-602

SUMMARY

cannot be applied on hierarchical instance . The constraint will be ignored.

DESCRIPTION

This warning is issued when the constraint mentioned in summary message is applied to a hierarchical pin.

.sp .5

This constraint is not supported on hierarchical pins and is ignored.

.P

In order to avoid this warning, please fix the constraint by applying it on an instance pin.

TCLCMD-603

NAME

TCLCMD-603

SUMMARY

Illegal restructuring effort level . Accepted levels { full | partial | none }

DESCRIPTION

TCLCMD-604

NAME

TCLCMD-604

SUMMARY

Ignoring -restructure_aware option with effort level low

DESCRIPTION

TCLCMD-605

NAME

TCLCMD-605

SUMMARY

GCF does not support different rise and fall numbers for pin based clock uncertainties, using maximum of rise and fall numbers

DESCRIPTION

TCLCMD-606

NAME

TCLCMD-606

SUMMARY

Default pin based clock uncertainty (without -to option) can not be converted to GCF, treating it as a -to unceratinty.

DESCRIPTION

In PKS by default, an uncertainty specification on pin X is honored only at those registers or latches where there is a path from pin X to both the data pin and the clock pin of the register or latch. However, the -to option lifts this restriction and makes it applicable to all registers and latches whose clock pin is in the transitive fanout of pin X. Since the default uncertainty can not be converted to GCF, it treats default uncertainty in the same way as -to uncertainty.

TCLCMD-607

NAME

TCLCMD-607

SUMMARY

Ambiguous option . Assumed as if it were

DESCRIPTION

TCLCMD-608

NAME

TCLCMD-608

SUMMARY

pvt_late_path is not equal to pvt_early_path but -min_lib option was not specified. Creating a new .lib file corresponding to operating condition for pvt_early_path. will have data corresponding to operating condition for pvt_late_path. Use read_dotlib -min -max to load both pvt data simultaneously after model extraction.

DESCRIPTION

Model extractor creates two sets of data when pvt_late_path differs from pvt_early_path. One data corresponds to the operating condition for pvt_early_path and the other for pvt_late_path. The liberty format does not allow for specifying two sets of data in one file. Use -min_lib option to specify the library file to store the data corresponding to pvt_early_path.

TCLCMD-609

NAME

TCLCMD-609

SUMMARY

Without specifying the -to option, the -rise or -fall option will be ignored. Edge specification is supported if the path endpoint is specified.

DESCRIPTION

TCLCMD-610

NAME

TCLCMD-610

SUMMARY

mode is already associated with SDC file , discarding the old mode to SDC file mapping

DESCRIPTION

TCLCMD-611

NAME

TCLCMD-611

SUMMARY

option expects integer value instead of . Truncating it to integer value.

DESCRIPTION

TCLCMD-612

NAME

TCLCMD-612

SUMMARY

Option requires option . Ignoring option.

DESCRIPTION

TCLCMD-613

NAME

TCLCMD-613

SUMMARY

The -nworst and -max_paths parameters of the report_timing command are mutually exclusive and should not be used together. The timing engine will honor only the -max_paths parameter. If you want to use the -nworst parameter, remove the -max_paths parameter from the script.

DESCRIPTION

The -nworst parameter specifies the number of paths to be enumerated for each endpoint and the -max_paths parameter reports the specified number of worst paths in the design regardless of the endpoint. Therefore, the -nworst parameter cannot be used with the -max_paths parameter.

TCLCMD-614

NAME

TCLCMD-614

SUMMARY

Ignoring write_sdf_force_calculation global in stage based delay calculator mode.

DESCRIPTION

TCLCMD-615

NAME

TCLCMD-615

SUMMARY

Ignoring -forceComputation option in stage based delay calculator mode.

DESCRIPTION

TCLCMD-616

NAME

TCLCMD-616

SUMMARY

No wire load model has been associated with . Use command set_wire_load_model first.

DESCRIPTION

TCLCMD-617

NAME

TCLCMD-617

SUMMARY

No wire load mode has been specified. Use command set_wire_load_mode first.

DESCRIPTION

TCLCMD-618

NAME

TCLCMD-618

SUMMARY

Unknown option

DESCRIPTION

TCLCMD-619

NAME

TCLCMD-619

SUMMARY

-gain option cannot be used with -blackbox option; ignored

DESCRIPTION

TCLCMD-620

NAME

TCLCMD-620

SUMMARY

Design doesnot contain any ports. Aborting model extraction

DESCRIPTION

TCLCMD-621

NAME

TCLCMD-621

SUMMARY

The output of the command is large, and will be dumped to file

DESCRIPTION

TCLCMD-700

NAME

TCLCMD-700

SUMMARY

Currently no constraints are written to adb file

DESCRIPTION

TCLCMD-701

NAME

TCLCMD-701

SUMMARY

Library was loaded from file

DESCRIPTION

TCLCMD-702

NAME

TCLCMD-702

SUMMARY

Netlist is acyclic

DESCRIPTION

TCLCMD-703

NAME

TCLCMD-703

SUMMARY

Netlist contains combinational feedback loop(s)

DESCRIPTION

TCLCMD-704

NAME

TCLCMD-704

SUMMARY

Top level module, no parent

DESCRIPTION

TCLCMD-705

NAME

TCLCMD-705

SUMMARY

Bound instance to

DESCRIPTION

TCLCMD-706

NAME

TCLCMD-706

SUMMARY

Two modules to be compared with are the same, verification succeeded

DESCRIPTION

TCLCMD-707

NAME

TCLCMD-707

SUMMARY

Verification failed between modules and

DESCRIPTION

TCLCMD-708

NAME

TCLCMD-708

SUMMARY

Verification succeeded between modules and

DESCRIPTION

TCLCMD-709

NAME

TCLCMD-709

SUMMARY

violations fixed

DESCRIPTION

TCLCMD-710

NAME

TCLCMD-710

SUMMARY

violations still exist

DESCRIPTION

TCLCMD-711

NAME

TCLCMD-711

SUMMARY

Constraints are met

DESCRIPTION

TCLCMD-712

NAME

TCLCMD-712

SUMMARY

Constraints are not met

DESCRIPTION

TCLCMD-713

NAME

TCLCMD-713

SUMMARY

Defined clock : leading edge %5.2f, trailing edge %5.2f, period %5.2f

DESCRIPTION

TCLCMD-714

NAME

TCLCMD-714

SUMMARY

Verification passed for library

DESCRIPTION

TCLCMD-715

NAME

TCLCMD-715

SUMMARY

Verification failed for library

DESCRIPTION

TCLCMD-716

NAME

TCLCMD-716

SUMMARY

Updating using the contents of compiled library

DESCRIPTION

TCLCMD-717

NAME

TCLCMD-717

SUMMARY

Setting user defined attribute with value

DESCRIPTION

TCLCMD-718

NAME

TCLCMD-718

SUMMARY

Global override: option

DESCRIPTION

TCLCMD-719

NAME

TCLCMD-719

SUMMARY

Renaming module from to

DESCRIPTION

TCLCMD-720

NAME

TCLCMD-720

SUMMARY

Renaming net from to in module

DESCRIPTION

TCLCMD-721

NAME

TCLCMD-721

SUMMARY

Renaming instance from to in module

DESCRIPTION

TCLCMD-722

NAME

TCLCMD-722

SUMMARY

wire load asserted on an instance in TOP wire load mode is no longer hierarchically applied to all lower level modules (unlike in v2.2 and before). Please use -hier switch to get equivalent behavior

DESCRIPTION

TCLCMD-723

NAME

TCLCMD-723

SUMMARY

fast slew propagation mode in command set_global slew_propagation_mode fast is no longer supported. Setting slew_propagation_mode to more accurate worst mode

DESCRIPTION

TCLCMD-724

NAME

TCLCMD-724

SUMMARY

Renaming bus from to in module

DESCRIPTION

TCLCMD-725

NAME

TCLCMD-725

SUMMARY

Switching slew propagation mode to

DESCRIPTION

TCLCMD-726

NAME

TCLCMD-726

SUMMARY

Renaming port from to in module

DESCRIPTION

TCLCMD-727

NAME

TCLCMD-727

SUMMARY

Behavior of `set_cycle_addition -early` has changed since v2.3.4 release. The command now relaxes the hold check by %f number of cycles. To revert back to old behavior, please use `-%f` as the number of cycles

DESCRIPTION

TCLCMD-728

NAME

TCLCMD-728

SUMMARY

Setting default Wireload Selection table on the library . Note that Timing Analysis always picks up the default Wireload Selection table from the first target technology library

DESCRIPTION

TCLCMD-729

NAME

TCLCMD-729

SUMMARY

Extracting shell-core for module

DESCRIPTION

TCLCMD-730

NAME

TCLCMD-730

SUMMARY

Un-extracting shell-core for module

DESCRIPTION

TCLCMD-731

NAME

TCLCMD-731

SUMMARY

Suggest setting aware_adder_architecture to architecture

DESCRIPTION

TCLCMD-732

NAME

TCLCMD-732

SUMMARY

option specified without option

DESCRIPTION

TCLCMD-733

NAME

TCLCMD-733

SUMMARY

Could not open file to write. Proceeding without writing to file

DESCRIPTION

TCLCMD-734

NAME

TCLCMD-734

SUMMARY

Could not find file to open. Proceeding without writing to file

DESCRIPTION

TCLCMD-735

NAME

TCLCMD-735

SUMMARY

Timing-driven structuring module ...

DESCRIPTION

TCLCMD-736

NAME

TCLCMD-736

SUMMARY

Running

DESCRIPTION

TCLCMD-737

NAME

TCLCMD-737

SUMMARY

Setting up default floorplan parameters. Timing system will be reset

DESCRIPTION

TCLCMD-738

NAME

TCLCMD-738

SUMMARY

Multi-edge waveform specification is not supported. Honoring just the first two edges { } for the clock

DESCRIPTION

create_clock does not support multi-edge waveform specification. You can solve this problem by specifying multiple create_clock -add assertions which describe the occurrence of rising and falling edges in one period on the same pins.

TCLCMD-739

NAME

TCLCMD-739

SUMMARY

Moving constraint on hierarchical pin to leaf pins.

DESCRIPTION

TCLCMD-740

NAME

TCLCMD-740

SUMMARY

will be set to default: , for -machine_readable

DESCRIPTION

TCLCMD-742

NAME

TCLCMD-742

SUMMARY

Property is not supported for . Command get_property ignored

DESCRIPTION

TCLCMD-744

NAME

TCLCMD-744

SUMMARY

report_timing option is not compatible with -collection. Continuing with -collection option

DESCRIPTION

TCLCMD-745

NAME

TCLCMD-745

SUMMARY

The property is not available in analysis mode. To get the property value, switch analysis mode using the setAnalysisMode -checkType command, and run the get_property command again.

DESCRIPTION

TCLCMD-746

NAME

TCLCMD-746

SUMMARY

Empty collection specified as argument to

DESCRIPTION

TCLCMD-747

NAME

TCLCMD-747

SUMMARY

The type has no property associated with it. Supported property types are .

DESCRIPTION

TCLCMD-748

NAME

TCLCMD-748

SUMMARY

Command sort_collection is not supported for .

DESCRIPTION

TCLCMD-749

NAME

TCLCMD-749

SUMMARY

Structural collection specified as argument to

DESCRIPTION

TCLCMD-799

NAME

TCLCMD-799

SUMMARY

Cannot use -reference_pin and -source_latency_included/-network_latency_included together in command

DESCRIPTION

TCLCMD-800

NAME

TCLCMD-800

SUMMARY

Attempt to set property on library failed

DESCRIPTION

TCLCMD-803

NAME

TCLCMD-803

SUMMARY

Module name must be specified in command

DESCRIPTION

TCLCMD-804

NAME

TCLCMD-804

SUMMARY

Invalid cell id

DESCRIPTION

TCLCMD-805

NAME

TCLCMD-805

SUMMARY

Cell is not a sum-of-products logic cell

DESCRIPTION

TCLCMD-806

NAME

TCLCMD-806

SUMMARY

-lead/-trail options can only be used to specify clock waveform insertion delays.

DESCRIPTION

TCLCMD-807

NAME

TCLCMD-807

SUMMARY

-rise/-fall/-clock options can only be used to specify clock pin insertion delays.

DESCRIPTION

TCLCMD-808

NAME

TCLCMD-808

SUMMARY

The option cannot be specified without in command .

DESCRIPTION

TCLCMD-810

NAME

TCLCMD-810

SUMMARY

-clock option can only be used with -source.

DESCRIPTION

TCLCMD-811

NAME

TCLCMD-811

SUMMARY

-clock_from/-clock_to/-edge_from/-edge_to options can only be used to specify clock waveform uncertainty.

DESCRIPTION

TCLCMD-812

NAME

TCLCMD-812

SUMMARY

Cannot specify -from/-to options with pin object list. -from/-to can only be used for inter-clock uncertainty.

DESCRIPTION

TCLCMD-813

NAME

TCLCMD-813

SUMMARY

-min/-typ/-max library option is not supported for read_ctlf

DESCRIPTION

TCLCMD-814

NAME

TCLCMD-814

SUMMARY

Attempt to load tlf library failed

DESCRIPTION

TCLCMD-815

NAME

TCLCMD-815

SUMMARY

Can not set constraint on non-IO port

DESCRIPTION

TCLCMD-816

NAME

TCLCMD-816

SUMMARY

Attempt to read from file failed

DESCRIPTION

TCLCMD-817

NAME

TCLCMD-817

SUMMARY

Attempt to write into file failed

DESCRIPTION

TCLCMD-818

NAME

TCLCMD-818

SUMMARY

Illegal combination of options and for an unplaced design

DESCRIPTION

TCLCMD-819

NAME

TCLCMD-819

SUMMARY

No clock gating check was asserted on

DESCRIPTION

TCLCMD-820

NAME

TCLCMD-820

SUMMARY

set_disable_timing was not asserted before

DESCRIPTION

TCLCMD-821

NAME

TCLCMD-821

SUMMARY

Option -latch is used with a non-latch cell. Or option -flip_flop is used with a non-flip-flop cell

DESCRIPTION

TCLCMD-822

NAME

TCLCMD-822

SUMMARY

No matching assertion is found on pin

DESCRIPTION

TCLCMD-823

NAME

TCLCMD-823

SUMMARY

Unknown assertion type

DESCRIPTION

TCLCMD-824

NAME

TCLCMD-824

SUMMARY

Conflicting options and

DESCRIPTION

TCLCMD-825

NAME

TCLCMD-825

SUMMARY

: Only one clock allowed

DESCRIPTION

TCLCMD-826

NAME

TCLCMD-826

SUMMARY

Command syntax error - :

DESCRIPTION

TCLCMD-827

NAME

TCLCMD-827

SUMMARY

Invalid slew threshold value %.4f(must be between 0 and 100 percent)

DESCRIPTION

TCLCMD-828

NAME

TCLCMD-828

SUMMARY

Upper slew threshold value %.4f must be greater than the lower slew threshold value %.4f

DESCRIPTION

TCLCMD-829

NAME

TCLCMD-829

SUMMARY

Invalid slew threshold range. Specify slew threshold in percentage. The slew threshold range must be greater than one percent. For example, set_slew_threshold -lower 10 -upper 90

DESCRIPTION

TCLCMD-830

NAME

TCLCMD-830

SUMMARY

Unable to set Grow Parameters

DESCRIPTION

Check if current top module is set

TCLCMD-831

NAME

TCLCMD-831

SUMMARY

Unable to get Grow Parameters

DESCRIPTION

Check if current top module is set

TCLCMD-832

NAME

TCLCMD-832

SUMMARY

Unable to set Grow Anchors

DESCRIPTION

Check if current top module is set

TCLCMD-833

NAME

TCLCMD-833

SUMMARY

Unable to get Grow Anchors

DESCRIPTION

Check if current top module is set

TCLCMD-834

NAME

TCLCMD-834

SUMMARY

The -edges option cannot be combined with -multiply_by or -divide_by

DESCRIPTION

TCLCMD-835

NAME

TCLCMD-835

SUMMARY

Must specify one of the -edges, -divide_by, -multiply_by or -combinational options for the create_generated_clock command.

DESCRIPTION

TCLCMD-836

NAME

TCLCMD-836

SUMMARY

Cannot specify the -edge_shift option without -edges option.

DESCRIPTION

TCLCMD-837

NAME

TCLCMD-837

SUMMARY

The -duty_cycle option is only valid with the -multiply_by option.

DESCRIPTION

TCLCMD-838

NAME

TCLCMD-838

SUMMARY

-pos, -neg or -invert option can only be used with a pin list.

DESCRIPTION

TCLCMD-839

NAME

TCLCMD-839

SUMMARY

-first_parent_edge option value must be lead or trail

DESCRIPTION

TCLCMD-840

NAME

TCLCMD-840

SUMMARY

Invalid :

DESCRIPTION

TCLCMD-841

NAME

TCLCMD-841

SUMMARY

Invalid argument in list

DESCRIPTION

TCLCMD-842

NAME

TCLCMD-842

SUMMARY

option must be used with option

DESCRIPTION

TCLCMD-843

NAME

TCLCMD-843

SUMMARY

One of the following options must be specified: ,

DESCRIPTION

TCLCMD-844

NAME

TCLCMD-844

SUMMARY

Option can only be used with option

DESCRIPTION

TCLCMD-845

NAME

TCLCMD-845

SUMMARY

The -duty_cycle option value must be in range between 0 and 100.

DESCRIPTION

TCLCMD-846

NAME

TCLCMD-846

SUMMARY

Three edge numbers must be given with -edges option.

DESCRIPTION

TCLCMD-847

NAME

TCLCMD-847

SUMMARY

Three real numbers must be given with -edge_shift

DESCRIPTION

TCLCMD-848

NAME

TCLCMD-848

SUMMARY

Only one pin can be specified with -source option in create_generated_clock command.

DESCRIPTION

TCLCMD-849

NAME

TCLCMD-849

SUMMARY

-from option must be present and be one of the input pins of black box, for generated clocks asserted at the output of black-box models

DESCRIPTION

TCLCMD-850

NAME

TCLCMD-850

SUMMARY

No generated clock was asserted on pin

DESCRIPTION

TCLCMD-851

NAME

TCLCMD-851

SUMMARY

This command cannot reference a generated clock

DESCRIPTION

TCLCMD-852

NAME

TCLCMD-852

SUMMARY

Could not open file to execute

DESCRIPTION

TCLCMD-853

NAME

TCLCMD-853

SUMMARY

Could not find cadence hierarchy.

DESCRIPTION

TCLCMD-854

NAME

TCLCMD-854

SUMMARY

pkgnsrw returned as version. Expecting

DESCRIPTION

TCLCMD-855

NAME

TCLCMD-855

SUMMARY

Cell has no timing model associated with

DESCRIPTION

TCLCMD-856

NAME

TCLCMD-856

SUMMARY

Unable to delete timing model for mappable cell

DESCRIPTION

TCLCMD-857

NAME

TCLCMD-857

SUMMARY

Failed to delete timing model associated with cell

DESCRIPTION

TCLCMD-858

NAME

TCLCMD-858

SUMMARY

The timing model of cell has been deleted

DESCRIPTION

TCLCMD-859

NAME

TCLCMD-859

SUMMARY

Failed to delete library

DESCRIPTION

TCLCMD-860

NAME

TCLCMD-860

SUMMARY

Library has been deleted

DESCRIPTION

TCLCMD-861

NAME

TCLCMD-861

SUMMARY

Both -gns_library, -gns_lib_path and -gns_techlib must be specified

DESCRIPTION

TCLCMD-862

NAME

TCLCMD-862

SUMMARY

Conflicting options -delete_inst_pins and -create_ports

DESCRIPTION

TCLCMD-863

NAME

TCLCMD-863

SUMMARY

Module for instance does not have unique timing context

DESCRIPTION

TCLCMD-864

NAME

TCLCMD-864

SUMMARY

This command cannot be run on an unplaced design

DESCRIPTION

TCLCMD-865

NAME

TCLCMD-865

SUMMARY

Error encountered in option

DESCRIPTION

TCLCMD-866

NAME

TCLCMD-866

SUMMARY

File cant be opened for writing

DESCRIPTION

TCLCMD-867

NAME

TCLCMD-867

SUMMARY

Timing model filename must be specified

DESCRIPTION

TCLCMD-868

NAME

TCLCMD-868

SUMMARY

Cannot process -generate_top_level_budgets option for do_time_budget command because top level budgets are applied

DESCRIPTION

Use set_top_level_budgets -remove command to remove top level budgets

TCLCMD-869

NAME

TCLCMD-869

SUMMARY

Top level budgets are already applied

DESCRIPTION

TCLCMD-870

NAME

TCLCMD-870

SUMMARY

No top level budgets are applied

DESCRIPTION

TCLCMD-871

NAME

TCLCMD-871

SUMMARY

Inconsistency in top level budgets files

DESCRIPTION

The user may have removed manually .top_level_budgets or .remove_top_level_budgets files

TCLCMD-872

NAME

TCLCMD-872

SUMMARY

Please provide a positive argument with the option

DESCRIPTION

TCLCMD-873

NAME

TCLCMD-873

SUMMARY

No top level budgets are defined

DESCRIPTION

Use `do_time_budget -generate_top_level_budgets` to generate top level budgets

TCLCMD-874

NAME

TCLCMD-874

SUMMARY

Only one module can be specified

DESCRIPTION

TCLCMD-875

NAME

TCLCMD-875

SUMMARY

Only one net can be specified

DESCRIPTION

TCLCMD-876

NAME

TCLCMD-876

SUMMARY

Please specify a flexibility factor

DESCRIPTION

TCLCMD-877

NAME

TCLCMD-877

SUMMARY

Invalid budget modification value specified

DESCRIPTION

Expecting a positive floating-point number

TCLCMD-878

NAME

TCLCMD-878

SUMMARY

invalid command name

DESCRIPTION

TCLCMD-879

NAME

TCLCMD-879

SUMMARY

Value encountered in option where is expected

DESCRIPTION

TCLCMD-880

NAME

TCLCMD-880

SUMMARY

: Command disabled when abstraction is done

DESCRIPTION

TCLCMD-882

NAME

TCLCMD-882

SUMMARY

is neither valid bus nor pin id

DESCRIPTION

TCLCMD-883

NAME

TCLCMD-883

SUMMARY

Inconsistent object types of two object ids and

DESCRIPTION

TCLCMD-884

NAME

TCLCMD-884

SUMMARY

Cannot use option for a design

DESCRIPTION

TCLCMD-885

NAME

TCLCMD-885

SUMMARY

option requires a PKS license

DESCRIPTION

TCLCMD-886

NAME

TCLCMD-886

SUMMARY

Option only valid during optimization on a design. Option ignored.

DESCRIPTION

TCLCMD-887

NAME

TCLCMD-887

SUMMARY

Option requires a physical library to be loaded

DESCRIPTION

TCLCMD-888

NAME

TCLCMD-888

SUMMARY

does not match any in library

DESCRIPTION

Regular expression that user specify does not match to any property in the database. This may happen while searching library, cells, pins, etc.

TCLCMD-889

NAME

TCLCMD-889

SUMMARY

Pin given with -from option must be different from on which assertion is applied

DESCRIPTION

TCLCMD-890

NAME

TCLCMD-890

SUMMARY

There is no snipped arc due to a loop

DESCRIPTION

TCLCMD-891

NAME

TCLCMD-891

SUMMARY

-gns_library, -gns_view and -gns_tech options must be specified

DESCRIPTION

TCLCMD-892

NAME

TCLCMD-892

SUMMARY

Floorplan has no site area

DESCRIPTION

TCLCMD-893

NAME

TCLCMD-893

SUMMARY

This command cannot reference an internally generated clock

DESCRIPTION

TCLCMD-894

NAME

TCLCMD-894

SUMMARY

Wrong use of set_dcl_level. Performance level 1 is ignored, not to be used with Silicon Metrics SSM library.

DESCRIPTION

This warning message is issued when users try to set the dcl performance level to 1 while setting the global dcl_use_ssm_library to true. Setting the global to true forces BG to compute reduced model while relying on OLA library to compute net delay and slew DCL performance level of 1 instructs BG to pass parasitics data and have the OLA library reduce it. The two settings are conflicting and thus the performance level setting is ignored

TCLCMD-895

NAME

TCLCMD-895

SUMMARY

Globals, clock propagation mode, ideal clocks and clock to clock assertions (insertion delay/path exceptions/uncertainty etc.) will not be reset/removed

DESCRIPTION

Exit the tool and restart it or use appropriate reset commands to remove/reset these settings

TCLCMD-896

NAME

TCLCMD-896

SUMMARY

Model extractor requires that the top timing module and the current module point to the same module.

DESCRIPTION

TCLCMD-897

NAME

TCLCMD-897

SUMMARY

Update to Library using for pvt is ignored. Library does not have the pvt to update.

DESCRIPTION

When users update a library for specific pvt corner using -pvt, the pvt corner should have existed for the library to update. BG will not update non-existing pvt

TCLCMD-898

NAME

TCLCMD-898

SUMMARY

When lef_multiplier is specified, -all option should be specified. It is not supported with a net or a group of nets.

DESCRIPTION

TCLCMD-899

NAME

TCLCMD-899

SUMMARY

The design is routed: auto implying -ipo option. Use -no_ipo if you dont want to run optimization in ipo mode

DESCRIPTION

TCLCMD-900

NAME

TCLCMD-900

SUMMARY

Error in removing path group assertions

DESCRIPTION

TCLCMD-901

NAME

TCLCMD-901

SUMMARY

Wrong usage of command when reading min,typ,max library. You must specify at least two libraries for different pvt corners e.g. read_[alf|tlf] -min min.alf -max max.alf

DESCRIPTION

This error message is to stop users from using the read library command wrongly. When reading mtm library, users will have to specify more than one corner. For example, read_alf -min min.alf -max max.alf

TCLCMD-902

NAME

TCLCMD-902

SUMMARY

No cells matching specified pattern in library

DESCRIPTION

TCLCMD-903

NAME

TCLCMD-903

SUMMARY

-name option is used only for merging min/typ/max library

DESCRIPTION

Renaming the library is not allowed when reading single library. This is to prevent confusion.

TCLCMD-904

NAME

TCLCMD-904

SUMMARY

This command requires a license

DESCRIPTION

TCLCMD-905

NAME

TCLCMD-905

SUMMARY

Option requires a license

DESCRIPTION

TCLCMD-906

NAME

TCLCMD-906

SUMMARY

Unknown option

DESCRIPTION

TCLCMD-907

NAME

TCLCMD-907

SUMMARY

set_path_group does not allow the use of the options <pin_list> <clk_name_list> together. Try specifying the pin list as a -through <pin_list> in conjunction with the clock specifier to achieve the desired results.

DESCRIPTION

TCLCMD-908

NAME

TCLCMD-908

SUMMARY

Reading ALF library is not allowed in FE ASIC mode

DESCRIPTION

When using FE ASIC license, you should only use the library that is provided by specific ASIC vendor. Reading ALF library is prohibited. Contact your ASIC vendor to provide you the dot-lib or tlf libraries with proper FE ASIC license string and use read_dot_lib or read_tlf command

TCLCMD-909

NAME

TCLCMD-909

SUMMARY

Cannot write timing windows while global lib_cell_thresholds_for_reporting is on. Please turn off the global before writing timing windows.

DESCRIPTION

When global lib_cell_thresholds_for_reporting is turned on, users expect to see slews of each instance pin based on threshold points that are defined in each respective library. Timing Windows File contains its own slew threshold which conflicts the library slew thresholds if the global is on. To be able to write out timing windows file, you must turn off the global using set_global command.

TCLCMD-910

NAME

TCLCMD-910

SUMMARY

No clock waveforms specified

DESCRIPTION

TCLCMD-911

NAME

TCLCMD-911

SUMMARY

is an invalid list of clocks

DESCRIPTION

TCLCMD-912

NAME

TCLCMD-912

SUMMARY

No valid clock found in list

DESCRIPTION

TCLCMD-913

NAME

TCLCMD-913

SUMMARY

No pins specified

DESCRIPTION

TCLCMD-914

NAME

TCLCMD-914

SUMMARY

is an invalid list of pins

DESCRIPTION

TCLCMD-915

NAME

TCLCMD-915

SUMMARY

No valid pin found in list

DESCRIPTION

TCLCMD-916

NAME

TCLCMD-916

SUMMARY

Illegal option: (may be incompatible with other options)

DESCRIPTION

TCLCMD-917

NAME

TCLCMD-917

SUMMARY

Cannot find that match

DESCRIPTION

TCLCMD-918

NAME

TCLCMD-918

SUMMARY

The argument passed to command should resolve to a single unique object. Multiple objects were found that matched the name . Make the argument unique by modifying the argument such that it maps to a single unique object by use of collection commands (eg: [get_pins] instead of) or by reorganizing wildcards.

DESCRIPTION

TCLCMD-919

NAME

TCLCMD-919

SUMMARY

Options -rise/-rise_from/-rise_to and -fall/-fall_from/-fall_to can only be used to specify/remove inter-clock uncertainty.

DESCRIPTION

TCLCMD-920

NAME

TCLCMD-920

SUMMARY

The option is unsupported

DESCRIPTION

TCLCMD-921

NAME

TCLCMD-921

SUMMARY

The command is ignored because the option is not supported.

DESCRIPTION

TCLCMD-923

NAME

TCLCMD-923

SUMMARY

Specified argument is not a valid collection

DESCRIPTION

This error is displayed when the argument specified with sizeof_collection command is not a collection. This error might also be reported due to a typographical error. For example: if there are unmatched pair(s) of [or] characters.

TCLCMD-924

NAME

TCLCMD-924

SUMMARY

index is greater than size of the collection

DESCRIPTION

TCLCMD-925

NAME

TCLCMD-925

SUMMARY

Valid index values range from 0 to sizeof_collection - 1

DESCRIPTION

TCLCMD-926

NAME

TCLCMD-926

SUMMARY

is not a valid class type for query_objects

DESCRIPTION

TCLCMD-927

NAME

TCLCMD-927

SUMMARY

Cannot find for specified

DESCRIPTION

TCLCMD-928

NAME

TCLCMD-928

SUMMARY

Value is invalid for option . Specified value should be ≥ 0

DESCRIPTION

TCLCMD-929

NAME

TCLCMD-929

SUMMARY

Command foreach_in_collection exited:

DESCRIPTION

TCLCMD-930

NAME

TCLCMD-930

SUMMARY

Option -nocase can not be specified without using option -regexp

DESCRIPTION

TCLCMD-931

NAME

TCLCMD-931

SUMMARY

Options -exact and -regexp can not be specified together. They are mutually exclusive

DESCRIPTION

TCLCMD-932

NAME

TCLCMD-932

SUMMARY

Only hierarchical instances can be specified with -inst option. Specified instance(s) are not valid.

DESCRIPTION

TCLCMD-933

NAME

TCLCMD-933

SUMMARY

Option -of_objects and pattern can not be specified together. They are mutually exclusive

DESCRIPTION

TCLCMD-934

NAME

TCLCMD-934

SUMMARY

The command is not supported for hierarchical instance

DESCRIPTION

TCLCMD-935

NAME

TCLCMD-935

SUMMARY

File already exists.

DESCRIPTION

TCLCMD-956

NAME

TCLCMD-956

SUMMARY

The -add option requires both -name and the -master_clock option to be specified

DESCRIPTION

TCLCMD-957

NAME

TCLCMD-957

SUMMARY

The -combinational option cannot be specified with -divide_by or -multiply_by or -edges

DESCRIPTION

TCLCMD-958

NAME

TCLCMD-958

SUMMARY

Previously defined source objects for clock have been overwritten. To avoid overwrite, ensure that all source objects are passed to the create_clock command at once.

DESCRIPTION

This warning is seen because clock with the same name <clkname> is defined more than once.

.P

Multiple clock definitions with same name are not allowed and the previous definition for <clkname> is overwritten with last definition. If these multiple clock definitions with same name have same waveform, period definition but have different source objects, all desired source objects can be provided as a list to single create_clock command.

.P

In order to avoid this warning, make sure that Multiple clock definitions with same name are not defined.

TCLCMD-961

NAME

TCLCMD-961

SUMMARY

You must specify the -name option with the -add option for the command.

DESCRIPTION

TCLCMD-962

NAME

TCLCMD-962

SUMMARY

Must specify -name or sources option for the command.

DESCRIPTION

TCLCMD-963

NAME

TCLCMD-963

SUMMARY

Invalid clock period value . Value must be greater than or equal to zero.

DESCRIPTION

TCLCMD-964

NAME

TCLCMD-964

SUMMARY

does not support multiple-edge waveform specification. It only supports two-edge waveform specification. Solve this problem by specifying multiple assertions on the same pin.

DESCRIPTION

TCLCMD-965

NAME

TCLCMD-965

SUMMARY

Invalid waveform. The duration between edges is greater than one period.

DESCRIPTION

TCLCMD-966

NAME

TCLCMD-966

SUMMARY

on is ignored. You must specify the -clock option.

DESCRIPTION

TCLCMD-967

NAME

TCLCMD-967

SUMMARY

Cannot map pin to its primitive pins.

DESCRIPTION

TCLCMD-968

NAME

TCLCMD-968

SUMMARY

Attempt to load .lib library failed

DESCRIPTION

TCLCMD-969

NAME

TCLCMD-969

SUMMARY

Illegal operation on collection object (%x).

DESCRIPTION

TCLCMD-971

NAME

TCLCMD-971

SUMMARY

Invalid resistance value %f. Value specified with must be greater than or equal to zero.

DESCRIPTION

TCLCMD-972

NAME

TCLCMD-972

SUMMARY

invalid delay value. Specify a value greater than or equal to 0 with the command.

DESCRIPTION

TCLCMD-975

NAME

TCLCMD-975

SUMMARY

Mixing options is not allowed

DESCRIPTION

TCLCMD-976

NAME

TCLCMD-976

SUMMARY

Invalid path description. Please specify an argument for the option

DESCRIPTION

TCLCMD-977

NAME

TCLCMD-977

SUMMARY

Gray box models are not supported for liberty format. Please use -blackbox or -blackbox_2d option to extract black box models.

DESCRIPTION

TCLCMD-978

NAME

TCLCMD-978

SUMMARY

Option can not be used with cpr analysis enabled. To use this option, disable cpr analysis by setting global timing_remove_clock_reconvergence_pessimism to false

DESCRIPTION

TCLCMD-979

NAME

TCLCMD-979

SUMMARY

Cannot apply the command to port because it is an invalid port for this command. Fix the timing constraint command so that it is applied to the correct type of port.

DESCRIPTION

This error is displayed when either the `set_output_delay` command or the `set_input_delay` command is applied to an invalid port. The `set_output_delay` command can only be applied to output or inout ports. Similarly, the `set_input_delay` command can only be applied to input and inout ports. Timing constraint commands that are applied to invalid ports are ignored. To avoid this error, fix the timing constraint command so that it is applied to the correct type of port.

TCLCMD-980

NAME

TCLCMD-980

SUMMARY

Invalid Values is provided for option of command , for clock . Please use Value greater than 0

DESCRIPTION

TCLCMD-981

NAME

TCLCMD-981

SUMMARY

Unsupported extra argument in command .

DESCRIPTION

The related command does not recognize the extra argument. Doing a help on the command name will give you the proper syntax of the command.

TCLCMD-982

NAME

TCLCMD-982

SUMMARY

Missing required argument in command

DESCRIPTION

TCLCMD-983

NAME

TCLCMD-983

SUMMARY

unknown analysis_type

DESCRIPTION

TCLCMD-984

NAME

TCLCMD-984

SUMMARY

Cannot specify option without specifying the option in command .

DESCRIPTION

TCLCMD-985

NAME

TCLCMD-985

SUMMARY

data can not be reported in timing analysis mode, to report data use option or use to run timing analysis

DESCRIPTION

TCLCMD-986

NAME

TCLCMD-986

SUMMARY

Clock waveform cannot be propagated as this is a virtual clock.

DESCRIPTION

TCLCMD-987

NAME

TCLCMD-987

SUMMARY

Import of procedures in namespace into current namespace failed : timing analysis commands will not be available to the application user

DESCRIPTION

TCLCMD-989

NAME

TCLCMD-989

SUMMARY

cannot open SDC file for mode

DESCRIPTION

TCLCMD-990

NAME

TCLCMD-990

SUMMARY

there is no SDC file for mode

DESCRIPTION

TCLCMD-991

NAME

TCLCMD-991

SUMMARY

Cannot specify option without using option.

DESCRIPTION

TCLCMD-994

NAME

TCLCMD-994

SUMMARY

Can not find object with the name . This can happen when the specified name of any MMMC object is incorrectly spelled or has been removed from the running session.

DESCRIPTION

TCLCMD-995

NAME

TCLCMD-995

SUMMARY

Can not open file for

DESCRIPTION

TCLCMD-996

NAME

TCLCMD-996

SUMMARY

`` is invalid view name.

DESCRIPTION

TCLCMD-997

NAME

TCLCMD-997

SUMMARY

`` is invalid view for analysis mode.

DESCRIPTION

TCLCMD-998

NAME

TCLCMD-998

SUMMARY

Skipping pin . Could not find sequential device clock pins in its transitive fanout.

DESCRIPTION

TCLCMD-999

NAME

TCLCMD-999

SUMMARY

Could not find any matching the values specified for

DESCRIPTION

TCLCMD-1000

NAME

TCLCMD-1000

SUMMARY

Invalid option. This option may only be used in the bc_wc analysis.

DESCRIPTION

The -bc/-wc option in the write_timing_windows command may only be used in the bc_wc analysis. Use the set_operating_condition command to set the timing analysis type. When the -bc option is specified, the delays and slews will be computed using the best-case library. Similarly, when the -wc option is specified, the delays and slews will be computed using the worst-case library.

TCLCMD-1001

NAME

TCLCMD-1001

SUMMARY

Current instance is the top-level of design

DESCRIPTION

TCLCMD-1002

NAME

TCLCMD-1002

SUMMARY

Could not find source pin for clock

DESCRIPTION

TCLCMD-1003

NAME

TCLCMD-1003

SUMMARY

Could not find master for generated clock in view

DESCRIPTION

TCLCMD-1004

NAME

TCLCMD-1004

SUMMARY

The software has detected multiple master clocks: specified on the source pin for generated clock in view and will choose one of them for the derivation of the generated clock. You can use the -master_clock option of create_generated_clock to explicitly specify which of the master clocks to use when there is a conflict.

DESCRIPTION

This message is issued when the option -master_clock is not specified in create_generated_clock command and the corresponding source pin has multiple clocks on it. In such case, software picks one of the clock at source pin as master clock to perform the timing analysis. To fix the issue, specify -master_clock in create_generated_clock command.

TCLCMD-1005

NAME

TCLCMD-1005

SUMMARY

Using master clock for generated clock in view

DESCRIPTION

TCLCMD-1006

NAME

TCLCMD-1006

SUMMARY

Empty list of clock pins specified to -group option of set_clock_group. Constraint ignored.

DESCRIPTION

TCLCMD-1007

NAME

TCLCMD-1007

SUMMARY

Clock group name was already defined. New specification has overwritten previous specification.

DESCRIPTION

TCLCMD-1008

NAME

TCLCMD-1008

SUMMARY

No name is specified with -name option.

DESCRIPTION

TCLCMD-1009

NAME

TCLCMD-1009

SUMMARY

Conflicting options asynchronous and exclusive

DESCRIPTION

TCLCMD-1010

NAME

TCLCMD-1010

SUMMARY

Same clock cannot be specified twice in -group option of set_clock_groups command.

DESCRIPTION

TCLCMD-1011

NAME

TCLCMD-1011

SUMMARY

Wrong Usage: No other arguments accepted along with all_clocks or *

DESCRIPTION

TCLCMD-1012

NAME

TCLCMD-1012

SUMMARY

The specified view is not currently an active analysis view - so it cannot be used as default view.
The set_default_view command is ignored.

DESCRIPTION

TCLCMD-1013

NAME

TCLCMD-1013

SUMMARY

The SDC `set_operating_conditions` assertion is not supported. Please use the Innovus `setOpCond` command to specify library and operating condition information. Use the `setAnalysisMode` command to control single vs. bestCase/worstCase vs. on-chip variation timing analysis.

DESCRIPTION

SDC `set_operating_conditions` assertion is not supported and will be ignored. Control on operating condition can be done using `set_op_cond` command. Analysis type can be set using `set_analysis_mode/setAnalysisMode` command.

TCLCMD-1014

NAME

TCLCMD-1014

SUMMARY

The SDC `set_operating_conditions` assertion is not supported. Please use the `create_delay_corner` command to specify the desired operating environment. Use the `setAnalysisMode` command to control single vs. bestCase/worstCase vs. on-chip variation timing analysis.

DESCRIPTION

TCLCMD-1015

NAME

TCLCMD-1015

SUMMARY

The command is not currently supported as a constraint which can be loaded as part of your SDC

file. You can use the command interactively or in a Tcl command script.

DESCRIPTION

TCLCMD-1016

NAME

TCLCMD-1016

SUMMARY

Found existing constraint, overriding it.

DESCRIPTION

TCLCMD-1017

NAME

TCLCMD-1017

SUMMARY

Could not parse input file due to wrong inputs.

DESCRIPTION

TCLCMD-1018

NAME

TCLCMD-1018

SUMMARY

Ignoring -locv option as . Doing report_timing without locv

DESCRIPTION

TCLCMD-1019

NAME

TCLCMD-1019

SUMMARY

Cannot find an active view for analysis.

DESCRIPTION

TCLCMD-1020

NAME

TCLCMD-1020

SUMMARY

Filter option not supported. Command Ignored

DESCRIPTION

TCLCMD-1021

NAME

TCLCMD-1021

SUMMARY

A of name already exists.

DESCRIPTION

TCLCMD-1023

NAME

TCLCMD-1023

SUMMARY

The command is not supported while the SOCE system is operating in multi-mode / multi-corner (MMMC). MMMC functionality is not available for hierarchical implementation flow at this time.

DESCRIPTION

TCLCMD-1024

NAME

TCLCMD-1024

SUMMARY

Cannot perform set_clock_latency on pin as create_clock or create_generated_clock is not asserted on this pin in view .

DESCRIPTION

TCLCMD-1025

NAME

TCLCMD-1025

SUMMARY

Option value for command is obsolete. Use the command to define your own objects instead. The obsolete option still works in this release, but to avoid this warning and to ensure compatibility with future releases, remove from your script.

DESCRIPTION

TCLCMD-1026

NAME

TCLCMD-1026

SUMMARY

Cannot apply on input pin .

DESCRIPTION

TCLCMD-1027

NAME

TCLCMD-1027

SUMMARY

Option does not match object list .

DESCRIPTION

TCLCMD-1028

NAME

TCLCMD-1028

SUMMARY

-delay_corner <delay_corner_name> option is required in multi-mode/multi-corner

DESCRIPTION

TCLCMD-1029

NAME

TCLCMD-1029

SUMMARY

-min or -max option should not be used with -delay_corner in multi-mode/multi-corner

DESCRIPTION

TCLCMD-1030

NAME

TCLCMD-1030

SUMMARY

Cannot find driver pin for the generated clock hierarchical source pin

DESCRIPTION

TCLCMD-1031

NAME

TCLCMD-1031

SUMMARY

Specifying multiple ECSM .lib files in the create_library_set command to be used for an interpolated SignalStorm delay calculation (TriLib) flow is a beta feature. This feature is not available using default settings to ensure the features and limitations of this feature are understood. Contact your Cadence representative if you have any questions.

DESCRIPTION

TCLCMD-1032

NAME

TCLCMD-1032

SUMMARY

No timing library definition in conf file. MMMC mode is disallowed.

DESCRIPTION

TCLCMD-1033

NAME

TCLCMD-1033

SUMMARY

Ignoring option . Current analysis mode is .

DESCRIPTION

TCLCMD-1034

NAME

TCLCMD-1034

SUMMARY

Invalid : specified.

DESCRIPTION

TCLCMD-1035

NAME

TCLCMD-1035

SUMMARY

No clock reconvergence found for pins and .

DESCRIPTION

TCLCMD-1036

NAME

TCLCMD-1036

SUMMARY

not specified. Use option.

DESCRIPTION

TCLCMD-1037

NAME

TCLCMD-1037

SUMMARY

CPPR computation is disabled. Set global timing_remove_clock_reconvergence_pessimism to true.

DESCRIPTION

TCLCMD-1038

NAME

TCLCMD-1038

SUMMARY

Multiple values supplied for the option . Specify single value.

DESCRIPTION

TCLCMD-1039

NAME

TCLCMD-1039

SUMMARY

Required object_list argument not specified.

DESCRIPTION

TCLCMD-1040

NAME

TCLCMD-1040

SUMMARY

-library option of set_wire_load_model accepts only a timing library.

DESCRIPTION

TCLCMD-1041

NAME

TCLCMD-1041

SUMMARY

current_design should use the top cell as a argument

DESCRIPTION

TCLCMD-1042

NAME

TCLCMD-1042

SUMMARY

cannot open IR drop file

DESCRIPTION

TCLCMD-1043

NAME

TCLCMD-1043

SUMMARY

is not a valid argument to .

DESCRIPTION

TCLCMD-1044

NAME

TCLCMD-1044

SUMMARY

The -edge_shifts option is used without -edges, it will be dropped and will be used to derive the waveform for generated clock

DESCRIPTION

TCLCMD-1045

NAME

TCLCMD-1045

SUMMARY

checks can only be reported in analysis mode, or simultaneous setup/hold analysis mode. Use either or simultaneous setup/hold analysis mode, and re-run the command.

DESCRIPTION

The software maintains different analysis modes which at a given time can provide either Setup check related data, Hold check related data, or the combined data for both Setup and Hold style checks. The default and normal operation in Innovus is to have either the Setup or Hold data active at any given time. In Tempus, the normal operation is to have both Setup and Hold active at the same time. The general recommendation is for users to run both of the tools in their respective default modes. As such, users are more likely to see this error message while running Innovus rather than Tempus.

The type of analysis being performed by the timer is controlled by:

```
> setAnalysisMode -checkType setup | hold (Innovus Legacy UI)
> set_analysis_mode -checkType setup| hold (Tempus Legacy UI)
> set_db timing_analysis_mode setup | hold (Stylus UI)
```

When Setup and Hold analysis data are not simultaneously present in the timer, some types of reporting queries will generate an error - indicating the timer is not in the correct analysis mode to provide the requested data.

Example:

```
innovus> setAnalysisMode -checkType hold
```

```
innovus> report_timing -check_type setup
```

```
**ERROR: (TCLCMD-1045): setup checks can only be reported in setup analysis mode, or
simultaneous setup/hold analysis mode. Use either setup or simultaneous setup/hold analysis
mode, and re-run the command.
```

In Innovus, you could resolve this issue by switching the analysis type back to setup.

```
innovus> setAnalysisMode -checkType setup
```

You should be aware that switching the check type back-and-forth between setup and hold will cause timing resets to occur. You will need to wait for delay calculation and timing analysis to be regenerated.

Tempus is almost always run in a mode where both Setup and Hold data are present at the same time - so this type of error condition is rarely encountered.

TCLCMD-1046

NAME

TCLCMD-1046

SUMMARY

Command is not supported. Command ignored.

DESCRIPTION

TCLCMD-1047

NAME

TCLCMD-1047

SUMMARY

Constraint mode being specified is not related to any active analysis view. To know the list of all active view and their corresponding modes, use command `report_analysis_views -type active`

DESCRIPTION

TCLCMD-1048

NAME

TCLCMD-1048

SUMMARY

constraints are specified but no constraint mode is enabled interactively. Use `man TCLCMD-1048` to check the correct usage example.

DESCRIPTION

Example: >set_interactive_constraint_modes {func1 test} >set_propagated_clock [all_clocks]
>set_interactive_constraint_modes {} This example applies the interactive constraint editing to modes func1 and test. The set_propagated_clock [all_clocks] constraint is applied to both of these modes. To exit interactive constraint editing, specify set_interactive_constraint_modes with an empty list {}.

TCLCMD-1049

NAME

TCLCMD-1049

SUMMARY

Resetting all latency settings from fanout cone of

DESCRIPTION

TCLCMD-1050

NAME

TCLCMD-1050

SUMMARY

Executing command

DESCRIPTION

TCLCMD-1051

NAME

TCLCMD-1051

SUMMARY

Clock not found.

DESCRIPTION

TCLCMD-1052

NAME

TCLCMD-1052

SUMMARY

View option is only supported for objects. Option Ignored

DESCRIPTION

TCLCMD-1053

NAME

TCLCMD-1053

SUMMARY

Command `set_lib_pin` is obsolete and may not be supported in future release. The command still works, but to avoid this warning and to ensure compatibility with future releases, update your script to use `set_max_transition`/`set_max_capacitance`/`set_max_fanout` commands as needed on library pin objects. `set_lib_pin` command information is not preserved during saving the design. Re-apply this command if you wish to duplicate the effects in a later database.

DESCRIPTION

TCLCMD-1054

NAME

TCLCMD-1054

SUMMARY

not found for instance .

DESCRIPTION

TCLCMD-1055

NAME

TCLCMD-1055

SUMMARY

is not a generated clock. `reset_generated_clock` can remove only generated clocks.

DESCRIPTION

TCLCMD-1057

NAME

TCLCMD-1057

SUMMARY

Option -restructure_only can only be used with -clock_mesh.

DESCRIPTION

TCLCMD-1059

NAME

TCLCMD-1059

SUMMARY

Modes and cannot be activated at the same time for mode group on instance . set_mode command is ignored.

DESCRIPTION

TCLCMD-1060

NAME

TCLCMD-1060

SUMMARY

Threshold options without edge (rise/fall) cannot be specified with threshold options with edge (rise/fall). Use either of them.

DESCRIPTION

TCLCMD-1061

NAME

TCLCMD-1061

SUMMARY

set_fanout_load is supported for output and bidir ports only. Ignoring on input port .

DESCRIPTION

TCLCMD-1062

NAME

TCLCMD-1062

SUMMARY

Using the -min and -max options to the set_timing_derate command is recommended only for best-case / worst-case analysis mode. If these options are used in OCV mode, the system will preserve the -min -early and the -max -late derating factors.

DESCRIPTION

This warning is encountered when the -min or -max parameter of the set_timing_derate command is specified while the design is in on-chip variation mode. The -min or -max parameter is relevant in best-case worst-case analysis mode, but not in the on-chip variation mode. The -min, -max, -early, and -late parameters of the set_timing_derate command can be used to specify upto four derating factors. But in on-chip variation mode, out of the four derating factors, the software uses only -max -late (used for late paths) and -min -early (used for early paths) factors. However, the -min -late and -max -early derating factors are ignored. If the -min or -max parameter is not specified, then both the late and early derating factors are used. To avoid this warning, do not use the -min or -max parameter of the set_timing_derate command while the design is in on-chip variation mode. To specify the required derating factors, use the -early and -late parameters of the set_timing_derate command.

TCLCMD-1063

NAME

TCLCMD-1063

SUMMARY

A timing license is required to run this Timing Analysis command. Checkout a timing license and rerun.

DESCRIPTION

TCLCMD-1064

NAME

TCLCMD-1064

SUMMARY

set_timing_derate command can not be used simultaneously with locv pessimistic latch analysis. So the global timing_enable_locv_pessimistic_latch_analysis is being turned off.

DESCRIPTION

TCLCMD-1065

NAME

TCLCMD-1065

SUMMARY

The clock waveform(s) previously defined on pin have been overwritten by a new constraint for clock waveform on the same pin. Use the -add option to retain existing clocks.

DESCRIPTION

This warning is issued when multiple clocks with different names are being defined on same source object.

.P

If multiple clocks with different names are defined on same source object , the tool will overwrite the previously defined clock(s) on that source object by last clock definition. If you need to specify multiple clocks on the same source object, you must specify the -add parameter to create_clock/create_generated_clock commands, to add more clock(s) to the existing source object.

TCLCMD-1066

NAME

TCLCMD-1066

SUMMARY

Hierarchical pin specified as -source with generated clock could not be uniquely resolved to its flat pin. The tool has mapped it to its first available source/sink pin . If this is not the intended pin, modify the constraint accordingly to use the correct flat pin.

DESCRIPTION

TCLCMD-1067

NAME

TCLCMD-1067

SUMMARY

No valid clock found for option .

DESCRIPTION

TCLCMD-1068

NAME

TCLCMD-1068

SUMMARY

One or more of the target pins given with `create_clock/create_generated_clock` for clock are hierarchical pins. These hierarchical pins have been mapped to their respective flat pin(s). `report_clocks <clk_signame_list>` can be used to report the mapped flat pins for a given clock.

DESCRIPTION

This warning is issued when one or more of the target pins, specified using the `create_clock` or `create_generated_clock` constraint, for clock `<clkname>` are hierarchical pins.

.P

Even though constraints on hierarchical pins are supported, but in some cases (for example, floating hierarchical pins or hierarchical pin on power/ground nets), the hierarchical pins are mapped to the best possible driver or are connected to sink flat pins. You can use the `report_clocks <clk_signame_list>` command to report mapped flat pins for a given clock.

.P

In order to avoid this warning and subsequent mapping of such hierarchical pins to flat pins, you must redefine the clocks, specified using the above constraints, by specifying flat pins.

TCLCMD-1069

NAME

TCLCMD-1069

SUMMARY

`-level_sensitive` and `-edge_triggered` cannot be specified at the same time.

DESCRIPTION

TCLCMD-1070

NAME

TCLCMD-1070

SUMMARY

Option valid only for properties on objects of type

DESCRIPTION

TCLCMD-1071

NAME

TCLCMD-1071

SUMMARY

Cannot find that match in view

DESCRIPTION

TCLCMD-1072

NAME

TCLCMD-1072

SUMMARY

Options -clock_path/-data_path or -rise/-fall cannot be specified with non-clock objects.

DESCRIPTION

TCLCMD-1073

NAME

TCLCMD-1073

SUMMARY

Must specify one of , and

DESCRIPTION

TCLCMD-1074

NAME

TCLCMD-1074

SUMMARY

and cannot be specified with

DESCRIPTION

TCLCMD-1075

NAME

TCLCMD-1075

SUMMARY

No valid objects specified

DESCRIPTION

TCLCMD-1076

NAME

TCLCMD-1076

SUMMARY

No matching check arc delay assertion found

DESCRIPTION

TCLCMD-1077

NAME

TCLCMD-1077

SUMMARY

Cannot specify same pin with -from and -to options

DESCRIPTION

TCLCMD-1078

NAME

TCLCMD-1078

SUMMARY

create_dc_corner is obsolete, use create_delay_corner

DESCRIPTION

TCLCMD-1079

NAME

TCLCMD-1079

SUMMARY

update_dc_corner is obsolete, use update_delay_corner

DESCRIPTION

TCLCMD-1080

NAME

TCLCMD-1080

SUMMARY

option -dc_corner is obsolete, use -delay_corner

DESCRIPTION

TCLCMD-1081

NAME

TCLCMD-1081

SUMMARY

get_dc_corner is obsolete, use get_delay_corner

DESCRIPTION

TCLCMD-1082

NAME

TCLCMD-1082

SUMMARY

all_dc_corners is obsolete, use all_delay_corners

DESCRIPTION

TCLCMD-1084

NAME

TCLCMD-1084

SUMMARY

Command is no longer supported. Use instead.

DESCRIPTION

TCLCMD-1085

NAME

TCLCMD-1085

SUMMARY

The set_timing_derate option must be specified with -net_delay.

DESCRIPTION

TCLCMD-1086

NAME

TCLCMD-1086

SUMMARY

Clock group has already been defined with an identical specification. This assertion will be ignored.

DESCRIPTION

TCLCMD-1087

NAME

TCLCMD-1087

SUMMARY

Incorrect Number of Clock Sense Types specified.

DESCRIPTION

TCLCMD-1089

NAME

TCLCMD-1089

SUMMARY

-allow_path can only be used with -asynchronous.

DESCRIPTION

TCLCMD-1090

NAME

TCLCMD-1090

SUMMARY

Hierarchical pin is not supported in set_clock_sense.

DESCRIPTION

TCLCMD-1091

NAME

TCLCMD-1091

SUMMARY

Clock group option need to be specified.

DESCRIPTION

TCLCMD-1092

NAME

TCLCMD-1092

SUMMARY

More than one options are specified.

DESCRIPTION

TCLCMD-1093

NAME

TCLCMD-1093

SUMMARY

The -exclusive option is obsolete. It is being automatically mapped to -logically_exclusive.

DESCRIPTION

TCLCMD-1094

NAME

TCLCMD-1094

SUMMARY

The system is switching to MMMC mode. Existing timing, extraction and delay/sdf information from single mode will be reset. Please respecify spef for specific RC corners and/or sdf information for specific views.

DESCRIPTION

TCLCMD-1095

NAME

TCLCMD-1095

SUMMARY

Multiple objects not supported with .

DESCRIPTION

TCLCMD-1096

NAME

TCLCMD-1096

SUMMARY

Wrong is used in create_delay_corner -name . Please specify internal library name for the library in which the operating condition is defined. This is not the library file name.

DESCRIPTION

TCLCMD-1097

NAME

TCLCMD-1097

SUMMARY

The specified in create_delay_corner -name not found

DESCRIPTION

TCLCMD-1098

NAME

TCLCMD-1098

SUMMARY

Same clock cannot be specified multiple times in -group options of command.

DESCRIPTION

TCLCMD-1099

NAME

TCLCMD-1099

SUMMARY

Pin must be specified in command.

DESCRIPTION

TCLCMD-1100

NAME

TCLCMD-1100

SUMMARY

Constraints specified through option with command in unflattened ILM state are not usable. Use for specifying constraints for flattened state.

DESCRIPTION

TCLCMD-1101

NAME

TCLCMD-1101

SUMMARY

Constraints cannot be specified through option with command in flattened ILM state. Use option.

DESCRIPTION

TCLCMD-1102

NAME

TCLCMD-1102

SUMMARY

report_clock_timing -type jitter is supported in ocv mode only

DESCRIPTION

TCLCMD-1103

NAME

TCLCMD-1103

SUMMARY

Option value for command set_interactive_constraint_modes is not allowed. Interactive constraints cannot be preserved through the flow for default_mode_setup or default_mode_hold. Use the create_constraint_mode command to define your own constraint mode objects instead.

DESCRIPTION

TCLCMD-1104

NAME

TCLCMD-1104

SUMMARY

Both -late and -early checks can only be reported in simultaneous setup/hold analysis mode. Use simultaneous setup/hold analysis mode, and re-run the command.

DESCRIPTION

TCLCMD-1105

NAME

TCLCMD-1105

SUMMARY

Option is specified more than once

DESCRIPTION

TCLCMD-1106

NAME

TCLCMD-1106

SUMMARY

Option is ignored as it cannot be specified in analysis mode. Use `setAnalysisMode -checkType` to change the analysis mode in order to honor the option.

DESCRIPTION

TCLCMD-1107

NAME

TCLCMD-1107

SUMMARY

Option is obsolete and shall be removed in future. This is mapped to .

DESCRIPTION

TCLCMD-1108

NAME

TCLCMD-1108

SUMMARY

Option must be specified along with .

DESCRIPTION

TCLCMD-1109

NAME

TCLCMD-1109

SUMMARY

Could not find source for command

DESCRIPTION

TCLCMD-1110

NAME

TCLCMD-1110

SUMMARY

Invalid argument supplied to option . Valid arguments are .

DESCRIPTION

TCLCMD-1111

NAME

TCLCMD-1111

SUMMARY

argument supplied to option in mode. Valid arguments in this mode are .

DESCRIPTION

TCLCMD-1112

NAME

TCLCMD-1112

SUMMARY

Option is obsolete for this command.

DESCRIPTION

TCLCMD-1113

NAME

TCLCMD-1113

SUMMARY

Option is not supported any more, and will be removed in a future release.

DESCRIPTION

TCLCMD-1114

NAME

TCLCMD-1114

SUMMARY

Invalid field in list. Valid fields are :

DESCRIPTION

TCLCMD-1115

NAME

TCLCMD-1115

SUMMARY

No derating factor specified with command . Command will be ignored.

DESCRIPTION

TCLCMD-1116

NAME

TCLCMD-1116

SUMMARY

Support for multiple objects with option will be discontinued in a future release. This function still works in this release, but to avoid this warning and to ensure compatibility with future releases, specify a single object with this option.

DESCRIPTION

TCLCMD-1117

NAME

TCLCMD-1117

SUMMARY

The constraint is not allowed on output ports. Constraint is ignored on output port .

DESCRIPTION

TCLCMD-1118

NAME

TCLCMD-1118

SUMMARY

The constraint is not allowed on internal pins. In this release, the constraint is applied as as is an internal pin. To avoid this warning and to ensure compatibility with future releases, make sure this constraint is applied only on input or bidirectional ports.

DESCRIPTION

TCLCMD-1119

NAME

TCLCMD-1119

SUMMARY

The warning type is obsolete for command . This type of warnings are still issued when requested in this release. However, this will be phased out in a future release.

DESCRIPTION

TCLCMD-1120

NAME

TCLCMD-1120

SUMMARY

Mandatory option not specified for command .

DESCRIPTION

TCLCMD-1121

NAME

TCLCMD-1121

SUMMARY

Analysis view is not active.

DESCRIPTION

TCLCMD-1122

NAME

TCLCMD-1122

SUMMARY

The `update_constraint_mode` command will reset all interactive constraints for all views. Any interactive constraints entered before this command will be lost.

DESCRIPTION

TCLCMD-1123

NAME

TCLCMD-1123

SUMMARY

Ignoring clock mesh specific locv settings as .

DESCRIPTION

TCLCMD-1124

NAME

TCLCMD-1124

SUMMARY

Must specify one of , with command .

DESCRIPTION

TCLCMD-1125

NAME

TCLCMD-1125

SUMMARY

The software is currently configured with the timing global `timing_defer_mmmc_object_updates` set to true. In this mode, you are required to issue the `set_analysis_view` command with the `-update_timing` option after you are done with constraint updates to enable the new configuration.

DESCRIPTION

TCLCMD-1126

NAME

TCLCMD-1126

SUMMARY

object converted from to mode

DESCRIPTION

TCLCMD-1127

NAME

TCLCMD-1127

SUMMARY

The directory cannot be created, writing to the current directory.

DESCRIPTION

TCLCMD-1128

NAME

TCLCMD-1128

SUMMARY

The warning type requested through is not among the types specified with -type. This type of warnings will not be included in the check_timing report.

DESCRIPTION

TCLCMD-1129

NAME

TCLCMD-1129

SUMMARY

Design must be in memory before running .

DESCRIPTION

TCLCMD-1130

NAME

TCLCMD-1130

SUMMARY

The -late and -early options to the command can only be specified together when the timing system is in simultaneous setup and hold mode. You can use `set_global timing_enable_simultaneous_setup_hold_mode true` to enable this mode for timing analysis only. All non-timing commands are disabled while the system is in simultaneous setup/hold mode. Ignoring and using alone.

DESCRIPTION

TCLCMD-1132

NAME

TCLCMD-1132

SUMMARY

and cannot be specified together with command

DESCRIPTION

TCLCMD-1133

NAME

TCLCMD-1133

SUMMARY

No matching delay arc assertion found

DESCRIPTION

TCLCMD-1134

NAME

TCLCMD-1134

SUMMARY

The -name and -all options of the reset_clock_groups command are mutually exclusive options. You may only specify one of these options at a time.

DESCRIPTION

TCLCMD-1135

NAME

TCLCMD-1135

SUMMARY

The clock group named , specified by the -name option of the reset_clock_groups command, could not be found. You must specify a preexisting clock group name.

DESCRIPTION

TCLCMD-1136

NAME

TCLCMD-1136

SUMMARY

The -asynchronous, -logically_exclusive, and -physically_exclusive options of the reset_clock_groups command are mutually exclusive options. You may only specify one of these options at a time.

DESCRIPTION

TCLCMD-1137

NAME

TCLCMD-1137

SUMMARY

`` is not a view. checks can only be reported for a view.

DESCRIPTION

TCLCMD-1138

NAME

TCLCMD-1138

SUMMARY

Could not use option with

DESCRIPTION

TCLCMD-1139

NAME

TCLCMD-1139

SUMMARY

Clock mesh and local region locv libraries have different number of derating tables. Local region library will be used to decide the single/dual locv analysis.

DESCRIPTION

TCLCMD-1140

NAME

TCLCMD-1140

SUMMARY

The set_timing_derate -power_domain option should not be used along with object lists. The -power_domain option is ignored.

DESCRIPTION

TCLCMD-1141

NAME

TCLCMD-1141

SUMMARY

Locv analysis is not possible as .

DESCRIPTION

TCLCMD-1142

NAME

TCLCMD-1142

SUMMARY

Virtual clock is being created with no source objects.

DESCRIPTION

This message is issued when no object list is provided with `create_clock` command. In such case, virtual clock is created. A virtual clock is a clock that has no source object associated to it. To create a clock with specified source, the object list in `create_clock` has to be specified.

TCLCMD-1143

NAME

TCLCMD-1143

SUMMARY

Invalid clock latency delay value

DESCRIPTION

TCLCMD-1144

NAME

TCLCMD-1144

SUMMARY

The warning type is obsolete for command . Use instead.

DESCRIPTION

TCLCMD-1145

NAME

TCLCMD-1145

SUMMARY

Invalid scale factor {}. The RC scale factor - can only support up to 3 values.

DESCRIPTION

TCLCMD-1146

NAME

TCLCMD-1146

SUMMARY

not found.

DESCRIPTION

TCLCMD-1147

NAME

TCLCMD-1147

SUMMARY

Required argument -group was not found. Constraint ignored.

DESCRIPTION

TCLCMD-1148

NAME

TCLCMD-1148

SUMMARY

Invalid scale factor %f supplied to option .

DESCRIPTION

TCLCMD-1149

NAME

TCLCMD-1149

SUMMARY

command ignored implicit arguments in object_collection_or_list as type of the base collection is unknown

DESCRIPTION

TCLCMD-1150

NAME

TCLCMD-1150

SUMMARY

Obsolete option -generated. This is now turned on by default

DESCRIPTION

TCLCMD-1151

NAME

TCLCMD-1151

SUMMARY

Obsolete option -clocks. Clocks can now be specified even without giving this option.

DESCRIPTION

TCLCMD-1152

NAME

TCLCMD-1152

SUMMARY

Option is ignored as it is not supported with object for command

DESCRIPTION

TCLCMD-1153

NAME

TCLCMD-1153

SUMMARY

The -min and -max options of the set_timing_derate command are not supported when specifying timing check derating factors. The -early -cell_check derating will be applied to hold checks, and the -late -cell_check derating will be applied to setup checks. The derating factors will be applied - ignoring any -min or -max specification. If per delay corner derating of timing checks are required, you should utilize the multi-mode/multi-corner (MMMC) environment.

DESCRIPTION

TCLCMD-1154

NAME

TCLCMD-1154

SUMMARY

Different LOCV interface path derate offsets have been specified for Power Domains tied up with same Power Supply Net .

DESCRIPTION

TCLCMD-1155

NAME

TCLCMD-1155

SUMMARY

Command does not work correctly until timing graph is built.

DESCRIPTION

TCLCMD-1156

NAME

TCLCMD-1156

SUMMARY

Options in command are mutually exclusive and cannot be combined together. Specify only one of these options.

DESCRIPTION

TCLCMD-1157

NAME

TCLCMD-1157

SUMMARY

No derating factor specified with option. You must specify derating values when both -early and -late options are specified. Command ignored.

DESCRIPTION

TCLCMD-1158

NAME

TCLCMD-1158

SUMMARY

No arc between pin and pin.

DESCRIPTION

TCLCMD-1159

NAME

TCLCMD-1159

SUMMARY

No arc from pin.

DESCRIPTION

TCLCMD-1160

NAME

TCLCMD-1160

SUMMARY

No arc to pin.

DESCRIPTION

TCLCMD-1161

NAME

TCLCMD-1161

SUMMARY

No arc found on between pin and pin

DESCRIPTION

TCLCMD-1162

NAME

TCLCMD-1162

SUMMARY

Option -nosplit is obsolete and will be removed in a future release. Use command `set_table_style -no_frame_fix_width -nosplit` before calling this command to get nosplit style reports.

DESCRIPTION

TCLCMD-1163

NAME

TCLCMD-1163

SUMMARY

Invalid scale factor supplied to option . Only single scale factor value should be given.

DESCRIPTION

TCLCMD-1164

NAME

TCLCMD-1164

SUMMARY

Previously defined source objects for generated clock have been overwritten. To avoid overwrite, ensure that all source objects are passed to the `create_generated_clock` -source option in a single command call.

DESCRIPTION

TCLCMD-1165

NAME

TCLCMD-1165

SUMMARY

Must specify two threshold values with option `-voltage_threshold`.

DESCRIPTION

TCLCMD-1166

NAME

TCLCMD-1166

SUMMARY

Invalid value specified. It should be in the range (0-100).

DESCRIPTION

TCLCMD-1167

NAME

TCLCMD-1167

SUMMARY

lowerThreshold value must be less than or equal to upperThreshold.

DESCRIPTION

TCLCMD-1168

NAME

TCLCMD-1168

SUMMARY

cannot be applied on sub design . The constraint will be ignored.

DESCRIPTION

TCLCMD-1169

NAME

TCLCMD-1169

SUMMARY

Option cannot be specified with command . Command Ignored.

DESCRIPTION

TCLCMD-1170

NAME

TCLCMD-1170

SUMMARY

Invalid path description specified for command

DESCRIPTION

TCLCMD-1171

NAME

TCLCMD-1171

SUMMARY

The constraint is not allowed on input pins. Constraint is ignored on input pin .

DESCRIPTION

TCLCMD-1172

NAME

TCLCMD-1172

SUMMARY

Option for command is obsolete and has been replaced by . The command will be ignored.

DESCRIPTION

TCLCMD-1173

NAME

TCLCMD-1173

SUMMARY

Backward compatibility supported with the global will disappear in a future release. Update your scripts to .

DESCRIPTION

TCLCMD-1174

NAME

TCLCMD-1174

SUMMARY

command ignored collection in object_collection_or_list as its type differs from other collections in object_collection_or_list

DESCRIPTION

TCLCMD-1175

NAME

TCLCMD-1175

SUMMARY

Base collection for command is empty. At least one collection should be specified in object_collection_or_list

DESCRIPTION

TCLCMD-1176

NAME

TCLCMD-1176

SUMMARY

Hierarchical pin is not supported in as -reference_pin.

DESCRIPTION

TCLCMD-1177

NAME

TCLCMD-1177

SUMMARY

Multiple clocks specified with -clock option in command

DESCRIPTION

TCLCMD-1178

NAME

TCLCMD-1178

SUMMARY

set_output_delay command specified without -clock option. Command will be ignored for timing analysis.

DESCRIPTION

TCLCMD-1179

NAME

TCLCMD-1179

SUMMARY

Option -group_path with set_output_delay command cannot be specified in MMMC analysis mode. Command will be used by timing analysis as if no -group_path option was specified.

DESCRIPTION

TCLCMD-1180

NAME

TCLCMD-1180

SUMMARY

Clocks are not added to default group path in MMMC mode.

DESCRIPTION

TCLCMD-1181

NAME

TCLCMD-1181

SUMMARY

Option cannot be specified without .

DESCRIPTION

TCLCMD-1182

NAME

TCLCMD-1182

SUMMARY

Multiple pins are specified as -reference_pin with command . The tool has used first available pin . If this is not the intended pin, modify the constraint accordingly to use the correct pin.

DESCRIPTION

TCLCMD-1185

NAME

TCLCMD-1185

SUMMARY

LOCV library not specified.

DESCRIPTION

TCLCMD-1186

NAME

TCLCMD-1186

SUMMARY

Option should have one of values.

DESCRIPTION

TCLCMD-1187

NAME

TCLCMD-1187

SUMMARY

Option is ignored when specified with option in command .

DESCRIPTION

TCLCMD-1188

NAME

TCLCMD-1188

SUMMARY

Cannot specify without the option.

DESCRIPTION

TCLCMD-1189

NAME

TCLCMD-1189

SUMMARY

Incompatible options: and

DESCRIPTION

TCLCMD-1190

NAME

TCLCMD-1190

SUMMARY

Option cannot be specified with -clock, -data, -cell_check, -cell_delay, -early, -retain_delay, -net_delay

DESCRIPTION

TCLCMD-1191

NAME

TCLCMD-1191

SUMMARY

ObjectList cannot be specified with

DESCRIPTION

TCLCMD-1192

NAME

TCLCMD-1192

SUMMARY

Option -view is mandatory as view specific tables were read.

DESCRIPTION

TCLCMD-1193

NAME

TCLCMD-1193

SUMMARY

Cant specify net objects without using option -no_propagate. Ignoring all net objects in command set_ideal_network.

DESCRIPTION

TCLCMD-1194

NAME

TCLCMD-1194

SUMMARY

Command is obsolete and no longer supported.

DESCRIPTION

TCLCMD-1195

NAME

TCLCMD-1195

SUMMARY

Current instance is

DESCRIPTION

TCLCMD-1197

NAME

TCLCMD-1197

SUMMARY

Only one pin is allowed to be specified in command.

DESCRIPTION

TCLCMD-1198

NAME

TCLCMD-1198

SUMMARY

Specifying with objects is not supported. Ignoring objects in command .

DESCRIPTION

TCLCMD-1199

NAME

TCLCMD-1199

SUMMARY

Cannot specify with as direction does not match. Use option .

DESCRIPTION

TCLCMD-1200

NAME

TCLCMD-1200

SUMMARY

Global timing_path_based_recompute_use_incr_slew has been set true, but there is no effect of this global. In order to see the effect, global timing_path_based_recompute_noise must also be set to true.

DESCRIPTION

TCLCMD-1201

NAME

TCLCMD-1201

SUMMARY

Global timing_path_groups_for_clocks is not supported in MMMC analysis mode. Clocks created will not be added to default path group automatically.

DESCRIPTION

TCLCMD-1202

NAME

TCLCMD-1202

SUMMARY

Multiple object collection detected in command get_property. First object will be used, other objects will be ignored. Use command foreach_in_collection to iterate on each element of collection and use get_property on them.

DESCRIPTION

TCLCMD-1203

NAME

TCLCMD-1203

SUMMARY

The set_timing_derate -add and -multiply options should not be used together. This set_timing_derate is ignored.

DESCRIPTION

TCLCMD-1204

NAME

TCLCMD-1204

SUMMARY

The set_timing_derate -add or -multiply options should not be used along with -power_domain, -pulse_width_check or -clock_period. This set_timing_derate command is ignored.

DESCRIPTION

TCLCMD-1205

NAME

TCLCMD-1205

SUMMARY

Cannot find that match .

DESCRIPTION

TCLCMD-1206

NAME

TCLCMD-1206

SUMMARY

Option -qx_lib_file is obsolete and will be removed in a future release. This option is ignored in the current release, to avoid this warning and to ensure compatibility with future releases, update your script to remove it and use the default LEF flow to run QRC extraction.

DESCRIPTION

TCLCMD-1207

NAME

TCLCMD-1207

SUMMARY

Option -qx_conf_file is obsolete and will be removed in a future release. The QX config file is independent of the RC corner and should be specified with the setExtractRCMode -qrcCmdType partial -qrcCmdFile <cmdFile> with a CCL syntax based command file. This option is ignored in the current release , to avoid this warning and to ensure compatibility with future releases, update your

script to use setExtractRCMode -qrcCmdType partial -qrcCmdFile <cmdFile>.

DESCRIPTION

TCLCMD-1208

NAME

TCLCMD-1208

SUMMARY

AOCV analysis mode has been selected via setAnalysisMode -aocv. Since no AOCV libraries have been read, this analysis cannot be performed . The analysis will continue without AOCV based derating applied. AOCV libraries can be loaded via the .conf file or by including them in an MMMC library_set.

DESCRIPTION

TCLCMD-1209

NAME

TCLCMD-1209

SUMMARY

Can not set global timing_enable_aocv_analysis to true. The current analysisType is . AOCV can only run in OCV mode. OCV mode has to be set before setting timing_enable_aocv_analysis to true. The analysis will continue with the global timing_enable_aocv_analysis off without AOCV analysis.

DESCRIPTION

TCLCMD-1210

NAME

TCLCMD-1210

SUMMARY

The value for preset distance in WID variation for parameter is too small. Unrealistically small value of distance may result in large usage of memory. The value of %f is being used in the current run.

DESCRIPTION

TCLCMD-1211

NAME

TCLCMD-1211

SUMMARY

Property is not defined on .

DESCRIPTION

TCLCMD-1212

NAME

TCLCMD-1212

SUMMARY

Options -from/-to is not supported with library arc collection.

DESCRIPTION

TCLCMD-1213

NAME

TCLCMD-1213

SUMMARY

Option is obsolete. Use global instead.

DESCRIPTION

TCLCMD-1214

NAME

TCLCMD-1214

SUMMARY

The global can only be used in SSTA mode. Please use setAnalysisMode -timingEngine statistical before setting the global. The timing analysis will continue with the global off.

DESCRIPTION

TCLCMD-1215

NAME

TCLCMD-1215

SUMMARY

The option is unsupported

DESCRIPTION

TCLCMD-1216

NAME

TCLCMD-1216

SUMMARY

The set_timing_derate -add and -multiply options should be used either without objects or leaf instance objects only. This set_timing_derate is ignored.

DESCRIPTION

TCLCMD-1218

NAME

TCLCMD-1218

SUMMARY

Negative derating factor can be specified with -add or incremental_adjust option only for command set_timing_derate. Command ignored.

DESCRIPTION

TCLCMD-1219

NAME

TCLCMD-1219

SUMMARY

Cannot find clock that matches . set_clock_exclusivity constraint is ignored.

DESCRIPTION

TCLCMD-1220

NAME

TCLCMD-1220

SUMMARY

Option -retime is ignored when specified with check_type pulse_width, clock_period, skew, pulse_clock_max_width or pulse_clock_min_width

DESCRIPTION

TCLCMD-1225

NAME

TCLCMD-1225

SUMMARY

Cell derate property names excluding rise/fall edges are obsolete and have been replaced by rise/fall cell derate properties such as late_rise_data_cell_derate_factor, late_fall_data_cell_derate_factor etc. The obsolete properties still work in this release, but to avoid this warning and to ensure compatibility with future releases, please update your scripts.

DESCRIPTION

TCLCMD-1226

NAME

TCLCMD-1226

SUMMARY

Attribute is not supported on generated clocks.

DESCRIPTION

TCLCMD-1227

NAME

TCLCMD-1227

SUMMARY

Options -corner and -statistical are exclusive, do not specify both at the same time.

DESCRIPTION

TCLCMD-1228

NAME

TCLCMD-1228

SUMMARY

The option -variation can only be used in SSTA mode. The timing analysis will continue with the option -scalar.

DESCRIPTION

TCLCMD-1229

NAME

TCLCMD-1229

SUMMARY

Option -comment is not supported by the tool. The option will be parsed but will not be saved and so subsequent constraint writing will not dump them.

DESCRIPTION

TCLCMD-1230

NAME

TCLCMD-1230

SUMMARY

The command set_analysis_view is called before the design is initialized and not from init_design.

DESCRIPTION

TCLCMD-1231

NAME

TCLCMD-1231

SUMMARY

You must specify global interface path derate offset for power domain before applying derate offset on specific cell(s).

DESCRIPTION

TCLCMD-1232

NAME

TCLCMD-1232

SUMMARY

is not supported in budget slack stealing mode. Please use max-borrow slack stealing mode using `set_global latch_time_borrow_mode max_borrow`.

DESCRIPTION

Model extraction traverse across the latches in borrow mode to find a non-borrowing latch, flop or output port. This is a max-borrow slack stealing methodology. So model extraction can not be matched with the budget stealing mode. Please use `set_global latch_time_borrow_mode max_borrow` and then extract the model.

TCLCMD-1233

NAME

TCLCMD-1233

SUMMARY

The -view option should be specified for the command write_timing_windows in MMMC mode. For now The tool will assume the view specified as and write the timing windows corresponding to this view. For compatibility with future releases, update your scripts to provide -view option.

DESCRIPTION

TCLCMD-1234

NAME

TCLCMD-1234

SUMMARY

In multi-mode, multi-corner analysis mode, you should include the -view option with the read_sdf command to annotate the SDF file to specific view(s). Without the -view option, the SDF file will only be annotated to the current default view.

DESCRIPTION

TCLCMD-1236

NAME

TCLCMD-1236

SUMMARY

AOCV mode for timing analysis is set to false, but AOCV mode for timing model extraction is set to true. If you want to do AOCV analysis during timing model extraction then you need to also turn on AOCV mode in timing analysis. Otherwise you can turn off AOCV analysis for timing model extraction by setting global timing_extract_model_aocv_mode to none.

DESCRIPTION

TCLCMD-1237

NAME

TCLCMD-1237

SUMMARY

AOCV analysis could not be done as either AOCV mode in timing analysis is not turned on or AOCV libraries are not read. To proceed further, please load AOCV library if it is not read and set the analysis mode to aocv.

DESCRIPTION

TCLCMD-1238

NAME

TCLCMD-1238

SUMMARY

model extractor only supports the extraction of models in dotlib format

DESCRIPTION

TCLCMD-1239

NAME

TCLCMD-1239

SUMMARY

The software has been initialized without timing library information into a physical-only mode of operation. Access to timing analysis related commands and functionality is not available in this mode. To operate the software with full functionality you must exit or issue the freeDesign command - then reinitialize the system with a design configuration that includes a valid timing library configuration.

DESCRIPTION

TCLCMD-1240

NAME

TCLCMD-1240

SUMMARY

INFO: Path Based Analysis (PBA) performed on total paths

DESCRIPTION

TCLCMD-1241

NAME

TCLCMD-1241

SUMMARY

Could not find for

DESCRIPTION

TCLCMD-1242

NAME

TCLCMD-1242

SUMMARY

Option cannot be specified with command . Option Ignored.

DESCRIPTION

TCLCMD-1243

NAME

TCLCMD-1243

SUMMARY

AOCV mode for timing analysis is set to false, but -include_aocv_weights has been used in model extraction. If you want to include aocv_weight in extracted model then you need to also turn on AOCV mode in timing analysis using setAnalysisMode -aocv option.

DESCRIPTION

TCLCMD-1244

NAME

TCLCMD-1244

SUMMARY

AOCV analysis mode has been selected via setAnalysisMode -aocv. Since no AOCV libraries have been read, aocv_weight cannot be modeled in extracted model. The model extraction will continue ignoring the -include_aocv_weights option.

DESCRIPTION

TCLCMD-1245

NAME

TCLCMD-1245

SUMMARY

is not hierarchical instance. accepts only hierarchical instance

DESCRIPTION

TCLCMD-1246

NAME

TCLCMD-1246

SUMMARY

Pin does not belong to specified hierarchical instance .

DESCRIPTION

TCLCMD-1247

NAME

TCLCMD-1247

SUMMARY

Option is being ignored since it has been specified without option.

DESCRIPTION

TCLCMD-1248

NAME

TCLCMD-1248

SUMMARY

Since no -max_slack has been specified with exhaustive -retime_mode, tool will assume its default value in this case which is 0

DESCRIPTION

TCLCMD-1250

NAME

TCLCMD-1250

SUMMARY

The number of paths analyzed are equal to specified max_paths limit for some of the clock groups. Refer to verbose log file of current session for more details.

DESCRIPTION

TCLCMD-1252

NAME

TCLCMD-1252

SUMMARY

-early_irdrop_file and -late_irdrop_file options are now supported in set_instance_voltage command.

DESCRIPTION

TCLCMD-1253

NAME

TCLCMD-1253

SUMMARY

-path option with leaf instance is not supported in write_sdf command.

DESCRIPTION

TCLCMD-1254

NAME

TCLCMD-1254

SUMMARY

The master clock not found for the given clock:

DESCRIPTION

TCLCMD-1255

NAME

TCLCMD-1255

SUMMARY

Only hierarchical instances, ports or top module can be specified. Specified value(s) are not valid.

DESCRIPTION

TCLCMD-1256

NAME

TCLCMD-1256

SUMMARY

Option is ignored since NPass Path Based Analysis (PBA) flow is not enabled. To enable the NPass Path Based Analysis (PBA) flow, use set timing_path_based_enable_multi_pass true.

DESCRIPTION

TCLCMD-1257

NAME

TCLCMD-1257

SUMMARY

INFO: N-Pass Path Based Analysis (PBA) filtered paths after first AOCV based pass

DESCRIPTION

TCLCMD-1258

NAME

TCLCMD-1258

SUMMARY

INFO: Efficient Path Based Analysis (PBA): Actual Calls: , Total Calls:

DESCRIPTION

TCLCMD-1259

NAME

TCLCMD-1259

SUMMARY

INFO: Path Based Analysis (PBA) Stage Based Filtering: countAOCVFilter: , countPathSlewFilter:

DESCRIPTION

TCLCMD-1260

NAME

TCLCMD-1260

SUMMARY

The write_sdf -map_file option as been specified without a required filename argument. The SDF file will be written out without any mapping being performed.

DESCRIPTION

TCLCMD-1261

NAME

TCLCMD-1261

SUMMARY

The software has detected a master/generated clock topology where the propagated state of the clocks is not uniform. is in propagated mode while its master is in ideal mode. Since the `timing_generated_clocks_inherit_ideal_latency` is set to true, the generated clock will inherit latency of its master clock.

DESCRIPTION

TCLCMD-1262

NAME

TCLCMD-1262

SUMMARY

INFO: Efficient WP Path Based Analysis (PBA): Waveform Created : , Waveform Deleted :

DESCRIPTION

TCLCMD-1263

NAME

TCLCMD-1263

SUMMARY

The `-frame` parameter is obsolete and will be removed from the software in a future release. To ensure compatibility with future releases, remove this parameter from your scripts.

DESCRIPTION

TCLCMD-1264

NAME

TCLCMD-1264

SUMMARY

The option is not supported with specified mode.

DESCRIPTION

TCLCMD-1265

NAME

TCLCMD-1265

SUMMARY

Setting zero category on path group is not allowed. Constraint is ignored!

DESCRIPTION

TCLCMD-1266

NAME

TCLCMD-1266

SUMMARY

Setting category on clock based path group is not allowed. Constraint is ignored!

DESCRIPTION

TCLCMD-1267

NAME

TCLCMD-1267

SUMMARY

INFO: The software has found an intermediate begin point between the specified -from and -to points to the report_timing command. To report paths between these points accurately, you should use the report_timing -view option to analyze one view at a time

DESCRIPTION

TCLCMD-1268

NAME

TCLCMD-1268

SUMMARY

Unsupported extra argument -of_objects in command get_clocks.

DESCRIPTION

TCLCMD-1269

NAME

TCLCMD-1269

SUMMARY

merge_timing_reports command can be used to merge reports that are generated with -nosplit and -no_frame parameters of the set_table_style command. The report is not generated with these parameters and will be ignored.

DESCRIPTION

TCLCMD-1270

NAME

TCLCMD-1270

SUMMARY

Another clock group has already been defined with an identical specification. This assertion will be ignored.

DESCRIPTION

TCLCMD-1271

NAME

TCLCMD-1271

SUMMARY

-retime waveform_propagation/aocv_waveform_propagation is not supported with AAE as delay calculation engine

DESCRIPTION

TCLCMD-1272

NAME

TCLCMD-1272

SUMMARY

The argument specified for the constraint is not valid for analysis view in the current context of design. If these arguments are not established as valid by future assertions, the constraint will be ignored.

DESCRIPTION

The path exceptions set_false_path and set_multicycle_path require the arguments to the -from/-to options to be valid beginpoint/endpoint respectively. Valid beginpoints/endpoints can be determined structurally from the netlist and timing library - Examples of beginpoints are input ports and clock input of sequential elements. Similarly examples of endpoints are output ports and data

input of sequential elements. Other valid endpoints can be inferred in the design by the application of constraints. Examples of non-structural endpoints are the enable input of inferred clock gate and the segmentation points of path delay constraints. This warning is issued if at the time the path exception is processed, the specified -from/-to argument is not recognized as valid beginpoint/endpoint given the softwares current understanding of the design based on the netlist, timing library information, and previously specified constraints. This path exception constraint will be ignored unless future constraints establish the specified beginpoints/endpoints as valid. You can refer to the command reference documentation for `set_false_path` and `set_multicycle_path` for the complete list of valid -from and -to endpoints.

TCLCMD-1273

NAME

TCLCMD-1273

SUMMARY

Noise parameter `enable_pba_detailed_reports` is set. Enable group-based reporting by setting `set_global_timing_report_group_based_mode` true to generate delta delay reports during path-based analysis.

DESCRIPTION

TCLCMD-1274

NAME

TCLCMD-1274

SUMMARY

Noise parameter `enable_pba_detailed_reports` is set. Delta delay report for path-based analysis is supported only when `-max_paths` is set to 1. Modify the command to generate the reports.

DESCRIPTION

TCLCMD-1275

NAME

TCLCMD-1275

SUMMARY

The number of views used in `set_analysis_view` command has exceed the maximum limit of 256 views. Reduce to a subset of views that are most important for analysis and specify the `set_analysis_view` command again.

DESCRIPTION

TCLCMD-1277

NAME

TCLCMD-1277

SUMMARY

The `-si` option only accepts plain list of files format such as `[list file1 file2 file3 ...]` instead of `[list [list file1 file2 file3 ...]]`

DESCRIPTION

TCLCMD-1278

NAME

TCLCMD-1278

SUMMARY

During path based analysis found a path with common pin outside the Path. The endpoint of this path is and beginpoint is . Launch Clock is and capturing clock is .

DESCRIPTION

TCLCMD-1279

NAME

TCLCMD-1279

SUMMARY

The support for Combined Launch-Capture AOCV stage count mode is limited to path based analysis only (with -retime option). For Graph based analysis the tool will use the default launch capture mode (launch_capture).

DESCRIPTION

TCLCMD-1280

NAME

TCLCMD-1280

SUMMARY

The cell/pin is a single input pin cell, hence the set_disable_clock_gating_check is being ignored.

DESCRIPTION

TCLCMD-1281

NAME

TCLCMD-1281

SUMMARY

The pin is a output pin, hence the set_disable_clock_gating_check is being ignored.

DESCRIPTION

TCLCMD-1282

NAME

TCLCMD-1282

SUMMARY

Slack %f reported for pin at line was skipped due to more critical slack report being present. Specify option -nworst value higher than to include same in merged output

DESCRIPTION

TCLCMD-1283

NAME

TCLCMD-1283

SUMMARY

Report at line was skipped due to issue in fetching relevant pin name.

DESCRIPTION

TCLCMD-1284

NAME

TCLCMD-1284

SUMMARY

Switching to Single CPU.

DESCRIPTION

TCLCMD-1285

NAME

TCLCMD-1285

SUMMARY

Valus could not run properly

DESCRIPTION

TCLCMD-1286

NAME

TCLCMD-1286

SUMMARY

The generated ETM could not be checked with Valus. Valus not found in the path.

DESCRIPTION

TCLCMD-1287

NAME

TCLCMD-1287

SUMMARY

The write_sdf option has been specified without a required -exclude_cells option. The SDF file will be written out without any excluding cell performed.

DESCRIPTION

TCLCMD-1288

NAME

TCLCMD-1288

SUMMARY

CSV file name missing

DESCRIPTION

TCLCMD-1289

NAME

TCLCMD-1289

SUMMARY

The set_pll_timing constraint specified for target pin and clock will override the already asserted constraint for same clock.

DESCRIPTION

TCLCMD-1290

NAME

TCLCMD-1290

SUMMARY

The set_pll_timing constraint can not be specified for multiple pins (or clocks) simultaneously.

DESCRIPTION

TCLCMD-1291

NAME

TCLCMD-1291

SUMMARY

The given set_pll_timing could not be applied successfully on given pin as there is no generated clock applied on it.

DESCRIPTION

TCLCMD-1292

NAME

TCLCMD-1292

SUMMARY

The PLL defined on pin has reference pin specified as , but the corresponding generated clock definition has source

DESCRIPTION

TCLCMD-1293

NAME

TCLCMD-1293

SUMMARY

Property is not supported for . Command report_property ignored

DESCRIPTION

TCLCMD-1294

NAME

TCLCMD-1294

SUMMARY

Setting default super category is not allowed. Constraint is ignored!

DESCRIPTION

TCLCMD-1295

NAME

TCLCMD-1295

SUMMARY

Assigning existing path group category ID to super category is not allowed. Constraint is ignored!

DESCRIPTION

TCLCMD-1296

NAME

TCLCMD-1296

SUMMARY

Path group category is not defined. Constraint is ignored!

DESCRIPTION

TCLCMD-1297

NAME

TCLCMD-1297

SUMMARY

Path group super category is not defined. Constraint is ignored!

DESCRIPTION

TCLCMD-1298

NAME

TCLCMD-1298

SUMMARY

The existing has been replaced with new attributes.

DESCRIPTION

TCLCMD-1299

NAME

TCLCMD-1299

SUMMARY

The option is replaced with in delay corner

DESCRIPTION

TCLCMD-1300

NAME

TCLCMD-1300

SUMMARY

INFO: Path Based Analysis (PBA) found paths with Reconvergent clock network

DESCRIPTION

TCLCMD-1301

NAME

TCLCMD-1301

SUMMARY

The path adjust group named , specified by the -path_adjust_group option of the set_path_adjust command, could not be found. You must specify a preexisting path adjust group name.

DESCRIPTION

TCLCMD-1302

NAME

TCLCMD-1302

SUMMARY

The `update_io_latency` command cannot be run when the design is not in ideal mode. Check the setting: `#CUI#setAnalysisMode -clockPropagation sdcControl#set_db timing_analysis_clock_propagation_mode sdc_control#`

DESCRIPTION

TCLCMD-1303

NAME

TCLCMD-1303

SUMMARY

The `update_io_latency` command cannot be run when a clock is propagated. Check if there is any `set_propagated_clock` constraint on clock.

DESCRIPTION

TCLCMD-1304

NAME

TCLCMD-1304

SUMMARY

The `set_latch_loop_disable` command is skipped as specified input pin to output pin is not a latch flush arc of instance .

DESCRIPTION

TCLCMD-1305

NAME

TCLCMD-1305

SUMMARY

A default timing condition is needed for delay corner .

DESCRIPTION

TCLCMD-1306

NAME

TCLCMD-1306

SUMMARY

read_mmmc command should be called once before init_design.

DESCRIPTION

TCLCMD-1307

NAME

TCLCMD-1307

SUMMARY

Option is not supported with -by_pin/-by_net/-by_driver

DESCRIPTION

TCLCMD-1308

NAME

TCLCMD-1308

SUMMARY

write_sdf options -delta_delay and -base_delay were specified but they are mutually exclusive. They will be ignored.

DESCRIPTION

TCLCMD-1309

NAME

TCLCMD-1309

SUMMARY

option cannot be used without validation flow. Please specify -validate <dir name > to switch ON the validation flow

DESCRIPTION

TCLCMD-1310

NAME

TCLCMD-1310

SUMMARY

-validate option has to be used with user defined validation directory. Correct usage : -validate valDir

DESCRIPTION

TCLCMD-1311

NAME

TCLCMD-1311

SUMMARY

Timing reset occurred while multithreaded reporting commands (-from/-through/-to) mode was on. The reports are not reliable. Avoid commands which can reset timing under this mode.

DESCRIPTION

TCLCMD-1312

NAME

TCLCMD-1312

SUMMARY

Clock is defined in view but not defined in view . Check constraint specification or library generated clock specification for this clock.

DESCRIPTION

TCLCMD-1313

NAME

TCLCMD-1313

SUMMARY

Unsupported extra argument in command .Use command help for proper usage information.

DESCRIPTION

The related command does not recognize the extra argument. Doing a help on the command name will give you the proper syntax of the command.

TCLCMD-1314

NAME

TCLCMD-1314

SUMMARY

Missing required argument in command .Use command help for proper usage information.

DESCRIPTION

TCLCMD-1315

NAME

TCLCMD-1315

SUMMARY

The constraint, when used with the -combinational_from_to option, requires both -from and -to arguments to be provided. The constraint has been ignored.

DESCRIPTION

TCLCMD-1316

NAME

TCLCMD-1316

SUMMARY

write_sdf options -abstracted_model and -adjust_setuphold_for_zero_hold_slack were specified but they are mutually exclusive. -adjust_setuphold_for_zero_hold_slack option is ignored

DESCRIPTION

TCLCMD-1317

NAME

TCLCMD-1317

SUMMARY

The view is either inactive or undefined. set_path_adjust constraint will be ignored.

DESCRIPTION

TCLCMD-1318

NAME

TCLCMD-1318

SUMMARY

Cannot find that matches the defined delay corner . Check the MMMC configuration to see if it was created properly. You can also use the all_delay_corners command to report the set of currently defined corners.

DESCRIPTION

TCLCMD-1319

NAME

TCLCMD-1319

SUMMARY

The option is not currently available. Use -help to see the supported options.

DESCRIPTION

TCLCMD-1320

NAME

TCLCMD-1320

SUMMARY

The file provided to the read_mmmc command did not contain enough data to successfully initialize the system. The minimum requirement for this file is that it fully creates one analysis view and its sub-components successfully - and then activates that view. You should check the log file for issues related creation of the MMMC objects or the enabling of the active views via set_analysis_view.

DESCRIPTION

TCLCMD-1321

NAME

TCLCMD-1321

SUMMARY

The multi_drive_preferred_driver property is a user-defined property which behaves globally - affecting all analysis views. This is not allowed to be set in the constraint file. The existing definition has been ignored. Commands define_property, set_property, and reset_property for defining the multi_drive_preferred_driver property should be done in command script directly.

DESCRIPTION

TCLCMD-1322

NAME

TCLCMD-1322

SUMMARY

The constraint, when used with the -combinational_from_to option, requires both -from and -to arguments of the same type to be provided. The constraint has been ignored.

DESCRIPTION

TCLCMD-1323

NAME

TCLCMD-1323

SUMMARY

The value provided to the `set_clock_latency` is required to be positive. The provided value `%.4f` does not meet this requirement, so the constraint has been ignored.

DESCRIPTION

TCLCMD-1324

NAME

TCLCMD-1324

SUMMARY

The `set_clock_latency` constraint requires that the value provided for be less than the value specified for the total source latency. The jitter value of `%.4f` is greater than the total source latency value of `%.4f`. This constraint has been ignored.

DESCRIPTION

TCLCMD-1325

NAME

TCLCMD-1325

SUMMARY

Archiving variable

DESCRIPTION

TCLCMD-1326

NAME

TCLCMD-1326

SUMMARY

Restoring variable

DESCRIPTION

TCLCMD-1327

NAME

TCLCMD-1327

SUMMARY

Variable has collection invalidated and will not be saved. Please ensure that no timing update occurs post save_design for the variable to be archived successfully.

DESCRIPTION

TCLCMD-1328

NAME

TCLCMD-1328

SUMMARY

Variable has collection invalidated and will not be restored. Please ensure that no timing update occurs post save_design for the variable to be archived successfully.

DESCRIPTION

TCLCMD-1329

NAME

TCLCMD-1329

SUMMARY

Archiving path collection variables

DESCRIPTION

TCLCMD-1330

NAME

TCLCMD-1330

SUMMARY

Restoring path collection variables

DESCRIPTION

TCLCMD-1331

NAME

TCLCMD-1331

SUMMARY

End of Archiving path collection variables

DESCRIPTION

TCLCMD-1332

NAME

TCLCMD-1332

SUMMARY

End of Restoration of path collection variables

DESCRIPTION

TCLCMD-1333

NAME

TCLCMD-1333

SUMMARY

The combinational generated clock was specified without any of -divide_by, -multiply_by, or -edges options which are normally used to define the output waveform. A -divide_by 1 specification has been inferred for this clock.

DESCRIPTION

TCLCMD-1334

NAME

TCLCMD-1334

SUMMARY

Invalid field in list.

DESCRIPTION

TCLCMD-1335

NAME

TCLCMD-1335

SUMMARY

INFO: Processing level: with total pins, with uncovered (yellow) pins and green pins in level

DESCRIPTION

TCLCMD-1336

NAME

TCLCMD-1336

SUMMARY

INFO: Completed processing level: with pins covered by paths. pins are still uncovered (yellow) and pins are green

DESCRIPTION

TCLCMD-1337

NAME

TCLCMD-1337

SUMMARY

INFO: The number of pins:

DESCRIPTION

TCLCMD-1338

NAME

TCLCMD-1338

SUMMARY

INFO: GBA violating pins are fully covered by violating paths

DESCRIPTION

TCLCMD-1339

NAME

TCLCMD-1339

SUMMARY

INFO: The specified max slack for violation: %f

DESCRIPTION

TCLCMD-1340

NAME

TCLCMD-1340

SUMMARY

INFO: Comprehensive PBA analysis on pins with depth has with the following results:

DESCRIPTION

TCLCMD-1341

NAME

TCLCMD-1341

SUMMARY

INFO: The number of levels:

DESCRIPTION

TCLCMD-1342

NAME

TCLCMD-1342

SUMMARY

INFO: Processing uncovered nodes through step

DESCRIPTION

TCLCMD-1343

NAME

TCLCMD-1343

SUMMARY

INFO: step

DESCRIPTION

TCLCMD-1344

NAME

TCLCMD-1344

SUMMARY

INFO: Total pins tried for level id. Filtered pins.

DESCRIPTION

TCLCMD-1345

NAME

TCLCMD-1345

SUMMARY

INFO: Collecting the violating pins in the design

DESCRIPTION

TCLCMD-1346

NAME

TCLCMD-1346

SUMMARY

INFO: Collected all the violating pins in the design

DESCRIPTION

TCLCMD-1348

NAME

TCLCMD-1348

SUMMARY

The software does not currently allow dual-ported objects - such as those returned by `get_db`, to be used as arguments to timing constraint commands. You must provide these commands with either a collection or a Tcl list of name strings for the objects.

DESCRIPTION

TCLCMD-1349

NAME

TCLCMD-1349

SUMMARY

matched with given pattern, none qualified with given filter.

DESCRIPTION

TCLCMD-1350

NAME

TCLCMD-1350

SUMMARY

No matched with given pattern .

DESCRIPTION

TCLCMD-1351

NAME

TCLCMD-1351

SUMMARY

INFO: GBA violating pins have exhausted all GBA paths without PBA violations and are guaranteed to meet the slack criteria

DESCRIPTION

TCLCMD-1352

NAME

TCLCMD-1352

SUMMARY

INFO: Out of total path queries, paths filtered, paths added and finally total entries in path cache.

DESCRIPTION

TCLCMD-1353

NAME

TCLCMD-1353

SUMMARY

Innovus currently does not support -validate parameter of the model extraction.

DESCRIPTION

TCLCMD-1354

NAME

TCLCMD-1354

SUMMARY

Property is not supported. Command ignored. Supported properties are { }

DESCRIPTION

TCLCMD-1355

NAME

TCLCMD-1355

SUMMARY

Auto validation is ON. In auto-validation mode, the value of global `timing_extract_model_exhaustive_validation_mode = true` is being ignored since both auto-validation and exhaustive-validation cannot run simultaneously.

DESCRIPTION

TCLCMD-1356

NAME

TCLCMD-1356

SUMMARY

The specification of is not currently recognized by the software as a valid for . The constraint has been ignored.

DESCRIPTION

TCLCMD-1357

NAME

TCLCMD-1357

SUMMARY

Invalid variation factor value %f. It should be in the range 0.0 to 1.0

DESCRIPTION

TCLCMD-1358

NAME

TCLCMD-1358

SUMMARY

INFO: The total number of gba enumerated paths:

DESCRIPTION

TCLCMD-1359

NAME

TCLCMD-1359

SUMMARY

INFO: The total number of excluded paths after enumeration:

DESCRIPTION

TCLCMD-1360

NAME

TCLCMD-1360

SUMMARY

INFO: The number of useful paths for processing after gba enumeration:

DESCRIPTION

TCLCMD-1361

NAME

TCLCMD-1361

SUMMARY

INFO: The number of pins on which through path is attempted in step : , on this level: pins. Total skip pins , on this level :

DESCRIPTION

TCLCMD-1362

NAME

TCLCMD-1362

SUMMARY

The exhaustive retime mode is incompatible with path collection.

DESCRIPTION

TCLCMD-1364

NAME

TCLCMD-1364

SUMMARY

Invalid value %f of sigma multiplier. Value specified should be ≥ 0

DESCRIPTION

TCLCMD-1365

NAME

TCLCMD-1365

SUMMARY

INFO: The number of final non-violating(passing) or retime-filtered paths:

DESCRIPTION

TCLCMD-1366

NAME

TCLCMD-1366

SUMMARY

INFO: The number of rejected paths due to unsuccessful final marking:

DESCRIPTION

TCLCMD-1367

NAME

TCLCMD-1367

SUMMARY

INFO: The number of paths sent for final processing:

DESCRIPTION

TCLCMD-1368

NAME

TCLCMD-1368

SUMMARY

INFO: Path Based Analysis (PBA) performed on (, ,) paths during Path Based Coverage mode.

DESCRIPTION

TCLCMD-1369

NAME

TCLCMD-1369

SUMMARY

command is unsupported when specified inside sdc file.

DESCRIPTION

TCLCMD-1370

NAME

TCLCMD-1370

SUMMARY

INFO: GBA violating pins reached PBA path depth limit without violating. The PBA violation may be present in greater depth

DESCRIPTION

TCLCMD-1371

NAME

TCLCMD-1371

SUMMARY

Paths that are not inter-power domain related have been filtered at the time of reporting. Due to this filtering, number of paths reported may not correspond to -max_paths/-nworst value specified.

DESCRIPTION

TCLCMD-1372

NAME

TCLCMD-1372

SUMMARY

Inter-power-domain analysis can not be configured due to insufficient power domains. Correct power domains specification.

DESCRIPTION

TCLCMD-1373

NAME

TCLCMD-1373

SUMMARY

Constraints not defined before specifying configure_ipd_mode. It is recommended to read constraints before specifying configure_ipd_mode for accurate timing analysis.

DESCRIPTION

TCLCMD-1374

NAME

TCLCMD-1374

SUMMARY

INFO: The number of pins which are not fully covered but are covered by non-worst violating path :

DESCRIPTION

TCLCMD-1375

NAME

TCLCMD-1375

SUMMARY

Type is not supported for option. Constraint has been ignored

DESCRIPTION

TCLCMD-1376

NAME

TCLCMD-1376

SUMMARY

set_clock_sense -stop option default_data has been specified along with the -clock filter option. This condition can never be satisfied so default_data phases will not be stopped. To block default_data phases, you should not include a -clock specification.

DESCRIPTION

TCLCMD-1377

NAME

TCLCMD-1377

SUMMARY

The max paths limit for PBA paths has been exhausted and tool will fallback to GBA slacks for incomplete or leftover endpoints. To identify the paths for which tool has fallback on GBA slacks, please look for the following marker in final path report : Retime Analysis {GBA}

DESCRIPTION

TCLCMD-1378

NAME

TCLCMD-1378

SUMMARY

-merged_views option is only supported with -merged_groups option in report_analysis_summary command.

DESCRIPTION

TCLCMD-1379

NAME

TCLCMD-1379

SUMMARY

do_extract_model is invoked with option -include_power. However, the timing libraries were loaded without reading the power information. Therefore, power analysis cannot be performed. To enable power analysis, set the global variable set timing_library_read_without_power false and re-load timing libraries.

DESCRIPTION

TCLCMD-1380

NAME

TCLCMD-1380

SUMMARY

Command is valid only for Tempus.

DESCRIPTION

TCLCMD-1381

NAME

TCLCMD-1381

SUMMARY

-out_dir_cmmmc option has to be used with user defined directory where files related to concurrent ETM generation will be written. Correct usage : -out_dir_cmmmc <directory_name>

DESCRIPTION

TCLCMD-1382

NAME

TCLCMD-1382

SUMMARY

Info: Resetting Path Peeling properties on pins/ports.

DESCRIPTION

TCLCMD-1383

NAME

TCLCMD-1383

SUMMARY

INFO (INTERRUPT): User specified time limit exceeded for current command. The command will stop at next legal exit point.

DESCRIPTION

TCLCMD-1384

NAME

TCLCMD-1384

SUMMARY

INFO: Incremental depth mode. Setting effective depth:

DESCRIPTION

TCLCMD-1385

NAME

TCLCMD-1385

SUMMARY

The `set_timing_derate` `-mean` and `-sigma` options are only supported at the global or power domain levels for net objects. Mean or sigma derating for cell delays or cell checks should be done at the `library_cell` or instance-level. To avoid this warning, you can add the `-net_delay` option to clarify the intent.

DESCRIPTION

TCLCMD-1386

NAME

TCLCMD-1386

SUMMARY

INFO: paths filtered due to existence of higher node slack on the timing paths.

DESCRIPTION

TCLCMD-1387

NAME

TCLCMD-1387

SUMMARY

Invalid object found in the input

DESCRIPTION

TCLCMD-1388

NAME

TCLCMD-1388

SUMMARY

The command requires a valid directory name to be specified. You should review the command specification to ensure the output directory is properly specified and rerun the command. No timing context data has been .

DESCRIPTION

TCLCMD-1389

NAME

TCLCMD-1389

SUMMARY

The directory specified for the command cannot be due to the directory permissions. You should specify a different directory or modify the directory permissions and reissue the command. No timing context data has been .

DESCRIPTION

TCLCMD-1390

NAME

TCLCMD-1390

SUMMARY

The output directory specified for the `write_timing_context` command already exists. You should specify a different directory or reissue the command with the `-overwrite` option to overwrite the existing directory. No timing context data has been written.

DESCRIPTION

TCLCMD-1391

NAME

TCLCMD-1391

SUMMARY

Module specified for the `write_timing_context` contains no instances. You should specify a different module. Timing context cannot be written.

DESCRIPTION

TCLCMD-1392

NAME

TCLCMD-1392

SUMMARY

The current version of write_timing_context supports only instance per module. You should specify only one instance with the option -instances from the selected module. Timing context cannot be written.

DESCRIPTION

TCLCMD-1393

NAME

TCLCMD-1393

SUMMARY

The directory specified for the read_timing_context command does not exist. You should specify a different directory and reissue the command. No timing context data read.

DESCRIPTION

TCLCMD-1394

NAME

TCLCMD-1394

SUMMARY

The `write_timing_context` commands `-instances` option requires a valid list or collection of instance names to be provided. The specified instance is not a valid instance of module . You should review the design specification and reissue the command. No timing context data has been written.

DESCRIPTION

TCLCMD-1395

NAME

TCLCMD-1395

SUMMARY

Argument is required for command , either this option is not specified or an option prior to it is not specified correctly. No timing context data has been written.

DESCRIPTION

TCLCMD-1396

NAME

TCLCMD-1396

SUMMARY

The commands option requires a valid argument name to be provided. The specified name could not be found in the design. You should reissue the command with a valid argument. No timing context data has been written.

DESCRIPTION

TCLCMD-1397

NAME

TCLCMD-1397

SUMMARY

The write_timing_context commands -instances option requires a valid list or collection of instance names to be provided. You should review the command specification to ensure an instance list or collection is properly specified and rerun the command. No timing context data has been written.

DESCRIPTION

TCLCMD-1398

NAME

TCLCMD-1398

SUMMARY

Specify only one view as view. view supports only single view.

DESCRIPTION

TCLCMD-1399

NAME

TCLCMD-1399

SUMMARY

: Multiple cells for the pattern matches in library

DESCRIPTION

TCLCMD-1400

NAME

TCLCMD-1400

SUMMARY

The software has encountered unexpected internal reporting flow error condition TCLCMD-1400 - and may not perform reliably in this state. You should contact Cadence support for assistance in resolving this problem and refer to the error code mentioned above. There is no current user-level workaround for resolving this problem.

DESCRIPTION

TCLCMD-1401

NAME

TCLCMD-1401

SUMMARY

The maximum GBA path limit for doing exhaustive PBA analysis has been exhausted for some of the clock groups. The current results may not be based upon all the violating endpoints in design for these clock groups.

DESCRIPTION

TCLCMD-1402

NAME

TCLCMD-1402

SUMMARY

Ignoring command `configure_ipd_mode` as it is IPD master session.

DESCRIPTION

TCLCMD-1404

NAME

TCLCMD-1404

SUMMARY

The specified -nworst limit exhausted for endpoint:

DESCRIPTION

TCLCMD-1405

NAME

TCLCMD-1405

SUMMARY

The specified -nworst limit of has been exhausted for endpoints. Refer to verbose log file (logv) of the current run to get the endpoints for which this limit has been exhausted. You may increase the value of -nworst to report if more timing paths for each endpoint exist.

DESCRIPTION

TCLCMD-1406

NAME

TCLCMD-1406

SUMMARY

The specified -max_paths limit has been exhausted . You may increase the value of -max_paths to report if more timing paths exist.

DESCRIPTION

TCLCMD-1407

NAME

TCLCMD-1407

SUMMARY

The specified -max_paths limit of has been exhausted. You may increase the value of -max_paths to report if more timing paths exist.

DESCRIPTION

TCLCMD-1408

NAME

TCLCMD-1408

SUMMARY

Tool is not able to create valid clock group as few of the clocks used in constraint does not exist in design. The constraint will be dropped.

DESCRIPTION

TCLCMD-1409

NAME

TCLCMD-1409

SUMMARY

The write_twf command is not supported when signal integrity analysis is enabled. To generate a TWF file you must disable SI analysis using the set_delay_cal_mode -SIAware false option.

DESCRIPTION

TCLCMD-1410

NAME

TCLCMD-1410

SUMMARY

The delay corner is not associated with any active view. The can be applied to corners associated with active views only.

DESCRIPTION

TCLCMD-1411

NAME

TCLCMD-1411

SUMMARY

The `update_io_latency` command cannot be run when a clock is propagated. Check if there is any `set_propagated_clock` constraint on pin/port.

DESCRIPTION

TCLCMD-1412

NAME

TCLCMD-1412

SUMMARY

Option value provided in is not supported. Specify separator value from the predefined set { , # : <white space> }.

DESCRIPTION

TCLCMD-1413

NAME

TCLCMD-1413

SUMMARY

No option value specified for .Specify separator value from the predefined set { , # : <white space> }.

DESCRIPTION

TCLCMD-1414

NAME

TCLCMD-1414

SUMMARY

Resetting property last_pba_slack on pins and ports due to variable timing_path_based_coverage_assert_last_pba_slack being changed to false.

DESCRIPTION

TCLCMD-1415

NAME

TCLCMD-1415

SUMMARY

Setting the variable timing_path_based_coverage_enable_auto_incr_depth_mode to true will have no effect unless the variable timing_path_based_coverage_enable_incr_depth_mode is also set to true.

DESCRIPTION

TCLCMD-1416

NAME

TCLCMD-1416

SUMMARY

Ignoring the setting of true of variable
timing_path_based_coverage_enable_auto_incr_depth_mode since the variable is set to false.

DESCRIPTION

TCLCMD-1417

NAME

TCLCMD-1417

SUMMARY

Currently only -of_objects option is supported.

DESCRIPTION

TCLCMD-1418

NAME

TCLCMD-1418

SUMMARY

Bad index : must be integer? or end?[-]integer?.

DESCRIPTION

TCLCMD-1419

NAME

TCLCMD-1419

SUMMARY

mapped to is index, so it is changed to which is the last index of the collection

DESCRIPTION

TCLCMD-1420

NAME

TCLCMD-1420

SUMMARY

INFO: Max depth %lu reached for end point during Path Based Analysis

DESCRIPTION

TCLCMD-1421

NAME

TCLCMD-1421

SUMMARY

Minimum char limit for multi-line write sdc support should be at least equal to 100.

DESCRIPTION

TCLCMD-1422

NAME

TCLCMD-1422

SUMMARY

Command does not support -from/-through/-to/end point edge(s). Correct specification. Command is ignored.

DESCRIPTION

TCLCMD-1423

NAME

TCLCMD-1423

SUMMARY

Input instances provided are not of the same type

DESCRIPTION

TCLCMD-1424

NAME

TCLCMD-1424

SUMMARY

****DIAG:** Global timing update is performed in a worker thread!

DESCRIPTION

TCLCMD-1425

NAME

TCLCMD-1425

SUMMARY

****DIAG:** Timing query is performed without necessary timing update!

DESCRIPTION

TCLCMD-1426

NAME

TCLCMD-1426

SUMMARY

****DIAG:** Timing query is performed in a worker thread without necessary timing update!

DESCRIPTION

TCLCMD-1427

NAME

TCLCMD-1427

SUMMARY

Stack trace of the query:

DESCRIPTION

TCLCMD-1428

NAME

TCLCMD-1428

SUMMARY

INFO: paths filtered due to application of user slack_credit on the pins through these timing paths.

DESCRIPTION

TCLCMD-1429

NAME

TCLCMD-1429

SUMMARY

INFO: nodes filtered in due to application of user slack_credit on the pins.

DESCRIPTION

TCLCMD-1430

NAME

TCLCMD-1430

SUMMARY

The global timing_path_based_coverage_enable_user_slack_credit needs to be turned on before using slack_credit property.

DESCRIPTION

TCLCMD-1431

NAME

TCLCMD-1431

SUMMARY

Check type electromigration is only supported with -drv_violation_type
max_transition|max_capacitance. -check type electromigration is ignored in this command

DESCRIPTION

TCLCMD-1432

NAME

TCLCMD-1432

SUMMARY

Check type electromigration is only supported with settings
timing_report_drv_per_frequency/timing_report_drv_per_frequency_per_input_slew being true. -
check type electromigration is ignored in this command

DESCRIPTION

TCLCMD-1433

NAME

TCLCMD-1433

SUMMARY

INFO: C-PBA Endpoint Summary (Total Violating EP:) Green Endpoints: , Red Endpoints: , Yellow Endpoints:

DESCRIPTION

TCLCMD-1434

NAME

TCLCMD-1434

SUMMARY

INFO: GBA violating endpoints were marked violated by exhaustive coverage (EPBA) analysis. The software has marked these nodes as Red.

DESCRIPTION

TCLCMD-1435

NAME

TCLCMD-1435

SUMMARY

INFO: No violation found for GBA violating endpoints within the existing exhaustive coverage (EPBA) analysis depth and maximum paths limit. The software has marked these nodes as Yellow.

DESCRIPTION

TCLCMD-1436

NAME

TCLCMD-1436

SUMMARY

INFO: GBA violating endpoints have exhausted all GBA paths without any PBA violation and are guaranteed to meet the slack criteria. The software has marked these nodes as Green.

DESCRIPTION

TCLCMD-1437

NAME

TCLCMD-1437

SUMMARY

INFO: Found endpoints already marked to be Red and endpoints marked as Green.

DESCRIPTION

TCLCMD-1438

NAME

TCLCMD-1438

SUMMARY

INFO: Exhaustive coverage (EPBA) analysis invoked with depth of and max paths limit as .

DESCRIPTION

TCLCMD-1439

NAME

TCLCMD-1439

SUMMARY

The `set_disable_timing` command can only be applied to gate delay or gate check arcs. It cannot be used to disable interconnect arcs between instances. The arc with from-pin and to-pin is an interconnect arc - so the disable will not be applied. Please review the semantics of `set_disable_timing` and modify your constraints appropriately.

DESCRIPTION

TCLCMD-1440

NAME

TCLCMD-1440

SUMMARY

-order_dependent option will be ignored as one of the inputs is a dual ported objects.

DESCRIPTION

TCLCMD-1441

NAME

TCLCMD-1441

SUMMARY

Since the scope of command is not limited to iteration specific object and is across full design, invoking it within all threads simultaneously may lead to increase in peak memory. It is therefore recommended to run this command outside the parallel_body.

DESCRIPTION

TCLCMD-1442

NAME

TCLCMD-1442

SUMMARY

The set_global command is obsolete and should be removed from your scripts. It will continue to function in this release, but for compatibility with future releases you should now use the Tcl set command to change global variable settings.

DESCRIPTION

TCLCMD-1443

NAME

TCLCMD-1443

SUMMARY

all_connected command does not support Power or Ground Net

DESCRIPTION

TCLCMD-1444

NAME

TCLCMD-1444

SUMMARY

The software has found a pre-existing clock-gating check on pin in view with an inconsistent clock non-controlling level. The new non-controlling level will overwrite the previous non-controlling level . All previously applied clock gating constraints on the pin for the view will be reset.

DESCRIPTION

TCLCMD-1445

NAME

TCLCMD-1445

SUMMARY

Incorrect value specified for -precision option. Specify value between 1 and 10.

DESCRIPTION

TCLCMD-1446

NAME

TCLCMD-1446

SUMMARY

Incorrect value specified for -max_num_loads option. Specify value between 1 and 32.

DESCRIPTION

TCLCMD-1447

NAME

TCLCMD-1447

SUMMARY

Incorrect value specified for -max_num_slews option. Specify value between 1 and 32.

DESCRIPTION

TCLCMD-1448

NAME

TCLCMD-1448

SUMMARY

The specified view is currently an active analysis view - so it cannot be deleted. You should use the set_analysis_view command to exclude this view and then reissue the delete_analysis_view command.

DESCRIPTION

TCLCMD-1449

NAME

TCLCMD-1449

SUMMARY

Command is not supported when `timing_enable_zero_delay_analysis_mode` is set to true.

DESCRIPTION

TCLCMD-1450

NAME

TCLCMD-1450

SUMMARY

The clock(s) specified with `and` and the clock(s) specified with `do not overlap` in any active analysis view. The constraint will be ignored. You should correct the specification so that the `-from` and `-to` arguments refer to clocks that exist in a common view.

DESCRIPTION

TCLCMD-1451

NAME

TCLCMD-1451

SUMMARY

Maximum 32 float values can be specified in option of `do_extract_model`.

DESCRIPTION

TCLCMD-1452

NAME

TCLCMD-1452

SUMMARY

No mapping pair found for . The alias and property should be given in pairs like {x y} or {{x1 y1}{x2 y2}}

DESCRIPTION

TCLCMD-1453

NAME

TCLCMD-1453

SUMMARY

No valid alias property mapping found in create_property_alias.

DESCRIPTION

TCLCMD-1454

NAME

TCLCMD-1454

SUMMARY

Alias is repeated, there can be only one mapping for an alias to avoid ambiguity

DESCRIPTION

TCLCMD-1455

NAME

TCLCMD-1455

SUMMARY

Mapping for alias is ignored as its mapping property is not found

DESCRIPTION

TCLCMD-1456

NAME

TCLCMD-1456

SUMMARY

Replacing previous alias property pair { } with new pair { }

DESCRIPTION

TCLCMD-1457

NAME

TCLCMD-1457

SUMMARY

Option -retime_view of report_timing should be used only with full path PBA mode. Please set the global timing_disable_retime_clock_path_slew_propagation to false.

DESCRIPTION

TCLCMD-1458

NAME

TCLCMD-1458

SUMMARY

Option -retime is mandatory when using -retime_view option with report_timing command. Please provide both of these.

DESCRIPTION

TCLCMD-1459

NAME

TCLCMD-1459

SUMMARY

INFO: Path Based Analysis (PBA) filtered paths with the help of Learning Model

DESCRIPTION

TCLCMD-1460

NAME

TCLCMD-1460

SUMMARY

Incorrect option specified for equivalent waveform model propagation. Model extraction will continue without waveform propagation.

DESCRIPTION

TCLCMD-1461

NAME

TCLCMD-1461

SUMMARY

Skipped unsupported command:

DESCRIPTION

TCLCMD-1462

NAME

TCLCMD-1462

SUMMARY

INFO: Path Based Analysis (PBA) filtered paths with the help of real GBA slack

DESCRIPTION

TCLCMD-1463

NAME

TCLCMD-1463

SUMMARY

Option -greybox of command do_extract_model is obsolete and is ignored by the tool. To avoid this warning, remove this option from your script.

DESCRIPTION

TCLCMD-1464

NAME

TCLCMD-1464

SUMMARY

INFO: Total nodes marked GREEN

DESCRIPTION

TCLCMD-1465

NAME

TCLCMD-1465

SUMMARY

The `write_sdf` option `-recompute_parallel_arcs` has been deprecated and replaced by the `-recompute_delay_calc` option. It will continue to function in this release, but you should update your scripts to use the new option. Refer to the command reference for additional information on the new option.

DESCRIPTION

TCLCMD-1466

NAME

TCLCMD-1466

SUMMARY

The analysis view: is not currently enabled for both Setup and Hold analysis. As a result, some delay information has not been fully calculated for these views. You should rerun write_sdf with the -recompute_delay_calc option to ensure full coverage of all delay and check arcs in the design.

DESCRIPTION

TCLCMD-1467

NAME

TCLCMD-1467

SUMMARY

is an obsolete option of command compare_model_timing and will not be supported from next releases. Please use options instead.

DESCRIPTION

absolute_tolerance/percentage_tolerance is an obsolete option of command compare_model_timing and will not be supported from next releases. Please use absolute_tolerance_optimistic/pessimistic or percentage_tolerance_optimistic/pessimistic options instead.

TCLCMD-1468

NAME

TCLCMD-1468

SUMMARY

absolute_tolerance/percentage_tolerance options of compare_model_timing command will be ignored by the tool. Failures will be flagged as per default/user given values of absolute_tolerance_optimistic, percentage_tolerance_optimistic, absolute_tolerance_pessimistic and percentage_tolerance_pessimistic options.

DESCRIPTION

absolute_tolerance/percentage_tolerance options of compare_model_timing command will be ignored by the tool. Failures will be flagged as per default/user given values of absolute_tolerance_optimistic, percentage_tolerance_optimistic, absolute_tolerance_pessimistic and percentage_tolerance_pessimistic options.

TCLCMD-1469

NAME

TCLCMD-1469

SUMMARY

No timing paths found matching the specified criteria. Empty collection returned.

DESCRIPTION

TCLCMD-1470

NAME

TCLCMD-1470

SUMMARY

Pins/ports are only supported with -drv_violation_type option

DESCRIPTION

TCLCMD-1471

NAME

TCLCMD-1471

SUMMARY

The setting is obsolete and is being replaced with the `set_socv_reporting_nsigma_multiplier` command which allows configuring separate n-sigma multipliers for Setup and Hold - per analysis view. The global may continue to be used in this release by settings `timing_socv_view_based_nsigma_multiplier_mode` to false. For future compatibility, you should migrate your scripts to utilize the new functionality.

DESCRIPTION

TCLCMD-1472

NAME

TCLCMD-1472

SUMMARY

The setting is obsolete and has been replaced with the `set_socv_reporting_nsigma_multiplier` command. The new functionality allows configuring separate n-sigma multipliers for Setup and Hold - per analysis view. You should migrate your scripts to use the new functionality.

DESCRIPTION

TCLCMD-1473

NAME

TCLCMD-1473

SUMMARY

The setting is obsolete and is no longer required. The software will now automatically set the appropriate internal n-sigma values based on the Setup and Hold n-sigma values specified for each analysis view.

DESCRIPTION

TCLCMD-1474

NAME

TCLCMD-1474

SUMMARY

Previously defined value for port have been overwritten by a new set_model_extraction_mode command

DESCRIPTION

TCLCMD-1475

NAME

TCLCMD-1475

SUMMARY

option of command set_model_extraction_mode should be given with the valid port name list. Please provide the arguments in expected format and rerun the command. For more details, please refer to Tempus Text Command reference

DESCRIPTION

TCLCMD-1476

NAME

TCLCMD-1476

SUMMARY

Please specify using option with option of command set_model_extraction_mode and rerun

DESCRIPTION

TCLCMD-1477

NAME

TCLCMD-1477

SUMMARY

Resetting the previous depth, timing_path_based_coverage_enable_incr_depth_mode will have no effect as .

DESCRIPTION

TCLCMD-1478

NAME

TCLCMD-1478

SUMMARY

Incorrect type of collection passed to the option for command . The option expects a collection of

DESCRIPTION

TCLCMD-1479

NAME

TCLCMD-1479

SUMMARY

-delay option of command report_etm_arcs is supported only with -write_model_timing switch.
Please provide the arguments in expected format and rerun the command

DESCRIPTION

This message is issued when -delay option of report_etm_arcs is used without -write_model_timing switch

TCLCMD-1480

NAME

TCLCMD-1480

SUMMARY

INFO: The number of paths discarded after gba enumeration due to filtering on real gba slack :

DESCRIPTION

TCLCMD-1481

NAME

TCLCMD-1481

SUMMARY

Argument can have values , , or .

DESCRIPTION

TCLCMD-1482

NAME

TCLCMD-1482

SUMMARY

Argument can have values or .

DESCRIPTION

TCLCMD-1483

NAME

TCLCMD-1483

SUMMARY

The report_statistical_timing_derate_factors command is obsolete and will be removed from the software in a future release. The command will continue to work in the current release - but to ensure that your scripts will be compatible with future versions of the software - you should remove this command from your scripts.

DESCRIPTION

TCLCMD-1484

NAME

TCLCMD-1484

SUMMARY

When using the -combinational_from_to option to constrain paths from a clock waveform - to a clock waveform, the -through pin argument cannot be used to specify intermediate points on the path. The constraint has been ignored.

DESCRIPTION

TCLCMD-1485

NAME

TCLCMD-1485

SUMMARY

The write_timing_context commands -module option requires a valid hierarchical module to be provided. The specified module is not a valid hierarchical module. You should review the design specification and reissue the command. No timing context data has been written.

DESCRIPTION

TCLCMD-1486

NAME

TCLCMD-1486

SUMMARY

The internal default clock - represented as @ - should not be included in the clock mapping file. You should modify your clock mapping file to include only user-defined clocks.

DESCRIPTION

TCLCMD-1487

NAME

TCLCMD-1487

SUMMARY

The clock which was specified for the -block_clock option could not be found in the current block-level constraints. You should review the block-level constraints and update the clock mapping file with a valid clock.

DESCRIPTION

TCLCMD-1489

NAME

TCLCMD-1489

SUMMARY

The constraint, when used with the -combinational_from_to option, requires the -from and -to arguments to be pin or port objects. The argument does not match one of these objects types. The constraint has been ignored.

DESCRIPTION

TCLCMD-1490

NAME

TCLCMD-1490

SUMMARY

Last accessed el was and currently asking phdata for when access type is

DESCRIPTION

TCLCMD-1491

NAME

TCLCMD-1491

SUMMARY

-drv_abs_margin and -drv_ratio_margin options of the report_constraint command are mutually exclusive. You may only specify one of these options at a time.

DESCRIPTION

TCLCMD-1492

NAME

TCLCMD-1492

SUMMARY

No -drv_violation_type is set. must have -drv_violation_type option. Run report_constraint -help to see the basic syntax for this command. Run man report_constraint to see a complete description of the command and its options. Review the syntax and specify a legal value for option .

DESCRIPTION

TCLCMD-1493

NAME

TCLCMD-1493

SUMMARY

The value provided to is the wrong data type for this argument. This option requires a .

DESCRIPTION

TCLCMD-1494

NAME

TCLCMD-1494

SUMMARY

The value provided to is an incorrect enum type for this argument. This option requires one of: .

DESCRIPTION

TCLCMD-1495

NAME

TCLCMD-1495

SUMMARY

The value of retime_max_slack should not be less than max_slack.

DESCRIPTION

TCLCMD-1496

NAME

TCLCMD-1496

SUMMARY

The option specified for `timing_socv_statistical_min_max_mode` is obsolete and will be removed in a future release of the software. Cadence recommends using `mean_and_three_sigma_bounded` for the SOCV max operation.

DESCRIPTION

TCLCMD-1497

NAME

TCLCMD-1497

SUMMARY

Error while allocating memory for file read in phase data manager

DESCRIPTION

TCLCMD-1498

NAME

TCLCMD-1498

SUMMARY

The timing global is obsolete and should be removed from your scripts. Please use `global` instead

DESCRIPTION

TCLCMD-1499

NAME

TCLCMD-1499

SUMMARY

-start is not set yet in the profile Constraint command.

DESCRIPTION

TCLCMD-1500

NAME

TCLCMD-1500

SUMMARY

The -expand_views option of report_timing_summary does not provide additional information when running a single-mode/single-corner analysis. You can use this option in multi-mode/multi-corner analysis to generate a view-by-view summary. This option will be ignored.

DESCRIPTION

TCLCMD-1501

NAME

TCLCMD-1501

SUMMARY

Error while getting file size in phase data manager.

DESCRIPTION

TCLCMD-1504

NAME

TCLCMD-1504

SUMMARY

INFO: Updating delays for advanced GBA mode.

DESCRIPTION

TCLCMD-1505

NAME

TCLCMD-1505

SUMMARY

INFO: Worst outlier of value %f found at Path (Begin Pin: , End Pin:) at stage : -> -> . Gba Values : Cell Delay %f (%f:%f), Net Delay %f (%f:%f). Pba Values : Cell Delay %f (%f:%f), Net Delay %f (%f:%f)

DESCRIPTION

TCLCMD-1506

NAME

TCLCMD-1506

SUMMARY

INFO: of value %f found at Path (Begin Pin: , End Pin:) at stage : -> -> . Gba Values : Cell Delay %f (%f:%f), Net Delay %f (%f:%f). Pba Values : Cell Delay %f (%f:%f), Net Delay %f (%f:%f)

DESCRIPTION

TCLCMD-1507

NAME

TCLCMD-1507

SUMMARY

-rise/-fall/-rise_from/-rise_to/-fall_from/-fall_to/-from/-to options not supported with -half_cycle_jitter option.

DESCRIPTION

TCLCMD-1508

NAME

TCLCMD-1508

SUMMARY

set_clock_uncertainty can not be applied on generated clock. Clock in view .

DESCRIPTION

TCLCMD-1509

NAME

TCLCMD-1509

SUMMARY

Incremental constraints have been applied which may affect SI analysis. You can continue working in this mode, but you should eventually run update_timing -full to generate a new report with updated SI data.

DESCRIPTION

TCLCMD-1510

NAME

TCLCMD-1510

SUMMARY

Incremental constraints have been applied which may affect SI analysis. You can continue working in this mode, but you should ensure a full timing update occurs to generate new results with updated SI data.

DESCRIPTION

TCLCMD-1511

NAME

TCLCMD-1511

SUMMARY

The -view option should be specified for the property arrival_window in dict format in MMMC mode. For now The tool will assume the view specified as and write the arrival windows corresponding to this view. For compatibility with future releases, update your scripts to provide -view option.

DESCRIPTION

TCLCMD-1513

NAME

TCLCMD-1513

SUMMARY

Extra argument in command is already specified. Multiple occurrence of same argument is ignored.

DESCRIPTION

TCLCMD-1514

NAME

TCLCMD-1514

SUMMARY

INFO: Total %u endpoints found yellow (GBA bounded) during exhaustive path based analysis.

DESCRIPTION

TCLCMD-1515

NAME

TCLCMD-1515

SUMMARY

View needed in MMMC mode for running flow.

DESCRIPTION

TCLCMD-1516

NAME

TCLCMD-1516

SUMMARY

-yellow_only option can be used with EPBA iterative mode only. Please ensure that you are running EPBA with global timing_path_based_enable_exhaustive_iterative_flow turned ON

DESCRIPTION

TCLCMD-1517

NAME

TCLCMD-1517

SUMMARY

The specified path_group is not active in analysis view so could not be analyzed, You should rerun report_timing with a consistent view and path_group combination or use the setPathGroupOptions command to activate path_groups for the necessary views.

DESCRIPTION

TCLCMD-1518

NAME

TCLCMD-1518

SUMMARY

The pin is not a legal argument to the `set_clock_exclusivity` command so the assertion has been ignored. The pin argument provided is expected to be an output pin which has timing arcs from multiple input pins of the cell. If you are trying to specify general exclusivity between different clocks - you should use the `set_clock_groups` constraint instead.

DESCRIPTION

TCLCMD-1519

NAME

TCLCMD-1519

SUMMARY

Timing mem stats and aae memory report will be printed in file.

DESCRIPTION

TCLCMD-1520

NAME

TCLCMD-1520

SUMMARY

Resetting the CPBA graph reduction mode.

DESCRIPTION

TCLCMD-1521

NAME

TCLCMD-1521

SUMMARY

option is compatible with only slew based retime.

DESCRIPTION

TCLCMD-1522

NAME

TCLCMD-1522

SUMMARY

Yield collection passed to the option for command should have atleast and atmost one object, it will be overridden with new collection of single yield object.

DESCRIPTION

TCLCMD-1523

NAME

TCLCMD-1523

SUMMARY

Argument passed to the option for command is not a collection, it will be overridden with new collection of .

DESCRIPTION

TCLCMD-1524

NAME

TCLCMD-1524

SUMMARY

Iso delay corner should be different from the corner used in the primary view. Corner is ignored from iso_performance_delay_corners.

DESCRIPTION

TCLCMD-1525

NAME

TCLCMD-1525

SUMMARY

Command is not supported in unflattened view.

DESCRIPTION

TCLCMD-1526

NAME

TCLCMD-1526

SUMMARY

The collection object of type size : is freed as collection object went out of scope.

DESCRIPTION

TCLCMD-1527

NAME

TCLCMD-1527

SUMMARY

The property is not supported in fetching multiple attributes in single call

DESCRIPTION

TCLCMD-1528

NAME

TCLCMD-1528

SUMMARY

Only pin/lib_pin/cell/lib_cell is supported in fetching multiple attributes in single call

DESCRIPTION

TCLCMD-1529

NAME

TCLCMD-1529

SUMMARY

Source of clock is defined on an internal pin .

DESCRIPTION

WARNING - The -source option of generated clock is defined on an internal pin. The generated clock will be created normally, but design must have necessary model structure available to propagate the external master clock to the generating clock assertion point.

TCLCMD-1530

NAME

TCLCMD-1530

SUMMARY

Clock has clock waveform with rise & fall edges defined at the same time.

DESCRIPTION

TCLCMD-1531

NAME

TCLCMD-1531

SUMMARY

has been applied on hierarchical pin

DESCRIPTION

The timing constraint has been applied on hierarchical pin, but the tool actually maps the constraint to flat fanout/fanin pins, which is the actual recommendation for the constraint.

TCLCMD-1532

NAME

TCLCMD-1532

SUMMARY

The software has determined that the current temporary directory is running low on disk space. This area is used for the temporary storage of analysis data. You must increase the availability of temporary storage here

DESCRIPTION

TCLCMD-1533

NAME

TCLCMD-1533

SUMMARY

Iso performance delay corner must use same rc corner as the primary corner .

DESCRIPTION

TCLCMD-1534

NAME

TCLCMD-1534

SUMMARY

The instance specified by the read_timing_context -instances option could not be found in the design. Timing contexts have not been loaded. You should remedy this mismatch and reissue the read_timing_context command.

DESCRIPTION

TCLCMD-1535

NAME

TCLCMD-1535

SUMMARY

set_input_delay command specified without -clock option.

DESCRIPTION

TCLCMD-1536

NAME

TCLCMD-1536

SUMMARY

Cannot perform set_clock_latency on port as create_clock or create_generated_clock is not asserted on this port in view .

DESCRIPTION

TCLCMD-1537

NAME

TCLCMD-1537

SUMMARY

Either -retime option is not provided or provided with incompatible parameters while using -retime_iso_views option.

DESCRIPTION

TCLCMD-1538

NAME

TCLCMD-1538

SUMMARY

Reporting disk cache for analysis mode

DESCRIPTION

TCLCMD-1539

NAME

TCLCMD-1539

SUMMARY

The view which was specified for the -block_view option could not be found in the current block-level view definition. You should review the block-level view definition and update the view mapping file with a valid view.

DESCRIPTION

TCLCMD-1540

NAME

TCLCMD-1540

SUMMARY

This option cannot be used in Multi Instance Hierarchical Analysis as common pin may not be available to perform skew modification relative to common pin latencies

DESCRIPTION

TCLCMD-1541

NAME

TCLCMD-1541

SUMMARY

Port(s) are invalid for constraint in hierarchical context mode

DESCRIPTION

TCLCMD-1542

NAME

TCLCMD-1542

SUMMARY

When generating the timing contexts for multiple instances of the same module, you must include the `-skip_interface_paths` option when invoking `write_timing_context`. You should reissue `write_timing_context` with the corrected syntax.

DESCRIPTION

TCLCMD-1543

NAME

TCLCMD-1543

SUMMARY

The option to `report_constraint` can only be used when reporting DRV violations. The `-drv_violation_type` option must be specified. Please rerun the command with the corrected syntax. More information is available on the command syntax by running `man report_constraint`.

DESCRIPTION

TCLCMD-1544

NAME

TCLCMD-1544

SUMMARY

Argument can have values , , or .

DESCRIPTION

TCLCMD-1545

NAME

TCLCMD-1545

SUMMARY

Path type full/full_clock is not supported with IPBA retime flow.

DESCRIPTION

TCLCMD-1546

NAME

TCLCMD-1546

SUMMARY

This feature is only supported when report_path_exception command is executed with -all.

DESCRIPTION

TCLCMD-1547

NAME

TCLCMD-1547

SUMMARY

read_path_descriptions command doesnt allow option

DESCRIPTION

TCLCMD-1548

NAME

TCLCMD-1548

SUMMARY

exceeded report_timing call in read_path_descriptions command

DESCRIPTION

TCLCMD-1550

NAME

TCLCMD-1550

SUMMARY

read_path_descriptions will create collection of path type

DESCRIPTION

TCLCMD-1551

NAME

TCLCMD-1551

SUMMARY

Option -filter_view can be used only along with -filter option and is_exist property specified along with it. Please specify with correct usage.

DESCRIPTION

TCLCMD-1552

NAME

TCLCMD-1552

SUMMARY

ports are not valid arguments to option of set_model_extraction_mode command. Tool will ignore the attribute(s) specified by command set_model_extraction_mode on port. Please rerun the command with correct arguments.

DESCRIPTION

TCLCMD-1553

NAME

TCLCMD-1553

SUMMARY

As a result of netlist changes caused by ECO operations, any preexisting Tcl collections have been reset and will need to be regenerated if they are still required.

DESCRIPTION

TCLCMD-1554

NAME

TCLCMD-1554

SUMMARY

Command will be ignored as incorrect command options are specified by the user. Please rerun the command with correct command options and respective arguments. Refer Tempus Text Command Reference for correct command usage.

DESCRIPTION

TCLCMD-1555

NAME

TCLCMD-1555

SUMMARY

option of set_model_extraction_mode command is incompatible with . Tool will ignore the max specified by set_model_extraction_mode command on port . Please rerun the command with correct command options.

DESCRIPTION

TCLCMD-1556

NAME

TCLCMD-1556

SUMMARY

Unsupported option in command

DESCRIPTION

TCLCMD-1557

NAME

TCLCMD-1557

SUMMARY

The `set_clock_sense` command has been replaced by the `set_sense` command in SDC2.1 specifications. `set_clock_sense` command still works but it is recommended to switch `set_sense` as per SDC 2.1 guidelines.

DESCRIPTION

TCLCMD-1558

NAME

TCLCMD-1558

SUMMARY

`set_sense` command requires `-type` with both `-stop_propagation` and `-clocks`

DESCRIPTION

TCLCMD-1559

NAME

TCLCMD-1559

SUMMARY

Since no -max_slack has been specified with IPBA mode, tool will assume its default value in this case which is 0

DESCRIPTION

TCLCMD-1561

NAME

TCLCMD-1561

SUMMARY

Unsupported option in command

DESCRIPTION

TCLCMD-1562

NAME

TCLCMD-1562

SUMMARY

set_sense command requires -type with -stop_propagation

DESCRIPTION

TCLCMD-1570

NAME

TCLCMD-1570

SUMMARY

Derate factor applied on corner is %f, which is more than maximum limit of %f.

DESCRIPTION

TCLCMD-1571

NAME

TCLCMD-1571

SUMMARY

Derate factor applied is %f, which is more than maximum limit of %f.

DESCRIPTION

TCLCMD-1572

NAME

TCLCMD-1572

SUMMARY

Property is a collection property. It is not supported by report_property command

DESCRIPTION

TCLCMD-1573

NAME

TCLCMD-1573

SUMMARY

Property is not supported for object type.

DESCRIPTION

TCLCMD-1576

NAME

TCLCMD-1576

SUMMARY

Option -filter_view is supported only when filtering is done with is_exist property. This option will be ignored in the current usage.

DESCRIPTION

TCLCMD-1577

NAME

TCLCMD-1577

SUMMARY

Violating endpoints max limit reached for Path based analysis in mode.

DESCRIPTION

TCLCMD-1578

NAME

TCLCMD-1578

SUMMARY

Hierarchical pin is not supported in set_sense.

DESCRIPTION

TCLCMD-1579

NAME

TCLCMD-1579

SUMMARY

Invalid value given for global timing_inter_power_domain_scope_data. The possible values are max and min

DESCRIPTION

TCLCMD-1580

NAME

TCLCMD-1580

SUMMARY

The command requires a valid directory name to be specified. You should review the command specification and rerun the command.

DESCRIPTION

TCLCMD-1581

NAME

TCLCMD-1581

SUMMARY

The directory specified for the option in command does not exist. You should specify a different directory and reissue the command.

DESCRIPTION

TCLCMD-1582

NAME

TCLCMD-1582

SUMMARY

Found a pin with NULL slack for path with endpoint

DESCRIPTION

TCLCMD-1583

NAME

TCLCMD-1583

SUMMARY

Starting analysis in eco mode..

DESCRIPTION

TCLCMD-1584

NAME

TCLCMD-1584

SUMMARY

INFO: IPBA:

DESCRIPTION

TCLCMD-1585

NAME

TCLCMD-1585

SUMMARY

Path based analysis could not find valid begin points satisfying given command options

DESCRIPTION

TCLCMD-1586

NAME

TCLCMD-1586

SUMMARY

Running in New Eco Mode.

DESCRIPTION

TCLCMD-1587

NAME

TCLCMD-1587

SUMMARY

Argument cant be specified with option for .

DESCRIPTION

TCLCMD-1588

NAME

TCLCMD-1588

SUMMARY

When ipba mode is enabled, multiple pins cannot be specified with any -through option

DESCRIPTION

TCLCMD-1589

NAME

TCLCMD-1589

SUMMARY

INFO: Path Based Analysis (PBA) filtered paths during , as they were not found to be relevant for timing yield generation

DESCRIPTION

TCLCMD-1590

NAME

TCLCMD-1590

SUMMARY

IPBA mode can only be set if variable timing_path_based_enable_infinite_depth_mode is set to true before timing update

DESCRIPTION

TCLCMD-1591

NAME

TCLCMD-1591

SUMMARY

Command cannot find power domain name given with option . Power domain will be ignored.
Rerun command with correct the power_domain name.

DESCRIPTION

TCLCMD-1592

NAME

TCLCMD-1592

SUMMARY

Unsupported option for on DMMMC Master:

DESCRIPTION

TCLCMD-1593

NAME

TCLCMD-1593

SUMMARY

Both retime and retime_mode options are mandatory for running report_analysis_summary in IPBA mode.

DESCRIPTION

TCLCMD-1594

NAME

TCLCMD-1594

SUMMARY

A clock with name was already defined in the design. Previously defined clock definition of will be overwritten.

DESCRIPTION

TCLCMD-1595

NAME

TCLCMD-1595

SUMMARY

-half_cycle_jitter/-full_cycle_jitter cant be applied on pins or ports.

DESCRIPTION

TCLCMD-1596

NAME

TCLCMD-1596

SUMMARY

Views sharing same -iso_corner_group should have same constraint_mode. Different constraint_mode and are using for -iso_corner_group .

DESCRIPTION

TCLCMD-1597

NAME

TCLCMD-1597

SUMMARY

Views sharing same -iso_corner_group should have same rc_corner. Different rc_corner and are using for -iso_corner_group .

DESCRIPTION

TCLCMD-1598

NAME

TCLCMD-1598

SUMMARY

Views belonged to same iso_corner_group should be identical in -setup and -hold if specified in both.

DESCRIPTION

TCLCMD-1599

NAME

TCLCMD-1599

SUMMARY

The `-combinational_from_to` and `-ignore_clock_latency` options are mutually-exclusive. Your constraints may use only one methodology or the other. The option has already been asserted - so the current constraint which uses will be ignored. You should modify your constraints to use one of the options consistently - migrating to the SDC 2.1 standard `-ignore_clock_latency` is recommended. You may also set the global `timing_constraint_path_delay_migrate_combinational_from_to` to true to dynamically migrate `-combinational_from_to` usage to `-ignore_clock_latency` in session.

DESCRIPTION

TCLCMD-1600

NAME

TCLCMD-1600

SUMMARY

The `-combinational_from_to` option of `set_max_delay` and `set_min_delay` is being replaced by the SDC 2.1 `-ignore_clock_latency` option. The option will continue to work in this release, but for compatibility with future releases, you should migrate your scripts to use the standardized `-ignore_clock_latency` option. The semantics of the two options are slightly different, but can be aligned using the global variable: `timing_constraint_path_delay_exclude_check_delay_from_ignore_clock_latency`. Consult the

documentation or man pages for more information about configuring this setting.

DESCRIPTION

TCLCMD-1601

NAME

TCLCMD-1601

SUMMARY

****INFO:** The option `-combinational_from_to` option is obsolete and will be replaced with the SDC 2.1 `-ignore_clock_latency` option. You can control this automated translation with `timing_constraint_path_delay_migrate_combinational_from_to`. Consult the documentation for additional information on this global or the `-ignore_clock_latency` option.

DESCRIPTION

TCLCMD-1602

NAME

TCLCMD-1602

SUMMARY

The clock has been asserted with zero period

DESCRIPTION

TCLCMD-1603

NAME

TCLCMD-1603

SUMMARY

**INFO: Max time borrow set to a negative value for pin

DESCRIPTION

TCLCMD-1604

NAME

TCLCMD-1604

SUMMARY

-ptr option is mandatory with -retime_slack_sort_by ptr_slack

DESCRIPTION

TCLCMD-1605

NAME

TCLCMD-1605

SUMMARY

set_sense -type is not supported with -positive or -negative switch.

DESCRIPTION

TCLCMD-1607

NAME

TCLCMD-1607

SUMMARY

There is no hierarchical module in the design for instance .

DESCRIPTION

TCLCMD-1608

NAME

TCLCMD-1608

SUMMARY

Multiple instances of same module have been mentioned. This is not supported when -with_constraints option is specified.

DESCRIPTION

TCLCMD-1609

NAME

TCLCMD-1609

SUMMARY

Module having multiple instances has been provided. This is not supported when `-with_constraints` option is specified.

DESCRIPTION

TCLCMD-1610

NAME

TCLCMD-1610

SUMMARY

Only positive derating factor can be specified for command `set_design_rule_derate`. Command ignored.

DESCRIPTION

TCLCMD-1611

NAME

TCLCMD-1611

SUMMARY

Clock with zero period is created

DESCRIPTION

TCLCMD-1612

NAME

TCLCMD-1612

SUMMARY

has reached its max value %f, values greater than this may cause unexpected behaviour

DESCRIPTION

TCLCMD-1613

NAME

TCLCMD-1613

SUMMARY

-hpin option in is only supported with pins. In current command it is called with net. Ignoring -hpin. It will return flat pins only.

DESCRIPTION

TCLCMD-1614

NAME

TCLCMD-1614

SUMMARY

Retime option in PBA will impact only max_delay/setup and min_delay/hold.

DESCRIPTION

TCLCMD-1615

NAME

TCLCMD-1615

SUMMARY

Invalid usage of ,

DESCRIPTION

TCLCMD-1616

NAME

TCLCMD-1616

SUMMARY

The current system unit for is - which is not consistent with the specified set_units value of . Any constraints entered interactively or by using Tcl source should be with respect to the current system units. SDC constraints files loaded via read_sdc or MMMC constraints modes can define the units in the files being loaded by specifying the SDC set_units command at the top of the individual constraint file(s).

DESCRIPTION

TCLCMD-1617

NAME

TCLCMD-1617

SUMMARY

Option -internal_pin is a mandatory for asserting arcs in ETM via assert_etm_arcs.

DESCRIPTION

TCLCMD-1618

NAME

TCLCMD-1618

SUMMARY

Verilog module and library cell name cannot be same. Specify different arguments for -cell_name and -verilog_shell_module options and rerun the do_extract_model command.

DESCRIPTION

TCLCMD-1619

NAME

TCLCMD-1619

SUMMARY

set_sense -stop type option default_data has been specified along with the -clock filter option. This condition can never be satisfied so default_data phases will not be stopped. To block default_data phases, you should not include a -clock specification.

DESCRIPTION

TCLCMD-1620

NAME

TCLCMD-1620

SUMMARY

The command can only be applied to select clock phase on delay arcs. It cannot be applied on interconnect arcs between instances. The arc with from-pin and to-pin is an interconnect arc - so the clock phase selection will not be applied. Please review the semantics of the command and modify your constraints appropriately.

DESCRIPTION

TCLCMD-1621

NAME

TCLCMD-1621

SUMMARY

The command can only be applied to select clock phase on delay arcs. It cannot be applied on check arcs. The arc with from-pin and to-pin is a check arc - so the clock phase selection will not be applied. Please review the semantics of the command and modify your constraints appropriately.

DESCRIPTION

TCLCMD-1624

NAME

TCLCMD-1624

SUMMARY

INFO: Path Based Analysis (PBA) processed paths for timing yield generation

DESCRIPTION

TCLCMD-1625

NAME

TCLCMD-1625

SUMMARY

Option , of command is missing a valid argument, will be skipped. Run -help for more information.

DESCRIPTION

TCLCMD-1626

NAME

TCLCMD-1626

SUMMARY

Graph Based Analysis (GBA) is recommended to be run with derate distance mode chip_size when spatial analysis mode is active. Currently GBA spatial derate distance mode is set to bounding_box which can lead to optimism in Path Based Analysis (PBA) result. Use the global timing_spatial_derate_distance_mode to set GBA distance derate mode to chip_size. This change will not impact the PBA distance as PBA distance will still continue to be based on bounding box mode.

DESCRIPTION

TCLCMD-1627

NAME

TCLCMD-1627

SUMMARY

RPD detail: Command index: triggered batch number: which scheduled commands in MT

DESCRIPTION

TCLCMD-1628

NAME

TCLCMD-1628

SUMMARY

RPD summary: Total batch scheduled: ,Total command ran in MT: ,Total command ran:, Total command:

DESCRIPTION

TCLCMD-1629

NAME

TCLCMD-1629

SUMMARY

RPD status: processing command index:, %f% completed

DESCRIPTION

TCLCMD-1630

NAME

TCLCMD-1630

SUMMARY

The specified max path limit of for timing yield has been exhausted. You may increase this limit by using variable `timing_path_based_enable_max_paths_for_ptr`.

DESCRIPTION

TCLCMD-1631

NAME

TCLCMD-1631

SUMMARY

PBA analysis was skipped on some paths/stages because of GBA slack value beyond the high slack threshold limit. This threshold limit can be configured by using the variable `timing_path_based_enable_high_slack_threshold`.

DESCRIPTION

TCLCMD-1632

NAME

TCLCMD-1632

SUMMARY

is not a valid constraint for -commands option of write_sdc.

DESCRIPTION

TCLCMD-1633

NAME

TCLCMD-1633

SUMMARY

Argument can have values , or .

DESCRIPTION

TCLCMD-1634

NAME

TCLCMD-1634

SUMMARY

No option value specified for -delimiter. Specify delimiter value from the predefined set { , # : <white space> }.

DESCRIPTION

TCLCMD-1635

NAME

TCLCMD-1635

SUMMARY

This command supports path_type full and full_clock only.

DESCRIPTION

TCLCMD-1636

NAME

TCLCMD-1636

SUMMARY

When using report_timing with -gui option, software will generate necessary data for invoking graphical user interface. In this mode, user control of the data output format via the -output_format option is not supported. This option will be ignored. To generate specific output formats - you should reinvoke report_timing without the -gui option.

DESCRIPTION

TCLCMD-1637

NAME

TCLCMD-1637

SUMMARY

An unconstrained hierarchical pin has been specified in the path description.

DESCRIPTION

TCLCMD-1638

NAME

TCLCMD-1638

SUMMARY

The max endpoint limit for IPBA is reached. It can be configured using `set_pba_mode - ipba_endpoints_max_limit <value>`. Increasing this limit can have significant impact on runtime and memory. EndPoint Max Limit: , Total EndPoints: .

DESCRIPTION

TCLCMD-1639

NAME

TCLCMD-1639

SUMMARY

Setting global timing_constraint_view_based_filtering will remain ineffective as the constraint mode specified is shared by multiple views.

DESCRIPTION

TCLCMD-1640

NAME

TCLCMD-1640

SUMMARY

Ending Parallel Report Timing:

DESCRIPTION

TCLCMD-1641

NAME

TCLCMD-1641

SUMMARY

Infinite PBA mode is not supported in AOCV analysis.

DESCRIPTION

TCLCMD-1642

NAME

TCLCMD-1642

SUMMARY

-retime option with this command is only supported in SOCV mode.

DESCRIPTION

TCLCMD-1643

NAME

TCLCMD-1643

SUMMARY

-to option is needed for Required Time flow. Ignoring the required Time flow.

DESCRIPTION

TCLCMD-1645

NAME

TCLCMD-1645

SUMMARY

option is compatible with only exhaustive retime mode.

DESCRIPTION

TCLCMD-1646

NAME

TCLCMD-1646

SUMMARY

ipba_converged_nodes and ipba_reference_slack options are supported in ipba mode only.

DESCRIPTION

TCLCMD-1647

NAME

TCLCMD-1647

SUMMARY

ipba_converged_nodes and ipba_reference_slack options should be used together. Either both should be given or none.

DESCRIPTION

TCLCMD-1648

NAME

TCLCMD-1648

SUMMARY

-keep can only be used in conjunction with -gui to keep temporary machine readable files.

DESCRIPTION

TCLCMD-1650

NAME

TCLCMD-1650

SUMMARY

Global timing_enable_efficient_clock_sense_mode cannot be applied incrementally. Ignoring this setting.

DESCRIPTION

TCLCMD-1651

NAME

TCLCMD-1651

SUMMARY

Timing analysis related commands and functionality are not available in physical-only mode of operation. Try to replace with non-timing related commands instead.

DESCRIPTION

TCLCMD-1652

NAME

TCLCMD-1652

SUMMARY

Infinite depth PBA (IPBA) analysis does not re-evaluate the clock path stages. Ignoring the non-default setting for variable `timing_disable_retime_clock_path_slew_propagation`

DESCRIPTION

TCLCMD-1653

NAME

TCLCMD-1653

SUMMARY

Setting up the analysis mode for infinite depth mode.

DESCRIPTION

TCLCMD-1654

NAME

TCLCMD-1654

SUMMARY

The timing context loaded for instance has missing TWCPFR info on signal port (clock) and reference port (clock). No credit has been added for paths related to these ports.

DESCRIPTION

TCLCMD-1655

NAME

TCLCMD-1655

SUMMARY

: Invalid collection passed to option

DESCRIPTION

TCLCMD-1656

NAME

TCLCMD-1656

SUMMARY

: passed to option ,.

DESCRIPTION

TCLCMD-1657

NAME

TCLCMD-1657

SUMMARY

Since -all_violators has not been specified for report_constraint, tool will assume the max_slack value of 0 while running IPBA Analysis

DESCRIPTION

TCLCMD-1658

NAME

TCLCMD-1658

SUMMARY

IPBA2 settings: Bounded by GBA=, Group Based Reporting=, depth=, max paths limit=, endpoints limit=, effort level=%f

DESCRIPTION

TCLCMD-1659

NAME

TCLCMD-1659

SUMMARY

INFO: Paths were filtered from retiming in EPBA Bounded Mode

DESCRIPTION

TCLCMD-1660

NAME

TCLCMD-1660

SUMMARY

Unknown option

DESCRIPTION

TCLCMD-1661

NAME

TCLCMD-1661

SUMMARY

Total instance count: Inter-Power-Domain instance count: .

DESCRIPTION

TCLCMD-1662

NAME

TCLCMD-1662

SUMMARY

The command cannot be applied to select positiveegative clock phases when specified on delay arcs. Hence, the assertion will not be applied. Please review the semantics of the command and modify your constraints appropriately.

DESCRIPTION

TCLCMD-1663

NAME

TCLCMD-1663

SUMMARY

The command cannot be applied to select pulse-based clock phases when specified on delay arcs. Hence, the assertion will not be applied. Please review the semantics of the command and modify your constraints appropriately.

DESCRIPTION

TCLCMD-1664

NAME

TCLCMD-1664

SUMMARY

Unsuppressing the messages for command is not allowed using `set_ilm_verbose_commands` command

DESCRIPTION

TCLCMD-1665

NAME

TCLCMD-1665

SUMMARY

Incorrect `-rise/-fall` and `-input_transition_rise/-input_transition_fall` pair specification for `set_driving_cell` constraint, the driving cell doesn't have edge satisfying the specified transition, the constraint will be applied but slew specification will be ignored.

DESCRIPTION

TCLCMD-1666

NAME

TCLCMD-1666

SUMMARY

Globals

timing_constraints_path_exception_verbose_reporting/timing_constraints_enable_path_exception
_efficient_file_line_debug_mode cant be turned ON as the constraints have already been read.

DESCRIPTION

TCLCMD-1667

NAME

TCLCMD-1667

SUMMARY

Command does not accept negative values for option . This command will be ignored.

DESCRIPTION

TCLCMD-1668

NAME

TCLCMD-1668

SUMMARY

The `write_sdf -exclude_cells_keep_top_level_nets` and `-exclude_cells_keep_top_level_nets` are mutually exclusive arguments. They control different methods for annotating top-level nets to instances which are being excluded from the SDF due to `-exclude_cells`. This `write_sdf` command has been ignored. You should determine the best top-level net annotation strategy for your flow and reissue a new, corrected `write_sdf` command.

DESCRIPTION

TCLCMD-1669

NAME

TCLCMD-1669

SUMMARY

Collection space has been reset, no collections created before will be accessible further.

DESCRIPTION

TCLCMD-1670

NAME

TCLCMD-1670

SUMMARY

-check_type option can not be specified with pure drv view.

DESCRIPTION

TCLCMD-1671

NAME

TCLCMD-1671

SUMMARY

The specified view(s) is pure drv view only so only drv checks will be reported by report_timing_summary.

DESCRIPTION

TCLCMD-1672

NAME

TCLCMD-1672

SUMMARY

Unrecognizable unit .

DESCRIPTION

TCLCMD-1673

NAME

TCLCMD-1673

SUMMARY

The internal path limit for doing PBA analysis has been exhausted. The current results may not be based upon all the violating endpoints in design. It is recommended to first clean the endpoints with default PBA analysis before running exhaustive analysis.

DESCRIPTION

TCLCMD-1674

NAME

TCLCMD-1674

SUMMARY

The pattern returns a collection size greater than the threshold limit of

DESCRIPTION

TCLCMD-1675

NAME

TCLCMD-1675

SUMMARY

Constraint mode corresponding to has been already set, please set constraint mode corresponding to using command `set_interactive_constraint_mode <mode>`.

DESCRIPTION

TCLCMD-1676

NAME

TCLCMD-1676

SUMMARY

pass for efficient IPBA mode in Eco Db.

DESCRIPTION

TCLCMD-1677

NAME

TCLCMD-1677

SUMMARY

Unable to find scope in directory specified with read_ipd_scope command. Either rerun with correct bounded scope directory or use -max/-min options for specifying the max and min scope, respectively.

DESCRIPTION

TCLCMD-1678

NAME

TCLCMD-1678

SUMMARY

PG Net voltages are not properly defined. Command will be ignored

DESCRIPTION

TCLCMD-1679

NAME

TCLCMD-1679

SUMMARY

One of options is mandatory. Command will be ignored

DESCRIPTION

TCLCMD-1680

NAME

TCLCMD-1680

SUMMARY

No found for pg net voltages .

DESCRIPTION

TCLCMD-1681

NAME

TCLCMD-1681

SUMMARY

The software has performed a full/partial timing reset. All existing timing path and timing point collection references have been invalidated and may no longer be referenced.

DESCRIPTION

TCLCMD-1682

NAME

TCLCMD-1682

SUMMARY

The portion of the sorting key provided to sort_collection returned multiple elements and could not be used. You should adjust your script to ensure that either a single sorting property or a list of different properties is provided as the sorting key.

DESCRIPTION

TCLCMD-1683

NAME

TCLCMD-1683

SUMMARY

Command cannot find PG Net or PG Net Group name given with option . PG Net or PG Net Group Name will be ignored. Rerun command with correct the pg_net name.

DESCRIPTION

TCLCMD-1684

NAME

TCLCMD-1684

SUMMARY

Invalid PG Net or PG Net Group name specified

DESCRIPTION

TCLCMD-1685

NAME

TCLCMD-1685

SUMMARY

INFO: EPBA bounded mode analysis is performed on total %u endpoints. A total of %u worst paths are reported as per max_paths and max_slack criteria.

DESCRIPTION

TCLCMD-1686

NAME

TCLCMD-1686

SUMMARY

EPBA bounded Mode is currently run in session with AOCV enabled. This feature is still in beta phase and not in production.

DESCRIPTION

TCLCMD-1687

NAME

TCLCMD-1687

SUMMARY

No uncertainty value specified with option. You must specify the values when both -half_cycle_jitter and -full_cycle_jitter options are specified. Command ignored.

DESCRIPTION

TCLCMD-1688

NAME

TCLCMD-1688

SUMMARY

Need to specify option along with -retime option for .

DESCRIPTION

TCLCMD-1689

NAME

TCLCMD-1689

SUMMARY

The -incremental_adjust option of set_timing_derate is being replaced by the SDC 2.1 -increment option. This option is being renamed for compatibility with industry standards. There is no change in functionality. The -incremental_adjust option will continue to function in this release, but to ensure compatibility with upcoming releases, you should migrate your scripts to the new option.

DESCRIPTION

TCLCMD-1690

NAME

TCLCMD-1690

SUMMARY

The -incremental_adjust option of set_timing_derate is being replaced by the SDC 2.1 -increment option. This option is being renamed for compatibility with industry standards. There is no change in functionality. To write the constraints with new option -increment, please use the setting set_global/set_db timing_constraint_write_increment_derate true.

DESCRIPTION

TCLCMD-1700

NAME

TCLCMD-1700

SUMMARY

Tempus-PI:

DESCRIPTION

TCLCMD-1701

NAME

TCLCMD-1701

SUMMARY

Tempus-PI:

DESCRIPTION

TCLCMD-1702

NAME

TCLCMD-1702

SUMMARY

Tempus-PI:

DESCRIPTION

TCLCMD-1703

NAME

TCLCMD-1703

SUMMARY

Tempus-PI: Missing option of command analyze_ir_paths

DESCRIPTION

TCLCMD-1704

NAME

TCLCMD-1704

SUMMARY

Tempus-PI: Overriding option of set_power_analysis_mode, analyze_ir_paths option will be honored.

DESCRIPTION

TCLCMD-1705

NAME

TCLCMD-1705

SUMMARY

Exhaustive PBA analysis has been done in adaptive mode as some of the endpoints have been found to be reasonably violating. PBA slacks for some of the endpoints might not be real worst in this mode. It is recommended to first fix the existing violations and run exhaustive PBA analysis again before sign-off.

DESCRIPTION

TCLCMD-1706

NAME

TCLCMD-1706

SUMMARY

User defined path groups have through specifications, this will not get modified. Non-IPD paths can get reported for this path group.

DESCRIPTION

TCLCMD-1707

NAME

TCLCMD-1707

SUMMARY

The size of collection passed in the command is greater than the threshold limit of

DESCRIPTION

TCLCMD-1708

NAME

TCLCMD-1708

SUMMARY

The size is greater than the threshold limit of

DESCRIPTION

TCLCMD-1709

NAME

TCLCMD-1709

SUMMARY

option is not supported with command in .

DESCRIPTION

TCLCMD-1710

NAME

TCLCMD-1710

SUMMARY

the option -save_exhaustive_pba_data can only be run in IPBA mode.

DESCRIPTION

TCLCMD-1711

NAME

TCLCMD-1711

SUMMARY

The -view option should be specified for the command in MMMC mode. The tool will assume the view specified as and generate report corresponding to this view. For compatibility with future releases, incorporate -view option.

DESCRIPTION

TCLCMD-1712

NAME

TCLCMD-1712

SUMMARY

and are mutually exclusive options for . Specify either or option with

DESCRIPTION

TCLCMD-1713

NAME

TCLCMD-1713

SUMMARY

Nets having a fanout higher than found in the design while performing PBA. For these nets GBA delay will be used.

DESCRIPTION

TCLCMD-1714

NAME

TCLCMD-1714

SUMMARY

cannot be specified with

DESCRIPTION

TCLCMD-1715

NAME

TCLCMD-1715

SUMMARY

MultiDie-IC :

DESCRIPTION

TCLCMD-1716

NAME

TCLCMD-1716

SUMMARY

Wrong syntax used for in -name . Required syntax is net@voltage.

DESCRIPTION

TCLCMD-1717

NAME

TCLCMD-1717

SUMMARY

Setting global timing_constraint_enable_get_clocks_single_mode_filtering will remain ineffective as multiple constraint modes are active.

DESCRIPTION

TCLCMD-1718

NAME

TCLCMD-1718

SUMMARY

Object type port is not supported currently for multi-property. The supported object type are instance pin, lib pin, instance and lib cell. Issue separate commands instead of multi-property.

DESCRIPTION

TCLCMD-1719

NAME

TCLCMD-1719

SUMMARY

Clock in view is being defined on port after reading the hierarchical timing context. In hierarchical timing context analysis, it is mandatory to re-read the hierarchical timing context if a clock is defined on a port post context reading. Re-read the context before proceeding with timing analysis.

DESCRIPTION

TCLCMD-1720

NAME

TCLCMD-1720

SUMMARY

Command is not enabled. Use `set_analysis_mode -enable_ipd true` to enable Inter-Power Domain Analysis features.

DESCRIPTION

TCLCMD-1721

NAME

TCLCMD-1721

SUMMARY

INFO: total paths retimed in chip size mode having worst slack than `threshold_slack`

DESCRIPTION

TCLCMD-1722

NAME

TCLCMD-1722

SUMMARY

Since no -max_slack has been specified with , tool will assume its default value in this case which is 0

DESCRIPTION

TCLCMD-1723

NAME

TCLCMD-1723

SUMMARY

The view is either inactive or undefined. constraint will be ignored.

DESCRIPTION

TCLCMD-1724

NAME

TCLCMD-1724

SUMMARY

MultiDie-IC :

DESCRIPTION

TCLCMD-1725

NAME

TCLCMD-1725

SUMMARY

The `set_timing_derate` command has been issued without a `-delay_corner` specification. The derates will be applied across all of the current, active `delay_corners`. This may modify the interpretation of derates which are `delay_corner` specific that are currently applied or will be applied in the future. It is recommended to use the `-delay_corner` argument in MMC configurations to keep derates segregated properly among the corners.

DESCRIPTION

TCLCMD-1726

NAME

TCLCMD-1726

SUMMARY

set_sense -clock_leaf is not supported on arcs.

DESCRIPTION

TCLCMD-1727

NAME

TCLCMD-1727

SUMMARY

report_timing_context_data is only supported in context analysis. Read the hierarchical timing context before using this command.

DESCRIPTION

TCLCMD-1728

NAME

TCLCMD-1728

SUMMARY

The -view option should be specified for the command in MMMC mode.

DESCRIPTION

TCLCMD-1729

NAME

TCLCMD-1729

SUMMARY

is an invalid type, valid types are - latency, incr_delay, uncertainty.

DESCRIPTION

TCLCMD-1730

NAME

TCLCMD-1730

SUMMARY

INFO: EPBA bounded mode analysis is performed on total %u endpoints. A total of %u worst paths are reported as per max_paths and max_slack criteria. Total %u endpoints are reported in adaptive mode

DESCRIPTION

TCLCMD-1731

NAME

TCLCMD-1731

SUMMARY

Constraint will overwrite previous definitions on .

DESCRIPTION

TCLCMD-1732

NAME

TCLCMD-1732

SUMMARY

Constraint will overwrite applied previously on

DESCRIPTION

TCLCMD-1733

NAME

TCLCMD-1733

SUMMARY

Constraint applied previously is being reset.

DESCRIPTION

TCLCMD-1734

NAME

TCLCMD-1734

SUMMARY

The value of report_slack should not be greater than max_slack.

DESCRIPTION

TCLCMD-1735

NAME

TCLCMD-1735

SUMMARY

Since no -late/early option has been specified with -retime, tool will generate report based upon default analysis mode.

DESCRIPTION

TCLCMD-1736

NAME

TCLCMD-1736

SUMMARY

EPBA bounded Mode is not supported in AOCV analysis.

DESCRIPTION

TCLCMD-1737

NAME

TCLCMD-1737

SUMMARY

Library corresponding to path not found.

DESCRIPTION

TCLCMD-1738

NAME

TCLCMD-1738

SUMMARY

No options are supported with option -of_lib_pathsof get_libs command.

DESCRIPTION

TCLCMD-1740

NAME

TCLCMD-1740

SUMMARY

VT group for is already defined in library or set using set_threshold_voltage_group command, use -override option if you want to override the existing VT group.

DESCRIPTION

TCLCMD-1741

NAME

TCLCMD-1741

SUMMARY

-save_exhaustive_pba_data option is provided. Adaptive EPBA bounded mode will be ignored.

DESCRIPTION

TCLCMD-1742

NAME

TCLCMD-1742

SUMMARY

Period %f specified for clock corresponding to view may result in precision loss during analysis.

DESCRIPTION

TCLCMD-1743

NAME

TCLCMD-1743

SUMMARY

Nets having driver count higher than found in the design while performing PBA. For these nets GBA delay will be used.

DESCRIPTION

TCLCMD-1744

NAME

TCLCMD-1744

SUMMARY

Mismatch found in the content of target and reference directories, context comparison will abort.

DESCRIPTION

TCLCMD-1745

NAME

TCLCMD-1745

SUMMARY

No text files are found in the target and/or reference directories, context comparison will abort. Re-generate context with `-include_text_format` before comparing.

DESCRIPTION

TCLCMD-1746

NAME

TCLCMD-1746

SUMMARY

No context information is found for , context comparison will abort.

DESCRIPTION

TCLCMD-1747

NAME

TCLCMD-1747

SUMMARY

is not a recognized option of , valid options are .

DESCRIPTION

TCLCMD-1748

NAME

TCLCMD-1748

SUMMARY

Failed to read context text file .

DESCRIPTION

TCLCMD-1749

NAME

TCLCMD-1749

SUMMARY

Output file specified with option of command cannot be created. Specify a valid file path with write permissions.

DESCRIPTION

TCLCMD-1750

NAME

TCLCMD-1750

SUMMARY

Multiple lib cells found in given collection. Tool will consider the first one and ignore rest of these.

DESCRIPTION

TCLCMD-1751

NAME

TCLCMD-1751

SUMMARY

change_cell_pba call failed :

DESCRIPTION

TCLCMD-1752

NAME

TCLCMD-1752

SUMMARY

The combination specified in -vt_type is not valid

DESCRIPTION

TCLCMD-1753

NAME

TCLCMD-1753

SUMMARY

Reporting some IPD paths may have been missed for endpoints. Refer to verbose log file (logv) of the current run to get the list of such endpoints. You may increase the value of -nworst to report if more IPD paths for each endpoint exist.

DESCRIPTION

TCLCMD-1755

NAME

TCLCMD-1755

SUMMARY

File is not in the recognized .txt format, context comparison will abort.

DESCRIPTION

TCLCMD-1756

NAME

TCLCMD-1756

SUMMARY

INFO: VT-skew analysis was done on paths. Out of this, paths had a dominant VT group in the path. Total combinations were run for these VT-skew paths. Total combinations were actually retimed.

DESCRIPTION

TCLCMD-1757

NAME

TCLCMD-1757

SUMMARY

The software has detected an invalid value of cpr threshold for view . Please provide non-negative value of cpr threshold for this view.

DESCRIPTION

TCLCMD-1758

NAME

TCLCMD-1758

SUMMARY

context_data is not generated for view

DESCRIPTION

TCLCMD-1759

NAME

TCLCMD-1759

SUMMARY

Only single clock is supported with context_data properties, specify a single clock with -clock to fetch context_data properties.

DESCRIPTION

TCLCMD-1760

NAME

TCLCMD-1760

SUMMARY

context_data can be generated for a single port only.

DESCRIPTION

TCLCMD-1761

NAME

TCLCMD-1761

SUMMARY

context_data is only supported with port based clock mapping. Set global timing_context_port_based_clock_mapping to true and re-read the context before fetching the context_data.

DESCRIPTION

TCLCMD-1762

NAME

TCLCMD-1762

SUMMARY

The timing context loaded for instance has missing TWCPFR info on signal port (clock) and interface clock on reference. No credit has been added for paths related to these ports.

DESCRIPTION

TCLCMD-1763

NAME

TCLCMD-1763

SUMMARY

MMMC-1 options are used in while the design is initialized with MMMC-2 format. Use -help to get correct options.

DESCRIPTION

TCLCMD-1764

NAME

TCLCMD-1764

SUMMARY

Cannot copy constraints of this instance for ungroup

DESCRIPTION

TCLCMD-1765

NAME

TCLCMD-1765

SUMMARY

Cannot copy constraints of this instance for ungroup

DESCRIPTION

TCLCMD-1766

NAME

TCLCMD-1766

SUMMARY

****DIAG:** The software has encountered unexpected internal error while running this reporting command and timing reports generated by this command may not be reliable.

DESCRIPTION

TCLCMD-1767

NAME

TCLCMD-1767

SUMMARY

is defined with dangling reference pin

DESCRIPTION

TCLCMD-1776

NAME

TCLCMD-1776

SUMMARY

global cannot be used with global. To use , Please set to false.

DESCRIPTION

TCLCMD-1777

NAME

TCLCMD-1777

SUMMARY

timing_report_constraint_format is specified with some values which are not relevant to all drv_violation_type. Tool is reporting selectively only those fields which are relevant to drv_violation_type reported. Refer documentation for timing_report_constraint_format for further info.

DESCRIPTION

TCLCMD-1778

NAME

TCLCMD-1778

SUMMARY

Option -drv_fields is given priority over user specified globals

DESCRIPTION

TCLCMD-1779

NAME

TCLCMD-1779

SUMMARY

Option is required for

DESCRIPTION

TCLCMD-1780

NAME

TCLCMD-1780

SUMMARY

A positive integer is expected for -max_pins_per_drv

DESCRIPTION

TCLCMD-1781

NAME

TCLCMD-1781

SUMMARY

is not a legal value for -max_pins_per_drv

DESCRIPTION

TCLCMD-1782

NAME

TCLCMD-1782

SUMMARY

Field is not a legal option for

DESCRIPTION

TCLCMD-1783

NAME

TCLCMD-1783

SUMMARY

Field csv|txt is expected as an argument for -drv_output_format

DESCRIPTION

TCLCMD-1784

NAME

TCLCMD-1784

SUMMARY

Field in drv_output_format field is not supported in the current tool version. The -drv_output_format option will be set txt.

DESCRIPTION

TCLCMD-1785

NAME

TCLCMD-1785

SUMMARY

Invalid characters specified in constraints which is ignored in constraint application. Review constraint at this location.

DESCRIPTION

TCLCMD-1786

NAME

TCLCMD-1786

SUMMARY

INFO: Total paths were bounded by GBA during path based analysis.

DESCRIPTION

TCLCMD-1787

NAME

TCLCMD-1787

SUMMARY

CSV output for DRV report is supported only in single-line report format. -drv_output_format will default to txt.

DESCRIPTION

TCLCMD-1788

NAME

TCLCMD-1788

SUMMARY

-drv_output_format csv is supported exclusively with -drv_violation_type option. When setup/hold checks are enabled with DRV checks, -drv_output_format will default to txt.

DESCRIPTION

TCLCMD-1789

NAME

TCLCMD-1789

SUMMARY

INFO: EPBA bounded mode analysis is performed on total %u endpoints.

DESCRIPTION

TCLCMD-1790

NAME

TCLCMD-1790

SUMMARY

is sub-hierarchical instance. accepts only top level hierarchical instance

DESCRIPTION

TCLCMD-1791

NAME

TCLCMD-1791

SUMMARY

create_clock option -name does not support whitespace characters, use separate create_clock commands if intention is to create multiple clocks

DESCRIPTION

TCLCMD-1792

NAME

TCLCMD-1792

SUMMARY

The setting is only supported for flat STA runs.

DESCRIPTION

TCLCMD-1801

NAME

TCLCMD-1801

SUMMARY

CONST_HIER_EXC1: A block path without set_false_path is part of a top-level false path. Witness:

DESCRIPTION

TCLCMD-1802

NAME

TCLCMD-1802

SUMMARY

CONST_HIER_EXC2: A blocks false path is part of a top-level path without set_false_path.
Witness:

DESCRIPTION

TCLCMD-1803

NAME

TCLCMD-1803

SUMMARY

CONST_HIER_EXC3: A blocks false path is part of a top-level multicycle path. Witness:

DESCRIPTION

TCLCMD-1804

NAME

TCLCMD-1804

SUMMARY

CONST_HIER_EXC4: A blocks multicycle path is part of a top-level false path. Witness:

DESCRIPTION

TCLCMD-1805

NAME

TCLCMD-1805

SUMMARY

CONST_HIER_EXC5: A blocks single cycle path is part of a top-level multicycle path. Witness:

DESCRIPTION

TCLCMD-1806

NAME

TCLCMD-1806

SUMMARY

CONST_HIER_EXC6: A blocks multicycle path is part of a top-level single cycle path. Witness:

DESCRIPTION

TCLCMD-1807

NAME

TCLCMD-1807

SUMMARY

CONST_HIER_EXC7: A blocks multicycle path is part of a top-level multicycle path with a different cycle count. Witness:

DESCRIPTION

TCLCMD-1808

NAME

TCLCMD-1808

SUMMARY

CONST_HIER_EXC8: Witness:

DESCRIPTION

TCLCMD-1809

NAME

TCLCMD-1809

SUMMARY

CONST_HIER_EXC9: Witness:

DESCRIPTION

TCLCMD-1810

NAME

TCLCMD-1810

SUMMARY

CONST_HIER_EXC10: Witness:

DESCRIPTION

TCLCMD-1811

NAME

TCLCMD-1811

SUMMARY

CONST_HIER_EXC11: Witness:

DESCRIPTION

TCLCMD-1812

NAME

TCLCMD-1812

SUMMARY

CONST_HIER_EXC12: Witness:

DESCRIPTION

TCLCMD-1813

NAME

TCLCMD-1813

SUMMARY

CONST_HIER_EXC13: Witness:

DESCRIPTION

TCLCMD-1814

NAME

TCLCMD-1814

SUMMARY

CONST_HIER_EXC14: Witness:

DESCRIPTION

TCLCMD-1815

NAME

TCLCMD-1815

SUMMARY

CONST_HIER_EXC15: Witness:

DESCRIPTION

TCLCMD-1816

NAME

TCLCMD-1816

SUMMARY

CONST_HIER_EXC16: Witness:

DESCRIPTION

TCLCMD-1817

NAME

TCLCMD-1817

SUMMARY

CONST_HIER_EXC17: A block path without set_false_path is part of top-level paths and all such top-level paths are defined as false paths. Witness:

DESCRIPTION

TCLCMD-1818

NAME

TCLCMD-1818

SUMMARY

Either -retime option is not provided or provided with incompatible parameters while using -retime_vmin_views option.

DESCRIPTION

TCLCMD-1819

NAME

TCLCMD-1819

SUMMARY

incorrect configuration for vmin-view analysis, vmin-view analysis can not be done

DESCRIPTION

TCLCMD-1820

NAME

TCLCMD-1820

SUMMARY

Error encountered during Delta MIS Flow. Error Reason : .

DESCRIPTION

TCLCMD-1821

NAME

TCLCMD-1821

SUMMARY

View is invalid for reporting since it is not an active view.

DESCRIPTION

TCLCMD-1822

NAME

TCLCMD-1822

SUMMARY

No timing context is read at module using `read_timing_context` prior to writing timing context with `write_timing_context -skip_generation`.

DESCRIPTION

TCLCMD-1823

NAME

TCLCMD-1823

SUMMARY

incorrect include_only/exclude format specified

DESCRIPTION

TCLCMD-1824

NAME

TCLCMD-1824

SUMMARY

vtskew constraint :

DESCRIPTION

TCLCMD-1825

NAME

TCLCMD-1825

SUMMARY

Number of max_paths to be reported by report_timing in begin_end_pair has exhausted the limit of . Some paths may not be reported , max_paths limit for begin end pair can be increased by global timing_report_begin_end_pair_max_path_limit for more coverage.

DESCRIPTION

TCLCMD-1826

NAME

TCLCMD-1826

SUMMARY

data can not be reported for view, use or to run timing analysis for view

DESCRIPTION

TCLCMD-1827

NAME

TCLCMD-1827

SUMMARY

-setup is only supported when timing_report_check_timing_enable_clock_crossing_merge_polarity is set to true. Please set the global timing_report_check_timing_enable_clock_crossing_merge_polarity to true and re-run the command

DESCRIPTION

TCLCMD-1828

NAME

TCLCMD-1828

SUMMARY

No disk space available in temporary directory . Sufficient temporary storage is required when `timing_enable_low_memory_mode` is enabled. Please make disk space available in the current temporary directory - or, specify a new temporary directory with adequate space using the `TMPDIR` environment global and re-launch the session

DESCRIPTION

TCLCMD-1829

NAME

TCLCMD-1829

SUMMARY

Specifying view in both `and` and `-inactive` is not allowed. View is removed from `-inactive` list

DESCRIPTION

TCLCMD-1830

NAME

TCLCMD-1830

SUMMARY

timing_report_constraint_enable_verbose_drv_format will be set to false for max_cell_delay analysis internally.

DESCRIPTION

TCLCMD-1831

NAME

TCLCMD-1831

SUMMARY

The libcell could not be found for delay corner . The set_timing_derate command for this library-cell and delay-corner combination will be ignored.

DESCRIPTION

TCLCMD-1832

NAME

TCLCMD-1832

SUMMARY

Wild card pattern is specified in constraints file : without get_* commands.

DESCRIPTION

TCLCMD-1833

NAME

TCLCMD-1833

SUMMARY

Wild card pattern is specified in constraints without get_* commands.

DESCRIPTION

TCLCMD-1834

NAME

TCLCMD-1834

SUMMARY

INFO: Accurate spatial exhaustive post process adjust: paths retimed as chipsize as BoundingBox,
paths retimed as actual BoundingBox

DESCRIPTION

TCLCMD-1835

NAME

TCLCMD-1835

SUMMARY

is not a hierarchical instance. current_instance works on hierarchical instance only

DESCRIPTION

TCLCMD-1836

NAME

TCLCMD-1836

SUMMARY

Incorrect instance names given to -skip_inputs option for write_timing_context command. Please reissue write_timing_context command with the correct instance names.

DESCRIPTION

TCLCMD-1837

NAME

TCLCMD-1837

SUMMARY

The object type provided to the command is not currently supported by the timing Tcl API schema. You should use get_db to query DB object types which are not supported by .

DESCRIPTION

TCLCMD-1838

NAME

TCLCMD-1838

SUMMARY

There were interactive constraints previously set on `constraint_mode` . Interactive constraints applied in between appends will not be added to the sdc files list.

DESCRIPTION

TCLCMD-1839

NAME

TCLCMD-1839

SUMMARY

Incorrect setting of `global timing_enable_uncertainty_for_clock_checks` is applied. Please correct the `global timing_enable_uncertainty_for_clock_checks` value. Permissible values are `true` | `all` | `false` | `none` | `{pulse_width period skew}`

DESCRIPTION

TCLCMD-1840

NAME

TCLCMD-1840

SUMMARY

The constraint, when used with the -no_segment option, requires at least one pin/port to be specified in -from/-through/-to arguments. The constraint has been ignored.

DESCRIPTION

TCLCMD-1841

NAME

TCLCMD-1841

SUMMARY

Tool has found valid matching with given pattern so it will not search for any other object types matching this pattern. In general, it is recommended to add the explicit commands (e.g. get_cells/get_nets etc) with search pattern before passing to the command.

DESCRIPTION

TCLCMD-1842

NAME

TCLCMD-1842

SUMMARY

Dynamic delay is mean only. set_timing_derate -dynamic -sigma will be ignored.

DESCRIPTION

TCLCMD-1843

NAME

TCLCMD-1843

SUMMARY

set_sense -pulse cannot be applied on output pin of a library pulse generator cell, . The constraint will be ignored.

DESCRIPTION

This warning is issued when the constraint mentioned in summary message is applied to the output pin of a library pulse generator cell. Library pulse generator cell specification takes precedence, and the constraint is ignored. In order to avoid this warning, please remove the constraint from this pin.

TCLCMD-1844

NAME

TCLCMD-1844

SUMMARY

VMIN Error:

DESCRIPTION

TCLCMD-1845

NAME

TCLCMD-1845

SUMMARY

Max Violation Reporting limit of exhausted for clock/data check reporting . Increase the limit from global .

DESCRIPTION

TCLCMD-1846

NAME

TCLCMD-1846

SUMMARY

Max Violation Reporting limit of exhausted for DRV reporting. Increase the limit from global .

DESCRIPTION

TCLCMD-1847

NAME

TCLCMD-1847

SUMMARY

-max_pins_per_drv specified without -all_violators is ignored.

DESCRIPTION

TCLCMD-1848

NAME

TCLCMD-1848

SUMMARY

Could not create directory to dump context driving load sdcs

DESCRIPTION

TCLCMD-1849

NAME

TCLCMD-1849

SUMMARY

Hierarchical timing context data has been reset due to `reset_clock -all` command. User should re-load hierarchical timing context data via `read_timing_context` before next timing analysis.

DESCRIPTION

TCLCMD-1850

NAME

TCLCMD-1850

SUMMARY

Vmin view is not allowed in `set_analysis_view`. The command is ignored.

DESCRIPTION

TCLCMD-1851

NAME

TCLCMD-1851

SUMMARY

`update_delay_corner -vmin_pg_net_voltages` should not be specified before `set_analysis_view`. The command is ignored.

DESCRIPTION

TCLCMD-1852

NAME

TCLCMD-1852

SUMMARY

The `write_timing_context` command requires either the use of `-module/-instances` options, or set the modules/instances in `global timing_context_module_list`.

DESCRIPTION

TCLCMD-1853

NAME

TCLCMD-1853

SUMMARY

Both the `timing_context_module_list` global and `-module/-instances` options are used for the command `write_timing_context`. So, `-module/-instances` options will be ignored.

DESCRIPTION

TCLCMD-1854

NAME

TCLCMD-1854

SUMMARY

Vmin flow is not supported in Innvows and vmin view expanded will not happen. The command is ignored.

DESCRIPTION

TCLCMD-1855

NAME

TCLCMD-1855

SUMMARY

set_vt_skew_config -unique_vt_mode is incompatible with PBA analysis.

DESCRIPTION

TCLCMD-1856

NAME

TCLCMD-1856

SUMMARY

Heterogeneous VT-skew views not supported

DESCRIPTION

TCLCMD-1857

NAME

TCLCMD-1857

SUMMARY

INFO: timing_path collection reset for : .

DESCRIPTION

TCLCMD-1858

NAME

TCLCMD-1858

SUMMARY

is not a hierarchical instance. set_mim_instances works on hierarchical instance only

DESCRIPTION

TCLCMD-1859

NAME

TCLCMD-1859

SUMMARY

No sibling instances are specified, this specification of set_mim_instances will be ignored

DESCRIPTION

TCLCMD-1860

NAME

TCLCMD-1860

SUMMARY

All the MIM instances do not belong to the same module, this specification of set_mim_instances will be ignored.

DESCRIPTION

TCLCMD-1861

NAME

TCLCMD-1861

SUMMARY

Root instance is already configured with a different set of sibling instances, this specification of set_mim_instances will be ignored.

DESCRIPTION

TCLCMD-1862

NAME

TCLCMD-1862

SUMMARY

INVALID option provided for .

DESCRIPTION

TCLCMD-1863

NAME

TCLCMD-1863

SUMMARY

Non Bidirectional Input/Output port passed to set_bidir_loopback_disable/
reset_bidir_loopback_disable

DESCRIPTION

TCLCMD-1864

NAME

TCLCMD-1864

SUMMARY

Timing context has not been generated for instance , timing constraints generated may not be correct.

DESCRIPTION

TCLCMD-1865

NAME

TCLCMD-1865

SUMMARY

Total violations count exceed the max_paths limit %ld for report_analysis_summary for analysis mode: , and view: . Limit is controlled by global timing_path_based_report_analysis_summary_max_paths_limit and increasing this can lead to runtime and memory overhead.

DESCRIPTION

TCLCMD-1866

NAME

TCLCMD-1866

SUMMARY

Invalid VT Group specified in set_vt_skew_config include_only/exclude command

DESCRIPTION

TCLCMD-1867

NAME

TCLCMD-1867

SUMMARY

VT Group defined for library will be overridden by VT Group

DESCRIPTION

TCLCMD-1868

NAME

TCLCMD-1868

SUMMARY

No aggressor information found for pin

DESCRIPTION

TCLCMD-1869

NAME

TCLCMD-1869

SUMMARY

output_collection name is same as input collection, please specify different output_collection name

DESCRIPTION

TCLCMD-1870

NAME

TCLCMD-1870

SUMMARY

Two Die Analysis:

DESCRIPTION

Timing Constraint Hierarchical Signoff

- [CONST_HIER_CG](#)
- [CONST_HIER_CLK](#)
- [CONST_HIER_CTR](#)
- [CONST_HIER_DAT](#)
- [CONST_HIER_EXC](#)
- [CONST_HIER_GRP](#)
- [CONST_HIER_HFN](#)
- [CONST_HIER_IO](#)
- [CONST_HIER_ITR](#)
- [CONST_HIER_LAT](#)
- [CONST_HIER_OLD](#)
- [CONST_HIER_PLS](#)
- [CONST_HIER_SCA](#)
- [CONST_HIER_SDT](#)
- [CONST_HIER_TIM](#)
- [CONST_HIER_UNC](#)

CONST_HIER_CG

- [CONST_HIER_CG1](#)
- [CONST_HIER_CG2](#)

CONST_HIER_CG1

NAME

CONST_HIER_CG1 (CONSTRAINT_HIER_SIGNOFF)

MESSAGE

Inconsistent `set_clock_gating_check` at top-level vs. block.

DEFAULT SEVERITY

Warning

DESCRIPTION

Indicates that an inconsistency in `set_clock_gating_check` command is found between top level and block level constraints. The following are the possible causes for this violation:

- `set_clock_gating_check` command specified at either top level or block level
- `set_clock_gating_check` commands specified at both top level and block level but with different values

This rule is checked by `check_target_constraints` command and the results can be reported by `report_rule_check` command.

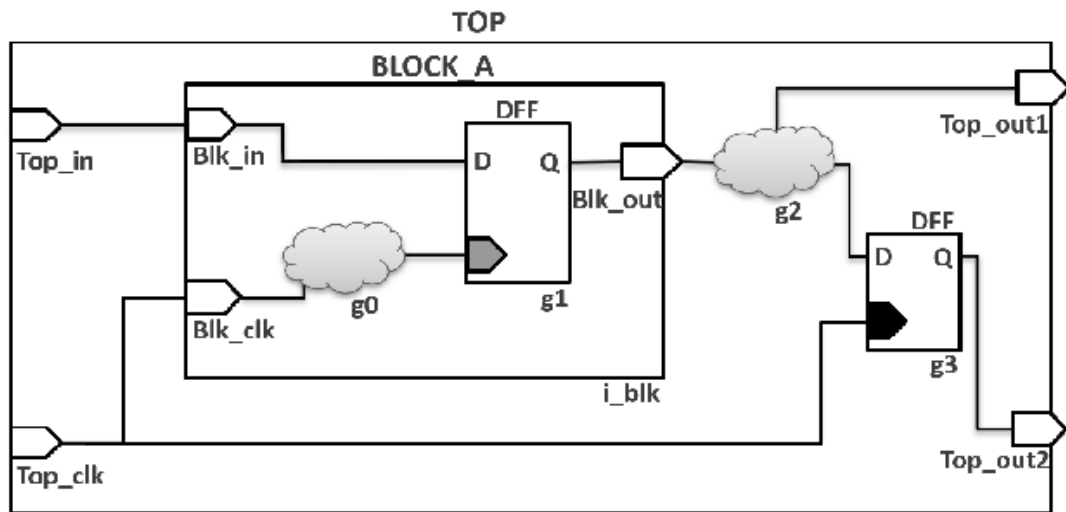
This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
block_view_name	string	Block view name
top_view_name	string	Top view name

scope	object	Scope
block_pin	object	Block pin
top_pin	object	Top pin
block_clock	object	Block clock
top_clock	object	Top clock
type	string	Type
block_value	object_list	block_value
top_value	object_list	top_value
block_file_line	object_list	Block file line number
block_file_name	object_list	Block file name
top_file_line	object_list	Top file line number
top_file_name	object_list	Top file name

EXAMPLE

- For the figure below, the following constraints contain `set_clock_gating_check` commands specified in both top level and block level constraints. This rule is reported for the `set_clock_gating_check` commands, because they are specified with different values.



Top level constraint

```
set_clock_gating_check -high -setup 0.4 -hold 0.2 [get_cells i_blk/g0]
```

Block level constraint (instance scope: i_blk)

```
set_clock_gating_check -high -setup 0.8 -hold 0.4 [get_cells g0]
```

CONST_HIER_CG2

NAME

CONST_HIER_CG2 (CONSTRAINT_HIER_SIGNOFF)

MESSAGE

Inconsistent `set_disable_clock_gating_check` between block-level and top-level

DEFAULT SEVERITY

Warning

DESCRIPTION

Indicates that an inconsistency in `set_disable_clock_gating_check` command is found between top level and block level constraints. The following are the possible causes for this violation:

- `set_disable_clock_gating_check` command specified at either top level or block level

This rule is checked by `check_target_constraints` command and the results can be reported by `report_rule_check` command.

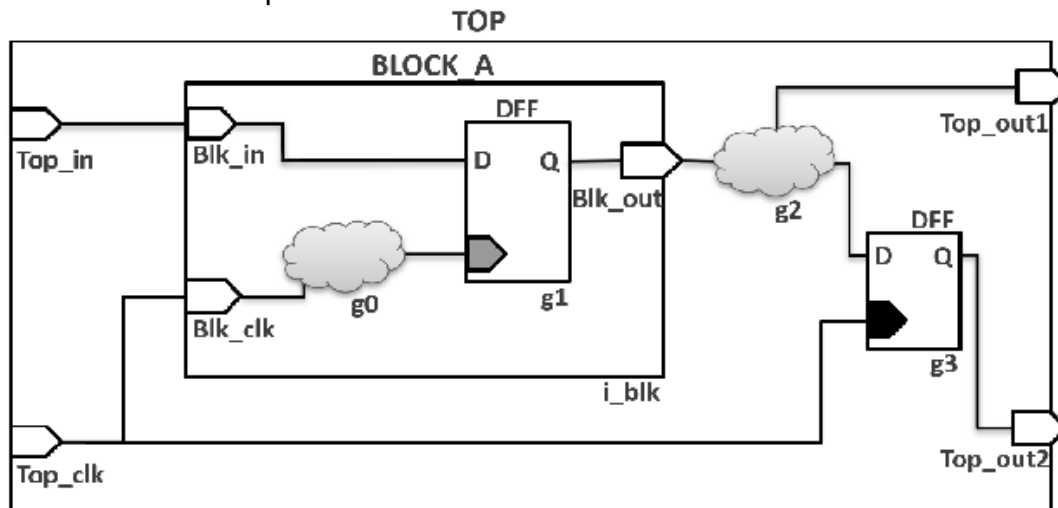
This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
block_view_name	string	Block view name
top_view_name	string	Top view name
scope	object	Scope
block_pin	object	Block pin
top_pin	object	Top pin

type	string	Type
block_file_line	unsigned	Block file line number
block_file_name	string	Block file name
top_file_line	unsigned	Top file line number
top_file_name	string	Top file name

EXAMPLE

- For the figure below, the following constraints contain `set_disable_clock_gating_check` command specified in top level constraints only. This rule is reported for the `set_disable_clock_gating_check` command, because it is not specified in block level constraints.



Top level constraint

```
set_disable_clock_gating_check [get_cells i_blk/g0]
```

Block level constraint (instance scope: i_blk)

No `set_disable_clock_gating_check` command specified on the instance g0

CONST_HIER_CLK

- [CONST_HIER_CLK1](#)
- [CONST_HIER_CLK2a](#)
- [CONST_HIER_CLK2b](#)
- [CONST_HIER_CLK2c](#)
- [CONST_HIER_CLK2d](#)
- [CONST_HIER_CLK2e](#)
- [CONST_HIER_CLK2f](#)
- [CONST_HIER_CLK3](#)
- [CONST_HIER_CLK4](#)
- [CONST_HIER_CLK6](#)
- [CONST_HIER_CLK7](#)
- [CONST_HIER_CLK8](#)

CONST_HIER_CLK1

NAME

CONST_HIER_CLK1 (CONSTRAINT_HIER_SIGNOFF)

MESSAGE

Multiple top-level clocks drive a block level clock

DEFAULT SEVERITY

Warning

DESCRIPTION

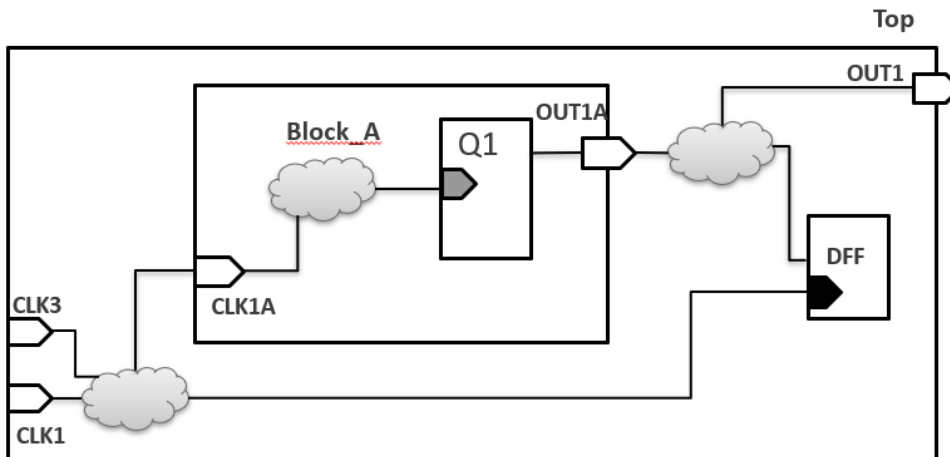
This rule reports that a block level clock constraint object has more than one top-level clocks in its fan-in cone. This rule is checked by [verify_hierarchical_timing_intent](#) command and results can be reported by [report_rule_check](#) command. Also, [report_hierarchical_clock_mapping](#) command can be used to understand how the clocks are mapped between the top level clocks and block level clocks to determine if this rule violation is due to an incomplete clock mapping.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
block_view_name	string	Block view name
top_view_name	string	Top view name
scope	object	Scope
block_clock_source	object	Block clock source
block_clocks	object_list	Block clocks
top_clocks	object_list	Top clocks

EXAMPLE

- For the figure below, this rule will be reported, if the block constraints file has a clock defined on its primary input like the following block level constraint and the top level constraints file has the following multiple clocks which are mapped to CLK1A and in the fanin cone of the CLK1A clock at the block level:



Top level constraints

```
create_clock -name top_c3 -period 10 [get_ports CLK3]
create_clock -name top_c1 -period 10 [get_ports CLK1]
```

Block level constraint

```
create_clock -name blk_c1 -period 8 [get_ports CLK1A]
```


CONST_HIER_CLK2a

NAME

CONST_HIER_CLK2a (CONSTRAINT_HIER_SIGNOFF)

MESSAGE

A block's clock is slower than a related top-level clock

DEFAULT SEVERITY

Warning

DESCRIPTION

This rule reports that the top-level clock's period is not greater than or equal to the period of the corresponding clock definition at the block. This rule is checked by [check_target_constraints](#) command and results can be reported by [report_rule_check](#) command. Also, [report_hierarchical_clock_mapping](#) command can be used to understand how the clocks are mapped between the top level clocks and block level clocks to determine if this rule violation is due to an incomplete clock mapping.

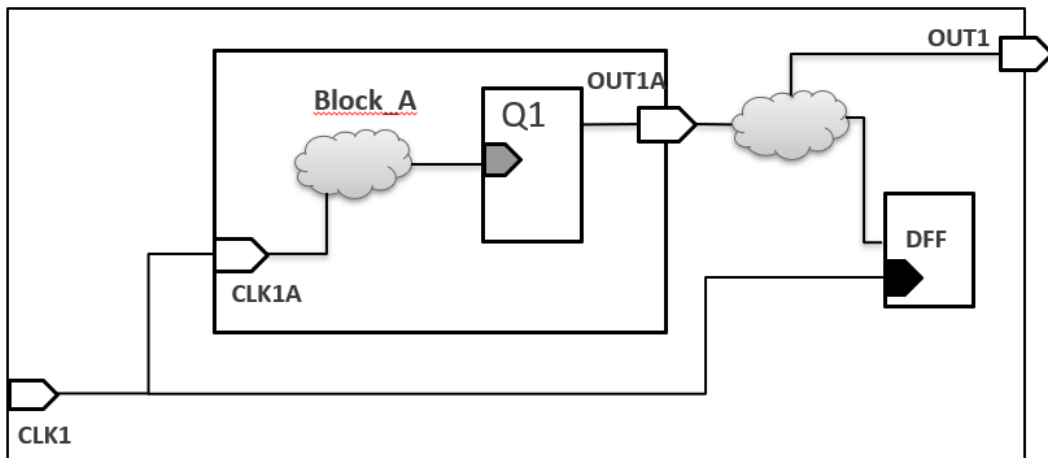
This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
block_view_name	string	Block view name
top_view_name	string	Top view name
scope	object	Scope
block_clock	object	Block clock
block_clock_source	object	Block clock source

top_clock	object	Top clock
block_file_line	unsigned	Block file line number
block_file_name	string	Block file name
top_file_line	unsigned	Top file line number
top_file_name	string	Top file name
block_time_period	double	Block time period
top_time_period	double	Top time period

EXAMPLE

- For the figure below, this rule will be reported, if the top level constraints file has the following clock which is mapped to CLK1A and in the fanin cone of the CLK1A clock at the block level and the block constraints file has a clock defined on its primary input like the following block level constraint. The Litmus software issues a warning because the clock's period at the top level is less than that at the block.



Top level constraint

```
create_clock -name top_c1 -period 3 [get_ports CLK1]
```

Block level constraint

```
create_clock -name blk_c1 -period 5 [get_ports CLK1A]
```

CONST_HIER_CLK2b

NAME

CONST_HIER_CLK2b (CONSTRAINT_HIER_SIGNOFF)

MESSAGE

A block's clock is faster than a related top-level clock

DEFAULT SEVERITY

Warning

DESCRIPTION

This rule reports that the top-level clock's period is not less than or equal to the period of the corresponding clock definition at the block. This rule is checked by [check_target_constraints](#) command and results can be reported by [report_rule_check](#) command.

Also, [report_hierarchical_clock_mapping](#) command can be used to understand how the clocks are mapped between the top level clocks and block level clocks to determine if this rule violation is due to an incomplete clock mapping.

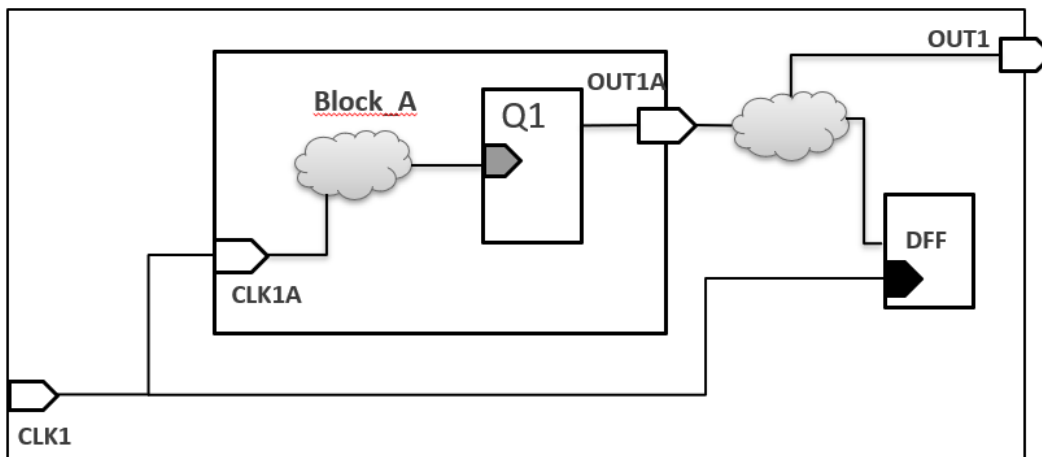
This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
block_view_name	string	Block view name
top_view_name	string	Top view name
scope	object	Scope
block_clock	object	Block clock
block_clock_source	object	Block clock source

top_clock	object	Top clock
block_file_line	unsigned	Block file line number
block_file_name	string	Block file name
top_file_line	unsigned	Top file line number
top_file_name	string	Top file name
block_time_period	double	Block time period
top_time_period	double	Top time period

EXAMPLE

- For the figure below, this rule will be reported, if the top level constraints file has the following clock which is mapped to CLK1A and the block level constraints file has a clock defined on its primary input like the following block level constraint. The Litmus software issues a warning because the clock's period at the top level is greater than that at the block.



Top level constraint

```
create_clock -name top_c1 -period 5 [get_ports CLK1]
```

Block level constraint

Litmus Rule Reference

Timing Constraint Hierarchical Signoff--CONST_HIER_CLK

```
create_clock -name blk_c1 -period 3 [get_ports CLK1A]
```

CONST_HIER_CLK2c

NAME

CONST_HIER_CLK2c (CONSTRAINT_HIER_SIGNOFF)

MESSAGE

A block's clock and a related top-level clock have different duty-cycle or waveform

DEFAULT SEVERITY

Warning

DESCRIPTION

This rule reports that a top-level clock's period is equal to the period of the corresponding clock definition at the block, but these two clocks have different duty cycle ((falling edge - rising edge)/period) or different waveform. This rule is checked by [check_target_constraints](#) command and results can be reported by [report_rule_check](#) command.

Also, [report_hierarchical_clock_mapping](#) command can be used to understand how the clocks are mapped between the top level clocks and block level clocks to determine if this rule violation is due to an incomplete clock mapping.

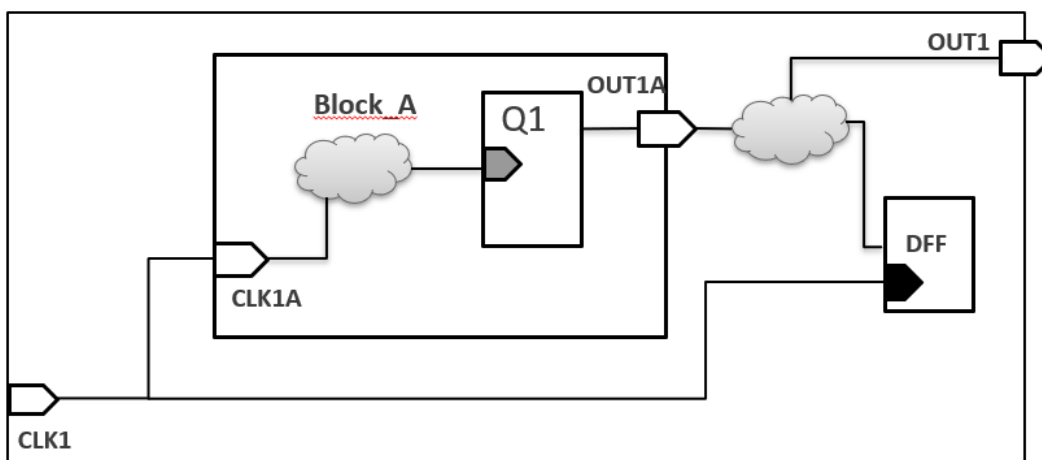
This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
block_view_name	string	Block view name
top_view_name	string	Top view name
scope	object	Scope
block_clock	object	Block clock

block_clock_source	object	Block clock source
top_clock	object	Top clock
block_file_line	unsigned	Block file line number
block_file_name	string	Block file name
top_file_line	unsigned	Top file line number
top_file_name	string	Top file name
block_time_period	double	Block time period
top_time_period	double	Top time period

EXAMPLE

- For the figure below, this rule will be reported, if the top level constraints file has the following clock which are mapped to CLK1A and the block level constraints file has a clock defined on its primary input like the following block level constraint. The Litmus software issues a warning because the clock's period at the top level is equal to that at the block, but their duty cycles are 50% and 40%, respectively.



Top level constraint

```
create_clock -period 10 [get_ports CLK1]
```


Block level constraint

```
create_clock -period 10 [get_ports CLK1A] -waveform {0 4}
```

CONST_HIER_CLK2d

NAME

CONST_HIER_CLK2d (CONSTRAINT_HIER_SIGNOFF)

MESSAGE

A block's clock and a related top-level clock have different phases

DEFAULT SEVERITY

Warning

DESCRIPTION

Indicates that an inconsistency in clock phase is found between mapped top level and block level clocks. The following are the possible causes for this violation:

- Mapped top level and block level clocks defined with different clock phases
- Block level clock is in the clock tree of mapped top level clock but with different clock phase

This rule is checked by `check_target_constraints` command and the results can be reported by `report_rule_check` command.

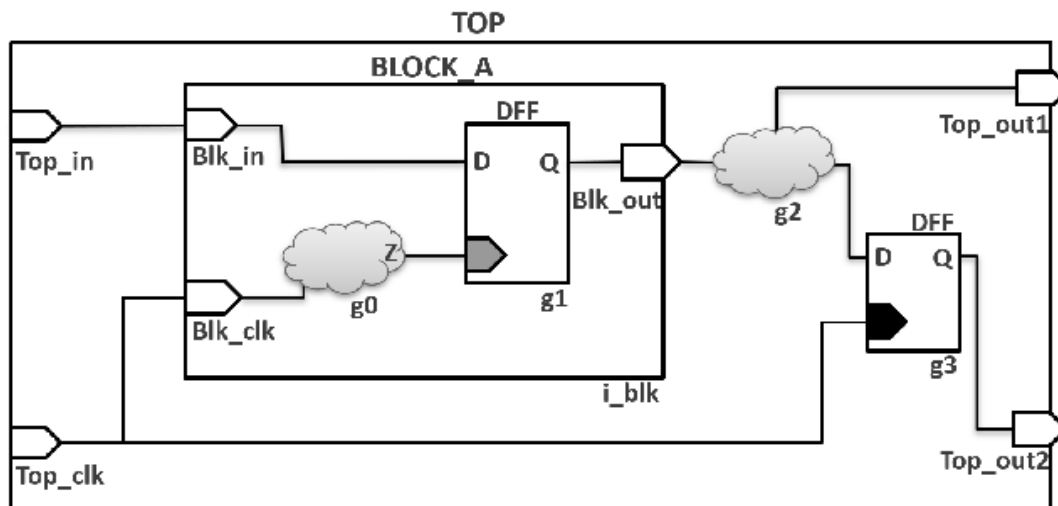
This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
block_view_name	string	Block view name
top_view_name	string	Top view name
scope	object	Scope
block_clock	object	Block clock

block_port	object	Block port
top_clock	object	Top clock
block_polarity	string	Block polarity
top_polarity	string	Top polarity
block_file_line	unsigned	Block file line number
block_file_name	string	Block file name
top_file_line	unsigned	Top file line number
top_file_name	string	Top file name

EXAMPLE

- For the figure below, the following contains generated clocks specified in both top level and block level constraints. This rule is reported for the generated clocks Top_gclk and Blk_gclk, because they are mapped but defined with different clock phases (`-invert`).



Top level constraints

```
create_clock -name Top_clk -period 2 [get_ports Top_clk]
create_generated_clock -name Top_gclk -source [get_ports Top_clk] -master_clock
[get_clocks Top_clk] -combinational [get_pins i_blk/g0/Z] -add
```

Block level constraints (instance scope: i_blk)

```
create_clock -name Blk_clk -period 2 [get_ports Blk_clk]
create_generated_clock -name Blk_gclk -source [get_ports Blk_clk] -master_clock
[get_clocks Blk_clk] -combinational [get_pins g0/Z] -add -invert
```

Hierarchical clock mapping

```
Top_clk is mapped to Blk_clk
Top_gclk is mapped to Blk_gclk
```

CONST_HIER_CLK2e

NAME

CONST_HIER_CLK2e (CONSTRAINT_HIER_SIGNOFF)

MESSAGE

A block's `create_generated_clock` is mapped to a top-level `create_clock`

DEFAULT SEVERITY

Warning

DESCRIPTION

Indicates that an inconsistency in clock type is found between mapped top level and block level clocks. The following are the possible causes for this violation:

- Block level generated clock is mapped to top level real clock (Not both of top-level clock and block level clock are defined inside a block)

This rule is checked by `check_target_constraints` command, and the results can be reported by `report_rule_check` command.

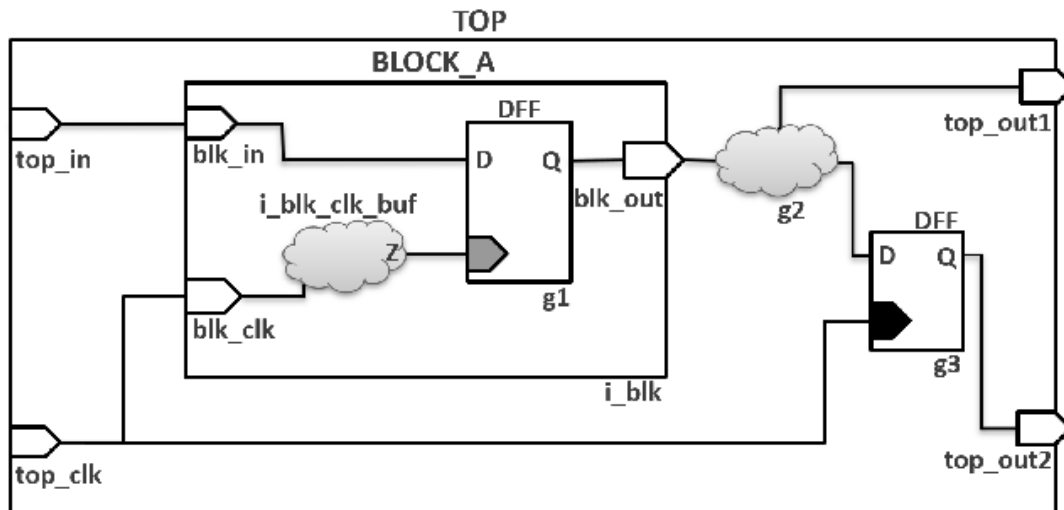
This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
block_view_name	string	Block view name
top_view_name	string	Top view name
scope	object	Scope
block_clock	object	Block clock

block_port	object	Block port
top_clock	object	Top clock
block_file_line	unsigned	Block file line number
block_file_name	string	Block file name
top_file_line	unsigned	Top file line number
top_file_name	string	Top file name

EXAMPLE

- For the figure below, the following example contains two clocks specified in top level constraints and two clocks specified in block level constraints. This rule is reported for the clocks top_gclk and blk_gclk, because they are mapped but defined with different clock types. Clock top_gclk is a real clock by `create_clock` command and clock blk_gclk is a generated clock by `create_generated_clock` command.



Top level constraints

```
create_clock -name top_clk -period 2 [get_ports top_clk]
create_clock -name top_gclk -period 2 [get_pins i_blk/blk_clk]
```

Block level constraints (instance scope: i_blk)

```
create_clock -name blk_clk -period 2 [get_ports blk_clk]
create_generated_clock -name blk_gclk -source [get_ports blk_clk] -master_clock
[get_clocks blk_clk] -combinational [get_ports blk_clk] -add
```

Hierarchical clock mapping

```
top_clk is mapped to blk_clk
top_gclk is mapped to blk_gclk
```

CONST_HIER_CLK2f

NAME

CONST_HIER_CLK2f (CONSTRAINT_HIER_SIGNOFF)

MESSAGE

Mapped internal top-level and block clocks are defined with different clock types

DEFAULT SEVERITY

Warning

DESCRIPTION

Indicates that an inconsistency in clock type is found between mapped internal top level and block level clocks. The following are the possible causes for this violation:

- Block level generated clock is mapped to top level real clock (Both of top-level clock and block level clock are defined inside a block)
- Block level real clock is mapped to top level generated clock (Both of top-level clock and block level clock are defined inside a block)

This rule is checked by `check_target_constraints` command, and the results can be reported by `report_rule_check` command.

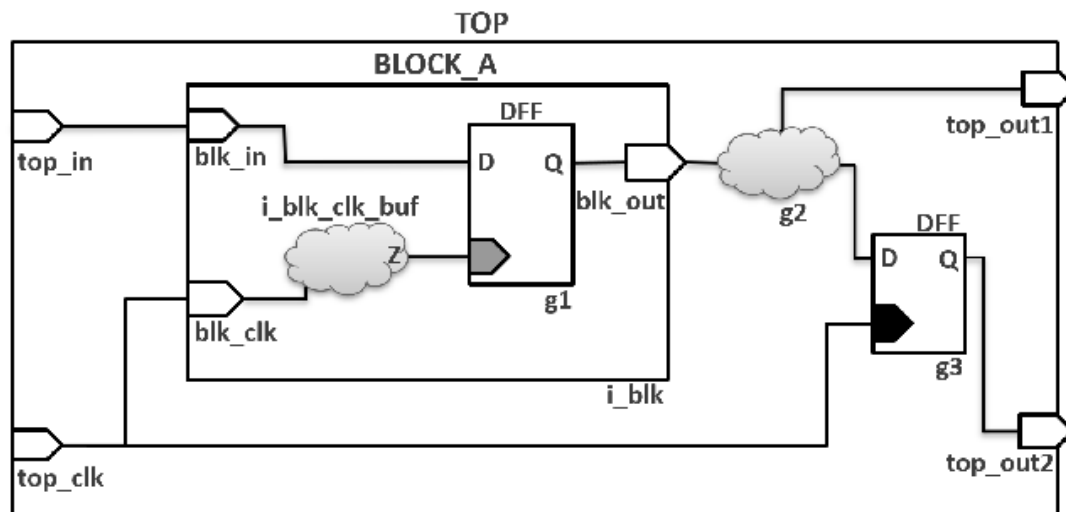
This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
block_view_name	string	Block view name
top_view_name	string	Top view name
scope	object	Scope

block_clock	object	Block clock
block_port	object	Block port
top_clock	object	Top clock
block_file_line	unsigned	Block file line number
block_file_name	string	Block file name
top_file_line	unsigned	Top file line number
top_file_name	string	Top file name

EXAMPLE

- For the figure below, the following example contains two clocks specified in top level constraints and two clocks specified in block level constraints. This rule is reported for the clocks top_gclk and blk_gclk, because they are mapped but defined with different clock types. Clock top_gclk is a real clock by `create_clock` command and clock blk_gclk is a generated clock by `create_generated_clock` command.



Top level constraints

```
create_clock -name top_clk -period 2 [get_ports top_clk]
create_clock -name top_gclk -period 2 [get_pins i_blk/i_blk_clk_buf/Z]
```

Block level constraints (instance scope: i_blk)

```
create_clock -name blk_clk -period 2 [get_ports blk_clk]
create_generated_clock -name blk_gclk -source [get_ports blk_clk] -master_clock
[get_clocks blk_clk] -combinational [get_pins i_blk_clk_buf/Z] -add
```

Hierarchical clock mapping

```
top_clk is mapped to blk_clk
top_gclk is mapped to blk_gclk
```

CONST_HIER_CLK3

NAME

CONST_HIER_CLK3 (CONSTRAINT_HIER_SIGNOFF)

MESSAGE

Missing top-level clock driving a block level clock source

DEFAULT SEVERITY

Warning

DESCRIPTION

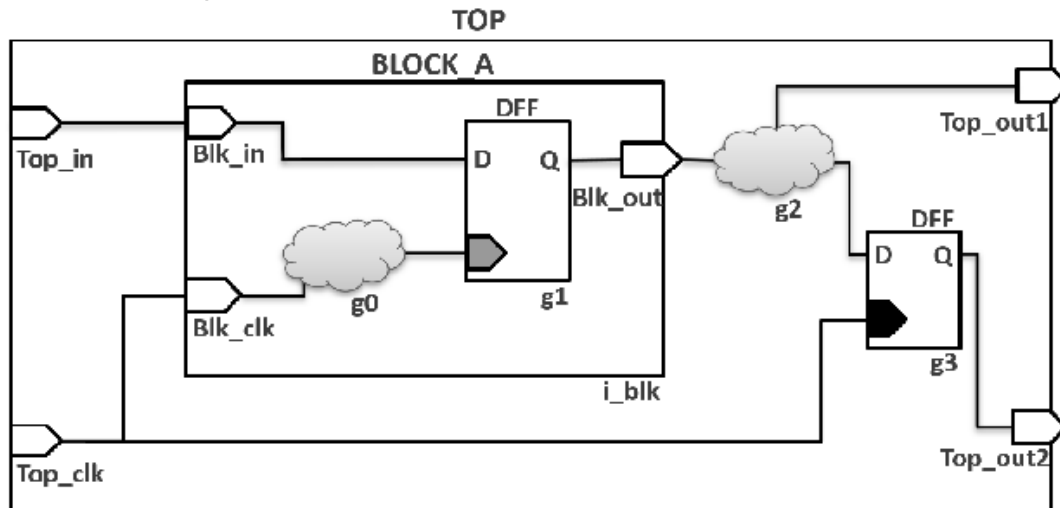
This rule reports that top level constraint does not have a clock defined in the fan-in cone of a block level clock source. This rule is checked by [check_target_constraints](#) command and results can be reported by [report_rule_check](#) command.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
block_view_name	string	Block view name
top_view_name	string	Top view name
scope	object	Scope
block_clock	object	Block clock
block_port	object	Block port
block_file_line	unsigned	Block file line number
block_file_name	string	Block file name

EXAMPLE

- For the figure below, if the block constraints file has a clock defined on its primary input Blk_clk like the following block level constraint, then this rule reports a violation if there is no clock from top level driving the BLOCK_A/Blk_clk to indicate that there is a potential missing clock in the top level constraints file.



Top level constraint (if missing)

```
create_clock -name Top_clk -period 10 [get_ports Top_clk]
```

Block level constraint

```
create_clock -name Blk_clk -period 8 [get_ports Blk_clk]
```

CONST_HIER_CLK4

NAME

CONST_HIER_CLK4 (CONSTRAINT_HIER_SIGNOFF)

MESSAGE

Missing block clock on block port/pin connected to a top-level clock source

DEFAULT SEVERITY

Warning

DESCRIPTION

This rule reports that a block's constraint file does not define a clock on a port or pin that is in the clock tree, or is a source, of a top-level clock object. In the case that such a block clock is defined, this rule reports if the block clock and the top-level clock are not defined as equivalent. Note that [report_hierarchical_clock_mapping](#) command can be used to understand how the clocks are mapped between the top level clocks and block level clocks to determine if this rule violation is due to incomplete clock mapping. This rule is checked by [check_target_constraints](#) command and results can be reported by [report_rule_check](#) command.

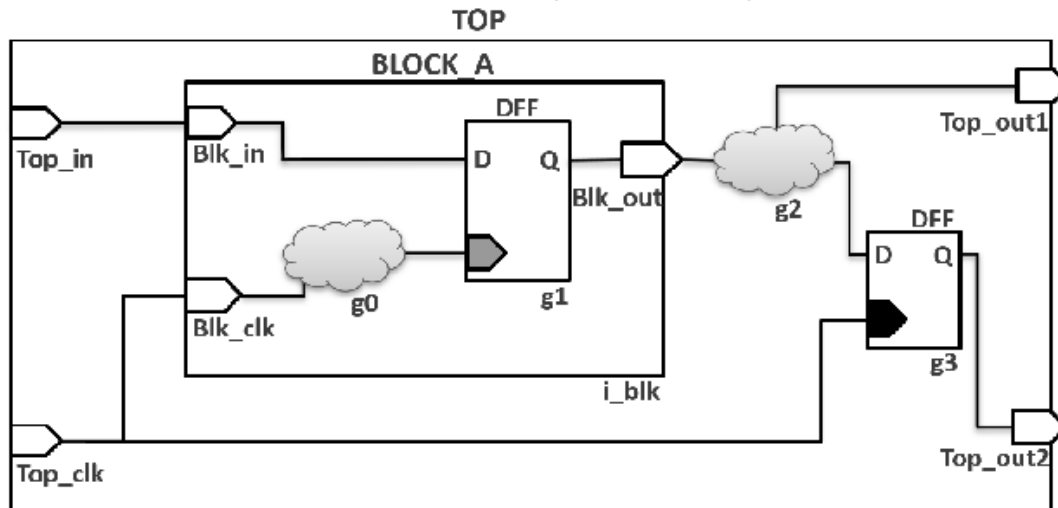
This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
block_view_name	string	Block view name
top_view_name	string	Top view name
scope	object	Scope
block_port	object	Block port

top_clock	object	Top clock
top_lib_gen_clk_file_line	unsigned	Top_Lib_Gen_Clk_File_Line
top_lib_gen_clk_file_name	string	Top_Lib_Gen_Clk_File_Name
top_file_line	unsigned	Top file line number
top_file_name	string	Top file name

EXAMPLE

- For the figure below, this rule will report a violation if the top level port Top_clk is connected to i_blk/Blk_clk, and the top level constraints file contains the following, but the block level constraints file of block i_blk is missing the following block level command:



Top level constraint

```
create_clock -period 10 [get_ports Top_clk] -name Top_clk
```

Block level constraint (instance scope: i_blk)

```
create_clock -period 10 [get_ports Blk_clk] -name Blk_clk
```

Hierarchical clock mapping

Top_clk is mapped to Blk_clk

CONST_HIER_CLK6

NAME

CONST_HIER_CLK6 (CONSTRAINT_HIER_SIGNOFF)

MESSAGE

Missing virtual clock at block level

DEFAULT SEVERITY

Warning

DESCRIPTION

This rule reports that a block's constraint file does not define a virtual clock for a top level equivalent clock which launches or captures timing path data through a block port. Note that [report_hierarchical_clock_mapping](#) command can be used to understand how the clocks are mapped between the top level clocks and block level clocks to determine if this rule violation is due to incomplete clock mapping. This rule is checked by [check_target_constraints](#) command and results can be reported by [report_rule_check](#) command.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
block_view_name	string	Block View Name
top_view_name	string	Top View Name
scope	object	Block Instance

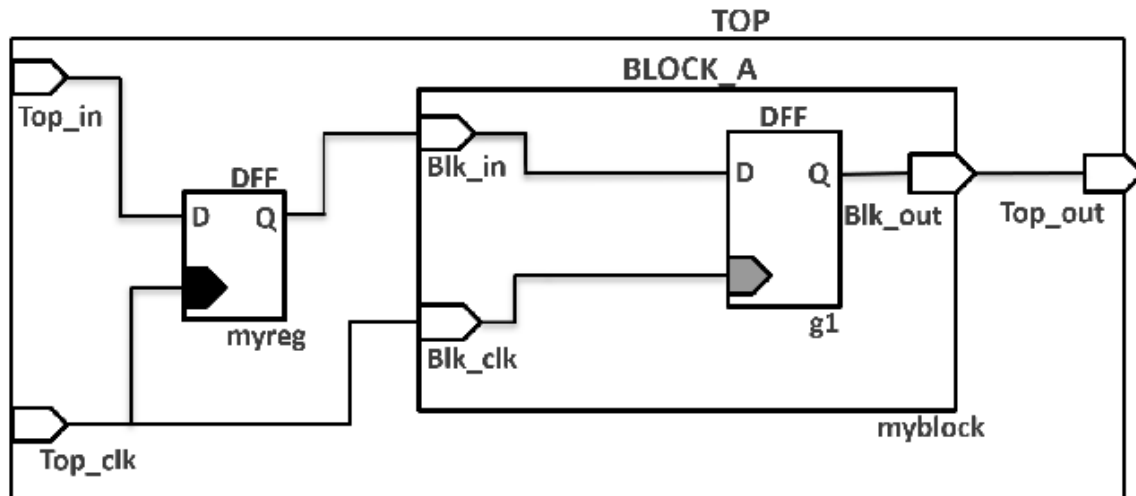
top_clock	object	Top-level clock definition
block_port_witnesses	object_list	Block ports witnesses
top_file_line	unsigned	Top file number
top_file_name	string	Top file name

In addition, common rule occurrence attribute `type` provides the possible causes for the violation.

Attribute	Value	Description
type	BLOCK_MISSING_CLOCK	Missing clock at block level

EXAMPLE

- For the figure below, top level port Top_clk drives the clock pin of top level flop myreg. The flop myreg is connected to block pin myblock/Blk_in. The top level constraints file contains the following constraint. This rule will report a violation if the block level constraints file of block myblock is missing a command like the following block level command for defining a virtual clock for the top level equivalent clock.



Top level constraint

```
create_clock -name TCK -add -period 4.0 -waveform {0.0 2.0} [get_ports Top_clk]
```

Block level constraint (instance scope: myblock)

```
create_clock -name BCK -period 4.0 -waveform {0.0 2.0}
```

CONST_HIER_CLK7

NAME

CONST_HIER_CLK7 (CONSTRAINT_HIER_SIGNOFF)

MESSAGE

Inconsistent `set_sense/set_clock_sense` at the top-level vs. block

DEFAULT SEVERITY

Warning

DESCRIPTION

Indicates that an inconsistency in `set_sense/set_clock_sense` command is found between top level and block level constraints. The following are the possible causes for this violation:

- `set_sense/set_clock_sense` command specified at either top level or block level
- `set_sense/set_clock_sense` commands specified at both top level and block level but with different values

This rule is checked by `check_target_constraints` command and the results can be reported by `report_rule_check` command.

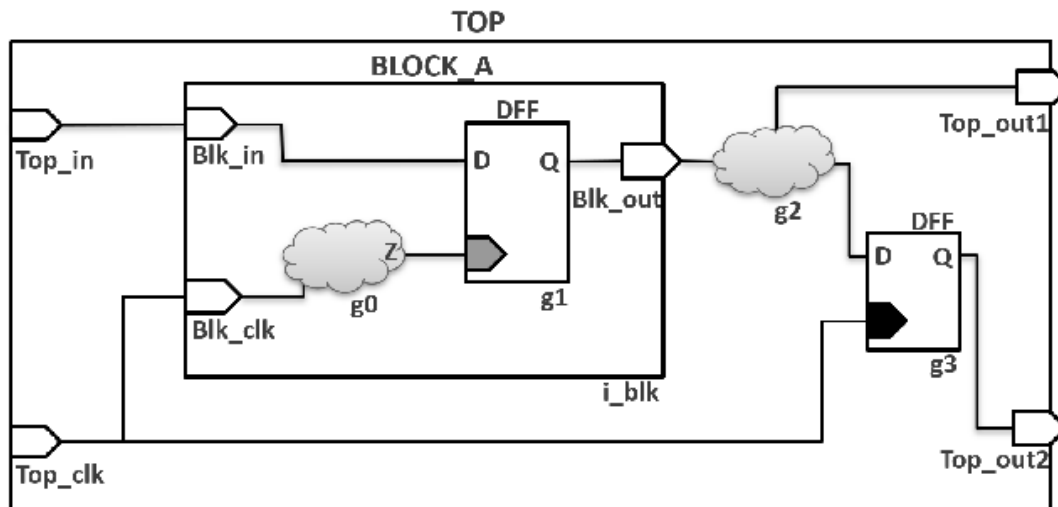
This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
block_view_name	string	Block view name
top_view_name	string	Top view name
scope	object	Scope

block_port	object	Block port
type	string	Type
block_file_line	unsigned	Block file line number
block_file_name	string	Block file name
top_file_line	unsigned	Top file line number
top_file_name	string	Top file name

EXAMPLE

- For the figure below, the following contains `set_sense` command specified in top level constraints only. This rule is reported for the `set_sense` command, because it is not specified in block level constraints.



Top level constraint

```
set_sense -stop_propagation [get_pins i_blk/g0/Z]
```

Block level constraint (instance scope: i_blk)

No `set_sense` command specified on the pin g0/Z.

CONST_HIER_CLK8

NAME

CONST_HIER_CLK8 (CONSTRAINT_HIER_SIGNOFF)

MESSAGE

Inconsistent `set_propagated_clock` at the top-level vs. block

DEFAULT SEVERITY

Warning

DESCRIPTION

Indicates that an inconsistency in propagated clock specification is found between mapped top level and block level clocks. The following are the possible causes for this violation:

- Top level clock is an ideal clock but mapped block level clock is a propagated clock
- Top level clock is a propagated clock but mapped block level clock is an ideal clock

This rule is checked by `check_target_constraints` command and the results can be reported by `report_rule_check` command.

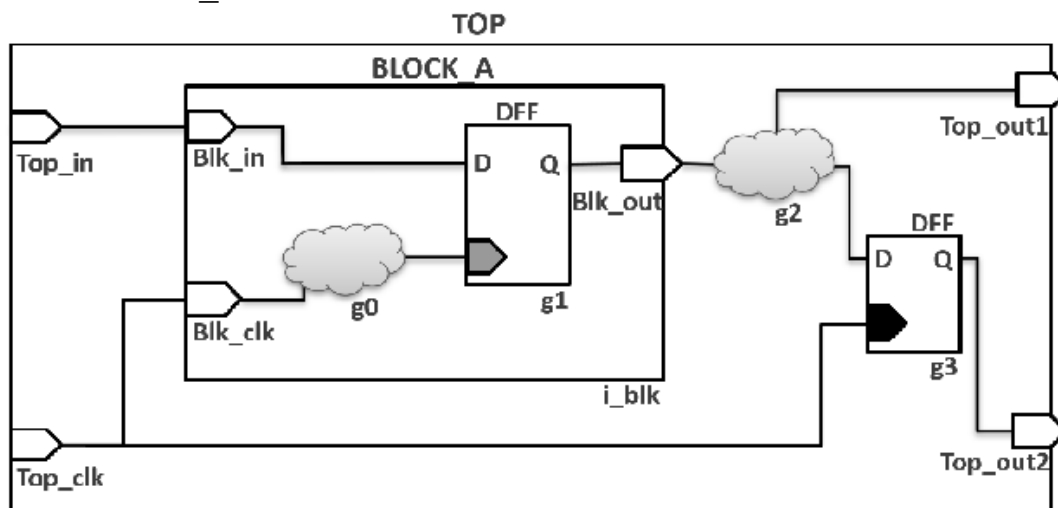
This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
block_view_name	string	Block view name
top_view_name	string	Top view name
scope	object	Scope
block_clock	object	Block clock

top_clock	object	Top clock
block_pin	object	Block Pin
top_pin	object	Top Pin
type	string	Type
block_file_line	unsigned	Block file line number
block_file_name	string	Block file name
top_file_line	unsigned	Top file line number
top_file_name	string	Top file name

EXAMPLE

- For the figure below, the following constraints contain a propagated clock specified in top level constraints and an ideal clock specified in block level constraints. This rule is reported for the clocks Top_clk and Blk_clk, because they are mapped but with different propagated clock specification. Clock Top_clk is a propagated clock by `set_propagated_clock` command and clock Blk_clk is an ideal clock.



Top level constraints

```
create_clock -name Top_clk -period 2 [get_ports Top_clk]  
set_propagated_clock [get_clocks Top_clk]
```

Block level constraint (instance scope: i_blk)

```
create_clock -name Blk_clk -period 2 [get_ports Blk_clk]
```

Hierarchical clock mapping

```
Top_clk is mapped to Blk_clk
```

CONST_HIER_CTR

- [CONST_HIER_CTR1](#)

CONST_HIER_CTR1

NAME

CONST_HIER_CTR1 (CONSTRAINT_HIER_SIGNOFF)

MESSAGE

Inconsistent `set_clock_transition` at the top-level vs. block

DEFAULT SEVERITY

Warning

DESCRIPTION

Indicates that an inconsistency in `set_clock_transition` command is found between top level and block level constraints. The following are the possible causes for this violation:

- `set_clock_transition` command specified at either top level or block level
- `set_clock_transition` commands specified at both top level and block level but with different values

This rule is checked by `check_target_constraints` command and the results can be reported by `report_rule_check` command.

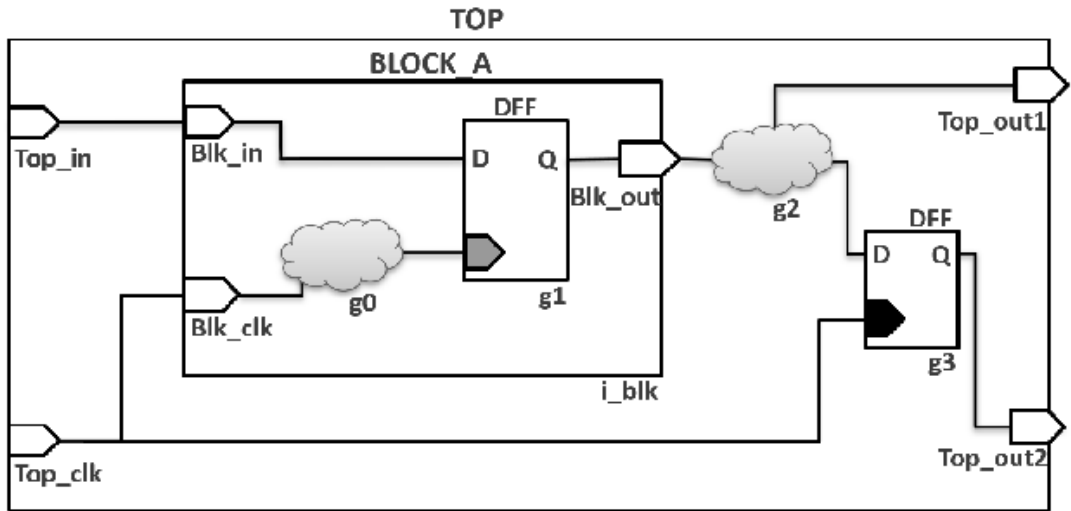
This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
block_view_name	string	Block view name
top_view_name	string	Top view name
scope	object	Scope

block_clock	object	Block clock
top_clock	object	Top clock
type	string	Type
block_file_line	unsigned	Block file line number
block_file_name	string	Block file name
top_file_line	unsigned	Top file line number
top_file_name	string	Top file name

EXAMPLE

- For the figure below, the following constraints contain `set_clock_transition` commands specified in both top level and block level constraints. This rule is reported for the `set_clock_transition` commands, because they are specified with different values.



Top level constraints

```
create_clock -name Top_clk -period 2 [get_ports Top_clk]
set_clock_transition 0.5 [get_clocks Top_clk]
```

Block level constraints (instance scope: i_blk)

```
create_clock -name Blk_clk -period 2 [get_ports Blk_clk]
set_clock_transition 1.0 [get_clocks Blk_clk]
```

Hierarchical clock mapping

```
Top_clk is mapped to Blk_clk
```

CONST_HIER_DAT

- [CONST_HIER_DAT1](#)

CONST_HIER_DAT1

NAME

CONST_HIER_DAT1 (CONSTRAINT_HIER_SIGNOFF)

MESSAGE

Inconsistent `set_data_check` at the top level vs. block

DEFAULT SEVERITY

Warning

DESCRIPTION

Indicates that an inconsistency in `set_data_check` command is found between top level and block level constraints. The following are the possible causes for this violation:

- `set_data_check` command specified at either top level or block level
- `set_data_check` commands specified at both top level and block level but with different values

This rule is checked by `check_target_constraints` command and the results can be reported by `report_rule_check` command.

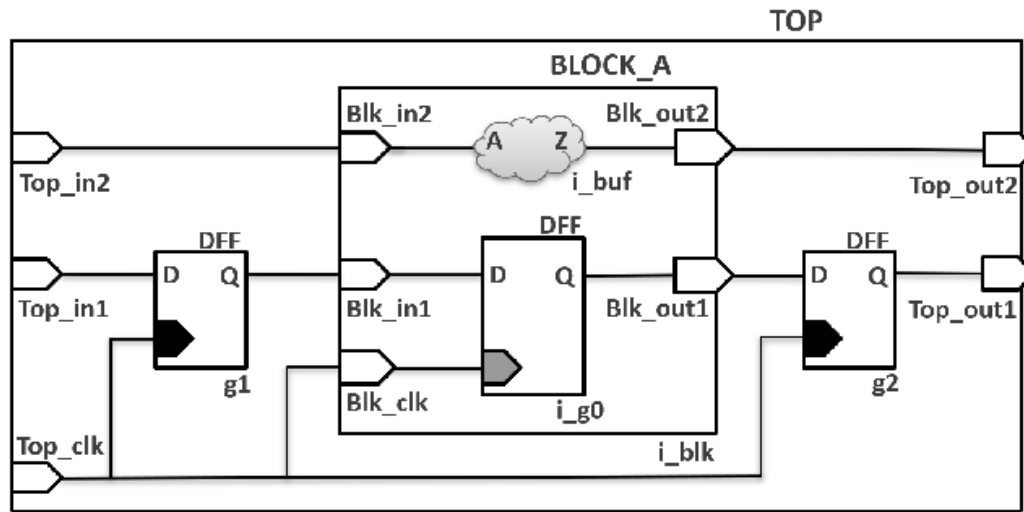
This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
block_view_name	string	Block view name
top_view_name	string	Top view name
scope	object	Scope
block_from_pin	object	block_from_pin

block_to_pin	object	block_to_pin
top_from_pin	object	top_from_pin
top_to_pin	object	top_to_pin
type	string	Type
top_clock	object	Top clock
block_clock	object	Block clock
block_value	float	Block value
top_value	float	Top value
block_file_line	object_list	Block file line number
block_file_name	object_list	Block file name
top_file_line	object_list	Top file line number
top_file_name	object_list	Top file name

EXAMPLE

- For the figure below, the following contains `set_data_check` commands specified in both top level and block level constraints. This rule is reported for the `set_data_check` commands, because they are specified with different values.



Top level constraint

```
set_data_check 0.5 -from [get_pins i_blk/i_buf/A] -to [get_pins i_blk/i_buf/Z]
```

Block level constraint (instance scope: i_blk)

```
set_data_check 1.0 -from [get_pins i_buf/A] -to [get_pins i_buf/Z]
```

CONST_HIER_EXC

- [CONST_HIER_EXC1](#)
- [CONST_HIER_EXC2](#)
- [CONST_HIER_EXC3](#)
- [CONST_HIER_EXC4](#)
- [CONST_HIER_EXC5](#)
- [CONST_HIER_EXC6](#)
- [CONST_HIER_EXC7](#)
- [CONST_HIER_EXC10](#)
- [CONST_HIER_EXC11](#)
- [CONST_HIER_EXC12](#)
- [CONST_HIER_EXC13](#)
- [CONST_HIER_EXC14](#)
- [CONST_HIER_EXC15](#)
- [CONST_HIER_EXC16](#)
- [CONST_HIER_EXC17](#)

CONST_HIER_EXC1

NAME

CONST_HIER_EXC1 (CONSTRAINT_HIER_SIGNOFF)

MESSAGE

A block path without `set_false_path` is part of a top-level false path

DEFAULT SEVERITY

Warning

DESCRIPTION

This rule reports that a block path, which was not declared as false path at the block level, has been affected by top level false path specification. This rule is checked by `check_target_constraints` command and results can be reported by `report_rule_check` command.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id 123
block_view_name	string	Block view name
top_view_name	string	Top view name
block_path	object	Block Path
top_path	object	Top Path
block_file_line	unsigned	Block file line number
block_file_name	string	Block file name
top_file_line	unsigned	Top file line number

Litmus Rule Reference
Timing Constraint Hierarchical Signoff--CONST_HIER_EXC

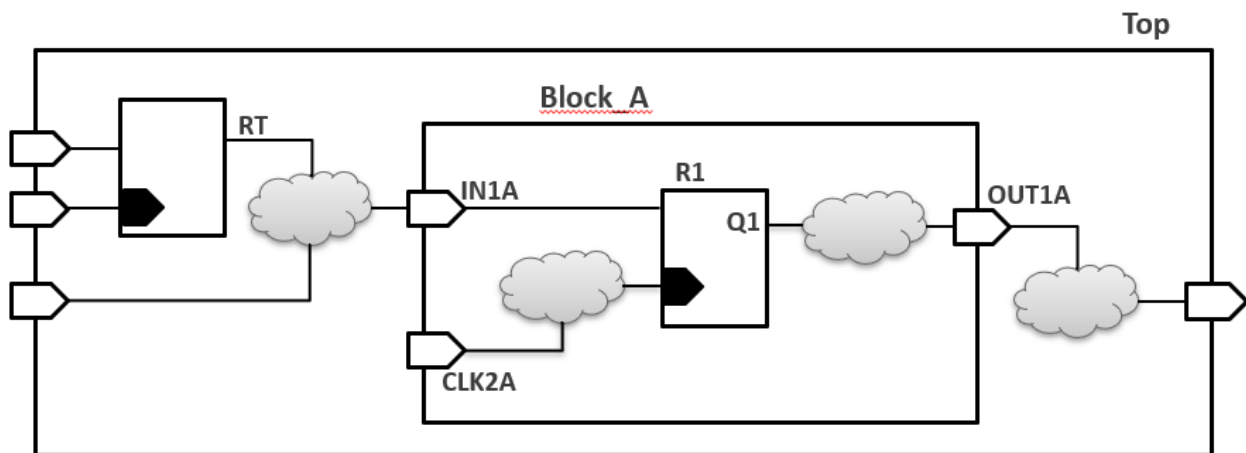
top_file_name	string	Top file name
block_path_startpoint	object	Block path startpoint
top_path_startpoint	object	Top path startpoint
block_path_endpoint	object	Block path endpoint
top_path_endpoint	object	Top path endpoint
block_corner	string	block_corner
top_corner	string	top_corner
block_exc_from	object_list	Block exception from
block_exc_to	object_list	Block exception to
block_exc_through	object_list	Block exception through
block_exc_type	string	Block exception type
top_exc_from	object_list	Top exception from
top_exc_to	object_list	Top exception to
top_exc_through	object_list	Top exception through
top_exc_type	string	Top exception type
block_exc_from_clock	object_list	Block exception from clock
block_exc_to_clock	object_list	Block exception to clock
top_exc_from_clock	object_list	Top exception from clock
top_exc_to_clock	object_list	Top exception to clock
block_path_startpoint_clock	object	block_path_startpoint_clock
block_path_endpoint_clock	object	block_path_endpoint_clock
top_path_startpoint_clock	object	top_path_startpoint_clock
top_path_endpoint_clock	object	top_path_endpoint_clock

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scope	object	Scope
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EXAMPLE

- For the figure below, if the block level constraints file has a false path defined from R1 as follows and the top level constraints file has defined a false path in Block_A like the following then the Litmus issues a warning because the block level did not consider the path from IN1A to R1 as false with a false path specification from the top level.



Top level constraint

```
set_false_path -through Block_A/IN1A
```

Block level constraint (instance scope: Block_A)

```
set_false_path -from R1
```

RELATED TOPICS

Commands

```
set_false_path
check_target_constraints
```

`report_rule_check`

Attributes

[Rule Attributes - rule_occurrence](#)

CONST_HIER_EXC2

NAME

CONST_HIER_EXC2 (CONSTRAINT_HIER_SIGNOFF)

MESSAGE

A block's false path is part of a top-level path without `set_false_path`

DEFAULT SEVERITY

Warning

DESCRIPTION

This rule reports that a block level false path is not declared as a false path at the top level. This rule is checked by `check_target_constraints` command and results can be reported by `report_rule_check` command.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
block_view_name	string	Block view name
top_view_name	string	Top view name
block_path	object	Block Path
top_path	object	Top Path
block_file_line	unsigned	Block file line number
block_file_name	string	Block file name
top_file_line	unsigned	Top file line number

Litmus Rule Reference
Timing Constraint Hierarchical Signoff--CONST_HIER_EXC

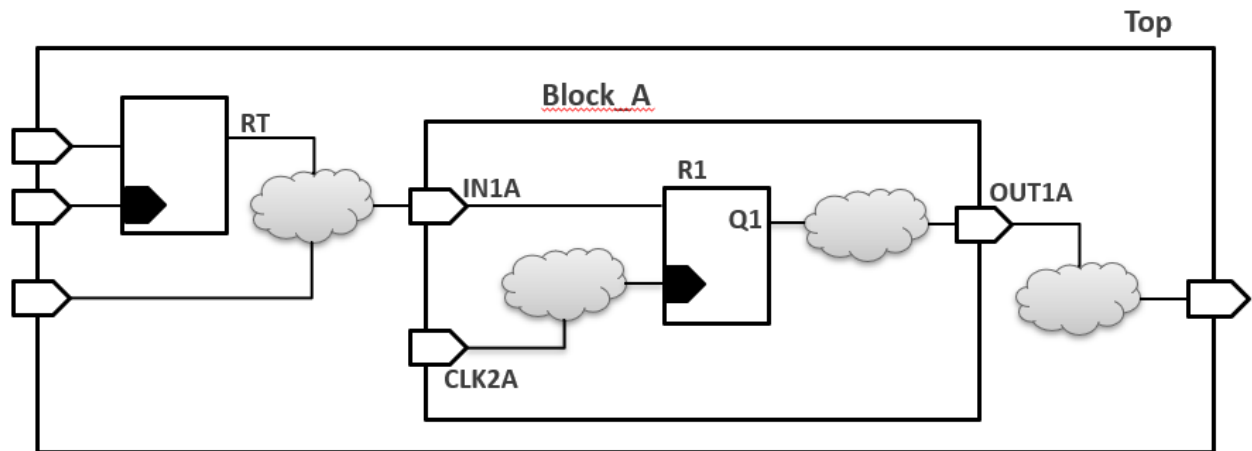
top_file_name	string	Top file name
block_path_startpoint	object	Block path startpoint
top_path_startpoint	object	Top path startpoint
block_path_endpoint	object	Block path endpoint
top_path_endpoint	object	Top path endpoint
block_corner	string	block_corner
top_corner	string	top_corner
block_exc_from	object_list	Block exception from
block_exc_to	object_list	Block exception to
block_exc_through	object_list	Block exception through
block_exc_type	string	Block exception type
top_exc_from	object_list	Top exception from
top_exc_to	object_list	Top exception to
top_exc_through	object_list	Top exception through
top_exc_type	string	Top exception type
block_exc_from_clock	object_list	Block exception from clock
block_exc_to_clock	object_list	Block exception to clock
top_exc_from_clock	object_list	Top exception from clock
top_exc_to_clock	object_list	Top exception to clock
block_path_startpoint_clock	object	block_path_startpoint_clock
block_path_endpoint_clock	object	block_path_endpoint_clock
top_path_startpoint_clock	object	top_path_startpoint_clock
top_path_endpoint_clock	object	top_path_endpoint_clock

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scope	object	Scope
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EXAMPLE

- For the figure below, if the block level constraints file has a false path defined from R1 like the following and the top level constraints file has defined a false path as the following in line #1 but missing line #2, then the Litmus issues a warning because the block level would consider the path from R1 to OUT1A as false, but the top level does not specify this as false.



Top level constraints

```
set_false_path -from RT
set_false_path -from Block_A/R1
```

Block level constraint (instance scope: Block_A)

```
set_false_path -from R1
```

RELATED TOPICS

- Commands

```
set_false_path  
check_target_constraints  
report_rule_check
```

- **Attributes**

Rule Attributes - rule_occurrence

CONST_HIER_EXC3

NAME

CONST_HIER_EXC3 (CONSTRAINT_HIER_SIGNOFF)

MESSAGE

A block's false path is part of a top-level multicycle path

DEFAULT SEVERITY

Warning

DESCRIPTION

This rule reports that a false path at the block level is declared as a multi-cycle path at the top level. This rule is checked by `check_target_constraints` command and results can be reported by `report_rule_check` command.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
block_view_name	string	Block view name
top_view_name	string	Top view name
block_path	object	Block Path
top_path	object	Top Path
block_file_line	unsigned	Block file line number
block_file_name	string	Block file name
top_file_line	unsigned	Top file line number

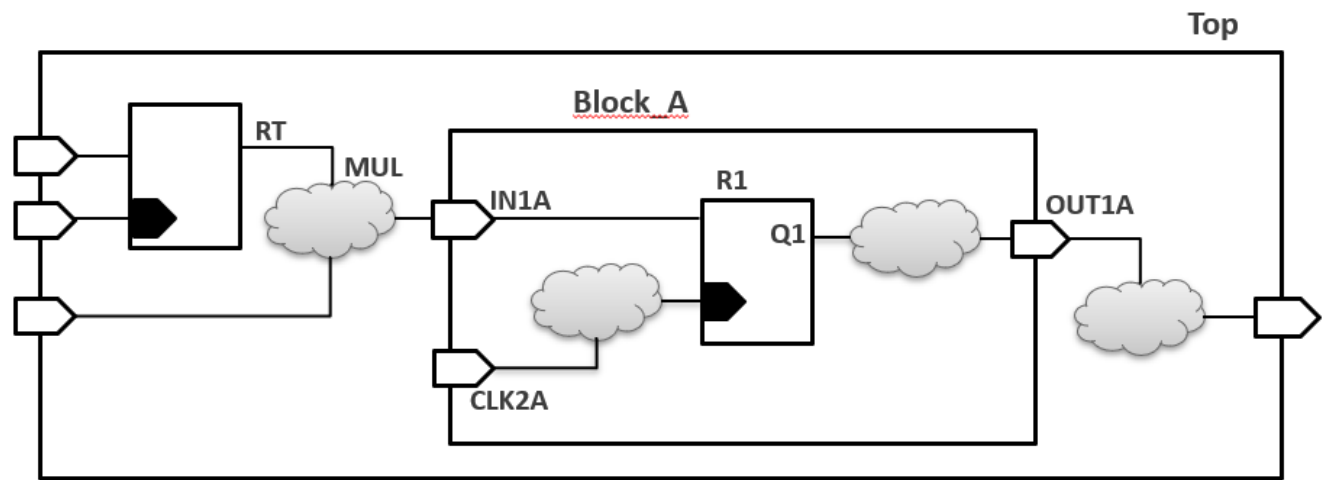
top_file_name	string	Top file name
block_path_startpoint	object	Block path startpoint
top_path_startpoint	object	Top path startpoint
block_path_endpoint	object	Block path endpoint
top_path_endpoint	object	Top path endpoint
top_exc_cycle	int	top_exc_cycle
top_exc_cycle_on_start	string	top_exc_cycle_on_start
block_corner	string	block_corner
top_corner	string	top_corner
block_exc_from	object_list	Block exception from
block_exc_to	object_list	Block exception to
block_exc_through	object_list	Block exception through
block_exc_type	string	Block exception type
top_exc_from	object_list	Top exception from
top_exc_to	object_list	Top exception to
top_exc_through	object_list	Top exception through
top_exc_type	string	Top exception type
block_exc_from_clock	object_list	Block exception from clock
block_exc_to_clock	object_list	Block exception to clock
top_exc_from_clock	object_list	Top exception from clock
top_exc_to_clock	object_list	Top exception to clock
block_path_startpoint_clock	object	block_path_startpoint_clock
block_path_endpoint_clock	object	block_path_endpoint_clock

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top_path_startpoint_clock	object	top_path_startpoint_clock
top_path_endpoint_clock	object	top_path_endpoint_clock
scope	object	Scope

EXAMPLE

- For the figure below, if the block level constraints file has a false path defined as follows and the top level constraints file has defined a multicycle path like the following, then the Litmus issues a warning because the FP from port IN1A is now considered an MCP because of the – through MUL specified in the top level constraints file.



Top level constraint

```
set_multicycle_path 2 -through MUL
```

Block level constraint (instance scope: Block_A)

```
set_false_path -through IN1A
```

RELATED TOPICS

- **Commands**

- `set_false_path`
 - `set_multicycle_path`
 - `check_target_constraints`
 - `report_rule_check`

- **Attributes**

- [Rule Attributes - rule_occurrence](#)

CONST_HIER_EXC4

NAME

CONST_HIER_EXC4 (CONSTRAINT_HIER_SIGNOFF)

MESSAGE

A block's multicycle path is part of a top-level false path

DEFAULT SEVERITY

Warning

DESCRIPTION

This rule reports that a block level multicycle path is declared as a false path at the top level. This rule is checked by `check_target_constraints` command and results can be reported by `report_rule_check` command.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
block_view_name	string	Block view name
top_view_name	string	Top view name
block_path	object	Block Path
top_path	object	Top Path
block_file_line	unsigned	Block file line number
block_file_name	string	Block file name
top_file_line	unsigned	Top file line number

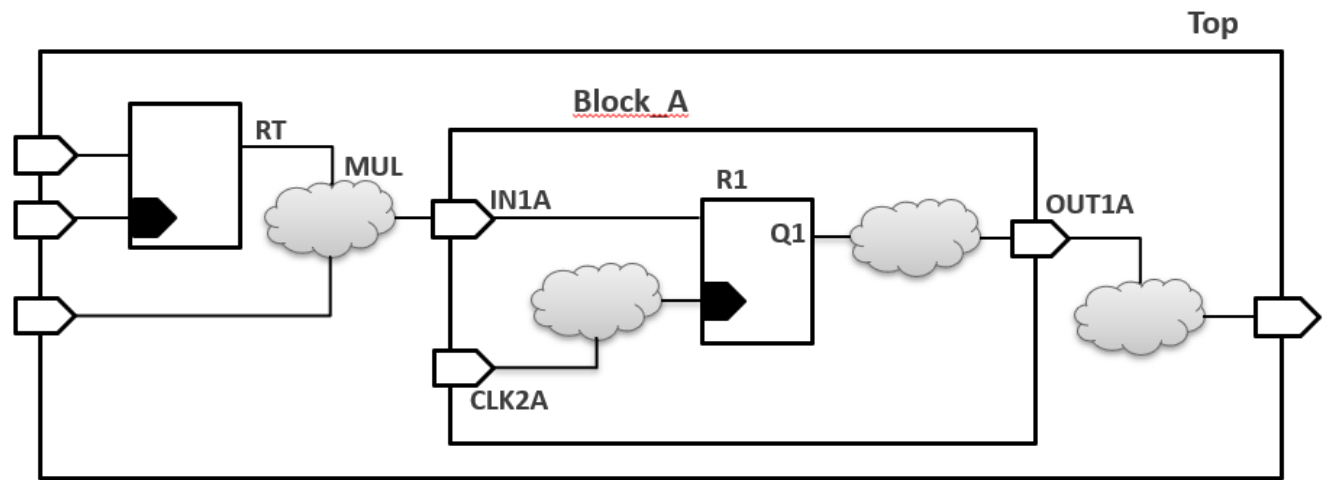
top_file_name	string	Top file name
block_path_startpoint	object	Block path startpoint
top_path_startpoint	object	Top path startpoint
block_path_endpoint	object	Block path endpoint
top_path_endpoint	object	Top path endpoint
block_exc_cycle	int	block_exc_cycle
block_exc_cycle_on_start	string	block_exc_cycle_on_start
block_corner	string	block_corner
top_corner	string	top_corner
block_exc_from	object_list	Block exception from
block_exc_to	object_list	Block exception to
block_exc_through	object_list	Block exception through
block_exc_type	string	Block exception type
top_exc_from	object_list	Top exception from
top_exc_to	object_list	Top exception to
top_exc_through	object_list	Top exception through
top_exc_type	string	Top exception type
block_exc_from_clock	object_list	Block exception from clock
block_exc_to_clock	object_list	Block exception to clock
top_exc_from_clock	object_list	Top exception from clock
top_exc_to_clock	object_list	Top exception to clock
block_path_startpoint_clock	object	block_path_startpoint_clock
block_path_endpoint_clock	object	block_path_endpoint_clock

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top_path_startpoint_clock	object	top_path_startpoint_clock
top_path_endpoint_clock	object	top_path_endpoint_clock
scope	object	Scope

EXAMPLE

- For the figure below, if the block level constraints file has a multicycle path defined as follows and the top level constraints file has defined a false path like the following, then the Litmus issues a warning because the MCP from port IN1A is now considered an FP because of the –
through MUL specified in the top level constraints file.



Top level constraint

```
set_false_path -through MUL
```

Block level constraint (instance scope: Block_A)

```
set_multicycle_path 2 -through IN1A
```

RELATED TOPICS

- **Commands**

- `set_false_path`
 - `set_multicycle_path`
 - `check_target_constraints`
 - `report_rule_check`

- **Attributes**

- [Rule Attributes - rule_occurrence](#)

CONST_HIER_EXC5

NAME

CONST_HIER_EXC5 (CONSTRAINT_HIER_SIGNOFF)

MESSAGE

A block's single cycle path is part of a top-level multicycle path

DEFAULT SEVERITY

Warning

DESCRIPTION

This rule reports that a block path that is not declared as a multi-cycle path is part of a top level path that is declared as a multi-cycle path at the top level. This rule is checked by `check_target_constraints` command and results can be reported by `report_rule_check` command.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
block_view_name	string	Block view name
top_view_name	string	Top view name
block_path	object	Block Path
top_path	object	Top Path
block_file_line	unsigned	Block file line number
block_file_name	string	Block file name

Litmus Rule Reference
Timing Constraint Hierarchical Signoff--CONST_HIER_EXC

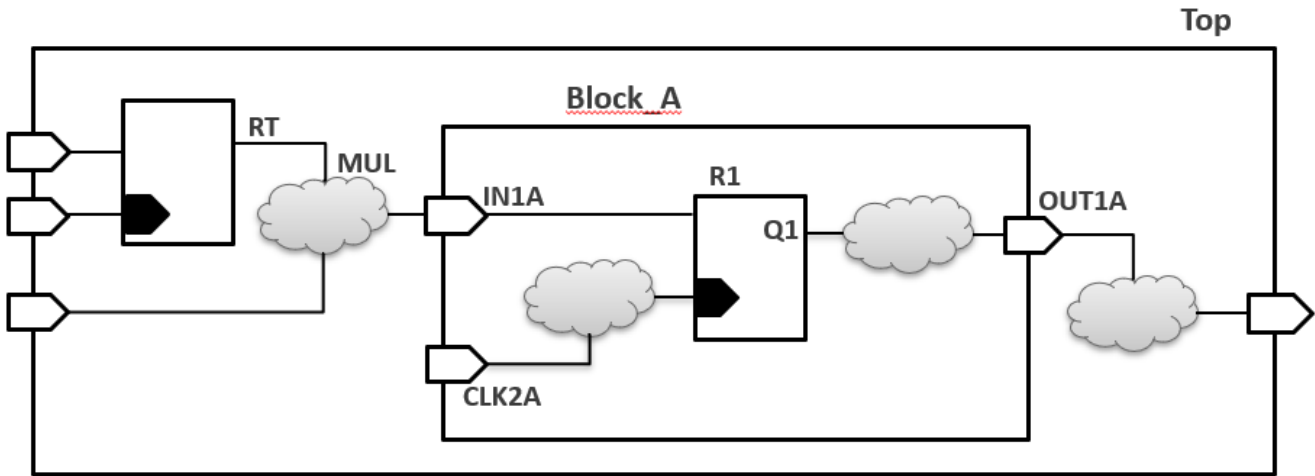
top_file_line	unsigned	Top file line number
top_file_name	string	Top file name
block_path_startpoint	object	Block path startpoint
top_path_startpoint	object	Top path startpoint
block_path_endpoint	object	Block path endpoint
top_path_endpoint	object	Top path endpoint
top_exc_cycle	int	top_exc_cycle
top_exc_cycle_on_start	string	top_exc_cycle_on_start
block_corner	string	block_corner
top_corner	string	top_corner
block_exc_from	object_list	Block exception from
block_exc_to	object_list	Block exception to
block_exc_through	object_list	Block exception through
block_exc_type	string	Block exception type
top_exc_from	object_list	Top exception from
top_exc_to	object_list	Top exception to
top_exc_through	object_list	Top exception through
top_exc_type	string	Top exception type
block_exc_from_clock	object_list	Block exception from clock
block_exc_to_clock	object_list	Block exception to clock
top_exc_from_clock	object_list	Top exception from clock
top_exc_to_clock	object_list	Top exception to clock
block_path_startpoint_clock	object	block_path_startpoint_clock

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block_path_endpoint_clock	object	block_path_endpoint_clock
top_path_startpoint_clock	object	top_path_startpoint_clock
top_path_endpoint_clock	object	top_path_endpoint_clock
scope	object	Scope

EXAMPLE

- For the figure below, if the block level constraints file has a multicycle path defined as follows and the top level constraints file has defined a multicycle path like the following, then the Litmus issues a warning because the block level path from port IN1A to R1 is now considered an MCP path because of the top level MCP constraint.



Top level constraint

```
set_multicycle_path 3 -through Block_A/R1
```

Block level constraint (instance scope: Block_A)

```
set_multicycle_path 3 -from R1
```

RELATED TOPICS

- **Commands**

- [set_multicycle_path](#)
 - [check_target_constraints](#)
 - [report_rule_check](#)

- **Attributes**

- [Rule Attributes - rule_occurrence](#)

CONST_HIER_EXC6

NAME

CONST_HIER_EXC6 (CONSTRAINT_HIER_SIGNOFF)

MESSAGE

A block's multicycle path is part of a top-level single cycle path

DEFAULT SEVERITY

Warning

DESCRIPTION

This rule reports that a multi-cycle path at the block level is not declared as a multi-cycle path at the top level. This rule is checked by `check_target_constraints` command and results can be reported by `report_rule_check` command.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
block_view_name	string	Block view name
top_view_name	string	Top view name
block_path	object	Block Path
top_path	object	Top Path
block_file_line	unsigned	Block file line number
block_file_name	string	Block file name
top_file_line	unsigned	Top file line number

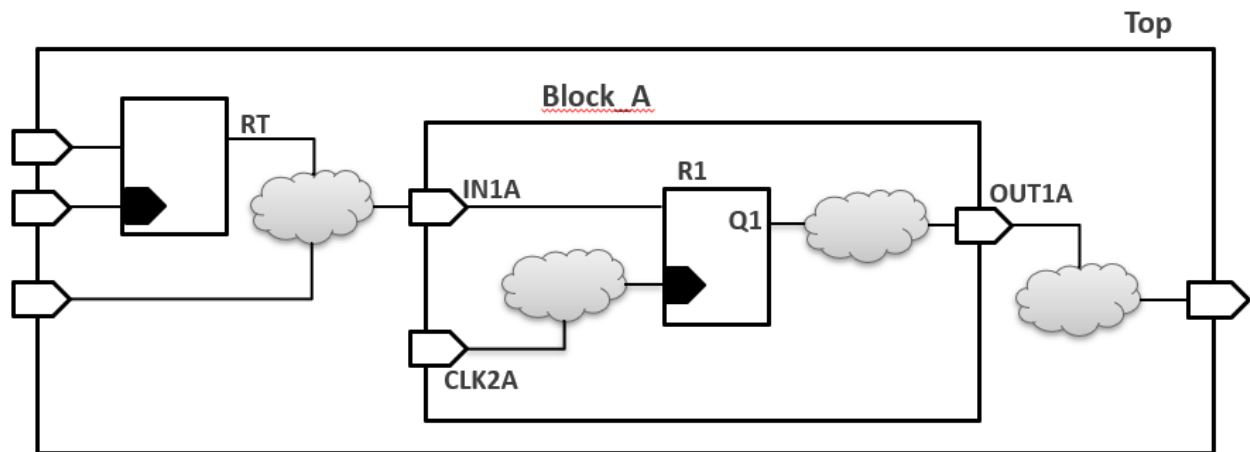
top_file_name	string	Top file name
block_path_startpoint	object	Block path startpoint
top_path_startpoint	object	Top path startpoint
block_path_endpoint	object	Block path endpoint
top_path_endpoint	object	Top path endpoint
block_exc_cycle	int	block_exc_cycle
block_exc_cycle_on_start	string	block_exc_cycle_on_start
block_corner	string	block_corner
top_corner	string	top_corner
block_exc_from	object_list	Block exception from
block_exc_to	object_list	Block exception to
block_exc_through	object_list	Block exception through
block_exc_type	string	Block exception type
top_exc_from	object_list	Top exception from
top_exc_to	object_list	Top exception to
top_exc_through	object_list	Top exception through
top_exc_type	string	Top exception type
block_exc_from_clock	object_list	Block exception from clock
block_exc_to_clock	object_list	Block exception to clock
top_exc_from_clock	object_list	Top exception from clock
top_exc_to_clock	object_list	Top exception to clock
block_path_startpoint_clock	object	block_path_startpoint_clock
block_path_endpoint_clock	object	block_path_endpoint_clock

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top_path_startpoint_clock	object	top_path_startpoint_clock
top_path_endpoint_clock	object	top_path_endpoint_clock
scope	object	Scope

EXAMPLE

- For the figure below, if the block level constraints file has a multicycle path defined as follows and the top level constraints file does not have a multicycle path defined like the following, then the Litmus issues a warning because the block level MCP path from port IN1A to R1 is now considered a single-cycle path.



Top level constraint

```
set_multicycle_path 2 -from Block_A/IN1A
```

Block level constraint (instance scope: Block_A)

```
set_multicycle_path 2 -through IN1A
```

RELATED TOPICS

- Commands

- [set_multicycle_path](#)
 - [check_target_constraints](#)
 - [report_rule_check](#)

- Attributes

- [Rule Attributes - rule_occurrence](#)

CONST_HIER_EXC7

NAME

CONST_HIER_EXC7 (CONSTRAINT_HIER_SIGNOFF)

MESSAGE

A block's multicycle path is part of a top-level multicycle path with a different cycle count

DEFAULT SEVERITY

Warning

DESCRIPTION

This rule reports that there are conflicting cycle counts between the block and top level multi-cycle paths. This rule is checked by `check_target_constraints` command and results can be reported by `report_rule_check` command.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
block_view_name	string	Block view name
top_view_name	string	Top view name
block_path	object	Block Path
top_path	object	Top Path
block_file_line	unsigned	Block file line number
block_file_name	string	Block file name
top_file_line	unsigned	Top file line number

Litmus Rule Reference
Timing Constraint Hierarchical Signoff--CONST_HIER_EXC

top_file_name	string	Top file name
block_path_startpoint	object	Block path startpoint
top_path_startpoint	object	Top path startpoint
block_path_endpoint	object	Block path endpoint
top_path_endpoint	object	Top path endpoint
block_exc_cycle	int	block_exc_cycle
block_exc_cycle_on_start	string	block_exc_cycle_on_start
top_exc_cycle	int	top_exc_cycle
top_exc_cycle_on_start	string	top_exc_cycle_on_start
block_corner	string	block_corner
top_corner	string	top_corner
block_exc_from	object_list	Block exception from
block_exc_to	object_list	Block exception to
block_exc_through	object_list	Block exception through
block_exc_type	string	Block exception type
top_exc_from	object_list	Top exception from
top_exc_to	object_list	Top exception to
top_exc_through	object_list	Top exception through
top_exc_type	string	Top exception type
block_exc_from_clock	object_list	Block exception from clock
block_exc_to_clock	object_list	Block exception to clock
top_exc_from_clock	object_list	Top exception from clock
top_exc_to_clock	object_list	Top exception to clock

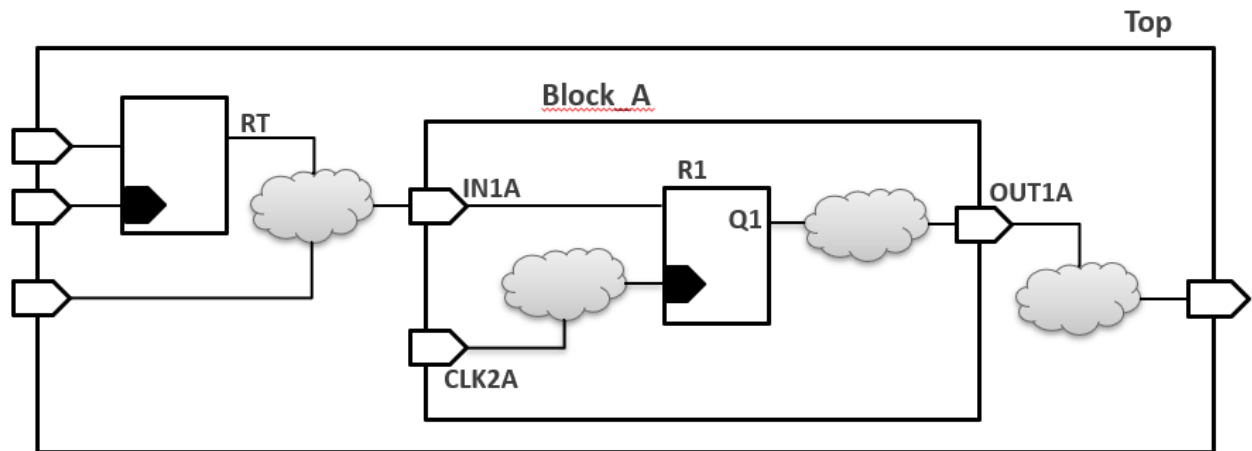
Litmus Rule Reference
Timing Constraint Hierarchical Signoff--CONST_HIER_EXC

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block_path_startpoint_clock	object	block_path_startpoint_clock
block_path_endpoint_clock	object	block_path_endpoint_clock
top_path_startpoint_clock	object	top_path_startpoint_clock
top_path_endpoint_clock	object	top_path_endpoint_clock
scope	object	Scope

EXAMPLE

- For the figure below, if the block level constraints file has a multicycle path defined as follows and the top level constraints file has a multicycle path defined like the following, then the Litmus issues a warning because of conflicting cycle counts in a multicycle path between the block level and top level constraints.



Top level constraint

```
set_multicycle_path 4 -through Block_A/R1
```

Block level constraint (instance scope: Block_A)

```
set_multicycle_path 3 -through R1
```

RELATED TOPICS

- Commands

```
set_multicycle_path  
check_target_constraints  
report_rule_check
```

- Attributes

[Rule Attributes - rule_occurrence](#)

CONST_HIER_EXC10

NAME

CONST_HIER_EXC10 (CONSTRAINT_HIER_SIGNOFF)

MESSAGE

A block path without `set_max_delay/set_min_delay` is part of a top-level path with max/min delay settings

DEFAULT SEVERITY

Warning

DESCRIPTION

This rule reports that a block path, which was not set any maximum or minimum delay at the block level, has been set by a top level maximum or minimum delay. This rule is checked by `check_target_constraints` command and results can be reported by `report_rule_check` command.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
block_view_name	string	Block view name
top_view_name	string	Top view name
block_path	object	Block Path
top_path	object	Top Path
block_file_line	unsigned	Block file line number
block_file_name	string	Block file name

Litmus Rule Reference
Timing Constraint Hierarchical Signoff--CONST_HIER_EXC

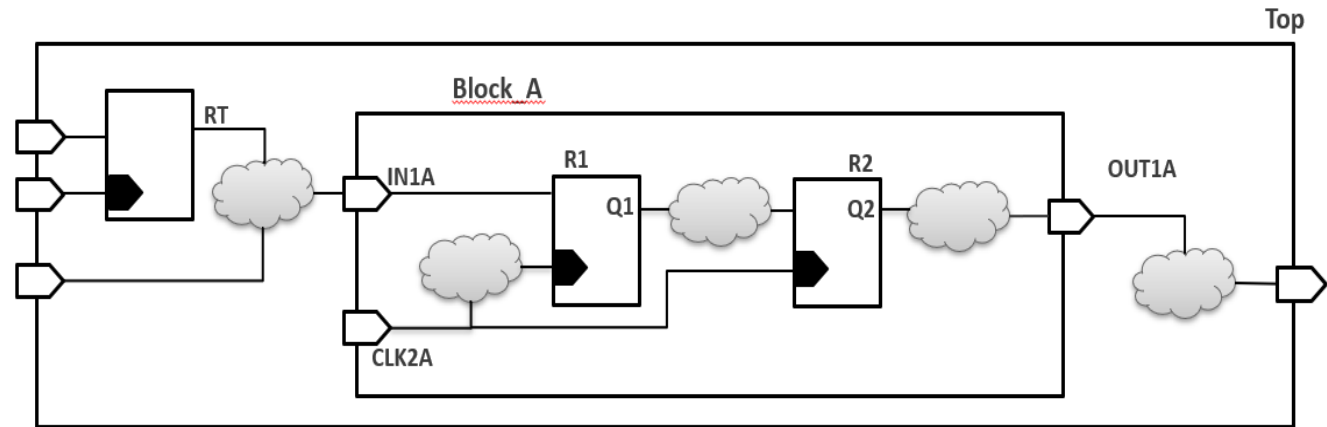
top_file_line	unsigned	Top file line number
top_file_name	string	Top file name
block_path_startpoint	object	Block path startpoint
top_path_startpoint	object	Top path startpoint
block_path_endpoint	object	Block path endpoint
top_path_endpoint	object	Top path endpoint
top_value	float	top_value
block_corner	string	block_corner
top_corner	string	top_corner
block_exc_from	object_list	Block exception from
block_exc_to	object_list	Block exception to
block_exc_through	object_list	Block exception through
block_exc_type	string	Block exception type
top_exc_from	object_list	Top exception from
top_exc_to	object_list	Top exception to
top_exc_through	object_list	Top exception through
top_exc_type	string	Top exception type
block_exc_from_clock	object_list	Block exception from clock
block_exc_to_clock	object_list	Block exception to clock
top_exc_from_clock	object_list	Top exception from clock
top_exc_to_clock	object_list	Top exception to clock
block_path_startpoint_clock	object	block_path_startpoint_clock
block_path_endpoint_clock	object	block_path_endpoint_clock

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top_path_startpoint_clock	object	top_path_startpoint_clock
top_path_endpoint_clock	object	top_path_endpoint_clock
scope	object	Scope

EXAMPLE

- For the figure below, the following command in the top level constraints file causes a rule check violation if the same path is not set maximum delay at the block level using the following command at the block level constraints file:



Top level constraint

```
set_max_delay 4 -from [get_cells Block_A/R1] -to [get_cells Block_A/R2]
```

Block level constraint (instance scope: Block_A)

```
set_max_delay 4 -from [get_cells R1] -to [get_cells R2]
```

RELATED TOPICS

- **Commands**

- `set_min_delay`
 - `set_max_delay`
 - `check_target_constraints`
 - `report_rule_check`

- **Attributes**

- [Rule Attributes - rule_occurrence](#)

CONST_HIER_EXC11

NAME

CONST_HIER_EXC11 (CONSTRAINT_HIER_SIGNOFF)

MESSAGE

A block's path with max/min delay settings is part of a top-level path without `set_max_delay/set_min_delay`

DEFAULT SEVERITY

Warning

DESCRIPTION

This rule reports that a block path, which was set a maximum or minimum delay at the block level, has not been set with any top level maximum or minimum delay. This rule is checked by `check_target_constraints` command and results can be reported by `report_rule_check` command.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
block_view_name	string	Block view name
top_view_name	string	Top view name
block_path	object	Block Path
top_path	object	Top Path
block_file_line	unsigned	Block file line number
block_file_name	string	Block file name

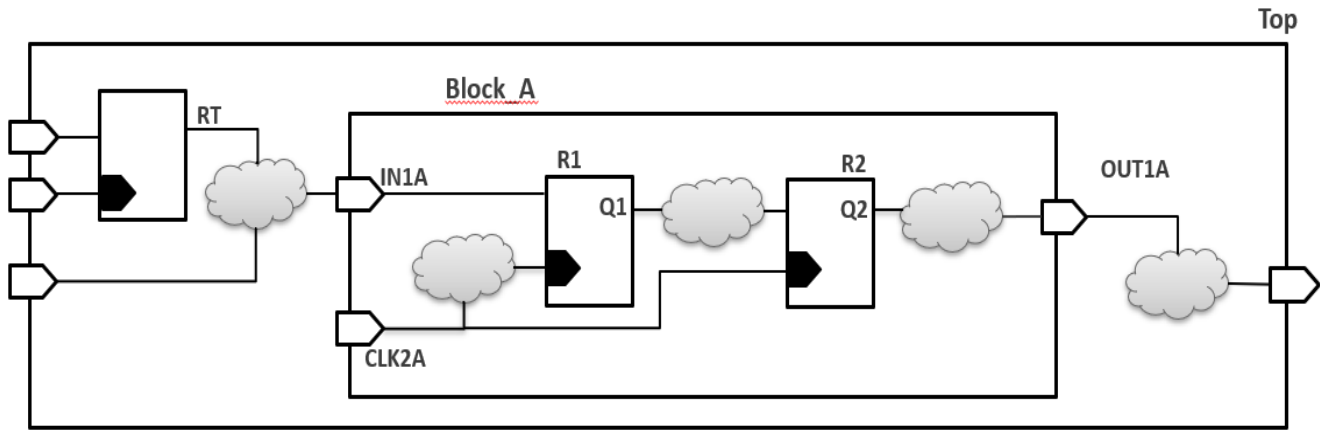
top_file_line	unsigned	Top file line number
top_file_name	string	Top file name
block_path_startpoint	object	Block path startpoint
top_path_startpoint	object	Top path startpoint
block_path_endpoint	object	Block path endpoint
top_path_endpoint	object	Top path endpoint
block_value	float	block_value
block_corner	string	block_corner
top_corner	string	top_corner
block_exc_from	object_list	Block exception from
block_exc_to	object_list	Block exception to
block_exc_through	object_list	Block exception through
block_exc_type	string	Block exception type
top_exc_from	object_list	Top exception from
top_exc_to	object_list	Top exception to
top_exc_through	object_list	Top exception through
top_exc_type	string	Top exception type
block_exc_from_clock	object_list	Block exception from clock
block_exc_to_clock	object_list	Block exception to clock
top_exc_from_clock	object_list	Top exception from clock
top_exc_to_clock	object_list	Top exception to clock
block_path_startpoint_clock	object	block_path_startpoint_clock
block_path_endpoint_clock	object	block_path_endpoint_clock

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top_path_startpoint_clock	object	top_path_startpoint_clock
top_path_endpoint_clock	object	top_path_endpoint_clock
scope	object	Scope

EXAMPLE

- For the figure below, the following command in the block level constraints file causes a rule check violation if it is not specified in the top level constraints file like the following:



Top level constraint

```
set_max_delay 4 -from [get_cells Block_A/R1] -to [get_cells Block_A/R2]
```

Block level constraint (instance scope: Block_A)

```
set_max_delay 4 -from [get_cells R1] -to [get_cells R2]
```

RELATED TOPICS

- Commands

```
set_min_delay  
set_max_delay  
check_target_constraints  
report_rule_check
```

- **Attributes**

Rule Attributes - rule_occurrence

CONST_HIER_EXC12

NAME

CONST_HIER_EXC12 (CONSTRAINT_HIER_SIGNOFF)

MESSAGE

A block's false path is part of a top-level path with max/min delay settings

DEFAULT SEVERITY

Warning

DESCRIPTION

This rule reports that a block path was declared as false at the block level, has been set by a top level maximum or minimum delay. This rule is checked by `check_target_constraints` command and results can be reported by `report_rule_check` command.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
block_view_name	string	Block view name
top_view_name	string	Top view name
block_path	object	Block Path
top_path	object	Top Path
block_file_line	unsigned	Block file line number
block_file_name	string	Block file name

Litmus Rule Reference
Timing Constraint Hierarchical Signoff--CONST_HIER_EXC

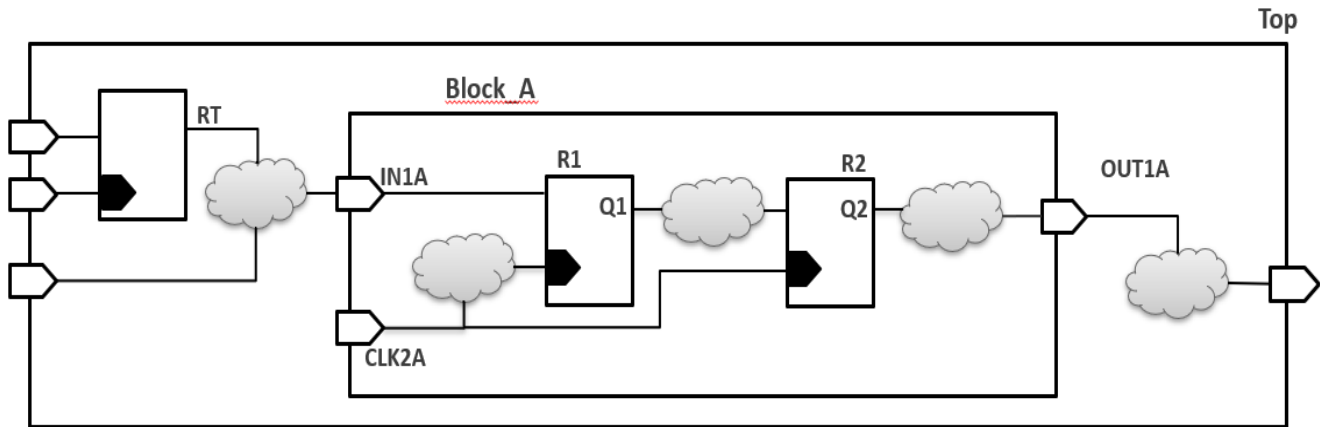
top_file_line	unsigned	Top file line number
top_file_name	string	Top file name
block_path_startpoint	object	Block path startpoint
top_path_startpoint	object	Top path startpoint
block_path_endpoint	object	Block path endpoint
top_path_endpoint	object	Top path endpoint
top_value	float	top_value
block_corner	string	block_corner
top_corner	string	top_corner
block_exc_from	object_list	block_exc_from
block_exc_to	object_list	block_exc_to
block_exc_through	object_list	block_exc_through
block_exc_type	string	block_exc_type
top_exc_from	object_list	top_exc_from
top_exc_to	object_list	top_exc_to
top_exc_through	object_list	top_exc_through
top_exc_type	string	top_exc_type
block_exc_from_clock	object_list	block_exc_fromclk
block_exc_to_clock	object_list	block_exc_toclk
top_exc_from_clock	object_list	top_exc_fromclk
top_exc_to_clock	object_list	top_exc_toclk
block_path_startpoint_clock	object	block_path_startpoint_clock
block_path_endpoint_clock	object	block_path_endpoint_clock

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top_path_startpoint_clock	object	top_path_startpoint_clock
top_path_endpoint_clock	object	top_path_endpoint_clock
scope	object	Scope

EXAMPLE

- For the figure below, the following commands in top level and block level constraints cause this rule check violation:



Top level constraint

```
set_max_delay 4 -from [get_cells Block_A/R1] -to [get_cells Block_A/R2]
```

Block level constraint (instance scope: Block_A)

```
set_false_path -from [get_cells R1] -to [get_cells R2]
```

RELATED TOPICS

- **Commands**

- `set_false_path`
 - `set_min_delay`
 - `set_max_delay`
 - `check_target_constraints`
 - `report_rule_check`

- **Attributes**

- [Rule Attributes - rule_occurrence](#)

CONST_HIER_EXC13

NAME

CONST_HIER_EXC13 (CONSTRAINT_HIER_SIGNOFF)

MESSAGE

A block's path with max/min delay settings is part of a top-level false path

DEFAULT SEVERITY

Warning

DESCRIPTION

This rule reports that a block path, which was set a maximum or minimum delay at the block level, has been affected by a false path specification at the top level. This rule is checked by `check_target_constraints` command and results can be reported by `report_rule_check` command.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
block_view_name	string	Block view name
top_view_name	string	Top view name
block_path	object	Block Path
top_path	object	Top Path
block_file_line	unsigned	Block file line number
block_file_name	string	Block file name

Litmus Rule Reference
Timing Constraint Hierarchical Signoff--CONST_HIER_EXC

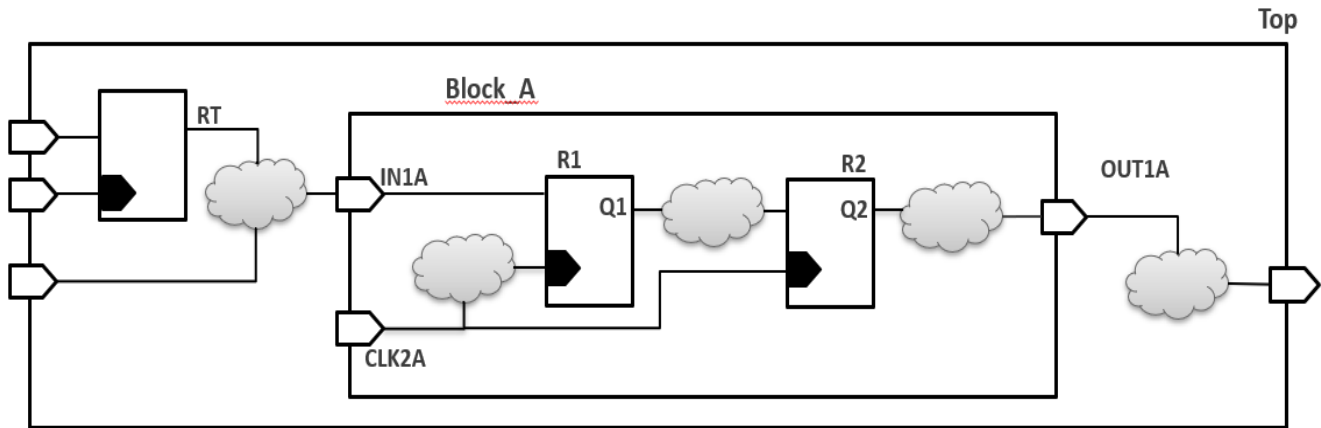
top_file_line	unsigned	Top file line number
top_file_name	string	Top file name
block_path_startpoint	object	Block path startpoint
top_path_startpoint	object	Top path startpoint
block_path_endpoint	object	Block path endpoint
top_path_endpoint	object	Top path endpoint
block_value	float	block_value
block_corner	string	block_corner
top_corner	string	top_corner
block_exc_from	object_list	block_exc_from
block_exc_to	object_list	block_exc_to
block_exc_through	object_list	block_exc_through
block_exc_type	string	block_exc_type
top_exc_from	object_list	top_exc_from
top_exc_to	object_list	top_exc_to
top_exc_through	object_list	top_exc_through
top_exc_type	string	top_exc_type
block_exc_from_clock	object_list	block_exc_fromclk
block_exc_to_clock	object_list	block_exc_tockl
top_exc_from_clock	object_list	top_exc_fromclk
top_exc_to_clock	object_list	top_exc_tockl
block_path_startpoint_clock	object	block_path_startpoint_clock
block_path_endpoint_clock	object	block_path_endpoint_clock

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top_path_startpoint_clock	object	top_path_startpoint_clock
top_path_endpoint_clock	object	top_path_endpoint_clock
scope	object	Scope

EXAMPLE

- For the figure below, the following commands in top level and block level constraints cause a rule check violation:



Top level constraint

```
set_false_path -from [get_cells Block_A/R1] -to [get_cells Block_A/R2]
```

Block level constraint (instance scope: Block_A)

```
set_max_delay 4 -from [get_cells R1] -to [get_cells R2]
```

RELATED TOPICS

- Commands

`set_false_path`

set_max_delay
set_min_delay
check_target_constraints
report_rule_check

- **Attributes**

Rule Attributes - rule_occurrence

CONST_HIER_EXC14

NAME

CONST_HIER_EXC14 (CONSTRAINT_HIER_SIGNOFF))

MESSAGE

A block's multicycle path is part of a top-level path with max/min delay settings

DEFAULT SEVERITY

Warning

DESCRIPTION

This rule reports that a block path, which was declared as a multi-cycle path at the block level, has been set by a top level maximum or minimum delay. This rule is checked by `check_target_constraints` command and results can be reported by `report_rule_check` command.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
block_view_name	string	Block view name
top_view_name	string	Top view name
block_path	object	Block Path
top_path	object	Top Path
block_file_line	unsigned	Block file line number
block_file_name	string	Block file name

Litmus Rule Reference
Timing Constraint Hierarchical Signoff--CONST_HIER_EXC

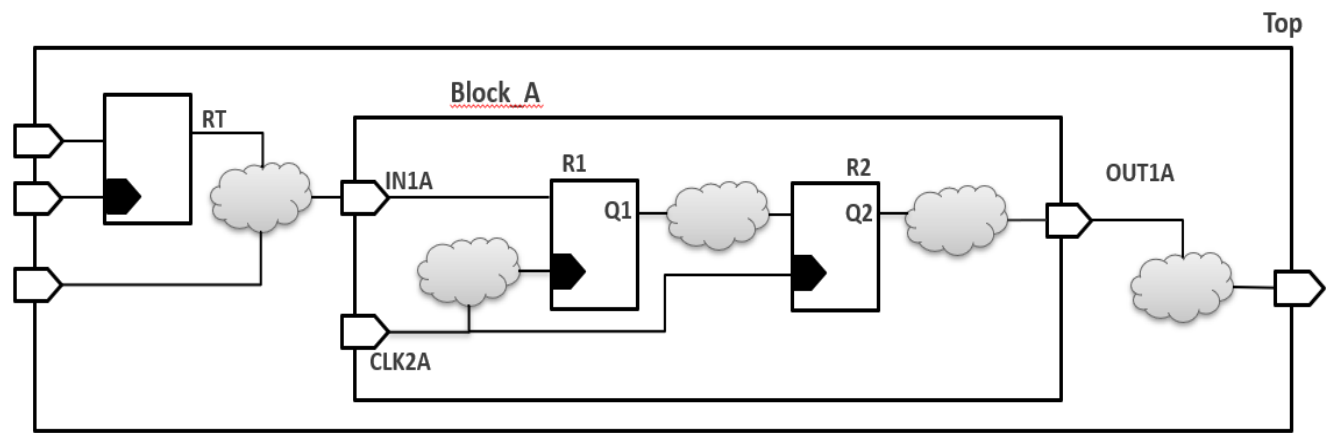
top_file_line	unsigned	Top file line number
top_file_name	string	Top file name
block_path_startpoint	object	Block path startpoint
top_path_startpoint	object	Top path startpoint
block_path_endpoint	object	Block path endpoint
top_path_endpoint	object	Top path endpoint
block_exc_cycle	int	block_exc_cycle
block_exc_cycle_on_start	string	block_exc_cycle_on_start
top_value	float	top_value
block_corner	string	block_corner
top_corner	string	top_corner
block_exc_from	object_list	block_exc_from
block_exc_to	object_list	block_exc_to
block_exc_through	object_list	block_exc_through
block_exc_type	string	block_exc_type
top_exc_from	object_list	top_exc_from
top_exc_to	object_list	top_exc_to
top_exc_through	object_list	top_exc_through
top_exc_type	string	top_exc_type
block_exc_from_clock	object_list	block_exc_fromclk
block_exc_to_clock	object_list	block_exc_tock
top_exc_from_clock	object_list	top_exc_fromclk
top_exc_to_clock	object_list	top_exc_tock

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block_path_startpoint_clock	object	block_path_startpoint_clock
block_path_endpoint_clock	object	block_path_endpoint_clock
top_path_startpoint_clock	object	top_path_startpoint_clock
top_path_endpoint_clock	object	top_path_endpoint_clock
scope	object	Scope

EXAMPLE

- For the figure below, the following commands in top level and block level constraints cause this rule check violation:



Top level constraint

```
set_max_delay 4 -from [get_cells Block_A/R1] -to [get_cells Block_A/R2]
```

Block level constraint (instance scope: Block_A)

```
set_multicycle_path 4 -from [get_cells R1] -to [get_cells R2]
```

RELATED TOPICS

- **Commands**

- `set_multicycle_path`
 - `set_max_delay`
 - `set_min_delay`
 - `check_target_constraints`
 - `report_rule_check`

- **Attributes**

- [Rule Attributes - rule_occurrence](#)

CONST_HIER_EXC15

NAME

CONST_HIER_EXC15 (CONSTRAINT_HIER_SIGNOFF)

MESSAGE

A block's path with max/min delay settings is part of a top-level multicycle path

DEFAULT SEVERITY

Warning

DESCRIPTION

This rule reports that a block path, which was set a maximum or minimum delay at the block level, has been affected by a multi-cycle path specification at the top level. This rule is checked by `check_target_constraints` command and results can be reported by `report_rule_check` command.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
block_view_name	string	Block view name
top_view_name	string	Top view name
block_path	object	Block Path
top_path	object	Top Path
block_file_line	unsigned	Block file line number
block_file_name	string	Block file name

Litmus Rule Reference
Timing Constraint Hierarchical Signoff--CONST_HIER_EXC

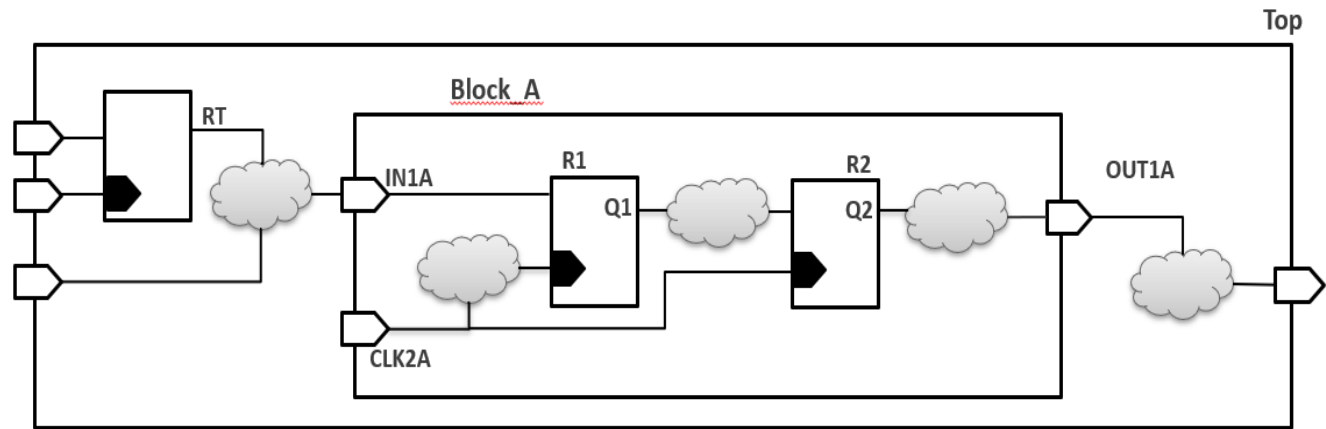
top_file_line	unsigned	Top file line number
top_file_name	string	Top file name
block_path_startpoint	object	Block path startpoint
top_path_startpoint	object	Top path startpoint
block_path_endpoint	object	Block path endpoint
top_path_endpoint	object	Top path endpoint
top_exc_cycle	int	top_exc_cycle
top_exc_cycle_on_start	string	top_exc_cycle_on_start
block_value	float	block_value
block_corner	string	block_corner
top_corner	string	top_corner
block_exc_from	object_list	block_exc_from
block_exc_to	object_list	block_exc_to
block_exc_through	object_list	block_exc_through
block_exc_type	string	block_exc_type
top_exc_from	object_list	top_exc_from
top_exc_to	object_list	top_exc_to
top_exc_through	object_list	top_exc_through
top_exc_type	string	top_exc_type
block_exc_from_clock	object_list	block_exc_fromclk
block_exc_to_clock	object_list	block_exc_tock
top_exc_from_clock	object_list	top_exc_fromclk
top_exc_to_clock	object_list	top_exc_tock

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block_path_startpoint_clock	object	block_path_startpoint_clock
block_path_endpoint_clock	object	block_path_endpoint_clock
top_path_startpoint_clock	object	top_path_startpoint_clock
top_path_endpoint_clock	object	top_path_endpoint_clock
scope	object	Scope

EXAMPLE

- For the figure below, the following commands in top level and block level constraints cause this rule check violation:



Top level constraint

```
set_multicycle_path 3 -from [get_cells Block_A/R1] -to [get_cells Block_A/R2]
```

Block level constraint (instance scope: Block_A)

```
set_max_delay 4 -from [get_cells R1] -to [get_cells R2]
```

RELATED TOPICS

- **Commands**

- `set_multicycle_path`
 - `set_max_delay`
 - `set_min_delay`
 - `check_target_constraints`
 - `report_rule_check`

- **Attributes**

- [Rule Attributes - rule_occurrence](#)

CONST_HIER_EXC16

NAME

CONST_HIER_EXC16 (CONSTRAINT_HIER_SIGNOFF)

MESSAGE

A block's path is part of a top-level path with different max/min delay settings

DEFAULT SEVERITY

Warning

DESCRIPTION

This rule reports that a block path is specified a minimum or maximum delay with a different value than that specified for a corresponding top level path. This rule is checked by `check_target_constraints` command and results can be reported by `report_rule_check` command.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
block_view_name	string	Block view name
top_view_name	string	Top view name
block_path	object	Block Path
top_path	object	Top Path
block_file_line	unsigned	Block file line number
block_file_name	string	Block file name

Litmus Rule Reference
Timing Constraint Hierarchical Signoff--CONST_HIER_EXC

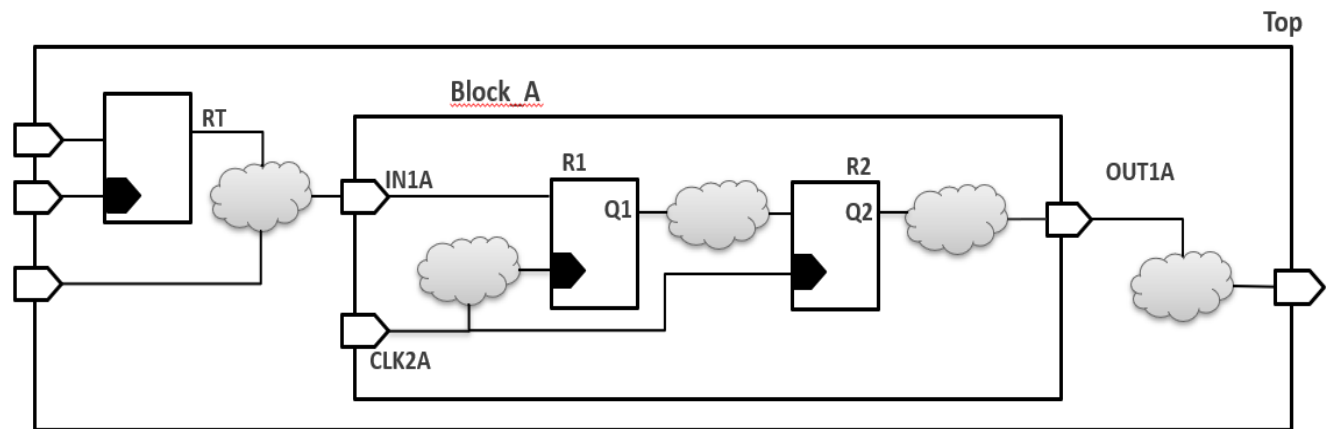
top_file_line	unsigned	Top file line number
top_file_name	string	Top file name
block_path_startpoint	object	Block path startpoint
top_path_startpoint	object	Top path startpoint
block_path_endpoint	object	Block path endpoint
top_path_endpoint	object	Top path endpoint
block_value	float	block_value
top_value	float	top_value
block_options	string	block_options
top_options	string	top_options
block_corner	string	block_corner
top_corner	string	top_corner
block_exc_from	object_list	Block exception from
block_exc_to	object_list	Block exception to
block_exc_through	object_list	Block exception through
block_exc_type	string	Block exception type
top_exc_from	object_list	Top exception from
top_exc_to	object_list	Top exception to
top_exc_through	object_list	Top exception through
top_exc_type	string	Top exception type
block_exc_from_clock	object_list	Block exception from clock
block_exc_to_clock	object_list	Block exception to clock
top_exc_from_clock	object_list	Top exception from clock

--	--	--

top_exc_to_clock	object_list	Top exception to clock
block_path_startpoint_clock	object	block_path_startpoint_clock
block_path_endpoint_clock	object	block_path_endpoint_clock
top_path_startpoint_clock	object	top_path_startpoint_clock
top_path_endpoint_clock	object	top_path_endpoint_clock
scope	object	Scope

EXAMPLE

- For the figure below, the following commands in top level and block level constraints cause this rule check violation:



Top level constraint

```
set_max_delay 8 -from [get_cells Block_A/R1] -to [get_cells Block_A/R2]
```

Block level constraint (instance scope: Block_A)

```
set_max_delay 4 -from [get_cells R1] -to [get_cells R2]
```

RELATED TOPICS

- **Commands**

- `set_max_delay`
 - `set_min_delay`
 - `check_target_constraints`
 - `report_rule_check`

- **Attributes**

- [Rule Attributes - rule_occurrence](#)

CONST_HIER_EXC17

NAME

CONST_HIER_EXC17 (CONSTRAINT_HIER_SIGNOFF)

MESSAGE

A block path without `set_false_path` is part of top-level paths and all such top-level paths are defined as false paths

DEFAULT SEVERITY

Warning

DESCRIPTION

This rule reports that a block path was not declared as false path at the block level and all top level paths containing this block path have been affected by top level false path specification. This rule is checked by `check_target_constraints` command and results can be reported by `report_rule_check` command.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
block_view_name	string	Block view name
top_view_name	string	Top view name
block_path	object	Block Path
top_path	object	Top Path
block_file_line	unsigned	Block file line number

Litmus Rule Reference
Timing Constraint Hierarchical Signoff--CONST_HIER_EXC

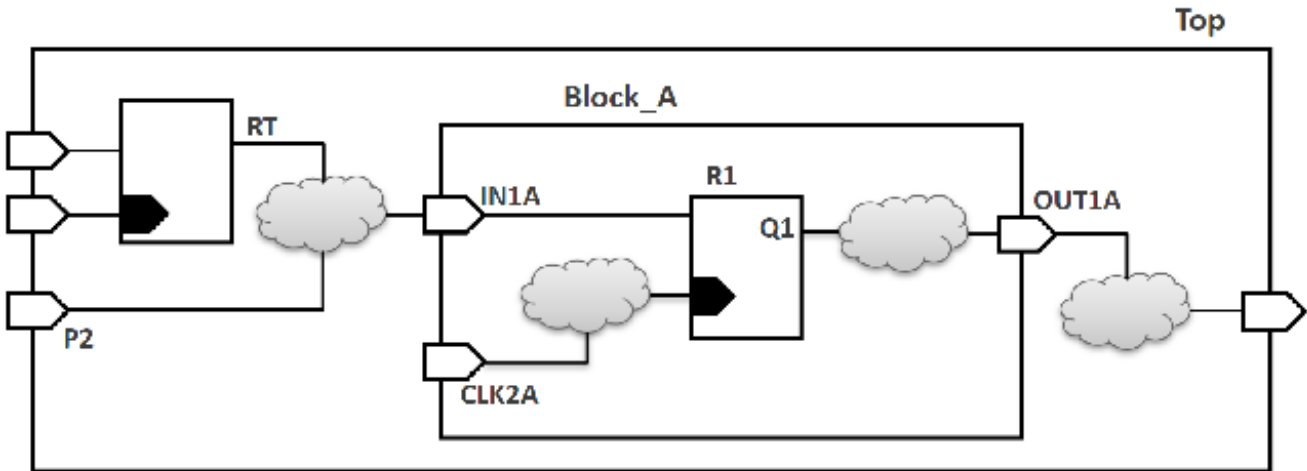
block_file_name	string	Block file name
top_file_line	unsigned	Top file line number
top_file_name	string	Top file name
block_path_startpoint	object	Block path startpoint
top_path_startpoint	object	Top path startpoint
block_path_endpoint	object	Block path endpoint
top_path_endpoint	object	Top path endpoint
block_corner	string	block_corner
top_corner	string	top_corner
block_exc_from	object_list	Block exception from
block_exc_to	object_list	Block exception to
block_exc_through	object_list	Block exception through
block_exc_type	string	Block exception type
top_exc_from	object_list	Top exception from
top_exc_to	object_list	Top exception to
top_exc_through	object_list	Top exception through
top_exc_type	string	Top exception type
block_exc_from_clock	object_list	Block exception from clock
block_exc_to_clock	object_list	Block exception to clock
top_exc_from_clock	object_list	Top exception from clock
top_exc_to_clock	object_list	Top exception to clock
block_path_startpoint_clock	object	block_path_startpoint_clock
block_path_endpoint_clock	object	block_path_endpoint_clock

--	--	--

top_path_startpoint_clock	object	top_path_startpoint_clock
top_path_endpoint_clock	object	top_path_endpoint_clock
scope	object	Scope

EXAMPLE

- For the figure below, if Block_A has no `set_false_path` defined in the block level constraints file and the top level constraints file has the following lines, then the Litmus issues a warning because the block level did not consider the path from R1 to OUT1A as false, even if all top paths containing this block path were affected by a false path specification. However, the path from IN1A to D1 was not reported by this rule check because the path from P2 through IN1A to D1 was not affected by any false path specification. However, such path will cause `CONST_HIER_EXC1` to be reported for the first top level exception.



Top level constraints

```
set_false_path -through RT
set_false_path -through Block_A/OUT1A
```

RELATED TOPICS

- Commands

- [set_false_path](#)
 - [check_target_constraints](#)
 - [report_rule_check](#)

- Attributes

- [Rule Attributes - rule_occurrence](#)

- Rules

- [CONST_HIER_EXC1](#)

CONST_HIER_GRP

- [CONST_HIER_GRP1](#)

CONST_HIER_GRP1

NAME

CONST_HIER_GRP1 (CONSTRAINT_HIER_SIGNOFF)

MESSAGE

Inconsistent clock relations between block-level clocks and top-level clocks

DEFAULT SEVERITY

Warning

DESCRIPTION

This rule reports that the relationship between the block-level clock and its corresponding top-level clock is inconsistent. The relationships being checked are: synchronous/asynchronous and mutually exclusive. The [report_hierarchical_clock_mapping](#) command can be used to understand how the clocks are mapped between the top level clocks and block level clocks to determine if this rule violation is due to incomplete clock mapping. This rule is checked by [check_target_constraints](#) command and results can be reported by [report_rule_check](#) command.

If multiple relationships are implied between the same clock pair, the following precedence is applied:

- Physically exclusive (highest)
- Asynchronous
- Logically exclusive (lowest)

RULE ATTRIBUTES

This hierarchical signoff rule check is customizable; its properties are specified through the attributes listed below. To view all the attributes for this rule check from the tool, the following command should be used:

```
report_rule_source "CONST_HIER_GRP1" -verbose
```

Attribute	Description
check_type	Specifies the types of clock relationships to be checked Possible Value: <all asynchronous logically_exclusive physically_exclusive> Where: all: Check the consistency of all types of clock relationships in block versus top. This is the default. asynchronous: Checks the consistency of only the synchronous/asynchronous clock relationship in block versus top. logically_exclusive: Checks the consistency of logically mutually exclusive clocks in block versus top. physically_exclusive: Checks the consistency of physically mutually exclusive clocks in block versus top.

check_connected	Specifies whether to check the clock connectivity before reporting an occurrence of this rule. This attribute will affect only the asynchronous check type. Possible Value: <yes no> Where: yes: Clock connectivity is checked. This is the default. no: Clock connectivity is not checked.
-----------------	--

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
block_view_name	string	Block view name
top_view_name	string	Top view name
scope	object	Scope
block_clock	object_list	Block clock
top_clock	object_list	Top clock
inconsistent_block_type	string	Inconsistent block type
inconsistent_top_type	string	Inconsistent top type
block_file_line	unsigned	Block file number
block_file_name	string	Block file name
top_file_line	unsigned	Top file number

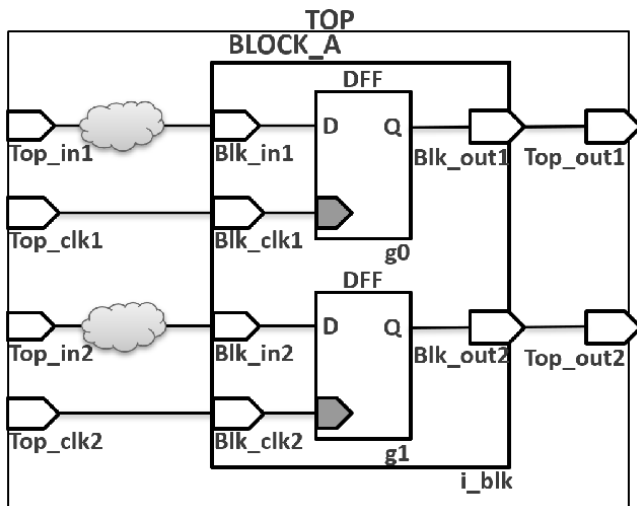
Litmus Rule Reference

Timing Constraint Hierarchical Signoff--CONST_HIER_GRP

top_file_name	string	Top file name

EXAMPLE

- For the figure below, the following top level constraints file defines Top_clk1 and Top_clk2 as physically exclusive, whereas the corresponding mapped clocks to block level, Blk_clk1 and Blk_clk2, respectively are synchronous to each other. This rule reports that the relationship between the top-level clocks (physically exclusive) is inconsistent to their corresponding block-level clocks (synchronous).



Top level constraints

```
create_clock -name Top_clk1 -period 8 [get_ports Top_clk1]
create_clock -name Top_clk2 -period 8 [get_ports Top_clk2]
set_clock_groups -name grp1 -group {Top_clk1} -group {Top_clk2} -
physically_exclusive
```

Block level constraints (instance scope: i_blk)

```
create_clock -name Blk_clk1 -period 8 [get_ports Blk_clk1]
create_clock -name Blk_clk2 -period 8 [get_ports Blk_clk2]
```

Hierarchical clock mapping

```
Top_clk1 is mapped to Blk_clk1
Top_clk2 is mapped to Blk_clk2
```

CONST_HIER_HFN

- [CONST_HIER_HFN2](#)

CONST_HIER_HFN2

NAME

CONST_HIER_HFN2 (CONSTRAINT_HIER_SIGNOFF)

MESSAGE

Inconsistent `set_dont_touch_network` at the top level vs. block

DEFAULT SEVERITY

Warning

DESCRIPTION

This rule reports that `set_dont_touch_network` is set on a block, but not at the top level or vice versa. The Litmus performs this check on a block's ports and pins. The attribute is propagated through the clock tree, which means the top level can set it on a port that drives a block's port, instead of directly on the block's port. If this attribute is set on a clock object, the Litmus treats it as if it was set on all of its source objects. This rule is checked by `check_target_constraints` command and the results can be reported by `report_rule_check` command.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
block_view_name	string	Block View Name
top_view_name	string	Top View Name
scope	object	Block Instance
block_object	object	Block object
top_object	object	Top object

type	string	Type

In addition, common rule occurrence attribute `type` provides the possible causes for the violation.

Attribute	Value	Description
type	BLOCK_NO_CONSTRAINT	Missing constraint at block level
	TOP_NO_CONSTRAINT	Missing constraint at top level
	DIFFERENT_CONSTRAINTS	Inconsistent constraints between top level and block level

CONST_HIER_IO

- [CONST_HIER_IO1](#)
- [CONST_HIER_IO2](#)
- [CONST_HIER_IO3](#)
- [CONST_HIER_IO4](#)
- [CONST_HIER_IO5i](#)
- [CONST_HIER_IO5o](#)
- [CONST_HIER_IO6i](#)
- [CONST_HIER_IO6o](#)
- [CONST_HIER_IO7i](#)
- [CONST_HIER_IO7o](#)
- [CONST_HIER_IO8i](#)
- [CONST_HIER_IO8o](#)
- [CONST_HIER_IO9](#)
- [CONST_HIER_IO10](#)
- [CONST_HIER_IO11i](#)
- [CONST_HIER_IO11o](#)
- [CONST_HIER_IO12i](#)
- [CONST_HIER_IO12o](#)
- [CONST_HIER_IO13i](#)
- [CONST_HIER_IO13o](#)

CONST_HIER_IO1

NAME

CONST_HIER_IO1 (CONSTRAINT_HIER_SIGNOFF)

MESSAGE

Block input port with delay vs. a clock is not in a timing path within the same clock group at the top level

DEFAULT SEVERITY

Warning

DESCRIPTION

For every block input port IN with input delay versus clock CLK, Litmus finds the top level clock groups of CLK's top level equivalent clock, and checks that there is a path between timing points in those clock groups that traverse IN. This rule is checked by [check_target_constraints](#) command and results can be reported by [report_rule_check](#) command.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
block_view_name	string	Block View Name
top_view_name	string	Top View Name
scope	object	Block Instance

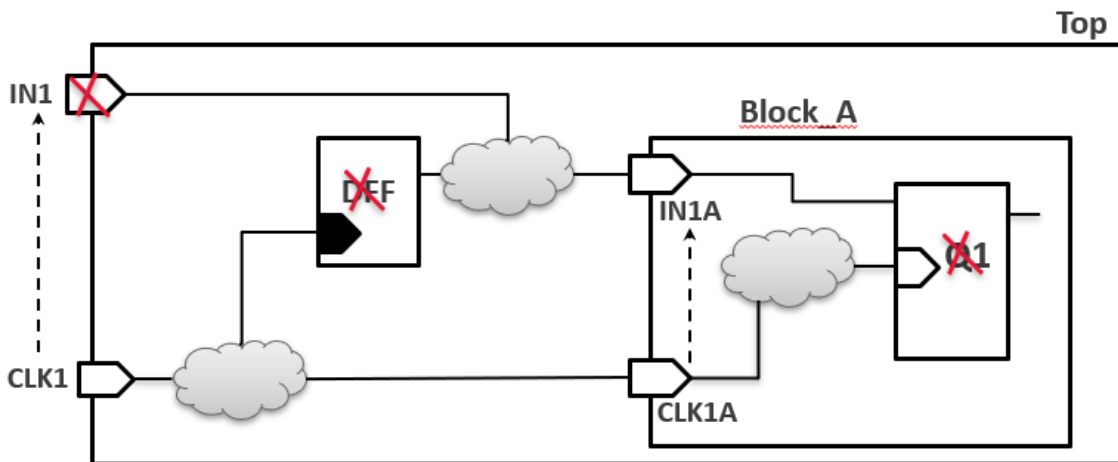
block_port	object	Block port
block_clock	object	Block-level clock
relative_top_clocks	object_list	Top clock
type	object_list	type
block_file_line	object_list	Block file number
block_file_name	object_list	Block file name

In addition, common rule occurrence attribute type provides the possible causes for the violation.

Attribute	Value	Description
type	TOP_NO_LAUNCH_CLOCK	Launch clock absent at top-level
	TOP_NO_CAPTURE_CLOCK	Capture clock absent at top-level
	TOP_UNCONSTRAINED	Top-level is unconstrained
	TOP_ASYNC	Top-level is asynchronous
	TOP_EXCLUSIVE	Top-level is mutually exclusive
	DIFFERENT_CLOCK_MAPPING	Inconsistent clock mapping

EXAMPLE

- The design in the figure below (where IN1A has an input delay versus CLK1A) would cause a rule check violation if the following were not present in the combinational fan-in of IN1A. This rule would also be flagged if no DFF (Q1) or port in the same clock group can be found in the fan-out of IN1A.
 - A DFF that is clocked by the top level clock CLK1
 - A primary input IN1 that has an input delay versus CLK1



- For the figure above, if a top level clock CLK2 is connected to the block level clock CLK1A and the top level DFF is clocked by another top level clock CLK1 from a different clock group other than CLK2 (asynchronous, defined like the following top level constraints), then this rule check violation will happen for the following block level constraints. Here, the block input port Block_A/IN1A has `set_input_delay` relative to block clock CLK1A which is getting top level equivalent clock CLK2. But at top level, the timing path is not in the same clock group of the top level equivalent clock.

Top level constraints

```
create_clock -name CLK1 -period 8 [get_ports CLK1] -waveform {0 4}
create_clock -name CLK2 -period 8 [get_ports CLK2] -waveform {0 4}
set_clock_groups -name group1 -group {CLK1} -group {CLK2} -asynchronous
```

Block level constraints (instance scope: Block_A)

```
create_clock -name CLK1A -period 8 [get_ports CLK1A] -waveform {0 4}
set_input_delay 1 -clock CLK1A [get_ports IN1A]
```

Litmus Rule Reference

Timing Constraint Hierarchical Signoff--CONST_HIER_IO

CONST_HIER_IO2

NAME

CONST_HIER_IO2 (CONSTRAINT_HIER_SIGNOFF)

MESSAGE

Block input port is in a timing path within a clock group for which it does not have input delay defined

DEFAULT SEVERITY

Warning

DESCRIPTION

For every block input port IN that has a timing startpoint in its fanin and a timing endpoint in its fanout, such as both belong to the same top level clock group, Litmus checks that IN has input delay defined versus a block clock whose top level equivalent clock is in the same clock group. This rule is checked by `check_target_constraints` command and results can be reported by `report_rule_check` command.

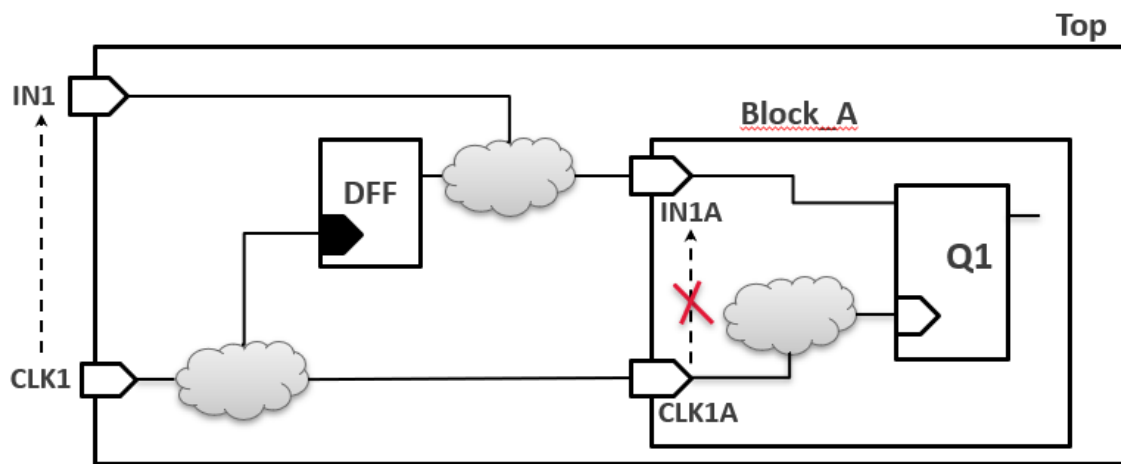
This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
block_view_name	string	Block view name
top_view_name	string	Top view name
scope	object	Scope
block_port	object	Block port
relative_top_clocks	object_list	Relative top clock

equivalent_block_clocks	object_list	Equivalent block clock

EXAMPLE

- The design in the following figure would cause a rule check violation, if IN1A did not have an input delay versus CLK1A like the following block level constraints and CLK1A is related to the top level clock CLK1, whose domain drives IN1A:



Top level constraint

```
create_clock -name CLK1 -period 8 [get_ports CLK1] -waveform {0 4}
```

Block level constraints (instance scope: Block_A)

```
create_clock -name CLK1A -period 8 [get_ports CLK1A] -waveform {0 4}
set_input_delay 1 -clock CLK1A [get_ports IN1A]
```

CONST_HIER_IO3

NAME

CONST_HIER_IO3 (CONSTRAINT_HIER_SIGNOFF)

MESSAGE

Block output port with delay vs. a clock is not in a timing path within the same clock group at the top level

DEFAULT SEVERITY

Warning

DESCRIPTION

For every block output port OUT with output delay versus clock CLK, Litmus finds the top level clock group of CLK's top level equivalent clock, and checks that there is a path between timing points in that clock group that traverses OUT. This rule is checked by [check_target_constraints](#) command and results can be reported by [report_rule_check](#) command.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
block_view_name	string	Block View Name
top_view_name	string	Top View Name
scope	object	Block Instance

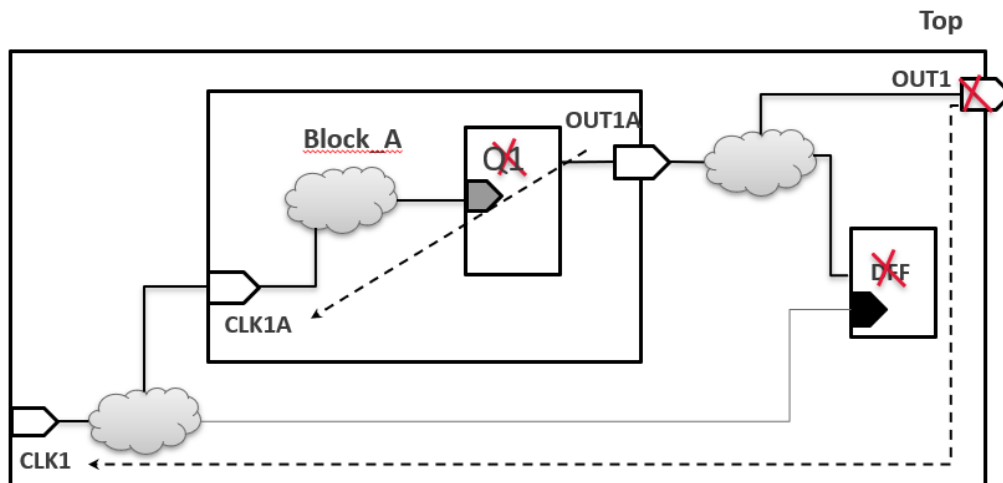
block_port	object	Output port of the block
block_clock	object	Block-level clock
relative_top_clocks	object_list	Top clock
type	object_list	type
block_file_line	object_list	Block file number
block_file_name	object_list	Block file name

In addition, common rule occurrence attribute type provides the possible causes for the violation.

Attribute	Value	Description
type	TOP_NO_LAUNCH_CLOCK	Launch clock is absent at top-level
	TOP_NO_CAPTURE_CLOCK	Capture clock is absent at top-level
	TOP_UNCONSTRAINED	top-level unconstrained
	TOP_ASYNC	Top-level is asynchronous
	TOP_EXCLUSIVE	Top-level is mutually exclusive
	DIFFERENT_CLOCK_MAPPING	Inconsistent clock mapping

EXAMPLE

- The design in the figure below (where OUT1A has a delay versus CLK1A) would cause a rule check violation if none of the following were present in the combinational fanout of OUT1A. This rule would also be flagged if no DFF/port in the same clock group can be found in the fanin of OUT1A.
 - A DFF that is clocked by the top level clock CLK1
 - A pin or primary output OUT1 that has an output delay versus CLK1



- For the figure above, if a top level clock CLK2 is connected to the block level clock CLK1A and the top level DFF is clocked by another top level clock CLK1 from a different clock group other than CLK2 (asynchronous, defined like the following top level constraints), then this rule check violation will happen for the following block level constraints. Here, the block output port Block_A/OUT1A has `set_output_delay` relative to block clock CLK1A which is getting top level equivalent clock CLK2. But at top level, the timing path is not in the same clock group of the top level equivalent clock.

Top level constraints

```
create_clock -name CLK1 -period 8 [get_ports CLK1] -waveform {0 4}
create_clock -name CLK2 -period 8 [get_ports CLK2] -waveform {0 4}
set_clock_groups -name group1 -group {CLK1} -group {CLK2} -asynchronous
```

Block level constraints (instance scope: Block_A)

```
create_clock -name CLK1A -period 8 [get_ports CLK1A] -waveform {0 4}
set_output_delay 1 -clock CLK1A [get_ports OUT1A]
```


CONST_HIER_IO4

NAME

CONST_HIER_IO4 (CONSTRAINT_HIER_SIGNOFF)

MESSAGE

Block output port is in a timing path within a clock group for which it does not have output delay defined

DEFAULT SEVERITY

Warning

DESCRIPTION

For every block output port OUT that has a timing startpoint in its fanin and a timing endpoint in its fanout, such as both belong to the same top level clock group, Litmus checks that OUT has output delay defined versus a block clock whose top level equivalent clock is in the same clock group. This rule is checked by `check_target_constraints` command and results can be reported by `report_rule_check` command.

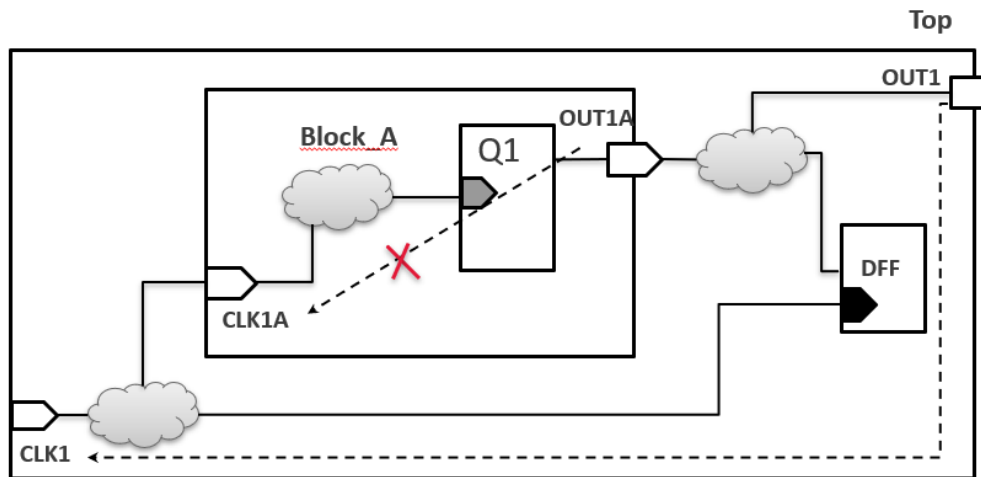
This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
block_view_name	string	Block view name
top_view_name	string	Top view name
scope	object	Scope
block_port	object	Block port
relative_top_clocks	object_list	Relative top clock

equivalent_block_clocks	object_list	Equivalent block clock

EXAMPLE

- The design in the following figure would cause a rule check violation, if OUT1A did not have an output delay versus CLK1A like the following block level constraints and CLK1A is related to the top level clock CLK1, whose domain is driven by block port OUT1A:



Top level constraint

```
create_clock -name CLK1 -period 8 [get_ports CLK1] -waveform {0 4}
```

Block level constraints (instance scope: Block_A)

```
create_clock -name CLK1A -period 8 [get_ports CLK1A] -waveform {0 4}
set_output_delay 1 -clock CLK1A [get_ports OUT1A]
```

CONST_HIER_IO5i

NAME

CONST_HIER_IO5i (CONSTRAINT_HIER_SIGNOFF)

MESSAGE

Missing `set_input_delay` in full-chip SDC

DEFAULT SEVERITY

Warning

DESCRIPTION

This rule reports that a `set_input_delay` in a block level constraints file has no corresponding `set_input_delay` at the top level constraints file. The corresponding delays are determined based on the clock equivalences and the presence or absence of the `-clock_fall` option. This rule is checked by `check_target_constraints` command and results can be reported by `report_rule_check` command.

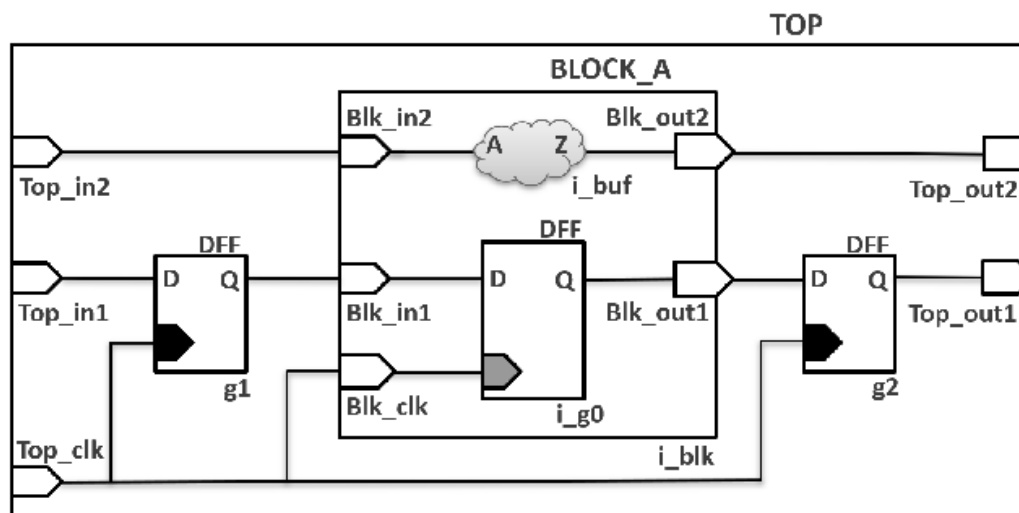
This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
block_view_name	string	Block view name
top_view_name	string	Top view name
scope	object	Scope
block_object	object	Block object
top_object	object	Top object

block_clock	object	Block clock
type	string	Type
block_value	float	block_value
block_file_line	object_list	Block file line number
block_file_name	object_list	Block file name

EXAMPLE

- For the figure below, if a port Top_in2 has no input delay vs. top level clock vtop_c1 like the following (with or without `-clock_fall1`), this rule will be reported if Top_in2 is connected to block port i_blk/Blk_in2 and the block level constraints file contains the following command:



Top level constraint

```
set_input_delay 1 -clock vtop_c1 [get_ports Top_in2]
```

Block level constraint (instance scope: i_blk)

```
set_input_delay 1 -clock vblk_c11 [get_ports Blk_in2]
```

- For the figure above, this rule will also be reported if the input delay at the block pin i_blk/i_buf/A is set by the block level constraint, but missing the top level constraint like the

following:

Top level constraint

```
set_input_delay 1 -clock vtop_c1 [get_pins i_blk/i_buf/A]
```

Block level constraint (instance scope: i_blk)

```
set_input_delay 1 -clock vblk_c11 [get_pins i_buf/A]
```

and the dofile contains the following:

```
set_timing_context_clock_mapping -name hier_clk_map -top_clock vtop_c1 -  
scope_name i_blk -scope_clocks vblk_c11 -force
```

CONST_HIER_IO5o

NAME

CONST_HIER_IO5o (CONSTRAINT_HIER_SIGNOFF)

MESSAGE

Missing `set_output_delay` in full-chip SDC

DEFAULT SEVERITY

Warning

DESCRIPTION

This rule reports that a `set_output_delay` in a block level constraints file has no corresponding `set_output_delay` at the top level constraints file. The corresponding delays are determined based on the clock equivalences and the presence or absence of the – `clock_fall` option. This rule is checked by `check_target_constraints` command and results can be reported by `report_rule_check` command.

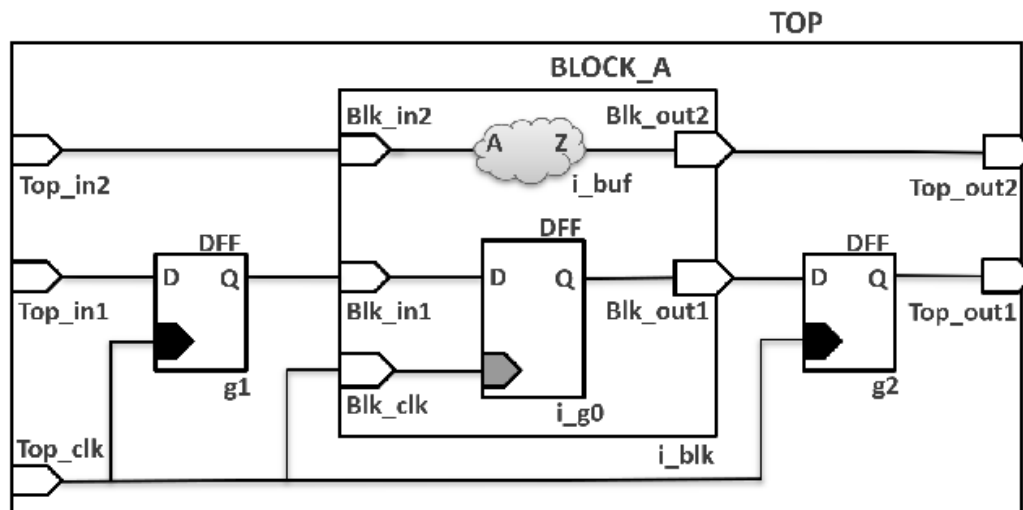
This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
block_view_name	string	Block view name
top_view_name	string	Top view name
scope	object	Scope
block_object	object	Block object
top_object	object	Top object

block_clock	object	Block clock
type	string	Type
block_value	float	block_value
block_file_line	object_list	Block file line number
block_file_name	object_list	Block file name

EXAMPLE

- For the figure below, if a port Top_out2 has no output delay vs. top level clock vtop_c1 like the following (with or without `-clock_fall`), this rule will be reported if Top_out2 is connected to block port i_blk/Blk_out2 and the block level constraints file contains the following command:



Top level constraint

```
set_output_delay 1 -clock vtop_c1 [get_ports Top_out2]
```

Block level constraint (instance scope: i_blk)

```
set_output_delay 1 -clock vblk_c11 [get_ports Blk_out2]
```

- For the figure above, this rule will also be reported if the output delay at the block pin

i_blk/i_buf/Z is set by the block level constraint, but missing the top level constraint like the following:

Top level constraint

```
set_output_delay 1 -clock vtop_c1 [get_pins i_blk/i_buf/Z]
```

Block level constraint (instance scope: i_blk)

```
set_output_delay 1 -clock vblk_c11 [get_pins i_buf/Z]
```

and the dofile contains the following:

```
set_timing_context_clock_mapping -name hier_clk_map -top_clock vtop_c1 -  
scope_name i_blk -scope_clocks vblk_c11 -force
```


CONST_HIER_IO6i

NAME

CONST_HIER_IO6i (CONSTRAINT_HIER_SIGNOFF)

MESSAGE

Missing `set_input_delay` in block/glue SDC

DEFAULT SEVERITY

Warning

DESCRIPTION

This rule reports that a `set_input_delay` in a top level constraints file has no corresponding `set_input_delay` at the block level constraints file. The corresponding delays are determined based on the clock equivalences and the presence or absence of the `-clock_fall` option. This rule is checked by `check_target_constraints` command and results can be reported by `report_rule_check` command.

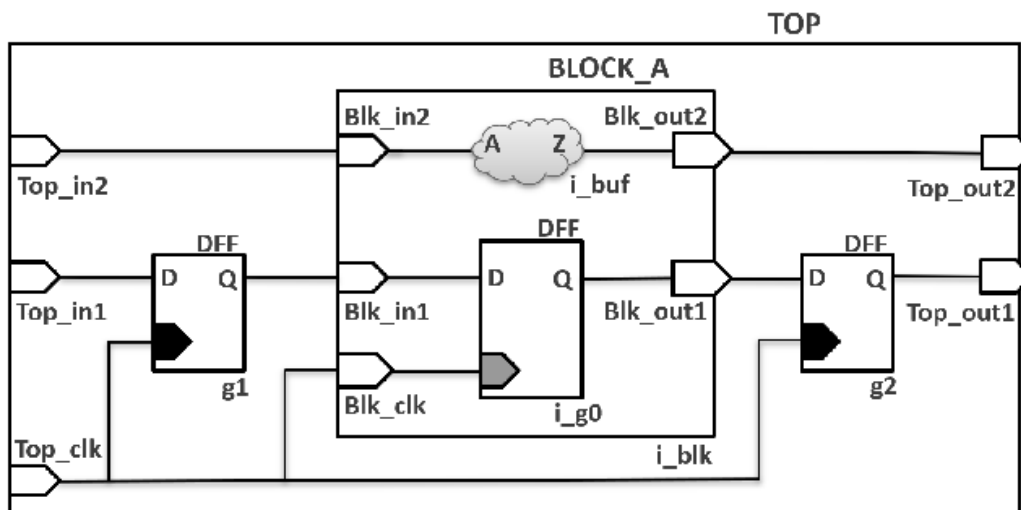
This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
block_view_name	string	Block view name
top_view_name	string	Top view name
scope	object	Scope
block_object	object	Block object
top_object	object	Top object

top_clock	object	Top clock
type	string	Type
top_value	float	top_value
top_file_line	object_list	Top file line number
top_file_name	object_list	Top file name

EXAMPLE

- For the figure below, if a port Top_in2 has input delay vs. top level clock vtop_c1 like the following (with or without -clock_fall), this rule will be reported if Top_in2 is connected to block port i_blk/Blk_in2 and the block level constraints file does not contain the following command:



Top level constraint

```
set_input_delay 1 -clock vtop_c1 [get_ports Top_in2]
```

Block level constraint (instance scope: i_blk)

```
set_input_delay 1 -clock vblk_c11 [get_ports Blk_in2]
```

- For the figure above, this rule will also be reported if the input delay at the block pin

i_blk/i_buf/A is set by the top level constraint, but missing the block level constraint like the following:

Top level constraint

```
set_input_delay 1 -clock vtop_c1 [get_pins i_blk/i_buf/A]
```

Block level constraint (instance scope: i_blk)

```
set_input_delay 1 -clock vblk_c11 [get_pins i_buf/A]
```

and the dofile contains the following:

```
set_timing_context_clock_mapping -name hier_clk_map -top_clock vtop_c1 -  
scope_name i_blk -scope_clocks vblk_c11 -force
```

CONST_HIER_IO6o

NAME

CONST_HIER_IO6o (CONSTRAINT_HIER_SIGNOFF)

MESSAGE

Missing `set_output_delay` in block/glue SDC

DEFAULT SEVERITY

Warning

DESCRIPTION

This rule reports that a `set_output_delay` in a top level constraints file has no corresponding `set_output_delay` at the block level constraints file. The corresponding delays are determined based on the clock equivalences and the presence or absence of the `-clock_fall` option. This rule is checked by `check_target_constraints` command and results can be reported by `report_rule_check` command.

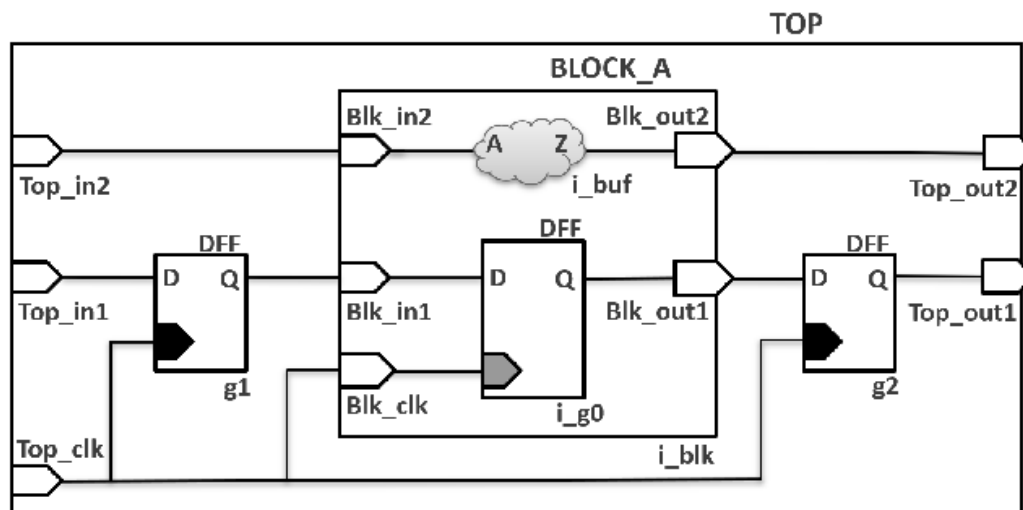
This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
block_view_name	string	Block view name
top_view_name	string	Top view name
scope	object	Scope
block_object	object	Block object
top_object	object	Top object

top_clock	object	Top clock
type	string	Type
top_value	float	top_value
top_file_line	object_list	Top file line number
top_file_name	object_list	Top file name

EXAMPLE

- For the figure below, if a port Top_out2 has output delay vs. top level clock vtop_c1 like the following (with or without `-clock_fall1`), this rule will be reported if Top_out2 is connected to block port i_blk/Blk_out2 and the block level constraints file does not contain the following command:



Top level constraint

```
set_output_delay 1 -clock vtop_c1 [get_ports Top_out2]
```

Block level constraint (instance scope: i_blk)

```
set_output_delay 1 -clock vblk_c11 [get_ports Blk_out2]
```

- For the figure above, this rule will also be reported if the output delay at the block pin

i_blk/i_buf/Z is set by the top level constraint, but missing the block level constraint like the following:

Top level constraint

```
set_output_delay 1 -clock vtop_c1 [get_pins i_blk/i_buf/Z]
```

Block level constraint (instance scope: i_blk)

```
set_output_delay 1 -clock vblk_c11 [get_pins i_buf/Z]
```

and the dofile contains the following:

```
set_timing_context_clock_mapping -name hier_clk_map -top_clock vtop_c1 -  
scope_name i_blk -scope_clocks vblk_c11 -force
```

CONST_HIER_IO7i

NAME

CONST_HIER_IO7i (CONSTRAINT_HIER_SIGNOFF)

MESSAGE

Block/glue `set_input_delay` has smaller value than in full-chip

DEFAULT SEVERITY

Warning

DESCRIPTION

This rule reports that a `set_input_delay` in a block level constraints file has a corresponding `set_input_delay` at the top level constraints file, which has a bigger value. Only delays with `-clock` are considered. The corresponding delays are determined based on the clock equivalences with or without using the `set_timing_context_clock_mapping` command in the dofile and the presence or absence of the `-clock_fall` option. This rule is checked by `check_target_constraints` command and results can be reported by `report_rule_check` command.

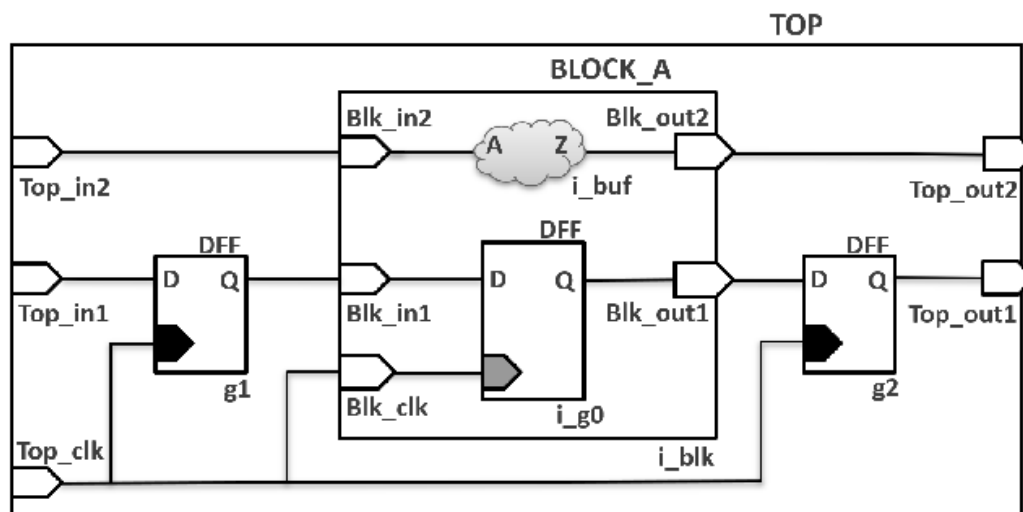
This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
block_view_name	string	Block view name
top_view_name	string	Top view name
scope	object	Scope
block_object	object	Block object

top_object	object	Top object
is_directly_connected	bool	Is directly connected
type	string	Type
block_value	float	block_value
top_value	float	top_value
block_file_line	object_list	Block file line number
block_file_name	object_list	Block file name
top_file_line	object_list	Top file line number
top_file_name	object_list	Top file name

EXAMPLE

- For the figure below, this rule will be reported if port Top_in2 is connected to block port i_blk/Blk_in2 and the top and block level constraints file have the following commands:



Top level constraint

```
set_input_delay 2 -clock vtop_c1 [get_ports Top_in2]
```


Block level constraint (instance scope: i_blk)

```
set_input_delay 1 -clock vblk_c11 [get_ports Blk_in2]
```

- For the figure above, this rule will also be reported if the input delay at some block pin i_blk/i_buf/A is set by the top and block level constraints file like the following:

Top level constraint

```
set_input_delay 2 -clock vtop_c1 [get_pins i_blk/i_buf/A]
```

Block level constraint (instance scope: i_blk)

```
set_input_delay 1 -clock vblk_c11 [get_pins i_buf/A]
```

and the dofile contains the following:

```
set_timing_context_clock_mapping -name hier_clk_map -top_clock vtop_c1 -  
scope_name i_blk -scope_clocks vblk_c11 -force
```

CONST_HIER_IO7o

NAME

CONST_HIER_IO7o (CONSTRAINT_HIER_SIGNOFF)

MESSAGE

Block/glue `set_output_delay` has smaller value than in full-chip

DEFAULT SEVERITY

Warning

DESCRIPTION

This rule reports that a `set_output_delay` in a block level constraints file has a corresponding `set_output_delay` at the top level constraints file, which has a bigger value. Only delays with `-clock` are considered. The corresponding delays are determined based on the clock equivalences with or without using the `set_timing_context_clock_mapping` command in the dofile and the presence or absence of the `-clock_fall` option. This rule is checked by `check_target_constraints` command and results can be reported by `report_rule_check` command.

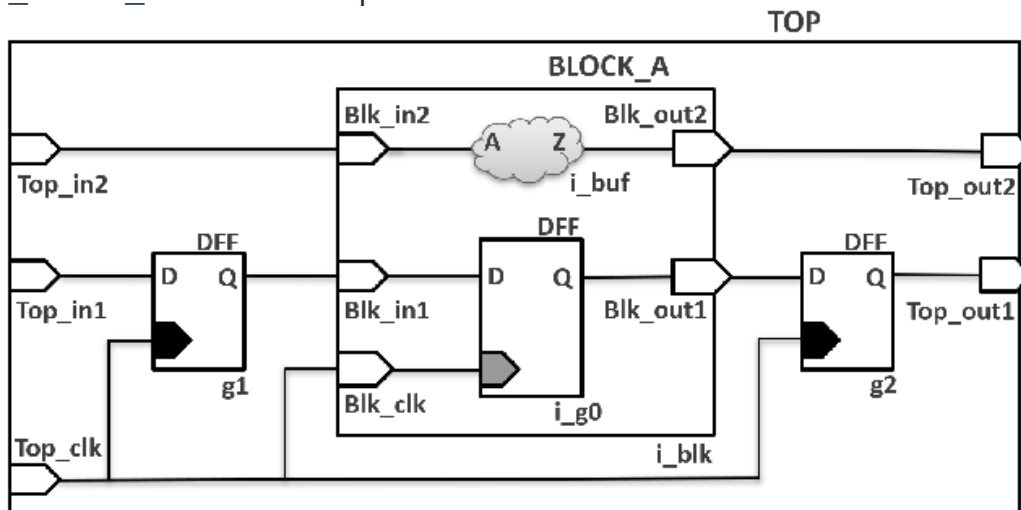
This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
block_view_name	string	Block view name
top_view_name	string	Top view name
scope	object	Scope
block_object	object	Block object
top_object	object	Top object

is_directly_connected	bool	Is directly connected
type	string	Type
block_value	float	block_value
top_value	float	top_value
block_file_line	object_list	Block file line number
block_file_name	object_list	Block file name
top_file_line	object_list	Top file line number
top_file_name	object_list	Top file name

EXAMPLE

- For the figure below, this rule will be reported if port Top_out2 is connected to block port i_blk/Blk_out2 and the top and block level constraints file have the following commands:



Top level constraint

```
set_output_delay 2 -clock vtop_cl [get_ports Top_out2]
```

Block level constraint (instance scope: i_blk)

```
set_output_delay 1 -clock vblk_c11 [get_ports Blk_out2]
```

- For the figure above, this rule will also be reported if the output delay at the block pin i_blk/i_buf/Z is set by the top and block level constraints file like the following:

Top level constraint

```
set_output_delay 2 -clock vtop_c1 [get_pins i_blk/i_buf/Z]
```

Block level constraint (instance scope: i_blk)

```
set_output_delay 1 -clock vblk_c11 [get_pins i_buf/Z]
```

and the dofile contains the following:

```
set_timing_context_clock_mapping -name hier_clk_map -top_clock vtop_c1 -  
scope_name i_blk -scope_clocks vblk_c11 -force
```

CONST_HIER_IO8i

NAME

CONST_HIER_IO8i (CONSTRAINT_HIER_SIGNOFF)

MESSAGE

Block/glue `set_input_delay` has greater value than in full-chip

DEFAULT SEVERITY

Warning

DESCRIPTION

This rule reports that a `set_input_delay` in a block level constraints file has a corresponding `set_input_delay` at the top level constraints file, which has a smaller value. Only delays with `-clock` are considered. The corresponding delays are determined based on the clock equivalences with or without using the `set_timing_context_clock_mapping` command in the dofile and the presence or absence of the `-clock_fall` option. This rule is checked by `check_target_constraints` command and results can be reported by `report_rule_check` command. Note that when this rule is reported for blocks, it does not necessarily mean there is an error, unless the blocks' ports are connected directly to top level ports. This rule is provided for completeness.

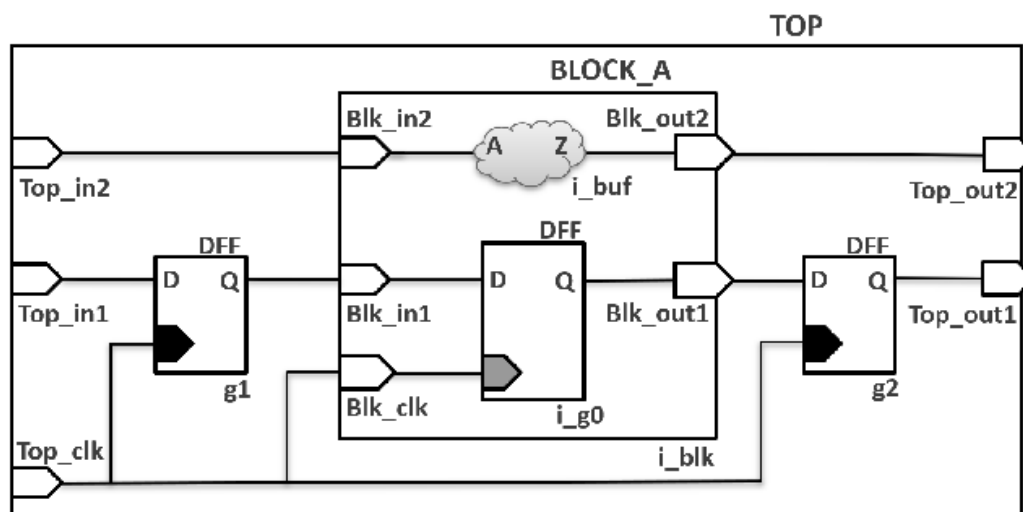
This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
block_view_name	string	Block view name
top_view_name	string	Top view name
scope	object	Scope

block_object	object	Block object
top_object	object	Top object
is_directly_connected	bool	Is directly connected
type	string	Type
block_value	float	block_value
top_value	float	top_value
block_file_line	object_list	Block file line number
block_file_name	object_list	Block file name
top_file_line	object_list	Top file line number
top_file_name	object_list	Top file name

EXAMPLE

- For the figure below, this rule will be reported if port Top_in2 is connected to block port i_blk/Blk_in2 and the top and block level constraints file have the following commands:



Top level constraint

```
set_input_delay 1 -clock vtop_c1 [get_ports Top_in2]
```

Block level constraint (instance scope: i_blk)

```
set_input_delay 2 -clock vblk_c11 [get_ports Blk_in2]
```

- For the figure above, this rule will also be reported if the input delay at some block pin i_blk/i_buf/A is set by the top and block level constraints file like the following:

Top level constraint

```
set_input_delay 1 -clock vtop_c1 [get_pins i_blk/i_buf/A]
```

Block level constraint (instance scope: i_blk)

```
set_input_delay 2 -clock vblk_c11 [get_pins i_buf/A]
```

and the dofile contains the following:

```
set_timing_context_clock_mapping -name hier_clk_map -top_clock vtop_c1 -  
scope_name i_blk -scope_clocks vblk_c11 -force
```

CONST_HIER_IO8o

NAME

CONST_HIER_IO8o (CONSTRAINT_HIER_SIGNOFF)

MESSAGE

Block/glue `set_output_delay` has greater value than in full-chip

DEFAULT SEVERITY

Warning

DESCRIPTION

This rule reports that a `set_output_delay` in a block level constraints file has a corresponding `set_output_delay` at the top level constraints file, which has a smaller value. Only delays with `-clock` are considered. The corresponding delays are determined based on the clock equivalences with or without using the `set_timing_context_clock_mapping` command in the dofile and the presence or absence of the `-clock_fall` option. This rule is checked by `check_target_constraints` command and results can be reported by `report_rule_check` command.

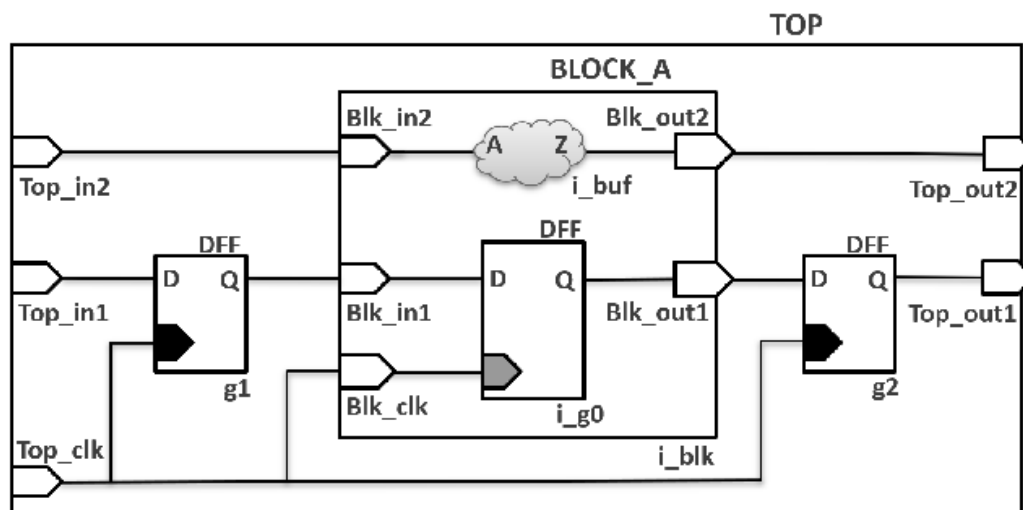
This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
block_view_name	string	Block view name
top_view_name	string	Top view name
scope	object	Scope
block_object	object	Block object

top_object	object	Top object
is_directly_connected	bool	Is directly connected
type	string	Type
block_value	float	block_value
top_value	float	top_value
block_file_line	object_list	Block file line number
block_file_name	object_list	Block file name
top_file_line	object_list	Top file line number
top_file_name	object_list	Top file name

EXAMPLE

- For the figure below, this rule will be reported if port Top_out2 is connected to block port i_blk/Blk_out2 and the top and block level constraints file have the following commands:



Top level constraint

```
set_output_delay 1 -clock vtop_cl [get_ports Top_out2]
```

Block level constraint (instance scope: i_blk)

```
set_output_delay 2 -clock vblk_c11 [get_ports Blk_out2]
```

- For the figure above, this rule will also be reported if the output delay at the block pin i_blk/i_buf/Z is set by the top and block level constraints file like the following:

Top level constraint

```
set_output_delay 1 -clock vtop_c1 [get_pins i_blk/i_buf/Z]
```

Block level constraint (instance scope: i_blk)

```
set_output_delay 2 -clock vblk_c11 [get_pins i_buf/Z]
```

and the dofile contains the following:

```
set_timing_context_clock_mapping -name hier_clk_map -top_clock vtop_c1 -  
scope_name i_blk -scope_clocks vblk_c11 -force
```

CONST_HIER_IO9

NAME

CONST_HIER_IO9 (CONSTRAINT_HIER_SIGNOFF)

MESSAGE

Inter-block connection output/input delays under-constrain the path

DEFAULT SEVERITY

Warning

DESCRIPTION

Indicates that a timing path crossing between two hierarchical blocks is budgeted more delay ($D1+D2$) than the total clock period (T).

At each point where the path crosses from one block to another one, the output (OD) and input delays (ID) at both sides represent the delay allocated ($(D1=T-OD)$, $(D2=T-ID)$, respectively) to the other side of the crossing. If these add up to less than the clock period ($OD+ID < T$), then the path is under-constrained ($D1+D2=(2T-OD-ID)=T++$).

The input and output delays to be correlated must be defined versus block clocks that have a common shared top level equivalent clock, whose period will be used for the check. This check requires the top-level constraints to have clocks defined and the block-top equivalences to be correctly extracted by the tool or specified in the dofile using the

`set_timing_context_clock_mapping` command.

To report clock equivalences `report_hierarchical_clock_mapping` command is used. This rule is checked by `check_target_constraints` command and results can be reported by `report_rule_check` command.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
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Litmus Rule Reference
Timing Constraint Hierarchical Signoff--CONST_HIER_IO

occurrence_id	unsigned	Occurrence id
block_view_name	string	Block View Name
top_view_name	string	Top View Name
source_scope	object	Source Block Instance
sink_scope	object	Sink Block Instance
source_block_port	object	Source Block Port
sink_block_port	object	Sink Block Port
top_clock	object	Top-level clock definition
source_delay_value	float	Source Delay Value
sink_delay_value	float	Sink Delay Value
top_time_period	double	Top time period
type	string	Type
source_file_line	unsigned	Source file number
source_file_name	string	Source file name
sink_file_line	unsigned	Sink file number

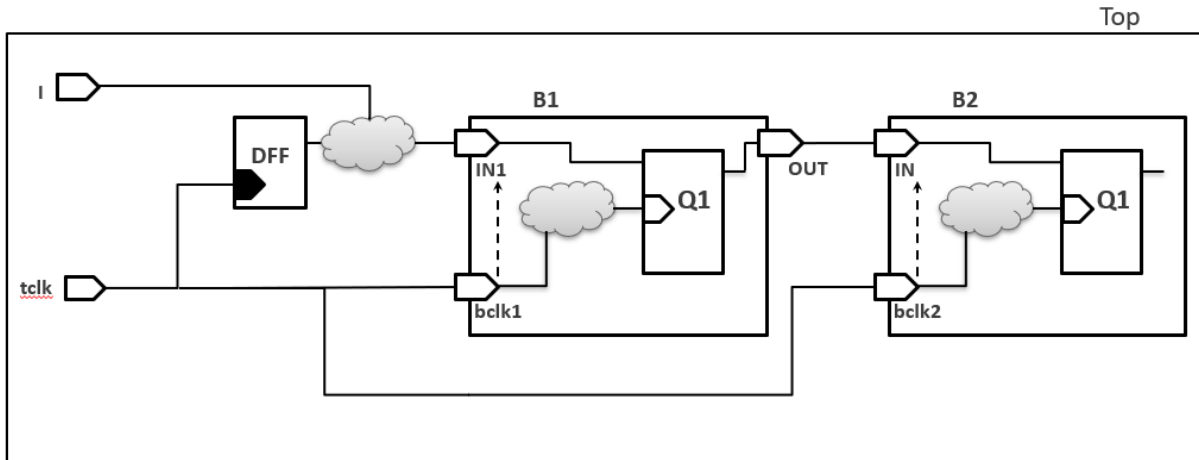
sink_file_name	string	Sink file name
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In addition, common rule occurrence attribute type provides the possible causes for the violation.

Attribute	Value	Description
type	UNDERCONSTRAINED_DELAYS	Inconsistent delays

EXAMPLE

- For the figure below, there is a path from B1/OUT to B2/IN and `tclk` is the top-level equivalent clock of B1/bclk1 and of B2/bclk2. For the following top and both block level constraints, the sum of the output delay of B1/OUT and the input delay of B2/IN is less than the clock period of clock `tclk` which will report this rule.



Top level constraint

```
create_clock -name tclk -period 4 [get_ports tclk]
```

Block level constraint (instance scope: B1)

```
set_output_delay 1 -clock bclk1 OUT
```

Block level constraint (instance scope: B2)

```
set_input_delay 2 -clock bclk2 IN
```

CONST_HIER_IO10

NAME

CONST_HIER_IO10 (CONSTRAINT_HIER_SIGNOFF)

MESSAGE

Inter-block connection output/input delays over-constrain the path

DEFAULT SEVERITY

Warning

DESCRIPTION

Indicates that a timing path crossing between two hierarchical blocks is budgeted less delay ($D1+D2$) than the total clock period (T). At each point where the path crosses from one block to another one, the output (OD) and input delays (ID) at both sides represent the delay allocated ($(D1=T-OD)$, $(D2=T-ID)$, respectively) to the other side of the crossing. If these add up to more than the clock period ($OD+ID > T$), then the path is over-constrained ($D1+D2=(2T-OD-ID)=T--$).

The input and output delays to be correlated must be defined versus block clocks that have a common shared top level equivalent clock, whose period will be used for the check. This check requires the top-level constraints to have clocks defined and the block-top equivalences to be correctly extracted by the tool or specified in the dofile using the `set_timing_context_clock_mapping` command.

To report clock equivalences `report_hierarchical_clock_mapping` command is used. This rule is checked by `check_target_constraints` command and results can be reported by `report_rule_check` command.

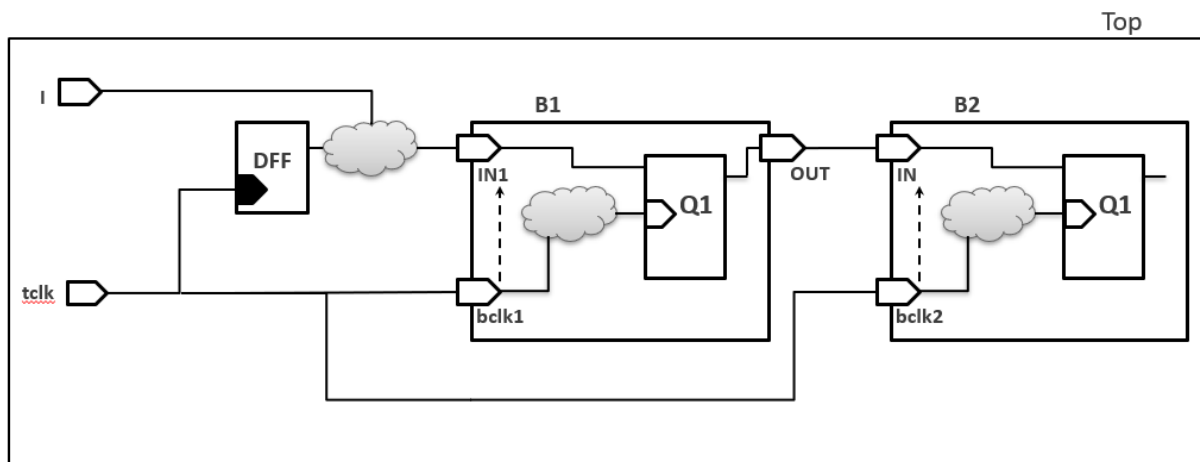
This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id

block_view_name	string	Block view name
top_view_name	string	Top view name
source_scope	object	Source Block Instance
sink_scope	object	Sink Block Instance
source_block_port	object	Source Block Port
sink_block_port	object	Sink Block Port
top_clock	object	Top clock
source_delay_value	float	Source Delay Value
sink_delay_value	float	Sink Delay Value
top_time_period	double	Top time period
type	string	Type
source_file_line	unsigned	Source file number
source_file_name	string	Source file name
sink_file_line	unsigned	Sink file number
sink_file_name	string	Sink file name

EXAMPLE

- For the figure below, there is a path from B1/OUT to B2/IN and `tclk` is the top-level equivalent clock of B1/bclk1 and of B2/bclk2. For the following top and both block level constraints, the sum of the output delay of B1/OUT and the input delay of B2/IN is larger than the clock period of clock `tclk` which will report this rule.



Top level constraint

```
create_clock -name tclk -period 4 [get_ports tclk]
```

Block level constraint (instance scope: B1)

```
set_output_delay 3 -clock bclk1 OUT
```

Block level constraint (instance scope: B2)

```
set_input_delay 2 -clock bclk2 IN
```

CONST_HIER_IO11i

NAME

CONST_HIER_IO11i (CONSTRAINT_HIER_SIGNOFF)

MESSAGE

Block input is undriven

DEFAULT SEVERITY

Warning

DESCRIPTION

Indicates that a block input is undriven. This rule is checked by `check_target_constraints` command and the results can be reported by `report_rule_check` command.

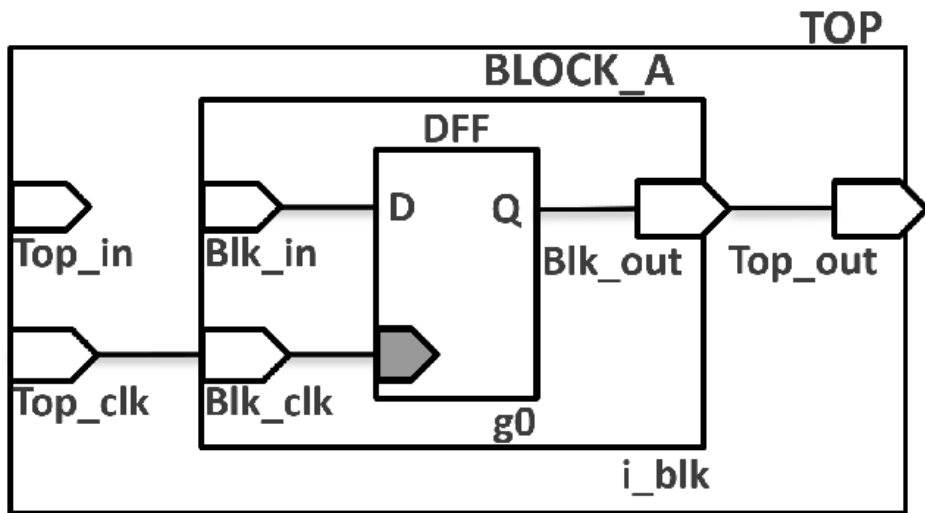
This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
block_view_name	string	Block view name
top_view_name	string	Top view name
scope	object	Scope
fstaref	object	Port/Pin in the block
output	bool	Indicates input/output in the block
found_obj	object	Start/end point
type	string	Type

is_found	bool	Indicates if the object is found

EXAMPLE

- For the figure below, the following source code contains an undriven block input. This rule is reported for the undriven block input i_blk/Blk_in.



```
module TOP (input Top_in, Top_clk, output Top_out);
    BLOCK_A i_blk (.Blk_in(), .Blk_clk(Top_clk), .Blk_out(Top_out));
endmodule
```

CONST_HIER_IO11o

NAME

CONST_HIER_IO11o (CONSTRAINT_HIER_SIGNOFF)

MESSAGE

Block output is floating

DEFAULT SEVERITY

Warning

DESCRIPTION

Indicates that a block output is floating. This rule is checked by `check_target_constraints` command and the results can be reported by `report_rule_check` command.

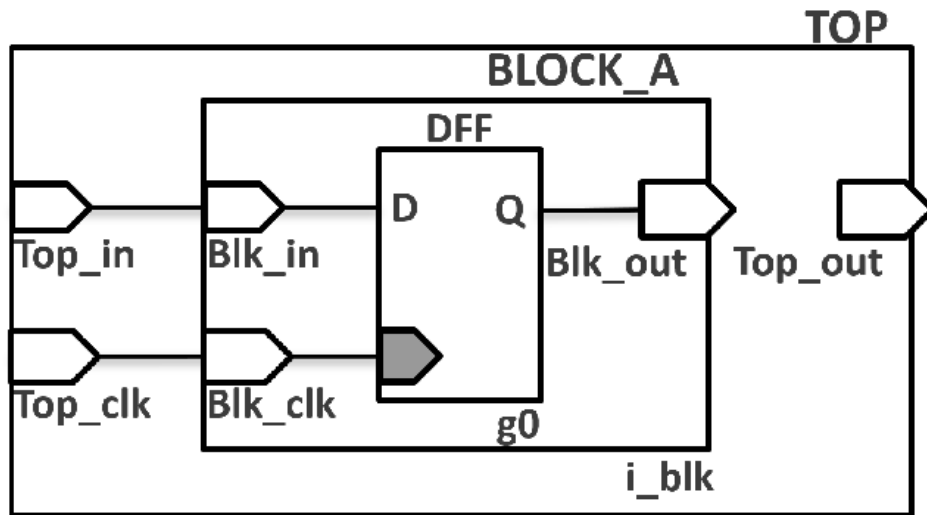
This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
block_view_name	string	Block view name
top_view_name	string	Top view name
scope	object	Scope
fstaref	object	Port/Pin in the block
output	bool	Indicates input/output in the block
found_obj	object	Start/end point
type	string	Type

is_found	bool	Indicates if the object is found

EXAMPLE

- For the figure below, the following source code contains a floating block output. This rule is reported for the floating block output i_blk/Blk_out.



```
module TOP (input Top_in, Top_clk, output Top_out);
    BLOCK_A i_blk (.Blk_in(Top_in), .Blk_clk(Top_clk), .Blk_out());
endmodule
```

CONST_HIER_IO12i

NAME

CONST_HIER_IO12i (CONSTRAINT_HIER_SIGNOFF)

MESSAGE

Inconsistent `set_input_delay` at the top-level vs. block

DEFAULT SEVERITY

Warning

DESCRIPTION

Indicates that an inconsistency in `set_input_delay` command is found between top level and block level constraints. The following are the possible causes for this violation:

- `set_input_delay` commands specified at both top level and block level but with different values

This rule is checked by `check_target_constraints` command and the results can be reported by `report_rule_check` command.

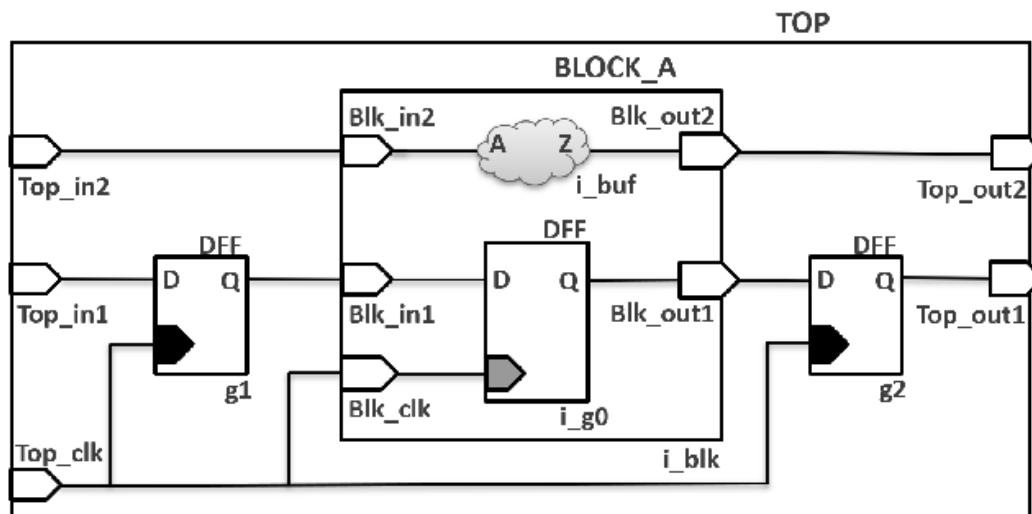
This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
block_view_name	string	Block view name
top_view_name	string	Top view name
scope	object	Scope
block_pin	object	Block pin

top_pin	object	Top pin
block_clock	object	Block clock
top_clock	object	Top clock
type	string	Type
block_file_line	object_list	Block file line number
block_file_name	object_list	Block file name
top_file_line	object_list	Top file line number
top_file_name	object_list	Top file name

EXAMPLE

- For the figure below, the following example contains `set_input_delay` commands specified in both top level and block level constraints. This rule is reported for the `set_input_delay` commands, because they are specified with different values (-reference_pin) provided the top level clock top_vclk and the block level clock blk_vclk are mapped using `set_timing_context_clock_mapping` command in the dofile.



Top level constraints

```
create_clock -name top_vclk -period 2
set_input_delay -clock [get_clocks top_vclk] -reference_pin [get_pins
i_blk/Blk_in1] 1.0 [get_pins i_blk/i_buf/Z]
```

Block level constraints (instance scope: i_blk)

```
create_clock -name blk_vclk -period 2
set_input_delay -clock [get_clocks blk_vclk] -reference_pin [get_pins i_buf/A]
1.0 [get_pins i_buf/Z]
```

Hierarchical clock mapping

```
top_vclk is mapped to blk_vclk
```


CONST_HIER_IO12o

NAME

CONST_HIER_IO12o (CONSTRAINT_HIER_SIGNOFF)

MESSAGE

Inconsistent `set_output_delay` at the top-level vs. block

DEFAULT SEVERITY

Warning

DESCRIPTION

Indicates that an inconsistency in `set_output_delay` command is found between top level and block level constraints. The following are the possible causes for this violation:

- `set_output_delay` commands specified at both top level and block level but with different values

This rule is checked by `check_target_constraints` command and the results can be reported by `report_rule_check` command.

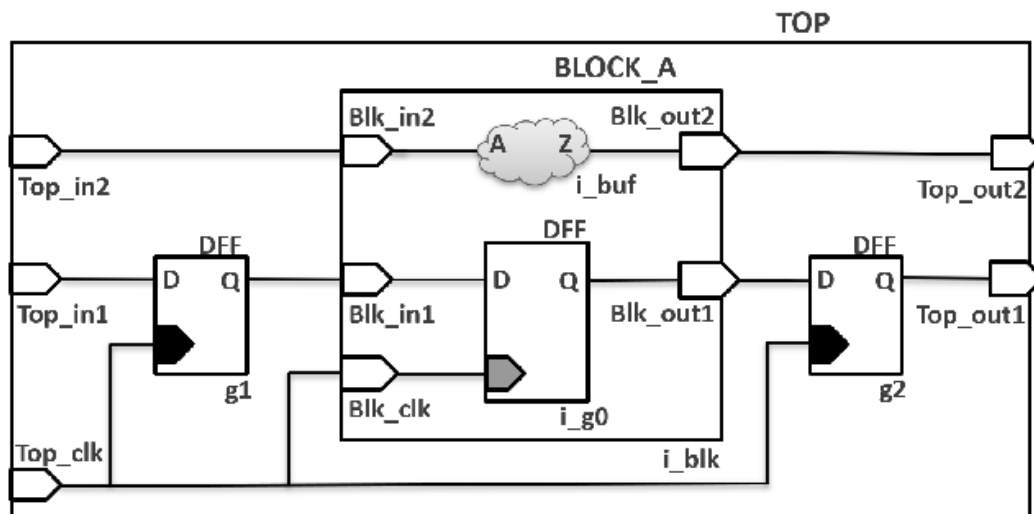
This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
block_view_name	string	Block view name
top_view_name	string	Top view name
scope	object	Scope
block_pin	object	Block pin

top_pin	object	Top pin
block_clock	object	Block clock
top_clock	object	Top clock
type	string	Type
block_file_line	object_list	Block file line number
block_file_name	object_list	Block file name
top_file_line	object_list	Top file line number
top_file_name	object_list	Top file name

EXAMPLE

- For the figure below, the following example contains `set_output_delay` commands specified in both top level and block level constraints. This rule is reported for the `set_output_delay` commands, because they are specified with different values (-reference_pin) provided the top level clock `top_vclk` and the block level clock `blk_vclk` are mapped using `set_timing_context_clock_mapping` command in the dofile.



Top level constraints

```
create_clock -name top_vclk -period 2
set_output_delay -clock [get_clocks top_vclk] -reference_pin [get_pins
i_blk/Blk_in1] 1.0 [get_pins i_blk/i_buf/Z]
```

Block level constraints (instance scope: i_blk)

```
create_clock -name blk_vclk -period 2
set_output_delay -clock [get_clocks blk_vclk] -reference_pin [get_pins i_buf/A]
1.0 [get_pins i_buf/Z]
```

Hierarchical clock mapping

```
top_vclk is mapped to blk_vclk
```

CONST_HIER_IO13i

NAME

CONST_HIER_IO13i (CONSTRAINT_HIER_SIGNOFF)

MESSAGE

Block `set_input_delay` reference clock edge does not match design trigger edge

DEFAULT SEVERITY

Warning

DESCRIPTION

Indicates that block level `set_input_delay` reference clock edge does not match design trigger edge. For a block input connected to a sequential cell in the fanin, a violation is reported when the following condition is broken:

- If the sequential cell is rising edge triggered or positive level enabled, the block level `set_input_delay` command on the block input should be specified without `-clock_fall` option
- If the sequential cell is falling edge triggered or negative level enabled, the block level `set_input_delay` command on the block input should be specified with `-clock_fall` option

This rule is checked by `check_target_constraints` command and the results can be reported by `report_rule_check` command.

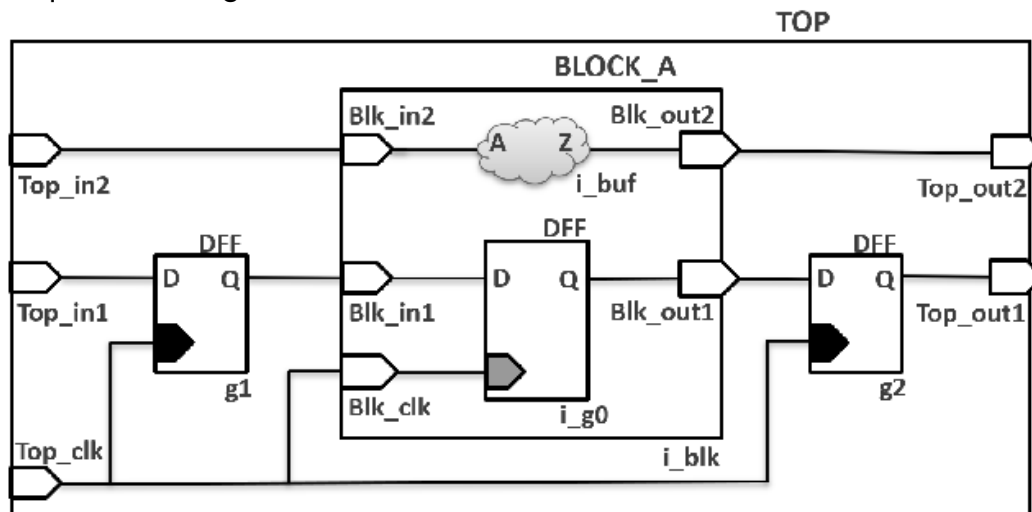
This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
block_view_name	string	Block view name

top_view_name	string	Top view name
scope	object	Scope
block_pin	object	Block pin
top_pin	object	Top pin
block_clock	object	Block clock
top_clock	object	Top clock
type	string	Type
block_file_line	object_list	Block file line number
block_file_name	object_list	Block file name

EXAMPLE

- The following example contains `set_input_delay` command specified in block level constraints. This rule is reported for the `set_input_delay` command, because it is specified with `-clock_fall` option, but the block input Blk_in1 is connected to rising edge triggered sequential cell g1.



Top level constraints

```
create_clock -name top_clk -period 2 [get_ports Top_clk]
```

Block level constraints (instance scope: i_blk)

```
create_clock -name blk_vclk -period 2  
set_input_delay -clock [get_clocks blk_vclk] 1.0 [get_ports Blk_in1] -clock_fall
```

Hierarchical clock mapping

```
blk_vclk is mapped to top_clk
```

CONST_HIER_IO13o

NAME

CONST_HIER_IO13o (CONSTRAINT_HIER_SIGNOFF)

MESSAGE

Block `set_output_delay` reference clock edge does not match design trigger edge

DEFAULT SEVERITY

Warning

DESCRIPTION

Indicates that block level `set_output_delay` reference clock edge does not match design trigger edge. For a block output connected to a sequential cell in the fanout, a violation is reported when the following condition is broken:

- If the sequential cell is rising edge triggered or positive level enabled, the block level `set_output_delay` command on the block output should be specified without `-clock_fall` option
- If the sequential cell is falling edge triggered or negative level enabled, the block level `set_output_delay` command on the block output should be specified with `-clock_fall` option

This rule is checked by `check_target_constraints` command and the results can be reported by `report_rule_check` command.

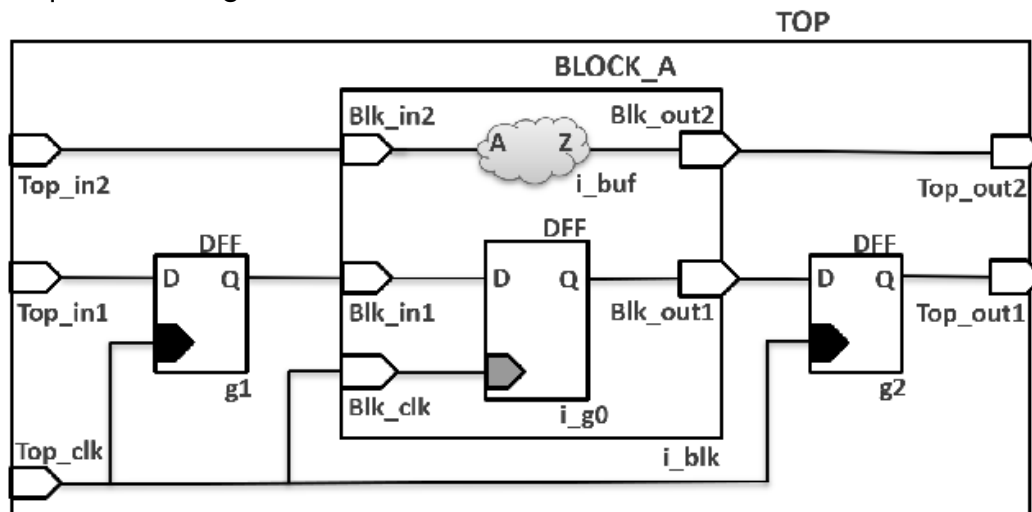
This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
block_view_name	string	Block view name
top_view_name	string	Top view name

scope	object	Scope
block_pin	object	Block pin
top_pin	object	Top pin
block_clock	object	Block clock
top_clock	object	Top clock
type	string	Type
block_file_line	object_list	Block file line number
block_file_name	object_list	Block file name

EXAMPLE

- The following example contains `set_output_delay` command specified in block level constraints. This rule is reported for the `set_output_delay` command, because it is specified with `-clock_fall` option, but the block output Blk_out1 is connected to rising edge triggered sequential cell g2.



Top level constraints

```
create_clock -name top_clk -period 2 [get_ports Top_clk]
```


Block level constraints (instance scope: i_blk)

```
create_clock -name blk_vclk -period 2  
set_output_delay -clock [get_clocks blk_vclk] 1.0 [get_ports Blk_out1] -  
clock_fall
```

Hierarchical clock mapping

```
blk_vclk is mapped to top_clk
```

CONST_HIER_ITR

- [CONST_HIER_ITR1](#)

CONST_HIER_ITR1

NAME

CONST_HIER_ITR1 (CONSTRAINT_HIER_SIGNOFF)

MESSAGE

Inconsistent input transition at the top-level vs. block

DEFAULT SEVERITY

Ignore (default Warning)

DESCRIPTION

Indicates that an inconsistency in input transition command ([set_input_transition](#) / [set_annotated_transition](#) / [set_driving_cell](#)) is found between top level and block level constraints. The following are the possible causes for this violation:

- Input transition command specified at either top level or block level
- Input transition commands specified at both top level and block level but with different or Incompatible values

The following attributes specify conditional checks on compatible transition values and parameters for [set_input_transition](#) / [set_annotated_transition](#):

- [Itms_const_hier_wire_transition_condition](#)
 - Specifies wire transition condition for block input connected to top input through wires
 - Default: $\text{block_transition} \geq \text{top_transition}$
- [Itms_const_hier_logic_transition_condition](#)
 - Specifies logic transition condition for block input connected to top input through logics
 - Default: $\text{block_transition} \leq \text{top_transition}$

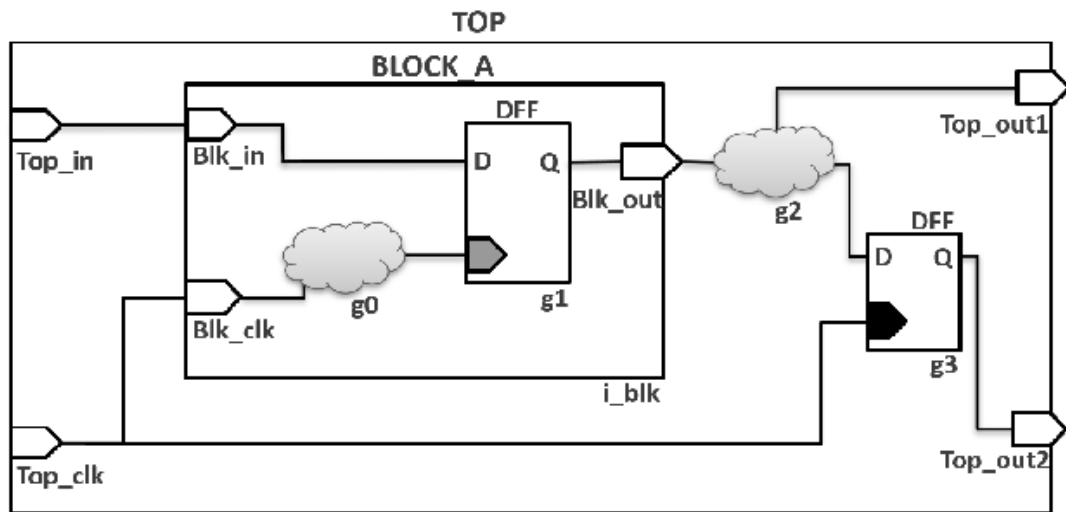
This rule is checked by [check_target_constraints](#) command and the results can be reported by [report_rule_check](#) command.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
block_view_name	string	Block view name
top_view_name	string	Top view name
scope	object	Scope
block_pin	object	Block port
top_pin	object	Top port
block_value	object_list	Block transition value
top_value	object_list	Top transition value
inconsistent_block_type	object_list	Inconsistent block type
inconsistent_top_type	object_list	Inconsistent top type
type	string	Type
block_file_line	object_list	Block file line number
block_file_name	object_list	Block file name
top_file_line	object_list	Top file line number
top_file_name	object_list	Top file name

EXAMPLE

- The following example contains `set_input_transition` commands specified in both top level and block level constraints. This rule is reported for the `set_input_transition` commands, because block level transition value is less than top level transition value.



Top level constraints

```
set_input_transition 1.0 [get_ports Top_in]
```

Block level constraints (instance scope: i_blk)

```
set_input_transition 0.5 [get_ports Blk_in]
```

CONST_HIER_LAT

- [CONST_HIER_LAT1](#)

CONST_HIER_LAT1

NAME

CONST_HIER_LAT1 (CONSTRAINT_HIER_SIGNOFF)

MESSAGE

Inconsistent `set_clock_latency` at the top-level vs. block

DEFAULT SEVERITY

Warning

DESCRIPTION

Indicates that an inconsistency in `set_clock_latency` command is found between top level and block level constraints. The following are the possible causes for this violation:

- `set_clock_latency` command specified at either top level or block level
- `set_clock_latency` commands specified at both top level and block level but with different values

This rule is checked by `check_target_constraints` command and the results can be reported by `report_rule_check` command.

This rule has specific rule occurrence attributes for the details about the violation.

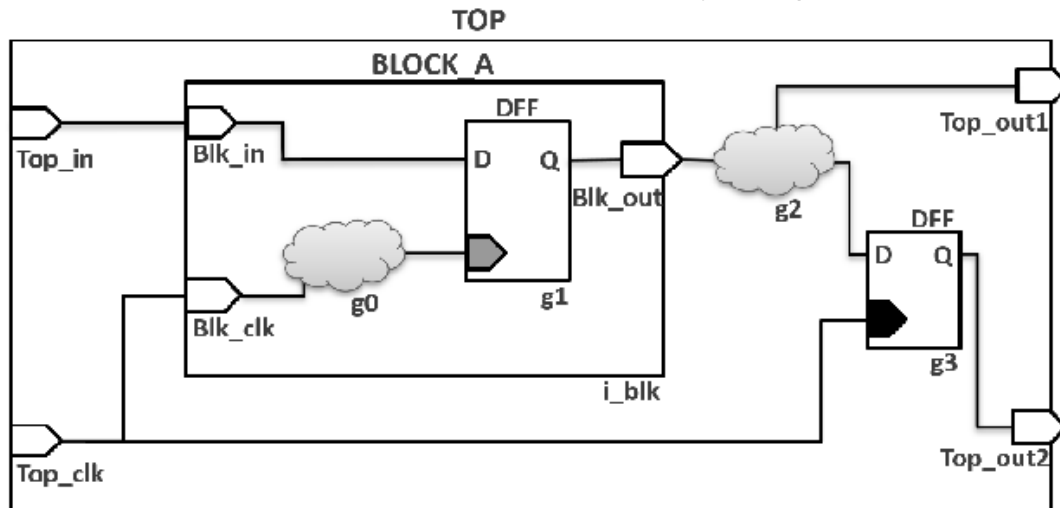
Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
block_view_name	string	Block view name
top_view_name	string	Top view name
scope	object	Scope

Litmus Rule Reference
Timing Constraint Hierarchical Signoff--CONST_HIER_LAT

block_pin	object	Block pin
top_pin	object	Top pin
block_clock	object	Block clock
top_clock	object	Top clock
is_clock_source_latency	bool	Is clock source latency
type	string	Type
block_file_line	unsigned	Block file line number
block_file_name	string	Block file name
top_file_line	unsigned	Top file line number
top_file_name	string	Top file name

EXAMPLE

- For the figure below, the following constraints contain `set_clock_latency` commands specified in both top level and block level constraints. This rule is reported for the `set_clock_latency` commands, because they are specified with different values.



Top level constraints

```
create_clock -name Top_clk -period 2 [get_ports Top_clk]
set_clock_latency 0.5 [get_clocks Top_clk]
```

Block level constraints (instance scope: i_blk)

```
create_clock -name Blk_clk -period 2 [get_ports Blk_clk]
set_clock_latency 1.0 [get_clocks Blk_clk]
```

Hierarchical clock mapping

```
Top_clk is mapped to Blk_clk
```

CONST_HIER_OLD

- [CONST_HIER_OLD1](#)

CONST_HIER_OLD1

NAME

CONST_HIER_OLD1 (CONSTRAINT_HIER_SIGNOFF)

MESSAGE

Inconsistent `set_load` at the top level vs. block

DEFAULT SEVERITY

Warning

DESCRIPTION

Indicates that an inconsistency in `set_load` command is found between top level and block level constraints. The following are the possible causes for this violation:

- `set_load` command specified at either top level or block level
- `set_load` commands specified at both top level and block level but with different values

This rule is checked by `check_target_constraints` command and the results can be reported by `report_rule_check` command.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
block_view_name	string	Block View Name
top_view_name	string	Top View Name
scope	object	Block Instance
block_object	object	Block object
top_object	object	Top object

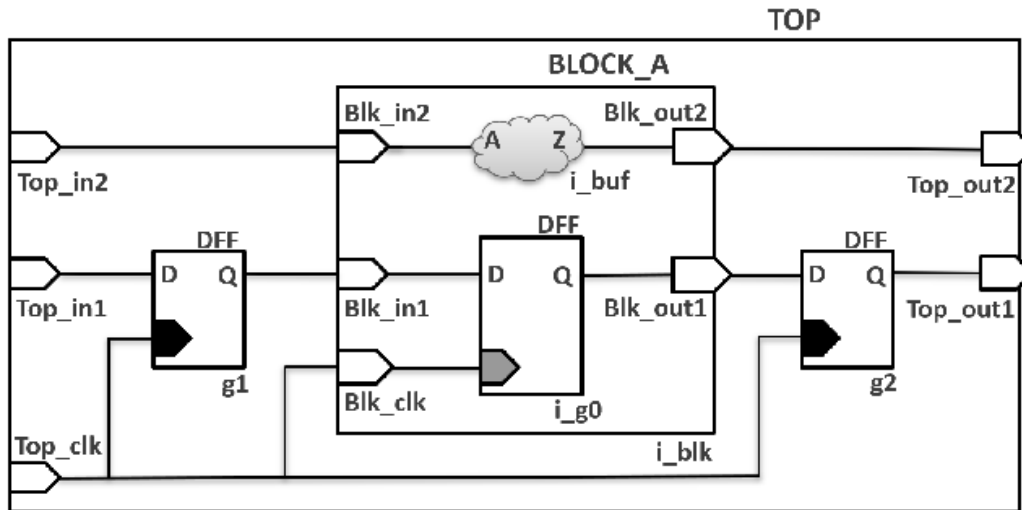
block_net	object	Block Net
top_net	object	Top Net
type	string	Type
block_value	object_list	block_value
top_value	object_list	top_value
block_file_line	object_list	Block file number
block_file_name	object_list	Block file name
top_file_line	object_list	Top file number
top_file_name	object_list	Top file name

In addition, common rule occurrence attribute `type` provides the possible causes for the violation.

Attribute	Value	Description
type	BLOCK_NO_CONSTRAINT	Missing constraint at block level
	TOP_NO_CONSTRAINT	Missing constraint at top level
	DIFFERENT_CONSTRAINTS	Inconsistent constraints between top level and block level

EXAMPLE

- For the figure below, the following contains `set_load` commands specified in both top level and block level constraints. This rule is reported for the `set_load` commands, because they are specified with different values.



Top level constraint

```
set_load 0.5 [get_nets i_blk/Blk_out2]
```

Block level constraint (instance scope: i_blk)

```
set_load 1.0 [get_nets Blk_out2]
```

CONST_HIER_PLS

- [CONST_HIER_PLS1](#)

CONST_HIER_PLS1

NAME

CONST_HIER_PLS1 (CONSTRAINT_HIER_SIGNOFF)

MESSAGE

Inconsistent `set_min_pulse_width` at the top-level vs. block

DEFAULT SEVERITY

Warning

DESCRIPTION

Indicates that an inconsistency in `set_min_pulse_width` command is found between top level and block level constraints. The following are the possible causes for this violation:

- `set_min_pulse_width` command specified at either top level or block level
- `set_min_pulse_width` commands specified at both top level and block level but with different values

This rule is checked by `check_target_constraints` command and the results can be reported by `report_rule_check` command.

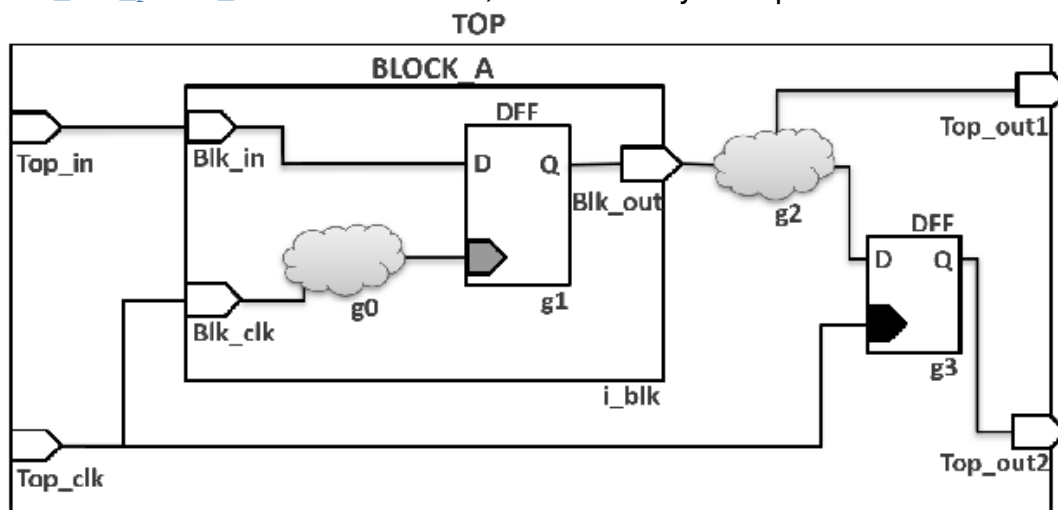
This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
block_view_name	string	Block view name
top_view_name	string	Top view name
scope	object	Scope

block_object	object	Block object
top_object	object	Top object
block_clock	object	Block clock
top_clock	object	Top clock
type	string	Type
block_file_line	unsigned	Block file line number
block_file_name	string	Block file name
top_file_line	unsigned	Top file line number
top_file_name	string	Top file name

EXAMPLE

- For the figure below, the following constraints contain `set_min_pulse_width` commands specified in both top level and block level constraints. This rule is reported for the `set_min_pulse_width` commands, because they are specified with different values.



Top level constraints

```
create_clock -name Top_clk -period 2 [get_ports Top_clk]
set_min_pulse_width 0.4 [get_clocks Top_clk]
```


Block level constraints (instance scope: i_blk)

```
create_clock -name Blk_clk -period 2 [get_ports Blk_clk]  
set_min_pulse_width 0.8 [get_clocks Blk_clk]
```

Hierarchical clock mapping

```
Top_clk is mapped to Blk_clk
```

CONST_HIER_SCA

- [CONST_HIER_SCA1](#)

CONST_HIER_SCA1

NAME

CONST_HIER_SCA1 (CONSTRAINT_HIER_SIGNOFF)

MESSAGE

Inconsistent `set_case_analysis` at the top-level vs. block

DEFAULT SEVERITY

Warning

DESCRIPTION

This rule reports that there is a `set_case_analysis` on a block object that does not have a tied value at the top level. Every block object with `set_case_analysis` at the top level, and every block port with a tied value at the top level, should have a consistent `set_case_analysis` at the block. This rule is checked by `check_target_constraints` command and results can be reported by `report_rule_check` command.

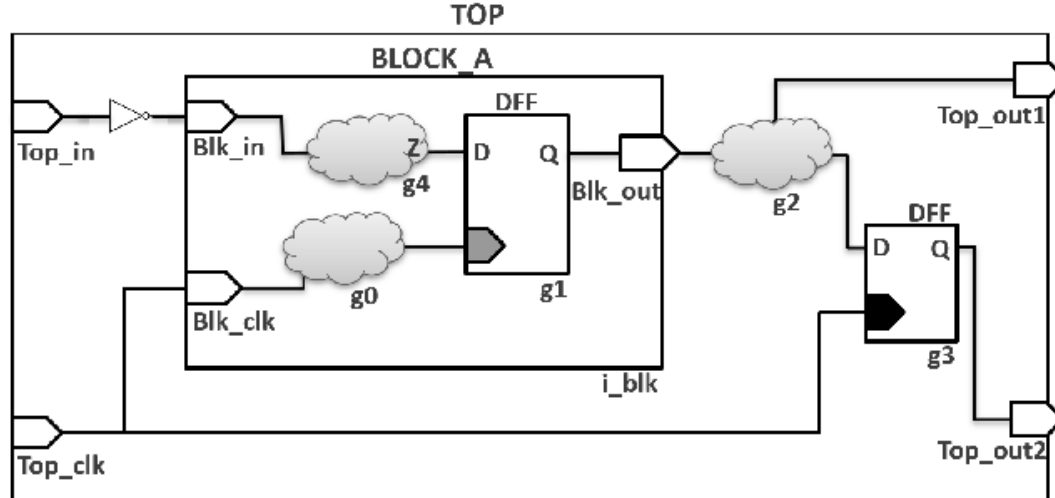
This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
block_view_name	string	Block view name
top_view_name	string	Top view name
scope	object	Scope
block_value	string	Block value
top_value	string	Top value

block_pin	object	Block pin
top_pin	object	Top pin
violation_type	string	Violation type
block_file_line	unsigned	Block file line number
block_file_name	string	Block file name
top_file_line	unsigned	Top file line number
top_file_name	string	Top file name

EXAMPLE

- For the figure below, the block level constraints file has this following constraint. The Litmus issues a warning if the top level constraints file does not have a `set_case_analysis` specified that causes pin `i_blk/g4/Z` to be tied to 1.



Block level constraint (instance scope: `i_blk`)

```
set_case_analysis 1 [get_pins g4/Z]
```

- For the above figure, the top and the block level constraints files have the following constraints, respectively, where `i_blk/Blk_in` gets the inverted value of `Top_in`. The Litmus issues a warning because `i_blk/Blk_in` is tied to 1 at the top level.

Top level constraint

```
set_case_analysis 0 [get_ports Top_in]
```

Block level constraint (instance scope: i_blk)

```
set_case_analysis 0 [get_ports Blk_in]
```

CONST_HIER_SDT

- [CONST_HIER_SDT1](#)

CONST_HIER_SDT1

NAME

CONST_HIER_SDT1 (CONSTRAINT_HIER_SIGNOFF)

MESSAGE

Inconsistent `set_disable_timing` at the top level vs. block

DEFAULT SEVERITY

Warning

DESCRIPTION

This rule reports that `set_disable_timing` is used at the block, but not at the top level, or vice versa. For `set_disable_timing` on library cell or library pin objects, the check is done only if the block instantiates the library cell. This rule is checked by `check_target_constraints` command and results can be reported by `report_rule_check` command.

This rule has specific rule occurrence attributes for the details about the violation.

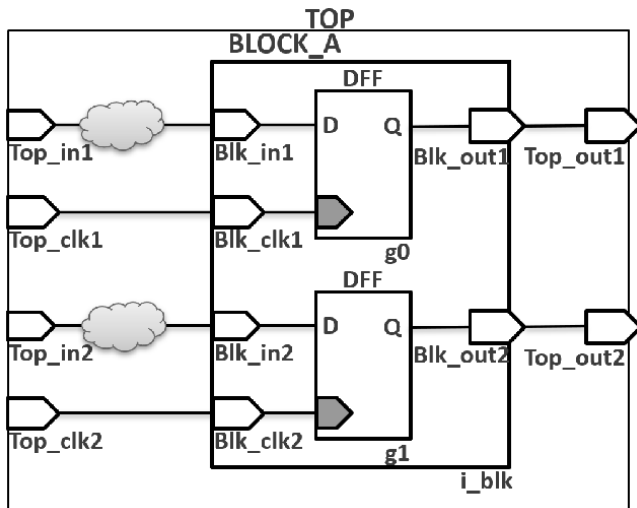
Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
block_view_name	string	Block view name
top_view_name	string	Top view name
scope	object	Block Instance
from_pin	object_list	from pin
to_pin	object	to pin
violation_type	string	Violation Type

Litmus Rule Reference
Timing Constraint Hierarchical Signoff--CONST_HIER_SDT

block_file_line	unsigned	Block file number
block_file_name	string	Block file name
top_file_line	unsigned	Top file number
top_file_name	string	Top file name

EXAMPLE

- For the figure below, if the block level constraints file has the line below but the top level constraints file does not have the line like the following, then the Litmus issues a warning because `set_disable_timing` is specified at the block level, but not at the top level:



Top level constraint

```
set_disable_timing [get_cells i_blk/g0]
```

Block level constraint (instance scope: i_blk)

```
set_disable_timing [get_cells g0]
```

- For the figure above, if the top level constraints file has the following line, but the block level constraints file does not have the same line, then the Litmus also issues a warning only if `i_blk` has an instance of cell DFF from library MYLIB:

Top level constraint

```
set_disable_timing [get_lib_pins MYLIB/DFF/CK]
```

CONST_HIER_TIM

- [CONST_HIER_TIM1](#)
- [CONST_HIER_TIM2](#)

CONST_HIER_TIM1

NAME

CONST_HIER_TIM1 (CONSTRAINT_HIER_SIGNOFF)

MESSAGE

A fragment of block's timing path cannot be matched to a corresponding fragment of top level timing path

DEFAULT SEVERITY

Ignore (default Warning)

DESCRIPTION

Indicates that the software was not able to identify a fragment of the path in the top level that includes, or is identical to, the fragment of the path in the block level. This rule can be reported as a consequence of inconsistent [set_disable_timing](#) between the top level and the block level designs. That problem would be reported by rule [CONST_HIER_SDT1](#). In this cases, making sure that there are sufficient [set_disable_timing](#) commands at both levels that are consistent and eliminating all occurrences of [CONST_HIER_SDT1](#) will solve the issue. If there are still occurrences of this rule, then that could be because of an incorrect definition of clock equivalences (using [set_timing_context_clock_mapping](#) and [report_hierarchical_clock_mapping](#)), or inconsistent application of SDC constraints that cause path splitting (such as [set_input_delay](#)) between the block and the top level. The occurrence of this rule points to an inconsistency between the timing graphs being compared, which can negatively affect the quality of results of the hierarchical checks. This rule is checked by [check_target_constraints](#) command and results can be reported by [report_rule_check](#) command.

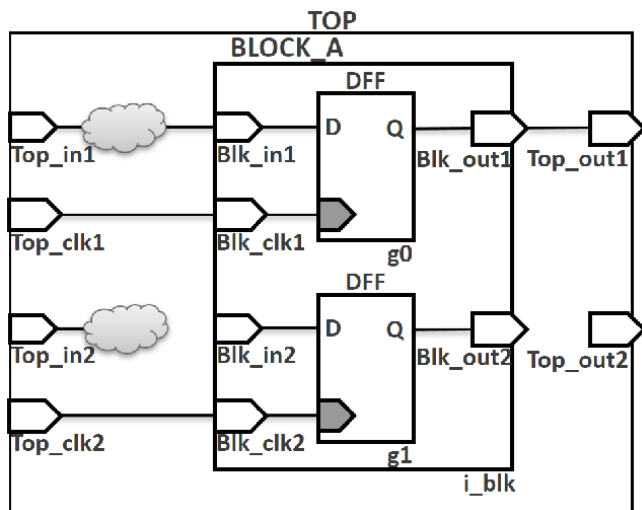
This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id

block_view_name	string	Block View Name
top_view_name	string	Top View Name
scope	object	Block Instance
block_pin	object	Block Pin
top_pin	object	Top Pin
block_clock	object	Block clock
top_clock	object	Top clock

EXAMPLE

- For the figure below, in the following source code, i_blk block's path starts/ends in an input/output port (Blk_in2 and Blk_out2, respectively) of that block, which is not connected to any timing start/end point in the top level design. These will trigger this rule violation.



```
module TOP ( input Top_in1, Top_clk1, Top_in2, Top_clk2, output Top_out1,
Top_out2 );
    assign w1 = ~Top_in1;
    BLOCK_A i_blk ( .Blk_in1(w1), .Blk_out1(Top_out1), .Blk_clk1(Top_clk1),
.Blk_in2(), .Blk_out2(), .Blk_clk2(Top_clk2) );
endmodule
```

- For the figure above, this rule will be reported if the block level clock Blk_clk1 that drives the start point of a block path has the following block level constraint and is not mapped to any top level clock, that is if the following top level command is missing at the top level constraints:

Top level constraint

```
create_clock -name Top_clk1 -period 8 [get_ports Top_clk1] -waveform {0 4}
```

Block level constraint (instance scope: i_blk)

```
create_clock -name Blk_clk1 -period 8 [get_ports Blk_clk1] -waveform {0 4}
```

- Also, for the figure above, this rule will be reported if there is a [set_disable_timing](#) only in the top level constraints file without one at the block level constraints file, which affects paths through the block:

Top level constraint

```
set_disable_timing [get_cells i_blk/g0]
```

CONST_HIER_TIM2

NAME

CONST_HIER_TIM2 (CONSTRAINT_HIER_SIGNOFF)

MESSAGE

Inconsistent `set_annotated_delay` at the top-level vs. block

DEFAULT SEVERITY

Warning

DESCRIPTION

Indicates that an inconsistency in `set_annotated_delay` command is found between top level and block level constraints. The following are the possible causes for this violation:

- `set_annotated_delay` command specified at either top level or block level
- `set_annotated_delay` commands specified at both top level and block level but with different values

This rule is checked by `check_target_constraints` command and the results can be reported by `report_rule_check` command.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
block_view_name	string	Block View Name
top_view_name	string	Top View Name
scope	object	Block Instance

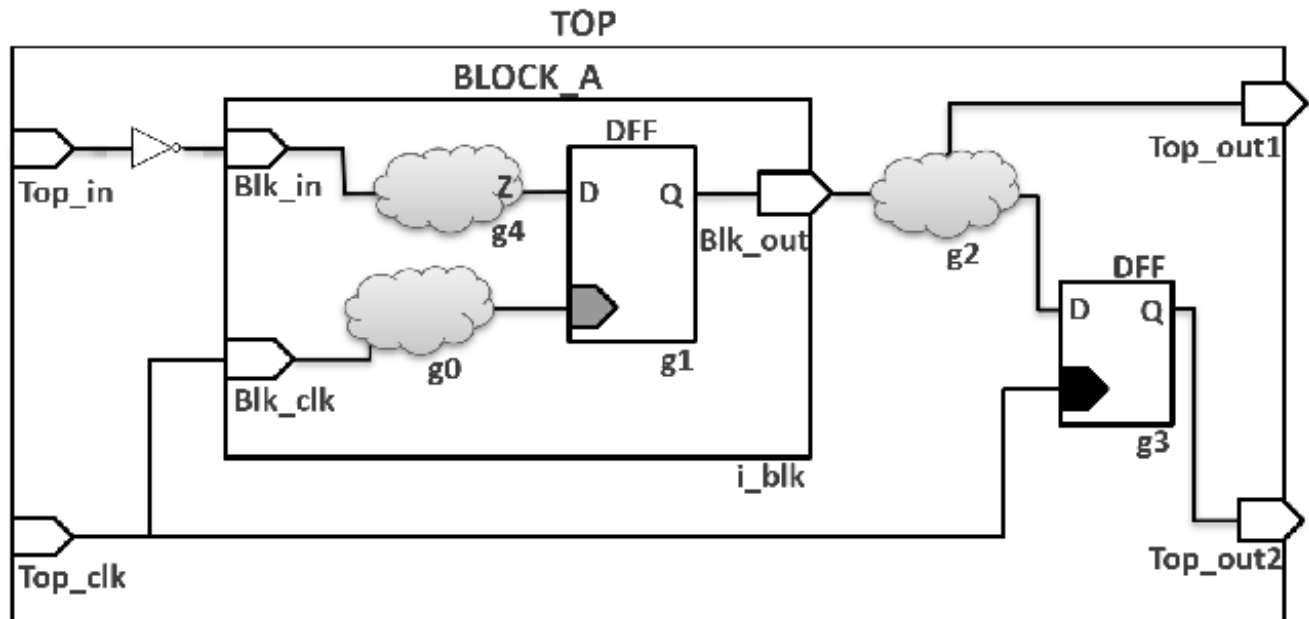
top_from	object	Top from object
top_to	object	Top to object
block_from	object	Block from object
block_to	object	Block to object
type	string	Type

In addition, common rule occurrence attribute `type` provides the possible causes for the violation.

Attribute	Value	Description
type	BLOCK_NO_CONSTRAINT	Missing constraint at block level
	TOP_NO_CONSTRAINT	Missing constraint at top level
	DIFFERENT_CONSTRAINTS	Inconsistent constraints between top level and block level

EXAMPLE

- For the figure below, the block level constraints file has this following constraint. The Litmus issues a warning if the top level constraints file does not have a `set_annotated_delay` specified for the net timing arc from `i_blk/g4/Z`.



Block level constraint (instance scope: `i_blk`)

```
set_annotated_delay 1 -net -from [get_pins g4/Z]
```

- For the above figure, the top and the block level constraints files have the following constraints, respectively, where the net timing arc from `i_blk/g4/Z` gets annotated delay value of 2 at top level. The Litmus issues a warning because the net timing arc gets annotated delay value of 1 at block level.

Top level constraint

```
set_annotated_delay 2 -net -from [get_pins i_blk/g4/Z]
```

Block level constraint (instance scope: `i_blk`)

```
set_annotated_delay 1 -net -from [get_pins g4/Z]
```


RELATED TOPICS

Commands

[set_annotated_delay](#)
[check_target_constraints](#)
[report_rule_check](#)

Attributes

[Rule Attributes - rule_occurrence](#)

CONST_HIER_UNC

- [CONST_HIER_UNC1](#)

CONST_HIER_UNC1

NAME

CONST_HIER_UNC1 (CONSTRAINT_HIER_SIGNOFF)

MESSAGE

Inconsistent `set_clock_uncertainty` at the top-level vs. block

DEFAULT SEVERITY

Warning

DESCRIPTION

Indicates that an inconsistency in `set_clock_uncertainty` command is found between top level and block level constraints. The following are the possible causes for this violation:

- `set_clock_uncertainty` command specified at either top level or block level
- `set_clock_uncertainty` commands specified at both top level and block level but with different values

This rule is checked by `check_target_constraints` command and the results can be reported by `report_rule_check` command.

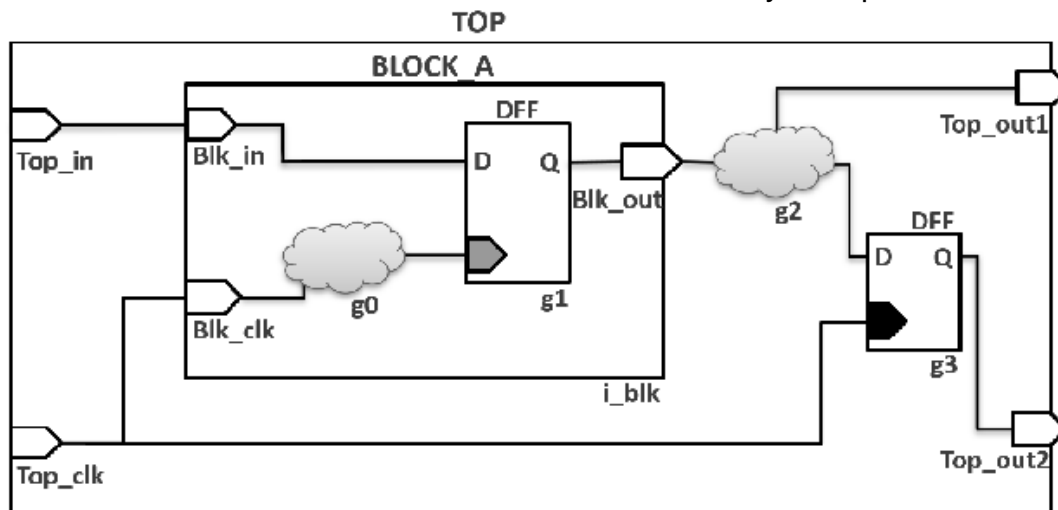
This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
block_view_name	string	Block view name
top_view_name	string	Top view name
scope	object	Scope

block_pin	object	Block pin
top_pin	object	Top pin
block_clock	object	Block clock
top_clock	object	Top clock
equivalent_block_clocks	object_list	Block clock Vec
relative_top_clocks	object_list	Top clock Vec
block_file_line	object_list	Block file line number
block_file_name	object_list	Block file name
top_file_line	object_list	Top file line number
top_file_name	object_list	Top file name

EXAMPLE

- For the figure below, the following constraints contain `set_clock_uncertainty` commands specified in both top level and block level constraints. This rule is reported for the `set_clock_uncertainty` commands, because they are specified with different values.



Top level constraints

```
create_clock -name Top_clk -period 2 [get_ports Top_clk]
set_clock_uncertainty 0.2 [get_clocks Top_clk]
```

Block level constraints (instance scope: i_blk)

```
create_clock -name Blk_clk -period 2 [get_ports Blk_clk]
set_clock_uncertainty 0.4 [get_clocks Blk_clk]
```

Hierarchical clock mapping

```
Top_clk is mapped to Blk_clk
```

Litmus Rule Reference

Timing Constraint Hierarchical Signoff--CONST_HIER_UNC

Timing Constraint Signoff

- [CONST_CG_CHK](#)
- [CONST_CLK_CTR](#)
- [CONST_CLK_DEF](#)
- [CONST_CLK_GRP](#)
- [CONST_CLK_LAT](#)
- [CONST_CLK_UNC](#)
- [CONST_DAT_CHK](#)
- [CONST_DGN_CMB](#)
- [CONST_DGN_TIM](#)
- [CONST_EXC_CLK](#)
- [CONST_EXC_FLP](#)
- [CONST_EXC_MCP](#)
- [CONST_EXC_OLP](#)
- [CONST_EXC_SMD](#)
- [CONST_HFN_CHK](#)
- [CONST_IO_DRC](#)
- [CONST_IO_IDL](#)
- [CONST_IO_ITR](#)
- [CONST_IO_ODL](#)
- [CONST_IO_OLD](#)
- [CONST_SCA_CHK](#)

CONST_CG_CHK

- [CONST_CG_CHK1](#)
- [CONST_CG_CHK2](#)
- [CONST_CG_CHK3](#)

CONST_CG_CHK1

NAME

CONST_CG_CHK1 (CONSTRAINT_SIGNOFF)

MESSAGE

A cell or pin specified in [set_clock_gating_check](#) is not on a clock tree or the cell has single input

DEFAULT SEVERITY

Warning

DESCRIPTION

Indicates that the [set_clock_gating_check](#) command has been applied to a cell or pin, and that none of that cell's inputs is reached by any clock. This rule is checked by [verify_timing_intent](#) command and results can be reported by [report_rule_check](#) command.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
scope	string	Scope
pin	object_list	Pin
file_line	unsigned	File line number
file_name	string	File name

EXAMPLE

- If the instance u_clock_gater of cell clock_gater is not reached by the clock tree of any clock, then this rule will be reported if a command like this is found:

```
set_clock_gating_check -setup 0.4 [get_cells {u_clock_gater}]
```

CONST_CG_CHK2

NAME

CONST_CG_CHK2 (CONSTRAINT_SIGNOFF)

MESSAGE

Incomplete clock gating check

DEFAULT SEVERITY

Warning

DESCRIPTION

Indicates that there is an object with clock gating check specified, but one or more of the combinations of the options `-setup/-hold` and `-rise/-fall` are missing. This rule is checked by `verify_timing_intent` command and results can be reported by `report_rule_check` command.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
scope	string	Scope
pin	object_list	Pin
clock	object	Clock
missing_corners	object_list	missing_corners
file_line	unsigned	File line number
file_name	string	File name

EXAMPLE

- In the following example, the Litmus issues a warning because the `-fall` clock gating check was not specified:

```
set_clock_gating_check -setup 1.0 -hold 0.5 -rise [get_cells mygate]
```

CONST_CG_CHK3

NAME

CONST_CG_CHK3 (CONSTRAINT_SIGNOFF)

MESSAGE

Clock gating check is missing on complex gate

DEFAULT SEVERITY

Warning

DESCRIPTION

Indicates that the non-controlling value of a clock for clock gating check on a complex gate, e.g. XOR, XNOR or MUX, is undetermined. The non-controlling value of the clock needs to be specified by `set_clock_gating_check` command with `-high` or `-low` option. By default, the non-controlling value of the clock is determined with the function of the gate. For example, clock gating check is performed on the high level for AND and NAND gates, and on the low level for OR and NOR gates. Also the rule is checked when the endpoint in the fanout of the gate contains clock pin of sequential cell, set/reset pin of sequential cell, or output/inout port. This rule is checked by `verify_timing_intent` command and the results can be reported by `report_rule_check` command.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
scope	string	Scope
clock_gating_clock_pin	object	Clock gating pin

clock_gating_output_pin	object	Clock gating output pin
clock_gating_enable_pin	object	Clock gating enable pin
sink_pin_types	object_list	Sink pin types

EXAMPLE

- The following example contains an XOR gate which gates a clock signal but does not have any clock gating check constraint. This rule is reported for the XOR gate i_xor, because `set_clock_gating_check` command is not specified with `-high` or `-low` option to determine the non-controlling value of the clock clk for clock gating check.

```
module top (input clk, input en, output gclk);
    XOR2_X2 i_xor (.A(clk), .B(en), .Z(gclk));
endmodule
```

Constraint that goes with above RTL:

```
create_clock -name clk -period 2 [get_ports clk]
```

CONST_CLK_CTR

- [CONST_CLK_CTR1](#)
- [CONST_CLK_CTR2](#)
- [CONST_CLK_CTR4](#)
- [CONST_CLK_CTR6](#)
- [CONST_CLK_CTR9](#)
- [CONST_CLK_CTR10](#)
- [CONST_CLK_CTR12](#)
- [CONST_CLK_CTR13](#)

CONST_CLK_CTR1

NAME

CONST_CLK_CTR1 (CONSTRAINT_SIGNOFF)

MESSAGE

Undefined clock transition for ideal real and generated clock : Clock without set_clock_transition

DEFAULT SEVERITY

Warning

DESCRIPTION

Indicates that `set_clock_transition` is not defined for an ideal real or generated clock. This rule is reported for ideal clocks. This rule is checked by `verify_timing_intent` command and results can be reported by `report_rule_check` command.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
scope	string	Scope
clock	object	Clock
file_line	unsigned	File line number
file_name	string	File name

EXAMPLE

- If there are three clocks CLK1, sysCLK, and Hclk and the following line:


```
set_clock_transition 0.75 -rise CLK1
```

then the Litmus issues a violation because clock transitions for sysClk and Hclk are not defined.

CONST_CLK_CTR2

NAME

CONST_CLK_CTR2 (CONSTRAINT_SIGNOFF)

MESSAGE

set_clock_transition is not recommended for propagated clock

DEFAULT SEVERITY

Warning

DESCRIPTION

Indicates that `set_clock_transition` is defined for a propagated clock. This rule is reported for propagated clocks. This rule is checked by `verify_timing_intent` command and results can be reported by `report_rule_check` command.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
scope	string	Scope
clock	object	Clock
max_min_option	string	Max/Min option
rise_fall_option	string	Rise/Fall option
file_line	unsigned	File line number
file_name	string	File name

e

EXAMPLE

- In the example, clock CLK is a propagated clock. This rule is reported because `set_clock_transition` is not recommended for propagated clocks.

```
create_clock -name CLK -period 2 [get_ports clk]
set_propagated_clock [get_clocks CLK]
set_clock_transition 0.75 -rise CLK1
```

CONST_CLK_CTR4

NAME

CONST_CLK_CTR4 (CONSTRAINT_SIGNOFF)

MESSAGE

`set_annotated_transition` not defined on the clock pin of a leaf cell

DEFAULT SEVERITY

Ignore (default Warning)

DESCRIPTION

Indicates that a pin of a liberty cell that has the clock attribute, does not have a transition defined using the `set_annotated_transition` command. This rule is checked by `verify_timing_intent` command and results can be reported by `report_rule_check` command. Note that the `set_annotated_transition` command is not included in SDC versions 1.6 or earlier.

 This rule is disabled by default. Use `set_rule_handling -unsuppress` command to enable the rule.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
scope	string	Design Name
clock_pin	object	Clock pin of leaf cell

EXAMPLE

- The Litmus issues a violation of this rule if there is an instance a_reg of a sequential cell whose clock pin is called CP, and the constraints file has no command like the following:

```
set_annotated_transition 0.1 [get_pins {a_reg/CP}]
```

CONST_CLK_CTR6

NAME

CONST_CLK_CTR6 (CONSTRAINT_SIGNOFF)

MESSAGE

Undefined clock transition for propagated real clock: Clock source input port without `set_input_transition`

DEFAULT SEVERITY

Warning

DESCRIPTION

Indicates that a real clock has an undefined clock transition. That is, PI/POs that are the source of a clock do not have a defined `set_input_transition`. This rule is reported for propagated clocks. This rule is checked by `verify_timing_intent` command and results can be reported by `report_rule_check` command.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
scope	string	Scope
clock_source	object	Clock source
file_line	unsigned	File line number
file_name	string	File name

EXAMPLE

- The Litmus issues a violation if the second line is missing, where CLK1 is the port name, not the clock name:

```
create_clock -period 10 -name CLK1_w CLK1  
set_input_transition 0.75 CLK1
```

CONST_CLK_CTR9

NAME

CONST_CLK_CTR9 (CONSTRAINT_SIGNOFF)

MESSAGE

`set_input_transition` is not recommended for ideal clock source input port

DEFAULT SEVERITY

Warning

DESCRIPTION

Indicates that `set_input_transition` is used on an ideal clock source input port. This rule is checked by `verify_timing_intent` command and results can be reported by `report_rule_check` command.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
scope	string	Scope
clock	object	Clock
clock_source	object	Clock source
max_min_option	string	Value of -max/ -min option

rise_fall_option	string	Value of - rise/ -fall option
file_line	unsigned	File line number
file_name	string	File name

EXAMPLE

- The following command causes a rule check violation because `set_input_transition` is used on a clock port:

```
create_clock -name clk1 -period 10 [get_ports clock]
set_input_transition 1 [get_ports clock]
```


CONST_CLK_CTR10

NAME

CONST_CLK_CTR10 (CONSTRAINT_SIGNOFF)

MESSAGE

`set_driving_cell` is not recommended for clock source input port

DEFAULT SEVERITY

Warning

DESCRIPTION

Indicates that `set_driving_cell` is defined on a clock source input port (either ideal or propagated clock). This rule is checked by `verify_timing_intent` command and results can be reported by `report_rule_check` command.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
scope	string	Scope
clock	object	Clock
clock_source	object	Clock source
file_line	unsigned	File line number
file_name	string	File name

EXAMPLE

- The following produces a warning because there should not be a `set_driving_cell` on the clock port:

```
set_driving_cell -lib_cell AND -from_pin B -pin Y CLK1
```

CONST_CLK_CTR12

NAME

CONST_CLK_CTR12 (CONSTRAINT_SIGNOFF)

MESSAGE

Incomplete clock transition options

DEFAULT SEVERITY

Warning

DESCRIPTION

Indicates that `set_input_transition` has incomplete clock transition options. The command needs to have values for both or none of the following pairs of options: `-rise/-fall` and `-min/-max`. This rule is checked by `verify_timing_intent` command and results can be reported by `report_rule_check` command.

This rule is enabled when root attribute `ltms_const_constraint_analysis_type` sets to post-layout (default: pre-layout).

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
scope	string	Scope
clock	object	Clock
clock_source	object	Clock source
type	string	Type
missing_corners	object_list	Missing corners

file_line	unsigned	File line number
file_name	string	File name

EXAMPLE

- The following produces a warning if the 1st line is there and missing the 2nd line:

```
set_input_transition 0.75 -rise CLK1  
set_input_transition <value> -fall CLK1
```

CONST_CLK_CTR13

NAME

CONST_CLK_CTR13 (CONSTRAINT_SIGNOFF)

MESSAGE

Inconsistent clock transition option values

DEFAULT SEVERITY

Warning

DESCRIPTION

Indicates that for some `set_input_transition` or `set_clock_transition`, the `-min` value is greater than the corresponding `-max` value. This rule is checked by `verify_timing_intent` command and results can be reported by `report_rule_check` command.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
scope	string	Scope
clock	object	Clock
clock_source	object	Clock source
rise_fall_option	string	Rise fall option
min_value	float	Min value
max_value	float	Max value
min_file_line	unsigned	Min file line number

min_file_name	string	Min file name
max_file_line	unsigned	Max file line number
max_file_name	string	Max file name

EXAMPLE

- In the following example, the minimum value of clock transition is 1 which is greater than the maximum value which is 0.5. Litmus identifies it as a warning.

```
set_clock_transition -min 1 [get_clocks CLK1]
set_clock_transition -max 0.5 [get_clocks CLK1]
```

CONST_CLK_DEF

- [CONST_CLK_DEF1](#)
- [CONST_CLK_DEF2](#)
- [CONST_CLK_DEF3](#)
- [CONST_CLK_DEF4](#)
- [CONST_CLK_DEF6](#)
- [CONST_CLK_DEF7](#)
- [CONST_CLK_DEF10](#)
- [CONST_CLK_DEF11](#)
- [CONST_CLK_DEF12](#)
- [CONST_CLK_DEF13](#)
- [CONST_CLK_DEF21](#)
- [CONST_CLK_DEF23](#)
- [CONST_CLK_DEF26](#)
- [CONST_CLK_DEF32](#)
- [CONST_CLK_DEF33](#)
- [CONST_CLK_DEF34](#)
- [CONST_CLK_DEF35](#)
- [CONST_CLK_DEF36](#)
- [CONST_CLK_DEF37](#)
- [CONST_CLK_DEF38](#)
- [CONST_CLK_DEF39](#)
- [CONST_CLK_DEF40](#)
- [CONST_CLK_DEF41](#)
- [CONST_CLK_DEF42](#)
- [CONST_CLK_DEF43](#)

- [CONST_CLK_DEF44](#)
- [CONST_CLK_DEF45](#)
- [CONST_CLK_DEF46](#)
- [CONST_CLK_DEF47](#)
- [CONST_CLK_DEF48](#)
- [CONST_CLK_DEF49](#)

CONST_CLK_DEF1

NAME

CONST_CLK_DEF1 (CONSTRAINT_SIGNOFF)

MESSAGE

Clock is missing

DEFAULT SEVERITY

Warning

DESCRIPTION

Indicates that there is an unconstrained clock pin on a sequential element. That is, this rule checks for clocks that do not belong to any clock tree. This can happen if there is a missing `create_clock` or `create_generated_clock` command, or if there is an incorrect `set_case_analysis`. Note that each clock tree driver cell is reported once, together with one sequential cell driven by it, and the number of additional such sequential cells. To help the debugging process (concentrating on the more important cases first), the detailed messages that appear in the verbose report and in the GUI are sorted by the number of driven sequential cells in descending order. This rule is checked by `verify_timing_intent` command and results can be reported by `report_rule_check` command.

The following are possible causes of this rule violation :

- Clock pin is not reached by any clock because of constant
- Clock pin is not reached by any clock because of blocked clock
- Clock pin is not reached by any clock because of missing clock
- Generated clock is missing because not match generated clock source path count

Attribute `ltms_const_generated_clock_enable_launch_clock_based_source_path_count` can specify generated clock source path count to be launch clock based or path based.

Attribute `ltms_const_report_all_witnesses` can specify whether to report all clock pins for one violation.

This rule has specific rule occurrence attributes for the details about the violation.

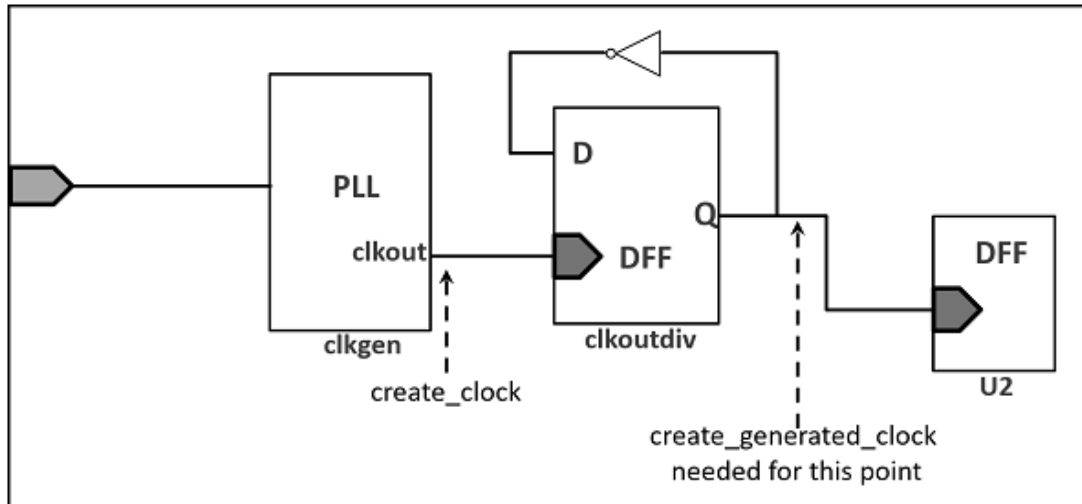
Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
scope	string	Scope
witness	object	Witness
clocks	object_list	clocks
count	unsigned	Count
clock_pin_witnesses	object_list	Clock pin witnesses
clock_pin_witnesses_filters	object_list	Filters for attribute 'clock_pin_witnesses'
type	string	Type
shared_driver	object	Shared driver
blocking_points	object	Blocking points
file_line	unsigned	File line number
file_name	string	File name

In addition, common rule occurrence attribute type provides the possible causes for the violation.

Attribute	Value	Description
type	MISSING_CLOCK	Clock is missing
	BLOCKED_CLOCK	Clock is blocked
	MISSING_GENERATED_CLOCK	Generated clock is missing

EXAMPLE

- For the figure below, the Litmus identifies a warning if the following SDC has only line #1 and missing line #2:



```
create_clock -name CLK -period 8 [get_pins clkgen/clkout]
create_generated_clock -name GCLK -divide_by 2 -source [get_pins clkgen/clkout]
[get_pins clkoutdiv/Q]
```

CONST_CLK_DEF2

NAME

CONST_CLK_DEF2 (CONSTRAINT_SIGNOFF)

MESSAGE

Unused clock constraint

DEFAULT SEVERITY

Warning

DESCRIPTION

Indicates an SDC statement that defines a clock, but that clock is not used in the design. A real or generated clock is considered used when:

- The port or pin that it drives is used to drive the clock port of a DLAT or DFF
- It is the master clock of a generated clock

A virtual clock is considered used when either of the SDC commands `set_input_delay` or `set_output_delay` is used to define a delay between the virtual clock and any port/pin in the design. This rule is checked by `verify_timing_intent` command and results can be reported by `report_rule_check` command.

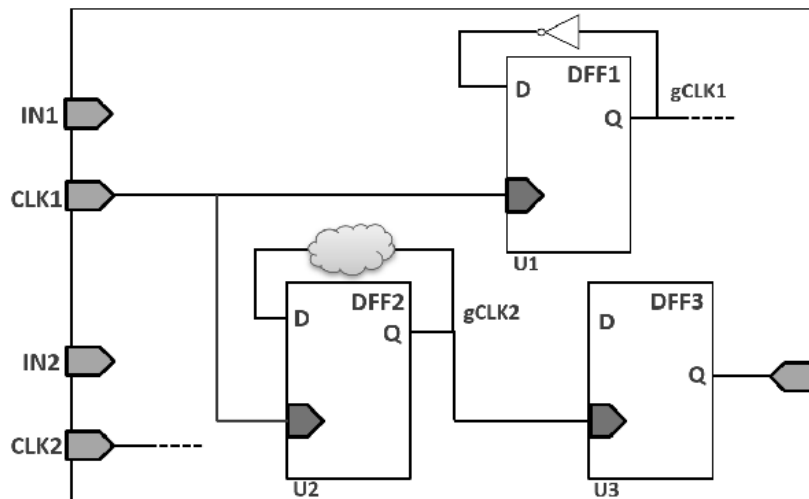
This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
scope	string	Scope
clock	object	Clock
file_line	unsigned	File line number

file_name	string	File name

EXAMPLE

- For the figure below, the following causes a violation, because rCLK2 and gCLK1 are not used within the block:



```
create_clock -name rCLK1 -period 8 CLK1
create_clock -name rCLK2 -period 10 CLK2
create_generated_clock -name gCLK1 -source CLK1 -divide_by 2 [get_pins U1/Q]
create_generated_clock -name gCLK2 -source CLK1 -divide_by 4 [get_pins U2/Q]
```

CONST_CLK_DEF3

NAME

CONST_CLK_DEF3 (CONSTRAINT_SIGNOFF)

MESSAGE

Clock pin is tied to a constant value

DEFAULT SEVERITY

Warning

DESCRIPTION

Indicates that a clock pin of a sequential cell has a constant value. Such a constant value can be a constant inside the design imposed on the clock pin, or a constant set by a [set_case_analysis](#) command. A clock pin can also have a constant value propagated from another port or pin. These can be found in port/pin attributes `ltms_constant_value` and `ltms_p_constant_value`. This rule is checked by [verify_timing_intent](#) command and results can be reported by [report_rule_check](#) command.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence Id
scope	string	Design Name
clock_pin	object	clock_pin
design_constant	object_list	design_constant
is_propagated_constant	bool	is_propagated_constant
constant_val	int	constant_val

EXAMPLE

- The following design causes this rule check violation because the clock pin of the sequential cell `i_ff/CK` is tied to a constant value 0:

```
module top (input clk, input din, output dout);  
    fflopd i_ff (.CK(1'b0), .D(din), .Q(dout));  
endmodule
```

To find the local constant value of the clock pin `i_ff/CK`, the following 1st Tcl command is used. To find the propagated constant value of the clock pin `i_ff/CK`, the following 2nd Tcl command will be used :

```
get_db [get_db pins i_ff/CK] .ltms_constant_value  
get_db [get_db pins i_ff/CK] .ltms_p_constant_value
```

CONST_CLK_DEF4

NAME

CONST_CLK_DEF4 (CONSTRAINT_SIGNOFF)

MESSAGE

Generated Clock is missing

DEFAULT SEVERITY

Warning

DESCRIPTION

Indicates that the generated clock is missing.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence Id
scope	string	Design Name
generated_clock_pin	object	generated_clock_pin
type	string	type
count	unsigned	count
combinational_source_path_clocks	object_list	combinational_source_path_clocks
sequential_source_path_clocks	object_list	sequential_source_path_clocks

CONST_CLK_DEF6

NAME

CONST_CLK_DEF6 (CONSTRAINT_SIGNOFF)

MESSAGE

Clock convergence or reconvergence is found

DEFAULT SEVERITY

Warning

DESCRIPTION

Indicates that two clock trees propagate to the same pin.

 Only clock tree overlaps on pins in the fanin of the clock pin of sequential elements are reported. This rule is checked by `verify_timing_intent` command and results can be reported by `report_rule_check` command.

Attribute `ltms_const_report_all_witnesses` can specify whether to report all clock pins for one violation.

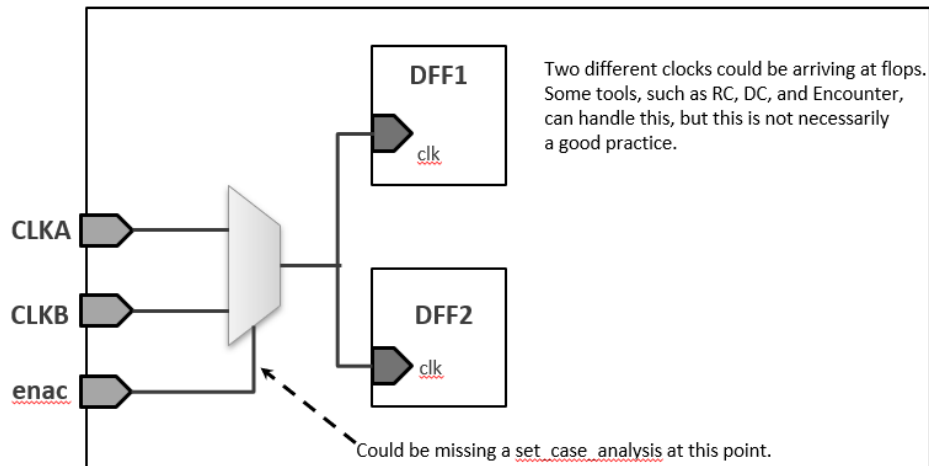
This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
scope	string	Scope
clock_point	object	File name
clocks	object_list	Clocks
is_clock_convergence	bool	Is clock convergence

clock_pin_witnesses	object_list	Clock pin witnesses
clock_pin_witnesses_filters	object_list	Filters for attribute 'clock_pin_witnesses'
convergence_clock_pairs	object_list	convergence_clock_pairs

EXAMPLE

- For the figure below, the following generates a warning because the Litmus detects converging clocks. A possible cause could be a missing `set_case_analysis` statement.



```
create_clock -name MCLK [get_ports CLKA]
create_clock -name TCLK [get_ports CLKB]
```

CONST_CLK_DEF7

NAME

CONST_CLK_DEF7 (CONSTRAINT_SIGNOFF)

MESSAGE

Unbuffered clock tree nets connected to output ports

DEFAULT SEVERITY

Warning

DESCRIPTION

Indicates that there is an output port in a clock tree driven by a combinational gate with more than one fanout. The rule checks for every output port in a clock tree that it is driven by any combinational gate with a single fanout. When there is no buffer that connects a branch of a clock tree net and an output port, the slopes and the delays after the clock tree net have to depend heavily on the load capacitance of the port. This rule is checked by `verify_timing_intent` command and results can be reported by `report_rule_check` command.

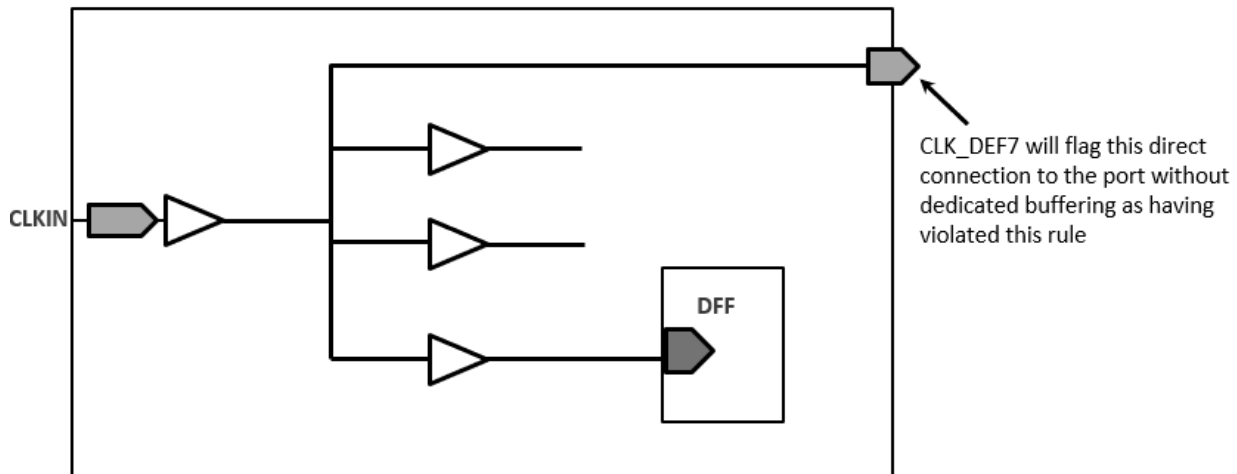
This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
scope	string	Design Name
clock	object_list	Clock source
clock_source	object	clock_source
output_pin	object	output_pin
fan_out_count	int	fan_out_count

cell_type	string	cell_type

EXAMPLE

- Any unbuffered direct connection to an output port like the following will cause this warning:



CONST_CLK_DEF10

NAME

CONST_CLK_DEF10 (CONSTRAINT_SIGNOFF)

MESSAGE

Clock is not propagated

DEFAULT SEVERITY

Ignore (default Warning)

DESCRIPTION

Indicates that there is a real or generated clock without a `set_propagated_clock` command. This rule is checked by `verify_timing_intent` command and results can be reported by `report_rule_check` command.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
scope	string	Scope
clock	object	Clock
file_line	unsigned	File line number
file_name	string	File name

EXAMPLE

- In the following case, this rule is reported for the ideal clock clk0, since no `set_propagated_clock` specified on clk0:

```
create_clock -name clk0 -period 2 [get_ports clk0]
```

CONST_CLK_DEF11

NAME

CONST_CLK_DEF11 (CONSTRAINT_SIGNOFF)

MESSAGE

Clock is propagated

DEFAULT SEVERITY

Ignore (default Warning)

DESCRIPTION

Indicates that a real or generated clock has [set_propagated_clock](#) specified. This rule is checked by [verify_timing_intent](#) command and results can be reported by [report_rule_check](#) command.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
scope	string	Scope
clock	object	Clock
file_line	unsigned	File line number
file_name	string	File name

EXAMPLE

- In the following case, this rule is reported for the propagated clock clk1:

```
create_clock -name clk1 -period 2 [get_ports clk1]  
set_propagated_clock [get_clocks clk1]
```


CONST_CLK_DEF12

NAME

CONST_CLK_DEF12 (CONSTRAINT_SIGNOFF)

MESSAGE

Mixed use of ideal and propagated clocks

DEFAULT SEVERITY

Warning

DESCRIPTION

Indicates that ideal and propagated clocks exist in the design at the same time. An ideal clock refers to a real or generated clock without `set_propagated_clock` command specified. A propagated clock refers to a real or generated clock with `set_propagated_clock` command specified. Mixed use of ideal and propagated clocks is not recommended. This rule is checked by `verify_timing_intent` command and the results can be reported by `report_rule_check` command.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
scope	string	Scope
ideal_clocks	object_list	Ideal clocks
propagated_clocks	object_list	Propagated clocks

EXAMPLE

- The following example contains an ideal clock and a propagated clock. This rule is reported for the ideal clock clk0 and the propagated clock clk1, because of mixed use of ideal and propagated clocks in the design at the same time.

```
create_clock -name clk0 -period 2 [get_ports clk0]
create_clock -name clk1 -period 2 [get_ports clk1]
set_propagated_clock [get_clocks clk1]
```

CONST_CLK_DEF13

NAME

CONST_CLK_DEF13 (CONSTRAINT_SIGNOFF)

MESSAGE

Virtual clock has no corresponding real clock with the same waveform and period

DEFAULT SEVERITY

Warning

DESCRIPTION

Indicates that there is a virtual clock without a corresponding real clock with the same period and waveform. This rule is checked by `verify_timing_intent` command and results can be reported by `report_rule_check` command. Note that this rule check applies to the block level only.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
scope	string	Scope
clock	object	Clock
file_line	unsigned	File line number
file_name	string	File name

EXAMPLE

- The following case reports this rule if the virtual clock in line #1 is defined without the corresponding real clock in line #2 with the same period and waveform:

```
create_clock -name VCLK1 -period 8 -waveform {0 5}  
create_clock -name CLK1 -period 8 -waveform {0 5} [get_ports {clk1}]
```

CONST_CLK_DEF21

NAME

CONST_CLK_DEF21 (CONSTRAINT_SIGNOFF)

MESSAGE

A clock does not belong to any of the user-defined clock groups

DEFAULT SEVERITY

Warning

DESCRIPTION

Indicates that there is a clock that does not belong to any of the clock groups created by a `set_clock_groups` command. A clock can become part of a user-defined clock group even if its name is not mentioned explicitly. Only clocks that remain unrelated to any user-defined groups will be reported. This rule is checked by `verify_timing_intent` command and results can be reported by `report_rule_check` command.

Note: If there are no user-defined clock groups, this rule will not be reported for any clock.

These are the recommended actions to take when this rule is reported:

- o Use the `report_clocks -groups` command to display the clock groups.
- o If the clock groups are incorrect, use the `set_clock_groups` or `reset_clock_groups` command to define the clock groups.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id

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Timing Constraint Signoff--CONST_CLK_DEF

scope	string	Design Name
clock	object_list	Clock definition

CONST_CLK_DEF23

NAME

CONST_CLK_DEF23 (CONSTRAINT_SIGNOFF)

MESSAGE

Clock redefined: used `create_clock` on an object that is in the transitive fanout of another clock

DEFAULT SEVERITY

Warning

DESCRIPTION

Indicates that there is a redefined clock. This happens when `create_clock` is used on an object that is in another clock's transitive fanout. When checking if a clock is defined in the fanout of another one, Litmus takes into account constant propagation. Also, a clock's fanout propagation is interrupted where other clocks are defined. If a clock GCLK is defined in the fanout of MCLK, then the logic in the fanout of GCLK is not considered to be in the fanout of MCLK unless it can be reached from MCLK without traversing the definition point of GCLK. This rule is checked by `verify_timing_intent` command and results can be reported by `report_rule_check` command.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
scope	string	Scope
original_clock	object	Original clock definition
clock_redefinition	object	Clock redefinition
original_clock_source	object	Original clock source

redefinition_clock_source	object	redefinition_clock_source
original_clock_cmd_file_line	unsigned	Original clock command file line number
original_clock_cmd_file_name	string	Original clock command file
clock_redefintion_cmd_file_line	unsigned	clock_redefinition file line
clock_redefinition_cmd_file_name	string	Clock redefinition command file name
path_type	string	path_type

EXAMPLE

- The following case reports this rule because clock GCLK is defined on the output pin g1/X of the g1 cell, since one of the inputs of the g1 cell is in the transitive fanout of another clock MCLK:

```
create_clock -name MCLK -period 10 -waveform {0 5} [get_ports {clk1}]
create_clock -name GCLK -period 20 -waveform {0 5} [get_pins {g1/X}]
```


CONST_CLK_DEF26

NAME

CONST_CLK_DEF26 (CONSTRAINT_SIGNOFF)

MESSAGE

Clock source is tied to a constant value

DEFAULT SEVERITY

Warning

DESCRIPTION

Indicates that a port or pin where a clock is defined has a constant logic value. Usually, such a constant value is set by a [set_case_analysis](#) SDC command. In some cases, clocks defined on pins inside the design can be tied to a constant value in the design, or can have a constant value propagated from another pin or port. Both ports and pins can have locally defined constant values, as well as propagated constant values. These can be found in port/pin

attributes `ltms_constant_value` and `ltms_p_constant_value`. This rule is checked by [verify_timing_intent](#) command and results can be reported by [report_rule_check](#) command.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
scope	string	Scope
clock	object	Clock
pin	object	Pin

is_tied_0	bool	Indicates if the clock pin is tied to zero
file_line	unsigned	File line number
file_name	string	File name

EXAMPLE

- The following commands cause this rule check violation because the second command ties the clock source defined in the first command to a constant:

```
create_clock -name clk -period 2 [get_ports clk]
set_case_analysis 0 [get_ports clk]
```

To find the locally defined constant value of the port clk, the following 1st Tcl command is used. To find the propagated constant value to any pin (e.g., g1/X), the following 2nd Tcl command will be used:

```
get_db [get_db ports clk] .ltms_constant_value
get_db [get_db pins g1/X] .ltms_p_constant_value
```

CONST_CLK_DEF32

NAME

CONST_CLK_DEF32 (CONSTRAINT_SIGNOFF)

MESSAGE

Clock tree drives a constrained non-clock pin of a sequential cell or a port but the clock is not used as clock at the point

DEFAULT SEVERITY

Warning

DESCRIPTION

This rule reports that there is a real or generated clock whose clock tree reaches a constrained non-clock pin of a sequential cell or a constrained primary output. This is normally considered an incorrect design practice. This rule also checks whether the clock tree end point is on `set_false_path` or `set_multicycle_path` matching path (not only clock-to-clock exceptions, but also all types of exceptions). In addition, checks whether the clock tree end point is on a generated clock feedback path. This rule is checked by `verify_timing_intent` command and results can be reported by `report_rule_check` command.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
scope	string	Scope
clock	object	Clock
pin	object	Pin
file_line	unsigned	File line number

file_name	string	File name
is_on_generated_clock_feedback_path	bool	is_on_generated_clock_feedback_path
capture_clocks	object_list	capture_clocks
exception_types	object_list	exception_types
exception_file_names	object_list	exception_file_names
exception_file_lines	object_list	exception_file_lines

EXAMPLE

- The following will report this rule if U1/data pin of the DFF is not the clock pin of the sequential cell:

```
create_clock -name CLK1 -period 10 -waveform {0 5} [get_pins {U1/data}]
```

CONST_CLK_DEF33

NAME

CONST_CLK_DEF33 (CONSTRAINT_SIGNOFF)

MESSAGE

Input/output delay set on port/pin on which a clock is defined

DEFAULT SEVERITY

Warning

DESCRIPTION

Indicates that a `set_input_delay` or `set_output_delay` has been applied to a port or pin on which a clock is defined. This rule is checked by `verify_timing_intent` command and results can be reported by `report_rule_check` command.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
scope	string	Scope
clock_source	object	Clock source
clocks	object_list	Clocks
file_line	unsigned	File line number
file_name	string	File name

EXAMPLE

- The following commands will cause this rule to be reported for port CLK in Litmus:

```
set_input_delay 1 [all_inputs]  
create_clock -period 10 [get_ports CLK]
```

- The following commands will cause this rule to be reported for pin B/GCLK in Litmus:

```
create_generated_clock -source [get_pins B/MCLK] -divide_by 2 [get_pins B/GCLK]  
set_output_delay 2 [get_pins B/GCLK]
```

CONST_CLK_DEF34

NAME

CONST_CLK_DEF34 (CONSTRAINT_SIGNOFF)

MESSAGE

Clock tree propagates to output port

DEFAULT SEVERITY

Warning

DESCRIPTION

This rule reports that there is an output port connected to a clock tree net. Output ports that are defined as source objects of clocks are not reported by this rule (refer to rule [CONST_CLK_DEF7](#)). This rule is checked by `verify_timing_intent` command and results can be reported by `report_rule_check` command.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
scope	string	Scope
clock	object	Clock
pin	object	Pin
file_line	unsigned	File line number
file_name	string	File name

CONST_CLK_DEF35

NAME

CONST_CLK_DEF35 (CONSTRAINT_SIGNOFF)

MESSAGE

Clock sense is undetermined

DEFAULT SEVERITY

Warning

DESCRIPTION

Indicates that one or more clock trees propagate to an exclusive-or (XOR) gate or XNOR gate and that gate has a side input that is not on any clock's tree and is not set to a constant value, directly or through constant propagation. This is not always a problem in the SDC or the design. It can be due to a missing `set_case_analysis` propagating to the side input being reported. This rule is checked by `verify_timing_intent` command and the results can be reported by `report_rule_check` command.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
scope	string	Design Name
pin	object	Pin with non-unate clock sense
non_clock_pins	object_list	Non-clock input pins

clocks	object_list	Clocks

EXAMPLE

- If the RTL has the following assignment:

```
assign clk_or_clkbar = clk ^ negate_clk;
```

And the SDC file has the following command in line #1 but without line #2, then the XOR gate will be reported by this rule:

```
create_clock -name CLK -period 10 [get_ports clk]  
set_case_analysis 1 [get_ports negate_clk]
```

CONST_CLK_DEF36

NAME

CONST_CLK_DEF36 (CONSTRAINT_SIGNOFF)

MESSAGE

Slower clock selected in functional mode

DEFAULT SEVERITY

Warning

DESCRIPTION

Indicates that a clock tree propagates to a cell where its propagation is interrupted in favor of another clock with a bigger period. This rule is mainly intended to report [set_case_analysis](#) selecting the wrong clock at a multiplexer, a problem that would otherwise be only detected much later in the flow. This rule is checked by [verify_timing_intent](#) command, and the results can be reported by [report_rule_check](#) command.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
scope	string	Design Name
faster_clock	object	Faster clock
slower_clock	object	Slower clock
pin	object	Pin

EXAMPLE

- If the RTL of the design has the instance of multiplexer cell. And the constraints have the following commands. Then this rule will report that scanclk is selected over funcclk in mux_i.

```
MYMUX mux_i(.Z(theclk), .I0(funcclk), .I1(scanclk), .SEL(scan_mode));
```

```
create_clock -name FCLK -period 5 [get_ports funcclk]  
create_clock -name SCLK -period 20 [get_ports scanclk]  
set_case_analysis 1 [get_ports scan_mode]
```

CONST_CLK_DEF37

NAME

CONST_CLK_DEF37 (CONSTRAINT_SIGNOFF)

MESSAGE

Multiple clocks are defined on the same pin

DEFAULT SEVERITY

Warning

DESCRIPTION

Indicates that more than one active real or generated clock was defined on the same pin, using the `-add` option of the `create_clock` or `create_generated_clock` commands. This rule is checked by `verify_timing_intent` command and results can be reported by `report_rule_check` command.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
scope	string	Scope
pin	object	Pin
clocks	object_list	Clocks

EXAMPLE

- The following commands will cause this rule to be reported for port CLK:

```
create_clock -period 10 -name fastclk [get_ports CLK]  
create_clock -period 20 -name slowclk [get_ports CLK] -add
```

In this case, fastclk will be reported as active and slowclk as not active.

CONST_CLK_DEF38

NAME

CONST_CLK_DEF38 (CONSTRAINT_SIGNOFF)

MESSAGE

`set_disable_timing` on clock tree

DEFAULT SEVERITY

Warning

DESCRIPTION

Indicates that a `set_disable_timing` command has been specified on a clock tree. This rule is checked by `verify_timing_intent` command and results can be reported by `report_rule_check` command. Note that this rule flags any `set_disable_timing` on clock tree, even if it does not affect any clock pin of a sequential element or if the object on the clock tree is in the fanout of another object in the same clock tree with `set_disable_timing` on it.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
scope	string	Scope
pin	object	Pin
blocking_points	object	Blocking points
clocks	object_list	Clocks
file_line	unsigned	File line number

file_name	string	File name

EXAMPLE

- If the SDC constraint file contains the following commands, then this rule would report a warning for the command in line #2 or for any object on the clock tree of CLK1 with `set_disable_timing`:

```
create_clock -name CLK1 -period 5000 -waveform {0 2500} [get_ports clk]
set_disable_timing [get_ports clk]
```


CONST_CLK_DEF39

NAME

CONST_CLK_DEF39 (CONSTRAINT_SIGNOFF)

MESSAGE

Multiple clocks are defined on the same pin, with same period and waveform.

DEFAULT SEVERITY

Warning

DESCRIPTION

Indicates that more than one active real or generated clock was defined on the same source pin, using the `-add` option of the `create_clock` or `create_generated_clock` commands and these clocks have the same period and waveform.

 This rule is checked by `verify_timing_intent` command and results can be reported by `report_rule_check` command.

This rule has the following specific rule occurrence attributes for the details about the violation:

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
scope	string	Design Name
pin	object	Pin
clocks	object_list	List of clocks

CONST_CLK_DEF40

NAME

CONST_CLK_DEF40 (CONSTRAINT_SIGNOFF)

MESSAGE

Generated clock source path is undetermined

DEFAULT SEVERITY

Warning

DESCRIPTION

This rule indicates that the source latency path of a generated clock is not found. The following are possible causes of this rule violation.

- A source latency path does not exist
- A source latency path is broken by a real clock (through the `create_clock` command)
- A source latency path is broken by disabled timing arc (through the `set_disable_timing` command)
- A source latency path is broken by a constant (through the `set_case_analysis` command or a hard constant)
- A source latency path is broken by a clock stop (through the `set_clock_sense` command or the `set_sense -stop_propagation` command)

This rule is checked by `verify_timing_intent` command and results can be reported by `report_rule_check` command.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id

scope	string	Scope
generated_clock	object	Generated clock
generated_clock_pin	object	Generated clock pin
file_line	unsigned	File line number
file_name	string	File name
is_blocked	bool	Indicates if the source latency path is blocked
blocking_command_file_names	object_list	Blocking command file names
blocking_command_file_lines	object_list	Blocking command file line numbers
path_type	string	Path type
path_count	unsigned	path_count
levels	object_list	levels
type	object_list	type

EXAMPLE

In the following case there are no physical connections between input port `clk` and output port `gclk`. This rule is reported because the source latency path of generated clock `gclk` does not exist.

```
create_clock -name clk -period 2 [get_ports clk]create_generated_clock -name gclk -
source [get_ports clk] -divide_by 2 [get_ports gclk]
```

WHAT NEXT

The generated clock definition and source latency path should be reviewed first and then the `create_generated_clock` timing constraint or the design structure should be corrected accordingly.

CONST_CLK_DEF41

NAME

CONST_CLK_DEF41 (CONSTRAINT_SIGNOFF)

MESSAGE

Generated clock is reached by multiple master clocks

DEFAULT SEVERITY

Warning

DESCRIPTION

Indicates that a generated clock is created with master clock source point (`-source` option) which is reached by multiple master clocks but does not specify the associated master clock. The following are the approaches for explicitly determining the associated master clock:

- `create_generated_clock` command with `-master_clock` option
- `set_active_clocks` command

By default, the associated master clock is determined with the 1st one in the list of master clocks sorted by clock name. This rule is checked by `verify_timing_intent` command and the results can be reported by `report_rule_check` command.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
scope	string	Scope
master_source_pin	object	Master source pin
clock	object	Clock

master_clock	object	Master clock
master_clocks	object_list	Master clocks
file_line	unsigned	File line number
file_name	string	File name

EXAMPLE

- The following example contains a generated clock which is reached by multiple master clocks. This rule is reported for the generated clock gclk, because the master clock source point (`-source` option) is reached by two master clocks clk0 and clk1 but the associated master clock is not explicitly determined.

```
create_clock -name clk0 -period 2 [get_ports clk] -add
create_clock -name clk1 -period 2 [get_ports clk] -add
create_generated_clock -name gclk -source [get_ports clk] -divide_by 1
[get_ports gclk]
```

CONST_CLK_DEF42

NAME

CONST_CLK_DEF42 (CONSTRAINT_SIGNOFF)

MESSAGE

Reference object is not reached by correct clock

DEFAULT SEVERITY

Warning

DESCRIPTION

Indicates that an object referenced in a command is not reached by any clock or the clock specified in the command. The following are the possible causes for this violation:

- `set_sense` command specified without `-clock` option but the reference object is not reached by any clock
- `set_sense` command specified with `-clock` option but the reference object is not reached by the clock
- `set_input_delay` command specified with `-clock` and `-reference_pin` options but the reference object is not reached by the clock
- `set_output_delay` command specified with `-clock` and `-reference_pin` options but the reference object is not reached by the clock

This rule is checked by `verify_timing_intent` command and the results can be reported by `report_rule_check` command.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id

scope	string	Design Name
source_object	object	Source object
reference_object	object	Reference object

EXAMPLE

- The following example contains a dummy clock and constraints. The dummy clock `clk_dummy` does not have any influence on the constraints. This rule is reported for the `set_sense`, `set_input_delay`, and `set_output_delay` commands, because the reference object is not reached by any clock or the dummy clock `clk_dummy` specified in the commands.

```
create_clock -name clk_dummy -period 2 [get_ports clk_dummy]
set_sense -stop_propagation [get_pins i_clk_buf/Z]
set_sense -stop_propagation -clocks [get_clocks clk_dummy] [get_pins
i_clk_buf/Z]
set_input_delay -clock [get_clocks clk_dummy] -reference_pin [get_pins
i_clk_buf/A] 1 [get_ports din]
set_output_delay -clock [get_clocks clk_dummy] -reference_pin [get_pins
i_clk_buf/A] 1 [get_ports dout]
```

CONST_CLK_DEF43

NAME

CONST_CLK_DEF43 (CONSTRAINT_SIGNOFF)

MESSAGE

Generated clock polarity is mismatched

DEFAULT SEVERITY

Warning

DESCRIPTION

Indicates that the polarity at a generated clock is mismatched between design and constraint specified by `create_generated_clock` command. The following are the targets for this check:

- `create_generated_clock` command specified with `-combinational`, `-divide_by` or `-multiply_by` option against with or without `-invert` option
- `create_generated_clock` command specified with `-edges { 1 2 3 }`, `-edges { 2 3 4 }`, `-edges { 1 3 5 }` or `-edges { 3 5 7 }`, and likewise with or without `-invert` option

This rule is checked by `verify_timing_intent` command and the results can be reported by `report_rule_check` command.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
scope	string	Scope
clock_file_location	string	Clock file location
clock	object	Clock

generated_clock_pin	object	Generated clock pin
generated_clock_source	object	Generated clock source
master_clock	object	Master clock
type	string	Type
pos_source	object	Positive source
neg_source	object	Negative source

EXAMPLE

- The following example contains a master clock and generated clocks. There is an INV gate between the master clock source point clk (`-source` option) and the generated clock definition point gclk. This rule is reported for the generated clocks gclk0 and gclk1, because they are not created by `create_generated_clock` command with `-invert` option according to the polarity.

```
module top (input clk, output gclk);
    INV_X2 i_clk_inv (.A(clk), .ZN(gclk));
endmodule
```

Constraints that go with the above source code:

```
create_clock -name clk -period 2 [get_ports clk]
create_generated_clock -name gclk0 -source [get_ports clk] -master_clock
[get_clocks clk] -combinational [get_ports gclk] -add
create_generated_clock -name gclk1 -source [get_ports clk] -master_clock
[get_clocks clk] -edges {1 2 3} [get_ports gclk] -add
```

CONST_CLK_DEF44

NAME

CONST_CLK_DEF44 (CONSTRAINT_SIGNOFF)

MESSAGE

Clock reaches a multiplexer select pin

DEFAULT SEVERITY

Warning

DESCRIPTION

Indicates that the select input of a multiplexer is reached by a clock. This rule is checked by `verify_timing_intent` command and the results can be reported by `report_rule_check` command.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
scope	string	Scope
pin	object	Pin
clocks	object_list	Clocks
file_line_info	object_list	File line information

EXAMPLE

- The following example contains a multiplexer and a clock. This rule is reported for the select input of the multiplexer, because it is reached by the clock `clk_sel`.

```
module top (input [1:0] din, output dout, input sel);
    mux i_mux (.DI0(din[0]), .DI1(din[1]), .DO(dout), .S(sel));
endmodule

module mux (input DI0, input DI1, output DO, input S);
    assign DO = S? DI1: DI0;
endmodule
```

Constraint that goes with the above source code:

```
create_clock -name clk_sel -period 2 [get_ports sel]
```

CONST_CLK_DEF45

NAME

CONST_CLK_DEF45 (CONSTRAINT_SIGNOFF)

MESSAGE

Multiple clocks drive a clock pin

DEFAULT SEVERITY

Warning

DESCRIPTION

Indicates that a clock pin is reached by multiple clocks. Setting the attribute `ltms_const_max_clock_limit_on_clock_pin` (default: 1) configures the max limit on clocks for this check. This rule is checked by `verify_timing_intent` command and the results can be reported by `report_rule_check` command.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
scope	string	Scope
clocks	object_list	Clocks
clock_pins	object_list	Clock pins

EXAMPLE

- The following example contains a flip-flop and clocks. This rule is reported for the clock input of the flip-flop `dout_reg/CK`, because it is reached by two clocks `clk0` and `clk1`.

```
module top (input [1:0] clk, input sel, input din, output reg dout);  
    assign gclk = sel? clk[0]: clk[1];  
    always @(posedge gclk)  
        dout <= din;  
endmodule
```

Constraints that go with the above source code:

```
create_clock -name clk0 -period 2 [get_ports {clk[0]}]  
create_clock -name clk1 -period 2 [get_ports {clk[1]}]
```

CONST_CLK_DEF46

NAME

CONST_CLK_DEF46 (CONSTRAINT_SIGNOFF)

MESSAGE

Missing generated clock on the clock generator circuit

DEFAULT SEVERITY

Warning

DESCRIPTION

This rule flags a violation for each case where Q pin of any flop is connected to the clock pin of another flop or to an output port (without [set_output_delay](#)), through any combinational logic and [create_generated_clock](#) statement is not defined on the output pin of the generator flop.

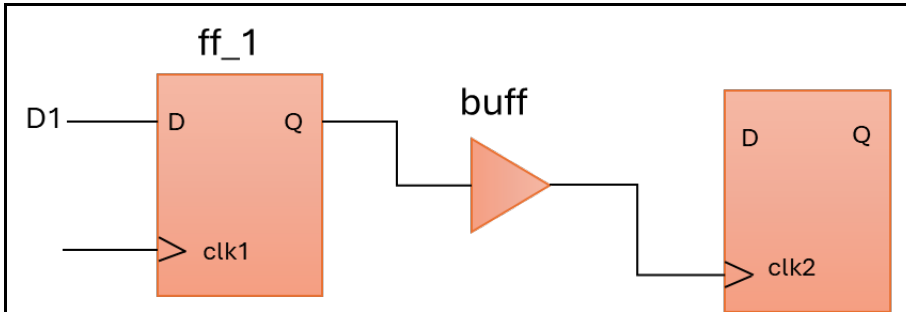
Reporting (message and attribute): *Missing generated clock on the output pin (Q pin) of the flop driving a clock pin ([clock_pin](#)) /output port ([output_port](#))*

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence ID
scope	string	Design Name
output_pin	object	Clock gating enable pin
clock_pin_or_output_ports	object_list	clock_pin_or_output_ports

EXAMPLES

Scenario 1: The Q pin of the flop drives a clock pin



Criteria for Q pin selection:

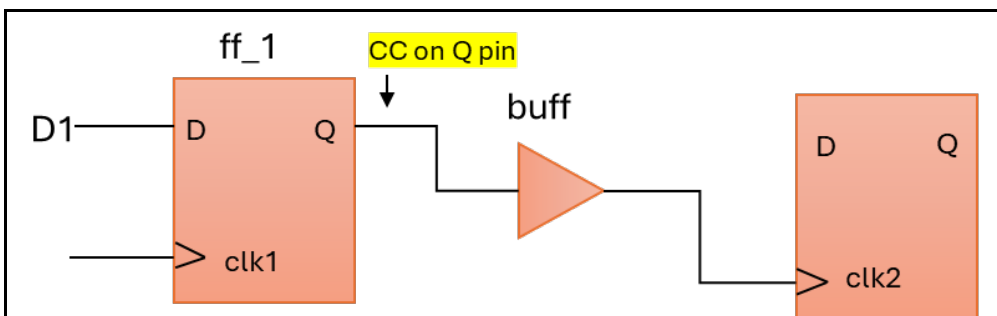
- Consider all Q pins that drive clock pin
- Ignore Q pins that do not drive any clock pin

Sample report:

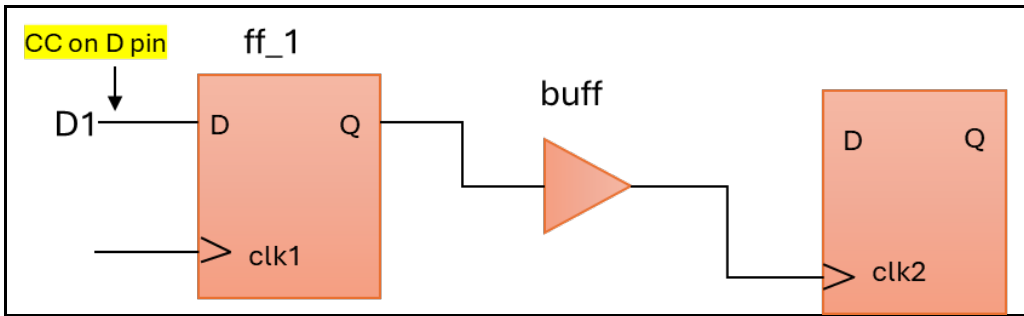
CONST_CLK_DEF46 : Missing generated clock on the clock generator circuitSeverity: Warning
Occurrence: 1Fail: Missing generated clock on the output pin (ff_1/Q) of the flop driving a clock pin (ff_2/clk2)

Reporting Scenarios:

- **Case 1:** No clock at the fanin of ff_1, but CC at the Q pin of ff_1 (dofile.cta1)
 - **SDC:** create_clock -name clk_ff1_q -period 10 [get_pins ff_1/Q]

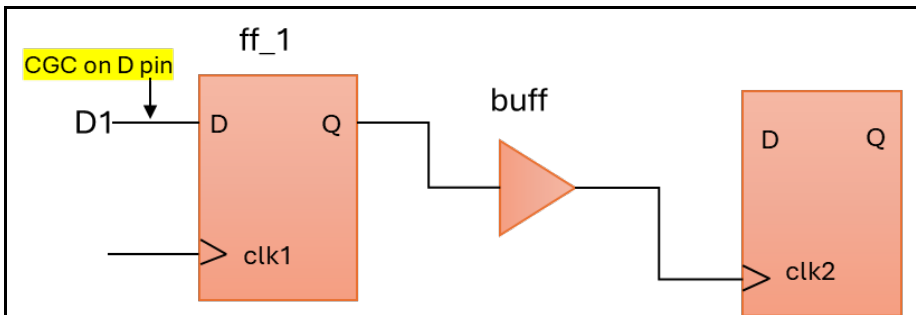


- **Case 2:** CC at the fanin (D) of ff_1, no CC/CGC at the Q pin of ff_1 (dofile.cta2)
 - **SDC:** create_clock -name clk_ff1_d -period 5 [get_pins ff_1/D]



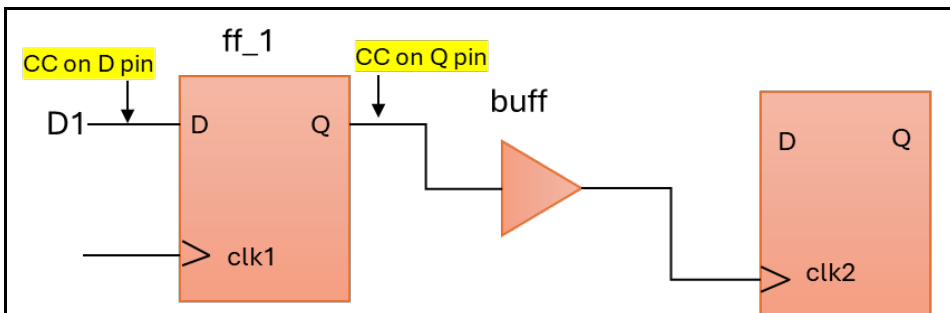
- **Case 3:** CGC at the fanin (D) of ff_1, no CC/CGC at the Q pin of ff_1 (dofile.cta2)

- **SDC:** `create_generated_clock -name gclk1 -source [get_pins ff_1/CK] -divide_by 1 [get_pins ff_1/D]`



- **Case 4:** CC at the fanin (D) of ff_1, CC at the Q pin of ff_1 (dofile.cta4)

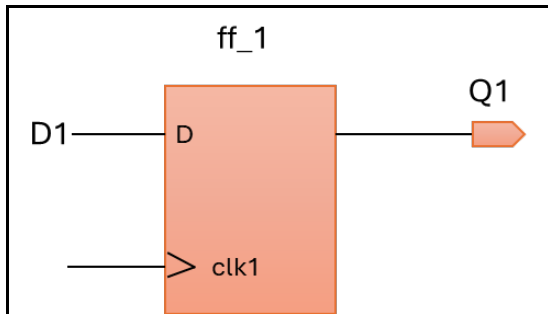
- **SDC:** `create_clock -name clk_ff1_q -period 10 [get_pins ff_1/Q] create_clock -name clk2 -period 5 [get_pins ff_1/D]`



- **Case 5:** CGC at the fanin (D) of ff_1, CC at the Q pin of ff_1 (dofile.cta5)

- **SDC:** `create_clock -name clk_ff1_q -period 10 [get_pins ff_1/Q]`
`create_generated_clock -name gclk1 -source [get_pins ff_1/CK] -divide_by 1 [get_pins ff_1/D]`

Scenario 2: The Q pin of the flop drives an output port



Criteria for output port selection:

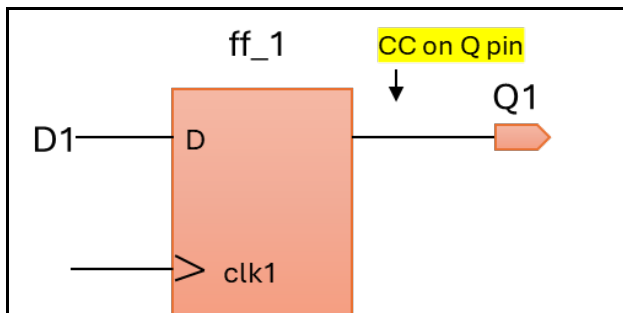
- Consider all output ports that do not have Set Output Direct (SOD) defined on it
- Ignore output ports that have SOD defined.

Sample report:

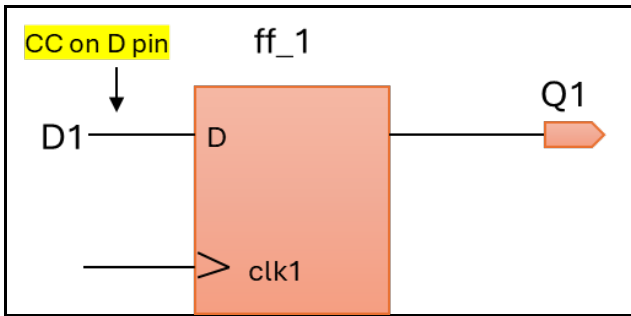
CONST_CLK_DEF46 : Missing generated clock on the clock generator circuitSeverity:
Warning Occurrence: 1Fail: Missing generated clock on the
output pin (ff_1/Q) of the flop driving a output port (Q1)

Reporting Scenarios:

- **Case 6:** No clock at the fanin of ff_1, but CC at the Q pin of ff_1 (dofile.cta6)
 - **SDC:** `create_clock -name clk_ff1_q -period 10 [get_pins ff_1/Q]`

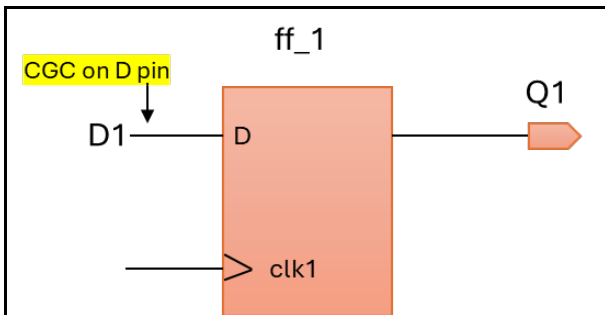


- **Case 7:** CC at the fanin (D) of ff_1, no CC/CGC at the Q pin of ff_1 (dofile.cta7)
 - **SDC:** `create_clock -name clk_ff1_d -period 5 [get_pins ff_1/D]`



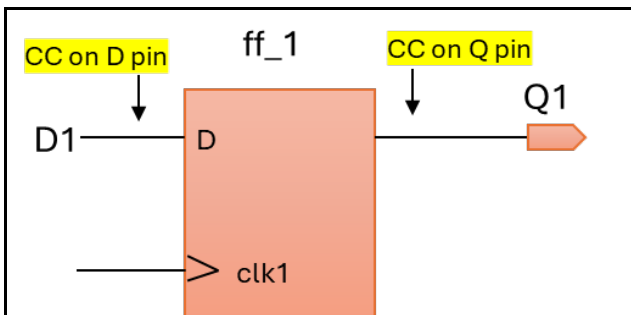
- **Case 8:** CGC at the fanin (D) of ff_1, no CC/CGC at the Q pin of ff_1 (dofile.cta8)

- **SDC:** `create_generated_clock -name gclk1 -source [get_pins ff_1/CK] -divide_by 1 [get_pins ff_1/D]`



- **Case 9:** CC at the fanin (D) of ff_1, CC at the Q pin of ff_1 (dofile.cta9)

- **SDC:** `create_clock -name clk_ff1_q -period 10 [get_pins ff_1/Q] create_clock -name clk2 -period 5 [get_pins ff_1/D]`



- **Case 10:** CGC at the fanin (D) of ff_1, CC at the Q pin of ff_1 (dofile.cta10)

- **SDC:** `create_clock -name clk_ff1_q -period 10 [get_pins ff_1/Q]`
`create_generated_clock -name gclk1 -source [get_pins ff_1/CK] -divide_by 1 [get_pins ff_1/D]`

CONST_CLK_DEF47

NAME

CONST_CLK_DEF47 (CONSTRAINT_SIGNOFF)

MESSAGE

Generated clock defined with -edge_shift option, the last edge shift value is not equal to the first edge shift value.

DEFAULT SEVERITY

Warning

DESCRIPTION

Indicates that for a generated clock defined with `-edge_shift` option, the last edge shift value is not equal to the first edge shift value. Failure of this rule may lead to distorted waveforms of clocks.

This rule has the following specific rule occurrence attributes for the details about the violation:

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
scope	string	Design Name
generated_clock	object	generated_clock
master_clock	object	master_clock
edges	string	Edges specified with -edges option
edge_shift	string	Edge shift specified with -edge_shift option

CONST_CLK_DEF48

NAME

CONST_CLK_DEF48 (CONSTRAINT_SIGNOFF)

MESSAGE

Generated clock source is same as that of its master clock, is not defined with combinational.

DEFAULT SEVERITY

Warning

DESCRIPTION

Indicates that a generated clock whose source is same as that of its master clock, is not defined with combinational.

This rule has the following specific rule occurrence attributes for the details about the violation:

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
scope	string	Design Name
source_object	object	source_object
generated_clock	object	generated_clock
master_clock	object	master_clock
file_line	unsigned	File Line
file_name	string	File Name

CONST_CLK_DEF49

NAME

CONST_CLK_DEF49 (CONSTRAINT_SIGNOFF)

MESSAGE

Generated clock has the same source as that of its master clock (which is also a generated clock).

DEFAULT SEVERITY

Warning

DESCRIPTION

Indicates that a generated clock has the same source as that of its master clock (which is also a generated clock). This leads to generated clock source path to be undetermined.

This rule has the following specific rule occurrence attributes for the details about the violation:

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
scope	string	Design Name
source_object	object	source_object
generated_clock	object	Generated clock object
master_clock	object	master_generated_clock

CONST_CLK_GRP

- [CONST_CLK_GRP1](#)
- [CONST_CLK_GRP2](#)
- [CONST_CLK_GRP3](#)
- [CONST_CLK_GRP4](#)
- [CONST_CLK_GRP5](#)
- [CONST_CLK_GRP6](#)
- [CONST_CLK_GRP7](#)
- [CONST_CLK_GRP8](#)
- [CONST_CLK_GRP9](#)
- [CONST_CLK_GRP10](#)

CONST_CLK_GRP1

NAME

CONST_CLK_GRP1 (CONSTRAINT_SIGNOFF)

MESSAGE

Asynchronous or physically exclusive clock group is defined among master clock and its generated clocks.

DEFAULT SEVERITY

Warning

DESCRIPTION

Master clock and its generated clocks are usually considered to be correlated. This rule indicates that an asynchronous or physically exclusive clock group is defined among a master clock and its generated clocks, making them no longer correlated. This rule is checked by `verify_timing_intent` command and results can be reported by `report_rule_check` command. The clock groups can be reviewed using the `report_clock_group` command and can be corrected using the `set_clock_groups` timing constraint command by removing any extra clock group definitions.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
scope	string	Scope
master_clock	object	Master clock
generated_clocks	object_list	Generated clocks
violation_type	string	Violation type

file_names	object_list	file_names
file_lines	object_list	file_lines

EXAMPLE

- The following case reports this rule because generated clock gclk1 is asynchronous to the master clock clk and correlated generated clock gclk0:

```
create_clock -name clk -period 2 [get_ports clk]
create_generated_clock -name gclk0 -source [get_ports clk] -divide_by 1
[get_ports gclk0]
create_generated_clock -name gclk1 -source [get_ports clk] -divide_by 1
[get_ports gclk1]
set_clock_groups -name async_cg0 -asynchronous -group [get_clocks "clk gclk0"] -
group [get_clocks gclk1]
```


CONST_CLK_GRP2

NAME

CONST_CLK_GRP2 (CONSTRAINT_SIGNOFF)

MESSAGE

Asynchronous or physically exclusive clock group is not defined for the clocks in the transitive fanout of asynchronous or physically exclusive source clocks.

DEFAULT SEVERITY

Warning

DESCRIPTION

Generated clocks in the transitive fanout of uncorrelated source clocks are usually considered to be uncorrelated as well. This rule indicates that a clock group inconsistency is found between generated clocks in the transitive fanout of asynchronous or physically exclusive source clocks. This rule is checked by `verify_timing_intent` command and results can be reported by `report_rule_check` command. The clock groups can be reviewed using the `report_clock_group` command and can be corrected using the `set_clock_groups` timing constraint command, by adding any missing clock group definitions.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
scope	string	Scope
first_clock	object	First clock source
second_clock	object	Second clock source

is_asynchronous	bool	Indicates if the clock is asynchronous
is_physically_exclusive	bool	Indicates if the clock is physically exclusive
asynchronous_groups_file_names	object_list	asynchronous_groups_file_names
asynchronous_groups_file_lines	object_list	asynchronous_groups_file_lines
physically_exclusive_file_names	object_list	physically_exclusive_file_names
physically_exclusive_file_lines	object_list	physically_exclusive_file_lines
first_clock_list	object_list	First clock list
second_clock_list	object_list	Second clock list

EXAMPLE

- The following case reports this rule because source clock clk0 and source clock clk1 are asynchronous, but their generated clocks gclk0 and gclk1 are synchronous:

```
create_clock -name clk0 -period 2 [get_ports clk0]
create_clock -name clk1 -period 2 [get_ports clk1]
create_generated_clock -name gclk0 -source [get_ports clk0] -divide_by 1
[get_ports gclk0]
create_generated_clock -name gclk1 -source [get_ports clk1] -divide_by 1
[get_ports gclk1]
set_clock_groups -name async_cg0 -asynchronous -group [get_clocks clk0] -group
[get_clocks clk1]
```

CONST_CLK_GRP3

NAME

CONST_CLK_GRP3 (CONSTRAINT_SIGNOFF)

MESSAGE

The harmonic cycle of a pair of connected and synchronous clocks exceeds the threshold

DEFAULT SEVERITY

Warning

DESCRIPTION

Indicates the number of harmonic cycles of a set of correlated clocks exceeds the threshold. The number of cycles is calculated based on the clock with the highest frequency amongst its correlated clocks.

The threshold can be specified by the attribute `ltms_const_harmonic_cycle_threshold`, which has default value of 100 cycles. This rule is checked by `verify_timing_intent` command and results can be reported by `report_rule_check` command.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
scope	string	Scope
first_clock	object	First clock
second_clock	object	Second clock

lcm_cycle	long	LCM harmonic cycle of clocks in clock group

EXAMPLE

- If there are three clocks in the same clock group specified by the following SDC statements:

```
create_clock -name clk1 -period 5 clk1
create_clock -name clk2 -period 15 clk2
create_generated_clock -name gclk -source clk2 -master_clock clk2 -divide_by 2
FF1/Q
set_clock_groups -name grp1 -asynchronous -group {clk1 clk2 gclk}
```

The periods of `clk1`, `clk2`, and `gclk` are 5, 15, and 30, respectively. So, the harmonic cycle will be 6 because the LCM of their periods is 30 and it is 6 cycles of `clk1`, which is the clock with highest frequency among correlated clocks. This number of cycles, 6, will be checked against the threshold.

CONST_CLK_GRP4

NAME

CONST_CLK_GRP4 (CONSTRAINT_SIGNOFF)

MESSAGE

A synchronous clock group has more than one root clock

DEFAULT SEVERITY

Warning

DESCRIPTION

Indicates that an instance where a given clock group has more than one root clock source object is defined. A clock group's root clock is any real clock or any generated clock whose master clock does not belong to the same group. This rule check ensures that clock tree synthesis can align all the registers in a clock group at the block level. In case of a violation for this rule check, the `report_clock_group` command is used to display the clock groups. This rule is checked by `verify_timing_intent` command and results can be reported by `report_rule_check` command.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
scope	string	Scope
clocks	object_list	Clocks
root_clocks	object_list	Root clocks

EXAMPLE

- The following generates this warning because a clock group has more than one root clock

source:

```
create_clock -name CLK_1 -period 5.0 -waveform {0.0 2.5} [get_ports CLK10]  
create_clock -name CLK_2 -period 5.0 -waveform {0.0 2.5} [get_ports CLK11]
```

CONST_CLK_GRP5

NAME

CONST_CLK_GRP5 (CONSTRAINT_SIGNOFF)

MESSAGE

Generated clocks are asynchronous but their master clocks are synchronous.

DEFAULT SEVERITY

Warning

DESCRIPTION

Generated clocks of synchronous master clocks are usually synchronous as well. This rule indicates that a clock group inconsistency is found between master clocks and their generated clocks. The clock groups can be reviewed using the `report_clock_group` command and can be corrected using the `set_clock_groups` timing constraint by adding missing clock group definitions. This rule is checked by `verify_timing_intent` command and results can be reported by `report_rule_check` command.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
scope	string	Design Name
first_create_generated_clock	object	first_create_generated_clock
second_create_generated_clock	object	second_create_generated_clock
file_line	unsigned	File Line
file_name	string	File Name

first_master_clock	object	First master clock
second_master_clock	object	Second master clock

EXAMPLE

- The following example results in this warning because master clock clk0 and master clock clk1 are synchronous, but their generated clocks gclk0 and gclk1 are asynchronous:

```
create_clock -name clk0 -period 2 [get_ports clk0]
create_clock -name clk1 -period 2 [get_ports clk1]
create_generated_clock -name gclk0 -source [get_ports clk0] -divide_by 1
[get_ports gclk0]
create_generated_clock -name gclk1 -source [get_ports clk1] -divide_by 1
[get_ports gclk1]
set_clock_groups -name async_cg0 -asynchronous -group [get_clocks gclk0] -group
[get_clocks gclk1]
```


CONST_CLK_GRP6

NAME

CONST_CLK_GRP6 (CONSTRAINT_SIGNOFF)

MESSAGE

Two clocks are physically exclusive by design, but not specified as physically exclusive by `set_clock_groups`

DEFAULT SEVERITY

Warning

DESCRIPTION

Indicates that clocks defined on the same definition point are not specified as physically exclusive to each other. The clock relationship of the clocks needs to be physically exclusive. Physically exclusive clocks cannot exist in the design at the same time. The clock relationship is specified by `set_clock_groups` command with `-physically_exclusive` option. Attribute `ltms_const_consider_bidirectional_clock_exception_as_exclusive` can specify whether to consider the bidirectional inter-clock false paths as mutually exclusive clocks. This rule is checked during design initialization by `init_design` command and the results can be reported by `report_sdc` command.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
scope	string	Design Name
first_clock	object	First clock in no-set_clock_groupspair

second_clock	object	Second clock in no - set_clock_groupspair

EXAMPLE

- The following example contains clocks defined on the same definition point. This rule is reported for the clocks clk0 and clk1, because no `set_clock_groups` command with `-physically_exclusive` option specifies the clock relationship as physically exclusive.

```
create_clock -name clk0 -period 2 [get_ports clk]
create_clock -name clk1 -period 4 [get_ports clk] -add
```

CONST_CLK_GRP7

NAME

CONST_CLK_GRP7 (CONSTRAINT_SIGNOFF)

MESSAGE

Identifies the clock(s)/ generated clock(s) which are not generated directly or indirectly from either of the clock(s), in an asynchronous pair.

DEFAULT SEVERITY

Warning

DESCRIPTION

Identifies the clocks which are not generated directly or indirectly from either of the clocks in asynchronous pair do not have an asynchronous relationship with at least one of the clocks of the async pair.

This rule has the following specific rule occurrence attributes for the details about the violation:

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
scope	string	Design Name
clock	object	Generated clock object
master_clock	object	master_clock
clocks	object_list	clocks

CONST_CLK_GRP8

NAME

CONST_CLK_GRP8 (CONSTRAINT_SIGNOFF)

MESSAGE

Logically exclusive clock group is defined among master clock and its generated clock.

DEFAULT SEVERITY

Warning

DESCRIPTION

Master clock and its generated clocks are usually considered to be correlated. This rule indicates that a logically exclusive clock group is defined among a master clock and its generated clocks. This rule is checked by `verify_timing_intent` command and results can be reported by `report_rule_check` command. The clock groups can be reviewed using the `report_clock_group` command and can be corrected using the `set_clock_groups` timing constraint command by removing any extra clock group definitions. It reports violations in the following scenarios:

1. **Master vs. Generated Clocks:** A logically exclusive relationship is defined between a master clock and any of its generated clocks.
2. **Generated Clocks with Same Master:** A logically exclusive relationship is defined between generated clocks that originate from the same master clock.

This rule has the following specific rule occurrence attributes for the details about the violation:

Attribute	Type	Description
occurrence_id	unsigned	Occurrence ID
scope	string	Scope
master_clock	object	Master clock

Attribute	Type	Description
occurrence_id	int	2
scope	string	scope
master_clock	clock*	CLKA
generated_clocks	clock*	CLKA_slow , CLKA_div2
file_name	string	Constraint file name
file_line	int	Constraint line number
violation_type	string	GENERATED_GENERATED

CONST_CLK_GRP9

NAME

CONST_CLK_GRP9 (CONSTRAINT_SIGNOFF)

MESSAGE

Two clocks are physically exclusive by design, but specified as logically exclusive by `set_clock_groups`

DEFAULT SEVERITY

Warning

DESCRIPTION

Indicates that clocks defined on the same definition point are specified as logically exclusive to each other. The clock relationship of the clocks needs to be physically exclusive instead of logically exclusive. Physically exclusive clocks cannot exist in the design at the same time. The clock relationship is specified by `set_clock_groups` command with `-physically_exclusive` option. Logically exclusive clocks can exist in the design at the same time, but do not have any functional path between them. The clock relationship is specified by `set_clock_groups` command with `-logically_exclusive` option. This rule is checked during design initialization by `init_design` command and the results can be reported by `report_sdc` command.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
scope	string	Design Name
first_clock	object	First clock inset_clock_groups- logically_exclusive clockpair

second_clock	object	Second clock inset_clock_groups- logically_exclusiveclock pair
file_line	unsigned	File Line
file_name	string	File Name

EXAMPLE

- The following example contains clocks defined on the same definition point and specified as logically exclusive to each other. This rule is reported for the clocks clk0 and clk1, because the `set_clock_groups` command with `-logically_exclusive` option specifies the clock relationship as logically exclusive.

```
create_clock -name clk0 -period 2 [get_ports clk]
create_clock -name clk1 -period 4 [get_ports clk] -add
set_clock_groups -name clk_grp_log_excl -logically_exclusive -group [get_clocks
clk0] -group [get_clocks clk1]
```


CONST_CLK_GRP10

NAME

CONST_CLK_GRP10 (CONSTRAINT_SIGNOFF)

MESSAGE

Two clocks are not physically exclusive by design, but specified as physically exclusive by `set_clock_groups`

DEFAULT SEVERITY

Warning

DESCRIPTION

Indicates that clocks defined on different definition points are specified as physically exclusive to each other. The clock relationship of the clocks is not to be physically exclusive. Physically exclusive clocks cannot exist in the design at the same time. The clock relationship is specified by `set_clock_groups` command with `-physically_exclusive` option. This rule is checked during design initialization by `init_design` command and the results can be reported by `report_sdc` command.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
scope	string	Design Name
first_clock	object	First clock inset_clock_groups- physcally_eclusiveclocks pair

second_clock	object	Second clock inset_clock_groups- physcially_exclusiveclocks pair
file_line	unsigned	File Line
file_name	string	File Name

EXAMPLE

- The following example contains clocks defined on different definition points and specified as physically exclusive to each other. This rule is reported for the clocks clk0 and clk1, because the `set_clock_groups` command with `-physically_exclusive` option specifies the clock relationship as physically exclusive.

```
create_clock -name clk0 -period 2 [get_ports {clk[0]}]
create_clock -name clk1 -period 4 [get_ports {clk[1]}]
set_clock_groups -name clk_grp_phy_excl -physically_exclusive -group [get_clocks
clk0] -group [get_clocks clk1]
```

CONST_CLK_LAT

- [CONST_CLK_LAT1](#)
- [CONST_CLK_LAT3](#)
- [CONST_CLK_LAT4](#)
- [CONST_CLK_LAT6](#)
- [CONST_CLK_LAT7](#)
- [CONST_CLK_LAT8](#)
- [CONST_CLK_LAT11](#)
- [CONST_CLK_LAT12](#)
- [CONST_CLK_LAT13](#)

CONST_CLK_LAT1

NAME

CONST_CLK_LAT1 (CONSTRAINT_SIGNOFF)

MESSAGE

Undefined clock network latency for real and generated clocks

DEFAULT SEVERITY

Warning

DESCRIPTION

Indicates that a real or generated clock has undefined clock latency. Clock latency is the time it takes a clock signal to propagate from the clock definition to the register clock pin. This rule is reported for ideal clocks. This rule is checked by [verify_timing_intent](#) command and results can be reported by [report_rule_check](#) command.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
scope	string	Scope
clock	object	Clock
file_line	unsigned	File line number
file_name	string	File name

EXAMPLE

- If line #1 is in the SDC Constraints without line #2, then the Litmus issues a rule check

violation if CLK3 is created using `create_clock` and CLK3 is not a virtual clock:

```
set_clock_latency -source 0.5 CLK3  
set_clock_latency <value> CLK3
```

- The Litmus does not issue a rule check violation for the following, if CLK3 is a virtual clock:

```
set_clock_latency 1.0 CLK3
```

CONST_CLK_LAT3

NAME

CONST_CLK_LAT3 (CONSTRAINT_SIGNOFF)

MESSAGE

Source latency for a generated clock is less than or equal to the source latency of its master clock

DEFAULT SEVERITY

Warning

DESCRIPTION

Indicates that the source latency on a generated clock is less than or equal to the latency of its master clock. This rule is reported for ideal clocks. This rule is checked by `verify_timing_intent` command and results can be reported by `report_rule_check` command. Note that the Litmus checks every combination of `-rise/-fall`, `-min/-max`, and `-early/-late` separately.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
scope	string	Scope
clock	object	Clock
latency_master_clock	object	Master clock latency
file_line	unsigned	File line number
file_name	string	File name
latency_master_clock_file_line	unsigned	latency_master_clock_file_line

latency_master_clock_file_name	string	Master clock latency file name

EXAMPLE

- For the following lines:

```
create_clock -period <value> clk2
create_generated_clock -name gCLK1 -source clk2 -divide_by 2 [get_pins FF1/Q]
create_generated_clock -name gCLK2 -source [get_pins FF1/Q] -divide_by 2
[get_pins FF2/Q]
set_clock_latency <value0> -source clk2
set_clock_latency <value1> -source gCLK1
set_clock_latency <value2> -source gCLK2
```

The Litmus issues a rule check violation if:

- clk2 generates gCLK1 and value0 >= value1, or
- gCLK1 generates gCLK2 and value1 >= value2

CONST_CLK_LAT4

NAME

CONST_CLK_LAT4 (CONSTRAINT_SIGNOFF)

MESSAGE

Clocks in the same clock group have different (latency + source latency)

DEFAULT SEVERITY

Warning

DESCRIPTION

Indicates that clocks in the same group have different values for (latency + source latency). This ensures that all the register clock pins in a group can see the simultaneous arrival of the clock signals. Note that for `set_clock_latency`, the Litmus checks every combination of `-rise/-fall`, `-min/-max`, and `-early/-late` separately. This rule is reported for ideal clocks and not include virtual clocks. This rule is checked by `verify_timing_intent` command and results can be reported by `report_rule_check` command.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
scope	string	Design Name
first_clock	object	First clock source
second_clock	object	Second clock source

max_min_option	string	Value of -max/ -min option
early_late_option	string	Value of -early/ -late option
rise_fall_option	string	Value of -rise/ -fall option

EXAMPLE

- For the following lines:

```
create_clock -period 10 clk2
create_generated_clock -name gCLK1 -source clk2 -divide_by 2 [get_pins FF1/Q]
create_generated_clock -name gCLK2 -source [get_pins FF1/Q] -divide_by 2
[get_pins FF2/Q]
set_clock_latency 2.5 clk2
set_clock_latency 2.0 gCLK1
set_clock_latency 0.5 -source gCLK1
set_clock_latency 1.5 gCLK2
set_clock_latency 1.0 -source gCLK2
```

There is one group that contains three clocks: {clk2, gCLK1, gCLK2}. This does not cause a rule check violation because the (latency + source latency) is always 2.5:

- clk2: 2.5
- gCLK1: 2.0+0.5
- gCLK2: 1.5+1.0

CONST_CLK_LAT6

NAME

CONST_CLK_LAT6 (CONSTRAINT_SIGNOFF)

MESSAGE

Undefined clock source latency for generated clocks

DEFAULT SEVERITY

Warning

DESCRIPTION

Indicates that there is a generated clock with undefined source clock latency. This rule is reported for ideal clocks. This rule is checked by `verify_timing_intent` command and results can be reported by `report_rule_check` command.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
scope	string	Scope
clock	object	Clock
file_line	unsigned	File line number
file_name	string	File name

EXAMPLE

- For the following lines:

```
create_clock -period 10 clk2
create_generated_clock -name gCLK1 -source clk2 -divide_by 2 [get_pins FF1/Q]
create_generated_clock -name gCLK2 -source [get_pins FF1/Q] -divide_by 2
[get_pins FF2/Q]
set_clock_latency 2.5 clk2
set_clock_latency 2.0 gCLK1
set_clock_latency 1.5 gCLK2
set_clock_latency 1.0 -source gCLK2
```

The Litmus identifies this as a warning because gCLK1 does not have `-source` defined.

CONST_CLK_LAT7

NAME

CONST_CLK_LAT7 (CONSTRAINT_SIGNOFF)

MESSAGE

Incomplete clock latency options

DEFAULT SEVERITY

Warning

DESCRIPTION

Indicates that incomplete clock source latency and clock network latency options are used where `set_clock_latency` needs to have both or none of the options in these pairs : `-rise/-fall`, `-min/-max` and `-early/-late` where `-early/-late` is for clock source latency only. This rule is checked by `verify_timing_intent` command and results can be reported by `report_rule_check` command.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
scope	string	Scope
pin	object	Pin
clock	object	Clock
missing_corners	object_list	Missing corners
file_line	unsigned	File line number
file_name	string	File name

is_source_latency	bool	Is source latency

EXAMPLE

- The following commands cause rule check violations in Litmus because the clock latency for `-fall` and `-late` options for line #1 and `-rise` option for line #2 are Incomplete:

```
set_clock_latency -rise -early 1.3 -source gCLK1  
set_clock_latency -fall 1.5 -source gCLK1
```

CONST_CLK_LAT8

NAME

CONST_CLK_LAT8 (CONSTRAINT_SIGNOFF)

MESSAGE

Inconsistent clock latency option values

DEFAULT SEVERITY

Warning

DESCRIPTION

Indicates that there are inconsistent clock latency option values. The Litmus checks `set_clock_latency` for instances where `min > max` and `early > late`. This rule is checked by `verify_timing_intent` command and results can be reported by `report_rule_check` command.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
scope	string	Scope
pin	object	Pin
clock	object	Clock
max_min_option	string	Max/min option
early_late_option	string	Early late option
rise_fall_option	string	Value of -rise/ -fall option
min_file_line	unsigned	Min file line number

min_file_name	string	Min file name
max_file_line	unsigned	Max file line number
max_file_name	string	Max file name
min_value	float	Min value
max_value	float	Max value
is_source_latency	bool	Indicates source latency if present
has_min_max	bool	Indicates min/max if present

EXAMPLE

- The following causes a rule check violation because $\text{min} > \text{max}$ for CLK1:

```
set_clock_latency 1.5 -max CLK1
set_clock_latency 2.0 -min CLK1
```

CONST_CLK_LAT11

NAME

CONST_CLK_LAT11 (CONSTRAINT_SIGNOFF)

MESSAGE

A port/pin with `set_clock_latency` defined for a specific clock is not in that clock's tree

DEFAULT SEVERITY

Warning

DESCRIPTION

The `-clock` option is used to specify latencies on objects that are not clocks, when applying different latency values to paths that are constrained by different clocks. It is expected that if the latency is applied to a port/pin, using the `-clock` option, then the specified clock's tree should propagate to that port/pin. This rule is checked by `verify_timing_intent` command and results can be reported by `report_rule_check` command.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
scope	string	Scope
clock	object	Clock
pin	object	Pin
file_line	unsigned	File line number
file_name	string	File name

EXAMPLE

- The Litmus issues a warning if pin clkgate/Z is not in the tree of clock CK1 and the following command is used:

```
set_clock_latency 2.0 -clock CK1 [get_pins {clkgate/Z}]
```

CONST_CLK_LAT12

NAME

CONST_CLK_LAT12 (CONSTRAINT_SIGNOFF)

MESSAGE

Undefined or zero source clock latency for virtual clock

DEFAULT SEVERITY

Warning

DESCRIPTION

Indicates that a virtual clock does not have any source latency defined. Source clock latency is defined with the `set_clock_latency -source` command. This rule is checked by `verify_timing_intent` command and results can be reported by `report_rule_check` command.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
scope	string	Scope
clock	object	Clock
file_line	unsigned	File line number
file_name	string	File name

EXAMPLE

- The following lines cause rule check violations for vclk2 and vclk3:

```
create_clock -period 10 -name vclk1
create_clock -period 20 -name vclk2
create_clock -period 30 -name vclk3
set_clock_latency 2.5 -source vclk1
set_clock_latency 0.0 -source vclk2
#set_clock_latency 1.0 -source vclk3
```

CONST_CLK_LAT13

NAME

CONST_CLK_LAT13 (CONSTRAINT_SIGNOFF)

MESSAGE

Clock network latency is missing

DEFAULT SEVERITY

Warning

DESCRIPTION

Indicates that a clock pin is reached by an ideal clock but without any clock network latency associated. The following are the approaches for specifying the clock network latency for the ideal clock at the clock pin:

- Clock network latency on clock
- Clock network latency on port or pin which has influence on the clock pin

This rule is reported for ideal clocks. This rule is checked by `verify_timing_intent` command and the results can be reported by `report_rule_check` command.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
scope	string	Scope
clock	object	Clock
clock_pin_witnesses	object_list	Clock pin witnesses
file_line	unsigned	File line number

file_name	string	File name

EXAMPLE

- The following example contains a flip-flop and an ideal clock. This rule is reported for the clock input of the flip-flop dout_reg/CK, because it is reached by the clock clk but without any clock network latency associated:

```
module top (input clk, input din, output reg dout);  
    always @(posedge clk)  
        dout <= din;  
endmodule
```

Constraint that goes with above RTL:

```
create_clock -name clk -period 2 [get_ports clk]
```

CONST_CLK_UNC

- [CONST_CLK_UNC1](#)
- [CONST_CLK_UNC3](#)
- [CONST_CLK_UNC4](#)
- [CONST_CLK_UNC5](#)
- [CONST_CLK_UNC6](#)
- [CONST_CLK_UNC7](#)
- [CONST_CLK_UNC9](#)
- [CONST_CLK_UNC10](#)

CONST_CLK_UNC1

NAME

CONST_CLK_UNC1 (CONSTRAINT_SIGNOFF)

MESSAGE

Undefined clock uncertainty

DEFAULT SEVERITY

Warning

DESCRIPTION

Indicates that a real or generated clock has an undefined clock uncertainty. This rule checks simple uncertainty only. This rule is checked by [verify_timing_intent](#) command and results can be reported by [report_rule_check](#) command. Note that this checks only per-clock uncertainty.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
scope	string	Scope
clock	object	Clock
file_line	unsigned	File line number
file_name	string	File name

EXAMPLE

- In the following example, the Litmus issues a warning if a command like the following is missing for clock CLK1:

```
set_clock_uncertainty 1.0 [get_clocks CLK1]
```


CONST_CLK_UNC3

NAME

CONST_CLK_UNC3 (CONSTRAINT_SIGNOFF)

MESSAGE

Inconsistent clock uncertainty vs. clock period

DEFAULT SEVERITY

Warning

DESCRIPTION

Indicates that a clock uncertainty value is bigger than the clock period of the relevant clock. For inter-clock uncertainty, the uncertainty value is compared with the clock period of the `-to` clock. This rule is checked by `verify_timing_intent` command and results can be reported by `report_rule_check` command.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
scope	string	Scope
clock	object	Clock
second_clock	object	Second clock
setup_hold_option	string	Setup hold option
rise_fall_from_option	string	Rise fall -from option
rise_fall_to_option	string	Rise fall -to option

uncertainty_value	float	Uncertainty value
period	float	period
file_line	unsigned	File line number
file_name	string	File name

EXAMPLE

- The following commands will cause this rule to be reported twice:

```
create_clock -period 5 -name clk11 [get_ports clk1]
create_clock -period 10 -name clk22 [get_ports clk2]
set_clock_uncertainty 8 [get_clocks clk11]
set_clock_uncertainty 9 -from [get_clocks clk22] -to [get_clocks clk11]
```

Following are the reasons for those:

- Clock uncertainty of clk11 in line #3 is greater than the clock period
- Inter-clock uncertainty in line #4 is greater than the `-to` clock's clock period which is 5

CONST_CLK_UNC4

NAME

CONST_CLK_UNC4 (CONSTRAINT_SIGNOFF)

MESSAGE

Incomplete clock uncertainty options for clock objects

DEFAULT SEVERITY

Warning

DESCRIPTION

Indicates that there is a clock object with incomplete clock-uncertainty options. The command must have both or none of the options for the following pairs: `-setup/-hold` and `-rise/-fall` (for inter-clocks). This rule is checked by `verify_timing_intent` command and results can be reported by `report_rule_check` command.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
scope	string	Scope
first_clock	object	First Clock
second_clock	object	Second clock
missing_corners	object_list	missing_corners
file_line	unsigned	File line number
file_name	string	File name

EXAMPLE

- In the following example, the Litmus issues a warning because the `-hold` uncertainty for CLK1 is missing:

```
set_clock_uncertainty 1.5 -setup CLK1
```

- In the following example, the `-fall` inter-clock uncertainty from CLK2 to CLK3 is missing:

```
set_clock_uncertainty 2.0 -rise -from CLK2 -to CLK3
```

CONST_CLK_UNC5

NAME

CONST_CLK_UNC5 (CONSTRAINT_SIGNOFF)

MESSAGE

Object of higher precedence had lesser clock uncertainty value

DEFAULT SEVERITY

Warning

DESCRIPTION

Indicates that clock uncertainty with smaller value takes precedence. Setting the attribute `timing_clock_uncertainty_from_to_precedence` to `true` gives the highest precedence to inter-clock uncertainty. The following is the precedence for clock uncertainty being applied:

- Attribute `timing_clock_uncertainty_from_to_precedence: false` (default)
 - Simple uncertainty on pin (highest)
 - Simple uncertainty on port
 - Inter-clock uncertainty
 - Simple uncertainty on clock (lowest)
- Attribute `timing_clock_uncertainty_from_to_precedence: true`
 - Inter-clock uncertainty (highest)
 - Simple uncertainty on pin
 - Simple uncertainty on port
 - Simple uncertainty on clock (lowest)

This rule is checked for the same clock. This rule is checked during design initialization by `init_design` command and the results can be reported by `report_sdc` command.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence Id
scope	string	Design Name
first_clock	object	first_clock
second_clock	object	second_clock
is_hold_corner	bool	is_hold_corner
is_setup_corner	bool	is_setup_corner
file_line	unsigned	File Line
file_name	string	File Name

EXAMPLE

- The following example contains clock uncertainty constraints. This rule is reported for the inter-clock uncertainty, because it takes precedence but has smaller value than the simple uncertainty on clock.

```
set_clock_uncertainty 3 -from [get_clocks clk] -to [get_clocks clk]
set_clock_uncertainty 4 [get_clocks clk]
```

CONST_CLK_UNC6

NAME

CONST_CLK_UNC6 (CONSTRAINT_SIGNOFF)

MESSAGE

Inter-clock uncertainty not defined between synchronous clocks that drive connected sequential elements

DEFAULT SEVERITY

Warning

DESCRIPTION

Indicates that inter-clock uncertainty between synchronous clocks (real or generated) that drive connected sequential elements are not defined. This rule will not report missing inter-clock uncertainties to a clock that has simple clock uncertainty defined because it affects all the paths ending in that clock. However, if a clock has no uncertainties defined, only [CONST_CLK_UNC1](#) will be reported to avoid repetition. In summary, this rule reports only clocks that do not have simple uncertainty defined, and have inter-clock uncertainty defined from at least one clock. This rule is checked by [verify_timing_intent](#) command and results can be reported by [report_rule_check](#) command.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
scope	string	Scope
pin	object	Pin
first_clock	object	First Clock

second_clock	object	Second clock
file_line	unsigned	File line number
file_name	string	File name

EXAMPLE

- For the following case, CLK1 and CLK2 are driving the clock pins of DFFs U1 and U2, respectively, and there are paths connected from U1 to U2 and U2 to U1. Then the following reports this rule because simple clock uncertainty is not defined on the clocks, inter-clock uncertainty is defined from CLK1 to CLK2, but missing from CLK2 to CLK1:

```
create_clock -name CLK1 -period 4 -waveform {0 2} {clk1}
create_clock -name CLK2 -period 5 -waveform {0 2} {clk2}
set_clock_uncertainty 1 -from [get_clocks CLK1] -to [get_clocks CLK2]
```


CONST_CLK_UNC7

NAME

CONST_CLK_UNC7 (CONSTRAINT_SIGNOFF)

MESSAGE

Inter-clock uncertainty defined between clocks that are not connected

DEFAULT SEVERITY

Warning

DESCRIPTION

Indicates that for two clocks (for example, CLK1 and CLK2), there is a command: `set_clock_uncertainty` defined from CLK1 to CLK2, but there is no path from clock groups containing CLK1 to the clock groups containing CLK2. This rule is checked by `verify_timing_intent` command and results can be reported by `report_rule_check` command.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
scope	string	Design Name
from_clock	object	From clock option
to_clock	object	To clock option
file_line	unsigned	File Line
file_name	string	File Name

EXAMPLE

- The following case reports this rule because inter-clock uncertainty is defined from clock CLK1 to clock CLK2, but they are not connected:

```
create_clock -name CLK1 -period 4 -waveform {0 2} [get_ports {clk1}]
create_clock -name CLK2 -period 5 -waveform {0 2} [get_ports {clk2}]
set_clock_uncertainty 1 -from [get_clocks CLK1] -to [get_clocks CLK2]
```

CONST_CLK_UNC9

NAME

CONST_CLK_UNC9 (CONSTRAINT_SIGNOFF)

MESSAGE

Inter-clock uncertainty not defined between a virtual clock and a connected non-virtual clock with the same waveform and period

DEFAULT SEVERITY

Warning

DESCRIPTION

Indicates that inter-clock uncertainty is not defined between a virtual clock and a non-virtual clock with the same period and waveform, that are connected (constrain a timing path). This rule will not report missing inter-clock uncertainties to a clock that has simple clock uncertainty defined because it affects all the paths ending in that clock. Note that clock pairs that are not related (for example, asynchronous clocks) will not be reported. This rule is checked by `verify_timing_intent` command and results can be reported by `report_rule_check` command.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
scope	string	Design Name
first_clock	object	Virtual clock definition

second_clock	object	Non-virtual clock definition

EXAMPLE

- If there is a path from DATA_IN to MY_REG/D, the clock tree of RCLK reaches MY_REG/CK, and virtual clock VCLK and real clock RCLK are synchronous, then this rule will be reported by Litmus if the following SDC commands are given:

```
create_clock -name VCLK -period 10.0 -waveform {0.0 5.0}
create_clock -name RCLK -period 10.0 -waveform {0.0 5.0} [get_ports clk]
set_input_delay -clock VCLK 1.0 [get_ports DATA_IN]
```

CONST_CLK_UNC10

NAME

CONST_CLK_UNC10 (CONSTRAINT_SIGNOFF)

MESSAGE

Inter-clock uncertainty is defined between asynchronous or mutually exclusive clocks.

DEFAULT SEVERITY

Warning

DESCRIPTION

The paths between asynchronous or mutually exclusive clocks are not considered during timing analysis (that is, they are considered false paths). This rule indicates that unnecessary inter-clock uncertainty is defined between asynchronous or mutually exclusive clocks. The inter-clock uncertainty defined between the asynchronous or mutually exclusive clocks should be removed to improve the timing constraint quality. This rule is checked by `verify_timing_intent` command and results can be reported by `report_rule_check` command.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
scope	string	Design Name
is_asynchronous	bool	Indicates if clocks are asynchronous
is_physically_exclusive	bool	Indicates if clocks are physically exclusive

is_logically_exclusive	bool	Indicates id clocks are logically exclusive
set_clock_groups	string	List ofset_clock_groups
file_line	unsigned	File Line
file_name	string	File Name

EXAMPLE

- The following case reports this rule because inter-clock uncertainty is defined from clock clk0 to clock clk1, which are asynchronous:

```
create_clock -name clk0 -period 2 [get_ports clk0]
create_clock -name clk1 -period 3 [get_ports clk1]
set_clock_groups -name async_cg0 -asynchronous -group [get_clocks clk0] -group
[get_clocks clk1]
set_clock_uncertainty 1 -from [get_clocks clk0] -to [get_clocks clk1]
```

CONST_DAT_CHK

- [CONST_DAT_CHK1](#)
- [CONST_DAT_CHK2](#)
- [CONST_DAT_CHK3](#)
- [CONST_DAT_CHK4](#)

CONST_DAT_CHK1

NAME

CONST_DAT_CHK1 (CONSTRAINT_SIGNOFF)

MESSAGE

Data check signal is undetermined

DEFAULT SEVERITY

Warning

DESCRIPTION

Indicates that data check reference point (`-from` option) or constrained point (`-to` option) specified by `set_data_check` command is undetermined. The following are the possible causes for this violation:

- Data check reference point or constrained point is not reached by any clocked signal
- Data check reference point or constrained point is reached by clocked signals which are associated with multiple clocks

This rule is checked by `verify_timing_intent` command and the results can be reported by `report_rule_check` command.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
scope	string	Design Name
pin	object	Data check pin
clocks	object_list	List of clocks

unconstrained_pins	object_list	List of unconstrained pins
corner_type	object_list	corner_type
file_names	object_list	file_names
file_lines	object_list	file_lines

EXAMPLE

- The following example contains AND gates and data check constraints. This rule is reported for the reference point i_and0/A1 and the constrained point i_and0/A2, because they are not reached by any clocked signal. Besides, this rule is also reported for the reference point i_and1/A1 and the constrained point i_and1/A2, because they are reached by the clocked signals associated with the clocks vclk0 and vclk1.

```
module top (input [1:0] din0, input [1:0] din1, output [1:0] dout);
    AND2_X2 i_and0 (.A1(din0[0]), .A2(din1[0]), .ZN(dout[0]));
    AND2_X2 i_and1 (.A1(din0[1]), .A2(din1[1]), .ZN(dout[1]));
endmodule
```

Constraints that go with the above RTL:

```
create_clock -name vclk0 -period 2
create_clock -name vclk1 -period 2
set_input_delay -clock [get_clocks vclk0] 1 [get_ports {din0[1] din1[1]}] -add
set_input_delay -clock [get_clocks vclk1] 1 [get_ports {din0[1] din1[1]}] -add
set_data_check 0.5 -from [get_pins i_and0/A1] -to [get_pins i_and0/A2]
set_data_check 0.5 -from [get_pins i_and1/A1] -to [get_pins i_and1/A2]
```

CONST_DAT_CHK2

NAME

CONST_DAT_CHK2 (CONSTRAINT_SIGNOFF)

MESSAGE

Pin is unconstrained

DEFAULT SEVERITY

Ignore (default Warning)

DESCRIPTION

Indicates that an end point of timing path or a segmentation point created by constraints, that becomes a timing path end point, is unconstrained. The following are the possible causes for this violation:

- Missing launch clock or capture clock AND missing max path delay or min path delay
- All the matching timing paths to the end point are constrained by [set_false_path](#)

This rule is disabled by default. Using [set_message](#) command the severity can be modified to enable the rule. This rule is checked by [verify_timing_intent](#) command and the results can be reported by [report_rule_check](#) command.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
scope	string	Design Name
pin	object	Pin

EXAMPLE

- The following example contains unconstrained end points. This rule is reported for the end points i_ff0/D and dout[0], because of missing capture clock and launch clock which is caused by missing clock at the clock input of the flip-flop i_ff0/CK. Besides, this rule is also reported for the end points i_ff1/D and dout[1], because all the matching timing paths are false paths which are caused by the `set_false_path` commands.

```
module top (input [1:0] clk, input [1:0] din, output [1:0] dout);
    DFF_X2 i_ff0 (.CK(clk[0]), .D(din[0]), .Q(dout[0]));
    DFF_X2 i_ff1 (.CK(clk[1]), .D(din[1]), .Q(dout[1]));
endmodule
```

Constraints that go with the above RTL:

```
create_clock -name clk1 -period 2 [get_ports {clk[1]}]
create_clock -name vclk -period 2
set_input_delay -clock [get_clocks vclk] 1 [get_ports {din[0] din[1]}]
set_output_delay -clock [get_clocks vclk] 1 [get_ports {dout[0] dout[1]}]
set_false_path -to [get_pins i_ff1/D]
set_false_path -to [get_ports {dout[1]}]
```

CONST_DAT_CHK3

NAME

CONST_DAT_CHK3 (CONSTRAINT_SIGNOFF)

MESSAGE

Incomplete `set_data_check`

DEFAULT SEVERITY

Warning

DESCRIPTION

Indicates that a `set_data_check` has been specified for `-setup` but not for `-hold`, or vice versa. This rule is checked by `verify_timing_intent` command and results can be reported by `report_rule_check` command.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
scope	string	Design Name
setup_hold_option	string	Value of -setup/ -hold option
rise_fall_from_option	string	Value of -rise_from/ -fall_from option

rise_fall_to_option	string	Value of - rise_to/- fall_to option
file_line	unsigned	File Line
file_name	string	File Name

EXAMPLE

- If the following constraints are specified:

```
set_data_check 1 -setup -from A -to B -clock CLK1
set_data_check 2 -hold -rise_from C -to D
set_data_check 3 -setup -from C -rise_to D
```

then this rule will report the following:

- `set_data_check -hold` is missing for (1)
- `set_data_check -setup` is missing for `-rise_from/-fall_to` for (2)
- `set_data_check -hold` is missing for `-fall_from/-rise_to` for (3)

CONST_DAT_CHK4

NAME

CONST_DAT_CHK4 (CONSTRAINT_SIGNOFF)

MESSAGE

`set_max_time_borrow` is not set on a latch or on a data/enable pin of a latch, or its value is bigger than half the clock period

DEFAULT SEVERITY

Warning

DESCRIPTION

The `set_max_time_borrow` command can be applied to clocks, cells or pins. For other object types, rule [TCLCMD-917](#) and [TCLCMD-513](#) will be reported. This rule is reported when the command is applied to clock objects, cells that are not level-sensitive, or pins that are not data or enable pins of level-sensitive sequential cells. It is also reported if the affected object is a level-sensitive cell or its data or enable pin, but the value specified is bigger than half the clock period of any clock propagating to the cell's enable pin. This rule is checked by `verify_timing_intent` command, and the results can be reported by `report_rule_check` command.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
scope	string	instance
scope	object	instance
pin	object	pin
clock	object	Clock definition

description	string	Description
value	float	value
period	float	period
file_line	unsigned	File Line
file_name	string	File Name

EXAMPLE

- If the enable pin of latch L1 is reached by clock CLK whose period is 10, this rule will be reported for all the following commands:

```
set_max_time_borrow 7 [get_cells L1]
set_max_time_borrow 7 [get_pins L1/D]
set_max_time_borrow 3 [get_clocks CLK]
set_max_time_borrow 3 [get_cells MUX1]
set_max_time_borrow 3 [get_pins L1/Q]
```

CONST_DGN_CMB

- [CONST_DGN_CMB1](#)
- [CONST_DGN_CMB2](#)
- [CONST_DGN_CMB3](#)
- [CONST_DGN_CMB5](#)
- [CONST_DGN_CMB6](#)

CONST_DGN_CMB1

NAME

CONST_DGN_CMB1 (CONSTRAINT_SIGNOFF)

MESSAGE

Unconstrained combinational path

DEFAULT SEVERITY

Warning

DESCRIPTION

Indicates that there is an unconstrained input/output delay value for a primary input and primary output connected combinational path. Every pair of primary inputs and primary outputs that are connected to a combinational path must have a complete set of input/output delays with respect to the same clock (or independent of any clock), or a matching set of `set_max_delay` and `set_min_delay` specified.

In addition, it checks whether a matching set of `set_max_delay` and `set_min_delay` is specified (not only for a primary input and primary output connected combinational path). You can check the rule using `verify_timing_intent` command and the results can be reported by `report_rule_check` command.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
scope	string	Scope
primary_input	object	Primary input
primary_output	object	Primary output
type	string	Type

missing_corners	object_list	Missing corners
file_line	unsigned	File line number
file_name	string	File name

EXAMPLE

In the following example, a combinational path exists between IN1 and OUT2, and a maximum delay is defined:

```
set_max_delay 3 -from IN1 -to OUT2
```

Litmus issues a warning because `set_min_delay` is missing.

CONST_DGN_CMB2

NAME

CONST_DGN_CMB2 (CONSTRAINT_SIGNOFF)

MESSAGE

Inconsistent set_min_delay/set_max_delay

DEFAULT SEVERITY

Warning

DESCRIPTION

Indicates that there is an inconsistent specification for `set_max_delay` and `set_min_delay`. The rule check detects whether $\text{min} > \text{max}$ for each pair of `set_max_delay` and `set_min_delay` with the same path specification.

This rule is checked by `verify_timing_intent` command and results can be reported by `report_rule_check` command.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
scope	string	Scope
min_file_line	unsigned	Min file line number
min_file_name	string	Min file name
max_file_line	unsigned	Max file line number
max_file_name	string	Max file name

min_delay	float	Min delay
max_delay	float	Max delay
rise_fall_option	string	Rise fall option

EXAMPLE

In the following example, Litmus issues a warning because the minimum delay is greater than the maximum delay for the same combinational path:

```
set_max_delay 3 -through U1/Z -to OUT2
set_min_dealy 3.5 -through U1/Z -to OUT2
```

CONST_DGN_CMB3

NAME

CONST_DGN_CMB3 (CONSTRAINT_SIGNOFF)

MESSAGE

Inconsistent input/output delay value vs. clock period for combinational path

DEFAULT SEVERITY

Warning

DESCRIPTION

Indicates that there is a combinational path with inconsistencies in input and output delay values versus clock periods. It has a rule check violation if the expression defined in the attribute [ltms_const_combinational_path_delay_condition](#) is not satisfied.

The default value of attribute [ltms_const_combinational_path_delay_condition](#) is `input_delay+output_delay<period`. It is required for input delay and output delay relative to same clock. This rule will not be reported for an input/output port pair if all the combinational paths connecting them are matched by one or more `set_false_path` commands.

This rule is checked by `verify_timing_intent` command and the results can be reported by `report_rule_check` command.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
scope	string	Scope
primary_input	object	Primary input
primary_output	object	Primary output

clock	object	Clock
first_cmd_file_line	unsigned	First command file line number
first_cmd_file_name	string	First command file name
second_cmd_file_line	unsigned	Second command file line number
second_cmd_file_name	string	Second command file name
analysis_type	object_list	analysis_type
transition_type	object_list	transition_type
input_delay	float	Input delay
output_delay	float	output_delay
delay_condition	string	delay_condition

EXAMPLE

The following example defines a virtual clock vCLK3 with a period of 8, and input and output delays of 5.00:

```
create_clock -period 8 -name vCLK3
set_input_delay 5.00 {IN1 IN2} -clock vCLK3
set_output_delay 5.00 {OUT1 OUT2} -clock vCLK3
```

In the dofile, the attribute `ltms_const_combinational_path_delay_condition` is defined as:

```
set_db ltms_const_combinational_path_delay_condition

```

The Litmus issues a warning because the sum of the input and output delays is greater than 90% of the clock period.

CONST_DGN_CMB5

NAME

CONST_DGN_CMB5 (CONSTRAINT_SIGNOFF)

MESSAGE

Combinational constraints present on elements belonging to internal sequential path

DEFAULT SEVERITY

Warning

DESCRIPTION

Indicates that combinational constraints are imposed on elements that belong to internal sequential paths. This rule checks whether a `-from` or `-to` point of a `set_max_delay` or `set_min_delay` is in a path between sequential elements.

This rule is checked by `verify_timing_intent` command and the results can be reported by `report_rule_check` command.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
scope	string	Scope
from_objects	object_list	-from objects
to_objects	object_list	-to objects
through_objects	object_list	-through objects
file_line	unsigned	File line number

file_name	string	File name
exc_type	string	Exc type

EXAMPLE

Litmus issues this message for the following `set_max_delay` and `set_min_delay` constraints, since they are present on elements belonging to internal sequential path:

```
module TOP (in1, clk1, out1);
  input in1, clk1;
  output out1;
  wire w1, w2;
  DFF_X1 g1 ( .D(in1), .Q(w1), .CK(clk1) );
  BLOCK1 g2 ( .in(w1), .out(w2), .clk(clk1) );
  DFF_X1 g3 ( .D(w2), .Q(out1), .CK(clk1) );
endmodule

module BLOCK1 (input in, clk, output out);
  wire w1;
  DFF_X1 ig1 ( .D(in), .Q(w1), .CK(clk) );
  DFF_X1 ig2 ( .D(w1), .Q(out), .CK(clk) );
endmodule
```

Constraints that go with the above RTL:

```
create_clock -period 10 -waveform {0 5} -name rCLK1 [get_ports clk1]
set_min_delay 1 -from [get_pins g2/ig1/Q] -to [get_pins g2/ig2/D]
set_max_delay 2 -from [get_pins g2/ig1/Q] -to [get_pins g2/ig2/D]
```


CONST_DGN_CMB6

NAME

CONST_DGN_CMB6 (CONSTRAINT_SIGNOFF)

MESSAGE

A timing loop was found in the design.

DEFAULT SEVERITY

Warning

DESCRIPTION

This rule has the following specific rule occurrence attributes for the details about the violation:

Attribute	Type	Description
occurrence_id	unsigned	Occurrence ID
scope	string	Design Name
from_arc	object	From arc point
to_arc	object	To arc point
ltms_diagnosis_path	string	Timing loop information from TA-112 reports
loop_breaking_point	object	First pin from the timing loop table reported by TA-112

CONST_DGN_TIM

- [CONST_DGN_TIM1](#)

CONST_DGN_TIM1

NAME

CONST_DGN_TIM1 (CONSTRAINT_SIGNOFF)

MESSAGE

Multiplexer inputs are on data and clock networks

DEFAULT SEVERITY

Warning

DESCRIPTION

Indicates that a multiplexer is on both clock and data networks, i.e. it is reached by both clock and data signals. This rule is checked by `verify_timing_intent` command and the results can be reported by `report_rule_check` command.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
scope	string	Scope
mux	object	Mux instance
is_tied	bool	Is tied
clock_sources	object_list	Clock sources
clock	object_list	Clock

EXAMPLE

- The following example contains a multiplexer. This rule is reported for the multiplexer i_mux, because it is reached by the clock signal clk and data signal.

```
module top (input clk, input din, output dout, input sel);  
    MUX2_X2 i_mux (.A(clk), .B(din), .S(sel), .Z(dout));  
endmodule
```

Constraints that go with the above RTL:

```
create_clock -name clk -period 2 [get_ports clk]  
create_clock -name vclk -period 2  
set_input_delay -clock [get_clocks vclk] 1 [get_ports din]  
set_output_delay -clock [get_clocks vclk] 1 [get_ports dout]
```

CONST_EXC_CLK

- [CONST_EXC_CLK1](#)

CONST_EXC_CLK1

NAME

CONST_EXC_CLK1 (CONSTRAINT_SIGNOFF)

MESSAGE

Connected master clock and generated clocks are declared as set_false_path

DEFAULT SEVERITY

Warning

DESCRIPTION

A master clock and its generated child clocks are usually considered to be correlated. This rule indicates that two connected and synchronous clocks (either a master clock and a generated child clock, or two generated child clocks from same master clock) are declared as false path by `set_false_path` command, making them no longer correlated. This rule is checked by `verify_timing_intent` command and results can be reported by `report_rule_check` command.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
scope	string	Design Name
from_clock	object	From clock
from_clock_file_name	string	from_clock_file_name
from_clock_file_line	unsigned	from_clock_file_line
to_clock	object	To clock
to_clock_file_name	string	to_clock_file_name

to_clock_file_line	unsigned	to_clock_file_line
false_path_file_names	object_list	false_path_file_names
false_path_file_lines	object_list	false_path_file_lines

This rule is customizable. The specific rule instance attribute `check_type` configures the rule checking behavior.

Attribute	Description
check_type	Specifies the type of clocks to be checked. Possible value: <all master_and_generated_clocks generated_clocks_only> Where: all: Checks all below types. This is the default. master_and_generated_clocks: Checks a master clock and its generated child clocks. generated_clocks_only: Checks generated child clocks from same master clock.

EXAMPLE

- The following case reports this rule when the master clock clk and the generated clock gclk are connected and synchronous (there is a data path from clock clk to clock gclk) because they are declared as false path by set_false_path command.

```
create_clock -name clk -period 2 [get_ports clk]
create_generated_clock -name gclk -source [get_ports clk] -divide_by 1 [get_pins
i_clk_buf/Z]
set_false_path -from [get_clocks clk] -to [get_clocks gclk]
```

RELATED TOPICS

- Commands

[set_false_path](#)
[verify_timing_intent](#)
[report_rule_check](#)

- Attributes

[Rule Attributes - rule_instance](#)

CONST_EXC_FLP

- [CONST_EXC_FLP1](#)
- [CONST_EXC_FLP5](#)
- [CONST_EXC_FLP7](#)
- [CONST_EXC_FLP8](#)
- [CONST_EXC_FLP9](#)
- [CONST_EXC_FLP10](#)
- [CONST_EXC_FLP11](#)

CONST_EXC_FLP1

NAME

CONST_EXC_FLP1 (CONSTRAINT_SIGNOFF)

MESSAGE

False path exception does not match any timing path

DEFAULT SEVERITY

Warning

DESCRIPTION

Indicates that a [set_false_path](#) command does not match any timing path, or any object specified in [set_false_path](#) command has no matching path. This rule is checked by [verify_timing_intent](#) command and the results can be reported by [report_rule_check](#) command.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
scope	string	Scope
exc_type	string	Exception type
exc_from	object_list	Exception from
exc_to	object_list	Exception to
exc_through	object_list	Exception through
exc_from_clock	object_list	exc_from_clock
exc_to_clock	object_list	exc_to_clock

no_matching_path_from_edge	string	No matching clock -from edge
no_matching_path_from	object_list	No matching clock -from
no_matching_path_to_edge	string	No matching path -to edge
no_matching_path_to	object_list	No matching path -to
no_matching_path_through_edges	object_list	No matching path -through edges
no_matching_path_through	object_list	No matching path -through
no_launch_clock_EDGE	string	no_launch_clock_edge
no_launch_clock	object_list	No launch clock
no_capture_clock_edge	string	No capture clock edge
no_capture_clock	object_list	No capture lock
start_clocks	object_list	start clocks
end_clocks	object_list	end clocks
file_line	unsigned	File line number
file_name	string	File name
type	object_list	Type

In addition, common rule occurrence attribute `type` provides the possible causes for the violation.

Attribute	Value	Description
type	NO_LAUNCH_CLOCK	No launch clock for the matching path for the path spec
	NO_CAPTURE_CLOCK	No capture clock for the matching path for the path spec

	NO_MATCHING_PATH	No matching path for the path spec
	PARTIALLY_NO_LAUNCH_CLOCK	No launch clock for the matching path for part of the objects in the path spec
	PARTIALLY_NO_CAPTURE_CLOCK	No capture clock for the matching path for part of the objects in the path spec
	PARTIALLY_NO_MATCHING_PATH	No matching path for part of the objects in the path spec

EXAMPLE

- The following example contains two timing paths and false path constraints. This rule is reported for the 1st `set_false_path` command, because it does not match any timing path. This rule is also reported for input port `din[0]` specified in the 2nd `set_false_path` command, because it has no matching path.

```
module top (input [1:0] din, output [1:0] dout);
    assign dout = din;
endmodule
```

Constraints that go with the above RTL:

```
create_clock -name vclk -period 2
set_input_delay -clock [get_clocks vclk] 1 [get_ports din]
set_output_delay -clock [get_clocks vclk] 1 [get_ports dout]
set_false_path -from [get_ports {din[1]}] -to [get_ports {dout[0]}]
set_false_path -from [get_ports {din[0] din[1]}] -to [get_ports {dout[1]}]
```

CONST_EXC_FLP5

NAME

CONST_EXC_FLP5 (CONSTRAINT_SIGNOFF)

MESSAGE

Incomplete false path options

DEFAULT SEVERITY

Warning

DESCRIPTION

Indicates that any combination of `-setup/-hold` with `-rise/-fall` is not covered by the `set_false_path` commands. This rule is checked by `verify_timing_intent` command, and the results can be reported by `report_rule_check` command.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
scope	string	Scope
exc_type	string	Exception type
exc_from	object_list	Exception from port / pin
exc_to	object_list	Exception to port / pin
exc_through	object_list	Exception through pin
exc_from_clock	object_list	Exception from clock
exc_to_clock	object_list	Exception to clock

missing_corners	object_list	Missing corners
file_line	unsigned	File line number
file_name	string	File name

EXAMPLE

- The following example contains false path constraints. This rule is reported for the 2nd false path, because of missing constraint for hold corner.

```
set_false_path -from [get_ports {din[0]]} -to [get_ports {dout[1]]}  
set_false_path -setup -from [get_ports {din[1]]} -to [get_ports {dout[1]]}
```

CONST_EXC_FLP7

NAME

CONST_EXC_FLP7 (CONSTRAINT_SIGNOFF)

MESSAGE

Clocks declared as `set_false_path -from clock -to clock` (in both directions) are in the same clock group

DEFAULT SEVERITY

Warning

DESCRIPTION

Indicates that a clock pair is in the same clock group (i.e. synchronous to each other), but declared as false paths in both directions specified by `set_false_path` commands. This rule is not checked if any combination of `-setup/-hold` with `-rise/-fall` is not covered by the `set_false_path` commands, or if the `set_false_path` commands have `-through` option. This rule is checked by `verify_timing_intent` command and the results can be reported by `report_rule_check` command.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
scope	string	Design Name
first_clock	object	First clock definition
first_clock_file_name	string	first_clock_file_name
first_clock_file_line	unsigned	first_clock_file_line
second_clock	object	Second clock definition

second_clock_file_name	string	second_clock_file_name
second_clock_file_line	unsigned	second_clock_file_line
false_path_file_names	object_list	false_path_file_names
false_path_file_lines	object_list	false_path_file_lines

EXAMPLE

- The following contains false path constraints. All of the clocks are in the same clock group. This rule is reported for the clock pair (clk1,clk2), but not for the clock pairs (clk1,clk3) and (clk2,clk3).

```
set_false_path -from clk1 -to clk2
set_false_path -from clk2 -to clk1
set_false_path -from clk1 -to clk3
set_false_path -from clk2 -to clk3
set_false_path -setup -from clk3 -to clk2
```


CONST_EXC_FLP8

NAME

CONST_EXC_FLP8 (CONSTRAINT_SIGNOFF)

MESSAGE

False path without clock specified overlaps with clock tree

DEFAULT SEVERITY

Warning

DESCRIPTION

Indicates that a false path specified by `set_false_path` command overlaps with a clock tree. A false path specified with a clock is skipped from this check. This rule is checked by `verify_timing_intent` command and the results can be reported by `report_rule_check` command.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
scope	string	Design Name
overlapped_clocks	object_list	List of overlapped clocks
file_line	unsigned	File Line
file_name	string	File Name

EXAMPLE

- The following contains a flip-flop and false path constraints. This rule is reported for the 1st `set_false_path` command, because it overlaps with the clock tree of clock `clk`. This check is skipped for the 2nd `set_false_path` command, because it is specified with a clock.

```
module top (input clk, input din, output reg dout);  
    always @(posedge clk)  
        dout <= din;  
endmodule
```

Constraints that go with the above RTL:

```
create_clock -name clk -period 2 [get_ports clk]  
set_output_delay 1 -clock [get_clocks clk] [get_pins dout_reg/CK]  
set_false_path -from [get_ports clk]  
set_false_path -from [get_clocks clk]
```

CONST_EXC_FLP9

NAME

CONST_EXC_FLP9 (CONSTRAINT_SIGNOFF)

MESSAGE

Start/end point is specified in false path `-through`

DEFAULT SEVERITY

Ignore (default Warning)

DESCRIPTION

Indicates that a start point or an end point is specified with `-through`, `-rise_through` or `-fall_through` option of `set_false_path` command. It is recommended to use `-from`, `-rise_from` or `-fall_from` option for a start point and `-to`, `-rise_to` or `-fall_to` for an end point. The false path specified with `from` (`-from`, `-rise_from` or `-fall_from`) or `to` (`-to`, `-rise_to` or `-fall_to`) option associated with a clock is skipped from this check. This rule is disabled by default. Using `set_message` command the severity can be modified to enable the rule. This rule is checked by `verify_timing_intent` command and the results can be reported by `report_rule_check` command.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
scope	string	Scope
exc_type	string	Exception type
exc_from	object_list	Exception from
exc_to	object_list	Exception to

exc_through	object_list	Exception through
from_ports	object_list	From ports
to_ports	object_list	To ports
from_to_ports	object_list	From to ports
file_line	unsigned	File line number
file_name	string	File name

EXAMPLE

- The following contains false path constraints. This rule is reported for the 1st `set_false_path` command, because the start point `din` is specified with `-through` option. This check is skipped for the 2nd `set_false_path` command, because it is specified with `-from` option associated with a clock.

```
create_clock -name vclk -period 2
set_input_delay -clock [get_clocks vclk] 1 [get_ports din]
set_output_delay -clock [get_clocks vclk] 1 [get_ports dout]
set_false_path -through [get_ports din]
set_false_path -from [get_clocks vclk] -through [get_ports din]
```

CONST_EXC_FLP10

NAME

CONST_EXC_FLP10 (CONSTRAINT_SIGNOFF)

MESSAGE

False path without both `-from` and `-to` specified matches multiple clock crossings

DEFAULT SEVERITY

Warning

DESCRIPTION

Indicates that a false path specified by `set_false_path` command without both from (`-from`, `-rise_from` or `-fall_from`) and to (`-to`, `-rise_to` or `-fall_to`) options matches multiple clock crossings. This rule is checked by `verify_timing_intent` command and the results can be reported by `report_rule_check` command.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
scope	string	Scope
exc_type	string	Exception type
exc_through	object_list	Exception through
start_clocks	object_list	Start clocks
end_clocks	object_list	End clocks
file_line	unsigned	File line number

file_name	string	File name

EXAMPLE

- The following contains two timing paths. This rule is reported for the `set_false_path` command, because it matches two clock crossings but not specified with both from and to options.

```
module top (input [1:0] din, output dout);
    AND2_X2 i_and (.A1(din[0]), .A2(din[1]), .ZN(dout));
endmodule
```

Constraints that go with the above RTL:

```
create_clock -name vclk_in0 -period 2
create_clock -name vclk_in1 -period 2
create_clock -name vclk_out -period 2
set_input_delay -clock [get_clocks vclk_in0] 1 [get_ports {din[0]}]
set_input_delay -clock [get_clocks vclk_in1] 1 [get_ports {din[1]}]
set_output_delay -clock [get_clocks vclk_out] 1 [get_ports dout]
set_false_path -through [get_pins i_and/ZN]
```

The two timing paths are followings:

- From the primary input `din[0]` (launch clock: `vclk_in0`), through the AND gate `i_and`, to the primary output `dout` (capture clock: `vclk_out`)
- From the primary input `din[1]` (launch clock: `vclk_in1`), through the AND gate `i_and`, to the primary output `dout` (capture clock: `vclk_out`)

CONST_EXC_FLP11

NAME

CONST_EXC_FLP11 (CONSTRAINT_SIGNOFF)

MESSAGE

Clock period ratio of connected inter-clock false path is not an integer

DEFAULT SEVERITY

Warning

DESCRIPTION

Indicates that neither "launch / capture" nor "capture / launch" clock period ratio of a connected inter-clock false path is an integer. Inter-clock false path can be specified as following:

- `set_false_path -from <clock> -to <clock>`
- `set_false_path -from <clock>`
- `set_false_path -to <clock>`

 This rule is checked by [verify_timing_intent](#) command and the results can be reported by [report_rule_check](#) command.

This rule has the following specific rule occurrence attributes for the details about the violation:

Attribute	Type	Description
occurrence_id	unsigned	Occurrence ID
scope	string	Design Name
start_clocks	object_list	List of start clocks

end_clocks	object_list	List of end clocks
file_line	unsigned	File Line
file_name	string	File Name

EXAMPLE

- The following case contains RTL design and false path constraints. This rule is reported for the `set_false_path` command, because neither `clk_0 (3) / clk_1 (7)` nor `clk_1 (7) / clk_0 (3)` clock period ratio is an integer.

```
module top (input [1:0] clk, input din, output dout);
    reg [1:0] data;
    always @(posedge clk[0])
        data[0] <= din;
    always @(posedge clk[1])
        data[1] <= data[0];
    assign dout = data[1];
endmodule
```

```
create_clock -name clk_0 -period 3 [get_ports {clk[0]}]
create_clock -name clk_1 -period 7 [get_ports {clk[1]}]
set_false_path -from [get_clocks clk_0] -to [get_ports clk_1]
```


CONST_EXC_MCP

- [CONST_EXC_MCP1](#)
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CONST_EXC_MCP1

NAME

CONST_EXC_MCP1 (CONSTRAINT_SIGNOFF)

MESSAGE

Multi-cycle path exception does not match any timing path

DEFAULT SEVERITY

Warning

DESCRIPTION

Indicates that a `set_multicycle_path` command does not match any timing path, or any object specified in `set_multicycle_path` command has no matching path. This rule is checked by `verify_timing_intent` command, and the results can be reported by `report_rule_check` command.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
scope	string	Scope
exc_type	string	Exception type
exc_from	object_list	Exception from
exc_to	object_list	Exception to
exc_through	object_list	Exception through
exc_from_clock	object_list	exc_from_clock

exc_to_clock	object_list	exc_to_clock
no_matching_path_from_edge	string	No matching clock -from edge
no_matching_path_from	object_list	No matching clock -from
no_matching_path_to_edge	string	No matching path -to edge
no_matching_path_to	object_list	No matching path -to
no_matching_path_through_edges	object_list	No matching path -through edges
no_matching_path_through	object_list	No matching path -through
no_launch_clock_EDGE	string	no_launch_clock_edge
no_launch_clock	object_list	No launch clock
no_capture_clock_edge	string	No capture clock edge
no_capture_clock	object_list	No capture lock
start_clocks	object_list	start clocks
end_clocks	object_list	end clocks
file_line	unsigned	File line number
file_name	string	File name
type	object_list	Type

In addition, common rule occurrence attribute `type` provides the possible causes for the violation.

Attribute	Value	Description
type	NO_LAUNCH_CLOCK	No launch clock for the matching path for the path spec

	NO_CAPTURE_CLOCK	No capture clock for the matching path for the path spec
	NO_MATCHING_PATH	No matching path for the path spec
	PARTIALLY_NO_LAUNCH_CLOCK	No launch clock for the matching path for part of the objects in the path spec
	PARTIALLY_NO_CAPTURE_CLOCK	No capture clock for the matching path for part of the objects in the path spec
	PARTIALLY_NO_MATCHING_PATH	No matching path for part of the objects in the path spec

EXAMPLE

- The following example contains two timing paths and multi-cycle path constraints. This rule is reported for the 1st `set_multicycle_path` command, because it does not match any timing path. This rule is also reported for input port `din[0]` specified in the 2nd `set_multicycle_path` command, because it has no matching path.

```
module top (input [1:0] din, output [1:0] dout);
    assign dout = din;
endmodule
```

Constraints that go with the above RTL:

```
create_clock -name vclk -period 2
set_input_delay -clock [get_clocks vclk] 1 [get_ports din]
set_output_delay -clock [get_clocks vclk] 1 [get_ports dout]
set_multicycle_path 2 -from [get_ports {din[1]}] -to [get_ports {dout[0]}]
set_multicycle_path 2 -from [get_ports {din[0] din[1]}] -to [get_ports {dout[1]}]
```

CONST_EXC_MCP5

NAME

CONST_EXC_MCP5 (CONSTRAINT_SIGNOFF)

MESSAGE

Incomplete multi-cycle path options

DEFAULT SEVERITY

Warning

DESCRIPTION

Indicates that any combination of `-setup/-hold` with `-rise/-fall` is not covered by the `set_multicycle_path` commands. This rule is checked by `verify_timing_intent` command, and the results can be reported by `report_rule_check` command.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
scope	string	Scope
exc_type	string	Exception type
exc_from	object_list	Exception from port / pin
exc_to	object_list	Exception to port / pin
exc_through	object_list	Exception through pin
exc_from_clock	object_list	Exception from clock
exc_to_clock	object_list	Exception to clock

missing_corners	object_list	Missing corners
file_line	unsigned	File line number
file_name	string	File name

EXAMPLE

- The following example contains multi-cycle path constraints. This rule is reported for the 2nd multi-cycle path, because of missing constraint for hold corner.

```
set_multicycle_path 2 -from [get_ports {din[0]}] -to [get_ports {dout[1]}]
set_multicycle_path 2 -setup -from [get_ports {din[1]}] -to [get_ports
{dout[1]}]
```

CONST_EXC_MCP6

NAME

CONST_EXC_MCP6 (CONSTRAINT_SIGNOFF)

MESSAGE

Multi-cycle path setup/hold multiplier is over/under defined.

DEFAULT SEVERITY

Warning

DESCRIPTION

Indicates the multiplier inconsistency of a multi-cycle path between setup and hold corners. Setting the attribute [ltms_const_multicycle_path_multiplier_difference](#) configures the difference between setup and hold multipliers of a multi-cycle path.

- `at_least_one` (default) : Setup multiplier is at least one greater than hold multiplier
- `one` : Setup multiplier is exactly one greater than hold multiplier

This rule is checked by [verify_timing_intent](#) command and the results can be reported by [report_rule_check](#) command.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
scope	string	Scope
exc_type	string	Exception type
setup_exc_from	object_list	Setup exception from

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setup_exc_to	object_list	Setup exception to
setup_exc_through	object_list	setup_exc_through
setup_exc_from_clock	object_list	setup_exc_from_clock
setup_exc_to_clock	object_list	setup_exc_to_clock
hold_exc_from	object_list	Hold exception from
hold_exc_to	object_list	Hold exception to
hold_exc_through	object_list	hold_exc_through
hold_exc_from_clock	object_list	hold_exc_from_clock
hold_exc_to_clock	object_list	hold_exc_to_clock
setup_cmd_file_line	unsigned	Setup command file line number
setup_cmd_file_name	string	Setup command file name
hold_cmd_file_line	unsigned	Hold command file line number
hold_cmd_file_name	string	hold_cmd_file_name
setup_val	float	Setup value
hold_val	float	Hold value
is_threshold_one	bool	is_threshold_one
rise_fall_option	string	Rise fall option

EXAMPLE

- The following example contains multi-cycle path constraints. This rule is reported for the multi-cycle path, because the setup multiplier is not at least one greater than the hold multiplier.

```
set_multicycle_path 2 -setup -from [get_ports din] -to [get_ports dout]  
set_multicycle_path 2 -hold -from [get_ports din] -to [get_ports dout]
```

CONST_EXC_MCP7

NAME

CONST_EXC_MCP7 (CONSTRAINT_SIGNOFF)

MESSAGE

Multi-cycle path start/end is on the lower frequency clock

DEFAULT SEVERITY

Warning

DESCRIPTION

Indicates that the multiplier of a multi-cycle path specified by a `set_multicycle_path` command with either one of `from` (`-from`, `-rise_from` or `-fall_from`) and `to` (`-to`, `-rise_to` or `-fall_to`) options is associated with a slower clock in a multi-frequency design. This rule is checked by `verify_timing_intent` command and the results can be reported by `report_rule_check` command.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
scope	string	Scope
exc_type	string	Exception type
exc_from	object_list	Exception from
exc_to	object_list	Exception to
exc_through	object_list	Exception through
exc_from_clock	object_list	Exception from clock

exc_to_clock	object_list	Exception to clock
start_clocks	object_list	Start clocks
end_clocks	object_list	End clocks
file_line	unsigned	File line number
file_name	string	File name

EXAMPLE

- The following example contains a multi-cycle path constraint. This rule is reported for the `set_multicycle_path` command, because the multiplier is associated with the start clock `vclk0` which is slower than the end clock `vclk1`.

```
create_clock -name vclk0 -period 10
create_clock -name vclk1 -period 5
set_input_delay -clock [get_clocks vclk0] 1 [get_ports din]
set_output_delay -clock [get_clocks vclk1] 1 [get_ports dout]
set_multicycle_path 5 -start -from [get_ports din] -to [get_ports dout]
```

CONST_EXC_MCP8

NAME

CONST_EXC_MCP8 (CONSTRAINT_SIGNOFF)

MESSAGE

Start/end point is specified in multi-cycle path `-through`

DEFAULT SEVERITY

Ignore (default Warning)

DESCRIPTION

Indicates that a start point or an end point is specified with `-through`, `-rise_through` or `-fall_through` option of `set_multicycle_path` command. It is recommended to use `-from`, `-rise_from` or `-fall_from` option for a start point and `-to`, `-rise_to` or `-fall_to` for an end point. The multi-cycle path specified with `from` (`-from`, `-rise_from` or `-fall_from`) or `to` (`-to`, `-rise_to` or `-fall_to`) option associated with a clock is skipped from this check. This rule is disabled by default. Using `set_message` command the severity can be modified to enable the rule. This rule is checked by `verify_timing_intent` command and the results can be reported by `report_rule_check` command.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
scope	string	Scope
exc_type	string	Exception type
exc_from	object_list	Exception from
exc_to	object_list	Exception to

exc_through	object_list	Exception through
from_ports	object_list	From ports
to_ports	object_list	To ports
from_to_ports	object_list	From to ports
file_line	unsigned	File line number
file_name	string	File name

EXAMPLE

- The following example contains multi-cycle path constraints. This rule is reported for the 1st `set_multicycle_path` command, because the start point `din` is specified with `-through` option. This check is skipped for the 2nd `set_multicycle_path` command, because it is specified with `-from` option associated with a clock.

```
create_clock -name vclk -period 2
set_input_delay -clock [get_clocks vclk] 1 [get_ports din]
set_output_delay -clock [get_clocks vclk] 1 [get_ports dout]
set_multicycle_path 2 -through [get_ports din]
set_multicycle_path 2 -from [get_clocks vclk] -through [get_ports din]
```

CONST_EXC_OLP

- [CONST_EXC_OLP1a](#)
- [CONST_EXC_OLP1b](#)
- [CONST_EXC_OLP1c](#)
- [CONST_EXC_OLP1d](#)
- [CONST_EXC_OLP1e](#)
- [CONST_EXC_OLP1f](#)
- [CONST_EXC_OLP2](#)
- [CONST_EXC_OLP3](#)

CONST_EXC_OLP1a

NAME

CONST_EXC_OLP1a (CONSTRAINT_SIGNOFF)

MESSAGE

A `set_false_path` and a `set_multicycle_path` command match the same timing path

DEFAULT SEVERITY

Warning

DESCRIPTION

Reports any two timing exception commands, one `set_false_path` and one `set_multicycle_path`, that apply to a common timing path. Note that this message does not imply that any of the two reported exceptions are necessarily redundant. This rule is checked by `verify_timing_intent` command and results can be reported by `report_rule_check` command.

This rule has specific rule occurrence attributes for the details about the violation.

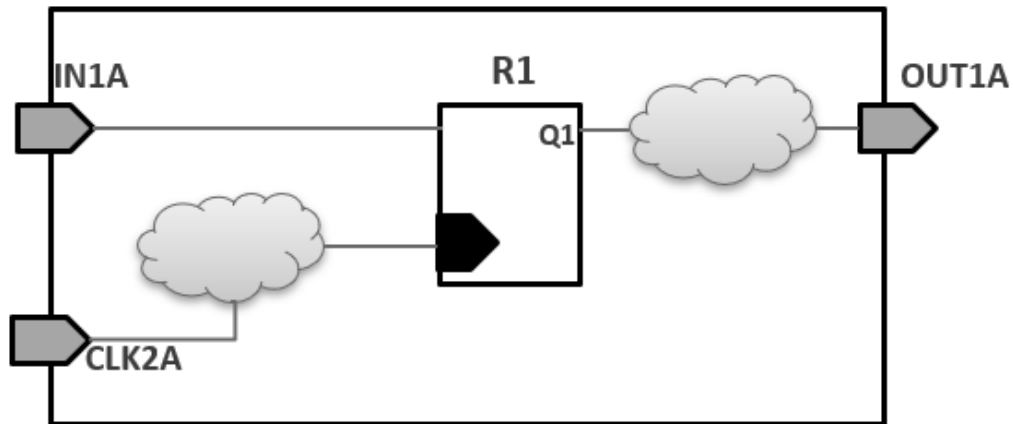
Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
scope	string	Scope
winning_exc_type	object_list	Winning exception type
winning_exc_file_line	object_list	Winning exception file line number
winning_exc_file_name	object_list	Winning exception file name
winning_exc_from	object_list	Winning exception from
winning_exc_to	object_list	Winning exception to

winning_exc_through	object_list	Winning exception through
ignored_exc_type	string	Ignored exception type
ignored_exc_file_line	unsigned	Ignored exception file line number
ignored_exc_file_name	string	Ignored exception file name
ignored_exc_from	object_list	Ignored exception from
ignored_exc_to	object_list	Ignored exception to
ignored_exc_through	object_list	Ignored exception through
path_startpoint	object_list	Path startpoint
path_endpoint	object_list	Path endpoint
launch_clocks	object_list	launch_clocks
capture_clocks	object_list	capture_clocks
winning_exc_from_clock	object_list	Winning exception from clock
winning_exc_to_clock	object_list	Winning exception to clock
ignored_exc_from_clock	object_list	Ignored exception from clock
ignored_exc_to_clock	object_list	Ignored exception to clock

EXAMPLE

- In the figure below, with the following constraints both `set_false_path` and `set_multicycle_path` are on the same path. Litmus identifies this as a warning.

Block A



```
create_clock -name rCLK1 -period 2 {CLK2A}
create_clock -name vCLK1 -period 2
set_input_delay -clock [get_clocks vCLK1] 1 [get_ports {IN1A}]
set_false_path -from [get_ports {IN1A}]
set_multicycle_path 2 -to [get_cells R1]
```

CONST_EXC_OLP1b

NAME

CONST_EXC_OLP1b (CONSTRAINT_SIGNOFF)

MESSAGE

A `set_false_path` and a `set_max/min_delay` command match the same timing path

DEFAULT SEVERITY

Warning

DESCRIPTION

Reports any two timing exception commands, one `set_false_path` and one `set_max_delay/set_min_delay`, that apply to a common timing path. Note that this message does not imply that any of the two reported exceptions are necessarily redundant. This rule is checked by `verify_timing_intent` command and results can be reported by `report_rule_check` command.

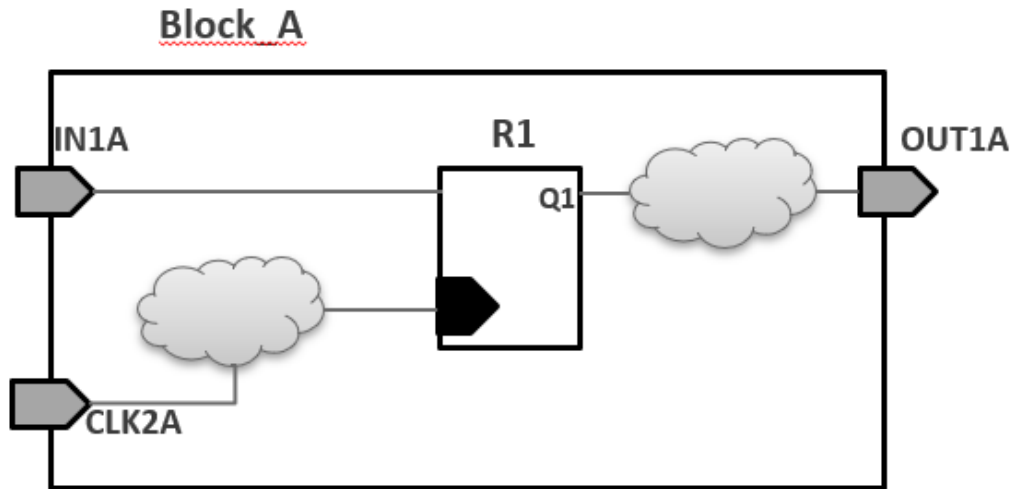
This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
scope	string	Design Name
winning_exc_type	object_list	winning_exc_type
winning_exc_file_line	object_list	winning_exc_file_line
winning_exc_file_name	object_list	winning_exc_file_name
winning_exc_from	object_list	winning_exc_from
winning_exc_to	object_list	winning_exc_to

winning_exc_through	object_list	winning_exc_through
ignored_exc_type	string	ignored_exc_type
ignored_exc_file_line	unsigned	ignored_exc_file_line
ignored_exc_file_name	string	ignored_exc_file_name
ignored_exc_from	object_list	ignored_exc_from
ignored_exc_to	object_list	ignored_exc_to
ignored_exc_through	object_list	ignored_exc_through
path_startpoint	object_list	Path start point
path_endpoint	object_list	Path end point
launch_clocks	object_list	launch_clocks
capture_clocks	object_list	capture_clocks
winning_exc_from_clock	object_list	winning_exc_from_clk
winning_exc_to_clock	object_list	winning_exc_to_clk
ignored_exc_from_clock	object_list	ignored_exc_from_clk
ignored_exc_to_clock	object_list	ignored_exc_to_clk

EXAMPLE

- In the figure below, with the following constraints both `set_false_path` and `set_max_delay` are on the same path. Litmus identifies this as a warning.



```
create_clock -name rCLK1 -period 2 {CLK2A}
create_clock -name vCLK1 -period 2
set_input_delay -clock [get_clocks vCLK1] 1 [get_ports {IN1A}]
set_false_path -from [get_ports {IN1A}]
set_max_delay 2 -to [get_cells R1]
```

CONST_EXC_OLP1c

NAME

CONST_EXC_OLP1c (CONSTRAINT_SIGNOFF)

MESSAGE

Delay path and multi-cycle path exceptions match the same timing path

DEFAULT SEVERITY

Warning

DESCRIPTION

Delay path and multi-cycle path exceptions match the same timing path.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence Id
scope	string	Design Name
winning_exc_type	object_list	winning_exc_type
winning_exc_file_line	object_list	winning_exc_file_line
winning_exc_file_name	object_list	winning_exc_file_name
winning_exc_from	object_list	winning_exc_from
winning_exc_to	object_list	winning_exc_to
winning_exc_through	object_list	winning_exc_through
ignored_exc_type	string	ignored_exc_type
ignored_exc_file_line	unsigned	ignored_exc_file_line

ignored_exc_file_name	string	ignored_exc_file_name
ignored_exc_from	object_list	ignored_exc_from
ignored_exc_to	object_list	ignored_exc_to
ignored_exc_through	object_list	ignored_exc_through
path_startpoint	object_list	start_point
path_endpoint	object_list	end_point
launch_clocks	object_list	launch_clocks
capture_clocks	object_list	capture_clocks
winning_exc_from_clock	object_list	winning_exc_from_clk
winning_exc_to_clock	object_list	winning_exc_to_clk
ignored_exc_from_clock	object_list	ignored_exc_from_clk
ignored_exc_to_clock	object_list	ignored_exc_to_clk

CONST_EXC_OLP1d

NAME

CONST_EXC_OLP1d (CONSTRAINT_SIGNOFF)

MESSAGE

Two `set_false_path` commands match the same timing path

DEFAULT SEVERITY

Warning

DESCRIPTION

Reports any two `set_false_path` commands that apply to a common timing path. Note that this message does not imply that any of the two reported exceptions are necessarily redundant. This rule is checked by `verify_timing_intent` command and results can be reported by `report_rule_check` command.

This rule has specific rule occurrence attributes for the details about the violation.

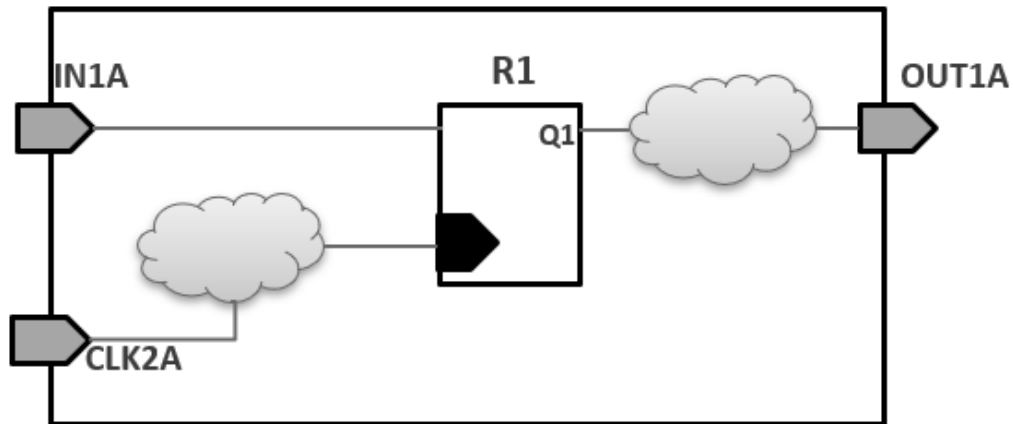
Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
scope	string	Scope
winning_exc_type	object_list	Winning exception type
winning_exc_file_line	object_list	Winning exception file line number
winning_exc_file_name	object_list	Winning exception file name
winning_exc_from	object_list	Winning exception from
winning_exc_to	object_list	Winning exception to

winning_exc_through	object_list	Winning exception through
ignored_exc_type	string	Ignored exception type
ignored_exc_file_line	unsigned	Ignored exception file line number
ignored_exc_file_name	string	Ignored exception file name
ignored_exc_from	object_list	Ignored exception from
ignored_exc_to	object_list	Ignored exception to
ignored_exc_through	object_list	Ignored exception through
path_startpoint	object_list	Path startpoint
path_endpoint	object_list	Path endpoint
launch_clocks	object_list	launch_clocks
capture_clocks	object_list	capture_clocks
winning_exc_from_clock	object_list	Winning exception from clock
winning_exc_to_clock	object_list	Winning exception to clock
ignored_exc_from_clock	object_list	Ignored exception from clock
ignored_exc_to_clock	object_list	Ignored exception to clock

EXAMPLE

- For the figure below, the following commands cause this rule check violation:

Block A



```
create_clock -name rCLK1 -period 2 {CLK2A}
create_clock -name vCLK1 -period 2
set_input_delay -clock [get_clocks vCLK1] 1 [get_ports {IN1A}]
set_false_path -from IN1A
set_false_path -to [get_cells R1]
```

CONST_EXC_OLP1e

NAME

CONST_EXC_OLP1e (CONSTRAINT_SIGNOFF)

MESSAGE

Two `set_multicycle_path` commands match the same timing path

DEFAULT SEVERITY

Warning

DESCRIPTION

Reports any two `set_multicycle_path` commands that apply to a common timing path. Note that this message does not imply that any of the two reported exceptions are necessarily redundant. This rule is checked by `verify_timing_intent` command and results can be reported by `report_rule_check` command.

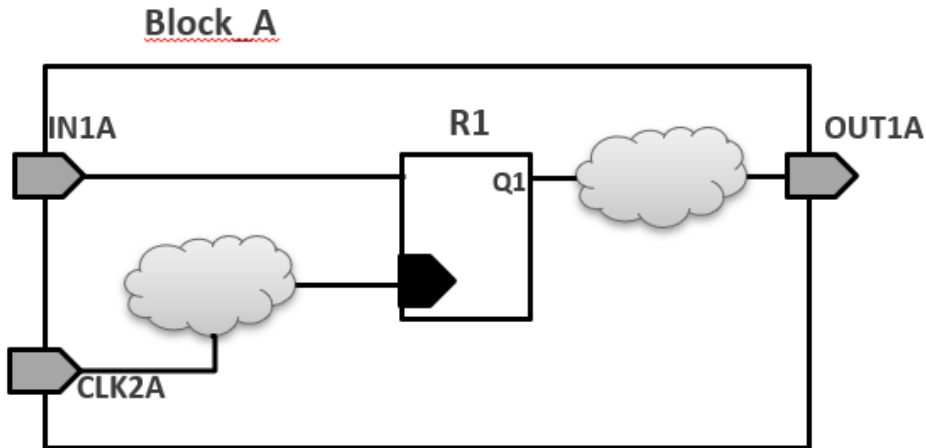
This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
scope	string	Scope
winning_exc_type	object_list	Winning exception type
winning_exc_file_line	object_list	Winning exception file line number
winning_exc_file_name	object_list	Winning exception file name
winning_exc_from	object_list	Winning exception from
winning_exc_to	object_list	Winning exception to

winning_exc_through	object_list	Winning exception through
ignored_exc_type	string	Ignored exception type
ignored_exc_file_line	unsigned	Ignored exception file line number
ignored_exc_file_name	string	Ignored exception file name
ignored_exc_from	object_list	Ignored exception from
ignored_exc_to	object_list	Ignored exception to
ignored_exc_through	object_list	Ignored exception through
path_startpoint	object_list	Path startpoint
path_endpoint	object_list	Path endpoint
launch_clocks	object_list	launch_clocks
capture_clocks	object_list	capture_clocks
winning_exc_from_clock	object_list	Winning exception from clock
winning_exc_to_clock	object_list	Winning exception to clock
ignored_exc_from_clock	object_list	Ignored exception from clock
ignored_exc_to_clock	object_list	Ignored exception to clock

EXAMPLE

- For the figure below, the following commands cause this rule check violation:



```
create_clock -name rCLK1 -period 2 {CLK2A}
create_clock -name vCLK1 -period 2
set_input_delay -clock [get_clocks vCLK1] 1 [get_ports {IN1A}]
set_multicycle_path 3 -from IN1A
set_multicycle_path 2 -to [get_cells R1]
```

CONST_EXC_OLP1f

NAME

CONST_EXC_OLP1f (CONSTRAINT_SIGNOFF)

MESSAGE

Two set_max/min_delay commands match the same timing path

DEFAULT SEVERITY

Warning

DESCRIPTION

Reports any two `set_max_delay/set_min_delay` commands that apply to a common timing path. Note that this message does not imply that any of the two reported exceptions are necessarily redundant. This rule is checked by `verify_timing_intent` command and results can be reported by `report_rule_check` command.

This rule has specific rule occurrence attributes for the details about the violation.

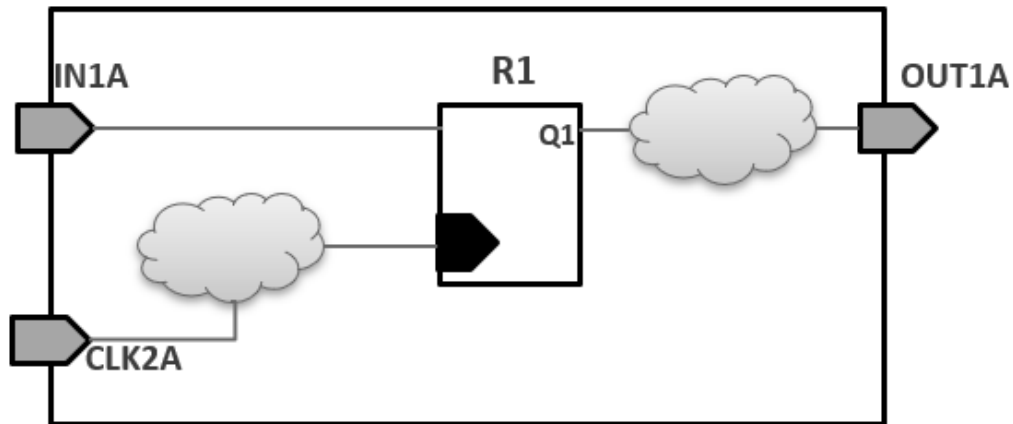
Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
scope	string	Scope
winning_exc_type	object_list	Winning exception type
winning_exc_file_line	object_list	Winning exception file line number
winning_exc_file_name	object_list	Winning exception file name
winning_exc_from	object_list	Winning exception from
winning_exc_to	object_list	Winning exception to

winning_exc_through	object_list	Winning exception through
ignored_exc_type	string	Ignored exception type
ignored_exc_file_line	unsigned	Ignored exception file line number
ignored_exc_file_name	string	Ignored exception file name
ignored_exc_from	object_list	Ignored exception from
ignored_exc_to	object_list	Ignored exception to
ignored_exc_through	object_list	Ignored exception through
path_startpoint	object_list	Path startpoint
path_endpoint	object_list	Path endpoint
launch_clocks	object_list	launch_clocks
capture_clocks	object_list	capture_clocks
winning_exc_from_clock	object_list	Winning exception from clock
winning_exc_to_clock	object_list	Winning exception to clock
ignored_exc_from_clock	object_list	Ignored exception from clock
ignored_exc_to_clock	object_list	Ignored exception to clock

EXAMPLE

- For the figure below, the following commands cause this rule check violation:

Block A



```
create_clock -name rCLK1 -period 2 {CLK2A}
create_clock -name vCLK1 -period 2
set_input_delay -clock [get_clocks vCLK1] 1 [get_ports {IN1A}]
set_max_delay 2 -from IN1A
set_max_delay 2 -to [get_cells R1]
```

CONST_EXC_OLP2

NAME

CONST_EXC_OLP2 (CONSTRAINT_SIGNOFF)

MESSAGE

An exception is overridden on all timing paths by exceptions of higher priority

DEFAULT SEVERITY

Warning

DESCRIPTION

Reports any timing exception command such that for every matching timing path that is not tied to a constant value, there is a different timing exception with higher priority that matches the same path. Note that this message is a strong indicator that the reported exception could be redundant. However, the results of this check are correct only if the design's timing paths are constrained properly, and if the check runs to completion. One should exercise care when deciding to remove a timing exception reported by this rule. This rule is checked by `verify_timing_intent` command and results can be reported by `report_rule_check` command.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
scope	string	Scope
overriding_exc_type	object_list	overriding_exc_type
overriding_exc_file_line	object_list	overriding_exc_file_line
overriding_exc_file_name	object_list	overriding_exc_file_name
overriding_exc_from	object_list	overriding_exc_from

overriding_exc_to	object_list	overriding_exc_to
overriding_exc_through	object_list	overriding_exc_through
overridden_exc_type	string	Overridden exception type
overridden_exc_file_line	unsigned	Overridden exception file line number
overridden_exc_file_name	string	Overridden exception file name
overridden_exc_from	object_list	Overridden exception from
overridden_exc_to	object_list	Overridden exception to
overridden_exc_through	object_list	Overridden exception through
path_startpoint	object_list	Path startpoint
path_endpoint	object_list	Path endpoint
launch_clocks	object_list	launch_clocks
capture_clocks	object_list	capture_clocks
winning_exc_from_clock	object_list	Winning exception from clock
winning_exc_to_clock	object_list	Winning exception to clock
ignored_exc_from_clock	object_list	Ignored exception from clock
ignored_exc_to_clock	object_list	Ignored exception to clock

EXAMPLE

- If there is a path from r_c_reg to x_reg that traverses the net mul, this rule reports the first of the following commands:

```
set_multicycle_path -through [get_nets mul] -setup 2
set_false_path -from [get_cells r_c_reg]
set_false_path -to [get_cells x_reg]
```

Note that the last two commands have the same meaning for the path mentioned above, so none of them will be reported by this rule.

CONST_EXC_OLP3

NAME

CONST_EXC_OLP3 (CONSTRAINT_SIGNOFF)

MESSAGE

A `set_max_delay | set_min_delay` exception overrides either a synchronous path or an asynchronous path explicitly timed with `-allow_paths`.

DEFAULT SEVERITY

Warning

DESCRIPTION

Reports `set_max_delay | set_min_delay` exception commands that override a valid constrained path. The path could either be a regular synchronous path. Note that in either cases it is possible that the `set_max_delay | set_min_delay` constraint was intentionally applied. Please review your constraints carefully before removing any `set_max_delay | set_min_delay` based on this rule violation. This rule is checked by `verify_timing_intent` command and results can be reported by `report_rule_check` command.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
scope	string	Scope
overriding_exc_type	string	overriding_exc_type
overriding_exc_file_line	unsigned	overriding_exc_file_line
overriding_exc_file_name	string	overriding_exc_file_name

Litmus Rule Reference
Timing Constraint Signoff--CONST_EXC_OLP

overriding_exc_from	object_list	overriding_exc_from
overriding_exc_to	object_list	overriding_exc_to
overriding_exc_through	object_list	overriding_exc_through
from_clock	object_list	from_clock
to_clock	object_list	to_clock
path_startpoint	object_list	Path startpoint
path_endpoint	object_list	Path endpoint
launch_clocks	object_list	launch_clocks
capture_clocks	object_list	capture_clocks

CONST_EXC_SMD

- [CONST_EXC_SMD1](#)
- [CONST_EXC_SMD2](#)
- [CONST_EXC_SMD3](#)
- [CONST_EXC_SMD4](#)
- [CONST_EXC_SMD5](#)
- [CONST_EXC_SMD6](#)
- [CONST_EXC_SMD7](#)

CONST_EXC_SMD1

NAME

CONST_EXC_SMD1 (CONSTRAINT_SIGNOFF)

MESSAGE

`set_max_delay` exception does not match any timing path

DEFAULT SEVERITY

Warning

DESCRIPTION

Indicates that a `set_max_delay` command does not match any timing path, or any object specified in `set_max_delay` command has no matching path. This rule is checked by `verify_timing_intent` command and results can be reported by `report_rule_check` command.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
scope	string	Scope
exc_type	string	Exception type
exc_from	object_list	Exception from
exc_to	object_list	Exception to
exc_through	object_list	Exception through
no_matching_path_from_edge	string	No matching clock -from edge
no_matching_path_from	object_list	No matching clock -from

no_matching_path_to_edge	string	No matching path -to edge
no_matching_path_to	object_list	No matching path -to
no_matching_path_through_edges	object_list	No matching path -through edges
no_matching_path_through	object_list	No matching path -through
file_line	unsigned	File line number
file_name	string	File name
type	string	Type

In addition, common rule occurrence attribute `type` provides the possible causes for the violation.

Attribute	Value	Description
type	NO_MATCHING_PATH	No matching path for the path spec
	PARTIALLY_NO_MATCHING_PATH	No matching path for part of the objects in the path spec

EXAMPLE

- The following example contains two timing paths and delay path constraints. This rule is reported for the 1st `set_max_delay` command, because it does not match any timing path. This rule is also reported for input port `din[0]` specified in the 2nd `set_max_delay` command, because it has no matching path.

```
module top (input [1:0] din, output [1:0] dout);  
    assign dout = din;  
endmodule
```

Constraints that go with the above RTL:

```
create_clock -name vclk -period 2  
set_input_delay -clock [get_clocks vclk] 1 [get_ports din]  
set_output_delay -clock [get_clocks vclk] 1 [get_ports dout]  
set_max_delay 1 -from [get_ports {din[1]}] -to [get_ports {dout[0]}]  
set_max_delay 1 -from [get_ports {din[0] din[1]}] -to [get_ports {dout[1]}]
```


CONST_EXC_SMD2

NAME

CONST_EXC_SMD2 (CONSTRAINT_SIGNOFF)

MESSAGE

`set_min_delay` exception does not match any timing path

DEFAULT SEVERITY

Warning

DESCRIPTION

Indicates that a `set_min_delay` command does not match any timing path, or any object specified in `set_min_delay` command has no matching path. This rule is checked by `verify_timing_intent` command and results can be reported by `report_rule_check` command.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
scope	string	Scope
exc_type	string	Exception type
exc_from	object_list	Exception from
exc_to	object_list	Exception to
exc_through	object_list	Exception through
no_matching_path_from_edge	string	No matching clock -from edge

no_matching_path_from	object_list	No matching clock -from
no_matching_path_to_edge	string	No matching path -to edge
no_matching_path_to	object_list	No matching path -to
no_matching_path_through_edges	object_list	No matching path -through edges
no_matching_path_through	object_list	No matching path -through
file_line	unsigned	File line number
file_name	string	File name
type	string	Type

In addition, common rule occurrence attribute `type` provides the possible causes for the violation.

Attribute	Value	Description
type	NO_MATCHING_PATH	No matching path for the path spec
	PARTIALLY_NO_MATCHING_PATH	No matching path for part of the objects in the path spec

EXAMPLE

- The following example contains two timing paths and delay path constraints. This rule is reported for the 1st `set_min_delay` command, because it does not match any timing path. This rule is also reported for input port `din[0]` specified in the 2nd `set_min_delay` command, because it has no matching path.

```
module top (input [1:0] din, output [1:0] dout);  
    assign dout = din;  
endmodule
```

Constraints that go with the above RTL:

```
create_clock -name vclk -period 2  
set_input_delay -clock [get_clocks vclk] 1 [get_ports din]  
set_output_delay -clock [get_clocks vclk] 1 [get_ports dout]  
set_min_delay 0.5 -from [get_ports {din[1]}] -to [get_ports {dout[0]}]  
set_min_delay 0.5 -from [get_ports {din[0] din[1]}] -to [get_ports {dout[1]}]
```

CONST_EXC_SMD3

NAME

CONST_EXC_SMD3 (CONSTRAINT_SIGNOFF)

MESSAGE

Incomplete set_min/max_delay options

DEFAULT SEVERITY

Warning

DESCRIPTION

Indicates that a `set_max_delay` or `set_min_delay` has been specified for `-rise` but not for `-fall`, or vice versa. This rule is checked by `verify_timing_intent` command and results can be reported by `report_rule_check` command.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
scope	string	Design Name
start_point	object	start_point
end_point	object	end_point
min_max_option	string	min_max_option
rise_fall_option	string	Value of -rise/ -fall option
file_line	unsigned	File Line

file_name	string	File Name

EXAMPLE

- For the following:

```
set_max_delay 5 -rise -from [get_ports in1] -to [get_ports out1]
set_min_delay 6 -fall -from [get_ports in2]
set_min_delay 5 -rise -to [get_ports out2]
```

The Litmus generates warnings if the following commands are missing:

```
set_max_delay <value> -fall -from [get_ports in1] -to [get_ports out1]
set_min_delay <value> -rise -from [get_ports in2]
set_min_delay <value> -fall -to [get_ports out2]
```

CONST_EXC_SMD4

NAME

CONST_EXC_SMD4

MESSAGE

set_max/min_delay exception is ignored

DEFAULT SEVERITY

Warning

DESCRIPTION

Indicates that a path delay specified by `set_max_delay` or `set_min_delay` command without `-to` option is ignored for the matching path ending at a clock pin. This rule is checked by `verify_timing_intent` command, and the results can be reported by `report_rule_check` command.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
scope	string	Scope
exc_type	string	Exception type
exc_from	object_list	Exception from
exc_to	object_list	Exception to
exc_through	object_list	Exception through
start_point	object	Start point
end_point	object	End point

type	string	type
file_line	unsigned	File line number
file_name	string	File name

EXAMPLE

- The following contains a path delay constraint. This rule is reported for the `set_max_delay` command specified without `-to` option, because it is ignored for the matching path ending at the clock pin `dout_reg/CK`.

```
module top (input clk, input din, output reg dout);
    always @(posedge clk)
        dout <= din;
endmodule
```

Constraints that go with the above RTL:

```
create_clock -name clk -period 2 [get_ports clk]
set_max_delay 4 -from [get_ports clk]
```

CONST_EXC_SMD5

NAME

CONST_EXC_SMD5 (CONSTRAINT_SIGNOFF)

MESSAGE

Start/end point is specified in `set_max_delay/set_min_delay -through`

DEFAULT SEVERITY

Ignore (default Warning)

DESCRIPTION

Indicates that a start point or an end point is specified with `-through`, `-rise_through` or `-fall_through` option of `set_max_delay` or `set_min_delay` command. It is recommended to use `-from`, `-rise_from` or `-fall_from` option for a start point and `-to`, `-rise_to` or `-fall_to` for an end point. The path delay specified with `from` or `to` option associated with a clock is skipped from this check. This rule is disabled by default. Using `set_message` command the severity can be modified to enable the rule. This rule is checked by `verify_timing_intent` command and the results can be reported by `report_rule_check` command.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
scope	string	Scope
exc_type	string	Exception type
exc_from	object_list	Exception from
exc_to	object_list	Exception to

exc_through	object_list	Exception through
from_ports	object_list	From ports
to_ports	object_list	To ports
from_to_ports	object_list	From to ports
file_line	unsigned	File line number
file_name	string	File name

EXAMPLE

- The following example contains path delay constraints. This rule is reported for the `set_max_delay` and `set_min_delay` commands, because the start point `din` is specified with `-through` option.

```
create_clock -name vclk -period 2
set_input_delay -clock [get_clocks vclk] 1 [get_ports din]
set_output_delay -clock [get_clocks vclk] 1 [get_ports dout]
set_max_delay 4 -through [get_ports din]
set_min_delay 2 -through [get_ports din]
```

CONST_EXC_SMD6

NAME

CONST_EXC_SMD6 (CONSTRAINT_SIGNOFF)

MESSAGE

Inconsistent input/output delay value vs. set_min_delay/set_max_delay delay value

DEFAULT SEVERITY

Warning

DESCRIPTION

Inconsistent input/output delay value vs. set_min_delay/set_max_delay delay value.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence Id
scope	string	Design Name
from_clock	object	from_clock
to_clock	object	to_clock
path_startpoint	object	path_startpoint
path_endpoint	object	path_endpoint
file_line	unsigned	File Line
file_name	string	File Name

CONST_EXC_SMD7

NAME

CONST_EXC_SMD7 (CONSTRAINT_SIGNOFF)

MESSAGE

Incomplete set_min_delay/set_max_delay exception

DEFAULT SEVERITY

Warning

DESCRIPTION

Inconsistent input/output delay value vs. set_min_delay/set_max_delay delay value

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence Id
scope	string	Design Name
start_point	object	start_point
end_point	object	end_point
missing_corner	object_list	missing_corner

CONST_HFN_CHK

- [CONST_HFN_CHK1a](#)
- [CONST_HFN_CHK1b](#)
- [CONST_HFN_CHK2](#)
- [CONST_HFN_CHK3](#)

CONST_HFN_CHK1a

NAME

CONST_HFN_CHK1a (CONSTRAINT_SIGNOFF)

MESSAGE

Clock is not set as `set_dont_touch_network`

DEFAULT SEVERITY

Warning

DESCRIPTION

Indicates that a real or generated clock does not have a `set_dont_touch_network` and it is also missing from one of its source ports/pins. This rule is checked by `verify_timing_intent` command and results can be reported by `report_rule_check` command.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
scope	string	Design Name
clock_source	object	Clock source without <code>set_dont_touch_network</code>
file_line	unsigned	File Line
file_name	string	File Name

EXAMPLE

- This rule will be reported if the following `set_dont_touch_network` command is missing for

clock port CLK3:

```
set_dont_touch_network CLK3
```

CONST_HFN_CHK1b

NAME

CONST_HFN_CHK1b (CONSTRAINT_SIGNOFF)

MESSAGE

Set/reset is not set as `set_dont_touch_network`

DEFAULT SEVERITY

Warning

DESCRIPTION

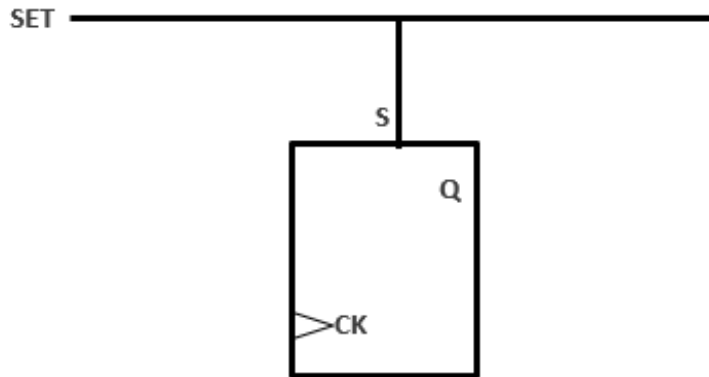
Indicates that a port or pin that drives sequential element set or reset pins does not have a `set_dont_touch_network` command. This rule is checked by `verify_timing_intent` command and results can be reported by `report_rule_check` command.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
scope	string	Design Name
pin	object	pin
set_reset	object	Set/reset pinset_dont_touch_network
file_line	unsigned	File Line
file_name	string	File Name

EXAMPLE

- For the figure below, this rule will be reported if the SDC file does not contain a command like the following:



```
set_dont_touch_network [get_ports SET]
```


CONST_HFN_CHK2

NAME

CONST_HFN_CHK2 (CONSTRAINT_SIGNOFF)

MESSAGE

Object with [set_dont_touch_network](#) is not a clock, set or reset

DEFAULT SEVERITY

Warning

DESCRIPTION

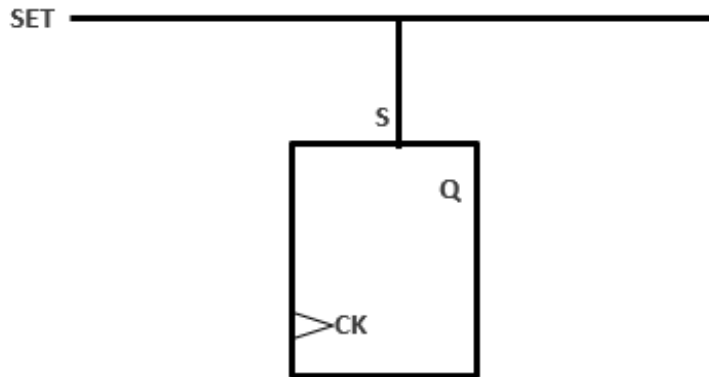
Indicates that a port or pin specified in [set_dont_touch_network](#) does not drive a sequential element clock, set or reset pins. Note that when checking if the specified port or pin drives a clock, set or reset pin, propagated constants are taking into account for paths leading to clock pins, and they are ignored for paths leading to set or reset pins. This rule is checked by [verify_timing_intent](#) command and results can be reported by [report_rule_check](#) command.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
scope	string	Design Name
pin	object	pin
file_line	unsigned	File Line
file_name	string	File Name

EXAMPLE

- For the figure below, this rule will be reported if the SDC file contains a command like the following where port IN is not connected to a sequential element clock, set or reset pins:



```
set_dont_touch_network [get_ports IN]
```

CONST_HFN_CHK3

NAME

CONST_HFN_CHK3 (CONSTRAINT_SIGNOFF)

MESSAGE

Inconsistent ideal transition values

DEFAULT SEVERITY

Warning

DESCRIPTION

Indicates that an ideal transition value specified in [set_ideal_transition](#) for maximum delay analysis is smaller than the corresponding value specified for minimum delay analysis. This rule is checked by [verify_timing_intent](#) command and results can be reported by [report_rule_check](#) command.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
scope	string	Design Name
port_pin	object	Port
rise_fall_option	string	Value of -rise/ -fall option
min_value	float	min_value
max_value	float	max_value

EXAMPLE

- In the following, the Litmus will report the two first commands as inconsistent for the `-fall` case:

```
set_ideal_transition 1.00 -max [get_ports {pi[3]}]
set_ideal_transition 2.00 -min -fall [get_ports {pi[3]}]
set_ideal_transition 0.50 -min -rise [get_ports {pi[3]}]
```

CONST_IO_DRC

- [CONST_IO_DRC1](#)
- [CONST_IO_DRC2](#)
- [CONST_IO_DRC3](#)

CONST_IO_DRC1

NAME

CONST_IO_DRC1 (CONSTRAINT_SIGNOFF)

MESSAGE

Undefined `set_max_capacitance` on input or inout port

DEFAULT SEVERITY

Warning

DESCRIPTION

Indicates that an input or inout port has not set `set_max_capacitance` . This rule is checked by `verify_timing_intent` command and results can be reported by `report_rule_check` command. The tool will report violation for both data ports as well as clock ports.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
scope	string	Scope
pin	object	Pin

EXAMPLE

- The rule issues this warning without something like the following for the input port in_port:

```
set_max_capacitance 0.5 [get_ports {in_port}]
```

CONST_IO_DRC2

NAME

CONST_IO_DRC2 (CONSTRAINT_SIGNOFF)

MESSAGE

Undefined `set_max_capacitance` on output or inout port

DEFAULT SEVERITY

Warning

DESCRIPTION

Indicates that an output or inout port has not set `set_max_capacitance` . This rule is checked by `verify_timing_intent` command and results can be reported by `report_rule_check` command. The tool will report violation for both data ports as well as clock ports.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
scope	string	Scope
pin	object	Pin

EXAMPLE

- The rule issues this warning without something like the following for the output port out_port:

```
set_max_capacitance 0.5 [get_ports {out_port}]
```

CONST_IO_DRC3

NAME

CONST_IO_DRC3 (CONSTRAINT_SIGNOFF)

MESSAGE

Total fanout load on a net that drives an output port exceeds the driver `max_fanout` attribute

DEFAULT SEVERITY

Warning

DESCRIPTION

Indicates that an output (or inout) port with `set_fanout_load` is driven by a port or pin whose `max_fanout` is smaller than the sum of the `fanout_load` attributes for all the loads connected to it. This rule is checked by `verify_timing_intent` command and results can be reported by `report_rule_check` command. Note that this rule can be diagnosed using the `diagnose_rule_check` command. The complete list of fanouts and their `fanout_load` attributes are displayed. A flattened schematic window shows the driver cell and expanding the displayed cell's fanout will show all the loads.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
scope	string	Design Name
driving_object	object	Driving port or pin
max_fanout_source	string	Maximum fanout source
max_fanout	float	Maximum fanout

loading_object	object	Loading pin/port
fanout_load_refs	object_list	List of fanout load references
fanout_loads	object_list	List of fanout loads
fanout_load_sources	object_list	List of fanout load sources
file_line	unsigned	File Line
file_name	string	File Name

EXAMPLE

- A cell pin I1/Z drives both the ports OUT and the cell pin I2/A. I1 is an instance of cell C1, whose Liberty description defines `max_fanout` of pin Z as 10. I2 is an instance of cell C2, that has no `fanout_load` defined in Liberty, but the library contains a default `fanout_load` of 1.0. If the SDC file contains the following command then this rule will be reported because the total fanout load is 11.0.:

```
set_fanout_load 10.0 [get_ports OUT]
```

CONST_IO_IDL

- [CONST_IO_IDL1](#)
- [CONST_IO_IDL4](#)
- [CONST_IO_IDL5](#)
- [CONST_IO_IDL6](#)
- [CONST_IO_IDL13](#)
- [CONST_IO_IDL14](#)
- [CONST_IO_IDL15](#)

CONST_IO_IDL1

NAME

CONST_IO_IDL1 (CONSTRAINT_SIGNOFF)

MESSAGE

Input is unconstrained

DEFAULT SEVERITY

Warning

DESCRIPTION

Indicates that there is an unconstrained input. Flags input ports that do not have `set_input_delay`. Only input ports that are connected to some constrained path end point (a clocked register or a pin with `set_output_delay`) are considered for this rule. However, a clock source will not be checked in this rule if it only drives clock pins of sequential elements. This rule is checked by `verify_timing_intent` command and results can be reported by `report_rule_check` command.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
scope	string	Scope
pin	object	Pin

EXAMPLE

- If the design has three input ports IN1, IN2, and IN3, the following produces a warning because the input delay for IN1 is missing:

```
set_input_delay 0.75 -rise -clock CLK1 {IN2}  
set_input_delay 0.75 -rise -clock CLK1 {IN3}
```

CONST_IO_IDL4

NAME

CONST_IO_IDL4 (CONSTRAINT_SIGNOFF)

MESSAGE

Incomplete input delay options

DEFAULT SEVERITY

Warning

DESCRIPTION

Indicates that there are incomplete input delay options. The rule checks `set_input_delay` to ensure that there are values for both or none of the following option combinations: `-rise/-fall` and `-min/-max`. This applies to every specified clock and for delays that are specified without a clock. This rule does not apply to ports that are specified in `set_false_path` and that have a single path specification switch, which is either `-from`, `-to`, or a single `-through`. Note that `set_false_path` should have both `-setup` and `-hold` specified, or neither. Otherwise, the port is still subject to this rule check. This rule is checked by `verify_timing_intent` command and results can be reported by `report_rule_check` command.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
scope	string	Scope
pin	object	Pin
clock	object	Clock
missing_corners	object_list	Missing corners

file_line	unsigned	File line number
file_name	string	File name

EXAMPLE

- The following command causes a rule check violation because it is missing the `-fall` option:

```
set_input_delay 0.75 -rise -clock CLK1 {IN2}
```

CONST_IO_IDL5

NAME

CONST_IO_IDL5 (CONSTRAINT_SIGNOFF)

MESSAGE

Inconsistent input delay vs. clock period, or `min>max`

DEFAULT SEVERITY

Warning

DESCRIPTION

Indicates one of the following conditions for `set_input_delay` values:

- `-min delay > -max delay`
- A delay value does not satisfy the formula specified in the attribute `ltms_const_input_delay_condition`

Note that the default formula is `delay<ratio*period`, where `delay` indicates `-min/-max delay`, `ratio` indicates the value of attribute `ltms_const_input_delay_ratio`, and `period` indicates the clock period. The attribute `ltms_const_input_delay_ratio` has a default value of 1.0. Also, if there is a `set_multicycle_path` with a single `-from` or `-through` list, in which the port is included, then the clock period is multiplied by the multi-cycle path value. Note that this rule does not apply to ports that are specified in `set_false_path` and that have a single path specification switch, which is either `-from` or a single `-through`. `set_false_path` should have both `-setup` and `-hold` specified, or neither. Otherwise, the port is still subject to this rule check. This rule is checked by `verify_timing_intent` command and the results can be reported by `report_rule_check` command.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
scope	string	Scope
pin	object	Pin
clock	object	Clock
is_clock_fall	bool	Is clock fall
max_min_option	string	Max/min option
rise_fall_option	string	Value of -rise/ -fall option
min_delay	float	Min delay value
max_delay	float	Max delay value
min_file_line	unsigned	Min file line number
min_file_name	string	Min file name
max_file_line	unsigned	Max file line number
max_file_name	string	Max file name
check_expression	string	Check expression

EXAMPLE

- In the following command, a violation occurs because 2.75 (minimum value) > 1.75 (maximum value):

```
set_input_delay 1.75 -max -clock CLK1_w {IN1}  
set_input_delay 2.75 -min -clock CLK1_w {IN1}
```

- In the following command, the Litmus issues a warning because the input delay does not satisfy the formula $\text{delay} < \text{ratio} * \text{period}$, where the attribute `ltms_const_input_delay_ratio` has its default value 1.0:


```
create_clock -period 8.0 -name vCLK3  
set_input_delay 10.0 [get_ports {IN1, IN2}] -clock vCLK3
```

CONST_IO_IDL6

NAME

CONST_IO_IDL6 (CONSTRAINT_SIGNOFF)

MESSAGE

Input delay constrained vs. wrong clock

DEFAULT SEVERITY

Warning

DESCRIPTION

For input ports that have input delay defined with respect to a clock, and that are connected to some constrained end point, this rule indicates that none of those end points is constrained by a clock that is synchronous to the one for which the input delay was defined. This rule does not apply to ports that are specified in `set_false_path` and that have a single path specification switch, which is either `-from`, `-to`, or a single `-through`. Note that `set_false_path` should have both `-setup` and `-hold` specified, or neither. Otherwise, the port is still subject to this rule check. This rule is checked by `verify_timing_intent` command and the results can be reported by `report_rule_check` command.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
scope	string	Design Name
clock	object	Clock definition
port_pin	object	ports
file_line	unsigned	File Line

file_name	string	File Name

EXAMPLE

- In the following example, if there is no path from IN1 to a sequential element constrained by CLK2, the Litmus issues a warning because `set_input_delay` has specified the wrong clock:

```
set_input_delay 2.75 -min -clock CLK2 {IN1}
```

CONST_IO_IDL13

NAME

CONST_IO_IDL13 (CONSTRAINT_SIGNOFF)

MESSAGE

An input port is in a timing path within a clock group for which it does not have input delay defined

DEFAULT SEVERITY

Warning

DESCRIPTION

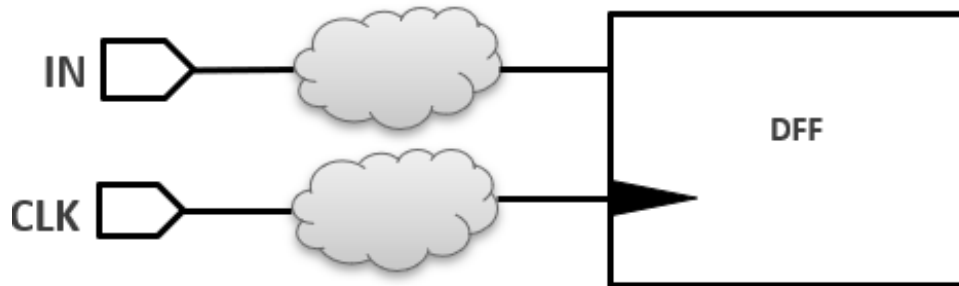
Indicates that there is an input port in whose fanout there are sequential elements or ports in a given clock group, but this input port has no input delay with respect to any clock in that clock group. This rule applies only to ports that are start points of a timing path that does not match any `set_false_path`. `set_false_path` should have both `-setup` and `-hold` specified, or neither. Otherwise, the port is still subject to this rule check. This rule does not check inputs that are clock sources. This rule is checked by `verify_timing_intent` command and the results can be reported by `report_rule_check` command. Note that this rule can be diagnosed using the `diagnose_rule_check` command.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
scope	string	Scope
port_pin	object	Port
capture_clocks	object_list	Capture clocks
sync_groups	object_list	sync_groups

EXAMPLE

- For the figure below, with the following commands, the Litmus checks that IN has an input delay with respect to CLK or any other clock in the same group. This example satisfies [CONST_IO_IDL1](#), however, this rule issues a warning because it requires a delay with respect to CLK or any clock synchronous to it.



```
create_clock -name rCLK1 -period 11 -waveform {0 5} {CLK}  
create_clock -name vCLK2 -period 10 -waveform {0 5}  
set_clock_groups -name grp1 -group {rCLK1} -group {vCLK2} -asynchronous  
set_input_delay -clock vCLK2 1.1 [get_ports {IN}]
```

CONST_IO_IDL14

NAME

CONST_IO_IDL14 (CONSTRAINT_SIGNOFF)

MESSAGE

An input port is relative to a non-virtual capture clock for which it does not have input delay defined.

DEFAULT SEVERITY

Warning

DESCRIPTION

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
scope	string	Design Name
port_pin	object	Port
capture_clocks	object_list	capture_clocks
clock_groups	object_list	clock_groups

CONST_IO_IDL15

NAME

CONST_IO_IDL15 (CONSTRAINT_SIGNOFF)

MESSAGE

An input delay is defined vs. a virtual clock on a port/pin, but no sequential element in its fanout is driven by a real/generated clock with the same waveform

DEFAULT SEVERITY

Warning

DESCRIPTION

Indicates that, for input ports or internal pins that have input delay defined with respect to a virtual clock, there are sequential elements in the fanout, but their clock pins are driven by real or generated clocks that have a different period or waveform edges than the virtual clock. This rule does not apply to ports that are specified in `set_false_path` and that have a single path specification switch, which is either `-from`, `-to`, or a single `-through`. Note that `set_false_path` should have both `-setup` and `-hold` specified, or neither. Otherwise, the port is still subject to this rule check. This rule is checked by `verify_timing_intent` command and the results can be reported by `report_rule_check` command.

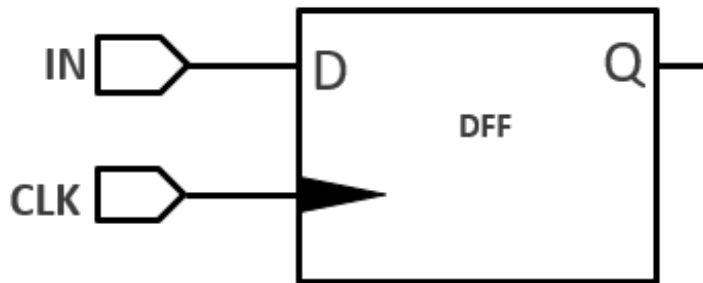
This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
scope	string	Design Name
clock	object	Clock definition
port_pin	object	Input port

file_line	unsigned	File Line
file_name	string	File Name

EXAMPLE

- For the figure below, with the following commands, the Litmus issues a warning because `set_input_delay` has specified a virtual clock whose frequency and waveform do not match that of any real clock driving a sequential element in the fanout of IN:



```
create_clock -name RCLK -period 10 -waveform {0 5} [get_ports CLK]
create_clock -name VCLK -period 20 -waveform {0 10}
set_input_delay 2.75 -clock VCLK {IN}
```


CONST_IO_ITR

- [CONST_IO_ITR1](#)
- [CONST_IO_ITR2](#)
- [CONST_IO_ITR3](#)
- [CONST_IO_ITR10](#)

CONST_IO_ITR1

NAME

CONST_IO_ITR1 (CONSTRAINT_SIGNOFF)

MESSAGE

Undefined input transition

DEFAULT SEVERITY

Warning

DESCRIPTION

Indicates that an input or inout port does not have `set_input_transition`, `set_driving_cell`, `set_drive`, or `set_load` specified. This rule check does not apply to ports that have `set_dont_touch_network` specified. This rule is checked by `verify_timing_intent` command and results can be reported by `report_rule_check` command.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
scope	string	Scope
pin	object	Pin

EXAMPLE

- This rule is reported for input port in1 if the following constraint is missing provided none of the `set_driving_cell`, `set_drive`, or `set_load` is specified on in1:

```
set_input_transition 1.0 [get_ports {in1}]
```

CONST_IO_ITR2

NAME

CONST_IO_ITR2 (CONSTRAINT_SIGNOFF)

MESSAGE

Incomplete input transition options

DEFAULT SEVERITY

Warning

DESCRIPTION

Indicates that there are incomplete input transition options for `set_input_transition`, `set_driving_cell` or `set_drive`. Both or none of the following option combinations: `-rise/-fall` and `-min/-max` should be specified. This rule is checked by `verify_timing_intent` command and results can be reported by `report_rule_check` command.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
scope	string	Scope
pin	object	Pin
missing_corners	object_list	Missing corners

EXAMPLE

- The following produces a warning because it specifies `-min`, but not `-max`:

```
set_input_transition 2.74 -min [get_ports {IN1}]
```

The -min and -max values can be specified in the same or separate commands.

CONST_IO_ITR3

NAME

CONST_IO_ITR3 (CONSTRAINT_SIGNOFF)

MESSAGE

Inconsistent input transition option values

DEFAULT SEVERITY

Warning

DESCRIPTION

Indicates that there are [set_input_transition](#) or [set_drive](#) options with inconsistent values. Specifically, the Litmus checks for instances where $\text{min} > \text{max}$. This rule is checked by [verify_timing_intent](#) command and results can be reported by [report_rule_check](#) command.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
scope	string	Scope
pin	object	Pin
rise_fall_option	string	Value of -rise/ -fall option
min_value	float	Min value
max_value	float	Max value
min_file_line	unsigned	Min file line number
min_file_name	string	Min file name

max_file_line	unsigned	Max file line number
max_file_name	string	Max file name

EXAMPLE

- The following generates a warning because `-min > -max` for IN1:

```
set_input_transition -rise -min 0.75 IN1
set_input_transition -rise -max 0.50 IN1
```

CONST_IO_ITR10

NAME

CONST_IO_ITR10 (CONSTRAINT_SIGNOFF)

MESSAGE

`set_load` and `set_driving_cell`/`set_drive` not used together

DEFAULT SEVERITY

Warning

DESCRIPTION

Indicates that a port has a driving cell (or a `set_drive`) but no load value, or a load value but no driving cell (or `set_drive`). This rule is checked by `verify_timing_intent` command and results can be reported by `report_rule_check` command.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
scope	string	Scope
port_pin	object_list	Port
missing_corners	string	Missing corners
file_names	object_list	file_names
file_lines	object_list	file_lines

EXAMPLE

- The following design has 4 input ports named pi0, pi1, pi2, and pi3.

```
set_load 0.5 [get_ports {pi0 pi1}]  
set_driving_cell -lib_cell $CELL_NAME [get_ports {pi0 pi2}]
```

- pi0 has both `set_load` and `set_driving_cell`. It does not flag ITR10.
- pi1 has only `set_load`. This port flags ITR10.
- pi2 has `set_driving_cell`, but lacks `set_load`. This port flags ITR10.
- pi3 has neither `set_driving_cell` or `set_load`. It does not flag ITR10.

CONST_IO_ODL

- [CONST_IO_ODL1](#)
- [CONST_IO_ODL4](#)
- [CONST_IO_ODL5](#)
- [CONST_IO_ODL6](#)
- [CONST_IO_ODL13](#)
- [CONST_IO_ODL15](#)
- [CONST_IO_ODL14](#)

CONST_IO_ODL1

NAME

CONST_IO_ODL1 (CONSTRAINT_SIGNOFF)

MESSAGE

Output is unconstrained

DEFAULT SEVERITY

Warning

DESCRIPTION

Indicates that there is an unconstrained output. Flags output ports that do not have `set_output_delay`. Only output ports that are connected to some constrained path start point (a clocked register or a pin with `set_input_delay`) are considered for this rule. However, an undriven output port will not be checked in this rule. This rule is checked by `verify_timing_intent` command and results can be reported by `report_rule_check` command.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
scope	string	Scope
pin	object	Pin

EXAMPLE

- If the design contains three output ports OUT1, OUT2, and OUT3, the following produces a warning because the output delay for OUT1 is missing:

```
set_output_delay 0.75 -rise -clock CLK1 {OUT2}  
set_output_delay 0.75 -rise -clock CLK1 {OUT3}
```

CONST_IO_ODL4

NAME

CONST_IO_ODL4 (CONSTRAINT_SIGNOFF)

MESSAGE

Incomplete output delay options

DEFAULT SEVERITY

Warning

DESCRIPTION

Indicates that there is an incomplete output delay option present. Values must be specified for both or none of the options in the following combinations: `-rise/-fall` and `-min/-max`. This applies for every clock specified and for delays specified without a clock. This rule does not apply to ports that are specified in `set_false_path` and that have a single path specification switch, which is either `-from`, `-to`, or a single `-through`. Note that `set_false_path` should have both `-setup` and `-hold` specified, or neither. Otherwise, the port is still subject to this rule check. This rule is checked by `verify_timing_intent` command and results can be reported by `report_rule_check` command.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
scope	string	Scope
pin	object	Pin
clock	object	Clock
missing_corners	object_list	Missing corners

file_line	unsigned	File line number
file_name	string	File name

EXAMPLE

- The following generates a warning because there is no corresponding `set_output_delay` for `-fall/-min`:

```
set_output_delay 0.75 -rise -clock CLK1 {OUT1}  
set_output_delay 0.75 -max -clock CLK1 {OUT1}
```

CONST_IO_ODL5

NAME

CONST_IO_ODL5 (CONSTRAINT_SIGNOFF)

MESSAGE

Inconsistent output delay vs. clock period, or `min>max`

DEFAULT SEVERITY

Warning

DESCRIPTION

Indicates one of the following conditions for `set_output_delay` values:

- `-min delay > -max delay`
- A delay value does not satisfy the formula specified in the attribute `ltms_const_output_delay_condition`

Note that the default formula is `delay < ratio * period`, where `delay` indicates `-min/-max delay`, `ratio` indicates the value of attribute `ltms_const_output_delay_ratio`, and `period` indicates the clock period. The attribute `ltms_const_output_delay_ratio` has a default value of 1.0. Also, if there is a `set_multicycle_path` with a single `-to` or `-through` list, in which the port is included, then the clock period is multiplied by the multi-cycle path value. Note that this rule does not apply to ports that are specified in `set_false_path` and that have a single path specification switch, which is either `-to` or a single `-through`. `set_false_path` should have both `-setup` and `-hold` specified, or neither. Otherwise, the port is still subject to this rule check. This rule is checked by `verify_timing_intent` command, and the results can be reported by `report_rule_check` command.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
scope	string	Scope
pin	object	Pin
clock	object	Clock
is_clock_fall	bool	Is clock fall
max_min_option	string	Max/min option
rise_fall_option	string	Value of -rise/ -fall option
min_delay	float	Min delay value
max_delay	float	Max delay value
min_file_line	unsigned	Min file line number
min_file_name	string	Min file name
max_file_line	unsigned	Max file line number
max_file_name	string	Max file name
check_expression	string	Check expression

EXAMPLE

- In the following command, the Litmus issues a warning because the minimum delay is greater than the maximum delay:

```
set_output_delay 1.74 -min -clock CLK1_w {OUT1}  
set_output_delay 0.74 -max -clock CLK1_w {OUT1}
```

- In the following command, the Litmus issues a warning because the output delay does not satisfy the formula $\text{delay} < \text{ratio} * \text{period}$, where the attribute `lms_const_output_delay_ratio` has its default value 1.0:

```
create_clock -period 8.0 -name vCLK3  
set_output_delay 10.0 [get_ports {IN1 IN2}] -clock vCLK3
```


CONST_IO_ODL6

NAME

CONST_IO_ODL6 (CONSTRAINT_SIGNOFF)

MESSAGE

Output delay constrained vs. wrong clock

DEFAULT SEVERITY

Warning

DESCRIPTION

For output ports that have output delay defined with respect to a clock, and that are connected to some constrained start point, this rule indicates that none of those start points is constrained by a clock that is synchronous to the one for which the output delay was defined. This rule does not apply to ports that are specified in `set_false_path` and that have a single path specification switch, which is either `-from`, `-to`, or a single `-through`. Note that `set_false_path` should have both `-setup` and `-hold` specified, or neither. Otherwise, the port is still subject to this rule check. This rule is checked by `verify_timing_intent` command, and the results can be reported by `report_rule_check` command.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
scope	string	Design Name
clock	object	Clock definition
port_pin	object	ports
file_line	unsigned	File Line

file_name	string	File Name

EXAMPLE

- In the following example, if there no path from a sequential element constrained by CLK2 to OUT1, the Litmus issues a warning because `set_output_delay` has specified the wrong clock:

```
set_output_delay 2.75 -min -clock CLK2 {OUT1}
```

CONST_IO_ODL13

NAME

CONST_IO_ODL13 (CONSTRAINT_SIGNOFF)

MESSAGE

An output port is in a timing path within a clock group for which it does not have output delay defined

DEFAULT SEVERITY

Warning

DESCRIPTION

Indicates that there is an output port in whose fan-in there are sequential elements or ports in a given clock group, but this output port has no output delay with respect to any clock in that clock group. This rule applies only to ports that are end points of a timing path that does not match any `set_false_path`. `set_false_path` should have both `-setup` and `-hold` specified, or neither.

Otherwise, the port is still subject to this rule check. This rule does not check output ports that have clock(s) reaching them. This rule is checked by `verify_timing_intent` command and the results can be reported by `report_rule_check` command. Note that this rule can be diagnosed using the `diagnose_rule_check` command.

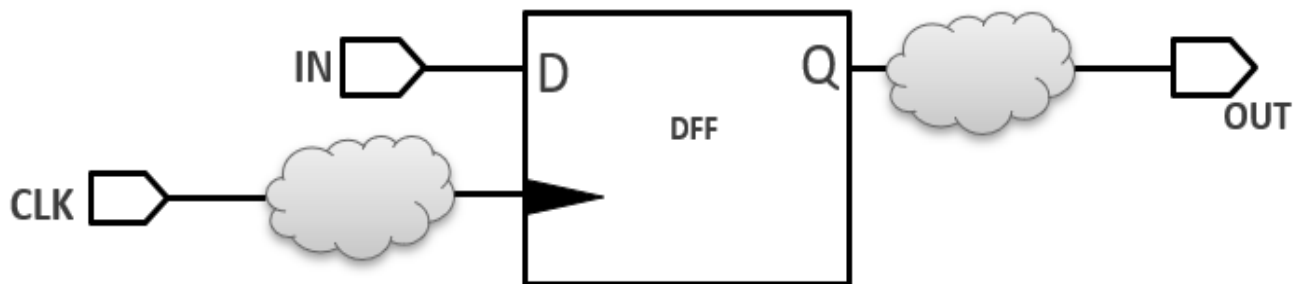
This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
scope	string	Scope
port_pin	object	Port
launch_clocks	object_list	Launch clocks

sync_groups	object_list	sync_groups

EXAMPLE

- For the figure below, the Litmus checks that whether OUT has an output delay defined with respect to CLK or any other clock in the same group. This example satisfies [CONST_IO_ODL1](#), however, this rule issues a warning because it requires a delay with respect to CLK or any clock synchronous to it.



```
create_clock -name rCLK1 -period 11 -waveform {0 5} {CLK}
create_clock -name vCLK2 -period 10 -waveform {0 5}
set_clock_groups -name grp1 -group {rCLK1} -group {vCLK2} -asynchronous
set_output_delay -clock vCLK2 1.1 [get_ports {OUT}]
```

CONST_IO_ODL15

NAME

CONST_IO_ODL15 (CONSTRAINT_SIGNOFF)

MESSAGE

An output delay is defined vs. a virtual clock on a port/pin, but no sequential element in its fanin is driven by a real/generated clock with the same waveform

DEFAULT SEVERITY

Warning

DESCRIPTION

Indicates that, for output ports or internal pins that have output delay defined with respect to a virtual clock, there are sequential elements in the fanin, but their clock pins are driven by real or generated clocks that have a different period or waveform edges than the virtual clock. This rule does not apply to ports that are specified in `set_false_path` and that have a single path specification switch, which is either `-from`, `-to`, or a single `-through`. Note that `set_false_path` should have both `-setup` and `-hold` specified, or neither. Otherwise, the port is still subject to this rule check. This rule is checked by `verify_timing_intent` command and the results can be reported by `report_rule_check` command.

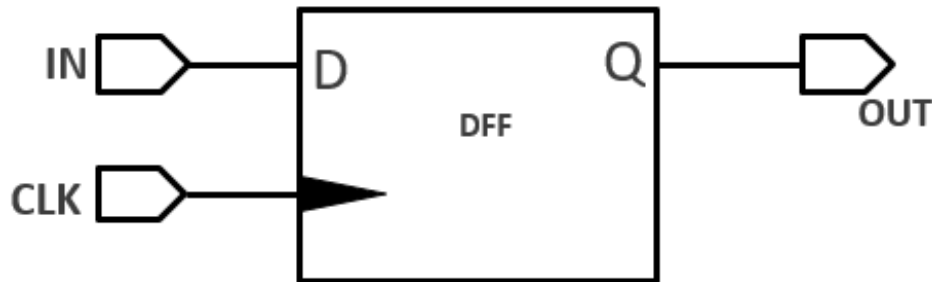
This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
scope	string	Design Name
clock	object	Clock definition
port_pin	object	Output port

file_line	unsigned	File Line
file_name	string	File Name

EXAMPLE

- For the figure below, with the following commands, the Litmus issues a warning because `set_output_delay` has specified a virtual clock whose frequency and waveform do not match that of any real clock driving a sequential element in the fanin of OUT:



```
create_clock -name RCLK -period 10 -waveform {0 5} [get_ports CLK]
create_clock -name VCLK -period 20 -waveform {0 10}
set_output_delay 2.75 -clock VCLK {OUT}
```

CONST_IO_ODL14

NAME

CONST_IO_ODL14 (CONSTRAINT_SIGNOFF)

MESSAGE

An output port is relative to a non-virtual launch clock for which it does not have output delay defined

DEFAULT SEVERITY

Warning

DESCRIPTION

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
scope	string	Design Name
port_pin	object	Port
launch_clocks	object_list	launch_clocks
clock_groups	object_list	clock_groups

CONST_IO_OLD

- [CONST_IO_OLD1](#)
- [CONST_IO_OLD3](#)
- [CONST_IO_OLD4](#)

CONST_IO_OLD1

NAME

CONST_IO_OLD1 (CONSTRAINT_SIGNOFF)

MESSAGE

Undefined load on output

DEFAULT SEVERITY

Warning

DESCRIPTION

Indicates that an undefined load on outputs (PO and PIO) is present, which is set using the `set_load` command. This rule does not trigger when only a `-min` or a `-max` is defined. This rule check does not apply to ports that have `set_dont_touch_network` specified. This rule is checked by `verify_timing_intent` command and results can be reported by `report_rule_check` command.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
scope	string	Scope
pin	object	Pin

EXAMPLE

- The rule will be reported if a design has three outputs OUT1, OUT2, and OUT3 and the following `set_load` command is set for OUT2 and OUT3 only:

```
set_load 5.36 {OUT2 OUT3}
```


CONST_IO_OLD3

NAME

CONST_IO_OLD3 (CONSTRAINT_SIGNOFF)

MESSAGE

Inconsistent load values ($-\text{min} > -\text{max}$)

DEFAULT SEVERITY

Warning

DESCRIPTION

Indicates that there is a port or net with a load value for $-\text{min}$ that is greater than the load value for $-\text{max}$. For ports, $-\text{pin_load}$ and $-\text{wire_load}$ values are checked independently.

This rule is checked by `verify_timing_intent` command and results can be reported by `report_rule_check` command.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
scope	string	Scope
pin	object	Pin
net	object	Net
load_option	string	Load option
min_value	float	Min value
max_value	float	Max value

min_file_line	unsigned	Min file line number
min_file_name	string	Min file name
max_file_line	unsigned	Max file line number
max_file_name	string	Max file name

EXAMPLE

The following two commands set inconsistent constraints on the port `OUT`:

```
set_load 0.5 [all_outputs]
set_load 0.75 -min [get_ports OUT] -pin_load
```

CONST_IO_OLD4

NAME

CONST_IO_OLD4 (CONSTRAINT_SIGNOFF)

MESSAGE

Incomplete load options

DEFAULT SEVERITY

Warning

DESCRIPTION

Indicates that there is a port or net with a load value for `-min` but not for `-max`, or vice-versa. For ports, `-pin_load` and `-wire_load` values are checked independently.

This rule is checked by `verify_timing_intent` command and results can be reported by `report_rule_check` command.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
scope	string	Scope
pin	object	Pin
net	object	Net
load_option	string	Load option
max_min_option	string	Max min option
file_line	unsigned	File line number
file_name	string	File name

EXAMPLE

An occurrence of this rule would be reported if these commands are specified because the `-max` wire load would be missing for port `OUT`:

```
set_load 0.75 -min [get_ports OUT] -pin_load  
set_load 1.0 -min [get_ports OUT] -wire_load  
set_load 2.0 -max [get_ports OUT]
```

CONST_SCA_CHK

- [CONST_SCA_CHK1](#)
- [CONST_SCA_CHK2](#)

CONST_SCA_CHK1

NAME

CONST_SCA_CHK1 (CONSTRAINT_SIGNOFF)

MESSAGE

`set_case_analysis` conflict

DEFAULT SEVERITY

Warning

DESCRIPTION

Indicates that a `set_case_analysis` command imposes a constant value on a pin that has a conflicting tied logic value, or that two `set_case_analysis` constants cannot be justified because that would require conflicting tied logic values at a given object in their fanin. Note that the conflicting tied value can originate from another `set_case_analysis` command, a pin constraint added in the dofile, or even a constant assignment in the design itself. This rule is checked by `verify_timing_intent` command and the results can be reported by `report_rule_check` command.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
scope	string	Scope
pin	object	Pin
pos_source	object	Positive source
conflicting_design_ref	object	Conflicting design reference
type	string	type

file_line	unsigned	File line number
file_name	string	File name
second_cmd_file_line	unsigned	Second command file line number
second_cmd_file_name	string	Second command file name

EXAMPLE

- If the DFF a_reg has its output pin Q connected to the S pin of the MUX select, the following commands will cause a conflict reported by this rule:

```
set_case_analysis 0 [get_pins a_reg/Q]
set_case_analysis 1 [get_pins select/S]
```

- If all scannable DFFs have their SE pins connected to the primary input SCAN_EN (perhaps through buffers or an even number of inverters), the following commands will cause a conflict reported by this rule:

```
set_case_analysis 0 [get_pins a_reg/SE]
set_case_analysis 1 [get_pins b_reg/SE]
```

CONST_SCA_CHK2

NAME

CONST_SCA_CHK2 (CONSTRAINT_SIGNOFF)

MESSAGE

Object with `set_case_analysis` is undriven or has an undriven pin in its effective fanin

DEFAULT SEVERITY

Warning

DESCRIPTION

Indicates that a `set_case_analysis` command is specified on a port or pin that either is undriven, or has some object in its effective fanin cone that has some undriven input. This rule is checked by `verify_timing_intent` command and the results can be reported by `report_rule_check` command.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
scope	string	Scope
file_line	unsigned	File line number
file_name	string	File name
undriven_pin	object_list	Undriven pin
type	string	Type

EXAMPLE

- If the input X of gate I1 is left floating, then this rule will report a command like the following:

```
set_case_analysis 0 [get_pins I1/X]
```

The same would happen if I1/X is driven by an AND gate with one undriven input pin, as long as the other input pin is not tied to 0.

Timing Constraint Comparison (1D2S)

- [CONST_COMP_CG](#)
- [CONST_COMP_CLK](#)
- [CONST_COMP_CTR](#)
- [CONST_COMP_DAT](#)
- [CONST_COMP_EXC](#)
- [CONST_COMP_GRP](#)
- [CONST_COMP_HFN](#)
- [CONST_COMP_IO](#)
- [CONST_COMP_ITR](#)
- [CONST_COMP_LAT](#)
- [CONST_COMP_OLD](#)
- [CONST_COMP_PLS](#)
- [CONST_COMP_SCA](#)
- [CONST_COMP_SDT](#)
- [CONST_COMP_TIM](#)
- [CONST_COMP_UNC](#)

CONST_COMP_CG

- [CONST_COMP_CG1](#)
- [CONST_COMP_CG2](#)

CONST_COMP_CG1

NAME

CONST_COMP_CG1 (CONSTRAINT_COMPARISON)

MESSAGE

Inconsistent `set_clock_gating_check` at reference vs. target.

DEFAULT SEVERITY

Warning

DESCRIPTION

Indicates that an inconsistency in `set_clock_gating_check` command is found between reference and target level constraints. The following are the possible causes for this violation:

- `set_clock_gating_check` command specified at either reference or target level
- `set_clock_gating_check` commands specified at both reference and target level but with different values

This rule is checked by `compare_constraints` command and the results can be reported by `report_rule_check` command.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
target_view_name	string	Target View Name
reference_view_name	string	Reference View Name

scope	object	Target Instance
target_pin	object	Target Pin
reference_pin	object	Reference Pin
target_clock	object	Target clock
reference_clock	object	Reference clock
type	string	Type
target_value	object_list	target_value
reference_value	object_list	reference_value
target_file_line	object_list	Target file number
target_file_name	object_list	Target file name
reference_file_line	object_list	Reference file number
reference_file_name	object_list	Reference file name

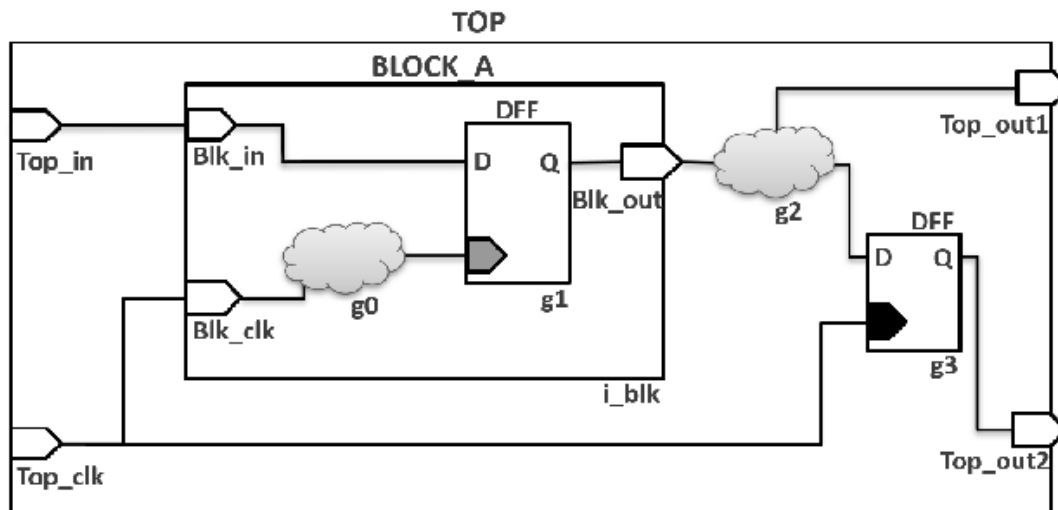
In addition, common rule occurrence attribute type provides the possible causes for the violation.

Attribute	Value	Description
type	BLOCK_NO_CONSTRAINT	Missing constraint at target level
	TOP_NO_CONSTRAINT	Missing constraint at the reference

	DIFFERENT_CONSTRAINTS	Inconsistent constraints between reference and target level
--	-----------------------	---

EXAMPLE

- For the figure below, the following constraints contain `set_clock_gating_check` commands specified in both reference and target level constraints. This rule is reported for the `set_clock_gating_check` commands, because they are specified with different values.



Top level constraint

```
set_clock_gating_check -high -setup 0.4 -hold 0.2 [get_cells i_blk/g0]
```

Block level constraint (instance scope: i_blk)

```
set_clock_gating_check -high -setup 0.8 -hold 0.4 [get_cells g0]
```

CONST_COMP_CG2

NAME

CONST_COMP_CG2 (CONSTRAINT_COMPARISON)

MESSAGE

Inconsistent `set_disable_clock_gating_check` between target-level and reference

DEFAULT SEVERITY

Warning

DESCRIPTION

Indicates that an inconsistency in `set_disable_clock_gating_check` command is found between reference and target level constraints. The following are the possible causes for this violation:

- `set_disable_clock_gating_check` command specified at either reference or target level

This rule is checked by `compare_constraints` command and the results can be reported by `report_rule_check` command.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
target_view_name	string	Target View Name
reference_view_name	string	Reference View Name
scope	object	Target Instance

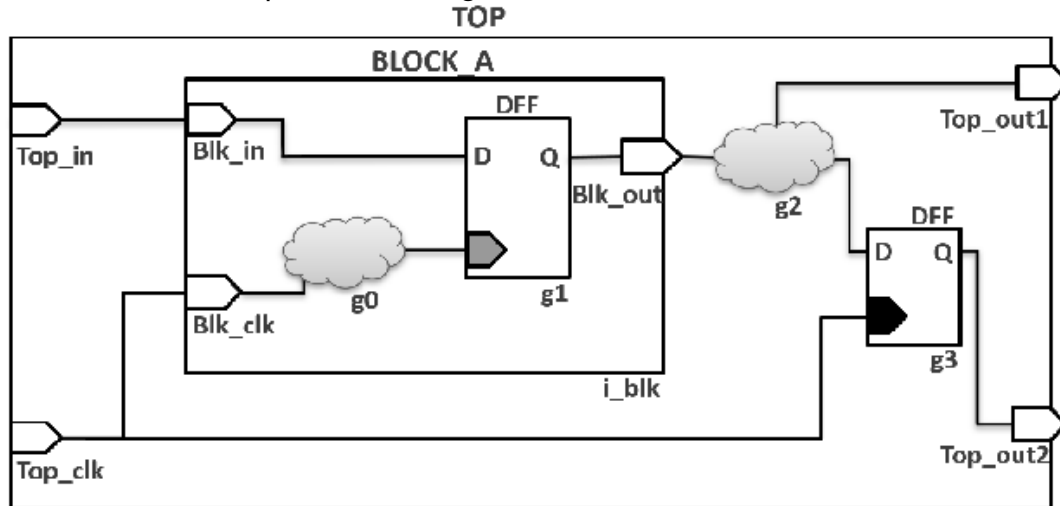
target_pin	object	Target Pin
reference_pin	object	Reference Pin
type	string	Type
target_file_line	unsigned	Target file number
target_file_name	string	Target file name
reference_file_line	unsigned	Reference file number
reference_file_name	string	Reference file name

In addition, common rule occurrence attribute type provides the possible causes for the violation.

Attribute	Value	Description
type	BLOCK_NO_CONSTRAINT	Missing constraint at target level
	TOP_NO_CONSTRAINT	Missing constraint at reference

EXAMPLE

- For the figure below, the following constraints contain `set_disable_clock_gating_check` command specified in reference constraints only. This rule is reported for the `set_disable_clock_gating_check` command, because it is not specified in target level constraints.



Top level constraint

```
set_disable_clock_gating_check [get_cells i_blk/g0]
```

Block level constraint (instance scope: i_blk)

No `set_disable_clock_gating_check` command specified on the instance g0

CONST_COMP_CLK

- [CONST_COMP_CLK1](#)
- [CONST_COMP_CLK2a](#)
- [CONST_COMP_CLK2b](#)
- [CONST_COMP_CLK2c](#)
- [CONST_COMP_CLK2d](#)
- [CONST_COMP_CLK2e](#)
- [CONST_COMP_CLK2f](#)
- [CONST_COMP_CLK3](#)
- [CONST_COMP_CLK4](#)
- [CONST_COMP_CLK6](#)
- [CONST_COMP_CLK7](#)
- [CONST_COMP_CLK8](#)

CONST_COMP_CLK1

NAME

CONST_COMP_CLK1 (CONSTRAINT_COMPARISON)

MESSAGE

Multiple reference clocks drive a target level clock

DEFAULT SEVERITY

Warning

DESCRIPTION

This rule reports that a target level clock constraint object has more than one reference clocks in its fan-in cone. This rule is checked by [compare_constraints](#) command and results can be reported by [report_rule_check](#) command. Also, [report_hierarchical_clock_mapping](#) command can be used to understand how the clocks are mapped between the reference clocks and target level clocks to determine if this rule violation is due to an incomplete clock mapping.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
target_view_name	string	Target View Name
reference_view_name	string	Reference View Name
scope	object	Block Instance

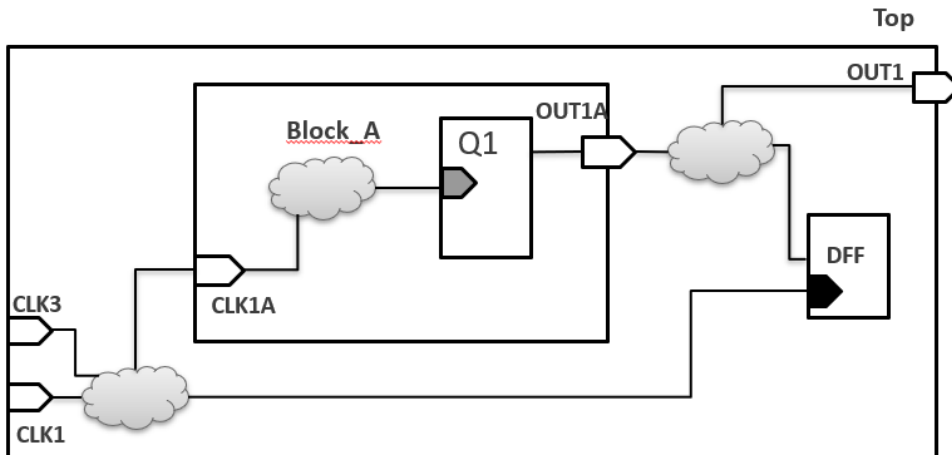
target_clock_source	object	Source of target's clock (hierarchical view)
target_clocks	object_list	Block clocks
reference_clocks	object_list	Top clock

In addition, common rule occurrence attribute `type` provides the possible causes for the violation.

Attribute	Value	Description
type	BLOCK_MISSING_CLOCK	Missing clock at the target

EXAMPLE

- For the figure below, this rule will be reported, if the target constraints file has a clock defined on its primary input like the following target level constraint and the reference constraints file has the following multiple clocks which are mapped to CLK1A and in the fanin cone of the CLK1A clock at the target level:



Top level constraints

```
create_clock -name top_c3 -period 10 [get_ports CLK3]
create_clock -name top_c1 -period 10 [get_ports CLK1]
```

Block level constraint

```
create_clock -name blk_c1 -period 8 [get_ports CLK1A]
```

CONST_COMP_CLK2a

NAME

CONST_COMP_CLK2a (CONSTRAINT_COMPARISON)

MESSAGE

A target's clock is slower than a related reference clock

DEFAULT SEVERITY

Warning

DESCRIPTION

This rule reports that the reference clock's period is not greater than or equal to the period of the corresponding clock definition at the target. This rule is checked by `compare_constraints` command and results can be reported by `report_rule_check` command. Also, `report_hierarchical_clock_mapping` command can be used to understand how the clocks are mapped between the reference clocks and target level clocks to determine if this rule violation is due to an incomplete clock mapping.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
target_view_name	string	Target View Name
reference_view_name	string	Reference View Name
scope	object	Target Instance

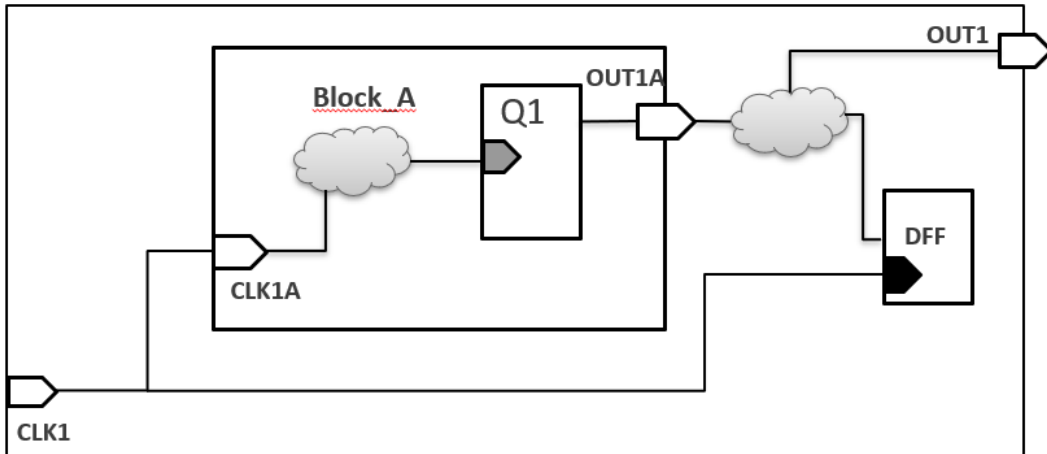
target_clock	object	Target-level clock definition
target_clock_source	object	Source of target's clock (hierarchical view)
reference_clock	object	Reference Clock
target_file_line	unsigned	Target file number
target_file_name	string	Target file name
reference_file_line	unsigned	Reference file number
reference_file_name	string	Reference file name
target_time_period	double	Target time period
reference_time_period	double	Reference time period

In addition, common rule occurrence attribute `type` provides the possible causes for the violation.

Attribute	Value	Description
type	DIFFERENT_CLOCK_ATTRIBUTES	Inconsistent clock attributes

EXAMPLE

- For the figure below, this rule will be reported, if the reference constraints file has the following clock which is mapped to CLK1A and in the fanin cone of the CLK1A clock at the target level and the target constraints file has a clock defined on its primary input like the following target level constraint. The Litmus software issues a warning because the clock's period at the reference is less than that at the target.



Top level constraint

```
create_clock -name top_c1 -period 3 [get_ports CLK1]
```

Block level constraint

```
create_clock -name blk_c1 -period 5 [get_ports CLK1A]
```


CONST_COMP_CLK2b

NAME

CONST_COMP_CLK2b (CONSTRAINT_COMPARISON)

MESSAGE

A target's clock is faster than a related reference clock

DEFAULT SEVERITY

Warning

DESCRIPTION

This rule reports that the reference clock's period is not less than or equal to the period of the corresponding clock definition at the target. This rule is checked by `compare_constraints` command and results can be reported by `report_rule_check` command.

Also, `report_hierarchical_clock_mapping` command can be used to understand how the clocks are mapped between the reference clocks and target level clocks to determine if this rule violation is due to an incomplete clock mapping.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
target_view_name	string	Target View Name
reference_view_name	string	Reference View Name
scope	object	Target Instance

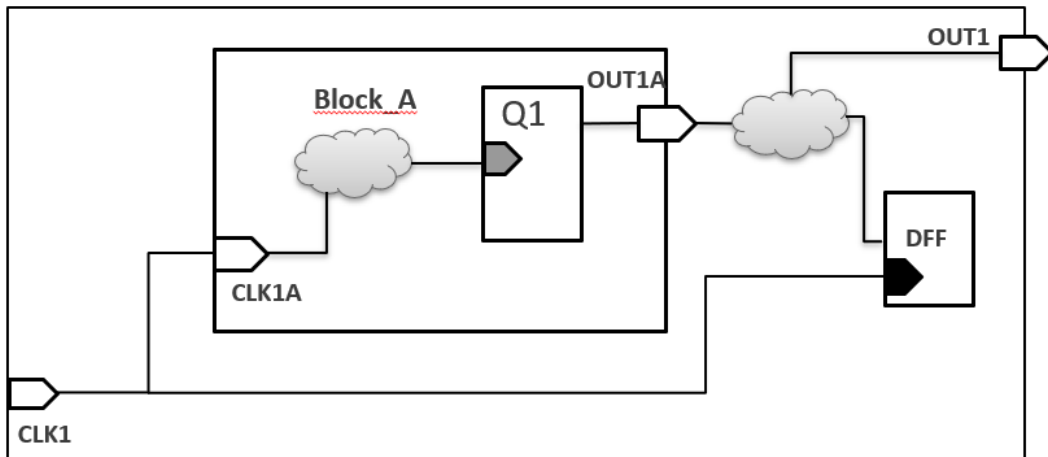
target_clock	object	Block's clock definition
target_clock_source	object	Source of target's clock (hierarchical view)
reference_clock	object	Reference Clock
target_file_line	unsigned	Target file number
target_file_name	string	Target file name
reference_file_line	unsigned	Reference file number
reference_file_name	string	Reference file name
target_time_period	double	Target time period
reference_time_period	double	Reference time period

In addition, common rule occurrence attribute `type` provides the possible causes for the violation.

Attribute	Value	Description
type	DIFFERENT_CLOCK_ATTRIBUTES	Inconsistent clock attributes

EXAMPLE

- For the figure below, this rule will be reported, if the reference constraints file has the following clock which is mapped to CLK1A and the target level constraints file has a clock defined on its primary input like the following target level constraint. The Litmus software issues a warning because the clock's period at the reference is greater than that at the target.



Top level constraint

```
create_clock -name top_c1 -period 5 [get_ports CLK1]
```

Block level constraint

```
create_clock -name blk_c1 -period 3 [get_ports CLK1A]
```

CONST_COMP_CLK2c

NAME

CONST_COMP_CLK2c (CONSTRAINT_COMPARISON)

MESSAGE

A target's clock and a related reference clock have different duty-cycle or waveform

DEFAULT SEVERITY

Warning

DESCRIPTION

This rule reports that a reference clock's period is equal to the period of the corresponding clock definition at the target, but these two clocks have different duty cycle ((falling edge - rising edge)/period) or different waveform. This rule is checked by `compare_constraints` command and results can be reported by `report_rule_check` command.

Also, `report_hierarchical_clock_mapping` command can be used to understand how the clocks are mapped between the reference clocks and target level clocks to determine if this rule violation is due to an incomplete clock mapping.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
target_view_name	string	Target View Name
reference_view_name	string	Reference View Name

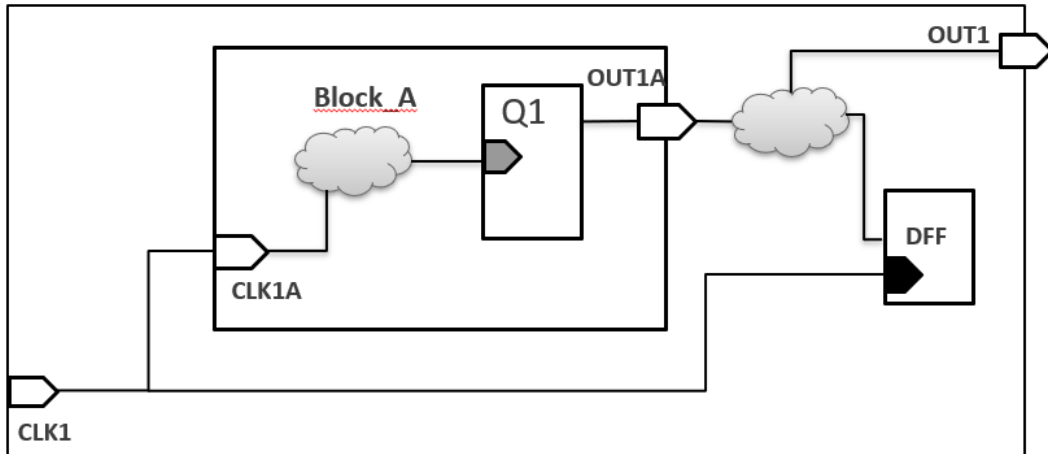
scope	object	Target Instance
target_clock	object	Block's clock definition
target_clock_source	object	Source of target's clock (hierarchical view)
reference_clock	object	Reference Clock
target_file_line	unsigned	Target file number
target_file_name	string	Target file name
reference_file_line	unsigned	Reference file number
reference_file_name	string	Reference file name
target_time_period	double	Target time period
reference_time_period	double	Reference time period

In addition, common rule occurrence attribute `type` provides the possible causes for the violation.

Attribute	Value	Description
type	DIFFERENT_CLOCK_ATTRIBUTES	Inconsistent clock attributes

EXAMPLE

- For the figure below, this rule will be reported, if the reference constraints file has the following clock which are mapped to CLK1A and the target level constraints file has a clock defined on its primary input like the following target level constraint. The Litmus software issues a warning because the clock's period at the reference is equal to that at the target, but their duty cycles are 50% and 40%, respectively.



Top level constraint

```
create_clock -period 10 [get_ports CLK1]
```

Block level constraint

```
create_clock -period 10 [get_ports CLK1A] -waveform {0 4}
```

CONST_COMP_CLK2d

NAME

CONST_COMP_CLK2d (CONSTRAINT_COMPARISON)

MESSAGE

A target's clock and a related reference clock have different phases

DEFAULT SEVERITY

Warning

DESCRIPTION

Indicates that an inconsistency in clock phase is found between mapped reference and target level clocks. The following are the possible causes for this violation:

- Mapped reference and target level clocks defined with different clock phases
- Block level clock is in the clock tree of mapped reference clock but with different clock phase

This rule is checked by `compare_constraints` command and the results can be reported by `report_rule_check` command.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
target_view_name	string	Target View Name
reference_view_name	string	Reference View Name

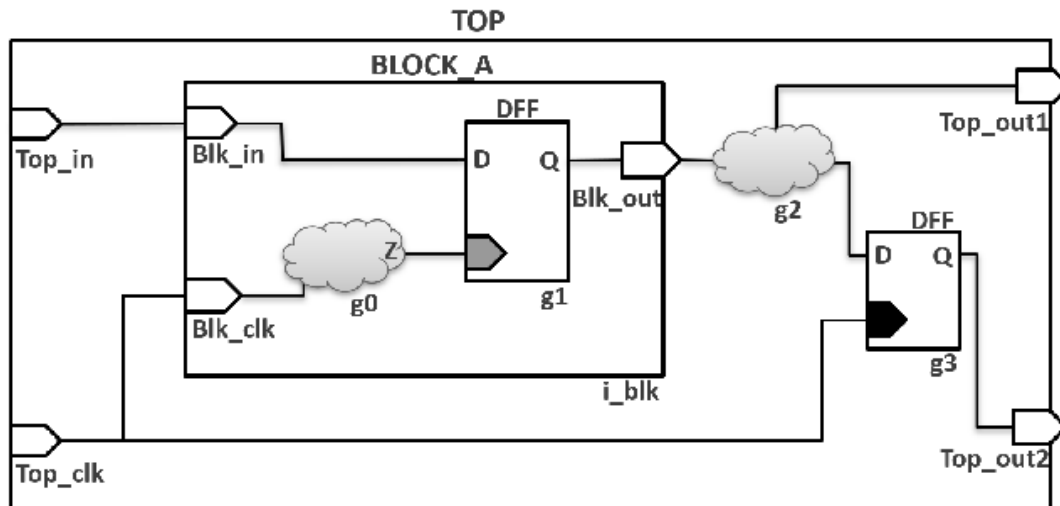
scope	object	Block Instance
target_clock	object	Block's clock definition
target_port	object	Target Port
reference_clock	object	Reference clock
target_polarity	string	Target Polarity
reference_polarity	string	Reference Polarity
target_file_line	unsigned	Target file number
target_file_name	string	Target file name
reference_file_line	unsigned	Reference file number
reference_file_name	string	Reference file name

In addition, common rule occurrence attribute `type` provides the possible causes for the violation.

Attribute	Value	Description
type	DIFFERENT_CLOCK_ATTRIBUTES	Inconsistent clock attributes

EXAMPLE

- For the figure below, the following contains generated clocks specified in both reference and target level constraints. This rule is reported for the generated clocks Top_gclk and Blk_gclk, because they are mapped but defined with different clock phases (`-invert`).



Top level constraints

```
create_clock -name Top_clk -period 2 [get_ports Top_clk]
create_generated_clock -name Top_gclk -source [get_ports Top_clk] -master_clock
[get_clocks Top_clk] -combinational [get_pins i_blk/g0/Z] -add
```

Block level constraints (instance scope: i_blk)

```
create_clock -name Blk_clk -period 2 [get_ports Blk_clk]
create_generated_clock -name Blk_gclk -source [get_ports Blk_clk] -master_clock
[get_clocks Blk_clk] -combinational [get_pins g0/Z] -add -invert
```

Hierarchical clock mapping

```
Top_clk is mapped to Blk_clk
Top_gclk is mapped to Blk_gclk
```

CONST_COMP_CLK2e

NAME

CONST_COMP_CLK2e (CONSTRAINT_COMPARISON)

MESSAGE

A target's `create_generated_clock` is mapped to a reference `create_clock`

DEFAULT SEVERITY

Warning

DESCRIPTION

Indicates that an inconsistency in clock type is found between mapped reference and target level clocks. The following are the possible causes for this violation:

- Block level generated clock is mapped to reference real clock (Not both of reference clock and target level clock are defined inside a target)

This rule is checked by `compare_constraints` command, and the results can be reported by `report_rule_check` command.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
target_view_name	string	Target View Name
reference_view_name	string	Reference View Name

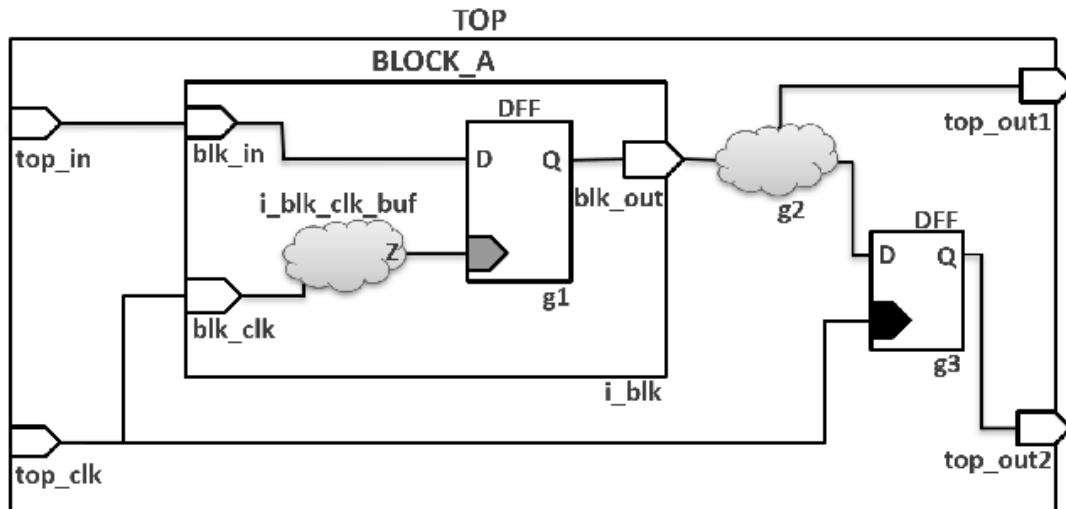
scope	object	Block Instance
target_clock	object	Block's clock definition
target_port	object	Target Port
reference_clock	object	Reference clock
target_file_line	unsigned	Target file number
target_file_name	string	Target file name
reference_file_line	unsigned	Reference file number
reference_file_name	string	Reference file name

In addition, common rule occurrence attribute `type` provides the possible causes for the violation.

Attribute	Value	Description
type	DIFFERENT_CLOCK_ATTRIBUTES	Inconsistent clock attributes

EXAMPLE

- For the figure below, the following example contains two clocks specified in reference constraints and two clocks specified in target level constraints. This rule is reported for the clocks top_gclk and blk_gclk, because they are mapped but defined with different clock types. Clock top_gclk is a real clock by `create_clock` command and clock blk_gclk is a generated clock by `create_generated_clock` command.



Top level constraints

```
create_clock -name top_clk -period 2 [get_ports top_clk]
create_clock -name top_gclk -period 2 [get_pins i_blk/blk_clk]
```

Block level constraints (instance scope: i_blk)

```
create_clock -name blk_clk -period 2 [get_ports blk_clk]
create_generated_clock -name blk_gclk -source [get_ports blk_clk] -master_clock
[get_clocks blk_clk] -combinational [get_ports blk_clk] -add
```

Hierarchical clock mapping

```
top_clk is mapped to blk_clk
top_gclk is mapped to blk_gclk
```

CONST_COMP_CLK2f

NAME

CONST_COMP_CLK2f (CONSTRAINT_COMPARISON)

MESSAGE

Mapped internal reference and target clocks are defined with different clock types

DEFAULT SEVERITY

Warning

DESCRIPTION

Indicates that an inconsistency in clock type is found between mapped internal reference and target level clocks. The following are the possible causes for this violation:

- Block level generated clock is mapped to reference real clock (Both of reference clock and target level clock are defined inside a target)
- Block level real clock is mapped to reference generated clock (Both of reference clock and target level clock are defined inside a target)

This rule is checked by `compare_constraints` command, and the results can be reported by `report_rule_check` command.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
target_view_name	string	Target View Name

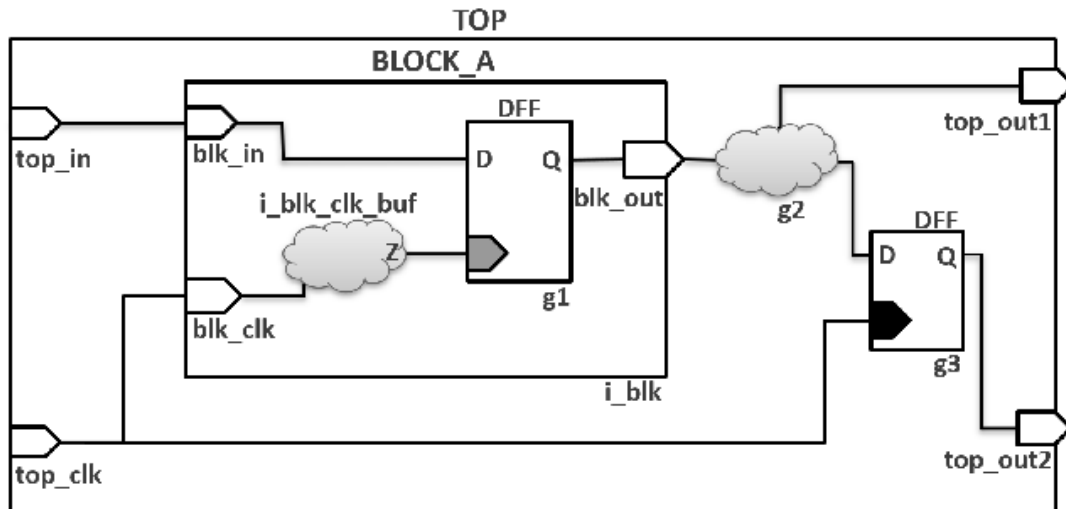
reference_view_name	string	Reference View Name
scope	object	Block Instance
target_clock	object	Block's clock definition
target_port	object	Target Port
reference_clock	object	Reference clock
target_file_line	unsigned	Target file number
target_file_name	string	Target file name
reference_file_line	unsigned	Reference file number
reference_file_name	string	Reference file name

In addition, common rule occurrence attribute `type` provides the possible causes for the violation.

Attribute	Value	Description
type	DIFFERENT_CLOCK_ATTRIBUTES	Inconsistent clock attributes

EXAMPLE

- For the figure below, the following example contains two clocks specified in reference constraints and two clocks specified in target level constraints. This rule is reported for the clocks top_gclk and blk_gclk, because they are mapped but defined with different clock types. Clock top_gclk is a real clock by `create_clock` command and clock blk_gclk is a generated clock by `create_generated_clock` command.



Top level constraints

```
create_clock -name top_clk -period 2 [get_ports top_clk]
create_clock -name top_gclk -period 2 [get_pins i_blk/i_blk_clk_buf/Z]
```

Block level constraints (instance scope: i_blk)

```
create_clock -name blk_clk -period 2 [get_ports blk_clk]
create_generated_clock -name blk_gclk -source [get_ports blk_clk] -master_clock
[get_clocks blk_clk] -combinational [get_pins i_blk_clk_buf/Z] -add
```

Hierarchical clock mapping

```
top_clk is mapped to blk_clk
top_gclk is mapped to blk_gclk
```

CONST_COMP_CLK3

NAME

CONST_COMP_CLK3 (CONSTRAINT_COMPARISON)

MESSAGE

Missing reference clock driving a target level clock source

DEFAULT SEVERITY

Warning

DESCRIPTION

This rule reports that reference constraint does not have a clock defined in the fan-in cone of a target level clock source. This rule is checked by `compare_constraints` command and results can be reported by `report_rule_check` command.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
target_view_name	string	Target View Name
reference_view_name	string	Reference View Name
scope	object	Block Instance

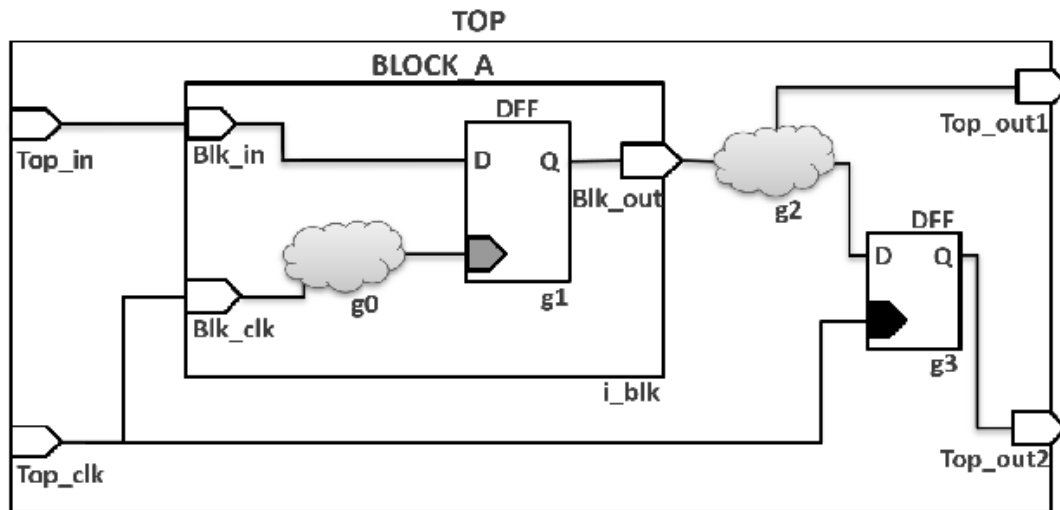
target_clock	object	Block's clock definition
target_port	object	Target Port
target_file_line	unsigned	Target file number
target_file_name	string	Target file name

In addition, common rule occurrence attribute `type` provides the possible causes for the violation.

Attribute	Value	Description
type	TOP_MISSING_CLOCK	Missing clock at reference

EXAMPLE

- For the figure below, if the target constraints file has a clock defined on its primary input Blk_clk like the following target level constraint, then this rule reports a violation if there is no clock from reference driving the BLOCK_A/Blk_clk to indicate that there is a potential missing clock in the reference constraints file.



Top level constraint (if missing)

```
create_clock -name Top_clk -period 10 [get_ports Top_clk]
```

Block level constraint

```
create_clock -name Blk_clk -period 8 [get_ports Blk_clk]
```

CONST_COMP_CLK4

NAME

CONST_COMP_CLK4 (CONSTRAINT_COMPARISON)

MESSAGE

Missing target clock on target port/pin connected to a reference clock source

DEFAULT SEVERITY

Warning

DESCRIPTION

This rule reports that a target's constraint file does not define a clock on a port or pin that is in the clock tree, or is a source, of a reference clock object. In the case that such a target clock is defined, this rule reports if the target clock and the reference clock are not defined as equivalent. Note that [report_hierarchical_clock_mapping](#) command can be used to understand how the clocks are mapped between the reference clocks and target level clocks to determine if this rule violation is due to incomplete clock mapping. This rule is checked by [compare_constraints](#) command and results can be reported by [report_rule_check](#) command.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
target_view_name	string	Target View Name
reference_view_name	string	Reference View Name
scope	object	Block Instance
target_port	object	Target Port

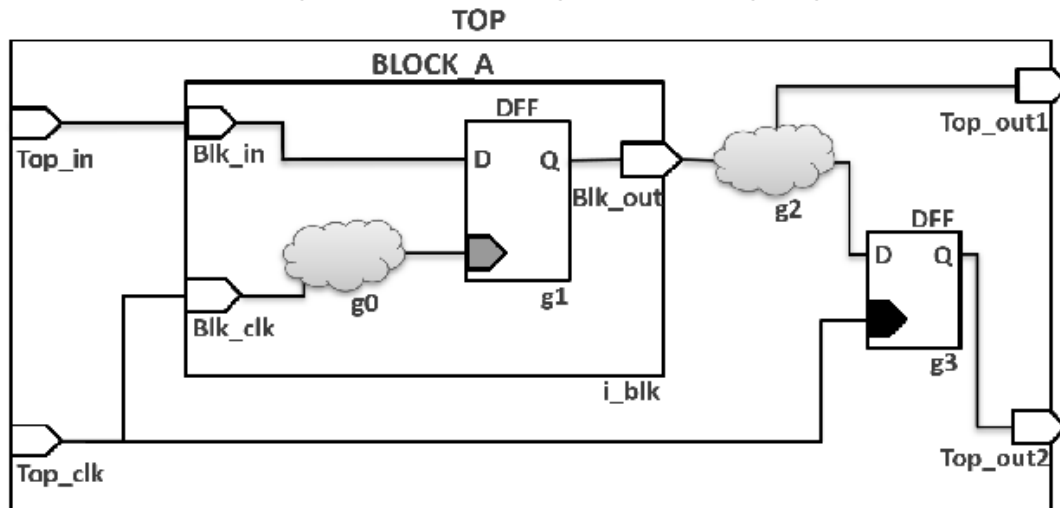
reference_clock	object	Reference clock
reference_lib_gen_clk_file_line	unsigned	Reference_Lib_Gen_Clk_File_Line
reference_lib_gen_clk_file_name	string	Reference_Lib_Gen_Clk_File_Name
reference_file_line	unsigned	Reference file number
reference_file_name	string	Reference file name

In addition, common rule occurrence attribute `type` provides the possible causes for the violation.

Attribute	Value	Description
type	BLOCK_MISSING_CLOCK	Missing clock at target level

EXAMPLE

- For the figure below, this rule will report a violation if the reference port Top_clk is connected to i_blk/Blk_clk, and the reference constraints file contains the following, but the target level constraints file of target i_blk is missing the following target level command:



Top level constraint

```
create_clock -period 10 [get_ports Top_clk] -name Top_clk
```

Block level constraint (instance scope: i_blk)

```
create_clock -period 10 [get_ports Blk_clk] -name Blk_clk
```

Hierarchical clock mapping

```
Top_clk is mapped to Blk_clk
```

CONST_COMP_CLK6

NAME

CONST_COMP_CLK6 (CONSTRAINT_COMP_SIGNOFF)

MESSAGE

Missing virtual clock at target level

DEFAULT SEVERITY

Warning

DESCRIPTION

This rule reports that a target's constraint file does not define a virtual clock for a reference equivalent clock which launches or captures timing path data through a target port.

 `report_hierarchical_clock_mapping` command can be used to understand how the clocks are mapped between the reference clocks and target level clocks to determine if this rule violation is due to incomplete clock mapping.

This rule is checked by `compare_constraints` command and results can be reported by `report_rule_check` command.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
target_view_name	string	Target View Name
reference_view_name	string	Reference View Name

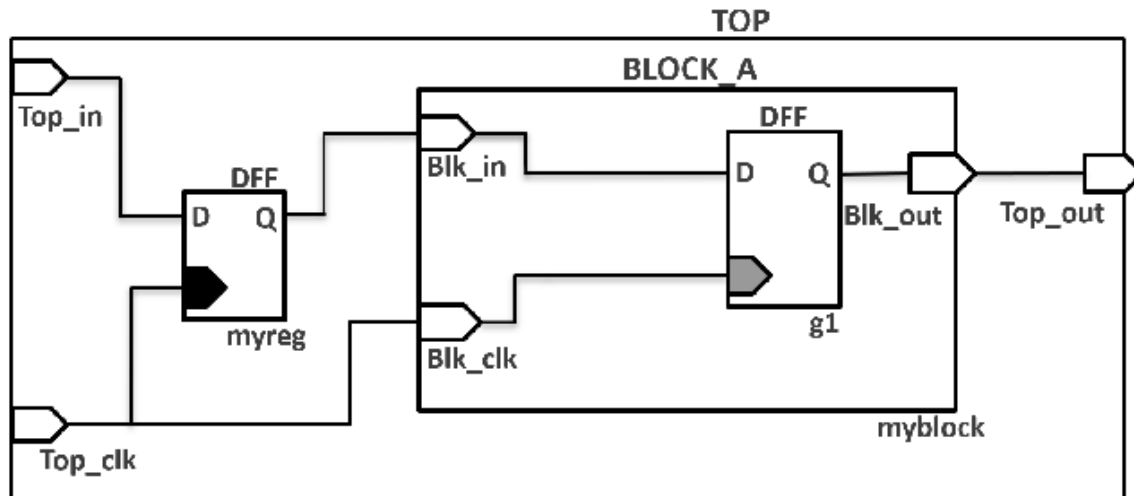
scope	object	Target Instance
reference_clock	object	Reference clock
target_port_witnesses	object_list	Block ports witnesses
reference_file_line	unsigned	Reference file number
reference_file_name	string	Reference file name

In addition, common rule occurrence attribute `type` provides the possible causes for the violation.

Attribute	Value	Description
type	BLOCK_MISSING_CLOCK	Missing clock at target level

EXAMPLE

- For the figure below, reference port Top_clk drives the clock pin of reference flop myreg. The flop myreg is connected to target pin mytarget/Blk_in. The reference constraints file contains the following constraint. This rule will report a violation if the target level constraints file of target mytarget is missing a command like the following target level command for defining a virtual clock for the reference equivalent clock.



Top level constraint

```
create_clock -name TCK -add -period 4.0 -waveform {0.0 2.0} [get_ports Top_clk]
```

Block level constraint (instance scope: mytarget)

```
create_clock -name BCK -period 4.0 -waveform {0.0 2.0}
```


CONST_COMP_CLK7

NAME

CONST_COMP_CLK7 (CONSTRAINT_COMPARISON)

MESSAGE

Inconsistent `set_sense/set_clock_sense` at the reference vs. target

DEFAULT SEVERITY

Warning

DESCRIPTION

Indicates that an inconsistency in `set_sense/set_clock_sense` command is found between reference and target level constraints. The following are the possible causes for this violation:

- `set_sense/set_clock_sense` command specified at either reference or target level
- `set_sense/set_clock_sense` commands specified at both reference and target level but with different values

This rule is checked by `compare_constraints` command and the results can be reported by `report_rule_check` command.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
target_view_name	string	Target View Name
reference_view_name	string	Reference View Name

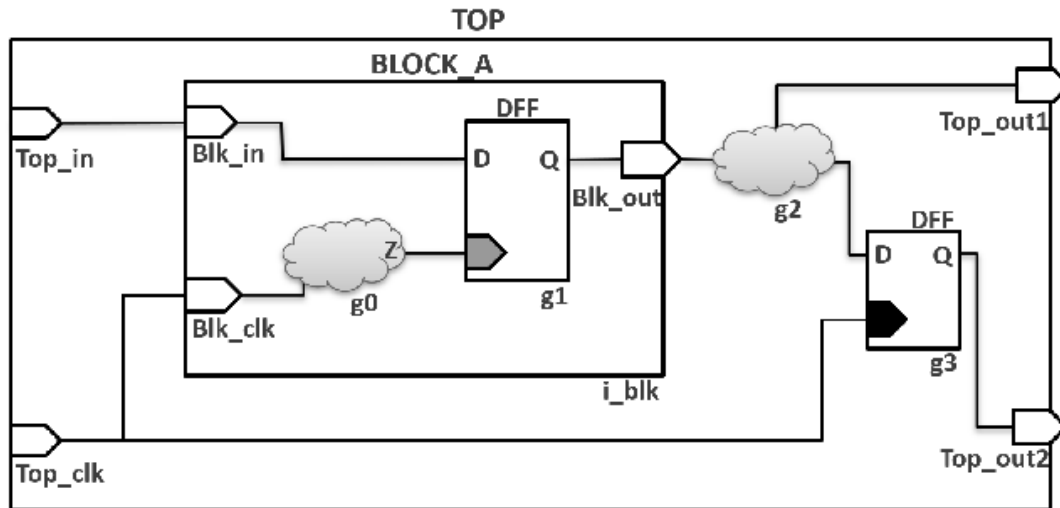
scope	object	Block Instance
target_port	object	Target Port
type	string	Type
target_file_line	unsigned	Target file number
target_file_name	string	Target file name
reference_file_line	unsigned	Reference file number
reference_file_name	string	Reference file name

In addition, common rule occurrence attribute `type` provides the possible causes for the violation.

Attribute	Value	Description
type	BLOCK_NO_CONSTRAINT	Missing constraint at target level
	TOP_NO_CONSTRAINT	Missing constraint at reference
	DIFFERENT_CONSTRAINTS	Inconsistent constraints between reference and target level

EXAMPLE

- For the figure below, the following contains `set_sense` command specified in reference constraints only. This rule is reported for the `set_sense` command, because it is not specified in target level constraints.



Top level constraint

```
set_sense -stop_propagation [get_pins i_blk/g0/Z]
```

Block level constraint (instance scope: i_blk)

No `set_sense` command specified on the pin g0/Z.

CONST_COMP_CLK8

NAME

CONST_COMP_CLK8 (CONSTRAINT_COMPARISON)

MESSAGE

Inconsistent `set_propagated_clock` at the reference vs. target

DEFAULT SEVERITY

Warning

DESCRIPTION

Indicates that an inconsistency in propagated clock specification is found between mapped reference and target level clocks. The following are the possible causes for this violation:

- Top level clock is an ideal clock but mapped target level clock is a propagated clock
- Top level clock is a propagated clock but mapped target level clock is an ideal clock

This rule is checked by `compare_constraints` command and the results can be reported by `report_rule_check` command.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
target_view_name	string	Target View Name
reference_view_name	string	Reference View Name
scope	object	Block Instance
target_clock	object	Target clock

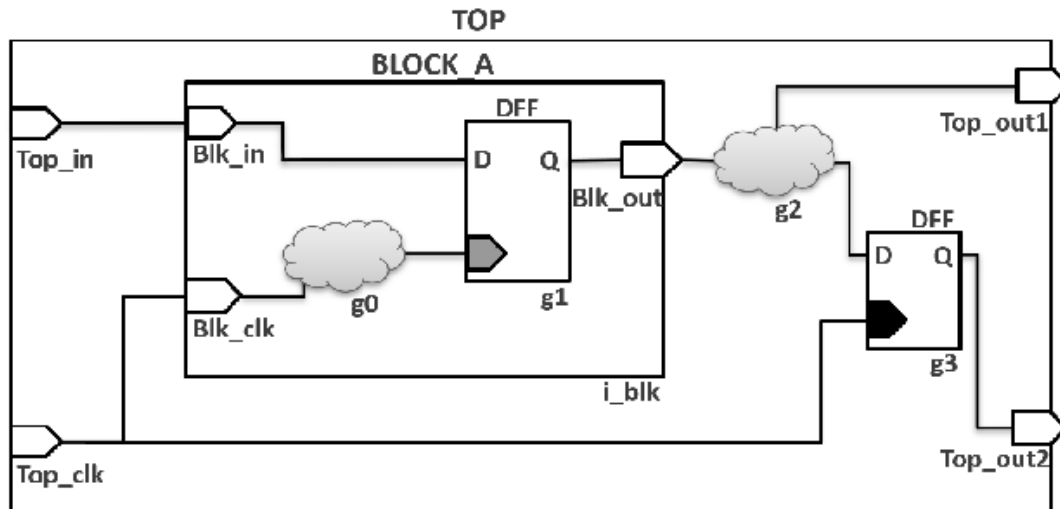
reference_clock	object	Reference clock
target_pin	object	Target Pin
reference_pin	object	Reference Pin
type	string	Type
target_file_line	unsigned	Target file number
target_file_name	string	Target file name
reference_file_line	unsigned	Reference file number
reference_file_name	string	Reference file name

In addition, common rule occurrence attribute `type` provides the possible causes for the violation.

Attribute	Value	Description
type	BLOCK_NO_CONSTRAINT	Missing constraint at target level
	TOP_NO_CONSTRAINT	Missing constraint at reference

EXAMPLE

- For the figure below, the following constraints contain a propagated clock specified in reference constraints and an ideal clock specified in target level constraints. This rule is reported for the clocks Top_clk and Blk_clk, because they are mapped but with different propagated clock specification. Clock Top_clk is a propagated clock by `set_propagated_clock` command and clock Blk_clk is an ideal clock.



Top level constraints

```
create_clock -name Top_clk -period 2 [get_ports Top_clk]
set_propagated_clock [get_clocks Top_clk]
```

Block level constraint (instance scope: i_blk)

```
create_clock -name Blk_clk -period 2 [get_ports Blk_clk]
```

Hierarchical clock mapping

```
Top_clk is mapped to Blk_clk
```

CONST_COMP_CTR

- [CONST_COMP_CTR1](#)

CONST_COMP_CTR1

NAME

CONST_COMP_CTR1 (CONSTRAINT_COMPARISON)

MESSAGE

Inconsistent `set_clock_transition` at the reference-level vs. target

DEFAULT SEVERITY

Warning

DESCRIPTION

Indicates that an inconsistency in `set_clock_transition` command is found between reference and target level constraints. The following are the possible causes for this violation:

- `set_clock_transition` command specified at either reference or target level
- `set_clock_transition` commands specified at both reference and target level but with different values

This rule is checked by `compare_constraints` command and the results can be reported by `report_rule_check` command.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
target_view_name	string	Target View Name
reference_view_name	string	Reference View Name

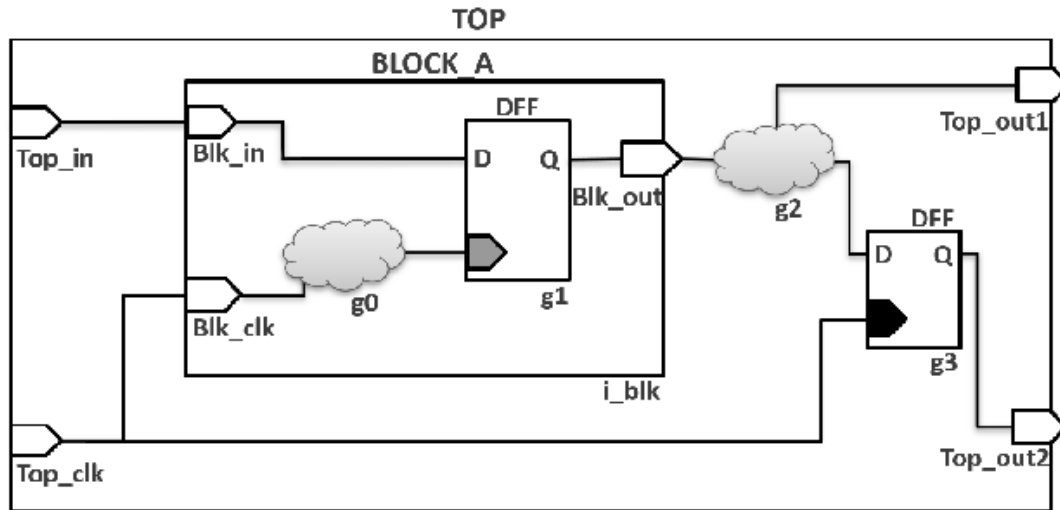
scope	object	Target Instance
target_clock	object	Target clock
reference_clock	object	Reference clock
type	string	Type
target_file_line	unsigned	Target file number
target_file_name	string	Target file name
reference_file_line	unsigned	Reference file number
reference_file_name	string	Reference file name

In addition, common rule occurrence attribute `type` provides the possible causes for the violation.

Attribute	Value	Description
type	BLOCK_NO_CONSTRAINT	Missing constraint at target level
	TOP_NO_CONSTRAINT	Missing constraint at reference
	DIFFERENT_CONSTRAINTS	Inconsistent constraints between reference and target level

EXAMPLE

- For the figure below, the following constraints contain `set_clock_transition` commands specified in both reference and target level constraints. This rule is reported for the `set_clock_transition` commands, because they are specified with different values.



Top level constraints

```
create_clock -name Top_clk -period 2 [get_ports Top_clk]
set_clock_transition 0.5 [get_clocks Top_clk]
```

Block level constraints (instance scope: i_blk)

```
create_clock -name Blk_clk -period 2 [get_ports Blk_clk]
set_clock_transition 1.0 [get_clocks Blk_clk]
```

Hierarchical clock mapping

```
Top_clk is mapped to Blk_clk
```

CONST_COMP_DAT

- [CONST_COMP_DAT1](#)

CONST_COMP_DAT1

NAME

CONST_COMP_DAT1 (CONSTRAINT_COMPARISON)

MESSAGE

Inconsistent `set_data_check` at the reference-level vs. target

DEFAULT SEVERITY

Warning

DESCRIPTION

Indicates that an inconsistency in `set_data_check` command is found between reference and target level constraints. The following are the possible causes for this violation:

- `set_data_check` command specified at either reference or target level
- `set_data_check` commands specified at both reference and target level but with different values

This rule is checked by `compare_constraints` command and the results can be reported by `report_rule_check` command.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
target_view_name	string	Target View Name
reference_view_name	string	Reference View Name
scope	object	Target Instance

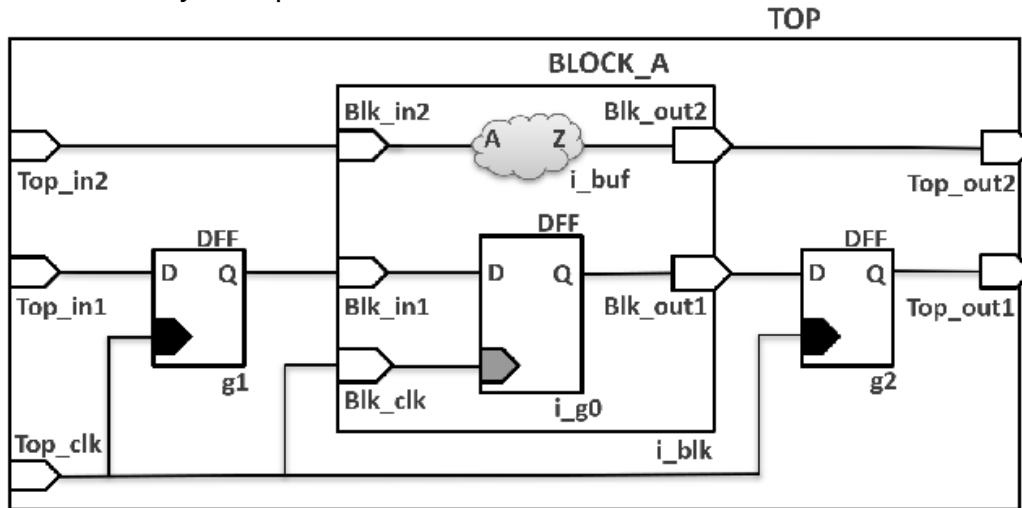
target_from_pin	object	target_from_pin
target_to_pin	object	target_to_pin
reference_from_pin	object	reference_from_pin
reference_to_pin	object	reference_to_pin
type	string	Type
reference_clock	object	Reference clock
target_clock	object	Target clock
target_value	float	target_value
reference_value	float	reference_value
target_file_line	object_list	Target file number
target_file_name	object_list	Target file name
reference_file_line	object_list	Reference file number
reference_file_name	object_list	Reference file name

In addition, common rule occurrence attribute `type` provides the possible causes for the violation.

Attribute	Value	Description
type	BLOCK_NO_CONSTRAINT	Missing constraint at target level
	TOP_NO_CONSTRAINT	Missing constraint at reference
	DIFFERENT_CONSTRAINTS	Inconsistent constraints between reference and target level

EXAMPLE

- For the figure below, the following contains `set_data_check` commands specified in both reference and target level constraints. This rule is reported for the `set_data_check` commands, because they are specified with different values.



Top level constraint

```
set_data_check 0.5 -from [get_pins i_blk/i_buf/A] -to [get_pins i_blk/i_buf/Z]
```

Block level constraint (instance scope: i_blk)

```
set_data_check 1.0 -from [get_pins i_buf/A] -to [get_pins i_buf/Z]
```

CONST_COMP_EXC

- [CONST_COMP_EXC1](#)
- [CONST_COMP_EXC2](#)
- [CONST_COMP_EXC3](#)
- [CONST_COMP_EXC4](#)
- [CONST_COMP_EXC5](#)
- [CONST_COMP_EXC6](#)
- [CONST_COMP_EXC7](#)
- [CONST_COMP_EXC10](#)
- [CONST_COMP_EXC11](#)
- [CONST_COMP_EXC12](#)
- [CONST_COMP_EXC13](#)
- [CONST_COMP_EXC14](#)
- [CONST_COMP_EXC15](#)
- [CONST_COMP_EXC16](#)
- [CONST_COMP_EXC17](#)

CONST_COMP_EXC1

NAME

CONST_COMP_EXC1 (CONSTRAINT_COMPARISON)

MESSAGE

A target path without `set_false_path` is part of a reference false path

DEFAULT SEVERITY

Warning

DESCRIPTION

This rule reports that a target path, which was not declared as false path at the target level, has been affected by reference false path specification. This rule is checked by `compare_constraints` command and results can be reported by `report_rule_check` command.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
target_view_name	string	Target View Name
reference_view_name	string	Reference View Name
target_path	object	Target-level path
reference_path	object	Reference Path
target_file_line	unsigned	Target file number
target_file_name	string	Target file name
reference_file_line	unsigned	Reference file number

Litmus Rule Reference
Timing Constraint Comparison (1D2S)--CONST_COMP_EXC

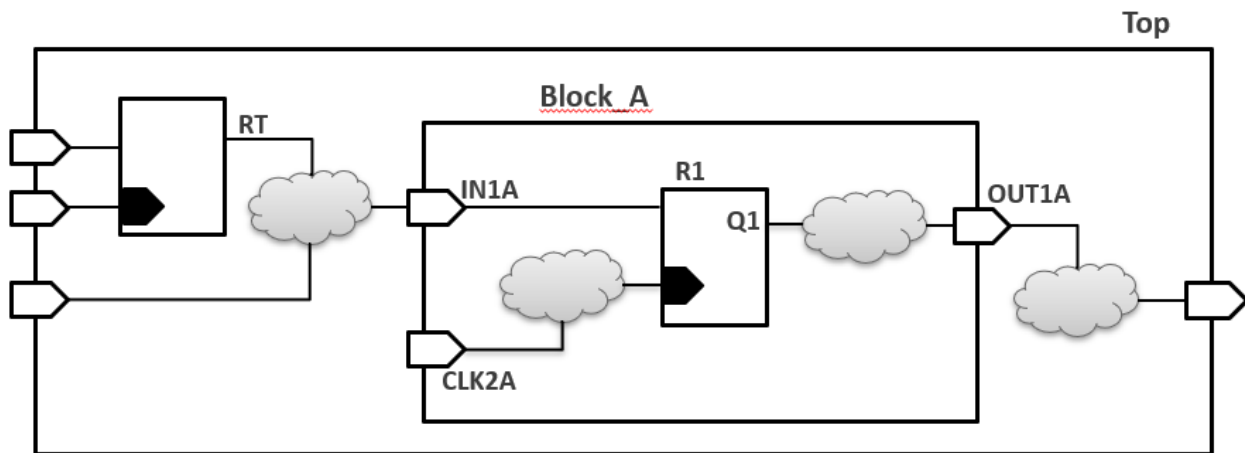
reference_file_name	string	Reference file name
target_path_startpoint	object	Target-level path startpoint
reference_path_startpoint	object	Reference-level path startpoint
target_path_endpoint	object	Target-level path endpoint
reference_path_endpoint	object	Reference-level path endpoint
target_corner	string	target_corner
reference_corner	string	reference_corner
target_exc_from	object_list	target_exc_from
target_exc_to	object_list	target_exc_to
target_exc_through	object_list	target_exc_through
target_exc_type	string	target_exc_type
reference_exc_from	object_list	reference_exc_from
reference_exc_to	object_list	reference_exc_to
reference_exc_through	object_list	reference_exc_through
reference_exc_type	string	reference_exc_type
target_exc_from_clock	object_list	target_exc_fromclk
target_exc_to_clock	object_list	target_exc_tockl
reference_exc_from_clock	object_list	reference_exc_fromclk
reference_exc_to_clock	object_list	reference_exc_tockl
target_path_startpoint_clock	object	target_path_startpoint_clock
target_path_endpoint_clock	object	target_path_endpoint_clock
reference_path_startpoint_clock	object	reference_path_startpoint_clock
reference_path_endpoint_clock	object	reference_path_endpoint_clock
scope	object	Target Instance

In addition, common rule occurrence attribute `type` provides the possible causes for the violation.

Attribute	Value	Description
type	DIFFERENT_PATH_EXCEPTIONS	False path at reference Single-cycle path at target level

EXAMPLE

- For the figure below, if the target level constraints file has a false path defined from R1 as follows and the reference constraints file has defined a false path in Block_A like the following then the Litmus issues a warning because the target level did not consider the path from IN1A to R1 as false with a false path specification from the reference.



Top level constraint

```
set_false_path -through Block_A/IN1A
```

Block level constraint (instance scope: Block_A)

```
set_false_path -from R1
```

RELATED TOPICS

Commands

[set_false_path](#)
[check_target_constraints](#)
[report_rule_check](#)

Attributes

[Rule Attributes - rule_occurrence](#)

CONST_COMP_EXC2

NAME

CONST_COMP_EXC2 (CONSTRAINT_COMPARISON)

MESSAGE

A target's false path is part of a reference path without `set_false_path`

DEFAULT SEVERITY

Warning

DESCRIPTION

This rule reports that a target level false path is not declared as a false path at the reference. This rule is checked by `compare_constraints` command and results can be reported by `report_rule_check` command.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	occurrence id
target_view_name	string	Target View Name
reference_view_name	string	Reference View Name
target_path	object	Block path
reference_path	object	Reference Path
target_file_line	unsigned	Target file number
target_file_name	string	Target file name
reference_file_line	unsigned	Reference file number

Litmus Rule Reference
Timing Constraint Comparison (1D2S)--CONST_COMP_EXC

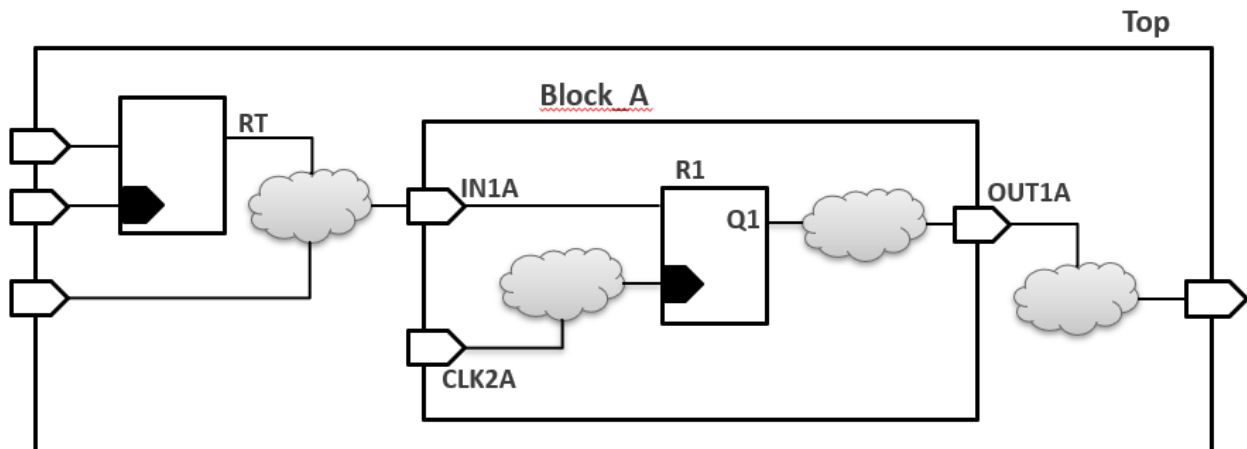
reference_file_name	string	Reference file name
target_path_startpoint	object	Block path startpoint
reference_path_startpoint	object	Reference-level path startpoint
target_path_endpoint	object	Block path endpoint
reference_path_endpoint	object	Reference-level path endpoint
target_corner	string	target_corner
reference_corner	string	reference_corner
target_exc_from	object_list	target_exc_from
target_exc_to	object_list	target_exc_to
target_exc_through	object_list	target_exc_through
target_exc_type	string	target_exc_type
reference_exc_from	object_list	reference_exc_from
reference_exc_to	object_list	reference_exc_to
reference_exc_through	object_list	reference_exc_through
reference_exc_type	string	reference_exc_type
target_exc_from_clock	object_list	target_exc_fromclk
target_exc_to_clock	object_list	target_exc_tock
reference_exc_from_clock	object_list	reference_exc_fromclk
reference_exc_to_clock	object_list	reference_exc_tock
target_path_startpoint_clock	object	target_path_startpoint_clock
target_path_endpoint_clock	object	target_path_endpoint_clock
reference_path_startpoint_clock	object	reference_path_startpoint_clock
reference_path_endpoint_clock	object	reference_path_endpoint_clock
scope	object	Target Instance

In addition, common rule occurrence attribute `type` provides the possible causes for the violation.

Attribute	Value	Description
type	DIFFERENT_PATH_EXCEPTIONS	Single-cycle path at reference False path at target level

EXAMPLE

- For the figure below, if the target level constraints file has a false path defined from R1 like the following and the reference constraints file has defined a false path as the following in line #1 but missing line #2, then the Litmus issues a warning because the target level would consider the path from R1 to OUT1A as false, but the reference does not specify this as false.



Top level constraints

```
set_false_path -from RT
set_false_path -from Block_A/R1
```

Block level constraint (instance scope: Block_A)

```
set_false_path -from R1
```

RELATED TOPICS

- Commands

```
set_false_path
check_target_constraints
report_rule_check
```

- Attributes

[Rule Attributes - rule_occurrence](#)

CONST_COMP_EXC3

NAME

CONST_COMP_EXC3 (CONSTRAINT_COMPARISON)

MESSAGE

A target's false path is part of a reference multicycle path

DEFAULT SEVERITY

Warning

DESCRIPTION

This rule reports that a false path at the target level is declared as a multi-cycle path at the reference. This rule is checked by `compare_constraints` command and results can be reported by `report_rule_check` command.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	occurrence id
target_view_name	string	Target View Name
reference_view_name	string	Reference View Name
target_path	object	Block path
reference_path	object	Reference Path
target_file_line	unsigned	Target file number
target_file_name	string	Target file name
reference_file_line	unsigned	Reference file number

Litmus Rule Reference
Timing Constraint Comparison (1D2S)--CONST_COMP_EXC

reference_file_name	string	Reference file name
target_path_startpoint	object	Block path startpoint
reference_path_startpoint	object	Reference-level path startpoint
target_path_endpoint	object	Block path endpoint
reference_path_endpoint	object	Reference-level path endpoint
reference_exc_cycle	int	reference_exc_cycle
reference_exc_cycle_on_start	string	reference_exc_cycle_on_start
target_corner	string	target_corner
reference_corner	string	reference_corner
target_exc_from	object_list	target_exc_from
target_exc_to	object_list	target_exc_to
target_exc_through	object_list	target_exc_through
target_exc_type	string	target_exc_type
reference_exc_from	object_list	reference_exc_from
reference_exc_to	object_list	reference_exc_to
reference_exc_through	object_list	reference_exc_through
reference_exc_type	string	reference_exc_type
target_exc_from_clock	object_list	target_exc_fromclk
target_exc_to_clock	object_list	target_exc_tock
reference_exc_from_clock	object_list	reference_exc_fromclk
reference_exc_to_clock	object_list	reference_exc_tock
target_path_startpoint_clock	object	target_path_startpoint_clock
target_path_endpoint_clock	object	target_path_endpoint_clock
reference_path_startpoint_clock	object	reference_path_startpoint_clock

Litmus Rule Reference
Timing Constraint Comparison (1D2S)--CONST_COMP_EXC

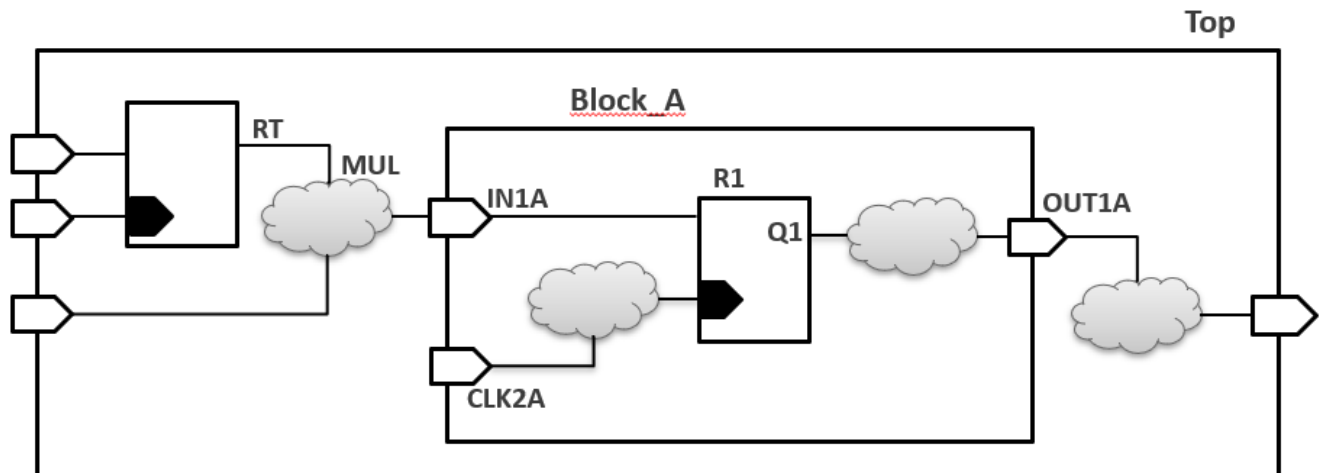
reference_path_endpoint_clock	object	reference_path_endpoint_clock
scope	object	Target Instance

In addition, common rule occurrence attribute `type` provides the possible causes for the violation.

Attribute	Value	Description
type	DIFFERENT_PATH_EXCEPTIONS	Multi-cycle path at reference False path at target level

EXAMPLE

- For the figure below, if the target level constraints file has a false path defined as follows and the reference constraints file has defined a multicycle path like the following, then the Litmus issues a warning because the FP from port IN1A is now considered an MCP because of the –
through MUL specified in the reference constraints file.



Top level constraint

```
set_multicycle_path 2 -through MUL
```

Block level constraint (instance scope: Block_A)

```
set_false_path -through IN1A
```

RELATED TOPICS

- Commands

```
set_false_path
set_multicycle_path
check_target_constraints
report_rule_check
```

- Attributes

[Rule Attributes - rule_occurrence](#)

CONST_COMP_EXC4

NAME

CONST_COMP_EXC4 (CONSTRAINT_COMPARISON)

MESSAGE

A target's multicycle path is part of a reference false path

DEFAULT SEVERITY

Warning

DESCRIPTION

This rule reports that a target level multicycle path is declared as a false path at the reference. This rule is checked by `compare_constraints` command and results can be reported by `report_rule_check` command.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	occurrence id
target_view_name	string	Target View Name
reference_view_name	string	Reference View Name
target_path	object	Block path
reference_path	object	Reference Path
target_file_line	unsigned	Target file number
target_file_name	string	Target file name
reference_file_line	unsigned	Reference file number

Litmus Rule Reference
Timing Constraint Comparison (1D2S)--CONST_COMP_EXC

reference_file_name	string	Reference file name
target_path_startpoint	object	Block path startpoint
reference_path_startpoint	object	Reference-level path startpoint
target_path_endpoint	object	Block path endpoint
reference_path_endpoint	object	Reference-level path endpoint
target_exc_cycle	int	target_exc_cycle
target_exc_cycle_on_start	string	target_exc_cycle_on_start
target_corner	string	target_corner
reference_corner	string	reference_corner
target_exc_from	object_list	target_exc_from
target_exc_to	object_list	target_exc_to
target_exc_through	object_list	target_exc_through
target_exc_type	string	target_exc_type
reference_exc_from	object_list	reference_exc_from
reference_exc_to	object_list	reference_exc_to
reference_exc_through	object_list	reference_exc_through
reference_exc_type	string	reference_exc_type
target_exc_from_clock	object_list	target_exc_fromclk
target_exc_to_clock	object_list	target_exc_tockl
reference_exc_from_clock	object_list	reference_exc_fromclk
reference_exc_to_clock	object_list	reference_exc_tockl
target_path_startpoint_clock	object	target_path_startpoint_clock
target_path_endpoint_clock	object	target_path_endpoint_clock
reference_path_startpoint_clock	object	reference_path_startpoint_clock

Litmus Rule Reference
Timing Constraint Comparison (1D2S)--CONST_COMP_EXC

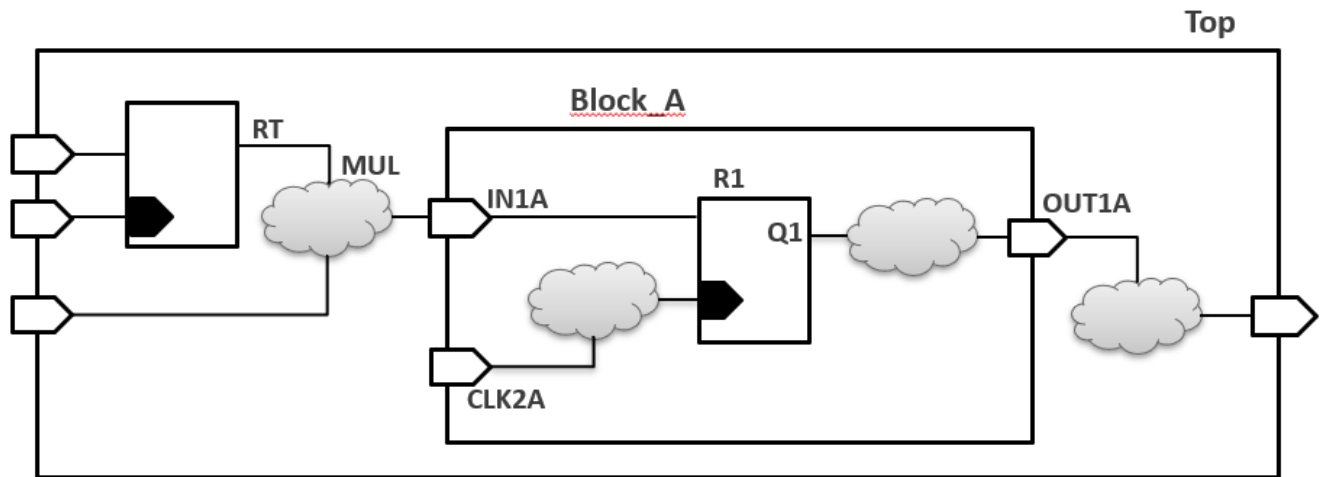
reference_path_endpoint_clock	object	reference_path_endpoint_clock
scope	object	Target Instance

In addition, common rule occurrence attribute `type` provides the possible causes for the violation.

Attribute	Value	Description
type	DIFFERENT_PATH_EXCEPTIONS	False path at reference Multi-cycle path at target level

EXAMPLE

- For the figure below, if the target level constraints file has a multicycle path defined as follows and the reference constraints file has defined a false path like the following, then the Litmus issues a warning because the MCP from port IN1A is now considered an FP because of the `-through MUL` specified in the reference constraints file.



Top level constraint

```
set_false_path -through MUL
```

Block level constraint (instance scope: Block_A)

```
set_multicycle_path 2 -through IN1A
```

RELATED TOPICS

- Commands

```
set_false_path
set_multicycle_path
check_target_constraints
report_rule_check
```

- Attributes

[Rule Attributes - rule_occurrence](#)

CONST_COMP_EXC5

NAME

CONST_COMP_EXC5 (CONSTRAINT_COMPARISON)

MESSAGE

A target's single cycle path is part of a reference multicycle path

DEFAULT SEVERITY

Warning

DESCRIPTION

This rule reports that a target path that is not declared as a multi-cycle path is part of a reference path that is declared as a multi-cycle path at the reference. This rule is checked by `compare_constraints` command and results can be reported by `report_rule_check` command.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	occurrence id
target_view_name	string	Target View Name
reference_view_name	string	Reference View Name
target_path	object	Block path
reference_path	object	Reference Path
target_file_line	unsigned	Target file number
target_file_name	string	Target file name
reference_file_line	unsigned	Reference file number

Litmus Rule Reference
Timing Constraint Comparison (1D2S)--CONST_COMP_EXC

reference_file_name	string	Reference file name
target_path_startpoint	object	Block path startpoint
reference_path_startpoint	object	Reference-level path startpoint
target_path_endpoint	object	Block path endpoint
reference_path_endpoint	object	Reference-level path endpoint
reference_exc_cycle	int	reference_exc_cycle
reference_exc_cycle_on_start	string	reference_exc_cycle_on_start
target_corner	string	target_corner
reference_corner	string	reference_corner
target_exc_from	object_list	target_exc_from
target_exc_to	object_list	target_exc_to
target_exc_through	object_list	target_exc_through
target_exc_type	string	target_exc_type
reference_exc_from	object_list	reference_exc_from
reference_exc_to	object_list	reference_exc_to
reference_exc_through	object_list	reference_exc_through
reference_exc_type	string	reference_exc_type
target_exc_from_clock	object_list	target_exc_fromclk
target_exc_to_clock	object_list	target_exc_tockl
reference_exc_from_clock	object_list	reference_exc_fromclk
reference_exc_to_clock	object_list	reference_exc_tockl
target_path_startpoint_clock	object	target_path_startpoint_clock
target_path_endpoint_clock	object	target_path_endpoint_clock
reference_path_startpoint_clock	object	reference_path_startpoint_clock

Litmus Rule Reference
Timing Constraint Comparison (1D2S)--CONST_COMP_EXC

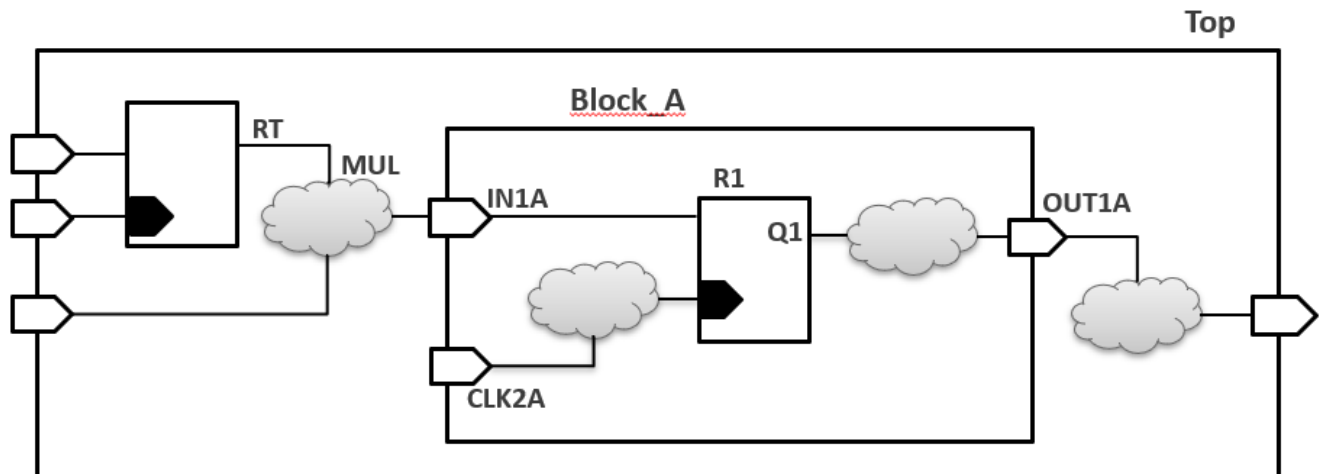
reference_path_endpoint_clock	object	reference_path_endpoint_clock
scope	object	Target Instance

In addition, common rule occurrence attribute `type` provides the possible causes for the violation.

Attribute	Value	Description
type	DIFFERENT_PATH_EXCEPTIONS	Multi-cycle path at reference Single-cycle path at target level

EXAMPLE

- For the figure below, if the target level constraints file has a multicycle path defined as follows and the reference constraints file has defined a multicycle path like the following, then the Litmus issues a warning because the target level path from port IN1A to R1 is now considered an MCP path because of the reference MCP constraint.



Top level constraint

```
set_multicycle_path 3 -through Block_A/R1
```

Block level constraint (instance scope: Block_A)

```
set_multicycle_path 3 -from R1
```

RELATED TOPICS

- Commands

```
set_multicycle_path
check_target_constraints
report_rule_check
```

- Attributes

[Rule Attributes - rule_occurrence](#)

CONST_COMP_EXC6

NAME

CONST_COMP_EXC6 (CONSTRAINT_COMPARISON)

MESSAGE

A target's multicycle path is part of a reference single cycle path

DEFAULT SEVERITY

Warning

DESCRIPTION

This rule reports that a multi-cycle path at the target level is not declared as a multi-cycle path at the reference. This rule is checked by `compare_constraints` command and results can be reported by `report_rule_check` command.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	occurrence id
target_view_name	string	Target View Name
reference_view_name	string	Reference View Name
target_path	object	Block path
reference_path	object	Reference Path
target_file_line	unsigned	Target file number
target_file_name	string	Target file name
reference_file_line	unsigned	Reference file number

Litmus Rule Reference
Timing Constraint Comparison (1D2S)--CONST_COMP_EXC

reference_file_name	string	Reference file name
target_path_startpoint	object	Block path startpoint
reference_path_startpoint	object	Reference-level path startpoint
target_path_endpoint	object	Block path endpoint
reference_path_endpoint	object	Reference-level path endpoint
target_exc_cycle	int	target_exc_cycle
target_exc_cycle_on_start	string	target_exc_cycle_on_start
target_corner	string	target_corner
reference_corner	string	reference_corner
target_exc_from	object_list	target_exc_from
target_exc_to	object_list	target_exc_to
target_exc_through	object_list	target_exc_through
target_exc_type	string	target_exc_type
reference_exc_from	object_list	reference_exc_from
reference_exc_to	object_list	reference_exc_to
reference_exc_through	object_list	reference_exc_through
reference_exc_type	string	reference_exc_type
target_exc_from_clock	object_list	target_exc_fromclk
target_exc_to_clock	object_list	target_exc_tockl
reference_exc_from_clock	object_list	reference_exc_fromclk
reference_exc_to_clock	object_list	reference_exc_tockl
target_path_startpoint_clock	object	target_path_startpoint_clock
target_path_endpoint_clock	object	target_path_endpoint_clock
reference_path_startpoint_clock	object	reference_path_startpoint_clock

Litmus Rule Reference
Timing Constraint Comparison (1D2S)--CONST_COMP_EXC

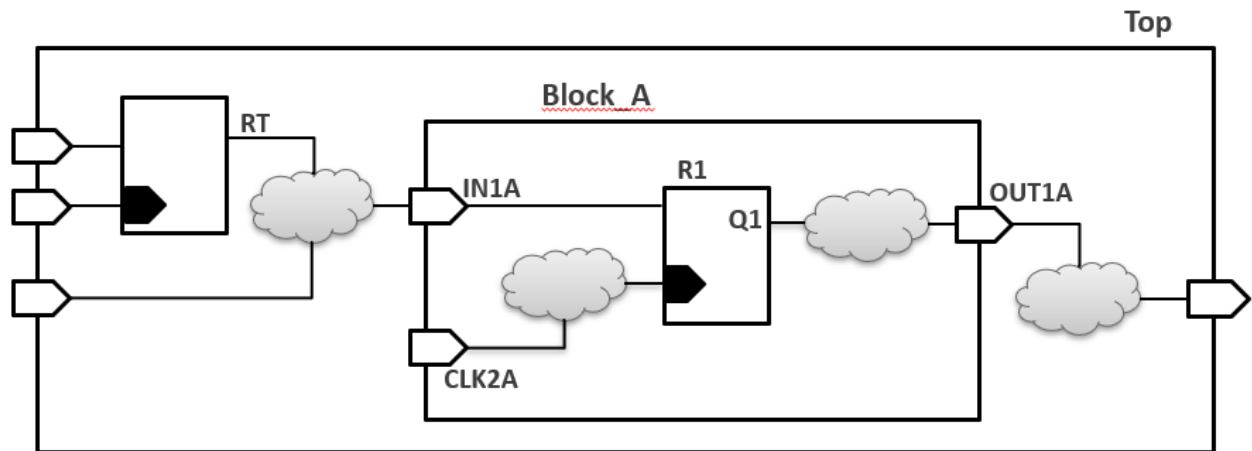
reference_path_endpoint_clock	object	reference_path_endpoint_clock
scope	object	Target Instance

In addition, common rule occurrence attribute `type` provides the possible causes for the violation.

Attribute	Value	Description
type	DIFFERENT_PATH_EXCEPTIONS	Single-cycle path at reference Multi-cycle path at target level

EXAMPLE

- For the figure below, if the target level constraints file has a multicycle path defined as follows and the reference constraints file does not have a multicycle path defined like the following, then the Litmus issues a warning because the target level MCP path from port IN1A to R1 is now considered a single-cycle path.



Top level constraint

```
set_multicycle_path 2 -from Block_A/IN1A
```

Block level constraint (instance scope: Block_A)

```
set_multicycle_path 2 -through IN1A
```

RELATED TOPICS

- Commands

```
set_multicycle_path  
check_target_constraints  
report_rule_check
```

- Attributes

[Rule Attributes - rule_occurrence](#)

CONST_COMP_EXC7

NAME

CONST_COMP_EXC7 (CONSTRAINT_COMPARISON)

MESSAGE

A target's multicycle path is part of a reference multicycle path with a different cycle count

DEFAULT SEVERITY

Warning

DESCRIPTION

This rule reports that there are conflicting cycle counts between the target and reference multi-cycle paths. This rule is checked by `compare_constraints` command and results can be reported by `report_rule_check` command.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	occurrence id
target_view_name	string	Target View Name
reference_view_name	string	Reference View Name
target_path	object	Block path
reference_path	object	Reference Path
target_file_line	unsigned	Target file number
target_file_name	string	Target file name
reference_file_line	unsigned	Reference file number

Litmus Rule Reference
Timing Constraint Comparison (1D2S)--CONST_COMP_EXC

reference_file_name	string	Reference file name
target_path_startpoint	object	Block path startpoint
reference_path_startpoint	object	Reference-level path startpoint
target_path_endpoint	object	Block path endpoint
reference_path_endpoint	object	Reference-level path endpoint
target_exc_cycle	int	target_exc_cycle
target_exc_cycle_on_start	string	target_exc_cycle_on_start
reference_exc_cycle	int	reference_exc_cycle
reference_exc_cycle_on_start	string	reference_exc_cycle_on_start
target_corner	string	target_corner
reference_corner	string	reference_corner
target_exc_from	object_list	target_exc_from
target_exc_to	object_list	target_exc_to
target_exc_through	object_list	target_exc_through
target_exc_type	string	target_exc_type
reference_exc_from	object_list	reference_exc_from
reference_exc_to	object_list	reference_exc_to
reference_exc_through	object_list	reference_exc_through
reference_exc_type	string	reference_exc_type
target_exc_from_clock	object_list	target_exc_fromclk
target_exc_to_clock	object_list	target_exc_tockl
reference_exc_from_clock	object_list	reference_exc_fromclk
reference_exc_to_clock	object_list	reference_exc_tockl
target_path_startpoint_clock	object	target_path_startpoint_clock

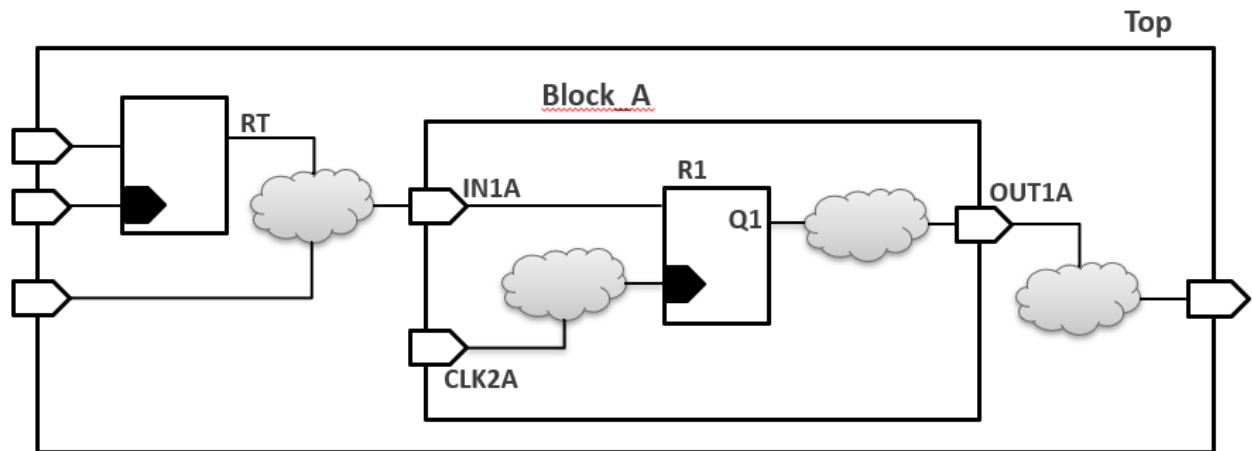
target_path_endpoint_clock	object	target_path_endpoint_clock
reference_path_startpoint_clock	object	reference_path_startpoint_clock
reference_path_endpoint_clock	object	reference_path_endpoint_clock
scope	object	Target Instance

In addition, common rule occurrence attribute `type` provides the possible causes for the violation.

Attribute	Value	Description
type	DIFFERENT_PATH_EXCEPTIONS	Multi-cycle path at reference (cycle count at reference is different from target level) Multi-cycle path at target level

EXAMPLE

- For the figure below, if the target level constraints file has a multicycle path defined as follows and the reference constraints file has a multicycle path defined like the following, then the Litmus issues a warning because of conflicting cycle counts in a multicycle path between the target level and reference constraints.



Top level constraint

```
set_multicycle_path 4 -through Block_A/R1
```

Block level constraint (instance scope: Block_A)

```
set_multicycle_path 3 -through R1
```

RELATED TOPICS

- Commands

```
set_multicycle_path  
check_target_constraints  
report_rule_check
```

- Attributes

[Rule Attributes - rule_occurrence](#)

CONST_COMP_EXC10

NAME

CONST_COMP_EXC10 (CONSTRAINT_COMPARISON)

MESSAGE

A target path without `set_max_delay/set_min_delay` is part of a reference path with max/min delay settings

DEFAULT SEVERITY

Warning

DESCRIPTION

This rule reports that a target path, which was not set any maximum or minimum delay at the target level, has been set by a reference maximum or minimum delay. This rule is checked by `compare_constraints` command and results can be reported by `report_rule_check` command.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	occurrence id
target_view_name	string	Target View Name
reference_view_name	string	Reference View Name
target_path	object	Block path
reference_path	object	Reference Path
target_file_line	unsigned	Target file number
target_file_name	string	Target file name

Litmus Rule Reference
Timing Constraint Comparison (1D2S)--CONST_COMP_EXC

reference_file_line	unsigned	Reference file number
reference_file_name	string	Reference file name
target_path_startpoint	object	Block path startpoint
reference_path_startpoint	object	Reference-level path startpoint
target_path_endpoint	object	Block path endpoint
reference_path_endpoint	object	Reference-level path endpoint
reference_value	float	reference_value
target_corner	string	target_corner
reference_corner	string	reference_corner
target_exc_from	object_list	target_exc_from
target_exc_to	object_list	target_exc_to
target_exc_through	object_list	target_exc_through
target_exc_type	string	target_exc_type
reference_exc_from	object_list	reference_exc_from
reference_exc_to	object_list	reference_exc_to
reference_exc_through	object_list	reference_exc_through
reference_exc_type	string	reference_exc_type
target_exc_from_clock	object_list	target_exc_fromclk
target_exc_to_clock	object_list	target_exc_tockl
reference_exc_from_clock	object_list	reference_exc_fromclk
reference_exc_to_clock	object_list	reference_exc_tockl
target_path_startpoint_clock	object	target_path_startpoint_clock
target_path_endpoint_clock	object	target_path_endpoint_clock
reference_path_startpoint_clock	object	reference_path_startpoint_clock

Litmus Rule Reference
Timing Constraint Comparison (1D2S)--CONST_COMP_EXC

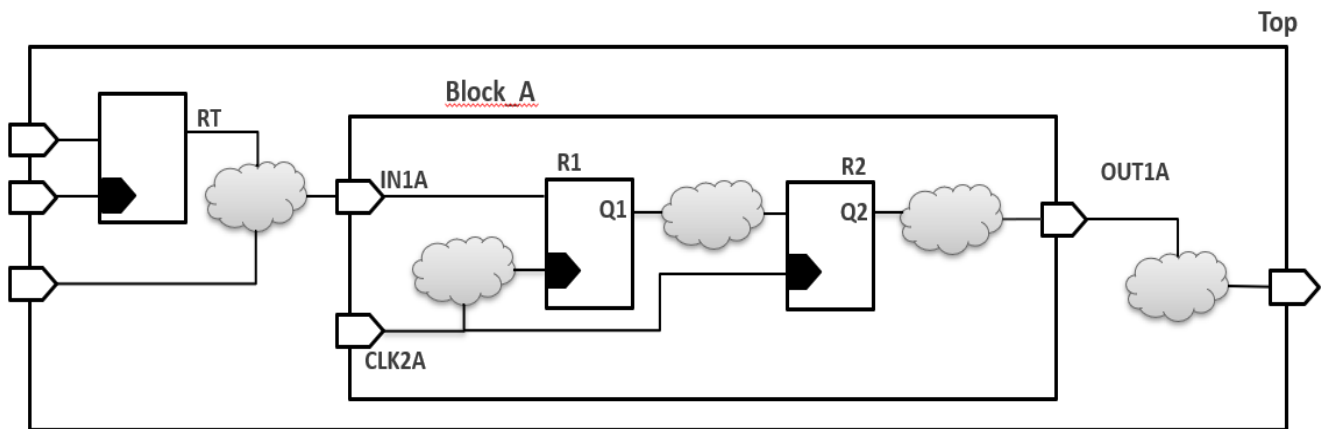
reference_path_endpoint_clock	object	reference_path_endpoint_clock
scope	object	Target Instance

In addition, common rule occurrence attribute `type` provides the possible causes for the violation.

Attribute	Value	Description
type	DIFFERENT_PATH_EXCEPTIONS	Delay path at reference Single-cycle path at target level

EXAMPLE

- For the figure below, the following command in the reference constraints file causes a rule check violation if the same path is not set maximum delay at the target level using the following command at the target level constraints file:



Top level constraint

```
set_max_delay 4 -from [get_cells Block_A/R1] -to [get_cells Block_A/R2]
```

Block level constraint (instance scope: Block_A)

```
set_max_delay 4 -from [get_cells R1] -to [get_cells R2]
```

RELATED TOPICS

- Commands

```
set_max_delay  
set_min_delay  
check_target_constraints  
report_rule_check
```

- Attributes

[Rule Attributes - rule_occurrence](#)

CONST_COMP_EXC11

NAME

CONST_COMP_EXC11 (CONSTRAINT_COMPARISON)

MESSAGE

A target's path with max/min delay settings is part of a reference path without `set_max_delay/set_min_delay`

DEFAULT SEVERITY

Warning

DESCRIPTION

This rule reports that a target path, which was set a maximum or minimum delay at the target level, has not been set with any reference maximum or minimum delay. This rule is checked by `compare_constraints` command and results can be reported by `report_rule_check` command.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	occurrence id
target_view_name	string	Target View Name
reference_view_name	string	Reference View Name
target_path	object	Block path
reference_path	object	Reference Path
target_file_line	unsigned	Target file number
target_file_name	string	Target file name

Litmus Rule Reference
Timing Constraint Comparison (1D2S)--CONST_COMP_EXC

reference_file_line	unsigned	Reference file number
reference_file_name	string	Reference file name
target_path_startpoint	object	Block path startpoint
reference_path_startpoint	object	Reference-level path startpoint
target_path_endpoint	object	Block path endpoint
reference_path_endpoint	object	Reference-level path endpoint
target_value	float	target_value
target_corner	string	target_corner
reference_corner	string	reference_corner
target_exc_from	object_list	target_exc_from
target_exc_to	object_list	target_exc_to
target_exc_through	object_list	target_exc_through
target_exc_type	string	target_exc_type
reference_exc_from	object_list	reference_exc_from
reference_exc_to	object_list	reference_exc_to
reference_exc_through	object_list	reference_exc_through
reference_exc_type	string	reference_exc_type
target_exc_from_clock	object_list	target_exc_fromclk
target_exc_to_clock	object_list	target_exc_tockl
reference_exc_from_clock	object_list	reference_exc_fromclk
reference_exc_to_clock	object_list	reference_exc_tockl
target_path_startpoint_clock	object	target_path_startpoint_clock
target_path_endpoint_clock	object	target_path_endpoint_clock
reference_path_startpoint_clock	object	reference_path_startpoint_clock

Litmus Rule Reference
Timing Constraint Comparison (1D2S)--CONST_COMP_EXC

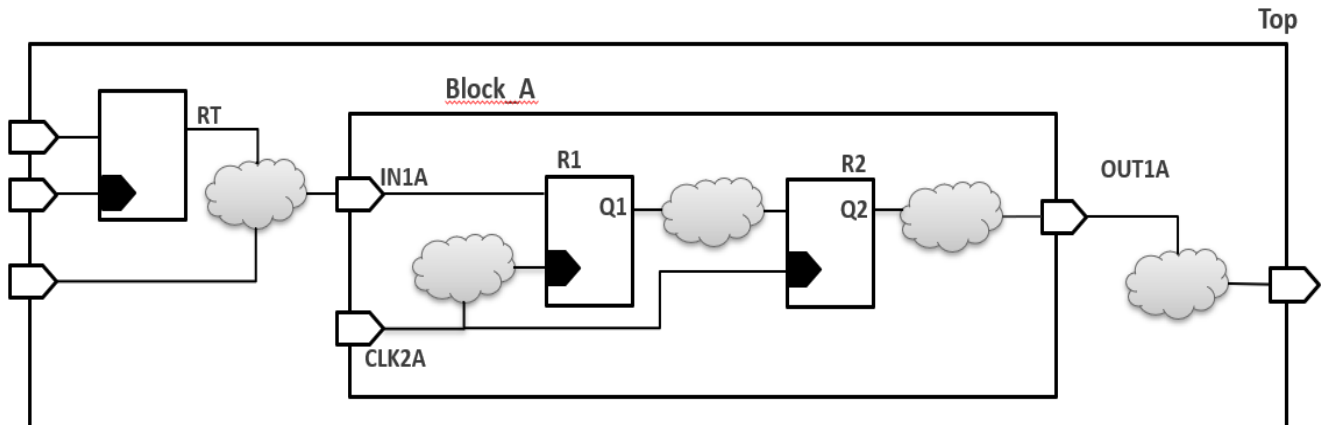
reference_path_endpoint_clock	object	reference_path_endpoint_clock
scope	object	Target Instance

In addition, common rule occurrence attribute `type` provides the possible causes for the violation.

Attribute	Value	Description
type	DIFFERENT_PATH_EXCEPTIONS	Single-cycle path at reference Delay path at target level

EXAMPLE

- For the figure below, the following command in the target level constraints file causes a rule check violation if it is not specified in the reference constraints file like the following:



Top level constraint

```
set_max_delay 4 -from [get_cells Block_A/R1] -to [get_cells Block_A/R2]
```

Block level constraint (instance scope: Block_A)

```
set_max_delay 4 -from [get_cells R1] -to [get_cells R2]
```

RELATED TOPICS

- Commands

```
set_max_delay  
set_min_delay  
check_target_constraints  
report_rule_check
```

- Attributes

[Rule Attributes - rule_occurrence](#)

CONST_COMP_EXC12

NAME

CONST_COMP_EXC12 (CONSTRAINT_COMPARISON)

MESSAGE

A target's false path is part of a reference path with max/min delay settings

DEFAULT SEVERITY

Warning

DESCRIPTION

This rule reports that a target path was declared as false at the target level, has been set by a reference maximum or minimum delay. This rule is checked by `compare_constraints` command and results can be reported by `report_rule_check` command.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	occurrence id
target_view_name	string	Target View Name
reference_view_name	string	Reference View Name
target_path	object	Block path
reference_path	object	Reference Path
target_file_line	unsigned	Target file number
target_file_name	string	Target file name

Litmus Rule Reference
Timing Constraint Comparison (1D2S)--CONST_COMP_EXC

reference_file_line	unsigned	Reference file number
reference_file_name	string	Reference file name
target_path_startpoint	object	Block path startpoint
reference_path_startpoint	object	Reference-level path startpoint
target_path_endpoint	object	Block path endpoint
reference_path_endpoint	object	Reference-level path endpoint
reference_value	float	reference_value
target_corner	string	target_corner
reference_corner	string	reference_corner
target_exc_from	object_list	target_exc_from
target_exc_to	object_list	target_exc_to
target_exc_through	object_list	target_exc_through
target_exc_type	string	target_exc_type
reference_exc_from	object_list	reference_exc_from
reference_exc_to	object_list	reference_exc_to
reference_exc_through	object_list	reference_exc_through
reference_exc_type	string	reference_exc_type
target_exc_from_clock	object_list	target_exc_fromclk
target_exc_to_clock	object_list	target_exc_tockl
reference_exc_from_clock	object_list	reference_exc_fromclk
reference_exc_to_clock	object_list	reference_exc_tockl
target_path_startpoint_clock	object	target_path_startpoint_clock
target_path_endpoint_clock	object	target_path_endpoint_clock
reference_path_startpoint_clock	object	reference_path_startpoint_clock

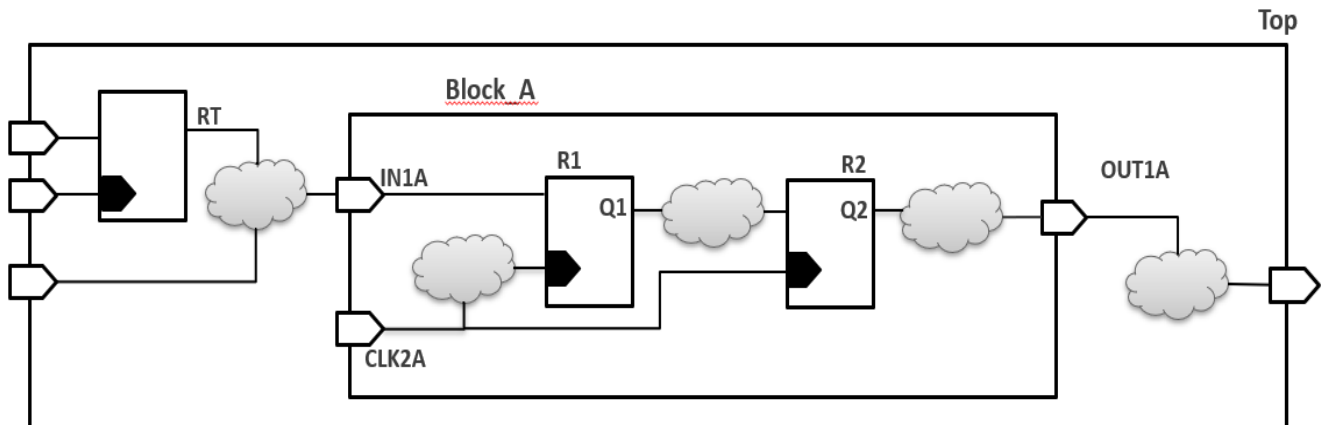
reference_path_endpoint_clock	object	reference_path_endpoint_clock
scope	object	Target Instance

In addition, common rule occurrence attribute `type` provides the possible causes for the violation.

Attribute	Value	Description
type	DIFFERENT_PATH_EXCEPTIONS	Delay path at reference False path at target level

EXAMPLE

- For the figure below, the following commands in reference and target level constraints cause this rule check violation:



Top level constraint

```
set_max_delay 4 -from [get_cells Block_A/R1] -to [get_cells Block_A/R2]
```

Block level constraint (instance scope: Block_A)

```
set_false_path -from [get_cells R1] -to [get_cells R2]
```

RELATED TOPICS

- **Commands**

- `set_false_pathset_max_delay`
 - `set_min_delay`
 - `check_target_constraints`
 - `report_rule_check`

- **Attributes**

- [Rule Attributes - rule_occurrence](#)

CONST_COMP_EXC13

NAME

CONST_COMP_EXC13 (CONSTRAINT_COMPARISON)

MESSAGE

A target's path with max/min delay settings is part of a reference false path

DEFAULT SEVERITY

Warning

DESCRIPTION

This rule reports that a target path, which was set a maximum or minimum delay at the target level, has been affected by a false path specification at the reference. This rule is checked by `compare_constraints` command and results can be reported by `report_rule_check` command.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	occurrence id
target_view_name	string	Target View Name
reference_view_name	string	Reference View Name
target_path	object	Block path
reference_path	object	Reference Path
target_file_line	unsigned	Target file number
target_file_name	string	Target file name
reference_file_line	unsigned	Reference file number

Litmus Rule Reference
Timing Constraint Comparison (1D2S)--CONST_COMP_EXC

reference_file_name	string	Reference file name
target_path_startpoint	object	Block path startpoint
reference_path_startpoint	object	Reference-level path startpoint
target_path_endpoint	object	Block path endpoint
reference_path_endpoint	object	Reference-level path endpoint
target_value	float	target_value
target_corner	string	target_corner
reference_corner	string	reference_corner
target_exc_from	object_list	target_exc_from
target_exc_to	object_list	target_exc_to
target_exc_through	object_list	target_exc_through
target_exc_type	string	target_exc_type
reference_exc_from	object_list	reference_exc_from
reference_exc_to	object_list	reference_exc_to
reference_exc_through	object_list	reference_exc_through
reference_exc_type	string	reference_exc_type
target_exc_from_clock	object_list	target_exc_fromclk
target_exc_to_clock	object_list	target_exc_tock
reference_exc_from_clock	object_list	reference_exc_fromclk
reference_exc_to_clock	object_list	reference_exc_tock
target_path_startpoint_clock	object	target_path_startpoint_clock
target_path_endpoint_clock	object	target_path_endpoint_clock
reference_path_startpoint_clock	object	reference_path_startpoint_clock
reference_path_endpoint_clock	object	reference_path_endpoint_clock

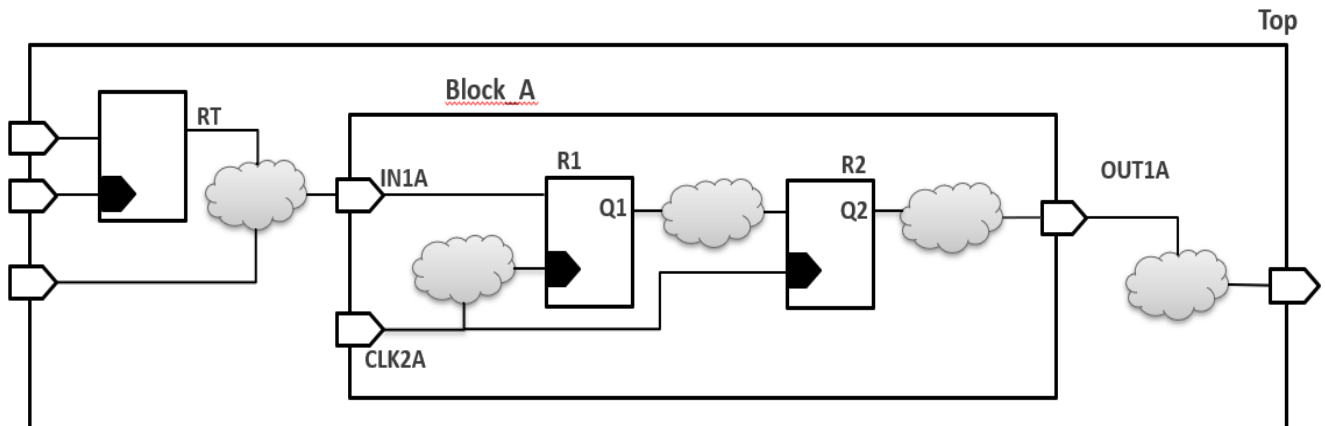
scope	object	Target Instance
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In addition, common rule occurrence attribute `type` provides the possible causes for the violation.

Attribute	Value	Description
type	DIFFERENT_PATH_EXCEPTIONS	False path at reference Delay path at target level

EXAMPLE

- For the figure below, the following commands in reference and target level constraints cause a rule check violation:



Top level constraint

```
set_false_path -from [get_cells Block_A/R1] -to [get_cells Block_A/R2]
```

Block level constraint (instance scope: Block_A)

```
set_max_delay 4 -from [get_cells R1] -to [get_cells R2]
```

RELATED TOPICS

- **Commands**

- `set_false_pathset_max_delay`
 - `set_min_delay`
 - `check_target_constraints`
 - `report_rule_check`

- **Attributes**

- [Rule Attributes - rule_occurrence](#)

CONST_COMP_EXC14

NAME

CONST_COMP_EXC14 (CONSTRAINT_COMPARISON))

MESSAGE

A target's multicycle path is part of a reference path with max/min delay settings

DEFAULT SEVERITY

Warning

DESCRIPTION

This rule reports that a target path, which was declared as a multi-cycle path at the target level, has been set by a reference maximum or minimum delay. This rule is checked by `compare_constraints` command and results can be reported by `report_rule_check` command.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	occurrence id
target_view_name	string	Target View Name
reference_view_name	string	Reference View Name
target_path	object	Block path
reference_path	object	Reference Path
target_file_line	unsigned	Target file number
target_file_name	string	Target file name
reference_file_line	unsigned	Reference file number

Litmus Rule Reference
Timing Constraint Comparison (1D2S)--CONST_COMP_EXC

reference_file_name	string	Reference file name
target_path_startpoint	object	Block path startpoint
reference_path_startpoint	object	Reference-level path startpoint
target_path_endpoint	object	Block path endpoint
reference_path_endpoint	object	Reference-level path endpoint
target_exc_cycle	int	target_exc_cycle
target_exc_cycle_on_start	string	target_exc_cycle_on_start
reference_value	float	reference_value
target_corner	string	target_corner
reference_corner	string	reference_corner
target_exc_from	object_list	target_exc_from
target_exc_to	object_list	target_exc_to
target_exc_through	object_list	target_exc_through
target_exc_type	string	target_exc_type
reference_exc_from	object_list	reference_exc_from
reference_exc_to	object_list	reference_exc_to
reference_exc_through	object_list	reference_exc_through
reference_exc_type	string	reference_exc_type
target_exc_from_clock	object_list	target_exc_fromclk
target_exc_to_clock	object_list	target_exc_tockl
reference_exc_from_clock	object_list	reference_exc_fromclk
reference_exc_to_clock	object_list	reference_exc_tockl
target_path_startpoint_clock	object	target_path_startpoint_clock
target_path_endpoint_clock	object	target_path_endpoint_clock

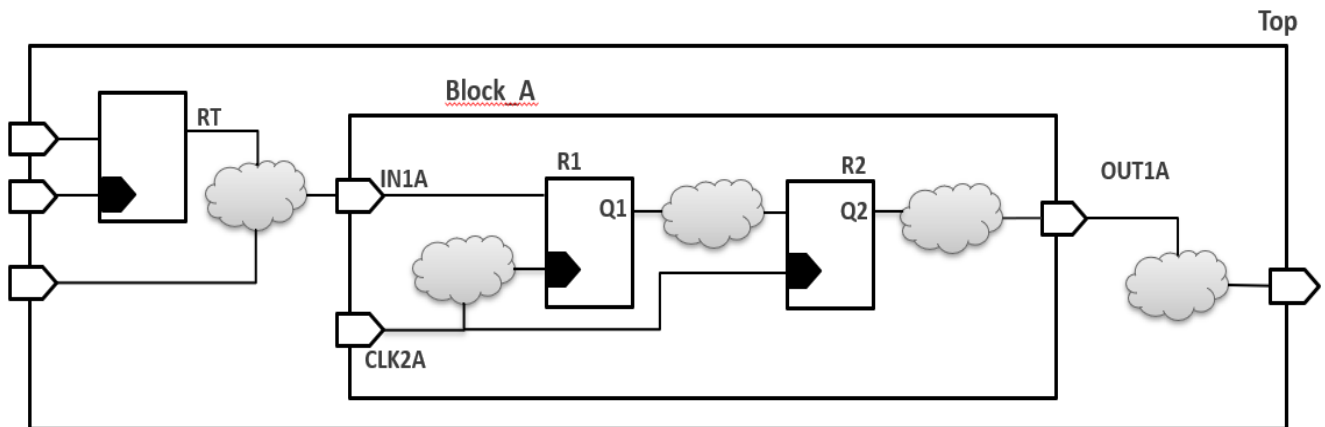
reference_path_startpoint_clock	object	reference_path_startpoint_clock
reference_path_endpoint_clock	object	reference_path_endpoint_clock
scope	object	Target Instance

In addition, common rule occurrence attribute `type` provides the possible causes for the violation.

Attribute	Value	Description
type	DIFFERENT_PATH_EXCEPTIONS	Delay path at reference Multi-cycle path at target level

EXAMPLE

- For the figure below, the following commands in reference and target level constraints cause this rule check violation:



Top level constraint

```
set_max_delay 4 -from [get_cells Block_A/R1] -to [get_cells Block_A/R2]
```

Block level constraint (instance scope: Block_A)

```
set_multicycle_path 4 -from [get_cells R1] -to [get_cells R2]
```

RELATED TOPICS

- Commands

```
set_multicycle_path  
set_max_delay  
set_min_delay  
check_target_constraints  
report_rule_check
```

- Attributes

[Rule Attributes - rule_occurrence](#)

CONST_COMP_EXC15

NAME

CONST_COMP_EXC15 (CONSTRAINT_COMPARISON)

MESSAGE

A target's path with max/min delay settings is part of a reference multicycle path

DEFAULT SEVERITY

Warning

DESCRIPTION

This rule reports that a target path, which was set a maximum or minimum delay at the target level, has been affected by a multi-cycle path specification at the reference. This rule is checked by `compare_constraints` command and results can be reported by `report_rule_check` command.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	occurrence id
target_view_name	string	Target View Name
reference_view_name	string	Reference View Name
target_path	object	Block path
reference_path	object	Reference Path
target_file_line	unsigned	Target file number
target_file_name	string	Target file name
reference_file_line	unsigned	Reference file number

Litmus Rule Reference
Timing Constraint Comparison (1D2S)--CONST_COMP_EXC

reference_file_name	string	Reference file name
target_path_startpoint	object	Block path startpoint
reference_path_startpoint	object	Reference-level path startpoint
target_path_endpoint	object	Block path endpoint
reference_path_endpoint	object	Reference-level path endpoint
reference_exc_cycle	int	reference_exc_cycle
reference_exc_cycle_on_start	string	reference_exc_cycle_on_start
target_value	float	target_value
target_corner	string	target_corner
reference_corner	string	reference_corner
target_exc_from	object_list	target_exc_from
target_exc_to	object_list	target_exc_to
target_exc_through	object_list	target_exc_through
target_exc_type	string	target_exc_type
reference_exc_from	object_list	reference_exc_from
reference_exc_to	object_list	reference_exc_to
reference_exc_through	object_list	reference_exc_through
reference_exc_type	string	reference_exc_type
target_exc_from_clock	object_list	target_exc_fromclk
target_exc_to_clock	object_list	target_exc_tockl
reference_exc_from_clock	object_list	reference_exc_fromclk
reference_exc_to_clock	object_list	reference_exc_tockl
target_path_startpoint_clock	object	target_path_startpoint_clock
target_path_endpoint_clock	object	target_path_endpoint_clock

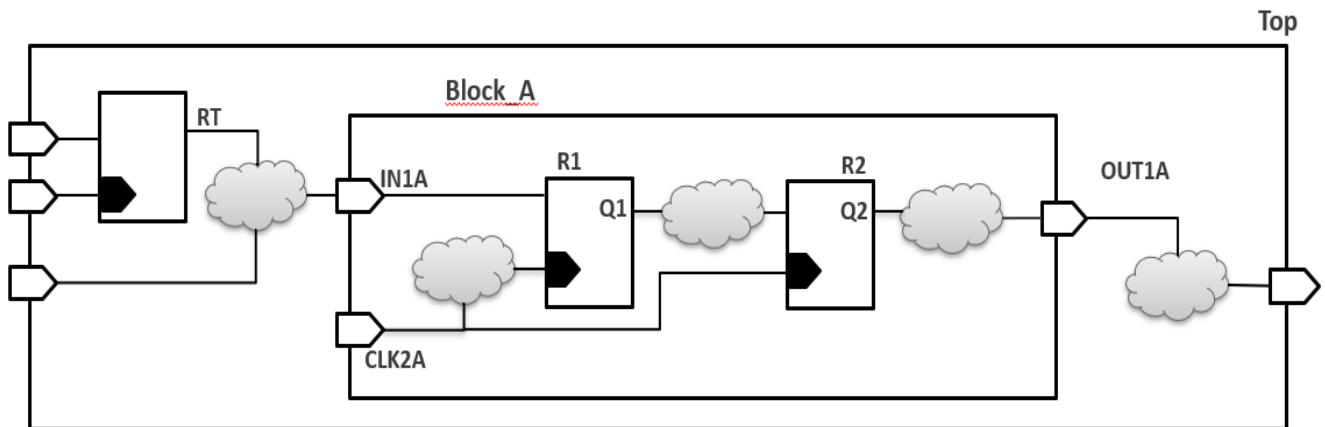
reference_path_startpoint_clock	object	reference_path_startpoint_clock
reference_path_endpoint_clock	object	reference_path_endpoint_clock
scope	object	Target Instance

In addition, common rule occurrence attribute `type` provides the possible causes for the violation.

Attribute	Value	Description
type	DIFFERENT_PATH_EXCEPTIONS	Multi-cycle path at reference Delay path at target level

EXAMPLE

- For the figure below, the following commands in reference and target level constraints cause this rule check violation:



Top level constraint

```
set_multicycle_path 3 -from [get_cells Block_A/R1] -to [get_cells Block_A/R2]
```

Block level constraint (instance scope: Block_A)

```
set_max_delay 4 -from [get_cells R1] -to [get_cells R2]
```

RELATED TOPICS

- Commands

```
set_multicycle_path set_max_delay  
set_min_delay  
check_target_constraints  
report_rule_check
```

- Attributes

[Rule Attributes - rule_occurrence](#)

CONST_COMP_EXC16

NAME

CONST_COMP_EXC16 (CONSTRAINT_COMPARISON)

MESSAGE

A target's path is part of a reference path with different max/min delay settings

DEFAULT SEVERITY

Warning

DESCRIPTION

This rule reports that a target path is specified a minimum or maximum delay with a different value than that specified for a corresponding reference path. This rule is checked by `compare_constraints` command and results can be reported by `report_rule_check` command.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	occurrence id
target_view_name	string	Target View Name
reference_view_name	string	Reference View Name
target_path	object	Block path
reference_path	object	Reference Path
target_file_line	unsigned	Target file number
target_file_name	string	Target file name
reference_file_line	unsigned	Reference file number

Litmus Rule Reference
Timing Constraint Comparison (1D2S)--CONST_COMP_EXC

reference_file_name	string	Reference file name
target_path_startpoint	object	Block path startpoint
reference_path_startpoint	object	Reference-level path startpoint
target_path_endpoint	object	Block path endpoint
reference_path_endpoint	object	Reference-level path endpoint
target_value	float	target_value
reference_value	float	reference_value
block_options	string	block_options
top_options	string	top_options
target_corner	string	target_corner
reference_corner	string	reference_corner
target_exc_from	object_list	target_exc_from
target_exc_to	object_list	target_exc_to
target_exc_through	object_list	target_exc_through
target_exc_type	string	target_exc_type
reference_exc_from	object_list	reference_exc_from
reference_exc_to	object_list	reference_exc_to
reference_exc_through	object_list	reference_exc_through
reference_exc_type	string	reference_exc_type
target_exc_from_clock	object_list	target_exc_fromclk
target_exc_to_clock	object_list	target_exc_tockl
reference_exc_from_clock	object_list	reference_exc_fromclk
reference_exc_to_clock	object_list	reference_exc_tockl
target_path_startpoint_clock	object	target_path_startpoint_clock

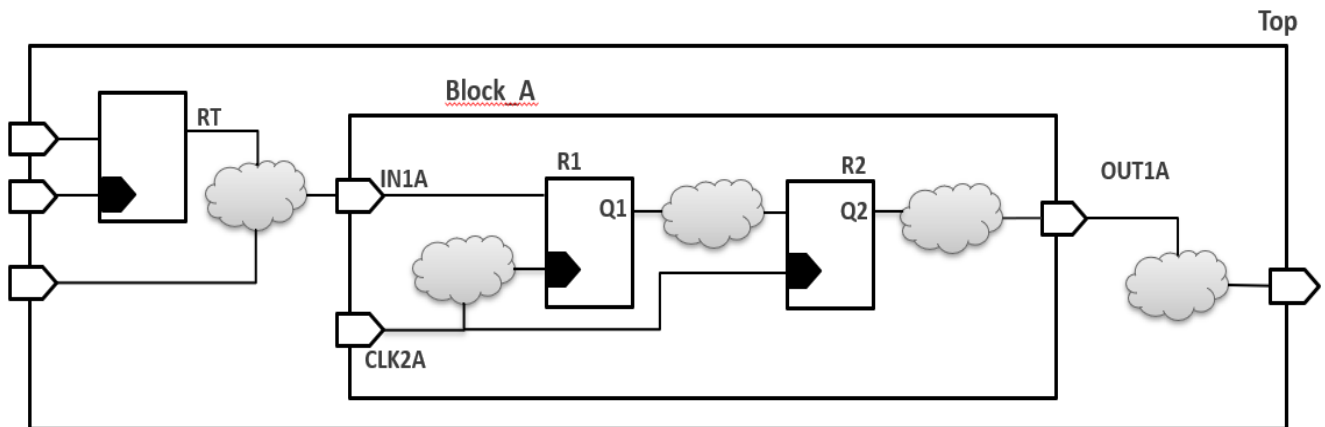
target_path_endpoint_clock	object	target_path_endpoint_clock
reference_path_startpoint_clock	object	reference_path_startpoint_clock
reference_path_endpoint_clock	object	reference_path_endpoint_clock
scope	object	Target Instance

In addition, common rule occurrence attribute `type` provides the possible causes for the violation.

Attribute	Value	Description
type	DIFFERENT_PATH_EXCEPTIONS	Delay path at reference (delay value at reference is different from target level) Delay path at target level

EXAMPLE

- For the figure below, the following commands in reference and target level constraints cause this rule check violation:



Top level constraint

```
set_max_delay 8 -from [get_cells Block_A/R1] -to [get_cells Block_A/R2]
```

Block level constraint (instance scope: Block_A)

```
set_max_delay 4 -from [get_cells R1] -to [get_cells R2]
```

RELATED TOPICS

- Commands

```
set_max_delay  
set_min_delay  
check_target_constraints  
report_rule_check
```

- Attributes

[Rule Attributes - rule_occurrence](#)

CONST_COMP_EXC17

NAME

CONST_COMP_EXC17 (CONSTRAINT_COMPARISON)

MESSAGE

A target path without `set_false_path` is part of reference paths and all such reference paths are defined as false paths

DEFAULT SEVERITY

Warning

DESCRIPTION

This rule reports that a target path was not declared as false path at the target level and all reference paths containing this target path have been affected by reference false path specification. This rule is checked by `compare_constraints` command and results can be reported by `report_rule_check` command.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence Id
target_view_name	string	Target View Name
reference_view_name	string	Reference View Name
target_path	object	Target Path
reference_path	object	Reference Path
target_file_line	unsigned	Target file number

Litmus Rule Reference
Timing Constraint Comparison (1D2S)--CONST_COMP_EXC

target_file_name	string	Target file name
reference_file_line	unsigned	Reference file number
reference_file_name	string	Reference file name
target_path_startpoint	object	Target path startpoint
reference_path_startpoint	object	Reference-level path startpoint
target_path_endpoint	object	Target path endpoint
reference_path_endpoint	object	Reference-level path endpoint
target_corner	string	target_corner
reference_corner	string	reference_corner
target_exc_from	object_list	target_exc_from
target_exc_to	object_list	target_exc_to
target_exc_through	object_list	target_exc_through
target_exc_type	string	target_exc_type
reference_exc_from	object_list	reference_exc_from
reference_exc_to	object_list	reference_exc_to
reference_exc_through	object_list	reference_exc_through
reference_exc_type	string	reference_exc_type
target_exc_from_clock	object_list	target_exc_fromclk
target_exc_to_clock	object_list	target_exc_tockl
reference_exc_from_clock	object_list	reference_exc_fromclk
reference_exc_to_clock	object_list	reference_exc_tockl
target_path_startpoint_clock	object	target_path_startpoint_clock
target_path_endpoint_clock	object	target_path_endpoint_clock
reference_path_startpoint_clock	object	reference_path_startpoint_clock

Litmus Rule Reference
Timing Constraint Comparison (1D2S)--CONST_COMP_EXC

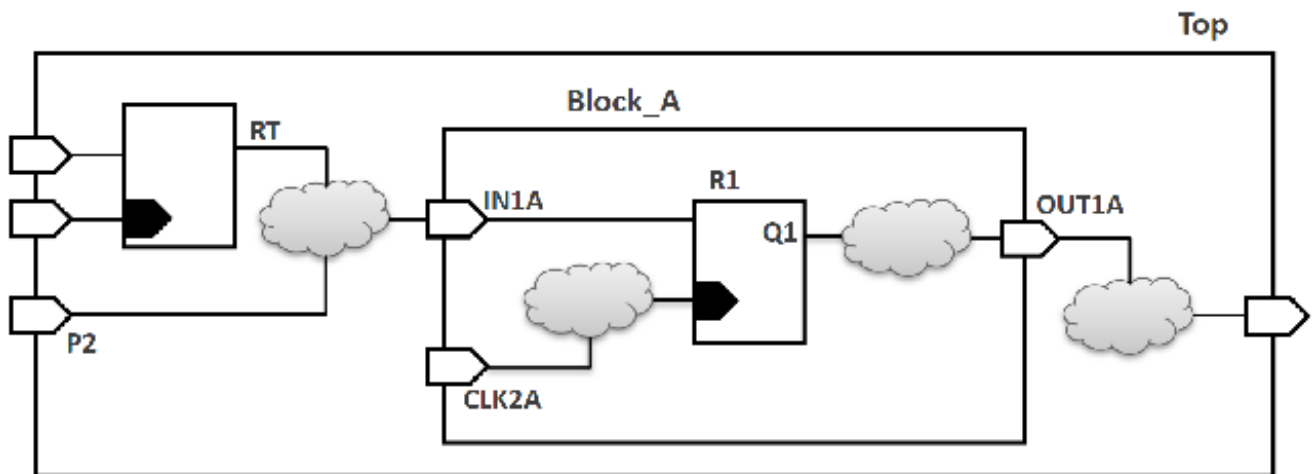
reference_path_endpoint_clock	object	reference_path_endpoint_clock
scope	object	Target Instance

In addition, common rule occurrence attribute `type` provides the possible causes for the violation.

Attribute	Value	Description
type	DIFFERENT_PATH_EXCEPTIONS	False path at reference (for all top paths containing the target path) Single-cycle path at target level

EXAMPLE

- For the figure below, if Block_A has no `set_false_path` defined in the target level constraints file and the reference constraints file has the following lines, then the Litmus issues a warning because the target level did not consider the path from R1 to OUT1A as false, even if all top paths containing this target path were affected by a false path specification. However, the path from IN1A to D1 was not reported by this rule check because the path from P2 through IN1A to D1 was not affected by any false path specification. However, such path will cause `CONST_COMP_EXC1` to be reported for the first reference exception.



Top level constraints

```
set_false_path -through RT
set_false_path -through Block_A/OUT1A
```

RELATED TOPICS

- Commands

```
set_false_path
check_target_constraints
report_rule_check
```

- Attributes

[Rule Attributes - rule_occurrence](#)

- Rules

[CONST_COMP_EXC1](#)

CONST_COMP_GRP

- [CONST_COMP_GRP1](#)

CONST_COMP_GRP1

NAME

CONST_COMP_GRP1 (CONSTRAINT_COMPARISON)

MESSAGE

Inconsistent clock relations between target-level clocks and reference clocks

DEFAULT SEVERITY

Warning

DESCRIPTION

This rule reports that the relationship between the target-level clock and its corresponding reference clock is inconsistent. The relationships being checked are: synchronous/asynchronous and mutually exclusive. The [report_hierarchical_clock_mapping](#) command can be used to understand how the clocks are mapped between the reference clocks and target level clocks to determine if this rule violation is due to incomplete clock mapping. This rule is checked by [compare_constraints](#) command and results can be reported by [report_rule_check](#) command.

RULE ATTRIBUTES

This hierarchical signoff rule check is customizable; its properties are specified through the attributes listed below. To view all the attributes for this rule check from the tool, the following command should be used:

```
report_rule_source "CONST_COMP_GRP1" -verbose
```

Attribute	Description
-----------	-------------

check_type	Specifies the types of clock relationships to be checked Possible Value: <all asynchronous logically_exclusive physically_exclusive> Where: all: Check the consistency of all types of clock relationships in target versus top. This is the default. asynchronous: Checks the consistency of only the synchronous/asynchronous clock relationship in target versus top. logically_exclusive: Checks the consistency of logically mutually exclusive clocks in target versus top. physically_exclusive: Checks the consistency of physically mutually exclusive clocks in target versus top.
------------	---

check_connected	Specifies whether to check the clock connectivity before reporting an occurrence of this rule. This attribute will affect only the asynchronous check type. Possible Value: <yes no> Where: yes: Clock connectivity is checked. This is the default. no: Clock connectivity is not checked.
-----------------	--

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
target_view_name	string	Target View Name
reference_view_name	string	Reference View Name
scope	object	Target Instance
target_clock	object_list	Target clock
reference_clock	object_list	Reference clock
inconsistent_target_type	string	Target Type
inconsistent_reference_type	string	Reference Type
target_file_line	unsigned	Target file number

target_file_name	string	Target file name
reference_file_line	unsigned	Reference file number
reference_file_name	string	Reference file name

In addition, common rule occurrence attribute `type` provides the possible causes for the violation.

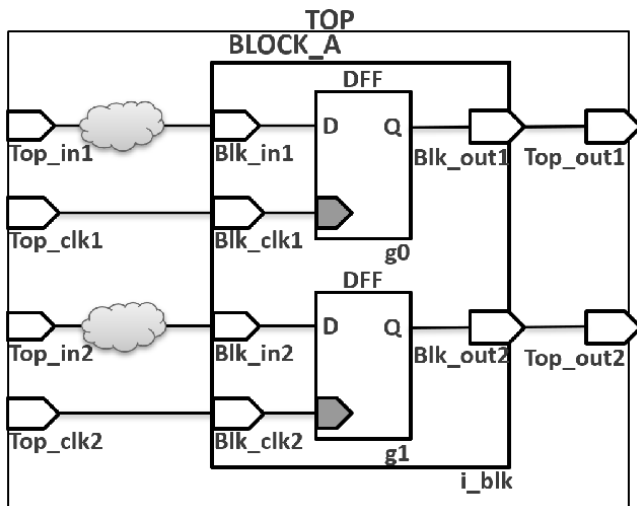
Attribute	Value	Description
type	BLOCK_NO_ASYNC	Not asynchronous at target level while it is at reference
	BLOCK_ASYNC_NO_ALLOW_PATH	Asynchronous without -allow_paths at target level while with -allow_paths at reference
	BLOCK_NO_LOG_EXCLUSIVE	Not logically exclusive at target level while it is at reference
	BLOCK_NO_PHY_EXCLUSIVE	Not physically exclusive at target level while it is at reference

Litmus Rule Reference
Timing Constraint Comparison (1D2S)--CONST_COMP_GRP

	TOP_NO_ASYNC	Not asynchronous at reference while it is at target level
	TOP_ASYNC_NO_ALLOW_PATH	Asynchronous without - allow_paths at reference while with - allow_paths at target level
	TOP_NO_LOG_EXCLUSIVE	Not logically exclusive at reference while it is at target level
	TOP_NO_PHY_EXCLUSIVE	Not physically exclusive at reference while it is at target level

EXAMPLE

- For the figure below, the following reference constraints file defines Top_clk1 and Top_clk2 as physically exclusive, whereas the corresponding mapped clocks to target level, Blk_clk1 and Blk_clk2, respectively are synchronous to each other. This rule reports that the relationship between the reference clocks (physically exclusive) is inconsistent to their corresponding target-level clocks (synchronous).



Top level constraints

```
create_clock -name Top_clk1 -period 8 [get_ports Top_clk1]
create_clock -name Top_clk2 -period 8 [get_ports Top_clk2]
set_clock_groups -name grp1 -group {Top_clk1} -group {Top_clk2} -
physically_exclusive
```

Block level constraints (instance scope: i_blk)

```
create_clock -name Blk_clk1 -period 8 [get_ports Blk_clk1]
create_clock -name Blk_clk2 -period 8 [get_ports Blk_clk2]
```

Hierarchical clock mapping

```
Top_clk1 is mapped to Blk_clk1
Top_clk2 is mapped to Blk_clk2
```

CONST_COMP_HFN

- [CONST_COMP_HFN2](#)

CONST_COMP_HFN2

NAME

CONST_COMP_HFN2 (CONSTRAINT_COMPARE)

MESSAGE

Inconsistent [set_dont_touch_network](#) at the reference vs. target

DEFAULT SEVERITY

Warning

DESCRIPTION

This rule reports that [set_dont_touch_network](#) is set on a target, but not at the reference or vice versa. The Litmus performs this check on a target's ports and pins. The attribute is propagated through the clock tree, which means the reference can set it on a port that drives a target's port, instead of directly on the target's port. If this attribute is set on a clock object, the Litmus treats it as if it was set on all of its source objects. This rule is checked by [compare_constraints](#) command and the results can be reported by [report_rule_check](#) command.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
target_view_name	string	Target View Name

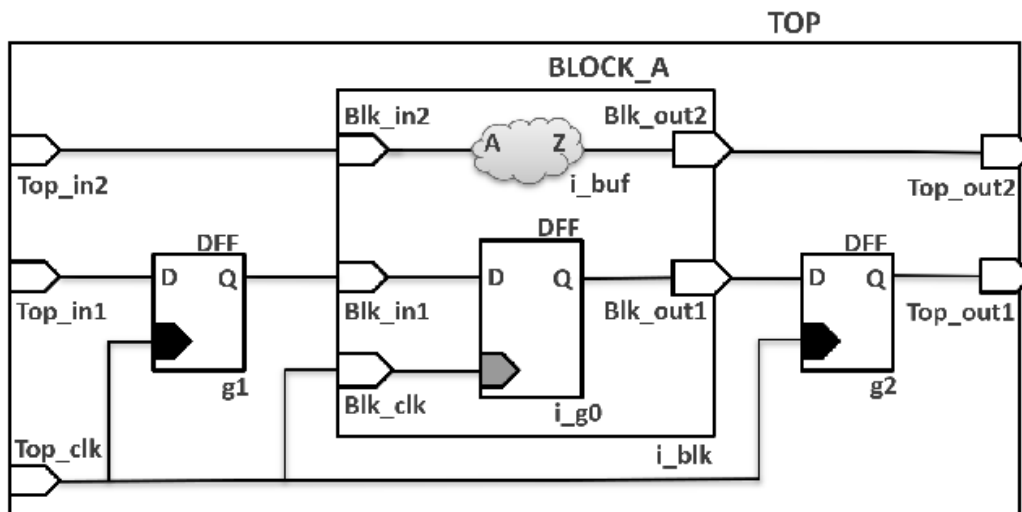
reference_view_name	string	Reference View Name
scope	object	Target Instance
target_object	object	Target object
reference_object	object	Reference object
type	string	Type

In addition, common rule occurrence attribute `type` provides the possible causes for the violation.

Attribute	Value	Description
type	BLOCK_NO_CONSTRAINT	Missing constraint at target level
	TOP_NO_CONSTRAINT	Missing constraint at reference
	DIFFERENT_CONSTRAINTS	Inconsistent constraints between reference and target level

EXAMPLE

- For the figure below, if the target level constraints file has these following lines, then the Litmus issues a warning if the reference constraints file does not have `set_dont_touch_network` specified on pin `i_blk/Blk_clk` or any clock source in its fanin:



Block level constraints (instance scope: `i_blk`)

```
create_clock -name Blk_clk -period 10 [get_ports Blk_clk]
set_dont_touch_network [get_clocks Blk_clk]
```

- For the figure above, if the reference constraints file has the following line, where `i_blk/Blk_clk` is in the clock tree of `Top_clk`, then the Litmus issues a warning if the target level constraints file does not have `set_dont_touch_network` specified for `Blk_clk`:

Top level constraint

```
set_dont_touch_network [get_clocks Top_clk]
```

CONST_COMP_IO

- [CONST_COMP_IO1](#)
- [CONST_COMP_IO2](#)
- [CONST_COMP_IO3](#)
- [CONST_COMP_IO4](#)
- [CONST_COMP_IO5i](#)
- [CONST_COMP_IO5o](#)
- [CONST_COMP_IO6i](#)
- [CONST_COMP_IO6o](#)
- [CONST_COMP_IO7i](#)
- [CONST_COMP_IO7o](#)
- [CONST_COMP_IO8i](#)
- [CONST_COMP_IO8o](#)
- [CONST_COMP_IO11i](#)
- [CONST_COMP_IO11o](#)
- [CONST_COMP_IO12i](#)
- [CONST_COMP_IO12o](#)

CONST_COMP_IO1

NAME

CONST_COMP_IO1 (CONSTRAINT_COMPARISON)

MESSAGE

Target input port with delay vs. a clock is not in a timing path within the same clock group at the reference level

DEFAULT SEVERITY

Warning

DESCRIPTION

For every block input port IN with input delay versus clock CLK, Litmus finds the top level clock groups of CLK's top level equivalent clock, and checks that there is a path between timing points in those clock groups that traverse IN. This rule is checked by `compare_constraints` command and results can be reported by `report_rule_check` command.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
target_view_name	string	Target View Name
reference_view_name	string	Reference View Name
scope	object	Target Instance
target_port	object	Target Port
target_clock	object	Target clock
relative_reference_clocks	object_list	Reference clock

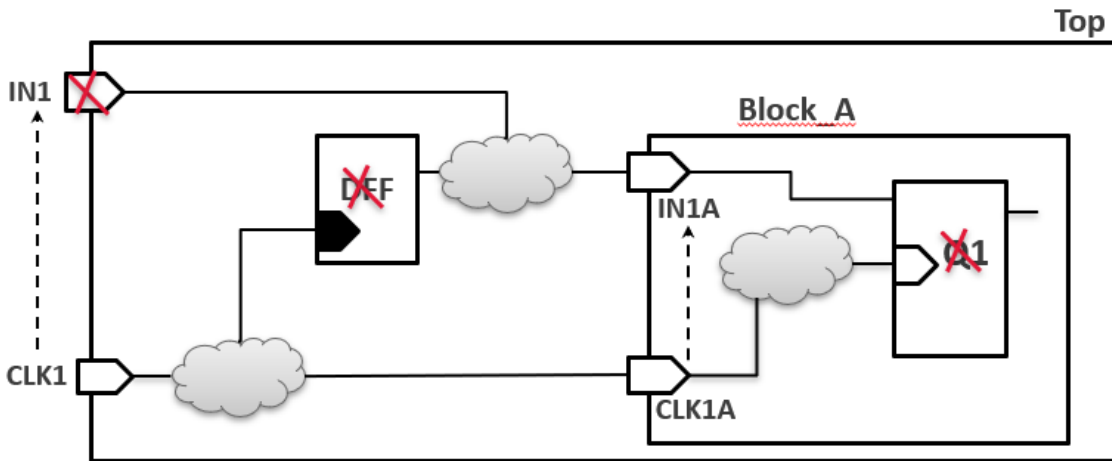
type	object_list	type
target_file_line	object_list	Target file number
target_file_name	object_list	Target file name

In addition, common rule occurrence attribute type provides the possible causes for the violation.

Attribute	Value	Description
type	TOP_NO_LAUNCH_CLOCK	Launch clock absent at top-level
	TOP_NO_CAPTURE_CLOCK	Capture clock absent at top-level
	TOP_UNCONSTRAINED	Top-level is unconstrained
	TOP_ASYNC	Top-level is asynchronous
	TOP_EXCLUSIVE	Top-level is mutually exclusive
	DIFFERENT_CLOCK_MAPPING	Inconsistent clock mapping

EXAMPLE

- The design in the figure below (where IN1A has an input delay versus CLK1A) would cause a rule check violation if the following were not present in the combinational fan-in of IN1A. This rule would also be flagged if no DFF (Q1) or port in the same clock group can be found in the fan-out of IN1A.
 - A DFF that is clocked by the top level clock CLK1
 - A primary input IN1 that has an input delay versus CLK1



- For the figure above, if a top level clock CLK2 is connected to the block level clock CLK1A and the the top level DFF is clocked by another top level clock CLK1 from a different clock group other than CLK2 (asynchronous, defined like the following top level constraints), then this rule check violation will happen for the following block level constraints. Here, the block input port Block_A/IN1A has `set_input_delay` relative to block clock CLK1A which is getting top level equivalent clock CLK2. But at top level, the timing path is not in the same clock group of the top level equivalent clock.

Top level constraints

```
1 create_clock -name CLK1 -period 8 [get_ports CLK1] -waveform {0 4}
2 create_clock -name CLK2 -period 8 [get_ports CLK2] -waveform {0 4}
3 set_clock_groups -name group1 -group {CLK1} -group {CLK2} -asynchronous
```

Block level constraints (instance scope: Block_A)

```
1 create_clock -name CLK1A -period 8 [get_ports CLK1A] -waveform {0 4}
2 set_input_delay 1 -clock CLK1A [get_ports IN1A]
```

CONST_COMP_IO2

NAME

CONST_COMP_IO2 (CONSTRAINT_COMPARISON)

MESSAGE

Block input port is in a timing path within a clock group for which it does not have input delay defined

DEFAULT SEVERITY

Warning

DESCRIPTION

For every target input port IN that has a timing startpoint in its fanin and a timing endpoint in its fanout, such as both belong to the same reference clock group, Litmus checks that IN has input delay defined versus a target clock whose reference equivalent clock is in the same clock group. This rule is checked by `compare_constraints` command and results can be reported by `report_rule_check` command.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
target_view_name	string	Target View Name
reference_view_name	string	Reference View Name
scope	object	Block Instance
target_port	object	Input port of the target
relative_reference_clocks	object_list	Reference clock

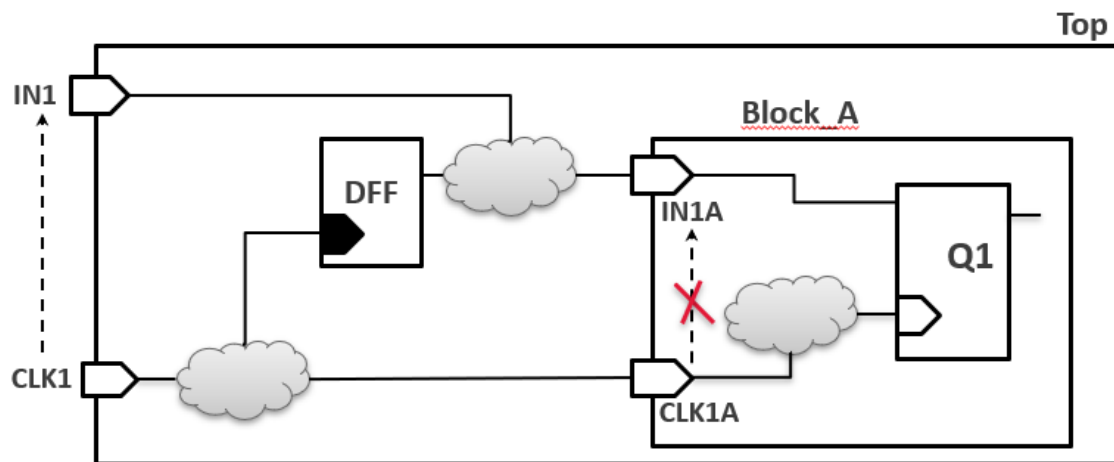
equivalent_target_clocks	object_list	Target clock
--------------------------	-------------	--------------

In addition, common rule occurrence attribute type provides the possible causes for the violation.

Attribute	Value	Description
type	BLOCK_MISSING_INPUT_DELAY	Missing input delay at the target

EXAMPLE

- The design in the following figure would cause a rule check violation, if IN1A did not have an input delay versus CLK1A like the following target level constraints and CLK1A is related to the reference clock CLK1, whose domain drives IN1A:



Top level constraint

```
create_clock -name CLK1 -period 8 [get_ports CLK1] -waveform {0 4}
```

Block level constraints (instance scope: Block_A)

```
create_clock -name CLK1A -period 8 [get_ports CLK1A] -waveform {0 4}
set_input_delay 1 -clock CLK1A [get_ports IN1A]
```

CONST_COMP_IO3

NAME

CONST_COMP_IO3 (CONSTRAINT_COMPARISON)

MESSAGE

Target output port with delay vs. a clock is not in a timing path within the same clock group at the reference level

DEFAULT SEVERITY

Warning

DESCRIPTION

For every block output port OUT with output delay versus clock CLK, Litmus finds the top level clock group of CLK's top level equivalent clock, and checks that there is a path between timing points in that clock group that traverses OUT. This rule is checked by `compare_constraints` command and results can be reported by `report_rule_check` command.

This rule has specific rule occurrence attributes for the details about the violation.

eAttribute	Type	Description
occurrence_id	unsigned	Occurrence Id
target_view_name	string	Target View Name
reference_view_name	string	Reference View Name
scope	object	Target Instance
target_port	object	Target Port
target_clock	object	Target clock
relative_reference_clocks	object_list	Reference clock

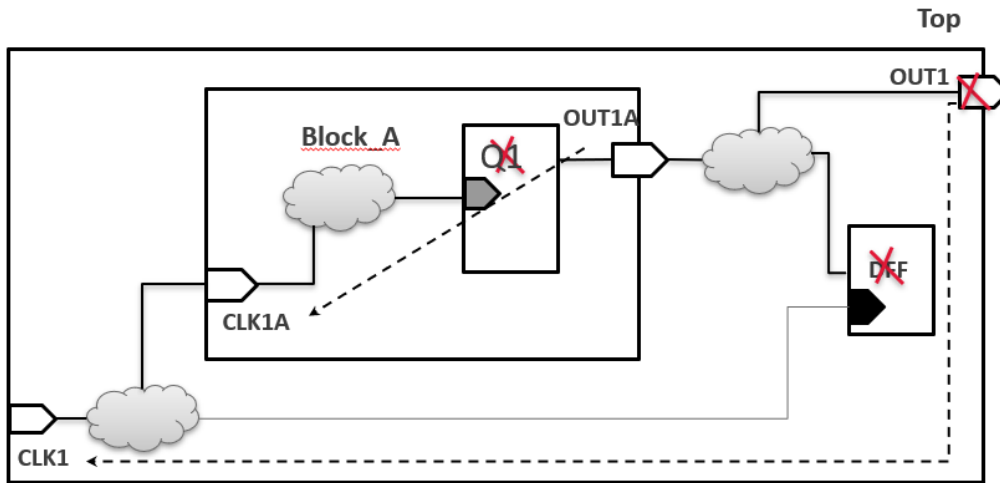
type	object_list	type
target_file_line	object_list	Target file number
target_file_name	object_list	Target file name

In addition, common rule occurrence attribute type provides the possible causes for the violation.

Attribute	Value	Description
type	TOP_NO_LAUNCH_CLOCK	Launch clock is absent at top-level
	TOP_NO_CAPTURE_CLOCK	Capture clock is absent at top-level
	TOP_UNCONSTRAINED	top-level unconstrained
	TOP_ASYNC	Top-level is asynchronous
	TOP_EXCLUSIVE	Top-level is mutually exclusive
	DIFFERENT_CLOCK_MAPPING	Inconsistent clock mapping

EXAMPLE

- The design in the figure below (where OUT1A has a delay versus CLK1A) would cause a rule check violation if none of the following were present in the combinational fanout of OUT1A. This rule would also be flagged if no DFF/port in the same clock group can be found in the fanin of OUT1A.
 - A DFF that is clocked by the top level clock CLK1
 - A pin or primary output OUT1 that has an output delay versus CLK1



- For the figure above, if a top level clock CLK2 is connected to the block level clock CLK1A and the the top level DFF is clocked by another top level clock CLK1 from a different clock group other than CLK2 (asynchronous, defined like the following top level constraints), then this rule check violation will happen for the following block level constraints. Here, the block output port Block_A/OUT1A has `set_output_delay` relative to block clock CLK1A which is getting top level equivalent clock CLK2. But at top level, the timing path is not in the same clock group of the top level equivalent clock.

Top level constraints

```

1 create_clock -name CLK1 -period 8 [get_ports CLK1] -waveform {0 4}
2 create_clock -name CLK2 -period 8 [get_ports CLK2] -waveform {0 4}
3 set_clock_groups -name group1 -group {CLK1} -group {CLK2} -asynchronous

```

Block level constraints (instance scope: Block_A)

```

1 create_clock -name CLK1A -period 8 [get_ports CLK1A] -waveform {0 4}
2 set_output_delay 1 -clock CLK1A [get_ports OUT1A]

```

CONST_COMP_IO4

NAME

CONST_COMP_IO4 (CONSTRAINT_COMPARISON)

MESSAGE

Block output port is in a timing path within a clock group for which it does not have output delay defined

DEFAULT SEVERITY

Warning

DESCRIPTION

For every target output port OUT that has a timing startpoint in its fanin and a timing endpoint in its fanout, such as both belong to the same reference clock group, Litmus checks that OUT has output delay defined versus a target clock whose reference equivalent clock is in the same clock group. This rule is checked by `compare_constraints` command and results can be reported by `report_rule_check` command.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
target_view_name	string	Target View Name
reference_view_name	string	Reference View Name
scope	object	Block Instance
target_port	object	Output port of the target
relative_reference_clocks	object_list	Reference clock

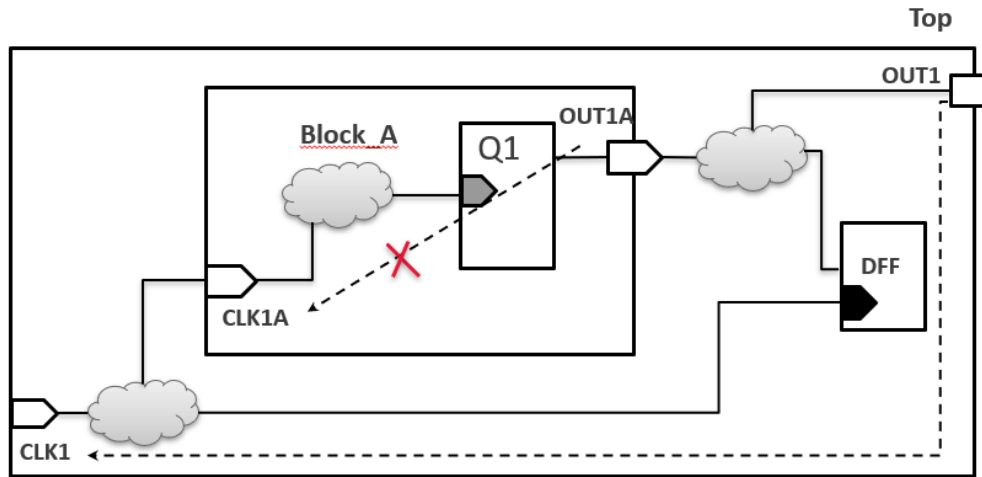
equivalent_target_clocks	object_list	Target clock

In addition, common rule occurrence attribute type provides the possible causes for the violation.

Attribute	Value	Description
type	BLOCK_MISSING_OUTPUT_DELAY	Missing output delay at the target

EXAMPLE

- The design in the following figure would cause a rule check violation, if OUT1A did not have an output delay versus CLK1A like the following target level constraints and CLK1A is related to the reference clock CLK1, whose domain is driven by target port OUT1A:



Top level constraint

```
create_clock -name CLK1 -period 8 [get_ports CLK1] -waveform {0 4}
```

Block level constraints (instance scope: Block_A)

```
create_clock -name CLK1A -period 8 [get_ports CLK1A] -waveform {0 4}  
set_output_delay 1 -clock CLK1A [get_ports OUT1A]
```

CONST_COMP_IO5i

NAME

CONST_COMP_IO5i (CONSTRAINT_COMPARISON)

MESSAGE

Missing `set_input_delay` in full-chip SDC

DEFAULT SEVERITY

Warning

DESCRIPTION

This rule reports that a `set_input_delay` in a target level constraints file has no corresponding `set_input_delay` at the reference constraints file. The corresponding delays are determined based on the clock equivalences and the presence or absence of the `-clock_fall` option. This rule is checked by `compare_constraints` command and results can be reported by `report_rule_check` command.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
target_view_name	string	Target View Name
reference_view_name	string	Reference View Name
scope	object	Target Instance

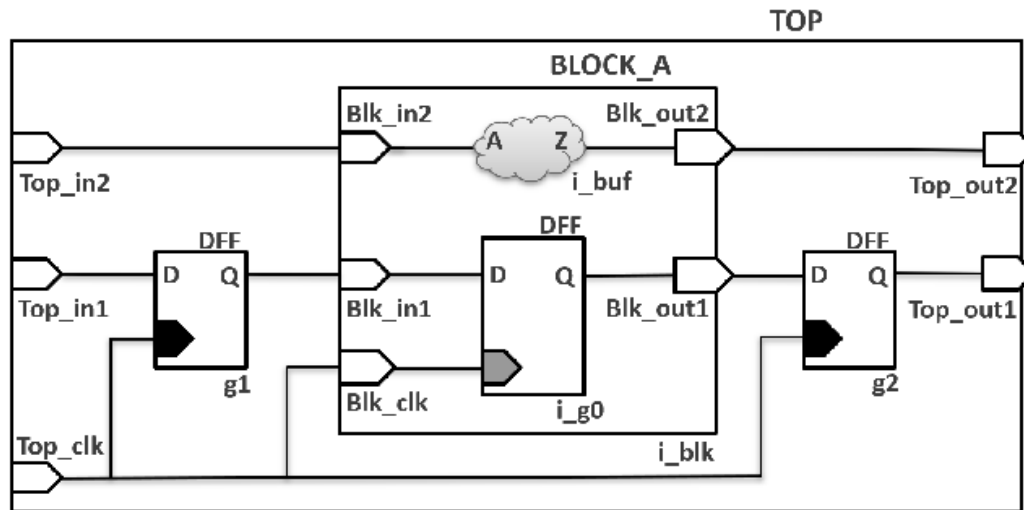
target_object	object	Port/pin withset_input_delayin target/glue constraints
reference_object	object	Reference object
target_clock	object	Target-level clock
type	string	Type
target_value	float	target_value
target_file_line	object_list	Target file number
target_file_name	object_list	Target file name

In addition, common rule occurrence attribute type provides the possible causes for the violation.

Attribute	Value	Description
type	TOP_MISSING_INPUT_DELAY	Missing input delay at the reference

EXAMPLE

- For the figure below, if a port Top_in2 has no input delay vs. reference clock vtop_c1 like the following (with or without `-clock_fall`), this rule will be reported if Top_in2 is connected to target port i_blk/Blk_in2 and the target level constraints file contains the following command:



Top level constraint

```
set_input_delay 1 -clock vtop_c1 [get_ports Top_in2]
```

Block level constraint (instance scope: i_blk)

```
set_input_delay 1 -clock vblk_c11 [get_ports Blk_in2]
```

- For the figure above, this rule will also be reported if the input delay at the target pin i_blk/i_buf/A is set by the target level constraint, but missing the reference constraint like the following:

Top level constraint

```
set_input_delay 1 -clock vtop_c1 [get_pins i_blk/i_buf/A]
```

Block level constraint (instance scope: i_blk)

```
set_input_delay 1 -clock vblk_c11 [get_pins i_buf/A]
```

and the dofile contains the following:

```
set_timing_context_clock_mapping -name hier_clk_map -top_clock vtop_c1 -  
scope_name i_blk -scope_clocks vblk_c11 -force
```

CONST_COMP_IO5o

NAME

CONST_COMP_IO5o (CONSTRAINT_COMPARISON)

MESSAGE

Missing `set_output_delay` in full-chip SDC

DEFAULT SEVERITY

Warning

DESCRIPTION

This rule reports that a `set_output_delay` in a target level constraints file has no corresponding `set_output_delay` at the reference constraints file. The corresponding delays are determined based on the clock equivalences and the presence or absence of the `clock_fall` option. This rule is checked by `compare_constraints` command and results can be reported by `report_rule_check` command.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
target_view_name	string	Target View Name
reference_view_name	string	Reference View Name
scope	object	Target Instance
target_object	object	Port/pin with <code>set_output_delay</code> in target/glue constraints

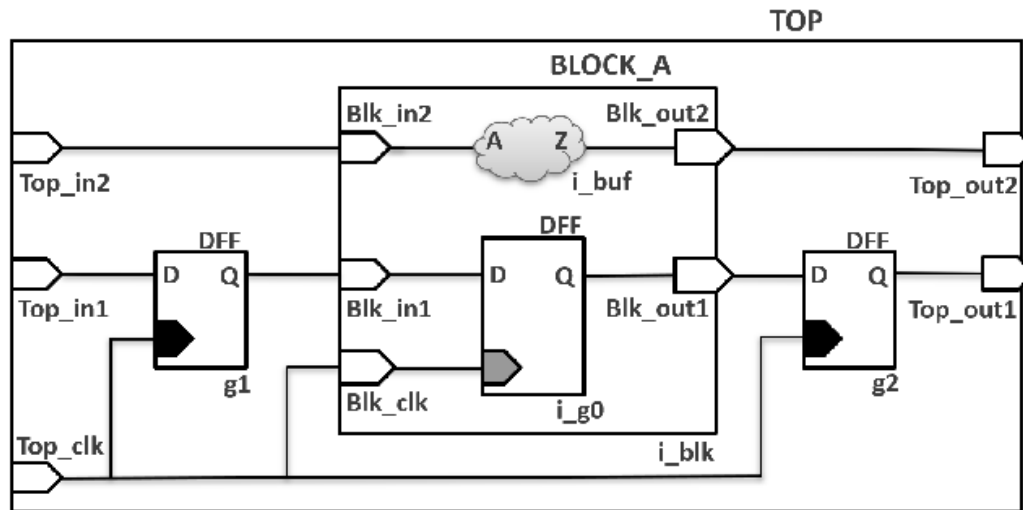
reference_object	object	Reference object
target_clock	object	Target-level clock
type	string	Type
target_value	float	target_value
target_file_line	object_list	Target file number
target_file_name	object_list	Target file name

In addition, common rule occurrence attribute type provides the possible causes for the violation.

Attribute	Value	Description
type	TOP_MISSING_OUTPUT_DELAY	Missing output delay at the reference

EXAMPLE

- For the figure below, if a port Top_out2 has no output delay vs. reference clock vtop_c1 like the following (with or without `-clock_fall`), this rule will be reported if Top_out2 is connected to target port i_blk/Blk_out2 and the target level constraints file contains the following command:



Top level constraint

```
set_output_delay 1 -clock vtop_c1 [get_ports Top_out2]
```

Block level constraint (instance scope: *i_blk*)

```
set_output_delay 1 -clock vblk_c11 [get_ports Blk_out2]
```

- For the figure above, this rule will also be reported if the output delay at the target pin `i_blk/i_buf/Z` is set by the target level constraint, but missing the reference constraint like the following:

Top level constraint

```
set_output_delay 1 -clock vtop_c1 [get_pins i_blk/i_buf/Z]
```

Block level constraint (instance scope: *i_blk*)

```
set_output_delay 1 -clock vblk_c11 [get_pins i_buf/Z]
```

and the dofile contains the following:

```
set_timing_context_clock_mapping -name hier_clk_map -top_clock vtop_c1 -
scope_name i_blk -scope_clocks vblk_c11 -force
```

CONST_COMP_IO6i

NAME

CONST_COMP_IO6i (CONSTRAINT_COMPARISON)

MESSAGE

Missing `set_input_delay` in target/glue SDC

DEFAULT SEVERITY

Warning

DESCRIPTION

This rule reports that a `set_input_delay` in a reference constraints file has no corresponding `set_input_delay` at the target level constraints file. The corresponding delays are determined based on the clock equivalences and the presence or absence of the `-clock_fall` option. This rule is checked by `compare_constraints` command and results can be reported by `report_rule_check` command.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
target_view_name	string	Target View Name
reference_view_name	string	Reference View Name
scope	object	Target Instance
target_object	object	Port/pin without <code>set_input_delay</code> in target/glue constraints

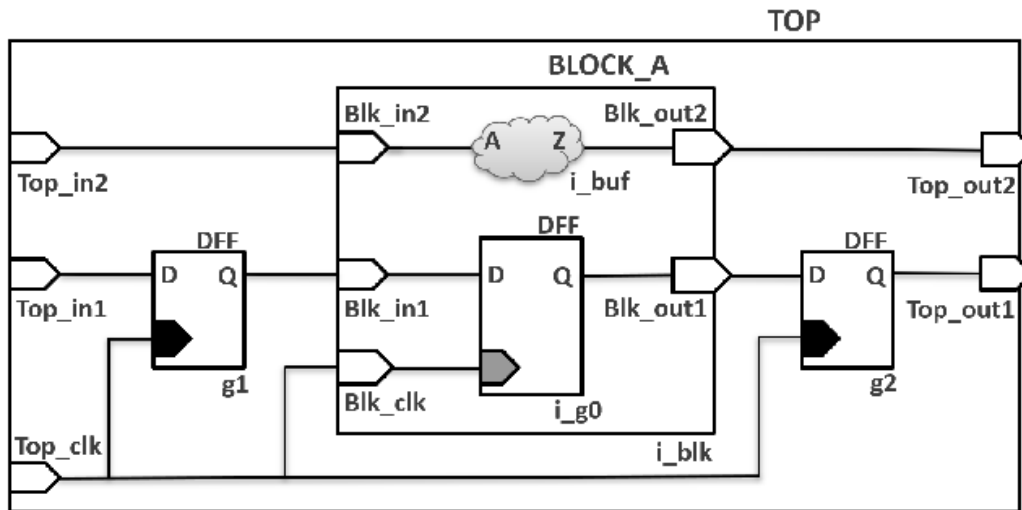
reference_object	object	Reference object
reference_clock	object	Reference clock
type	string	Type
reference_value	float	reference_value
reference_file_line	object_list	Reference file number
reference_file_name	object_list	Reference file name

In addition, common rule occurrence attribute type provides the possible causes for the violation.

Attribute	Value	Description
type	BLOCK_MISSING_INPUT_DELAY	Missing input delay at the target-level

EXAMPLE

- For the figure below, if a port Top_in2 has input delay vs. reference clock vtop_c1 like the following (with or without `-clock_fall`), this rule will be reported if Top_in2 is connected to target port i_blk/Blk_in2 and the target level constraints file does not contain the following command:



Top level constraint

```
set_input_delay 1 -clock vtop_c1 [get_ports Top_in2]
```

Block level constraint (instance scope: i_blk)

```
set_input_delay 1 -clock vblk_c11 [get_ports Blk_in2]
```

- For the figure above, this rule will also be reported if the input delay at the target pin i_blk/i_buf/A is set by the reference constraint, but missing the target level constraint like the following:

Top level constraint

```
set_input_delay 1 -clock vtop_c1 [get_pins i_blk/i_buf/A]
```

Block level constraint (instance scope: i_blk)

```
set_input_delay 1 -clock vblk_c11 [get_pins i_buf/A]
```

and the dofile contains the following:

```
set_timing_context_clock_mapping -name hier_clk_map -top_clock vtop_c1 -  
scope_name i_blk -scope_clocks vblk_c11 -force
```

CONST_COMP_IO6o

NAME

CONST_COMP_IO6o (CONSTRAINT_COMPARISON)

MESSAGE

Missing `set_output_delay` in target/glue SDC

DEFAULT SEVERITY

Warning

DESCRIPTION

This rule reports that a `set_output_delay` in a reference constraints file has no corresponding `set_output_delay` at the target level constraints file. The corresponding delays are determined based on the clock equivalences and the presence or absence of the `-clock_fall` option. This rule is checked by `compare_constraints` command and results can be reported by `report_rule_check` command.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
target_view_name	string	Target View Name
reference_view_name	string	Reference View Name
scope	object	Target Instance
target_object	object	Port/pin without <code>set_output_delay</code> in target/glue constraints

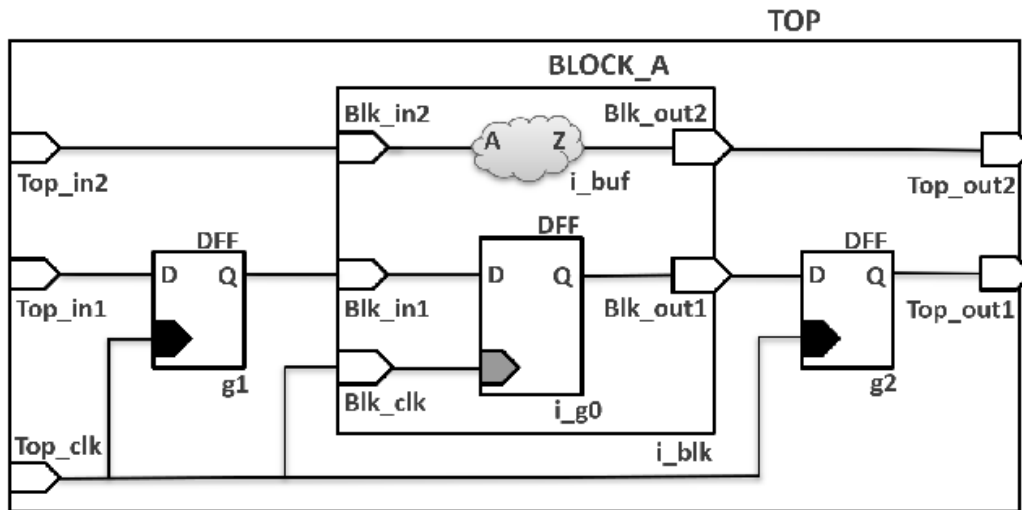
reference_object	object	Reference object
reference_clock	object	Reference clock
type	string	Type
reference_value	float	reference_value
reference_file_line	object_list	Reference file number
reference_file_name	object_list	Reference file name

In addition, common rule occurrence attribute type provides the possible causes for the violation.

Attribute	Value	Description
type	BLOCK_MISSING_OUTPUT_DELAY	Missing output delay at the target-level

EXAMPLE

- For the figure below, if a port Top_out2 has output delay vs. reference clock vtop_c1 like the following (with or without `-clock_fall`), this rule will be reported if Top_out2 is connected to target port i_blk/Blk_out2 and the target level constraints file does not contain the following command:



Top level constraint

```
set_output_delay 1 -clock vtop_cl [get_ports Top_out2]
```

Block level constraint (instance scope: i_blk)

```
set_output_delay 1 -clock vblk_cl1 [get_ports Blk_out2]
```

- For the figure above, this rule will also be reported if the output delay at the target pin i_blk/i_buf/Z is set by the reference constraint, but missing the target level constraint like the following:

Top level constraint

```
set_output_delay 1 -clock vtop_cl [get_pins i_blk/i_buf/Z]
```

Block level constraint (instance scope: i_blk)

```
set_output_delay 1 -clock vblk_cl1 [get_pins i_buf/Z]
```

and the dofile contains the following:

```
set_timing_context_clock_mapping -name hier_clk_map -top_clock vtop_cl -  
scope_name i_blk -scope_clocks vblk_cl1 -force
```

CONST_COMP_IO7i

NAME

CONST_COMP_IO7i (CONSTRAINT_COMPARISON)

MESSAGE

Block/glue `set_input_delay` has smaller value than in full-chip

DEFAULT SEVERITY

Warning

DESCRIPTION

This rule reports that a `set_input_delay` in a target level constraints file has a corresponding `set_input_delay` at the reference constraints file, which has a bigger value. Only delays with `-clock` are considered. The corresponding delays are determined based on the clock equivalences with or without using the `set_timing_context_clock_mapping` command in the dofile and the presence or absence of the `-clock_fall` option. This rule is checked by `compare_constraints` command and results can be reported by `report_rule_check` command.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
target_view_name	string	Target View Name
reference_view_name	string	Reference View Name
scope	object	Target Instance

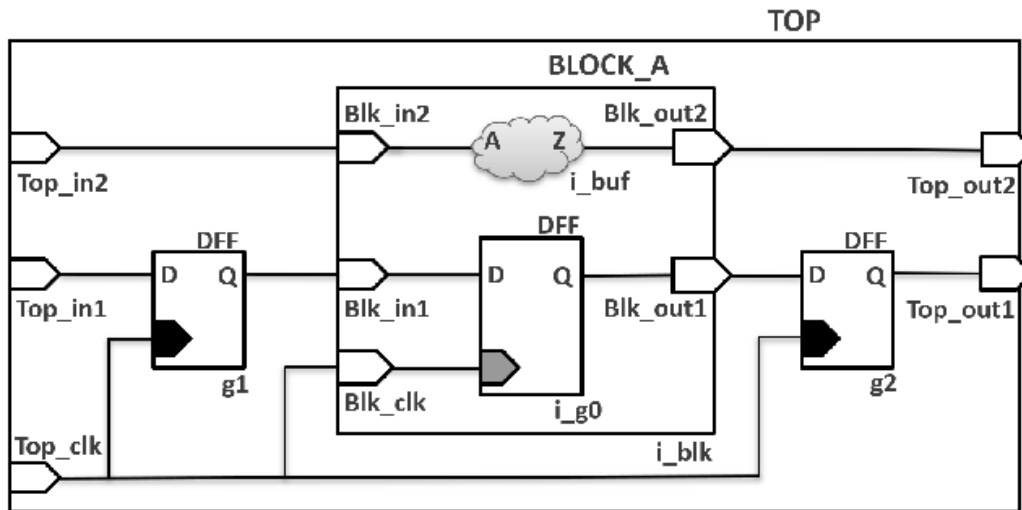
target_object	object	Port/pin in the target
reference_object	object	Reference object
is_directly_connected	bool	Indicates reference pin and target-level pin are directly connected
type	string	Type
target_value	float	target_value
reference_value	float	reference_value
target_file_line	object_list	Target file number
target_file_name	object_list	Target file name
reference_file_line	object_list	Reference file number
reference_file_name	object_list	Reference file name

In addition, common rule occurrence attribute type provides the possible causes for the violation.

Attribute	Value	Description
type	DIFFERENT_INPUT_DELAYS	Inconsistent input delays between reference and target level

EXAMPLE

- For the figure below, this rule will be reported if port Top_in2 is connected to target port i_blk/Blk_in2 and the top and target level constraints file have the following commands:



Top level constraint

```
set_input_delay 2 -clock vtop_c1 [get_ports Top_in2]
```

Block level constraint (instance scope: i_blk)

```
set_input_delay 1 -clock vblk_c11 [get_ports Blk_in2]
```

- For the figure above, this rule will also be reported if the input delay at some target pin i_blk/i_buf/A is set by the top and target level constraints file like the following:

Top level constraint

```
set_input_delay 2 -clock vtop_c1 [get_pins i_blk/i_buf/A]
```

Block level constraint (instance scope: i_blk)

```
set_input_delay 1 -clock vblk_c11 [get_pins i_buf/A]
```

and the dofile contains the following:

```
set_timing_context_clock_mapping -name hier_clk_map -top_clock vtop_c1 -  
scope_name i_blk -scope_clocks vblk_c11 -force
```


CONST_COMP_IO7o

NAME

CONST_COMP_IO7o (CONSTRAINT_COMPARISON)

MESSAGE

Block/glue `set_output_delay` has smaller value than in full-chip

DEFAULT SEVERITY

Warning

DESCRIPTION

This rule reports that a `set_output_delay` in a target level constraints file has a corresponding `set_output_delay` at the reference constraints file, which has a bigger value. Only delays with `-clock` are considered. The corresponding delays are determined based on the clock equivalences with or without using the `set_timing_context_clock_mapping` command in the dofile and the presence or absence of the `-clock_fall` option. This rule is checked by `compare_constraints` command and results can be reported by `report_rule_check` command.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
target_view_name	string	Target View Name
reference_view_name	string	Reference View Name
scope	object	Target Instance

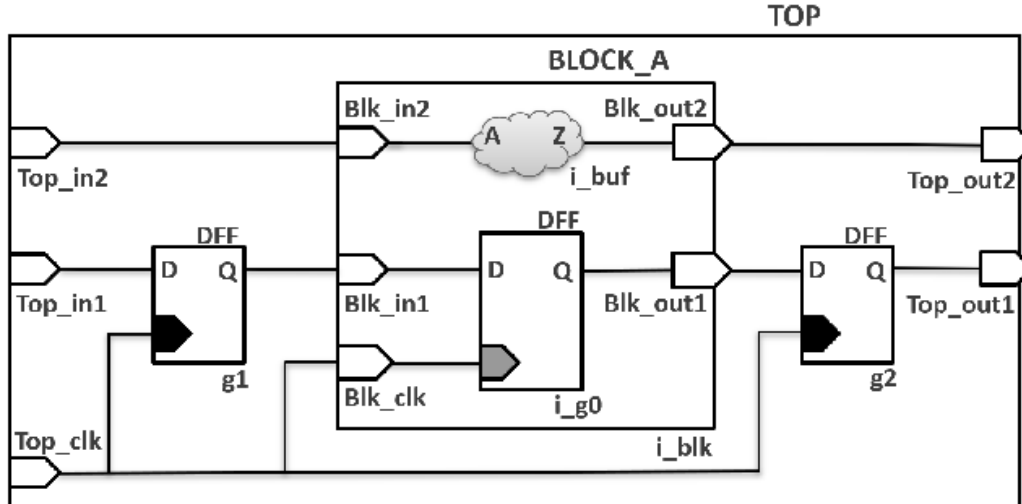
target_object	object	Port/pin in the target
reference_object	object	Reference object
is_directly_connected	bool	Indicates reference pin and target-level pin are directly connected
type	string	Type
target_value	float	target_value
reference_value	float	reference_value
target_file_line	object_list	Target file number
target_file_name	object_list	Target file name
reference_file_line	object_list	Reference file number
reference_file_name	object_list	Reference file name

In addition, common rule occurrence attribute type provides the possible causes for the violation.

Attribute	Value	Description
type	DIFFERENT_OUTPUT_DELAYS	Inconsistent output delays between reference and target level

EXAMPLE

- For the figure below, this rule will be reported if port Top_out2 is connected to target port i_blk/Blk_out2 and the top and target level constraints file have the following commands:



Top level constraint

```
set_output_delay 2 -clock vtop_c1 [get_ports Top_out2]
```

Block level constraint (instance scope: i_blk)

```
set_output_delay 1 -clock vblk_c11 [get_ports Blk_out2]
```

- For the figure above, this rule will also be reported if the output delay at the target pin `i_blk/i_buf/Z` is set by the top and target level constraints file like the following:

Top level constraint

```
set_output_delay 2 -clock vtop_c1 [get_pins i_blk/i_buf/Z]
```

Block level constraint (instance scope: *i_blk*)

```
set_output_delay 1 -clock vblk_c11 [get_pins i_buf/Z]
```

and the dofile contains the following:

```
set_timing_context_clock_mapping -name hier_clk_map -top_clock vtop_c1 -  
scope_name i_blk -scope_clocks vblk_c11 -force
```

CONST_COMP_IO8i

NAME

CONST_COMP_IO8i (CONSTRAINT_COMPARISON)

MESSAGE

Block/glue `set_input_delay` has greater value than in full-chip

DEFAULT SEVERITY

Warning

DESCRIPTION

This rule reports that a `set_input_delay` in a target level constraints file has a corresponding `set_input_delay` at the reference constraints file, which has a smaller value. Only delays with `-clock` are considered. The corresponding delays are determined based on the clock equivalences with or without using the `set_timing_context_clock_mapping` command in the dofile and the presence or absence of the `-clock_fall` option. This rule is checked by `compare_constraints` command and results can be reported by `report_rule_check` command. Note that when this rule is reported for targets, it does not necessarily mean there is an error, unless the targets' ports are connected directly to reference ports. This rule is provided for completeness.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
target_view_name	string	Target View Name
reference_view_name	string	Reference View Name
scope	object	Target Instance

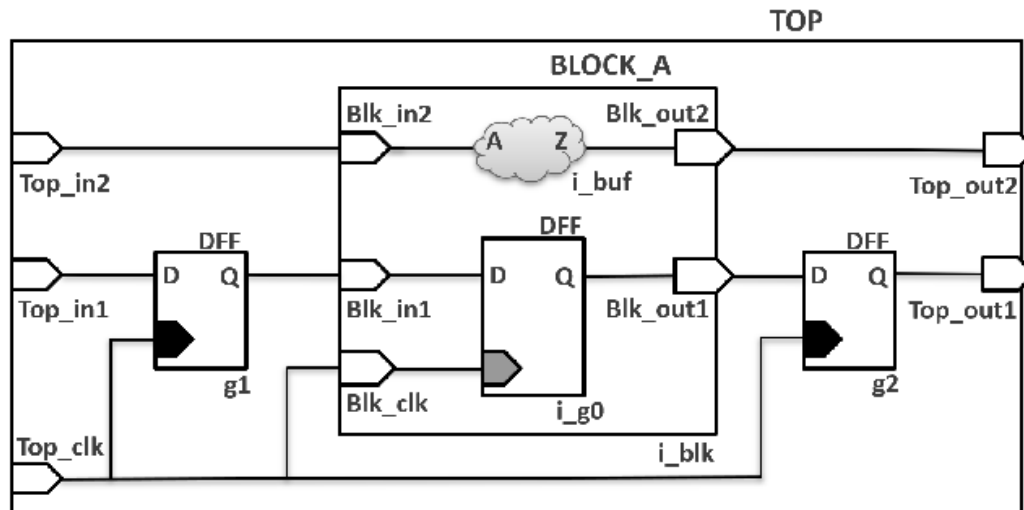
target_object	object	Port/pin in the target
reference_object	object	Reference object
is_directly_connected	bool	Indicates reference pin and target-level pin are directly connected
type	string	Type
target_value	float	target_value
reference_value	float	reference_value
target_file_line	object_list	Target file number
target_file_name	object_list	Target file name
reference_file_line	object_list	Reference file number
reference_file_name	object_list	Reference file name

In addition, common rule occurrence attribute type provides the possible causes for the violation.

Attribute	Value	Description
type	DIFFERENT_INPUT_DELAYS	Inconsistent input delays between reference and target level

EXAMPLE

- For the figure below, this rule will be reported if port Top_in2 is connected to target port i_blk/Blk_in2 and the top and target level constraints file have the following commands:



Top level constraint

```
set_input_delay 1 -clock vtop_c1 [get_ports Top_in2]
```

Block level constraint (instance scope: i_blk)

```
set_input_delay 2 -clock vblk_c11 [get_ports Blk_in2]
```

- For the figure above, this rule will also be reported if the input delay at some target pin i_blk/i_buf/A is set by the top and target level constraints file like the following:

Top level constraint

```
set_input_delay 1 -clock vtop_c1 [get_pins i_blk/i_buf/A]
```

Block level constraint (instance scope: i_blk)

```
set_input_delay 2 -clock vblk_c11 [get_pins i_buf/A]
```

and the dofile contains the following:

```
set_timing_context_clock_mapping -name hier_clk_map -top_clock vtop_c1 -  
scope_name i_blk -scope_clocks vblk_c11 -force
```

CONST_COMP_IO8o

NAME

CONST_COMP_IO8o (CONSTRAINT_COMPARISON)

MESSAGE

Block/glue `set_output_delay` has greater value than in full-chip

DEFAULT SEVERITY

Warning

DESCRIPTION

This rule reports that a `set_output_delay` in a target level constraints file has a corresponding `set_output_delay` at the reference constraints file, which has a smaller value. Only delays with `-clock` are considered. The corresponding delays are determined based on the clock equivalences with or without using the `set_timing_context_clock_mapping` command in the dofile and the presence or absence of the `-clock_fall` option. This rule is checked by `compare_constraints` command and results can be reported by `report_rule_check` command.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
target_view_name	string	Target View Name
reference_view_name	string	Reference View Name
scope	object	Target Instance

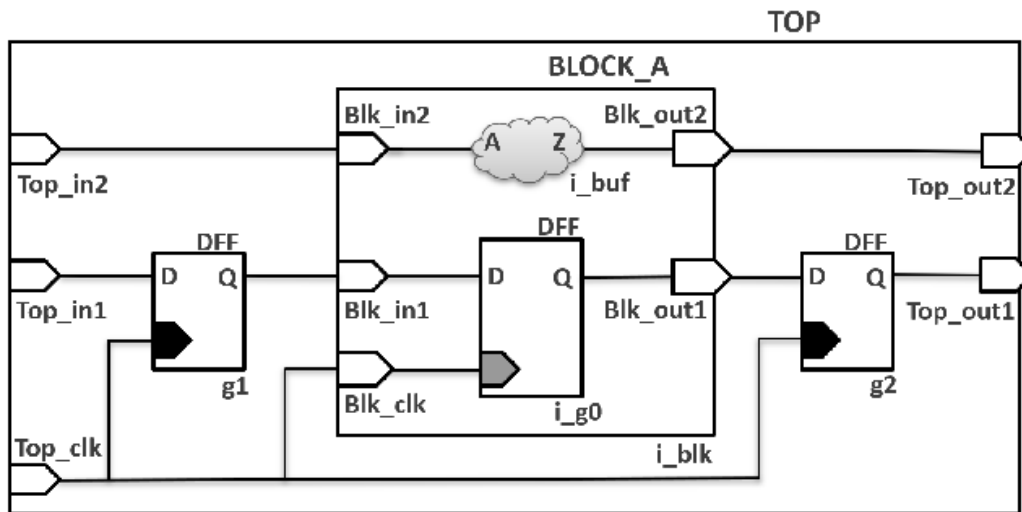
target_object	object	Port/pin in the target
reference_object	object	Reference object
is_directly_connected	bool	Indicates reference pin and target-level pin are directly connected
type	string	Type
target_value	float	target_value
reference_value	float	reference_value
target_file_line	object_list	Target file number
target_file_name	object_list	Target file name
reference_file_line	object_list	Reference file number
reference_file_name	object_list	Reference file name

In addition, common rule occurrence attribute type provides the possible causes for the violation.

Attribute	Value	Description
type	DIFFERENT_OUTPUT_DELAYS	Inconsistent output delays between reference and target level

EXAMPLE

- For the figure below, this rule will be reported if port Top_out2 is connected to target port i_blk/Blk_out2 and the top and target level constraints file have the following commands:



Top level constraint

```
set_output_delay 1 -clock vtop_cl [get_ports Top_out2]
```

Block level constraint (instance scope: i_blk)

```
set_output_delay 2 -clock vblk_cl1 [get_ports Blk_out2]
```

- For the figure above, this rule will also be reported if the output delay at the target pin i_blk/i_buf/Z is set by the top and target level constraints file like the following:

Top level constraint

```
set_output_delay 1 -clock vtop_cl [get_pins i_blk/i_buf/Z]
```

Block level constraint (instance scope: i_blk)

```
set_output_delay 2 -clock vblk_cl1 [get_pins i_buf/Z]
```

and the dofile contains the following:

Litmus Rule Reference
Timing Constraint Comparison (1D2S)--CONST_COMP_IO

```
set_timing_context_clock_mapping -name hier_clk_map -top_clock vtop_c1 -  
scope_name i_blk -scope_clocks vblk_c11 -force
```

CONST_COMP_IO11i

NAME

CONST_COMP_IO11i (CONSTRAINT_COMPARISON)

MESSAGE

Block input is undriven

DEFAULT SEVERITY

Warning

DESCRIPTION

Indicates that a target input is undriven. This rule is checked by `compare_constraints` command and the results can be reported by `report_rule_check` command.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
target_view_name	string	Target View Name
reference_view_name	string	Reference View Name
scope	object	Target Instance
fstaref	object	Port/pin in the target
output	bool	Input or output in the target

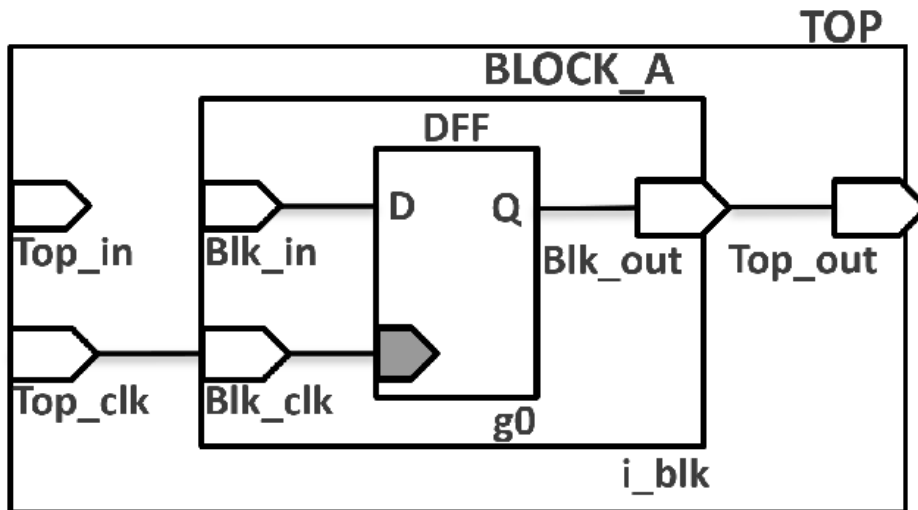
found_obj	object	Start/end point
type	string	Type
is_found	bool	Indicates if the object is found or not

In addition, common rule occurrence attribute type provides the possible causes for the violation.

Attribute	Value	Description
type	BLOCK_UNCONNECTED_INPUT	Reference-level input is not connected to the target-level

EXAMPLE

- For the figure below, the following source code contains an undriven target input. This rule is reported for the undriven target input i_blk/Blk_in.



```
module TOP (input Top_in, Top_clk, output Top_out);  
    BLOCK_A i_blk (.Blk_in(), .Blk_clk(Top_clk), .Blk_out(Top_out));  
endmodule
```

CONST_COMP_IO11o

NAME

CONST_COMP_IO11o (CONSTRAINT_COMPARISON)

MESSAGE

Block output is floating

DEFAULT SEVERITY

Warning

DESCRIPTION

Indicates that a target output is floating. This rule is checked by `compare_constraints` command and the results can be reported by `report_rule_check` command.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
target_view_name	string	Target View Name
reference_view_name	string	Reference View Name
scope	object	Target Instance
fstaref	object	Port/pin in the target
output	bool	Indicates input or output in the target
found_obj	object	Start/end point
type	string	Type

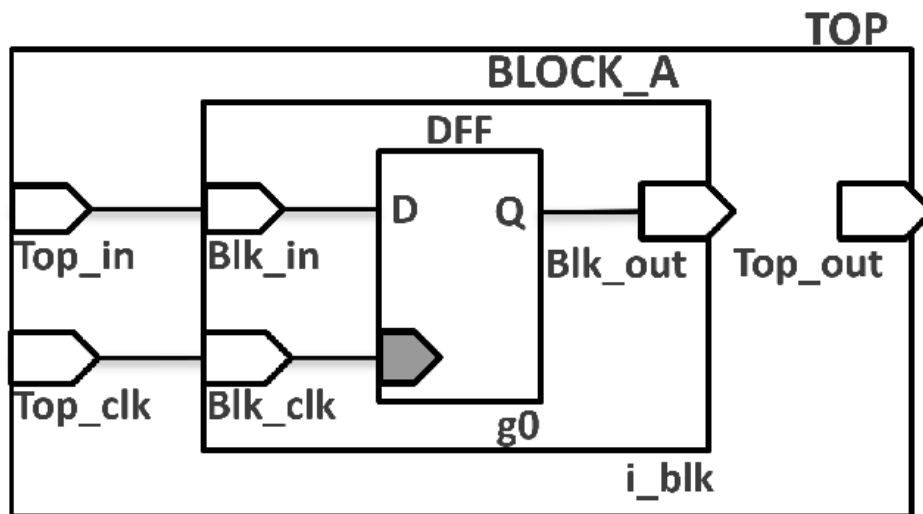
is_found	bool	Indicates if the object is found or not

In addition, common rule occurrence attribute type provides the possible causes for the violation.

Attribute	Value	Description
type	BLOCK_UNCONNECTED_OUTPUT	Block is not connected to the output at reference

EXAMPLE

- For the figure below, the following source code contains a floating target output. This rule is reported for the floating target output i_blk/Blk_out.



```
module TOP (input Top_in, Top_clk, output Top_out);
    BLOCK_A i_blk (.Blk_in(Top_in), .Blk_clk(Top_clk), .Blk_out());
endmodule
```

CONST_COMP_IO12i

NAME

CONST_COMP_IO12i (CONSTRAINT_COMPARISON)

MESSAGE

Inconsistent `set_input_delay` at the reference vs. target

DEFAULT SEVERITY

Warning

DESCRIPTION

Indicates that an inconsistency in `set_input_delay` command is found between reference and target level constraints. The following are the possible causes for this violation:

- `set_input_delay` commands specified at both reference and target level but with different values

This rule is checked by `compare_constraints` command and the results can be reported by `report_rule_check` command.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
target_view_name	string	Target View Name
reference_view_name	string	Reference View Name
scope	object	Target Instance

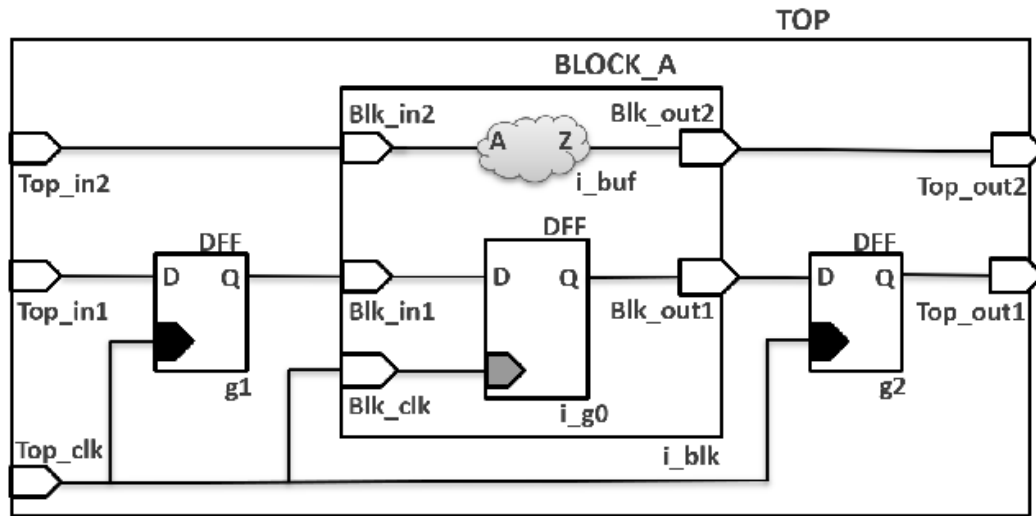
target_pin	object	Target Pin
reference_pin	object	Reference Pin
target_clock	object	Target clock
reference_clock	object	Reference clock
type	string	Type
target_file_line	object_list	Target file number
target_file_name	object_list	Target file name
reference_file_line	object_list	Reference file number
reference_file_name	object_list	Reference file name

In addition, common rule occurrence attribute `type` provides the possible causes for the violation.

Attribute	Value	Description
type	DIFFERENT_INPUT_DELAYS	Inconsistent input delays at reference and target-level

EXAMPLE

- For the figure below, the following example contains `set_input_delay` commands specified in both reference and target level constraints. This rule is reported for the `set_input_delay` commands, because they are specified with different values (-reference_pin) provided the reference clock `top_vclk` and the target level clock `blk_vclk` are mapped using `set_timing_context_clock_mapping` command in the dofile.



Top level constraints

```
create_clock -name top_vclk -period 2
set_input_delay -clock [get_clocks top_vclk] -reference_pin [get_pins
i_blk/Blk_in1] 1.0 [get_pins i_blk/i_buf/Z]
```

Block level constraints (instance scope: i_blk)

```
create_clock -name blk_vclk -period 2
set_input_delay -clock [get_clocks blk_vclk] -reference_pin [get_pins i_buf/A]
1.0 [get_pins i_buf/Z]
```

Hierarchical clock mapping

```
top_vclk is mapped to blk_vclk
```

CONST_COMP_IO12o

NAME

CONST_COMP_IO12o (CONSTRAINT_COMPARISON)

MESSAGE

Inconsistent `set_output_delay` at the reference vs. target

DEFAULT SEVERITY

Warning

DESCRIPTION

Indicates that an inconsistency in `set_output_delay` command is found between reference and target level constraints. The following are the possible causes for this violation:

- `set_output_delay` commands specified at both reference and target level but with different values

This rule is checked by `compare_constraints` command and the results can be reported by `report_rule_check` command.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
target_view_name	string	Target View Name
reference_view_name	string	Reference View Name
scope	object	Target Instance
target_pin	object	Target Pin

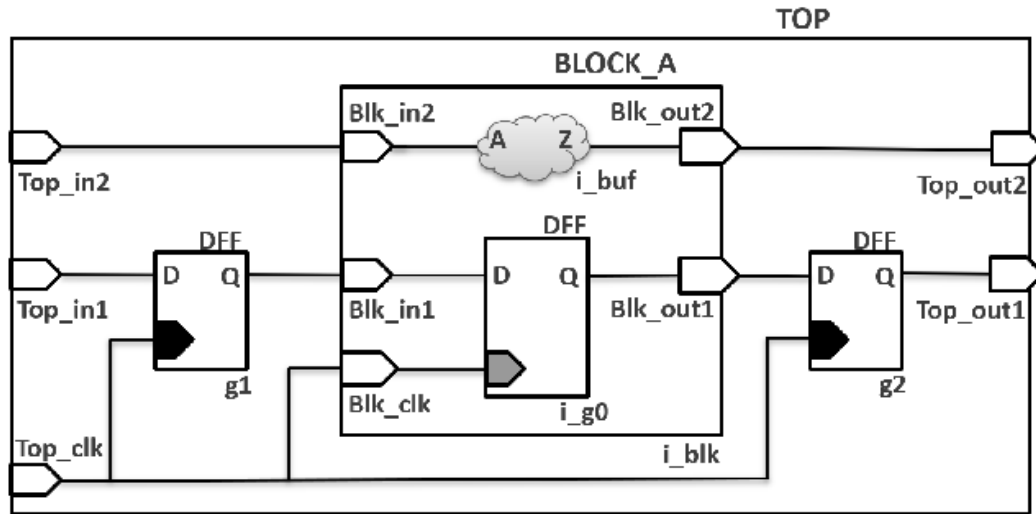
reference_pin	object	Reference Pin
target_clock	object	Target clock
reference_clock	object	Reference clock
type	string	Type
target_file_line	object_list	Target file number
target_file_name	object_list	Target file name
reference_file_line	object_list	Reference file number
reference_file_name	object_list	Reference file name

In addition, common rule occurrence attribute `type` provides the possible causes for the violation.

Attribute	Value	Description
type	DIFFERENT_OUTPUT_DELAYS	Inconsistent output delays

EXAMPLE

- For the figure below, the following example contains `set_output_delay` commands specified in both reference and target level constraints. This rule is reported for the `set_output_delay` commands, because they are specified with different values (-reference_pin) provided the reference clock `top_vclk` and the target level clock `blk_vclk` are mapped using `set_timing_context_clock_mapping` command in the dofile.



Top level constraints

```
create_clock -name top_vclk -period 2
set_output_delay -clock [get_clocks top_vclk] -reference_pin [get_pins
i_blk/Blk_in1] 1.0 [get_pins i_blk/i_buf/Z]
```

Block level constraints (instance scope: i_blk)

```
create_clock -name blk_vclk -period 2
set_output_delay -clock [get_clocks blk_vclk] -reference_pin [get_pins i_buf/A]
1.0 [get_pins i_buf/Z]
```

Hierarchical clock mapping

```
top_vclk is mapped to blk_vclk
```

CONST_COMP_ITR

- [CONST_COMP_ITR1](#)

CONST_COMP_ITR1

NAME

CONST_COMP_ITR1 (CONSTRAINT_COMPARISON)

MESSAGE

Inconsistent input transition at the reference-level vs. target

DEFAULT SEVERITY

Warning

DESCRIPTION

Indicates that an inconsistency in input transition command ([set_input_transition](#) / [set_annotated_transition](#) / [set_driving_cell](#)) is found between reference level and target level constraints. The following are the possible causes for this violation:

- Input transition command specified at either reference level or target level
- Input transition commands specified at both reference level and target level but with different values

This rule is checked by [compare_constraints](#) command and the results can be reported by [report_rule_check](#) command.

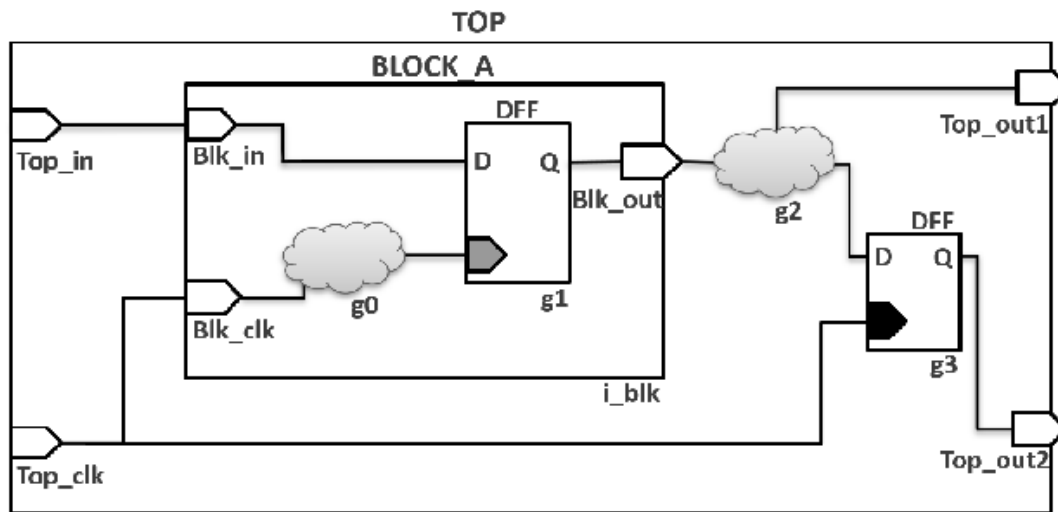
This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
target_view_name	string	Target view name
reference_view_name	string	Reference view name
scope	object	Scope

target_pin	object	Target port
reference_pin	object	Reference port
target_value	object_list	Target transition value
reference_value	object_list	Reference transition value
inconsistent_target_type	object_list	Inconsistent target type
inconsistent_reference_type	object_list	Inconsistent reference type
type	string	Type
target_file_line	object_list	Target file line number
target_file_name	object_list	Target file name
reference_file_line	object_list	Reference file line number
reference_file_name	object_list	Reference file name

EXAMPLE

- The following example contains `set_input_transition` commands specified in both reference level and target level constraints. This rule is reported for the `set_input_transition` commands, because target level transition value is different from reference level transition value.



Reference level constraints

```
set_input_transition 1.0 [get_ports Top_in]
```

Target level constraints

```
set_input_transition 0.5 [get_ports Top_in]
```

CONST_COMP_LAT

- [CONST_COMP_LAT1](#)

CONST_COMP_LAT1

NAME

CONST_COMP_LAT1 (CONSTRAINT_COMPARISON)

MESSAGE

Inconsistent `set_clock_latency` at the reference vs. target

DEFAULT SEVERITY

Warning

DESCRIPTION

Indicates that an inconsistency in `set_clock_latency` command is found between reference and target level constraints. The following are the possible causes for this violation:

- `set_clock_latency` command specified at either reference or target level
- `set_clock_latency` commands specified at both reference and target level but with different values

This rule is checked by `compare_constraints` command and the results can be reported by `report_rule_check` command.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
target_view_name	string	Target View Name
reference_view_name	string	Reference View Name
scope	object	Target Instance

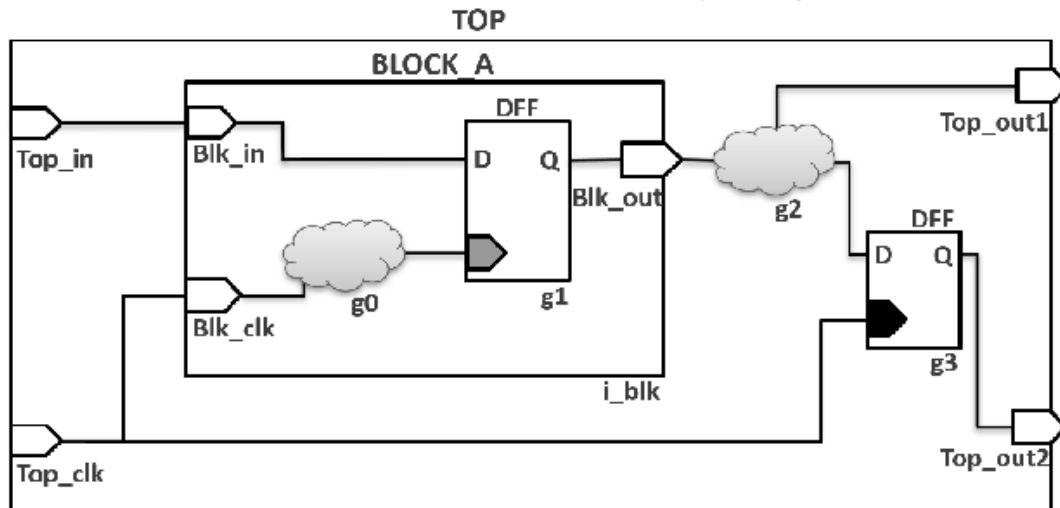
target_pin	object	Target Pin
reference_pin	object	Reference Pin
target_clock	object	Target clock
reference_clock	object	Reference clock
is_clock_source_latency	bool	is_clock_source_latency
type	string	Type
target_file_line	unsigned	Target file number
target_file_name	string	Target file name
reference_file_line	unsigned	Reference file number
reference_file_name	string	Reference file name

In addition, common rule occurrence attribute `type` provides the possible causes for the violation.

Attribute	Value	Description
type	BLOCK_NO_CONSTRAINT	Missing constraint at target level
	TOP_NO_CONSTRAINT	Missing constraint at reference
	DIFFERENT_CONSTRAINTS	Inconsistent constraints between reference and target level

EXAMPLE

- For the figure below, the following constraints contain `set_clock_latency` commands specified in both reference and target level constraints. This rule is reported for the `set_clock_latency` commands, because they are specified with different values.



Top level constraints

```
create_clock -name Top_clk -period 2 [get_ports Top_clk]
set_clock_latency 0.5 [get_clocks Top_clk]
```

Block level constraints (instance scope: i_blk)

```
create_clock -name Blk_clk -period 2 [get_ports Blk_clk]
set_clock_latency 1.0 [get_clocks Blk_clk]
```

Hierarchical clock mapping

```
Top_clk is mapped to Blk_clk
```

CONST_COMP_OLD

- [CONST_COMP_OLD1](#)

CONST_COMP_OLD1

NAME

CONST_COMP_OLD1 (CONSTRAINT_COMPARISON)

MESSAGE

Inconsistent [set_load](#) at the reference level vs. target

DEFAULT SEVERITY

Warning

DESCRIPTION

Indicates that an inconsistency in [set_load](#) command is found between reference level and target level constraints. The following are the possible causes for this violation:

- [set_load](#) command specified at either reference level or target level
- [set_load](#) commands specified at both reference level and target level but with different values

This rule is checked by [compare_constraints](#) command and the results can be reported by [report_rule_check](#) command.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
target_view_name	string	Target view name

reference_view_name	string	Reference view name
scope	object	Scope
target_object	object	Target object
reference_object	object	Reference object
target_net	object	Target Net
reference_net	object	Reference Net
type	string	Type
target_value	object_list	Target value
reference_value	object_list	Reference value
target_file_line	object_list	Target file line number
target_file_name	object_list	Target file name
reference_file_line	object_list	Reference file line number
reference_file_name	object_list	Reference file name

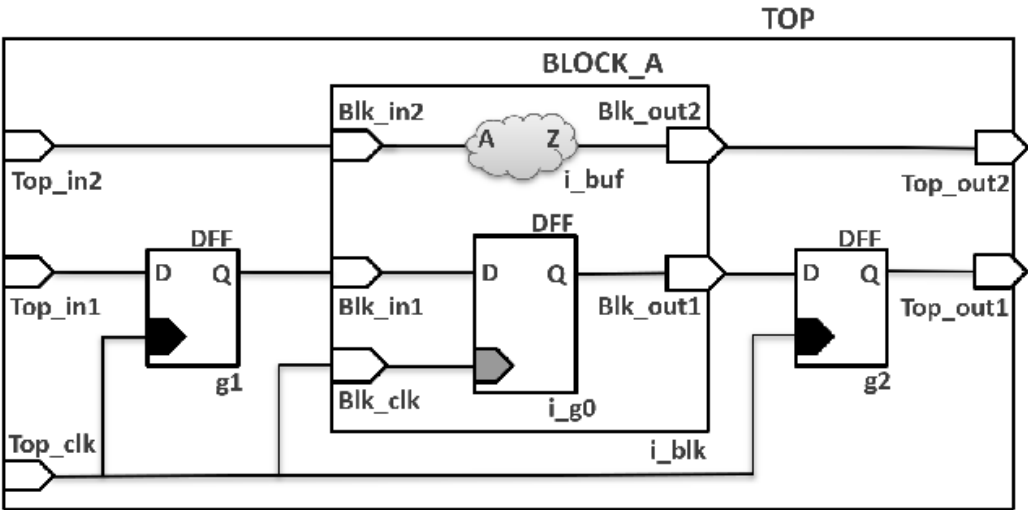
In addition, common rule occurrence attribute `type` provides the possible causes for the violation.

Attribute	Value	Description
type	TARGET_NO_CONSTRAINT	Missing constraint at target level
	REFERENCE_NO_CONSTRAINT	Missing constraint at reference level

	DIFFERENT_CONSTRAINTS	Inconsistent constraints between reference level and target level

EXAMPLE

- For the figure below, the following contains `set_load` commands specified in both reference level and target level constraints. This rule is reported for the `set_load` commands, because they are specified with different values.



Reference level constraint

```
set_load 0.5 [get_ports Top_out1]
```

Target level constraint

```
set_load 1.0 [get_ports Top_out1]
```


CONST_COMP_PLS

- [CONST_COMP_PLS1](#)

CONST_COMP_PLS1

NAME

CONST_COMP_PLS1 (CONSTRAINT_COMPARISON)

MESSAGE

Inconsistent `set_min_pulse_width` at the reference vs. target

DEFAULT SEVERITY

Warning

DESCRIPTION

Indicates that an inconsistency in `set_min_pulse_width` command is found between reference and target level constraints. The following are the possible causes for this violation:

- `set_min_pulse_width` command specified at either reference or target level
- `set_min_pulse_width` commands specified at both reference and target level but with different values

This rule is checked by `compare_constraints` command and the results can be reported by `report_rule_check` command.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
target_view_name	string	Target View Name
reference_view_name	string	Reference View Name
scope	object	Target Instance

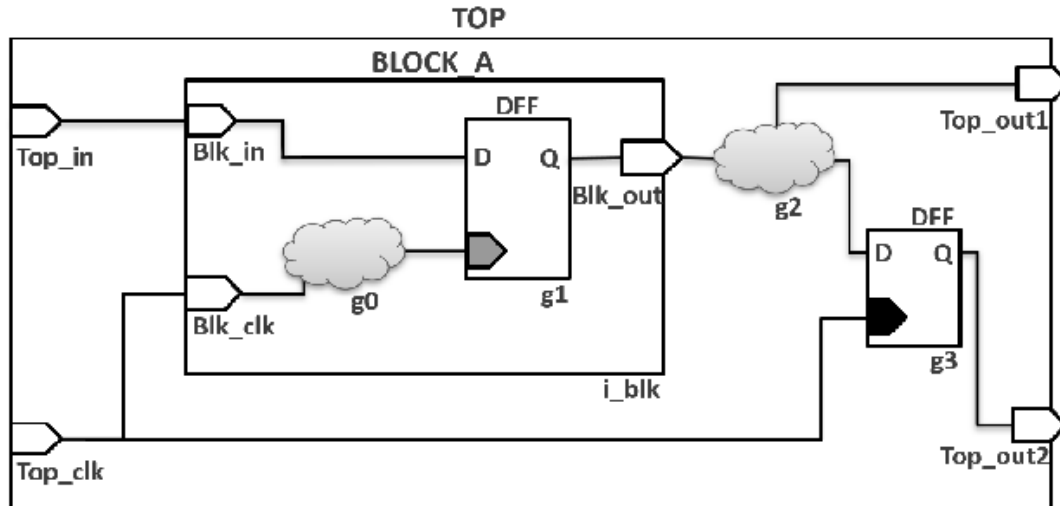
target_object	object	Block object with inconsistentset_min_pulse_width
reference_object	object	Reference object
target_clock	object	Target clock
reference_clock	object	Reference clock
type	string	Type
target_file_line	unsigned	Target file number
target_file_name	string	Target file name
reference_file_line	unsigned	Reference file number
reference_file_name	string	Reference file name

In addition, common rule occurrence attribute `type` provides the possible causes for the violation.

Attribute	Value	Description
type	BLOCK_NO_CONSTRAINT	Missing constraint at target level
	TOP_NO_CONSTRAINT	Missing constraint at reference
	DIFFERENT_CONSTRAINTS	Inconsistent constraints between reference and target level

EXAMPLE

- For the figure below, the following constraints contain `set_min_pulse_width` commands specified in both reference and target level constraints. This rule is reported for the `set_min_pulse_width` commands, because they are specified with different values.



Top level constraints

```
create_clock -name Top_clk -period 2 [get_ports Top_clk]
set_min_pulse_width 0.4 [get_clocks Top_clk]
```

Block level constraints (instance scope: i_blk)

```
create_clock -name Blk_clk -period 2 [get_ports Blk_clk]
set_min_pulse_width 0.8 [get_clocks Blk_clk]
```

Hierarchical clock mapping

```
Top_clk is mapped to Blk_clk
```

CONST_COMP_SCA

- [CONST_COMP_SCA1](#)

CONST_COMP_SCA1

NAME

CONST_COMP_SCA1 (CONSTRAINT_COMPARISON)

MESSAGE

Inconsistent `set_case_analysis` at the reference vs. target

DEFAULT SEVERITY

Warning

DESCRIPTION

This rule reports that there is a `set_case_analysis` on a target object that does not have a tied value at the reference. Every target object with `set_case_analysis` at the reference, and every target port with a tied value at the reference, should have a consistent `set_case_analysis` at the target. This rule is checked by `compare_constraints` command and results can be reported by `report_rule_check` command.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
target_view_name	string	Target View Name
reference_view_name	string	Reference View Name
scope	object	Target Instance
target_value	string	Internal value of <code>set_case_analysis</code> for target object

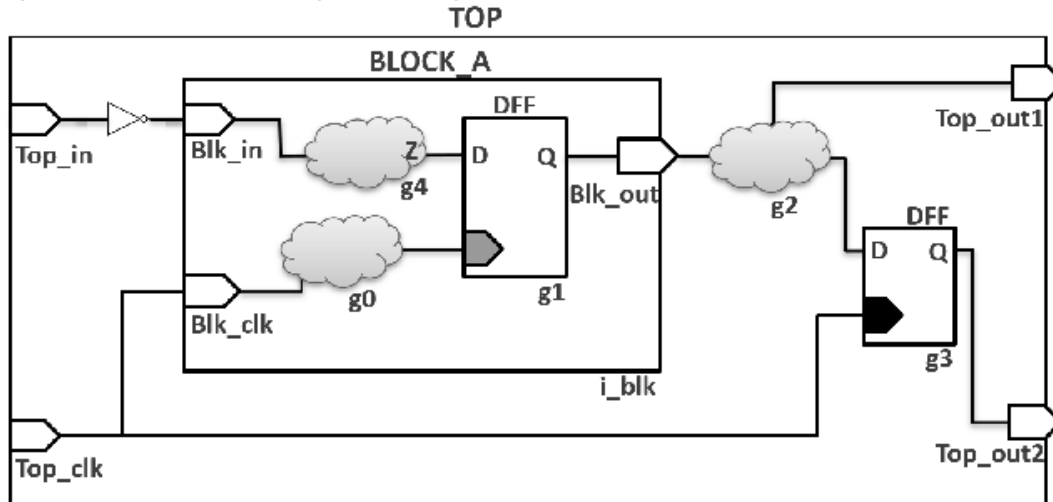
reference_value	string	Reference Value
target_pin	object	Block pin withset_case_analysis
reference_pin	object	Reference Pin
violation_type	string	Violation Type
target_file_line	unsigned	Target file number
target_file_name	string	Target file name
reference_file_line	unsigned	Reference file number
reference_file_name	string	Reference file name

In addition, common rule occurrence attribute `type` provides the possible causes for the violation.

Attribute	Value	Description
type	BLOCK_NO_CONSTRAINT	Missing constraint at target level
	TOP_NO_CONSTRAINT	Missing constraint at reference
	DIFFERENT_CONSTRAINTS	Inconsistent constraints between reference and target level

EXAMPLE

- For the figure below, the target level constraints file has this following constraint. The Litmus issues a warning if the reference constraints file does not have a `set_case_analysis` specified that causes pin `i_blk/g4/Z` to be tied to 1.



Block level constraint (instance scope: `i_blk`)

```
set_case_analysis 1 [get_pins g4/Z]
```

- For the above figure, the top and the target level constraints files have the following constraints, respectively, where `i_blk/Blk_in` gets the inverted value of `Top_in`. The Litmus issues a warning because `i_blk/Blk_in` is tied to 1 at the reference.

Top level constraint

```
set_case_analysis 0 [get_ports Top_in]
```

Block level constraint (instance scope: `i_blk`)

```
set_case_analysis 0 [get_ports Blk_in]
```


CONST_COMP_SDT

- [CONST_COMP_SDT1](#)

CONST_COMP_SDT1

NAME

CONST_COMP_SDT1 (CONSTRAINT_COMPARISON)

MESSAGE

Inconsistent `set_disable_timing` at the reference vs. target

DEFAULT SEVERITY

Warning

DESCRIPTION

This rule reports that `set_disable_timing` is used at the target, but not at the reference, or vice versa. For `set_disable_timing` on library cell or library pin objects, the check is done only if the target instantiates the library cell. This rule is checked by `compare_constraints` command and results can be reported by `report_rule_check` command.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
target_view_name	string	Target View Name
reference_view_name	string	Reference View Name
scope	object	Target Instance
from_pin	object_list	from pin
to_pin	object	to pin

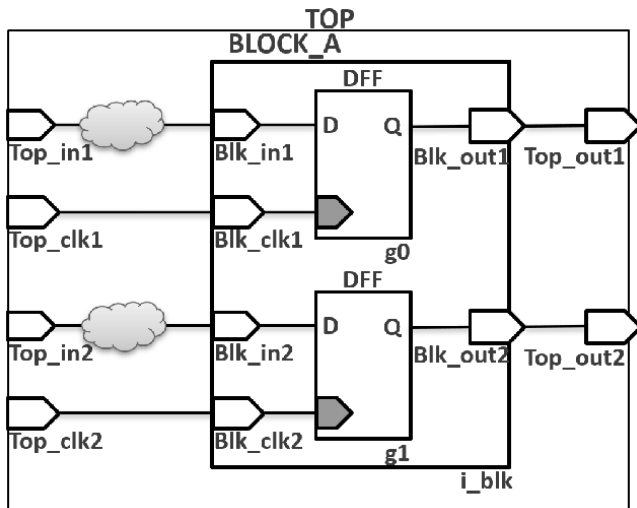
Litmus Rule Reference

Timing Constraint Comparison (1D2S)--CONST_COMP_SDT

violation_type	string	Violation Type
target_file_line	unsigned	Target file number
target_file_name	string	Target file name
reference_file_line	unsigned	Reference file number
reference_file_name	string	Reference file name

EXAMPLE

- For the figure below, if the target level constraints file has the line below but the reference constraints file does not have the line like the following, then the Litmus issues a warning because `set_disable_timing` is specified at the target level, but not at the reference:



Top level constraint

```
set_disable_timing [get_cells i_blk/g0]
```

Block level constraint (instance scope: i_blk)

```
set_disable_timing [get_cells g0]
```

- For the figure above, if the reference constraints file has the following line, but the target level constraints file does not have the same line, then the Litmus also issues a warning only if `i_blk` has an instance of cell DFF from library MYLIB:

Top level constraint

```
set_disable_timing [get_lib_pins MYLIB/DFF/CK]
```

CONST_COMP_TIM

- [CONST_COMP_TIM1](#)
- [CONST_COMP_TIM2](#)

CONST_COMP_TIM1

NAME

CONST_COMP_TIM1 (CONSTRAINT_COMPARISON)

MESSAGE

A fragment of target level timing path cannot be matched to a corresponding fragment of reference level timing path

DEFAULT SEVERITY

Warning

DESCRIPTION

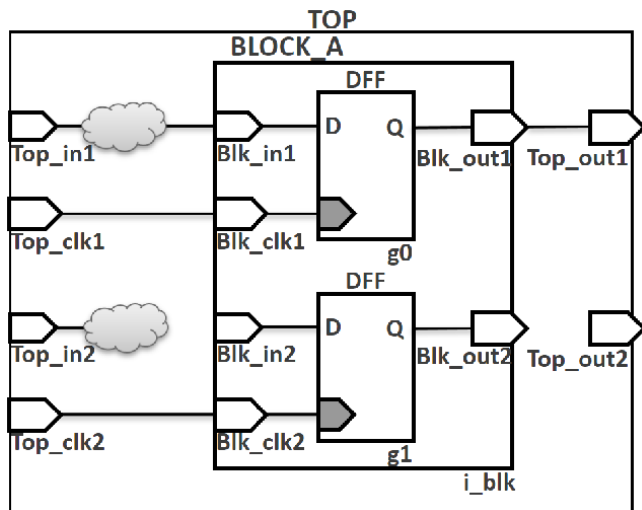
Indicates that the software was not able to identify a fragment of the path in the reference level that includes, or is identical to, the fragment of the path in the target level. This rule can be reported as a consequence of inconsistent [set_disable_timing](#) between the reference level and the target level designs. That problem would be reported by rule [CONST_COMP_SDT1](#). In this cases, making sure that there are sufficient [set_disable_timing](#) commands at both levels that are consistent and eliminating all occurrences of [CONST_COMP_SDT1](#) will solve the issue. If there are still occurrences of this rule, then that could be because of inconsistent application of SDC constraints that cause path splitting (such as [set_input_delay](#)) between the target and the reference level. The occurrence of this rule points to an inconsistency between the timing graphs being compared, which can negatively affect the quality of results of the hierarchical checks. This rule is checked by [compare_constraints](#) command and results can be reported by [report_rule_check](#) command.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
target_view_name	string	Target view name
reference_view_name	string	Reference view name
scope	object	Scope
target_pin	object	Target pin
reference_pin	object	Reference pin
target_clock	object	Target clock
reference_clock	object	Reference clock

EXAMPLE

- For the figure below, the following contains `set_disable_timing` command specified in reference level constraints. This rule is reported because the target level timing path from `i_blk/g0/CK` to `Top_out1` cannot be matched to corresponding reference level timing path.



Reference level constraint

```
set_disable_timing [get_cells i_blk/g0]
```

CONST_COMP_TIM2

NAME

CONST_COMP_TIM2 (CONSTRAINT_COMPARISON)

MESSAGE

Inconsistent `set_annotated_delay` at the top-level vs. block

DEFAULT SEVERITY

Warning

DESCRIPTION

Indicates that an inconsistency in `set_annotated_delay` command is found between top level and block level constraints. The following are the possible causes for this violation:

- `set_annotated_delay` command specified at either top level or block level
- `set_annotated_delay` commands specified at both top level and block level but with different values

This rule is checked by `compare_constraints` command and the results can be reported by `report_rule_check` command.

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
target_view_name	string	Target View Name
reference_view_name	string	Reference View Name

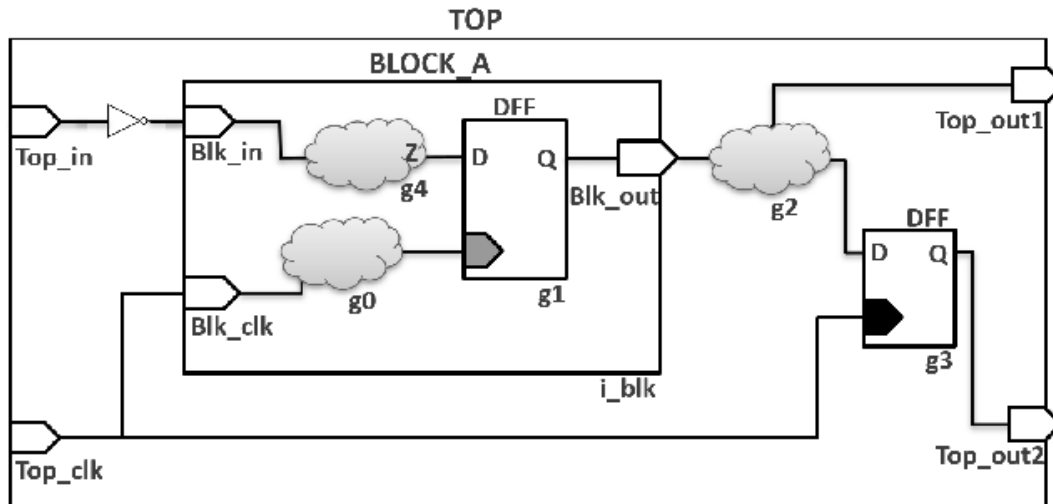
scope	object	Target Instance
reference_from	object	Reference from object
reference_to	object	Reference to object
target_from	object	Target from object
target_to	object	Target to object
reference_object	object	Reference object
type	string	Type

In addition, common rule occurrence attribute `type` provides the possible causes for the violation.

Attribute	Value	Description
type	BLOCK_NO_CONSTRAINT	Missing constraint at block level
	TOP_NO_CONSTRAINT	Missing constraint at top level
	DIFFERENT_CONSTRAINTS	Inconsistent constraints between top level and block level

EXAMPLE

- For the figure below, the block level constraints file has this following constraint. The Litmus issues a warning if the top level constraints file does not have a `set_annotated_delay` specified for the net timing arc from `i_blk/g4/Z`.



Block level constraint (instance scope: `i_blk`)

```
set_annotated_delay 1 -net -from [get_pins g4/Z]
```

- For the above figure, the top and the block level constraints files have the following constraints, respectively, where the net timing arc from `i_blk/g4/Z` gets annotated delay value of 2 at top level. The Litmus issues a warning because the net timing arc gets annotated delay value of 1 at block level.

Top level constraint

```
set_annotated_delay 2 -net -from [get_pins i_blk/g4/Z]
```

Block level constraint (instance scope: `i_blk`)

```
set_annotated_delay 1 -net -from [get_pins g4/Z]
```

RELATED TOPICS

Commands

`set_annotated_delay`

`check_target_constraints`
`report_rule_check`

Attributes

`Rule Attributes - rule_occurrence`

CONST_COMP_UNC

- [CONST_COMP_UNC1](#)

CONST_COMP_UNC1

NAME

CONST_COMP_UNC1 (CONSTRAINT_COMPARISON)

MESSAGE

Inconsistent `set_clock_uncertainty` at the reference vs. target

DEFAULT SEVERITY

Warning

DESCRIPTION

Indicates that an inconsistency in `set_clock_uncertainty` command is found between reference and target level constraints. The following are the possible causes for this violation:

- `set_clock_uncertainty` command specified at either reference or target level
- `set_clock_uncertainty` commands specified at both reference and target level but with different values

This rule is checked by `compare_constraints` command and the results can be reported by `report_rule_check` command.

This rule has specific rule occurrence attributes for the details about the violation.

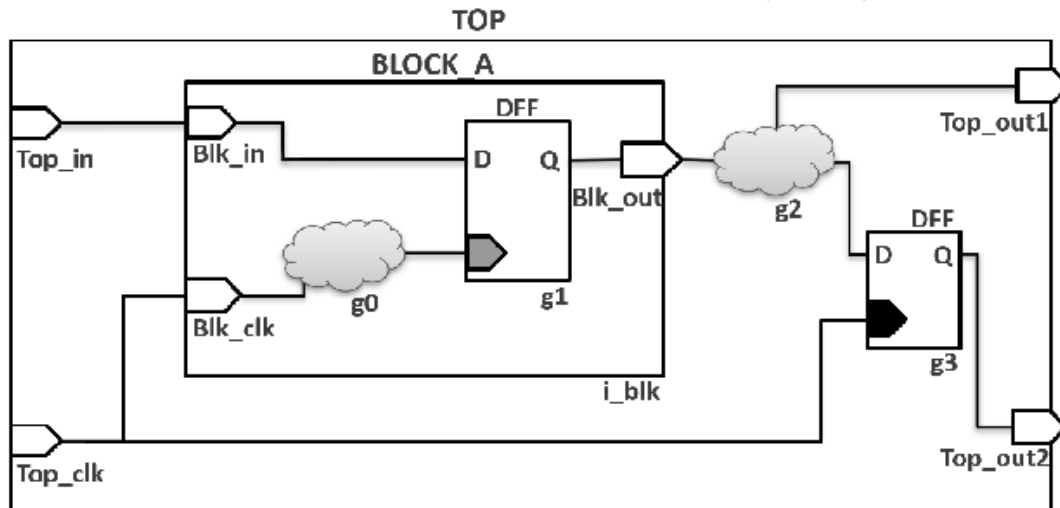
Attribute	Type	Description
occurrence_id	unsigned	Occurrence id
target_view_name	string	Target View Name
reference_view_name	string	Reference View Name

Litmus Rule Reference
Timing Constraint Comparison (1D2S)--CONST_COMP_UNC

scope	object	Target Instance
target_pin	object	Target Pin
reference_pin	object	Reference Pin
target_clock	object	Target clock
reference_clock	object	Reference clock
equivalent_target_clocks	object_list	Target clock
relative_reference_clocks	object_list	Reference clock
target_file_line	object_list	Target file number
target_file_name	object_list	Target file name
reference_file_line	object_list	Reference file number
reference_file_name	object_list	Reference file name

EXAMPLE

- For the figure below, the following constraints contain `set_clock_uncertainty` commands specified in both reference and target level constraints. This rule is reported for the `set_clock_uncertainty` commands, because they are specified with different values.



Top level constraints

```
create_clock -name Top_clk -period 2 [get_ports Top_clk]
set_clock_uncertainty 0.2 [get_clocks Top_clk]
```

Block level constraints (instance scope: i_blk)

```
create_clock -name Blk_clk -period 2 [get_ports Blk_clk]
set_clock_uncertainty 0.4 [get_clocks Blk_clk]
```

Hierarchical clock mapping

```
Top_clk is mapped to Blk_clk
```

Litmus Rule Reference

Timing Constraint Comparison (1D2S)--CONST_COMP_UNC

Constraint Coverage Signoff

- [CONST_COV_RR](#)
- [CONST_COV_CG](#)
- [CONST_COV_CP](#)
- [CONST_COV_CS](#)
- [CONST_COV_DATA](#)
- [CONST_COV_SH](#)
- [CONST_COV_PW](#)
- [CONST_COV_SKEW](#)

CONST_COV_RR

NAME

CONST_COV_RR (CONSTRAINT_COVERAGE_SIGNOFF)

MESSAGE

A pin is untested for recovery, removal or both timing checks

DEFAULT SEVERITY

Warning

DESCRIPTION

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence ID
scope	string	Scope
pin	object	Untested pin
reference_pin	object	Untested clock/reference pin
check_type	object_list	Check type
type	object_list	Reason for untested pin
command	object_list	command
constant_source_pin	object_list	constant_source_pin
constant_source_pin_value	object_list	constant_source_pin_value
constant_type	object_list	constant_type
when_condition	string	when_condition
constant_value	object_list	constant_value
blocking_points	object_list	blocking_points
bbox_cells	object_list	bbox_cells
start_pins	object_list	start_pins
start_ports	object_list	start_ports
launch_capture_clocks	object_list	launch_capture_clocks
constraint_cmd	object_list	constraint_cmd
file_lines	object_list	file_lines
file_names	object_list	file_names
ltms_diagnosis_path	object_list	ltms_diagnosis_path

EXAMPLES

CONST_COV_RR: A pin is untested for recovery, removal or both timing checks
Severity: Warning Occurrence: 21.Pin is untested for recovery
timing check due to no endpoint clock presentUntested Pin:
emu_ss_I0/usbd_dft_inst/bdkeyin_reg_cdr/datareg_reg_46_/CLRZ2.Pin is untested for
removal timing check due to constant on data pathUntested pin:
U0_dftss/I_ippseudostatic_ctrl_wrap/SE_gated_pipeline_mem_5_reg/CLRZ

CONST_COV_CG

NAME

CONST_COV_CG (CONSTRAINT_COVERAGE_SIGNOFF)

MESSAGE

A pin is untested for clock gating setup, clock gating hold or clock gating pulse width timing checks.

DEFAULT SEVERITY

Warning

DESCRIPTION

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence ID
scope	string	Scope
pin	object	Untested pin
reference_pin	object	Untested clock/reference pin
check_type	object_list	Check type
type	object_list	Reason for untested pin
command	object_list	command
constant_source_pin	object_list	constant_source_pin
constant_source_pin_value	object_list	constant_source_pin_value
constant_type	object_list	constant_type

when_condition	string	when_condition
constant_value	object_list	constant_value
blocking_points	object_list	blocking_points
bbox_cells	object_list	bbox_cells
start_pins	object_list	start_pins
start_ports	object_list	start_ports
launch_capture_clocks	object_list	launch_capture_clocks
constraint_cmd	object_list	constraint_cmd
file_lines	object_list	file_lines
file_names	object_list	file_names
ltms_diagnosis_path	object_list	ltms_diagnosis_path

EXAMPLES

CONST_COV_CP

NAME

CONST_COV_CP (CONSTRAINT_COVERAGE_SIGNOFF)

MESSAGE

A pin is untested for clock period timing checks.

DEFAULT SEVERITY

Warning

DESCRIPTION

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence ID
scope	string	Scope
pin	object	Untested pin
reference_pin	object	Untested clock/reference pin
check_type	object_list	Check type
type	object_list	Reason for untested pin
command	object_list	command
constant_source_pin	object_list	constant_source_pin
constant_source_pin_value	object_list	constant_source_pin_value
constant_type	object_list	constant_type

when_condition	string	when_condition
constant_value	object_list	constant_value
blocking_points	object_list	blocking_points
bbox_cells	object_list	bbox_cells
start_pins	object_list	start_pins
start_ports	object_list	start_ports
launch_capture_clocks	object_list	launch_capture_clocks
constraint_cmd	object_list	constraint_cmd
file_lines	object_list	file_lines
file_names	object_list	file_names
ltms_diagnosis_path	object_list	ltms_diagnosis_path

EXAMPLES

CONST_COV_CS

NAME

CONST_COV_CS (CONSTRAINT_COVERAGE_SIGNOFF)

MESSAGE

A pin is untested for clock separation timing checks.

DEFAULT SEVERITY

Warning

DESCRIPTION

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence ID
scope	string	Scope
pin	object	Untested pin
reference_pin	object	Untested clock/reference pin
check_type	object_list	Check type
type	object_list	Reason for untested pin
command	object_list	command
constant_source_pin	object_list	constant_source_pin
constant_source_pin_value	object_list	constant_source_pin_value
constant_type	object_list	constant_type

when_condition	string	when_condition
constant_value	object_list	constant_value
blocking_points	object_list	blocking_points
bbox_cells	object_list	bbox_cells
start_pins	object_list	start_pins
start_ports	object_list	start_ports
launch_capture_clocks	object_list	launch_capture_clocks
constraint_cmd	object_list	constraint_cmd
file_lines	object_list	file_lines
file_names	object_list	file_names
ltms_diagnosis_path	object_list	ltms_diagnosis_path

EXAMPLES

CONST_COV_DATA

NAME

CONST_COV_DATA (CONSTRAINT_COVERAGE_SIGNOFF)

MESSAGE

A pin is untested for data setup, data hold or both timing checks.

DEFAULT SEVERITY

Warning

DESCRIPTION

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence ID
scope	string	Scope
pin	object	Untested pin
reference_pin	object	Untested clock/reference pin
check_type	object_list	Check type
type	object_list	Reason for untested pin
command	object_list	command
constant_source_pin	object_list	constant_source_pin
constant_source_pin_value	object_list	constant_source_pin_value
constant_type	object_list	constant_type

when_condition	string	when_condition
constant_value	object_list	constant_value
blocking_points	object_list	blocking_points
bbox_cells	object_list	bbox_cells
start_pins	object_list	start_pins
start_ports	object_list	start_ports
launch_capture_clocks	object_list	launch_capture_clocks
constraint_cmd	object_list	constraint_cmd
file_lines	object_list	file_lines
file_names	object_list	file_names
ltms_diagnosis_path	object_list	ltms_diagnosis_path

EXAMPLES

CONST_COV_SH

NAME

CONST_COV_SH (CONSTRAINT_COVERAGE_SIGNOFF)

MESSAGE

A pin is untested for setup, hold or both timing checks.

DEFAULT SEVERITY

Warning

DESCRIPTION

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence ID
scope	string	Scope
pin	object	Untested pin
reference_pin	object	Untested clock/reference pin
check_type	object_list	Check type
type	object_list	Reason for untested pin
command	object_list	command
constant_source_pin	object_list	constant_source_pin
constant_source_pin_value	object_list	constant_source_pin_value
constant_type	object_list	constant_type

when_condition	string	when_condition
constant_value	object_list	constant_value
blocking_points	object_list	blocking_points
bbox_cells	object_list	bbox_cells
start_pins	object_list	start_pins
start_ports	object_list	start_ports
launch_capture_clocks	object_list	launch_capture_clocks
constraint_cmd	object_list	constraint_cmd
file_lines	object_list	file_lines
file_names	object_list	file_names
ltms_diagnosis_path	object_list	ltms_diagnosis_path

EXAMPLES

CONST_COV_SH: A pin is untested for setup, hold or both timing checks Severity: Warning
Occurrence: 21.Fail: Pin is untested for setup and hold timing check due to direct_false_path and clock_asyncUntested Pin:
U0_dftss/I_clock_obs/I_clkobs_freq_logic/UFREQ/COUNT_reg_13_/SD2.Fail: Pin is untested for hold timing check due to asynchronous clocksoUntested pin:
U0_dftss/I_clock_obs/I_clkobs_freq_logic/UFREQ/COUNT_reg_13_/SD

CONST_COV_PW

NAME

CONST_COV_PW (CONSTRAINT_COVERAGE_SIGNOFF)

MESSAGE

A pin is untested for pulse width timing checks.

DEFAULT SEVERITY

Warning

DESCRIPTION

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence ID
scope	string	Scope
pin	object	Untested pin
reference_pin	object	Untested clock/reference pin
check_type	object_list	Check type
type	object_list	Reason for untested pin
command	object_list	command
constant_source_pin	object_list	constant_source_pin
constant_source_pin_value	object_list	constant_source_pin_value
constant_type	object_list	constant_type

when_condition	string	when_condition
constant_value	object_list	constant_value
blocking_points	object_list	blocking_points
bbox_cells	object_list	bbox_cells
start_pins	object_list	start_pins
start_ports	object_list	start_ports
launch_capture_clocks	object_list	launch_capture_clocks
constraint_cmd	object_list	constraint_cmd
file_lines	object_list	file_lines
file_names	object_list	file_names
ltms_diagnosis_path	object_list	ltms_diagnosis_path

EXAMPLES

CONST_COV_SKEW

NAME

CONST_COV_SKEW (CONSTRAINT_COVERAGE_SIGNOFF)

MESSAGE

A pin is untested for skew timing checks.

DEFAULT SEVERITY

Warning

DESCRIPTION

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence ID
scope	string	Scope
pin	object	Untested pin
reference_pin	object	Untested clock/reference pin
check_type	object_list	Check type
type	object_list	Reason for untested pin
command	object_list	command
constant_source_pin	object_list	constant_source_pin
constant_source_pin_value	object_list	constant_source_pin_value
constant_type	object_list	constant_type

when_condition	string	when_condition
constant_value	object_list	constant_value
blocking_points	object_list	blocking_points
bbox_cells	object_list	bbox_cells
start_pins	object_list	start_pins
start_ports	object_list	start_ports
launch_capture_clocks	object_list	launch_capture_clocks
constraint_cmd	object_list	constraint_cmd
file_lines	object_list	file_lines
file_names	object_list	file_names
ltms_diagnosis_path	object_list	ltms_diagnosis_path

EXAMPLES

Constraint Enhanced Signoff

- [CONST_CG_CHK5](#)
- [CONST_CG_CHK6](#)
- [CONST_CG_CHK7](#)
- [CONST_DAT_CHK5](#)
- [CONST_DGN_TIM2](#)
- [CONST_EXC_CLK2](#)

CONST_CG_CHK5

NAME

CONST_CG_CHK5 (CONSTRAINT_ENHANCED_SIGNOFF)

MESSAGE

Clock gating enable pin is not reached by any constrained data

DEFAULT SEVERITY

Warning

DESCRIPTION

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence ID
file_lines	object_list	file_lines
file_names	object_list	file_names
blocking_commands	object_list	blocking_commands
clock_gating_enable_pin	object	Clock gating enable pin
reference_pin	object_list	reference_pins
type	string	Type

In addition, common rule occurrence attribute `type` provides the possible causes for the violation.

Attribute	Value	Description
type	NO_CLOCK	No clock reaching the data pin
	DATA_BLOCKED	Data phase blocked by “set_disable_timing” or “set_clock_groups” command

Checking: Leverage `check_timing -check_only clock_gating_no_data`. Thereafter all the multiple-input gates on a clock network are analyzed.

Reporting (message & attribute): Need to include the reason why no data phase reaches clock gating enable pin. For example, the fanin of clock gating enable pin is not constrained by SID w.r.t a clock. Hence data phase is missing. SDT stops the data phase in the fanin of clock gating enable pin. Hence data phase is blocked.

EXAMPLE

CONST_CG_CHK5: Clock gating enable pin is not reached by any constrained dataSeverity: Warning Occurrence: 11.Fail: No data signal reaching clock gating enable pin. Pin: common_digital/uh984_efuse_top_0/fuse_farm_0/I_sc_efuse_ctlr/I_efuse_ctlr_clkgen/g28/b1

CONST_CG_CHK6

NAME

CONST_CG_CHK6 (CONSTRAINT_ENHANCED_SIGNOFF)

MESSAGE

Clock gating causes a clipped clock

DEFAULT SEVERITY

Warning

DESCRIPTION

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence ID
file_lines	object_list	file_lines
file_names	object_list	file_names
pin	object	Pin
clock	string	Clock definition
side_clock	string	side_clock
gate	string	gate
input_pin	string	input_pin
command	string	command
type	object_list	type

Checking: Leverage `check_timing -check_only clock_clipping_gate`.

Reporting (message & attribute): Need to include the frequency and waveform of clock and data signals.

Message: Clock clipping due to wrong gate type or wrong trigger for input data.

EXAMPLE

```
CONST_CG_CHK6: Clock gating causes a clipped clock      Severity:
Warning          Occurrence: 11.Fail: Clock clipping due to wrong gate
('nand4_f2_s_svt') or wrong trigger for input data.
Pin:common_digital/uh98x_des_abuf_top_u1/uh98x_ser_abuf_ctrl_11_uh98x_ser_abuf_ctrl_u/g
24553/y      Clock: 'fpd4tx0_fclk_div2'   Input Pin:
'common_digital/uh98x_des_abuf_top_u1/uh98x_ser_abuf_ctrl_11_uh98x_ser_abuf_ctrl_u/g245
53/a
```

CONST_CG_CHK7

NAME

CONST_CG_CHK7 (CONSTRAINT_ENHANCED_SIGNOFF)

MESSAGE

Clock gating frequencies causes a clipped clock

DEFAULT SEVERITY

Warning

DESCRIPTION

Indicates clock gating frequencies causes a clipped clock

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence Id
file_lines	object_list	file_lines
file_names	object_list	file_names
pin	object	Pin
clock	string	clock
clock_pin	string	clock_pin
data_pin	string	data_pin
data_clock_name	string	data_clock_name

Checking: Leverage `check_timing -check_only clock_clipping_freq`.

Reporting (message & attribute): Need to include the frequency and waveform of clock and data

signals.

Message: Clock clipping due to incompatible clock signal and data signal frequencies.

EXAMPLE

```
CONST_CG_CHK7: Clock gating frequency causes a clipped clock          Severity:
Warning                      Occurrence: 11.Fail: Clock clipping due to incompatible
clock signal ('fpd3tx0_pclk0_div2_p0' ) and data signal ('fpd3tx0_fclk0') frequencies.
Pin: common_digital/fpd3_tx_datapath_u0/fpd3_clocks_0/pclk_mux/g63/y    Clock signal
pin: common_digital/fpd3_tx_datapath_u0/fpd3_clocks_0/pclk_mux/g63/b      Data
signal pin: common_digital/fpd3_tx_datapath_u0/fpd3_clocks_0/pclk_mux/g63/a
```

CONST_DAT_CHK5

NAME

CONST_DAT_CHK5 (CONSTRAINT_ENHANCED_SIGNOFF)

MESSAGE

Unconstrained internal end point is constrained by upstream delay path exception

DEFAULT SEVERITY

Warning

DESCRIPTION

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence ID
file_lines	object_list	file_lines
file_names	object_list	file_names
pin	object	Pin
max_delay_command	string	set_max_delay command

Checking: Leverage `check_timing -check_only max_delay_constrained_internal_endpoint`.

Reporting (message & attribute): Need to include the unconstrained internal end point and the SMD.

CONST_DGN_TIM2

NAME

CONST_DGN_TIM2 (CONSTRAINT_ENHANCED_SIGNOFF)

MESSAGE

Connected latches work on the same clock

DEFAULT SEVERITY

Warning

DESCRIPTION

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence ID
file_lines	object_list	file_lines
file_names	object_list	file_names
clock	object_list	Clock definition
pin	object	Pin
first_latch	string	Latch information
second_latch	string	Fanout latch

Checking: Leverage `check_timing -check_only latch_fanout_clock_connectivity`.

Reporting (message & attribute): Need to include the connected latches and the clock.

Message: Latch and its fanout wok on the same clock.

EXAMPLE

CONST_DGN_TIM2: Connected latches work on the same clock Severity:
Warning Occurrence:11.Fail: Latch and its fanout work on the
same clock. first_latch: tibox_criLatDRN
second_latch:svt/pd_svt_sw/hsm/I_eip130/u_eip130_tcm/u_eip130_engines_0/u_eip130_eip77/
u_eip130_eip77_crng/noiseTop_NO_SCAN/noiseSource/noiseRingb0c1/bit1/dt_latM/tibox_criLa
tDRN/Q

CONST_EXC_CLK2

NAME

CONST_EXC_CLK2 (CONSTRAINT_ENHANCED_SIGNOFF)

MESSAGE

Clock interactions between multiple clock domains

DEFAULT SEVERITY

Warning

DESCRIPTION

This rule has specific rule occurrence attributes for the details about the violation.

Attribute	Type	Description
occurrence_id	unsigned	Occurrence ID
file_lines	object_list	file_lines
file_names	object_list	file_names
from_clock	object	From clock source/definition
from_file_line	unsigned	File Line
from_file_name	string	File Name
to_clock	object	To clock source/definition
to_file_line	unsigned	File Line
to_file_name	string	File Name
is_asynchronous	bool	Indicates if the clocks are asynchronous

is_physically_exclusive	bool	Indicates if the clocks are physically exclusive
is_logically_exclusive	bool	Indicates if the clocks are logically exclusive
type	object_list	Type
command	object_list	Applied command (set_false_path or set_clock_groups)

In addition, common rule occurrence attribute `type` provides the possible causes for the violation.

Attribute	Value	Description
type	TRUE	Clock crossing is true path
	FALSE	Clock crossing is false path
	PARTIALLY_FALSE	Clock crossing is partially false path

Leverage `check_timing -check_only clock_crossing`.

EXAMPLE

CONST_EXC_CLK2: Clock interactions between multiple clock domains. Severity:
 Warning Occurrence: 11. Fail: Clock interactions between
 (dp0tx2_clk_qp_dsc1) and (dp1tx3_clk_qp_dsc1) which are
 (asynchronous/physically_exclusive/logically_exclusive).

